The Collected Mathematical Papers of Arthur Cayley
Vol. V

Arthur Cayley

Modern LaTeX reconstruction

MATHEMATICAL PAPERS.

London: C. J. CLAY and SONS,
CAMBRIDGE UNIVERSITY PRESS WAREHOUSE,
AVE MARIA LANE.

Cambridge: DEIGHTON, BELL AND CO.
Leipzig: F. A. BROCKHAUS.
New York: MACMILLAN AND CO.

THE COLLECTED MATHEMATICAL PAPERS

OF

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VOL. V.

CAMBRIDGE:
AT THE UNIVERSITY PRESS.

1892

CAMBRIDGE:
PRINTED BY C. J. CLAY, M.A. AND SONS,
AT THE UNIVERSITY PRESS.

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THE present volume contains 84 papers numbered 300 to 383 published for the most part in the years 1861 to 1866: No. 378, Report on Catalogue of Philosophical Memoirs, was however published in the British Association Report for 1856, and No. 379, Notices of Communications to the British Association, were published in the British Association Reports, 1854 to 1864: the concluding Paper 383, Problems and Solutions, contains problems for the most part geometrical ones proposed or solved by me in the Educational Times in the years 1863 to 1865.

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300.

NOTE RELATIVE AUX DROITES EN INVOLUTION DE M. SYLVESTER.

[From the *Comptes Rendus de l'Académie des Sciences de Paris*, t. LII. (Janvier—Juin, 1861), pp. 1039—1042.]

La courbe cubique dans l'espace, représentée par les Équations

$$yu - z^2 = 0, \quad zy - xu = 0, \quad xz - y^2 = 0,$$

passe par le point $A(x = y = z = 0)$ et le point $B(y = z = u = 0)$; le plan $x = 0$ est le plan osculant en A, le plan $y = 0$ le plan par la tangente en A et la droite AB; le plan $z = 0$ celui par la droite AB et la tangente en B; et enfin le plan $u = 0$ est le plan osculant en B. Réciproquement, pour une courbe cubique quelconque, en prenant les points A, B, sur la courbe à volonté, et en fixant comme ci-dessus les significations des coordonnées x, y, z, u, les facteurs constants que contiennent implicitement ces valeurs étant convenablement déterminés, les équations de la courbe cubique seront

$$yu - z^2 = 0, \quad zy - xu = 0, \quad xz - y^2 = 0.$$

Par un point quelconque de l'espace il passe une droite qui coupe deux fois la courbe cubique; et en prenant (x_1, y_1, z_1, u_1) pour les coordonnées du point dont il s'agit, et en écrivant

$$p_1 = y_1u_1 - z_1^2, \quad q_1 = z_1y_1 - x_1u_1, \quad r_1 = x_1z_1 - y_1^2,$$

les équations de la droite seront

$$p_1x + q_1y + r_1z = 0, \quad p_1y + q_1z + r_1u = 0.$$

Or, en considérant en général une droite représentée par les équations

$$\alpha x + \beta y + \gamma z + \delta u = 0, \quad \alpha' x + \beta' y + \gamma' z + \delta' u = 0,$$

les six quantités

$$\beta\gamma' - \beta'\gamma, \quad \gamma\alpha' - \gamma'\alpha, \quad \alpha\beta' - \alpha'\beta, \quad \alpha\delta' - \alpha'\delta, \quad \beta\delta' - \beta'\delta, \quad \gamma\delta' - \gamma'\delta,$$

sont ce que je nomine les *coordonnées* de la droite (en représentant par a, b, c, f, g, h ces coordonnées, on a l'équation identique $af + bg + ch = 0$, et les coordonnées d'une droite peuvent être des quantités quelconques qui satisfont à cette Équation). La condition pour l'involution de six droites est celle-ci, savoir: le déterminant formé avec les coordonnées des six droites est égal à zéro.

Je reviens à la droite qui coupe deux fois la courbe cubique. En écrivant les Équations sous la forme

$$p_1x + q_1y + r_1z = 0, \quad -p_1u + q_1x + r_1y = 0,$$

les coordonnées de cette droite seront

$$p_1^2, \quad q_1^2 - p_1r_1, \quad -p_1q_1, \quad p_1r_1, \quad q_1r_1, \quad r_1^2,$$

savoir, ces coordonnées seront des fonctions linéaires de $(p_1^2, q_1^2, r_1^2, q_1r_1, r_1p_1, p_1q_1)$.

Donc, en considérant six droites dont chacune coupe deux fois la courbe cubique, et en attribuant des significations analogues à (p_1, q_1, r_1), &c., la condition pour l'involution des six droites se trouve en égalant à zéro le déterminant dont les lignes sont

$$(p_1^2, q_1^2, r_1^2, q_1r_1, r_1p_1, p_1q_1), \quad (p_2^2, q_2^2, \text{etc.}), \quad \text{etc.};$$

condition qui exprime que les six droites

$$p_ix + q_iy + r_iz = 0,$$

dans le plan $u = 0$ (ou si l'on veut les six droites $p_iy + q_iz + r_iu = 0$ dans le plan $x = 0$) touchent une même conique. Or la droite

$$p_1x + q_1y + r_1z = 0$$

est la projection de l'une des six droites sur le plan osculant $u = 0$, avec le point $x = y = z = 0$ de la courbe cubique comme centre de projection; et si, en prenant un plan osculant quelconque et un point quelconque de la courbe cubique pour plan et centre de projection, nous appelons tout simplement *projection* une telle projection d'une droite quelconque (le plan osculant et le point de la cubique étant toujours les mêmes), on est conduit au théorème que voici, savoir:

Six droites dont chacune coupe deux fois la même courbe cubique seront en involution, si les projections de ces droites touchent une même conique.

Et de même, pour un nombre quelconque de droites, si les projections touchent une même conique, ces droites seront en involution, c'est-à-dire six quelconques des droites seront des droites en involution.

Il convient de remarquer qu'en considérant six droites quelconques, on peut en général trouver une courbe cubique coupée deux fois par chacune des droites: la condition du théorème est donc, comme cela doit être, une seule relation entre les six droites. Je remarque aussi que cette relation ne dépend nullement du plan osculant ni du point de la courbe cubique.

Réciproquement, en prenant dans un plan osculant quelconque de la courbe cubique un nombre quelconque (six ou plus) de tangentes d'une même conique, et en reprojetant ces tangentes sur la courbe cubique au moyen d'un point quelconque de la courbe comme centre de projection (de manière à obtenir pour reprojection de chaque tangente une droite qui coupe deux fois la courbe cubique), on obtient un système de droites en involution. Le lieu des droites dont chacune coupe deux fois la courbe cubique, et qui sont en involution, est une surface réglée du quatrième ordre qui a la courbe cubique pour courbe double. En effet, si l'équation en coordonnées tangentielles de la conique enveloppée par les droites $p_ix + q_iy + r_iz = 0$, &c. (ou, si l'on veut, par les droites $p_iy + q_iz + r_iu = 0$, &c.), est

$$(a, b, c, f, g, h \check{\rceil} p, q, r)^2 = 0,$$

cette même Équation, en y considérant p, q, r comme dénotant $yu - z^2$, $zy - xu$, $xz - y^2$, autrement dit, l'Équation,

$$(a, b, c, f, g, h \check{\rceil} yu - z^2,\ zy - xu,\ xz - y^2)^2 = 0,$$

sera celle d'une surface du quatrième ordre ayant la courbe cubique pour courbe double. Et cette surface sera une surface réglée; car en menant par un point quelconque de la surface une droite qui coupe deux fois la courbe cubique, chaque point d'intersection avec la courbe cubique doit compter pour deux points d'intersection avec la surface, et la droite coupe la surface en cinq points, c'est-à-dire que cette droite est située entièrement dans la surface.

J'ai remarqué ailleurs (Camb. and Dubl. Math. Journ., t. VII. (1852), p. 172, [107]) qu'il y a sur une surface réglée de l'ordre n une courbe double rencontrée par chaque génératrice en $(n-2)$ points. Cette courbe double sera de l'ordre $(n-2)$ au moins, et de l'ordre $\frac{1}{2}(n-1)(n-2)$ au plus; donc, pour $n = 4$, la courbe double sera de l'ordre 2 ou 3, et comme évidemment cette courbe n'est pas une courbe plane, elle sera: ou 1° deux droites qui ne se rencontrent pas; ou 2° une courbe cubique en espace. Cette seconde espèce des surfaces réglées du quatrième ordre est celle qui se présente dans la théorie des droites en involution.

* * *

301.

SUR LES CÔNES DU SECOND ORDRE QUI PASSENT PAR SIX POINTS DONNÉS.

[From the *Comptes Rendus de l'Académie des Sciences de Paris*, t. LII. (Janvier—Juin, 1861), pp. 1216—1218.]

Dans un Mémoire par feu M. Weddle "On the theorems in space analogous to those of Pascal and Brianchon in a plane" (Camb. and Dubl. Math. Journ., t. V. 1850, voir la Note p. 69), on trouve à propos d'un théorème de M. Chasles la remarque que le lieu du sommet d'un cône du second ordre qui passe par six points donnés est une surface du quatrième ordre qui contient la courbe cubique en espace par les six points. Voici comment je démontre ce théorème:

En prenant (X, Y, Z, U) pour les coordonnées courantes, $(\alpha_1, \beta_1, \gamma_1, \delta_1) \ldots (\alpha_6, \beta_6, \gamma_6, \delta_6)$ pour les coordonnées des six points donnés, et (x, y, z, u) pour ceux du sommet, je pose l'équation

$$\begin{vmatrix} X^2 & Y^2 & Z^2 & U^2 & YZ & ZX & XY & XU & YU & ZU \\ 2x & \cdot & \cdot & \cdot & z & \cdot & y & u & \cdot & \cdot \\ \cdot & 2y & \cdot & \cdot & z & \cdot & x & \cdot & u & \cdot \\ \cdot & \cdot & 2z & \cdot & y & x & \cdot & \cdot & \cdot & u \\ \cdot & \cdot & \cdot & 2u & \cdot & \cdot & \cdot & x & y & z \\ \alpha^2 & \beta^2 & \gamma^2 & \delta^2 & \beta\gamma & \gamma\alpha & \alpha\beta & \alpha\delta & \beta\delta & \gamma\delta \end{vmatrix} = 0,$$

où la dernière ligne dénote les six lignes qu'on obtient en écrivant successivement $(\alpha_1, \beta_1, \gamma_1, \delta_1) \ldots (\alpha_6, \beta_6, \gamma_6, \delta_6)$ au lieu de $(\alpha, \beta, \gamma, \delta)$, de manière que la fonction au côté gauche est un déterminant de l'ordre onze: les coefficients λ, μ, ν, ρ sont des quantités arbitraires et les points $(\cdot)$ dénotent des zéros.

Cette Équation est évidemment celle d'une surface du second ordre qui passe par les six points, et il ne faut qu'une seule condition pour que cette surface soit un cône: la condition sera

$$\begin{vmatrix} 2x & \cdot & \cdot & \cdot & z & \cdot & y & u & \cdot & \cdot \\ \cdot & 2y & \cdot & \cdot & z & \cdot & x & \cdot & u & \cdot \\ \cdot & \cdot & 2z & \cdot & y & x & \cdot & \cdot & \cdot & u \\ \cdot & \cdot & \cdot & 2u & \cdot & \cdot & \cdot & x & y & z \\ \alpha^2 & \beta^2 & \gamma^2 & \delta^2 & \beta\gamma & \gamma\alpha & \alpha\beta & \alpha\delta & \beta\delta & \gamma\delta \end{vmatrix} = 0,$$

ou la fonction à côté gauche est de même un déterminant de l'ordre dix; cette Équation, laquelle est de l'ordre quatre par rapport à (x, y, z, u), sera celle du lieu du sommet.

En effet, pour que la surface du second ordre soit un cône ayant pour sommet le point (x, y, z, u), il faut et il suffit que les équations dérivées par rapport à chacune des coordonnées (X, Y, Z, U), soient satisfaites en y écrivant (x, y, z, u) au lieu de (X, Y, Z, U); or en prenant la dérivée par rapport à X, on a les termes

$$\lambda(Xx + Zu + Yy) + \ldots,$$

et en y écrivant $X = x$, $Y = y$, $Z = z$, $U = u$, les termes entre parenthèses deviennent $2x$, z, y, u, savoir les termes de la première ligne du déterminant de l'ordre dix: ce déterminant sera donc zéro.

Cela étant, on aura

$$\frac{dD}{d\alpha_i} = \frac{dD}{d\beta_i} = \frac{dD}{d\gamma_i} = \frac{dD}{d\delta_i} = 0$$

pour l'un quelconque des points $(\alpha_i, \beta_i, \gamma_i, \delta_i), \ldots (\alpha_6, \beta_6, \gamma_6, \delta_6)$; et de là, au moyen des propriétés des déterminants, et en écrivant

$$D = 4(yu - z^2)(xz - y^2) - (zy - xu)^2,$$

on exprime l'équation de la surface comme fonction linéaire par rapport à x, y, z, u et par rapport à $\partial_x D$, $\partial_y D$, $\partial_z D$, $\partial_u D$; ces dernières fonctions se réduisent à zéro en vertu des équations

$$yu - z^2 = 0, \quad zy - xu = 0, \quad xz - y^2 = 0,$$

et ainsi, comme cela doit être, la surface passe par la courbe cubique.

Je prends l'occasion de remarquer que le théorème que j'ai donné par rapport aux six droites en involution de M. Sylvester [300], peut s'exprimer dans une forme encore plus simple comme suit:

Soit donnée une courbe cubique en espace, et prenons un point quelconque de la courbe pour sommet d'un cône du second ordre, d'ailleurs arbitraire; un plan tangent du cône rencontre la courbe en deux points, et par ces deux points on peut mener une droite: les droites qui correspondent de cette manière à six plans tangents quelconques du cône sont des droites en involution. Je dois remarquer que l'idée de rattacher ces droites à une surface du quatrième ordre est due à M. Sylvester.

À propos de ce sujet, j'ai considéré le problème de trouver le lieu du sommet d'un cône du second ordre qui touche à six droites données: ce lieu est une surface du huitième ordre; et en représentant les coordonnées de l'une quelconque des droites par (a, b, c, f, g, h), savoir les coordonnées de la première droite, &c., sont

$$(a_1, b_1, c_1, f_1, g_1, h_1), \ldots (a_6, b_6, c_6, f_6, g_6, h_6),$$

les coefficients de l'équation seront des fonctions linéaires des déterminants du sixième ordre formés au moyen de la matrice (a, b, c, f, g, h), à six lignes et vingt et une colonnes.

302.

CONSIDÉRATIONS GÉNÉRALES SUR LES COURBES EN ESPACE.

[From the *Comptes Rendus de l'Académie des Sciences de Paris*, t. LIV. (Janvier—Juin, 1862), pp. 55—60, 396—400, 672—678.]

Soit une courbe donnée du $m^{\text{ième}}$ ordre; je suppose toujours que cette courbe soit une courbe propre, savoir qu'elle n'est pas composée de courbes d'ordres inférieurs.

Si nous prenons pour sommet d'un cône qui passe par la courbe un point A quelconque qui n'est pas sur la courbe, ce cône sera de l'ordre m; cela est vrai en général quelle que soit la courbe; seulement si m est un nombre composé, alors pour de certaines courbes il peut y avoir des positions de A pour lesquelles le cône sera d'un ordre sous-multiple de m; mais en faisant abstraction de ces positions particulières, le cône sera de l'ordre m. Et, cela étant, une droite du cône ne contiendra en général qu'un seul point de la courbe. En employant quatre coordonnées (x, y, z, w) et en supposant qu'au point A on ait

$$x = 0, \quad y = 0, \quad z = 0,$$

l'équation du cône sera $U = 0$, où U est une fonction homogène de (x, y, z) de l'ordre m. On peut faire passer par la courbe une surface ayant pour Équation

$$Qw - P = 0 \quad \left[\text{ou } w = \frac{P}{Q}\right],$$

où P, Q sont des fonctions homogènes de (x, y, z) des ordres p, $p-1$ respectivement. Et on peut supposer que p soit égal tout au plus à $m-1$: en effet, en prenant $p = m-1$, l'équation contiendrait

$$\tfrac{1}{2}(m-1)m + m(m+1) - 1,$$

c'est-à-dire $m^2 - 1$ constantes arbitraires; et en déterminant convenablement $m^2 - m + 1$ de ces quantités, la surface de l'ordre $m-1$ passera par $m^2 - m + 1$ points de la courbe de l'ordre m, c'est-à-dire cette surface contiendra la courbe entière. De cette manière, on obtiendrait toujours une surface de l'ordre $m-1$; mais si les fonctions P, Q ainsi trouvées avaient un facteur commun, ce facteur devrait être écarté; il convient donc de supposer que les degrés de P, Q soient p, $p-1$ respectivement, p étant tout au plus égal à $m-1$. La surface $Qw-P=0$ a au point A un point conique du $(p-1)^{\text{ième}}$ ordre; en effet dans le voisinage de ce point l'équation se réduit à $Q=0$, laquelle appartient à un cône du $(p-1)^{\text{ième}}$ ordre. J'ajoute que la surface contient les $p(p-1)$ droites $P=0$, $Q=0$ qui passent chacune par le point A; toute autre droite par ce point rencontre la surface dans ce point (lequel compte pour $p-1$ points d'intersection) et encore dans un seul point donné par l'équation

$$w=\frac{P}{Q}.$$

On peut appeler *monoïde* une telle surface; le point A sera le *sommet*; le cône $P=0$ le cône supérieur; le cône $Q=0$, le cône inférieur; les droites d'intersection de ces deux cônes, les *droites de la monoïde.*

Or le cône circonscrit $U=0$ et la monoïde $Qw-P=0$ se coupent selon une courbe de l'ordre mp: si $p=1$, cette intersection des deux surfaces sera la courbe du $m^{\text{ième}}$ ordre, laquelle sera une courbe plane; mais, dans tout autre cas, la courbe d'intersection sera composée de la courbe du $m^{\text{ième}}$ ordre, et d'un autre système de l'ordre $m(p-1)$; ce système ne peut être autre chose que les droites d'intersection du cône circonscrit $U=0$, et du cône inférieur $Q=0$ de la monoïde; c'est-à-dire les Équations

$$U=0, \quad Q=0$$

doivent donner $P=0$; car, cela étant, les droites $U=0$, $Q=0$ seront situées sur la monoïde; et ces droites, lesquelles forment un système de l'ordre $m(p-1)$, seront partie de l'intersection de la monoïde et du cône circonscrit $U=0$. Et il est nécessaire que cela soit ainsi, car autrement chaque droite du cône $U=0$ ne contiendrait sur la monoïde que le point A, et le point déterminé par l'équation $w=\frac{P}{Q}$, lequel est un point sur la courbe du $m^{\text{ième}}$ ordre; donc cette autre partie de l'intersection de la monoïde et du cône $U=0$ serait, non pas une courbe quelconque, mais le seul point A; ce qui est absurde.

Le cône circonscrit $U=0$ ne peut pas être un cône quelconque à moins que $p=1$; en effet si $p>1$, il est nécessaire que le cône ait au moins $(p-1)m$ droites doubles (en comprenant dans cette locution le cas où le cône a des singularités qui équivalent à $(p-1)m$ droites doubles), car en supposant pour un moment que le cône $U = 0$ n'ait pas de singularités, le cône $P = 0$ de l'ordre p rencontre le cône $Q = 0$ de l'ordre $p - 1$ selon $p(p-1)$ droites, et le cône $Q = 0$ rencontre le cône $U = 0$ de l'ordre m selon $m(p-1)$ droites; de même le cône $P = 0$ rencontre le cône $U = 0$ selon mp droites; mais parmi ces mp droites il y a les m droites d'intersection du cône $Q = 0$ avec le cône $U = 0$; il ne reste donc que $m(p-1)$ droites d'intersection du cône $P = 0$ avec le cône $U = 0$, et ces $m(p-1)$ droites doivent être les droites d'intersection du cône $Q = 0$ avec le cône $U = 0$: c'est-à-dire les $m(p-1)$ droites d'intersection des cônes $Q = 0$, $U = 0$ doivent être situées sur le cône $P = 0$.

Je puis sans diminuer la généralité supposer que le cône $Q = 0$ passe par un certain nombre χ de droites doubles du cône $U = 0$; mais en faisant cette supposition, chacune de ces droites compte pour deux intersections des cônes $Q = 0$, $U = 0$; il y a encore $(p-1)m - 2\chi$ droites d'intersection; et les $\chi+[(p-1)m-2\chi]$, c'est-à-dire $(p-1)m-\chi$ droites peuvent être comprises parmi les $p(p-1)$ droites de la monoïde si χ est égal au moins à $(p-1)(m-p)$; c'est-à-dire le cône $U = 0$ doit avoir au moins ce nombre de droites doubles. Je remarque que pour m impair, et $p = \frac{1}{2}(m+1)$, le nombre sera $\frac{1}{4}(m^2 - 2m + 1)$, et pour m pair, et $p = \frac{1}{2}m$ ou $\frac{1}{2}m+1$, le nombre sera $\frac{1}{4}(m^2 - 2m)$; mais pour toute autre valeur de p, le nombre sera moins élevé.

Je résume comme suit:

Toute courbe du $m^{\text{ième}}$ *ordre est l'intersection d'un cône circonscrit* $U = 0$, *du* $m^{\text{ième}}$ *ordre, et d'une surface monoïde* $Qw - P$, *de l'ordre* $p = m - 1$ *au plus. L'intersection complète de deux surfaces est composée de la courbe du* $m^{\text{ième}}$ *ordre et des* $m(p-1)$ *droites d'intersection du cône circonscrit* $U = 0$, *et du cône inférieur* $Q = 0$ *de la monoïde. Ces droites seront* $(p-1)(m-p)+\alpha$ *droites, chacune répétée deux fois, et* $(p-1)(2p-m)-2\alpha$ *droites, où* α *peut être égal à zéro; chacune des* $(p-1)(m-p)+\alpha$ *droites sera une droite double du cône* $U = 0$*; et les* $(p-1)(m-p)+\alpha$ *droites et* $(p-1)(2p-m)-2\alpha$ *droites, ensemble* $p(p-1)-\alpha$ *droites, seront situées sur le cône supérieur* $P = 0$ *de la monoïde.*

Il y a deux circonstances qui empêchent que cette théorie ne conduise tout de suite à une classification des courbes en espace. D'abord, une droite double du cône $U = 0$ peut correspondre ou à un point double réel, ou à un point double apparent de la courbe; et de même en supposant que la droite double devienne une droite de rebroussement, cela peut correspondre à un point de rebroussement réel ou à un point de rebroussement apparent. Et ensuite il peut y avoir sur la courbe des points doubles ou de rebroussement qui ne correspondent pas à des droites doubles du cône circonscrit; car en représentant la courbe au moyen du cône circonscrit et d'une surface monoïde, un point double apparent de la courbe correspond à une droite double du cône, c'est-à-dire que deux branches de la courbe qui passent par le point double sont dans un même plan avec le sommet du cône; au lieu qu'un point double réel correspond à deux branches qui ne sont pas dans un même plan avec le sommet. Un point de rebroussement correspond toujours à une droite de rebroussement du cône; mais la droite de rebroussement peut ne pas être une droite double du cône (ce qui arrive lorsque le plan osculateur au point de rebroussement contient le sommet).

Je vais appliquer la théorie à l'énumération des points doubles et points de rebroussement d'une courbe de l'ordre m. Une courbe de l'ordre m est en général de la classe $m(m-1)$; car la classe est le nombre des tangentes par un point quelconque, et par un point quelconque on peut mener $m(m-1)$ tangentes à la surface polaire du point par rapport au cône circonscrit; or ces $m(m-1)$ tangentes sont les tangentes à la courbe par ce point. Mais supposons que la courbe ait δ points doubles et κ points de rebroussement; alors la classe est $n = m(m-1) - 2\delta - 3\kappa$: le cône circonscrit a $(p-1)(m-p)+\alpha$ droites doubles qui correspondent aux points doubles apparents, et $m(p-1) - 2(p-1)(m-p) - 2\alpha$, c'est-à-dire $(p-1)(3p-2m)+2\alpha$ droites de rebroussement qui correspondent aux points de rebroussement apparents. Cela étant, on a l'équation

$$m(m-1)-2\delta-3\kappa = m(m-1)-2[(p-1)(m-p)+\alpha]-3[(p-1)(3p-2m)+2\alpha]+\beta,$$

où β représente le nombre des tangentes qui passent par le sommet; car en ce qui regarde le sommet, chaque tangente par le sommet est comptée dans la détermination de la classe comme tangente double du cône circonscrit: ainsi les $m(m-1)$ tangentes par un point quelconque se réduisent au nombre $m(m-1)-2\delta-3\kappa$; mais parmi ces tangentes il y en a β qui passent par le sommet, et chacune de ces β tangentes est comptée pour deux; on a donc

$$n = m(m-1)-2\delta-3\kappa = m(m-1)-2[\text{droites doubles}]-3[\text{droites de rebroussement}]+$$

De plus, on a pour le nombre des points doubles apparents D et le nombre des points de rebroussement apparents R:

$$D = \tfrac{1}{2}(m-1)(m-2) - \delta - \kappa, \quad R = 3(m-2) - 6\delta - 8\kappa,$$

et de là

$$D - R = \tfrac{1}{2}(m^2 - 7m + 14) + 5\delta + 7\kappa.$$

Mais il convient d'abord de remarquer que dans le cas général d'une courbe du $m^{\text{ième}}$ ordre qui a δ points doubles et κ points de rebroussement, le cône circonscrit au moyen d'un point quelconque a $(p-1)(m-p)+\alpha$ droites doubles et $(p-1)(3p-2m)+2\alpha$ droites de rebroussement; le nombre des points doubles apparents est $\frac{1}{2}(m-1)(m-2)-\delta-\kappa$, et celui des points de rebroussement apparents est $3(m-2)-6\delta-8\kappa$. Ces nombres doivent être tous deux positifs ou nuls, ce qui donne des limites pour δ et κ.

En particulier, pour les courbes du quatrième ordre, $m = 4$, on a

$$D = 3 - \delta - \kappa, \quad R = 6 - 6\delta - 8\kappa.$$

Les valeurs possibles sont:

	δ	κ	D	R
Courbe plane	3	0	0	6
Courbe quadriquadrique	0	0	3	6
Courbe excubo-quartique	0	0	3	6

Mais pour la courbe excubo-quartique, qui est l'intersection de deux surfaces quadriques, on a une théorie plus simple. En effet, soit $U = 0$, $V = 0$ les équations des deux surfaces quadriques; la courbe est de l'ordre 4; elle a trois points doubles apparents. En effet, un plan quelconque coupe les deux surfaces selon deux coniques qui se rencontrent en quatre points; mais parmi ces quatre points il y a les points d'intersection de la courbe avec le plan; donc la courbe est de l'ordre 4. Le cône circonscrit d'un point quelconque coupe chaque surface quadrique selon deux droites; donc les quatre droites d'intersection du cône avec les deux surfaces sont les génératrices du cône qui passent par les points de la courbe; le cône a trois droites doubles qui correspondent aux trois points doubles apparents de la courbe.

Pour les courbes du cinquième ordre, $m = 5$, on a

$$D = 6 - \delta - \kappa, \quad R = 9 - 6\delta - 8\kappa.$$

Je vais à présent expliquer la théorie des cinq espèces de courbes quintiques.

Courbe plane, ou espèce 5. — Il va sans dire que cette courbe est l'intersection d'une surface quintique par un plan quelconque.

Courbe quadricubique, ou espèce $6 - 1$. — Cette courbe est l'intersection partielle d'une surface quadrique et d'une surface cubique qui ont en commun une droite. Soit $U = xw - yz = 0$ l'équation de la surface quadrique; on peut prendre pour l'équation de la surface cubique

$$V = \begin{vmatrix} x, & y \\ z, & w \end{vmatrix} \begin{pmatrix} a, & b \\ c, & d \end{pmatrix} \begin{pmatrix} x, & z \\ y, & w \end{pmatrix} = 0,$$

en représentant de cette manière la fonction $axz + byz + cxw + dyw$, linéaire par rapport à x, y et par rapport à z, w, avec des coefficients a, b, c, d, lesquels sont des fonctions linéaires quelconques de x, y, z, w.

En écrivant d'abord

$$V = (ax + by)z + (cx + dy)w,$$

et prenant

$$wV - (cx + dy)U = z[ax^2 + (b + c)xy + dy^2],$$

on voit que la surface cubique $wV - (cx + dy)U = 0$ (laquelle passe par la courbe quadriquadrique $U = 0$, $V = 0$) a en commun avec la surface quadrique la droite $z = 0$, $w = 0$ et la courbe quadriquadrique; de plus, en prenant un plan quelconque $wV - (cx + dy)U = 0$ qui passe par la droite $z = 0$, $w = 0$, ce plan coupe la surface quadrique selon la droite $z = 0$, $w = 0$ et une autre droite; et ces deux droites avec la courbe quadriquadrique forment l'intersection complète du plan et de la surface quadrique. La courbe quintique est donc l'intersection de la surface quadrique par un plan passant par une génératrice de la surface quadrique, moins cette génératrice: c'est la courbe $6 - 1$.

La courbe a 4 points doubles apparents; elle peut donc avoir 0, 1 ou 2 points doubles ou de rebroussement; cela donne les sous-espèces

$$V.1, \quad V.2, \quad V.3, \quad V.4, \quad V.5, \quad V.6,$$

de M. Salmon.

Je remarque en passant qu'en supposant que la surface cubique $wV - yV' = 0$ a en commun avec la surface quadrique $xw - yz = 0$, non-seulement la droite $x = 0$, $y = 0$, mais aussi une autre génératrice du même mode de génération, on aura, au lieu de la courbe quintique $6 - 1$, cette nouvelle droite, et une courbe excubo-quartique. C'est là le théorème qui donne une des constructions que M. Chasles a trouvées pour la courbe excubo-quartique.

J'ajoute que la courbe considérée comme courbe située sur une surface quadrique sera de l'espèce $(3, 2)$, ou, selon la notation de M. Chasles, $M(a^2\beta^2)$. On connaît ainsi un grand nombre des propriétés de cette courbe, et aussi de la courbe d'espèce $8 - 3$ dont nous allons parler, laquelle, considérée comme courbe située sur une surface quadrique, est de l'espèce $(4, 1)$ ou $M(a\epsilon^2)$.

Courbe quadriquartique, ou espèce $8 - 3$. — Une telle courbe est l'intersection partielle d'une surface quadrique et d'une surface quartique qui ont en commun trois droites qui ne se rencontrent pas: autrement dit, ces droites seront des génératrices du même mode de génération de la surface quadrique.

Soit $xw - yz = 0$ l'équation de la surface quadrique; on peut prendre pour les trois génératrices

$$(x-\lambda y = 0, \quad \lambda w-z = 0), \quad (x-\mu y = 0, \quad \mu w-z = 0), \quad (x-\nu y = 0, \quad \nu w-z = 0).$$

La surface quartique qui passe par ces trois droites peut être représentée sous une forme convenable; et la courbe quintique $8-3$ sera l'intersection résiduelle de la surface quadrique et de cette surface quartique.

La courbe a 6 points doubles apparents; il n'y a donc pas d'autre singularité: c'est l'espèce

$$V.10$$

de M. Salmon.

Courbe cubicubique, espèce $9-3-1$. — La courbe est l'intersection partielle de deux surfaces cubiques qui ont en commun une courbe cubique gauche et une droite qui ne rencontre pas la courbe cubique.

Soient p, q, r, s, t, u, P, Q des fonctions linéaires quelconques des coordonnées; α, β, γ, α', β', γ' des fonctions linéaires quelconques de P, Q (autrement dit, $\alpha=0$, $\beta=0$, &c., seront les équations de six plans quelconques qui passent par la droite $P=0$, $Q=0$).

Cela étant, les surfaces cubiques

$$\begin{vmatrix} p, & s, & \alpha \\ q, & t, & \beta \\ r, & u, & \gamma \end{vmatrix} = 0, \qquad \begin{vmatrix} p, & s, & \alpha' \\ q, & t, & \beta' \\ r, & u, & \gamma' \end{vmatrix} = 0,$$

ont en commun la courbe cubique

$$\begin{vmatrix} p, & s \\ q, & t \\ r, & u \end{vmatrix} = 0, \qquad \begin{vmatrix} p, & s, & \alpha \\ q, & t, & \beta \\ r, & u, & \gamma \end{vmatrix} = 0$$

(ainsi les surfaces quadriques $pt-sq=0$, $pu-sr=0$ se rencontrent selon la droite $p=0$, $s=0$ et selon la courbe cubique dont il s'agit) et la droite $P=0$, $Q=0$. Il y aura donc encore une intersection qui sera la courbe quintique $9-3-1$.

Or, en prenant α, β, γ, α', β', γ' des fonctions linéaires de P et Q, nous avons, en effet, réduit la courbe sextique à la droite $P = 0$, $Q = 0$ et à la courbe quintique $9 - 3 - 1$.

Courbe cubicubique, espèce $9-6+2$. — Cette courbe est l'intersection partielle de deux surfaces cubiques qui ont en commun une courbe excubo-quartique. En supposant que cette courbe excubo-quartique soit l'intersection partielle d'une surface quadrique et d'une surface cubique qui ont en commun les deux droites $(x = 0, y = 0)$ et $(z = 0, w = 0)$, on peut prendre pour équation de ces deux surfaces

$$U = xw - yz = 0,$$

$$V = \begin{vmatrix} x, & y \\ z, & w \end{vmatrix} \begin{pmatrix} a, & b \\ c, & d \end{pmatrix} \begin{pmatrix} x, & z \\ y, & w \end{pmatrix} = 0,$$

en représentant de cette manière la fonction $axz + byz + cxw + dyw$, linéaire par rapport à x, y et par rapport à z, w, avec des coefficients a, b, c, d, lesquels sont des fonctions linéaires quelconques de x, y, z, w.

En écrivant d'abord

$$\begin{aligned} V &= (ax + by)z + (cx + dy)w, \\ U &= \qquad\quad - yz + \qquad\quad xw, \end{aligned}$$

on obtient

$$wV - (cx + dy)U = z[ax^2 + (b + c)xy + dy^2].$$

Et de même en écrivant

$$\begin{aligned} V &= (ax + cw)x + (bz + dw)y, \\ U &= \qquad\quad wx - \qquad\quad zy, \end{aligned}$$

on obtient

$$zV + (bz + dw)U = x[az^2 + (b + c)zw + dw^2].$$

Or le premier de ces résultats fait voir qu'en supposant $U = 0$, $V = 0$, on obtient $ax^2 + (b + c)xy + dy^2 = 0$, et le second, qu'en supposant $U = 0$, $V = 0$, on obtient de même $az^2 + (b + c)zw + dw^2 = 0$. Les surfaces $U = 0$, $V = 0$ se coupent selon la courbe excubo-quartique et les droites $(x = 0, y = 0)$ et $(z = 0, w = 0)$; mais la surface $ax^2 + (b + c)xy + dy^2 = 0$ ne passe que par la première, et la surface $az^2 + (b + c)zw + dw^2 = 0$ ne passe que par la seconde de ces droites.

La courbe a cinq points doubles apparents; elle peut donc ne pas avoir d'autre singularité, ou avoir un point double ou de rebroussement: cela donne les trois sous-espèces

$$V.7, \quad V.8, \quad V.9$$

de M. Salmon.

On démontre sans peine que toute courbe quintique est plane, quadricubique, quadriquartique ou cubicubique; mais, pour faire voir qu'il n'existe que les cinq espèces ci-dessus mentionnées, il y a encore plusieurs cas à considérer. Par exemple, pour les courbes cubicubiques, on pourrait supposer que les deux surfaces cubiques avaient en commun une courbe quadriquadrique: si cela était, les Équations des deux surfaces seraient de la forme $Vx - Uy = 0$, $Vz - Uw = 0$ (surfaces qui ont en commun la courbe quadriquadrique $U = 0$, $V = 0$), mais dans ce cas la courbe quintique serait située sur la surface quadrique $xw - yz = 0$, et l'on ne fait que retrouver l'espèce quadricubique $6 - 1$. J'ai fait, après M. Salmon, cette revue des différents cas, et je me suis assuré qu'il n'y a que les cinq espèces. Il convient peut-être de remarquer que l'énumération des sous-espèces diffère un peu de celle de M. Salmon; car il y a des sous-espèces qui n'existent pas, à savoir les sous-espèces quadriquadriques analogues à V.7, V.8, V.9 (p. 42, où M. Salmon parle des courbes algébriques correspondantes à V.1, V.8, V.9, V.10, sans attacher des numéros à ces quatre sous-espèces).

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303.

SUR LE PROBLÈME DU POLYGONE INSCRIT ET CIRCONSCRIT.

LETTRE À M. PONCELET.

[From the *Comptes Rendus de l'Académie des Sciences de Paris*, t. LV. (Juillet—Décembre, 1862), pp. 700, 701.]

J'ose vous écrire par rapport aux remarques que vous faites p. 483 de l'ouvrage (*Applications d'Analyse etc.* [Paris, t. I. (1862)]) au sujet de mes recherches sur le problème du polygone inscrit et circonscrit [267 and the papers 113, 115, 116 and 128 therein referred to].

Je n'ai nullement voulu attribuer à Fuss le théorème pour les deux cercles. J'ai seulement dit, tout à fait en passant: "The case... of the two circles (the original case of the Porism as considered by Fuss)" et en effet Fuss a fait des recherches sur ce cas d'un polygone inscrit et circonscrit à deux cercles. Mais je n'ai jamais imaginé qu'il y eût un géomètre (algébriste ou non) qui ne connût pas tant votre ouvrage classique de 1822, que le mémoire de 1828 de Jacobi, où l'on voit précisément ce que Fuss a fait sur ce problème. Par rapport à mon dernier Mémoire (Phil. Trans. 1861) [267], que vous citez et qui résume quelques Notes que j'ai publiées en 1853, permettez-moi de vous mentionner la forme de ma solution: on a une fonction

$$a + b\xi + c\xi^2 + d\xi^3,$$

où ξ est une quantité indéterminée, et a, b, c, d sont des fonctions données très simples des paramètres qui déterminent les deux cercles (ou coniques). On développe la racine carrée de cette fonction dans la forme $A + B\xi + C\xi^2 + D\xi^3 + E\xi^4 + \ldots$, et, cela fait, on a tout de suite l'équation entre les paramètres pour un polygone d'ordre quelconque; savoir pour le triangle, le pentagone, l'heptagone, etc., ces conditions sont

$$C = 0, \quad \begin{vmatrix} C, & D \\ D, & E \end{vmatrix} = 0, \quad \begin{vmatrix} C, & D, & E \\ D, & E, & F \\ E, & F, & G \end{vmatrix} = 0, \quad \text{etc.},$$

tandis que pour le quadrangle, l'hexagone, l'octogone, etc., ces conditions sont

$$D = 0, \quad \begin{vmatrix} D, & E \\ E, & F \end{vmatrix} = 0, \quad \begin{vmatrix} D, & E, & F \\ E, & F, & G \\ F, & G, & H \end{vmatrix} = 0, \quad \text{etc.},$$

de manière que la condition est trouvée explicitement pour un polygone d'ordre quelconque sans passer par celles qui appartiennent aux polygones d'ordre inférieur.

Comme j'attache, je l'avoue, un peu d'importance à cette solution (laquelle selon l'explication que je viens de donner ne paraît pas mériter la critique que vous en faites) je serais bien aise si vous voulez bien communiquer cette lettre à l'Académie.

304.

SUR UN MÉMOIRE DE JACOBI.

EXTRAIT D'UNE LETTRE À M. J. BERTRAND.

[From the *Comptes Rendus de l'Académie des Sciences de Paris*, t. LVI. (Janvier—Juin, 1863), p. 43. See 298, foot-note to No. 117.]

Permettez-moi de vous soumettre une remarque que je viens de faire par rapport au Mémoire de Jacobi "Sur l'élimination des noeuds dans le problème des trois corps" (Compte Rendu 8 août, 1842 [and Crelle, t. XXVI. (1843), pp. 115—131]). Il me semble que quoique Jacobi dise qu'il a fait dépendre le problème d'un système de cinq équations du premier ordre et une seule du second ordre, il a réellement fait plus que cela, savoir qu'il l'a fait dépendre d'un système de six équations du premier ordre et qu'ainsi il est allé aussi loin que vous dans le "Mémoire sur l'intégration de quelques Équations différentielles de la Mécanique," Journal de M. Liouville, t. XVII. (1852). En effet si dans les équations i,…vi. de Jacobi, pour les réduire à un système d'équations du premier ordre, on écrit

$$\frac{di}{\eta} = \frac{di_1}{\eta_1} = \frac{du}{v} = \frac{du_1}{v_1} = \frac{dr}{\rho} = \frac{dr_1}{\rho_1} = \frac{d\theta}{\sigma} = dt,$$

le système peut évidemment se présenter sous la forme

$$\frac{di}{I} = \frac{di_1}{I_1} = \frac{du}{U} = \frac{du_1}{U_1} = \frac{dr}{R} = \frac{dr_1}{R_1} = \frac{d\theta}{\Theta} = dt,$$

et, cela étant, en remarquant que les fonctions I, I_1, U, …, ne contiennent pas t, et en omettant l'équation $(\Theta = d\theta)$, on a un système de six équations entre les quantités i, i_1, u, u_1, r, r_1, θ: en supposant que l'intégration soit effectuée, on obtient alors t au moyen d'une quadrature.

Je remarque en passant qu'il ne me paraît pas que Jacobi ait dû dire: "Par suite l'on a fait cinq intégrations;" les seules intégrations qu'il a faites sont: l'intégrale des forces vives, et les trois intégrales des aires: cela étant on obtient une quatrième équation (celle qui donne θ) au moyen d'une quadrature; mais ce n'est pas là ce qu'on entend par une intégration, et Jacobi, dans le passage cité, avait dû dire qu'on avait obtenu quatre intégrales seulement.

305.

CONSIDÉRATIONS GÉNÉRALES SUR LES COURBES EN ESPACE.

COURBES DU CINQUIÈME ORDRE.

[From the *Comptes Rendus de l'Académie des Sciences de Paris*, t. LVIII. (Janvier—Juin, 1864), pp. 994—1000. Continuation of 302.]

En considérant une courbe du $m^{\text{ième}}$ ordre représentée au moyen des Équations

$$U=0,\quad w=\frac{P}{Q},$$

qui dénotent respectivement un cône du $m^{\text{ième}}$ ordre et une surface monoïde du $p^{\text{ième}}$ ordre, le cône doit passer $m(p-1)$ fois par les $p(p-1)$ droites $(P=0, Q=0)$ de la monoïde. J'indique la manière de ce passage au moyen d'un symbole que je nomme la *signature* du système; ce symbole, composé ordinairement des numéros 2, 1, 0, ensemble $p(p-1)$ numéros, fait voir combien des $p(p-1)$ droites de la monoïde sont, par rapport au cône, des droites doubles, des droites simples, ou des droites qui ne sont pas situées sur le cône: par exemple, $m=5$, $p=3$, la signature 222211 fait voir qu'il y a quatre droites doubles, deux droites simples; la signature 222220, qu'il y a cinq droites doubles, une droite qui n'est pas située sur le cône.

Je reviens aux courbes du cinquième ordre; j'ai établi (t. LIV. p. 672) qu'il y a cinq espèces de ces courbes, à savoir:

		P.D.A.
Courbe plane ou espèce	5	
Courbe quadricubique	$6-1$	0
Courbe quadriquartique	$8-3$	4
Courbe cubicubique (deux espèces)	$9-3-1$	6
	$9-6+2$	6

où la colonne P. D. A. fait voir pour chaque espèce le nombre des points doubles apparents (voir le Mémoire de M. Salmon: "On the classification of Curves of double Curvature," Camb. et Dubl. Math. Journ., t. V. 1850). Cette classification est au fond celle du Mémoire cité; seulement M. Salmon a énuméré trois sous-espèces qui n'existent pas, à savoir les sous-espèces quadriquadriques analogues à V.7, V.8, V.9 (p. 42, où M. Salmon parle des courbes algébriques correspondantes à V.1, V.8, V.9, V.10, sans attacher des numéros à ces quatre sous-espèces). Je vais à présent expliquer la théorie des cinq espèces.

Courbe plane, ou espèce 5. — Il va sans dire que cette courbe est l'intersection d'une surface quintique par un plan quelconque.

Courbe quadricubique, ou espèce $6-1$. — Cette courbe est l'intersection partielle d'une surface quadrique et d'une surface cubique qui ont en commun une droite. Soit $U = xw - yz = 0$ l'équation de la surface quadrique; on peut prendre pour l'équation de la surface cubique

$$V = \begin{vmatrix} x, & y \\ z, & w \end{vmatrix} \begin{pmatrix} a, & b \\ c, & d \end{pmatrix} \begin{pmatrix} x, & z \\ y, & w \end{pmatrix} = 0,$$

en représentant de cette manière la fonction $axz + byz + cxw + dyw$, linéaire par rapport à x, y et par rapport à z, w, avec des coefficients a, b, c, d, lesquels sont des fonctions linéaires quelconques de x, y, z, w.

En écrivant d'abord

$$V = (ax + by)z + (cx + dy)w,$$

$$U = \qquad - yz + \qquad xw,$$

on obtient

$$wV - (cx + dy)U = z[ax^2 + (b + c)xy + dy^2].$$

Et de même en écrivant

$$V = (ax + cw)x + (bz + dw)y,$$

$$U = \qquad wx - \qquad zy,$$

on obtient

$$zV + (bz + dw)U = x[az^2 + (b + c)zw + dw^2].$$

Or le premier de ces résultats fait voir qu'en supposant $U = 0$, $V = 0$, on obtient $ax^2 + (b + c)xy + dy^2 = 0$, et le second, qu'en supposant $U = 0$, $V = 0$, on obtient de même $az^2 + (b + c)zw + dw^2 = 0$. Les surfaces $U = 0$, $V = 0$ se coupent selon la courbe excubo-quartique et les droites $(x = 0, y = 0)$ et $(z = 0, w = 0)$; mais la surface $ax^2 + (b + c)xy + dy^2 = 0$ ne passe que par la première, et la surface $az^2 + (b + c)zw + dw^2 = 0$ ne passe que par la seconde de ces deux droites. Cela fait voir qu'il passe par la courbe cette nouvelle surface cubique,

$$zV + vU = 0,$$

laquelle a en commun avec la première surface cubique la courbe quadriquadrique $U = 0$, $V = 0$.

La courbe a 4 points doubles apparents; elle peut donc avoir 0, 1 ou 2 points doubles ou de rebroussement; cela donne les sous-espèces

$$V.1, \quad V.2, \quad V.3, \quad V.4, \quad V.5, \quad V.6,$$

de M. Salmon.

Je remarque en passant qu'en supposant que la surface cubique $wV - yV' = 0$ a en commun avec la surface quadrique $xw - yz = 0$, non-seulement la droite $x = 0$, $y = 0$, mais aussi une autre génératrice du même mode de génération, on aura, au lieu de la courbe quintique $6 - 1$, cette nouvelle droite, et une courbe excubo-quartique. C'est là le théorème qui donne une des constructions que M. Chasles a trouvées pour la courbe excubo-quartique.

J'ajoute que la courbe considérée comme courbe située sur une surface quadrique sera de l'espèce $(3,2)$, ou, selon la notation de M. Chasles, $M(a^2\beta^2)$. On connaît ainsi un grand nombre des propriétés de cette courbe, et aussi de la courbe d'espèce $8-3$ dont nous allons parler, laquelle, considérée comme courbe située sur une surface quadrique, est de l'espèce $(4,1)$ ou $M(a\epsilon^2)$.

Courbe quadriquartique, ou espèce $8-3$. — Une telle courbe est l'intersection partielle d'une surface quadrique et d'une surface quartique qui ont en commun trois droites qui ne se rencontrent pas: autrement dit, ces droites seront des génératrices du même mode de génération de la surface quadrique.

Soit $xw-yz=0$ l'équation de la surface quadrique; on peut prendre pour les trois génératrices

$$(x-\lambda y=0,\quad \lambda w-z=0),\quad (x-\mu y=0,\quad \mu w-z=0),\quad (x-\nu y=0,\quad \nu w-z=0).$$

Les deux cônes $P = 0$, $Q = 0$ se coupent selon les six droites 7, 8, 9, $7'$, $8'$, $9'$, et selon six autres droites 1, 2, 3, 4, 5, 6: il suit de la théorie précédente (mais on peut aussi démontrer analytiquement) qu'il existe un cône quintique $U = 0$ qui satisfait aux conditions de passer deux fois par chacune des droites 1, 2, 3, 4, 5, 6 (avoir chacune de ces droites pour une droite double, 18 conditions) et une fois par chacune des droites 7, 8, 9 (3 conditions, en tout $18+3 = 21$ conditions). Et cela étant, on aura la courbe $8-3$ déterminée au moyen du cône $U = 0$ et la surface monoïde $w = \frac{P}{Q}$, à signature 222222111000 (c.à.d. les droites 1, 2, 3, 4, 5, 6 qui sont par rapport au cône des droites doubles, les droites 7, 8, 9 des droites simples, et les droites $7'$, $8'$, $9'$ des droites qui ne sont pas situées sur le cône). Le nombre des constantes est 21, mais au moyen de la transformation

$$w = \frac{P + \alpha P' + \beta P'' + \gamma P'''}{Q + \alpha Q' + \beta Q'' + \gamma Q'''}$$

on réduit comme auparavant ce nombre à $21 - 3 = 18$, ce qui est juste.

J'ajoute les considérations que voici: le cône $U = 0$ passe deux fois par chacune des droites 1, 2, 3, 4, 5, 6, une fois par chacune des droites 7, 8, 9. Soit $M = 0$ l'équation du système des trois plans qui contiennent les droites 7, $8'$, $9'$, les droites $7'$, 8, $9'$, et les droites $7'$, $8'$, 9 respectivement; le cône $M = 0$ contient chacune des droites 7, 8, 9 une fois, et chacune des droites $7'$, $8'$, $9'$ deux fois. Donc le cône $MU = 0$ contient chacune des droites 1, 2, 3, 4, 5, 6, 7, 8, 9, $7'$, $8'$, $9'$ deux fois; ces douze droites sont les droites d'intersection des cônes $P = 0$, $Q = 0$, et ainsi nous avons identiquement $MU = AP^2 + BPQ + CQ^2$, A, B, C étant des fonctions homogènes de (x, y, z) des ordres 0, 1, 2 respectivement. Cela étant, les équations

$$w = \frac{P}{Q}, \quad MU = AP^2 + BPQ + CQ^2 = 0$$

donnent

$$Aw^2 + Bw + C = 0,$$

Équation de la surface quadrique sur laquelle est située la courbe $8-3$.

Je passe à la théorie analytique. Soit, pour abréger,

$$\begin{array}{lll} \xi = by + cz, & X = \beta y + \gamma z, & \xi' = \lambda x + \mu y + \nu z, \\ \eta = a'x + c'z, & Y = \alpha' x + \gamma' z, & \\ \zeta = a''x + b''y, & Z = \alpha'' x + \beta'' y. & \end{array}$$

Je prends pour Équations des droites 7, 8, 9, 7′, 8′, 9′:

$$\begin{array}{lll} (\xi = 0,\ \zeta = 0), & (\eta = 0,\ \xi' = 0), & (\zeta = 0,\ \eta = 0), \\ (\xi = 0,\ \xi' = 0), & (\eta = 0,\ \eta' = 0), & (\zeta = 0,\ \zeta' = 0). \end{array}$$

et je forme les équations les plus générales pour le cône cubique et le cône quartique qui passent par ces droites; ces équations seront

$$Q = Z\zeta\xi\xi' + \Sigma X\eta\eta'\zeta' + \Sigma Y\zeta\zeta'\xi' + \Sigma Z\xi\xi'\eta' = 0,$$

$$-P = \Sigma\xi\eta' X + \Sigma\zeta\xi'\eta Y + \Sigma\eta\zeta'\xi Z + \Sigma\xi\eta\zeta\xi'\eta'\zeta' = 0.$$

On a de là la surface monoïde $w = \frac{P}{Q}$. En écrivant dans cette équation $x = 0$, on obtient $w = -\frac{\xi_0}{\eta_0}$; et de même, pour $y = 0$, on obtient $w = -\frac{\eta_0}{\zeta_0}$, et pour $z = 0$ on obtient $w = -\frac{\zeta_0}{\xi_0}$, c'est-à-dire qu'il y a sur la monoïde les trois transversales

$$(x = 0,\ Z + \xi' Y = 0), \quad (y = 0,\ Y + \zeta' X = 0), \quad (z = 0,\ Z + \xi' Y = 0),$$

ou, comme on peut écrire ces équations,

$$\begin{array}{l} (x = 0,\ \beta y + \gamma z + \xi'(\alpha'' x + \beta'' y) = 0), \\ (y = 0,\ \alpha' x + \gamma' z + \zeta'(\lambda x + \mu y + \nu z) = 0), \\ (z = 0,\ \alpha'' x + \beta'' y + \xi'(ax + by + cz) = 0). \end{array}$$

On trouve sans peine l'équation de la surface quadrique qui passe par les transversales; en écrivant, pour abréger,

$$\begin{aligned} A &= \beta\beta'\beta'', \\ B &= (\zeta'\alpha'' + \beta''\alpha')\xi x + (\zeta''\alpha + \zeta\alpha'')\beta y + \xi''(\beta\alpha' + \beta'\alpha)\beta'' z, \\ C &= \alpha\beta'\gamma''\xi^2 + \beta''\gamma\beta\xi'^2 + \gamma\nu'\xi'' z^2 + (\gamma\xi''\zeta' + \gamma'\beta\xi'')yz \\ &\quad + (\alpha'\gamma\xi'' + \alpha'\gamma\xi)\ zx + (\beta''\alpha'\xi + \beta\alpha''\zeta')\ xy, \end{aligned}$$

cette équation est

$$Aw^2 + Bw + C = 0,$$

et en eliminant w entre cette Équation et l'équation $w = \frac{P}{Q}$, on obtient l'équation

$$AP^2 + BPQ + CQ^2 = 0,$$

laquelle, en vertu de l'identité

$$AP^2 + BPQ + CQ^2 = xyz \cdot U,$$

se réduit à $U = 0$, équation d'un cône du cinquième ordre, ce qui donne le système $U = 0$, $w = \frac{P}{Q}$ de cône et monoïde à signature 222222111000. Pour démontrer l'identité dont il s'agit, il convient de remarquer qu'en substituant dans l'expression $AP^2 + BPQ + CQ^2$ les valeurs de P et Q, tous les termes contiennent explicitement le facteur xyz hormis les termes que voici:

$$A(\xi^2\zeta'^2X^2 + \zeta^2\xi'^2Y^2 + \eta^2\eta'^2Z^2),$$

$$-B(\xi\zeta'X\eta\xi'Y + \zeta\eta'Y\eta\zeta'Z + \eta\xi'Z\xi\eta'X),$$

et pour démontrer que ces termes exceptés contiennent aussi le facteur xyz, il suffit de faire voir que la fonction $AX^2 - BXB' + CB'^2$ contient le facteur x, car alors, par la symétrie, les fonctions $AY^2 - BYC' + CC'^2$ et $AZ^2 - BZC'' + CC''^2$ contiendront respectivement les facteurs y et z, et l'expression entière sera divisible par xyz. Mais en écrivant $x = 0$, on trouve

$$\begin{aligned} AX^2 &= \beta\beta'\beta''(\beta y + \gamma z)^2, \\ -BXB' &= -[B\beta'(\beta y + \gamma z) + B(\beta' B'' y + \gamma''\beta z)]\beta(\beta y + \gamma z) = 0, \\ CB'^2 &= 0, \end{aligned}$$

c'est-à-dire $AX^2 - BXB' + CB'^2$ contient le facteur x. Donc enfin

$$AP^2 + BPQ + CQ^2$$

contient le facteur xyz, ce qui était le théorème à démontrer.

* * *

310. Note on Mr Jerrard's Researches on the Equation of the Fifth Order.

[From the *Philosophical Magazine*, vol. xxi. (1861), pp. 348–350.]

The above property of rational expressibility, if true for W, will be true for any function homotypical with W; and conversely. I proceed to inquire into the form of the function W.

The function W is derived from the function P, which denotes any one of the quantities p_1, p_2, p_3. And if x_1, x_2, x_3, x_4, x_5 are the roots of the given equation of the fifth order, and if α, β, γ, δ, ϵ represent in an undetermined or arbitrary order of succession the five indices 1, 2, 3, 4, 5, and if t denote an imaginary fifth root of unity (I conform myself to Mr Jerrard's notation), then p_1, p_2, p_3, and the other auxiliary quantities t, u, are obtained from the system of equations:

$$\begin{aligned}
x_\alpha^3 + p_1 x_\alpha^2 + p_2 x_\alpha + p_3 &= t + u,\\
x_\beta^3 + p_1 x_\beta^2 + p_2 x_\beta + p_3 &= t^2 + t^4 u,\\
x_\gamma^3 + p_1 x_\gamma^2 + p_2 x_\gamma + p_3 &= t^3 + t^2 u,\\
x_\delta^3 + p_1 x_\delta^2 + p_2 x_\delta + p_3 &= t^4 + t u.
\end{aligned}$$

If from these equations we seek for the values of p_1, p_2, p_3, t, u, we have

$$1 : p_1 : p_2 : p_3 : -t : -u = \Pi_1 : \Pi_2 : \Pi_3 : \Pi_4 : \Pi_5 : \Pi_6,$$

where Π_1, Π_2, … denote the determinants formed out of the matrix

$$\begin{matrix}
x_\alpha^3, & x_\beta^3, & x_\gamma^3, & 1, & t, & t^5\\
x_\alpha^2, & x_\beta^2, & x_\gamma^2, & 1, & t^2, & t^4\\
x_\alpha, & x_\beta, & x_\gamma, & 1, & t^3, & t^3\\
x_\delta^3, & x_\epsilon^3, & x_\delta^2, & 1, & t^4, & t^2\\
x_\epsilon^2, & x_\delta, & x_\epsilon, & 1, & t^5, & t
\end{matrix}$$

i.e., denoting the columns of this matrix by 1, 2, 3, 4, 5, 6, we have $\Pi_1 = 23456$, $\Pi_2 = -34561$, $\Pi_3 = 45612$, &c. In particular, the value of Π_1 is

$$\begin{vmatrix}
x_\beta, & x_\gamma, & 1, & t, & t^5\\
x_\delta, & x_\epsilon, & 1, & t^2, & t^4\\
x_\delta^2, & x_\epsilon^2, & 1, & t^3, & t^3\\
x_\alpha, & x_\beta, & 1, & t^4, & t^2\\
x_\gamma, & x_\alpha, & 1, & t^5, & t
\end{vmatrix}$$

and developing, and putting for shortness $\{\alpha\beta\} = x_\alpha x_\beta(x_\alpha - x_\beta)$, &c., we have

$$\Pi_1 = (\{\alpha\beta\} + \{\beta\gamma\} + \{\gamma\delta\} + \{\delta\epsilon\} + \{\epsilon\alpha\})(1 + t + 2t^2 - 2t^3 - 2t^4);$$

and this is also the form of the other determinants, the only difference being as to the meaning of the symbol $\{\alpha\beta\}$, which, however, in each case denotes a function such that $\{\alpha\beta\} = -\{\beta\alpha\}$. Writing for greater shortness,

$$\{\alpha\beta\gamma\delta\epsilon\} = \{\alpha\beta\} + \{\beta\gamma\} + \{\gamma\delta\} + \{\delta\epsilon\} + \{\epsilon\alpha\},$$

and in like manner

$$\{\alpha\gamma\epsilon\beta\delta\} = \{\alpha\gamma\} + \{\gamma\epsilon\} + \{\epsilon\beta\} + \{\beta\delta\} + \{\delta\alpha\},$$

Π_1 is an unsymmetric linear function (without constant term) of $\{\alpha\beta\gamma\delta\epsilon\}$, $\{\alpha\gamma\epsilon\beta\delta\}$; or, what is all that is material, it is an unsymmetric function, containing only odd powers, of $\{\alpha\beta\gamma\delta\epsilon\}$, $\{\alpha\gamma\epsilon\beta\delta\}$.

If for

$$\alpha \quad \beta \quad \gamma \quad \delta \quad \epsilon$$

we substitute any one of the five arrangements

$$\begin{array}{ccccc}
\alpha & \beta & \gamma & \delta & \epsilon, \\
\beta & \gamma & \delta & \epsilon & \alpha, \\
\gamma & \delta & \epsilon & \alpha & \beta, \\
\delta & \epsilon & \alpha & \beta & \gamma, \\
\epsilon & \alpha & \beta & \gamma & \delta,
\end{array}$$

then $\{\alpha\beta\gamma\delta\epsilon\}$ and $\{\alpha\gamma\epsilon\beta\delta\}$ will in each case remain unaltered.

But if we substitute any one of the five arrangements

$$\begin{array}{ccccc}
\alpha & \epsilon & \delta & \gamma & \beta, \\
\epsilon & \delta & \gamma & \beta & \alpha, \\
\delta & \gamma & \beta & \alpha & \epsilon, \\
\gamma & \beta & \alpha & \epsilon & \delta, \\
\beta & \alpha & \epsilon & \delta & \gamma,
\end{array}$$

then in each case $\{\alpha\beta\gamma\delta\epsilon\}$ and $\{\alpha\gamma\epsilon\beta\delta\}$ will be changed into $-\{\alpha\beta\gamma\delta\epsilon\}$ and $-\{\alpha\gamma\epsilon\beta\delta\}$ respectively. Hence Π_1 remains unaltered by any one of the first five substitutions; and it is changed into $-\Pi_1$ by any one of the second five substitutions. And the like being the case as regards Π_2, &c., it follows that the quotient $\Pi_1 \div \Pi_2$, or say P, remains unaltered by any one of the ten substitutions. Now the 120 permutations of α, β, γ, δ, ϵ can be obtained as follows, viz. by forming the 12 different pentagons which can be formed with α, β, γ, δ, ϵ (treated as five points), and reading each of them off in either direction from any angle. To each of the 12 pentagons there corresponds a distinct value of P, but such value is not altered by the different modes of reading off the pentagon; P is consequently a 12-valued function.

But there is a more simple form of the analytical expression of such a 12-valued function; in fact, if $[\alpha\beta\gamma\delta\epsilon]$ be any function which is not altered by any one of the above ten substitutions—if, for instance, $[\alpha\beta]$ is a symmetrical function of x_α, x_β, and

$$[\alpha\beta\gamma\delta\epsilon] = [\alpha\beta] + [\beta\gamma] + [\gamma\delta] + [\delta\epsilon] + [\epsilon\alpha],$$

and therefore

$$[\alpha\gamma\epsilon\beta\delta] = [\alpha\gamma] + [\gamma\epsilon] + [\epsilon\beta] + [\beta\delta] + [\delta\alpha],$$

then any unsymmetrical function of $[\alpha\beta\gamma\delta\epsilon]$ and $[\alpha\gamma\epsilon\beta\delta]$ will be a 12-valued function homotypical with P.

Mr Jerrard's function W is the sum of two values of his function P; the substitution by which the second is derived from the first can only be that which interchanges the two functions $[\alpha\beta\gamma\delta\epsilon]$ and $[\alpha\gamma\epsilon\beta\delta]$; and hence any symmetrical function of $[\alpha\beta\gamma\delta\epsilon]$ and $[\alpha\gamma\epsilon\beta\delta]$ is a function homotypical with Mr Jerrard's W; such symmetric function is in fact a 6-valued function only. Indeed it is easy to see that the twelve pentagons correspond together in pairs, either pentagon of a pair being derived from the other one by stellation, and the six values of the function in question corresponding to the six pairs of pentagons respectively.

Writing with Mr Cockle and Mr Harley,

$$\tau = x_1x_2 + x_2x_3 + x_3x_4 + x_4x_5 + x_5x_1,$$
$$\tau' = x_1x_3 + x_3x_5 + x_5x_2 + x_2x_4 + x_4x_1,$$

then ($\tau + \tau'$ is a symmetrical function of all the roots, and it must be excluded; but) $(\tau - \tau')^2$ or $\tau\tau'$ are each of them 6-valued functions of the form in question, and either of these functions is linearly connected with the Resolvent Product. In Lagrange's general theory of the solution of equations, if

$$\Omega = x_1 + \omega x_2 + \omega^2 x_3 + \omega^3 x_4 + \omega^4 x_5,$$

then the coefficients of the equation the roots whereof are $(\Omega)^5$, $(\Omega^2)^5$, $(\Omega^3)^5$, $(\Omega^4)^5$, and in particular the last coefficient $(\Omega\Omega^2\Omega^3\Omega^4)^5$ is determined by an equation of the sixth degree; and this last coefficient is a perfect fifth power, and its fifth root, or $\Omega\Omega^2\Omega^3\Omega^4$, is the function just referred to as the Resolvent Product.

The conclusion from the foregoing remarks is that if the equation for W has the above property of the rational expressibility of its roots, the equation of the sixth order resulting from Lagrange's general theory has the same property.

I take the opportunity of adding a simple remark on cubic equations. The principle which furnishes what in a foregoing foot-note is called the *A priori* demonstration of Lagrange's theorem is that an equation need never contain extraneous roots; a quantity which has only one value will, if the investigation is properly conducted, be determined in the first instance by a linear equation; one which has two values by a quadratic equation, and so on; there is always enough, and not more than enough, to determine what is required.

Take Cardan's solution of the cubic equation $x^3 + qx - r = 0$, we have $x = a + b$, and thence $3ab = -q$, $a^3 + b^3 = r$; and to obtain the solution we write

$$a^3b^3 = -\frac{q^3}{27}, \quad a^3 + b^3 = r.$$

But these two equations are not enough to precisely determine x, they lead to the 9-valued function

$$\sqrt[3]{\frac{r}{2} + \sqrt{\frac{r^2}{4} + \frac{q^3}{27}}} + \sqrt[3]{\frac{r}{2} - \sqrt{\frac{r^2}{4} + \frac{q^3}{27}}};$$

in order to precisely determine x, it is (as everybody knows) necessary to use the original equation $ab = -\frac{q}{3}$. But seek for the solution as follows; viz. write $x = ab(a + b)$, which gives

$$3a^2b^2 = -q, \quad a^3b^3(a^3 + b^3) = r,$$

or what is the same thing,

$$a^3b^3 = -\frac{q}{3}, \quad a^3 + b^3 = -\frac{3r}{q};$$

these equations give $x = ab(a+b)$, where

$$a = \sqrt[3]{-\frac{3r}{2q} + \sqrt{\frac{9r^2}{4q^2} + \frac{q}{3}}}, \quad b = \sqrt[3]{-\frac{3r}{2q} - \sqrt{\frac{9r^2}{4q^2} + \frac{q}{3}}},$$

which is a 3-valued function only, ab in this case being not given.

2, Stone Buildings, W.C., January 28, 1861.

311. On a Theorem of Abel's Relating to Equations of the Fifth Order.

[From the *Philosophical Magazine*, vol. xxi. (1861), pp. 257–263.]

The following is given (Abel, Œuvres, vol. ii. p. 253 [Ed. 2, vol. II. p. 266]) as an extract of a letter to M. Crelle:

"Si une Equation du cinquieme degré, dont les coefficients sont des nombres rationnels, est resoluble algebriquement, on peut donner aux racines la forme suivante,

où

$$\begin{aligned} x &= c + A_1\sqrt{\Theta} + A_2\sqrt{\Theta_2} + A_3\sqrt{\Theta_3},\\ \Theta &= p + q\sqrt{R},\\ \Theta_2 &= p - q\sqrt{R},\\ \Theta_3 &= r + s\sqrt{\Theta} + t\sqrt{\Theta_2},\\ A_1 &= a + b\sqrt{\Theta_2} + (c + d\sqrt{\Theta_2})\sqrt{\Theta_3},\\ A_2 &= a - b\sqrt{\Theta_2} + (c - d\sqrt{\Theta_2})\sqrt{\Theta_3},\\ A_3 &= e + f\sqrt{\Theta_2}, \end{aligned}$$

R étant un nombre rationnel, p, q, r, s, t, a, b, c, d, e, f des nombres rationnels."

It is to be noticed that in the expressions for A, A_1, A_2, A_3, the radicals are such that

$$\sqrt{\Theta}\sqrt{p+q\sqrt{R}}\sqrt{p-q\sqrt{R}} = s, \text{ a rational number.}$$

The theorem is given as belonging to numerical equations; but considering it as belonging to literal equations, it will be convenient to change the notation; and in this point of view, and to avoid suffixes and accents, I write

$$\begin{aligned} \alpha &= m + n\sqrt{\Theta} + \sqrt{p + q\sqrt{\Theta}},\\ \beta &= m - n\sqrt{\Theta} + \sqrt{p - q\sqrt{\Theta}},\\ \gamma &= m + n\sqrt{\Theta} - \sqrt{p + q\sqrt{\Theta}},\\ \delta &= m - n\sqrt{\Theta} - \sqrt{p - q\sqrt{\Theta}}; \end{aligned}$$

the radicals being connected by

$$\sqrt{\Theta}\sqrt{p+q\sqrt{\Theta}}\sqrt{p-q\sqrt{\Theta}}=s,$$

and where

$$\begin{aligned}
A &= K+L\alpha+M\beta+N\alpha\beta,\\
B &= K+L\beta+M\gamma+N\beta\gamma,\\
C &= K+L\gamma+M\delta+N\gamma\delta,\\
D &= K+L\delta+M\alpha+N\delta\alpha,
\end{aligned}$$

in which equations Θ, m, n, p, q, s, K, L, M, N are rational functions of the elements of the given quintic equation.

The basis of the theorem is, that the expression for x has only the five values which it acquires by giving to the quintic radicals contained in it their five several values, and does not acquire any new value by substituting for the quadratic radicals their several values. For, this being so, x will be the root of a rational quintic; and conversely.

Now attending to the equation

$$\sqrt{\Theta}\sqrt{p+q\sqrt{\Theta}}\sqrt{p-q\sqrt{\Theta}}=s,$$

the different admissible values of the radicals are

$$\begin{array}{rcc}
\sqrt{\Theta}, & \sqrt{p+q\sqrt{\Theta}}, & \sqrt{p-q\sqrt{\Theta}},\\
-\sqrt{\Theta}, & \sqrt{p-q\sqrt{\Theta}}, & -\sqrt{p+q\sqrt{\Theta}},\\
\sqrt{\Theta}, & -\sqrt{p+q\sqrt{\Theta}}, & -\sqrt{p-q\sqrt{\Theta}},\\
-\sqrt{\Theta}, & -\sqrt{p-q\sqrt{\Theta}}, & \sqrt{p+q\sqrt{\Theta}},
\end{array}$$

corresponding to the systems

$$\begin{array}{cccc}
\alpha, & \beta, & \gamma, & \delta,\\
\beta, & \gamma, & \delta, & \alpha,\\
\gamma, & \delta, & \alpha, & \beta,\\
\delta, & \alpha, & \beta, & \gamma,
\end{array}$$

of the roots α, β, γ, δ; i.e. the effect of the alteration of the values of the quadratic radicals is merely to cyclically permute the roots α,

β, γ, δ; and observing that any such cyclical permutation gives rise to a like cyclical permutation of A, B, C, D, the alteration of the quadratic radicals produces no alteration in the expression for x.

The quantities α, β, γ, δ are the roots of a rational quartic. If, solving the quartic by Euler's method, we write

$$\begin{aligned}
\alpha &= m + \sqrt{F} + \sqrt{G} + \sqrt{H}, \quad \sqrt{F}\sqrt{G}\sqrt{H} = v, \text{ a rational function,}\\
\beta &= m - \sqrt{F} + \sqrt{G} - \sqrt{H},\\
\gamma &= m + \sqrt{F} - \sqrt{G} - \sqrt{H},\\
\delta &= m - \sqrt{F} - \sqrt{G} + \sqrt{H},
\end{aligned}$$

then the expressions for F, G, H in terms of the roots are

$$\tfrac{1}{4}(\alpha + \beta - \gamma - \delta)^2, \quad \tfrac{1}{4}(\alpha + \gamma - \beta - \delta)^2, \quad \tfrac{1}{4}(\beta + \delta - \beta - \gamma)^2,$$

which are the roots of a cubic equation

$$u^3 - \lambda u^2 + \mu u - \nu^2 = 0,$$

where λ, μ, ν are given rational functions of the coefficients of the quartic. We have so that, taking $\Theta = F$, the last-mentioned expressions for α, β, γ, δ will be of the assumed form

$$\alpha = m + \sqrt{\Theta} + \sqrt{p + q\sqrt{\Theta}}, \quad \&c.$$

The equation

$$\sqrt{\Theta}\sqrt{p + q\sqrt{\Theta}}\sqrt{p - q\sqrt{\Theta}} = s$$

thus becomes

$$\sqrt{F}\sqrt{(G - H)^2} = s, \quad \text{or} \quad F(G - H)^2 = s^2;$$

that is,

$$-F^2 + F(F^2 + G^2 + H^2) - 2FGH = s^2;$$

or, what is the same thing, and putting Θ for F,

$$-\lambda\Theta^2 + (\lambda^2 - \mu)\Theta - 3\nu^2 = s^2.$$

Hence in order that the roots of the quartic may be of the assumed form,

$$\alpha = m + \sqrt{\Theta} + \sqrt{p} + q\sqrt{\Theta}, \quad \text{\&c.},$$

where m, p, q, Θ are rational, and where also

$$\sqrt{\Theta}\sqrt{p + q\sqrt{\Theta}}\sqrt{p - q\sqrt{\Theta}} = s,\ \text{ a rational function,}$$

the necessary and sufficient conditions are that the quartic should be such that the reducing cubic

$$u^3 - \lambda u^2 + \mu u - \nu^2 = 0$$

roots are $(\alpha + \beta - \gamma - \delta)^2$, $(\alpha + \gamma - \beta - \delta)^2$, $(\alpha + \delta - \beta - \gamma)^2$ may have one rational root Θ, and moreover that the function

$$-\lambda\Theta^2 + (\lambda^2 - \mu)\Theta - 3\nu^2$$

shall be the square of a rational function s. This being so, the roots of the quartic will be of the assumed form

$$\alpha = m + \sqrt{\Theta} + \sqrt{p + q\sqrt{\Theta}}, \quad \text{\&c.};$$

and from what precedes, it is clear that any function of the roots of the quartic which remains unaltered by the cyclical substitution $\alpha\beta\gamma\delta$, or what is the same thing, any function of the form

$$\phi(\alpha, \beta, \gamma, \delta) + \phi(\beta, \gamma, \delta, \alpha) + \phi(\gamma, \delta, \alpha, \beta) + \phi(\delta, \alpha, \beta, \gamma)$$

will be a rational function of m, Θ, p, q, s, and consequently of the coefficients of the quartic. The above are the conditions in order that a quartic equation may be of the Abelian form.

It may be as well to remark that, assuming only the system of equations

$$\begin{aligned}
\alpha &= m + \sqrt{\Theta} + \sqrt{T}, \\
\beta &= m - \sqrt{\Theta} + \sqrt{T'}, \\
\gamma &= m + \sqrt{\Theta} - \sqrt{T}, \\
\delta &= m - \sqrt{\Theta} - \sqrt{T'},
\end{aligned}$$

then any rational function of α, β, γ, δ which remains unaltered by the cyclical substitution $\alpha\beta\gamma\delta$ will be a rational function of Θ, $T + T'$,

TT', $\sqrt{TT'}(T-T')$, $\sqrt{\Theta}(T-T')$, $\sqrt{\Theta}\sqrt{TT'}$. In fact, suppose such a function contains the term

$$(\sqrt{\Theta})^\alpha(\sqrt{T})^\beta(\sqrt{T'})^\gamma;$$

then it will contain the four terms

$$\begin{array}{rrr} (\sqrt{\Theta})^\alpha(& \sqrt{T})^\beta(& \sqrt{T'})^\gamma, \\ (-\sqrt{\Theta})^\alpha(& \sqrt{T})^\beta(& -\sqrt{T'})^\gamma, \\ (\sqrt{\Theta})^\alpha(- & \sqrt{T})^\beta(& -\sqrt{T'})^\gamma, \\ (-\sqrt{\Theta})^\alpha(- & \sqrt{T})^\beta(& \sqrt{T'})^\gamma, \end{array}$$

which together are

$$(\sqrt{\Theta})^\alpha\{(1+(-)^{\beta+\gamma})(\sqrt{T})^\beta(\sqrt{T'})^\gamma+(-)^\alpha[(-)^\gamma+(-)^\beta](\sqrt{T})^\gamma(\sqrt{T'})^\beta\},$$

an expression which vanishes unless $(-)^\beta$, $(-)^\gamma$ are both positive or both negative. The forms to be considered are therefore

α	β	γ
$+$	$+$	$+$
$-$	$+$	$+$
$+$	$-$	$-$
$-$	$-$	$-$

The first form is

$$(\sqrt{\Theta})^\alpha\{(\sqrt{T})^\beta(\sqrt{T'})^\gamma+(\sqrt{T})^\gamma(\sqrt{T'})^\beta\},$$

which, α, β, γ being each of them even, is a rational function of Θ, $T+T'$, TT'.

The second form is

$$(\sqrt{\Theta})^\alpha\{(\sqrt{T})^\beta(\sqrt{T'})^\gamma-(\sqrt{T})^\gamma(\sqrt{T'})^\beta\},$$

which, α being odd and β and γ each of them even, is the product of such a function into $\sqrt{\Theta}(T-T')$.

The third form is

$$(\sqrt{\Theta})^\alpha\{(\sqrt{T})^\beta(\sqrt{T'})^\gamma-(\sqrt{T})^\gamma(\sqrt{T'})^\beta\},$$

which, α being even and β and γ each of them odd, is the product of such a function into $\sqrt{TT'}(T-T')$.

And the fourth form is

$$(\sqrt{\Theta})^{\alpha}\{(\sqrt{T})^{\beta}(\sqrt{T'})^{\gamma}+(\sqrt{T})^{\gamma}(\sqrt{T'})^{\beta}\},$$

which, α, β, γ being each of them odd, is the product of such a function into $\sqrt{\Theta}(T-T')$.

Hence if $T = p + q\sqrt{\Theta}$, $T' = p - q\sqrt{\Theta}$, and $\sqrt{\Theta}\sqrt{p+q\sqrt{\Theta}}\sqrt{p-q\sqrt{\Theta}} = s$, then

$$\Theta,\quad T+T'(=2p),\quad TT'(=p^2-q^2\Theta),\quad \sqrt{TT'}(T-T')(=2qs),$$

$$\sqrt{\Theta}(T-T')(=2q\Theta),\quad \text{and}\quad \sqrt{\Theta}\sqrt{TT'}(=s)$$

are respectively rational functions. This is the *a posteriori* verification, that with the system of equations

$$\alpha = m+\sqrt{\Theta}+\sqrt{p+q\sqrt{\Theta}},\quad \&\text{c.},\quad \sqrt{\Theta}\sqrt{p+q\sqrt{\Theta}}\sqrt{p-q\sqrt{\Theta}} = s,$$

any function

$$\phi(\alpha,\beta,\gamma,\delta)+\phi(\beta,\gamma,\delta,\alpha)+\phi(\gamma,\delta,\alpha,\beta)+\phi(\delta,\alpha,\beta,\gamma)$$

is a rational function.

The coefficients of the quintic equation for x must of course be of the form just mentioned; that is, they must be functions of α, β, γ, δ, which remain unaltered by the cyclic substitution $\alpha\beta\gamma\delta$. To form the quintic equation, I write

$$\theta - x = a, \quad \sqrt[5]{\alpha^2\beta\gamma^3\delta^4} = b, \quad \sqrt[5]{\beta^2\gamma\delta^3\alpha^4} = c, \quad \sqrt[5]{\gamma^2\delta\alpha^3\beta^4} = d, \quad \sqrt[5]{\delta^2\alpha\beta^3\gamma^4} = e;$$

then we have

$$f\omega = a + b\omega + c\omega^2 + d\omega^3 + e\omega^4,$$

and the quintic equation is

$$0 = a + b + c + d + e,$$

where ω is an imaginary fifth root of unity, and

$$f\omega = a + b\omega + c\omega^2 + d\omega^3 + e\omega^4.$$

We have

$$\begin{aligned} f\omega \cdot f\omega^4 &= 2a^2 + (\omega + \omega^4)\Sigma' ab + (\omega^2 + \omega^3)\Sigma' ac, \\ f\omega^2 \cdot f\omega^3 &= 2a^2 + (\omega^2 + \omega^3)\Sigma' ab + (\omega + \omega^4)\Sigma' ac, \end{aligned}$$

where Σ' is Mr Harley's cyclical symbol, viz.

$$\Sigma' ab = ab + bc + cd + de + ea;$$

and so in other cases, the order of the cycle being always $abcde$. This gives

$$f\omega \cdot f\omega^4 + f\omega^2 \cdot f\omega^3 = 4a^2 + \Sigma' a^2b^2 - \Sigma' ab + 2\Sigma' a^2bc - \Sigma' abcd - 5\Sigma'(bc + cd);$$

and multiplying by $f\omega \cdot f\omega^4 = 2a$, and equating to zero, the result is found to be

$$2a^5 - abcde - 5\Sigma'(bc + cd) + a^2\Sigma'(bc^2 + cd^2) = 0;$$

or arranging in powers of a, this is

$$\begin{array}{ll} +\,a^5 \cdot & 2 \\ +\,a^4 \cdot & -5(be + cd) \\ +\,a^3 \cdot & 5(bc^2 + ce^2 + ed^2 + db^2) \\ +\,a^2 \cdot & 5(b^2c + c^2e + e^2d + d^2b) \\ +\,a^1 \cdot & b^5 + c^5 + e^5 + d^5 \\ & \quad - 5(b^2ce + c^2bd + e^2cb + d^2bc) \\ & \quad + 5(bd^2e^2 + cb^2d^2 + ec^2b^2 + de^2c^2) \\ +\,a^0 \cdot & -abcde = 0, \end{array}$$

the several coefficients being, it will be observed, cyclical functions to the cycle b, c, e, d.

Putting for a its value $-(x-\theta)$, and for b, c, d, e their values, the quintic equation in x is

$$
\begin{array}{ll}
(x-\theta)^5 & \\
+\,(x-\theta)^4\cdot & -5(AG+BD)\alpha\beta\gamma\delta \\
+\,(x-\theta)^3\cdot & -5(A^2B\gamma\delta+B^2C\delta\alpha+C^2D\alpha\beta+D^2A\beta\gamma) \\
+\,(x-\theta)^2\cdot & 5(AB^2\gamma\delta+BC^2\delta\alpha+CD^2\alpha\beta+DA^2\beta\gamma) \\
+\,(x-\theta)^1\cdot & A^5+B^5+C^5+D^5 \\
 & \qquad -\,5(A^2BC\gamma\delta+B^2CD\delta\alpha+C^2DA\alpha\beta+D^2AB\beta\gamma) \\
 & \qquad +\,5(AD^2B^2\gamma\delta+BA^2C^2\delta\alpha+CB^2D^2\alpha\beta+DC^2A^2\beta\gamma) \\
+\,(x-\theta)^0\cdot & -ABCDE\alpha\beta\gamma\delta=0,
\end{array}
$$

where A, B, C, D, E are the coefficients of the quintic equation.

C. V.

312. On the Partitions of a Close.

[From the *Philosophical Magazine*, vol. xxi. (1861), pp. 424–428.]

If F, S, E denote the number of faces, summits, and edges of a polyhedron, then, by Euler's well-known theorem,

$$F + S = E + 2;$$

and if we imagine the polyhedron projected on the plane of any one face in such manner that the projections of all the summits not belonging to the face fall within the face, then we have a partitioned polygon, in which (if P denote the number of component polygons, or say the number of parts) $F = P + 1$, or we have

$$P + S = E + 1,$$

where S is the number of summits and E the number of edges of the plane figure.

I retain for convenience the word edge, as having a different initial letter from summit.

The formula, however, excludes cases such as that of a polygon divided into two parts by means of an interior polygon wholly detached from it; and in order to extend it to such cases, the formula must be written under the form

$$P + S = E + 1 + B,$$

where B is the number of breaks of contour, as will be presently explained.

The edges of a polygon are right lines: it might at first sight appear that the theory would not be materially altered by removing this restriction, and allowing the edges to be curved lines; but the fact is otherwise.

Several definitions and explanations are required. The words line and curve are used indifferently to denote any path which can be described *currente calamo* without lifting the pen from the paper. A closed curve, not cutting or meeting itself(1), is called a *contour*. An enclosed space, such that no part of it is shut out from any other part of it, or, what is the same thing, such that any part can be joined with any other part by a line not cutting the boundary, is termed a *close*. The boundary of a close may be considered as the limit of a single contour, or of two or more contours lying wholly within the close. The reason for speaking of a limit will appear by an example. Consider a circle, and within it, but wholly detached from it, a figure of eight; the space interior to the circle but exterior to the figure of eight is a close: its boundary may be considered as the limit of two contours,—the first of them interior to the close, and indefinitely near the circle (in this case we might say the circle itself); the second of them an hour-glass-shaped curve, interior to the close (that is, exterior to the figure of eight) and indefinitely near to the figure of eight. The figure of eight, as being a line which cuts or meets itself, is not a contour, or mere contour, is not an edge. It may be added that an edge does not cut or meet itself or any other edge except at the summit or summits of the edge itself.

Consider now a close bounded by $\beta + 1$ mere contours: if for any partitioned close we have P the number of parts, S the number of summits, E the number of edges, B the number of breaks of contour; then, for the unpartitioned close, we have $P = 1$, $S = 0$, $E = 0$, $B = \beta$, and therefore

$$P + S = E + 1 + B;$$

and it is to be shown that this equation holds good in whatever manner the close is partitioned. The partitionment is effected by the addition, in any manner, of summits and mere contours, and by drawing edges, any edge from a summit to itself or to another summit. The effect of adding a summit is first to increase S by unity: if the summit added be on a contour, E will be thereby increased by unity; for if the contour is a mere contour, it is not an edge, but becomes so by the addition of the summit; if it is not a mere contour, but has upon it a summit or summits, the addition of the summit will increase by unity the number of edges of the contour. If, on the other hand, the summit added be an isolated one, then the addition

If we consider the surface of a plane as bounded by a mere contour at infinity, then for the infinite plane, $\beta = 0$, or we have $P + S = E + 1 + B$: in the case where the infinite plane is partitioned by a mere contour, $P = 2$, $S = 0$, $E = 0$, $B = 1$ (for the exterior part is bounded by the contour at infinity, and the partitioning contour, that is, for it, $B = 1$), and the equation is thus satisfied. And so for a contour having upon it n summits, $P = 2$, $S = n$, $E = n$, $B = 1$, and the equation is still satisfied: this is the case of the plane partitioned into two parts by means of a single polygon.

The case of a spherical surface is very interesting: the entire surface of the sphere must be considered as a close bounded by 0 contour, or we have $\beta = -1$, and the equation thus becomes $P+S = E+2+B$. Thus, if the sphere be divided into two parts by a mere contour, $P = 2$, $S = 0$, $E = 0$, $B = 0$, and the equation is satisfied. And in general, when $B = 0$, then $P + S = E + 2$; or writing F for P, then $F + S = E + 2$, which is Euler's equation for a polyhedron.

2, Stone Buildings, W.C., March 8, 1861.

C. V.

(1) The figure referred to will be at once understood by considering A, A' as the poles of an ellipsoid, or say of a sphere, $ABA'B'$ the meridian of projection, APA' any other meridian, BPB' the equator or any other great circle.

313. On a Surface of the Fourth Order.

[From the *Philosophical Magazine*, vol. xxi. (1861), pp. 491–495.]

Let A, B, C be fixed points; it is required to investigate the nature of the surface, the locus of a point P such that

$$\lambda AP + \mu BP + \nu CP = 0,$$

where λ, μ, ν are given coefficients; the equation depends, it is clear, on the ratios only of these quantities.

The surface is easily seen to be of the fourth order; it is obviously symmetrical in regard to the plane ABC; and the section by this plane, or say the principal section, is a curve of the fourth order, the locus of a point M such that

$$\lambda AM + \mu BM + \nu CM = 0.$$

The curve is considered incidentally by Mr Salmon, p. 125 of his *Higher Plane Curves* [Ed. 3, p. 126 and see also p. 240 *et seq.*]; and he has remarked that the two circular points at infinity are double points on the curve, which is therefore of the eighth class. Moreover, that there are two double foci, since at each of these circular points there are two tangents, each tangent of the one pair intersecting a tangent of the other pair in a double focus; hence, further, that there are four other foci, the points A, B, C, and a fourth point D lying in a circle with A, B, C, and which are such that, selecting any three of the points A, B, C, D, these are foci in regard to a pair of circles, each of them passing through the selected points and through the fourth point respectively.

It is easy to show that the principal section is in fact the locus of a point M such that the three spheres with the centres A, B, C and radii proportional to λ, μ, ν respectively, have their centres of similitude at M. In fact, if M is a point on the locus, and if from M we draw tangents to the three spheres, then the tangent to the first sphere is to the tangent to the second sphere as $\lambda : \mu$, and the tangent to the second sphere is to the tangent to the third sphere as $\mu : \nu$; hence the three tangents are as $\lambda : \mu : \nu$, and the point M is such that the tangents from it to the three spheres are proportional to λ, μ, ν.

The three spheres are such that, with regard to the first sphere, B and C are the images each of the other; that is, the centre of the sphere lies on the line BC, and the product of its distances from B and C is equal to the square of the radius; in like manner the second sphere is such that, with regard to it, C and A are the images each of the other; and the third sphere is such that, with regard to it, A and B are the images each of the other. The three spheres intersect in a circle through M at right angles to the principal plane (that is, the three spheres have a common circular section), and the equations of this circle may be taken to be

$$\frac{AP}{\lambda} = \frac{BP}{\mu} = \frac{CP}{\nu}.$$

It is clear that the circle of intersection lies wholly on the surface.

The spheres meet the principal plane in three circles, which are the diametral circles of the spheres; these circles are related to each other and to the points A, B, C, in like manner as the spheres are to each other and to the same points. The circles have thus a common chord; that is, they meet in the point M and in another point M': and MM' is the diameter of the circle, the intersection of the three spheres.

It may be shown that M, M' are the images each of the other in respect to the circle through A, B, C. In fact, considering the sphere through A, B, C, M, the points M and M' are the intersections of this sphere with the circle of intersection of the three spheres; and the circle through A, B, C is the intersection of the sphere through A, B, C, M with the principal plane; hence M and M' are inverse points in regard to the circle through A, B, C.

In my paper "On a Theorem in the Geometry of Position," *Camb. Math. Journ.* vol. ii. pp. 267—271 (1841), [1], I obtained this equation, the four points being there considered as lying in a plane, as the relation between the distances of four points in a circle, in addition to the relation

$$\begin{vmatrix} 0, & 1, & 1, & 1, & 1 \\ 1, & 0, & 12^2, & 13^2, & 14^2 \\ 1, & 21^2, & 0, & 23^2, & 24^2 \\ 1, & 31^2, & 32^2, & 0, & 34^2 \\ 1, & 41^2, & 42^2, & 43^2, & 0 \end{vmatrix} = 0,$$

which exists between the distances of any four points in a plane. The present investigation shows the signification of the equation $\Theta = 0$ between the distances of four points in space; viz. it expresses that the four points lie in a sphere radius zero, or cone-sphere. But the formula in question is in reality included in that given in the paper for the distances of five points in space. For calling the points 0, 1, 2, 3, 4, the relation between the distances of these five points is

$$\begin{vmatrix} 0, & 1, & 1, & 1, & 1, & 1 \\ 1, & 0, & 01^2, & 02^2, & 03^2, & 04^2 \\ 1, & 10^2, & 0, & 12^2, & 13^2, & 14^2 \\ 1, & 20^2, & 21^2, & 0, & 23^2, & 24^2 \\ 1, & 30^2, & 31^2, & 32^2, & 0, & 34^2 \\ 1, & 40^2, & 41^2, & 42^2, & 43^2, & 0 \end{vmatrix} = 0.$$

Hence if 1, 2, 3, 4 are the centres of spheres radii α, β, γ, δ, and if 0 is the centre of a tangent sphere radius r, we have

$$01 = r + \alpha, \quad 02 = r + \beta, \quad 03 = r + \gamma, \quad 04 = r + \delta;$$

so that, for any given combination of signs, it would at first sight appear that r is determined by a quartic equation; but by means of a simple transformation (indicated to me by Prof Sylvester) it may be shown that the equation for r is really a quadratic one; moreover, the equation remains unaltered if the signs of α, β, γ, δ and of r, are all reversed; and r^2 has thus in the whole sixteen values.

In the case where the four spheres reduce to points, the equation $\Theta = 0$ gives the relation between the distances of the four points, and the equation is in fact the condition that the four points may lie on a sphere.

C. V.

314. On the Curves Situate on a Surface of the Second Order.

[From the *Philosophical Magazine*, vol. xxii. (1861), pp. 35–38.]

A SURFACE of the second order has on it a double system of generating lines, real or imaginary; and any two generating lines of the first kind form with any two generating lines of the second kind a skew quadrangle. If the equations of the planes containing respectively the first and second, second and third, third and fourth, fourth and first sides of the quadrangle are $x = 0$, $y = 0$, $z = 0$, $w = 0$, and if the constant multipliers which are implicitly contained in x, y, z, w respectively are suitably determined, then the equation of the surface of the second order (or say for shortness the quadric surface) is $xw - yz = 0$.

Assume $\frac{x}{z} = \frac{\lambda}{\nu}$, $\frac{y}{w} = \frac{\mu}{\rho}$, then $\frac{x}{z}$, $\frac{y}{w}$, or say $(\lambda, \mu, \nu, \rho)$, may be regarded as the coordinates of a point on the quadric surface; we in fact have $x : y : z : w = \lambda\rho : \mu\rho : \nu\lambda : \mu\nu$. The four quantities $(\lambda, \mu, \nu, \rho)$ are for symmetry of notation used as coordinates; but it is to be throughout borne in mind that the absolute magnitudes of λ and μ, and of ν and ρ are essentially indeterminate; it is only the ratios $\lambda : \mu$ and $\nu : \rho$ that we are concerned with.

An equation of the form

$$(* \,\$\, \lambda, \mu)^p (\nu, \rho)^q = 0,$$

that is, an equation homogeneous of the degree p as regards (λ, μ), and homogeneous of the degree q as regards (ν, ρ), represents a curve on the quadric surface; and this curve is of the order $p + q$. In fact, combining with the equation of the curve the equation of an arbitrary plane

$$Ax + By + Cz + Dw = 0,$$

this equation, expressed in terms of the coordinates $(\lambda, \mu, \nu, \rho)$, is

$$A\lambda\rho + B\mu\rho + C\nu\lambda + D\mu\nu = 0;$$

or, as it is more conveniently written,

$$\begin{pmatrix} C, & D \\ A, & B \end{pmatrix} \$\ \lambda, \mu\backslash\backslash\nu, \rho) = 0;$$

and if from this and the equation of the curve we eliminate $\lambda : \mu$ or $\nu : \rho$, say the second of these quantities, we obtain

$$(*\ \$\ \lambda, \mu)^p(-C\lambda - B\mu, C\lambda + D\mu)^q = 0,$$

which is of the order $p + q$ in (λ, μ); and $\lambda : \mu$, being known, $\nu : \rho$ is linearly determined. There are thus $p + q$ systems of values of the coordinates, or the plane meets the curve in $p + q$ points; that is, the curve is of the order $p + q$.

A linear equation $A\lambda + B\mu = 0$ gives a generating line, say of the first kind, of the quadric surface, and a linear equation $C\nu + D\rho = 0$ gives a generating line of the second kind: and by combining the one or the other of these equations with the equation of the curve, it is at once seen that the curve meets each generating line of the first kind in q points, and each generating line of the second kind in p points.

Consider the curves of the order n: the different solutions of the equation $p + q = n$ give different species of curves. But the solution $(n, 0)$ gives only a system of n generating lines of the first kind, and the solution $(0, n)$ gives only a system of generating lines of the second kind. And in general the solutions (p, q) and (q, p) give species of curves which are related, the one of them to the generating lines of the first and second kinds, in the same way as the other of them to the generating lines of the second and first kinds; and they may be considered as corresponding to each other.

For $n = 4$, I take first the species $(2, 2)$ [the quadriquadric curve] which is represented by an equation of the form

$$(a, b, c\ \$\ \lambda, \mu)^2\nu^2 + 2(a', b', c'\ \$\ \lambda, \mu)^2\nu\rho + (a'', b'', c''\ \$\ \lambda, \mu)^2\rho^2 = 0,$$

which in fact belongs to a quartic curve, the intersection of two quadric surfaces.

For, reverting to the original coordinates, the equation becomes

$$(a, b, c\ \$\ x, y)^2w^2 + 2(a', b', c'\ \$\ x, y)^2wz + (a'', b'', c''\ \$\ x, y)^2z^2 = 0,$$

which by means of the equation $xw - yz = 0$ of the quadric surface is at once reduced to

$$(a, b, c \,\$ x, y)^2 w^2 + 2a' xwz + 4b' yz + 2c' yzw + a'' z^2 + 2b'' zw + c'' w^2 = 0,$$

which is the equation of a quadric surface intersecting the given quadric surface $xw - yz = 0$ in the curve in question.

Consider next the species $(3, 1)$ [the excubo-quartic curve] represented by an equation of the form

$$(a, b, c, d \;\$ \lambda, \mu)^3 \nu + (a', b', c', d' \;\$ \lambda, \mu)^3 \rho = 0,$$

which is the other species of quartic curve situate on only a single quadric surface. Reverting to the original coordinates, the equation becomes

$$(a, b, c, d \;\$ x, y)^3 x + (a', b', c', d' \;\$ x, y)^3 z = 0;$$

and by means of the equation $xw - yz = 0$ of the quadric surface this is reduced to

$$(a, b, c, d \;\$ x, y)^3 + a' x^2 z + 3b' xyz + 3c' y^2 z + d' y^3 w = 0,$$

which is the equation of a cubic surface containing the line $(x = 0, y = 0)$ twice, and therefore along this line touching the quadric surface $xw - yz = 0$; and consequently intersecting it besides in a quartic curve. And in like manner for the curves of the fifth and higher orders which lie upon a quadric surface.

The combination of the equations

$$\begin{aligned} (* \;\$ \; \lambda, \mu)^p (\nu, \rho)^q &= 0, \\ (*' \;\$ \; \lambda, \mu)^{p'} (\nu, \rho)^{q'} &= 0, \end{aligned}$$

shows at once that two curves of the species (p, q), (p', q') respectively meet in $pq' + p'q$ points.

315. On the Cubic Centres of a Line with Respect to Three Lines and a Line.

[From the *Philosophical Magazine*, vol. xxii. (1861), pp. 433–436.]

On referring to my Note on this subject (Phil. Mag. vol. xx. pp. 418–423, 1860 [257]), it will be seen that the cubic centres of the line

$$\frac{x}{\lambda}+\frac{y}{\mu}+\frac{z}{\nu}=0$$

in relation to the lines $x=0$, $y=0$, $z=0$, and the line $x+y+z=0$, are determined by the equations

$$\frac{1}{x}:\frac{1}{y}:\frac{1}{z}=\frac{1}{\theta+\lambda}:\frac{1}{\theta+\mu}:\frac{1}{\theta+\nu},$$

where θ is a root of the cubic equation

$$\frac{1}{\theta+\lambda}+\frac{1}{\theta+\mu}+\frac{1}{\theta+\nu}-\frac{1}{\theta}=0,$$

or as it may also be written,

$$\theta^3-\theta(\mu\nu+\nu\lambda+\lambda\mu)-2\lambda\mu\nu=0.$$

Two of the centres will coincide if the equation for θ has equal roots; and this will be the case if

$$(\mu\nu+\nu\lambda+\lambda\mu)^3=-27\lambda^2\mu^2\nu^2,$$

or, what is the same thing, if $\lambda, \mu, \nu=a^{-2}, b^{-2}, c^{-2}$, where $a+b+c=0$. In fact, if $a+b+c=0$, then $a^3+b^3+c^3=3abc$, and the equation for θ becomes

$$\theta^3-\theta\frac{a^3+b^3+c^3}{a^2b^2c^2}-\frac{2}{a^2b^2c^2}=0,$$

that is

$$(a^2b^2c^2\theta)^3-3abc(a^2b^2c^2\theta)-2=0,$$

which is

$$(a^2b^2c^2\theta+1)^2(a^2b^2c^2\theta-2)=0;$$

so that the values of θ are $-\frac{1}{a^2b^2c^2}$, $\frac{2}{a^2b^2c^2}$.

First, if $\theta = -\frac{1}{a^2b^2c^2}$, then x, y, z will be the coordinates of the double centre. And we have

$$\theta{+}\lambda = \frac{-1 + a^2\mu\nu}{a^2b^2c^2} = \frac{-1 + a^2 \cdot b^{-2}c^{-2}}{a^2b^2c^2} = \frac{-a^2b^2c^2 + a^2}{a^4b^4c^4} = \frac{a^2(1 - b^2c^2)}{a^4b^4c^4},$$

or putting for shortness $D = a^2 + b^2 + c^2$,

$$\theta + \lambda = \frac{-2a^2b^2c^2}{abc \cdot a^2} = \frac{D - 3a^2}{a^2 \cdot abc \cdot a^2},$$

with similar values for $\theta + \mu$, $\theta + \nu$. But $\frac{1}{x}$, $\frac{1}{y}$, $\frac{1}{z}$ are proportional to $\theta + \lambda$, $\theta + \mu$, $\theta + \nu$; and we may therefore write

$$\frac{P}{x} = \frac{D - 3a^2}{a^2}, \quad \frac{P}{y} = \frac{D - 3b^2}{b^2}, \quad \frac{P}{z} = \frac{D - 3c^2}{c^2},$$

whence, in virtue of the equation $a + b + c = 0$, we have for the locus of the double centre,

$$\sqrt{x} + \sqrt{y} + \sqrt{z} = 0;$$

or this locus is a conic touching the lines $x = 0$, $y = 0$, $z = 0$ harmonically in respect to the line $x + y + z = 0$, a result which was obtained somewhat differently in the paper above referred to.

Next, if $\theta = \frac{2}{a^2b^2c^2}$, x, y, z will be the coordinates of the single centre. And we now have

$$\theta + \lambda = \frac{2 + a^2\mu\nu}{a^2b^2c^2} = \frac{2a^2b^2c^2 + a^2}{a^4b^4c^4} = \frac{a^2(2b^2c^2+1)}{a^4b^4c^4} = \frac{2a^2b^2c^2 + a^2}{a^4b^4c^4},$$

with similar values for $\theta + \mu$, $\theta + \nu$. But $\frac{1}{x}$, $\frac{1}{y}$, $\frac{1}{z}$ are proportional to $\theta + \lambda$, $\theta + \mu$, $\theta + \nu$, and we may therefore write

$$\frac{P}{x} = \frac{D - 6a^2}{a^2}, \quad \frac{P}{y} = \frac{D - 6b^2}{b^2}, \quad \frac{P}{z} = \frac{D - 6c^2}{c^2},$$

from which equations, and the equation $a + b + c = 0$, the quantities P, a, b, c have to be eliminated. I at first effected the elimination as follows: viz., writing the equations under the form

$$\frac{x}{x+P} = \frac{a^2}{D}, \quad \frac{y}{y+P} = \frac{b^2}{D}, \quad \frac{z}{z+P} = \frac{c^2}{D},$$

we obtain

$$\frac{x}{x+P} + \frac{y}{y+P} + \frac{z}{z+P} = \frac{a^2+b^2+c^2}{D} = 1,$$

$$\frac{yz}{(y+P)(z+P)} + \frac{zx}{(z+P)(x+P)} + \frac{xy}{(x+P)(y+P)} = \frac{b^2c^2 + c^2a^2 + a^2b^2}{D^2},$$

which are easily transformed into

$$9P(P+x)(P+y)(P+z) - x(P+y)(P+z) - y(P+z)(P+x) - z(P+x)(P+y)$$
$$9(P+x)(P+y)(P+z) - yz(P+x) - zx(P+y) - xy(P+z) = 0,$$

which give

$$9P^3 + 9P^2(x+y+z) + 9P(yz+zx+xy) + 9xyz = 0,$$
$$9P^2 + 9P(x+y+z) + 9(yz+zx+xy) + 6xyz = 0;$$

or, multiplying the first equation by 2, and subtracting the second,

$$3P + (x+y+z) = 0:$$

and we thus obtain for the locus of the single centre the equation

$$\frac{x}{-2x+y+z} + \frac{y}{-2y+z+x} + \frac{z}{-2z+x+y} = 2,$$

or, what is the same thing,

$$x^3 + y^3 + z^3 - 2(yz^2 + zx^2 + xy^2 + y^2z + z^2x + x^2y) + 9xyz = 0,$$

which may also be written,

$$(-x + y + z)(x - y + z)(x + y - z) + xyz = 0.$$

The same result may also be obtained as follows: viz., observing that

$$D - 6a^2 = b^2 + c^2 - 5a^2 = -4a^2 - 2bc,$$

we have

$$\frac{P}{x} = \frac{-3a^2}{2a^2 + bc}, \quad \frac{P}{y} = \frac{-3b^2}{2b^2 + ca}, \quad \frac{P}{z} = \frac{-3c^2}{2c^2 + ab},$$

and then by means of the equation

$$\frac{a^2}{2a^2 + bc} + \frac{b^2}{2b^2 + ca} + \frac{c^2}{2c^2 + ab} - 1 = 0,$$

which is identically true in virtue of $a+b+c = 0$ (in fact, multiplying out, this gives

$$\begin{aligned}&12a^2b^2c^2 + 4(b^2c^4 + c^2a^4 + a^2b^4) + abc(a^6 + b^6 + c^6)\\ &\quad - 8a^2b^2c^2 - 4(b^2c^4 + c^2a^4 + a^2b^4) - 2abc(a^3 + b^3 + c^3) - a^2b^2c^2 = 0;\end{aligned}$$

that is

$$3a^2b^2c^2 - abc(a^3 + b^3 + c^3) = 0, \quad \text{or} \quad abc(a^3 + b^3 + c^3 - 3abc) = 0,$$

where the second factor divides by $a + b + c$), we find the above-mentioned equation.

We then have

$$\frac{-x + y + z}{P} = \frac{x + y + z}{P} - \frac{2x}{P} = -\frac{6a^2}{2a^2 + bc} = -\frac{6bc}{2a^2 + bc},$$

that is

$$\frac{-x + y + z}{P} = \frac{-3bc}{2a^2 + bc}, \quad \frac{x - y + z}{P} = \frac{-3ca}{2b^2 + ca}, \quad \frac{x + y - z}{P} = \frac{-3ab}{2c^2 + ab},$$

and forming the product of these functions, and that of the foregoing values of $\frac{P}{x}$, $\frac{P}{y}$, $\frac{P}{z}$, we find, as before,

$$(-x + y + z)(x - y + z)(x + y - z) + xyz = 0$$

for the equation of the locus of the single centre. The equation shows that the locus is a cubic curve which touches the lines $x = 0$, $y = 0$, $z = 0$ at the points where these lines are intersected by the lines $y - z = 0$, $z - x = 0$, $x - y = 0$ (that is, it touches the lines $x = 0$, $y = 0$, $z = 0$ harmonically in respect to the line $x + y + z = 0$), and besides meets the same lines $x = 0$, $y = 0$, $z = 0$ at the points in which they are respectively met by the line $x + y + z = 0$.

2, Stone Buildings, W.C., September 25, 1861.

316. Note on the Solution of an Equation of the Fifth Order.

[From the *Philosophical Magazine*, vol. xxiii. (1862), pp. 195, 196.]

This Note was in answer to Mr Jerrard's paper "Remarks on Mr Cayley's Note," Phil. Mag. vol. xxi. pp. 348–350, referring to the foregoing paper 310.

317. Note on the Transformation of a Certain Differential Equation.

[From the *Philosophical Magazine*, vol. xxiii. (1862), pp. 266, 267.]

The differential equation

$$(1+x^2)\frac{d^2y}{dx^2}+(1+3x^2)\frac{dy}{dx}-m(m+1)xy=0,$$

if we put therein $\theta=2x^2+1$ ($i=\sqrt{-1}$ as usual), becomes

$$(1+\theta^2)\frac{d^2y}{d\theta^2}+(1+3\theta^2)\frac{dy}{d\theta}-\tfrac{1}{4}m(m+1)\theta y=0.$$

In fact an integral of the second equation is $(\sqrt{1+x^2}+x)^m$; this is

$$=(\sqrt{(2x^2+1)^2-1}+2x^2+1)^{\frac{1}{2}m};$$

or putting $\theta=2x^2+1$, it is

$$=\{\sqrt{\theta^2-1}+\theta\}^{\frac{1}{2}m};$$

which is

$$=\{\theta(\sqrt{1+\frac{1}{\theta^2}}+1)\}^{\frac{1}{2}m};$$

so that an integral of the transformed equation in θ is

$$(\sqrt{1+\theta^2}+\theta)^{\frac{1}{2}m};$$

and writing in the second equation θ for x, and $\frac{1}{2}m$ for m, we see that the last-mentioned function, viz. $(\sqrt{1+\theta^2}+\theta)^{\frac{1}{2}m}$, is an integral of

$$(1+\theta^2)\frac{d^2y}{d\theta^2}+(1+3\theta^2)\frac{dy}{d\theta}-\tfrac{1}{4}m(m+1)\theta y=0;$$

whence the transformed equation in θ must be this very equation, that is, it must be the first equation. I have for shortness used the particular integral $(\sqrt{1+x^2}+x)^m$; but the reasoning should have been applied, and it is in fact applicable, without alteration, to the general integral

$$c(\sqrt{1+x^2}+x)^m+c'(\sqrt{1+x^2}-x)^m.$$

There is of course no difficulty in a direct verification. Thus, starting from the first equation, or equation in θ, the relation $i\theta = 2x^2+1$ gives

$$\frac{dy}{d\theta} = \frac{i}{4x}\frac{dy}{dx}, \quad \frac{d^2y}{d\theta^2} = \frac{i}{4x}\frac{d}{dx}\left(\frac{i}{4x}\frac{dy}{dx}\right) = -\frac{1}{16x^2}\left(\frac{d^2y}{dx^2} - \frac{1}{x}\frac{dy}{dx}\right),$$

$$1+\theta^2 = -4x^2(1+x^2);$$

so that the equation becomes

$$-4x^2(1+x^2)\cdot\left(-\frac{1}{16x^2}\right)\left(\frac{d^2y}{dx^2} - \frac{1}{x}\frac{dy}{dx}\right) - \frac{1}{4x}(1+3\cdot 2x^2+1)\frac{dy}{dx} - m(m+1)xy = 0,$$

or multiplying by 4,

$$(1+x^2)\left(\frac{d^2y}{dx^2} - \frac{1}{x}\frac{dy}{dx}\right) + \frac{1+6x^2}{x}\frac{dy}{dx} - 4m(m+1)xy = 0,$$

that is

$$(1+x^2)\frac{d^2y}{dx^2} + \frac{1+5x^2}{x}\frac{dy}{dx} - 4m(m+1)xy = 0,$$

the second equation. But the first method shows the reason why the two forms are thus connected together.

2, Stone Buildings, W.C., February 19, 1862.

318. On a Question in the Theory of Probabilities.

[From the *Philosophical Magazine*, vol. xxiii. (1862), pp. 361–365.]

The question referred to is that discussed in the paper 121; the remarks on that paper in the Notes and References to volume II. are in a great measure to the same effect as the present and next papers, 318 and 319, the existence of which I had entirely overlooked. In the first part (dated 2, Stone Buildings, W.C., March 1862) of the present paper 318, after referring to the two modes of statement which may be called the Causation statement and the Concomitance statement, I reproduce nearly as in the Notes and References first my own solution as completed by Dedekind, next Boole's solution of the problem, involving his logical probabilities; and the paper is then continued as follows.

The foregoing paper was submitted to Prof. Boole, who, in a letter dated March 26, 1862, writes:

"The observations which have occurred to me after studying your paper are the following.

"1st. I think that your solution is correct under conditions partly expressed and partly implied. The one to which you direct attention is the assumed independence of the causes denoted by A and B. Now I am not sure that I can state precisely what the others are; but one at least appears to me to be the assumed independence of the events of which the probabilities according to your hypothesis are $\alpha\lambda$, $\beta\mu$. Assuming the independence of the causes as to happening, I do not think that you are entitled on that ground to assume their independence as to acting; because, to confine our observations to common experience, we often notice that states of things apparently independent as to happening are not independent as to acting.

"2ndly. I am not sure that I understand your observation, that the probabilities of the causes acting efficiently should be part of the original data, and that they ought to be of such a kind that they can be established by experience in the same way as the other data are. For instance, if experience, as embodied in a sufficiently long series of statistical records, establish that

$$\text{Prob. } A = \alpha, \quad \text{Prob. } B = \beta,$$

the very same experience may, by establishing also that

$$\text{Prob. } AB = \alpha\beta,$$

whence in conjunction with the former it follows that

$$\text{Prob. } AB' = \alpha\beta', \quad \text{Prob. } A'B = \alpha'\beta, \quad \text{Prob. } A'B' = \alpha'\beta',$$

enable us to pronounce that A and B are in the long run, as to happening or not happening, in the position of mutually independent events.

"3rdly. I think it may be shown to demonstration, from the nature of the result, that the solution you have obtained does not apply simply and generally to the problem under the single modification of the assumption that A and B are independent. The completed data under this assumption are

$$\begin{gathered}\text{Prob. } A = \alpha, \quad \text{Prob. } B = \beta, \quad \text{Prob. } AB = \alpha\beta, \\ \text{Prob. } AE = \alpha p, \quad \text{Prob. } BE = \beta q.\end{gathered}$$

You may deduce all these from your Table of Probabilities of 'compound events' given in your paper. Now you may easily satisfy yourself that the sole necessary and sufficient conditions for the consistency of these data are the following:

$$\alpha p' + \beta q \geq \alpha\beta, \tag{1}$$

$$\alpha p + \beta q \leq \alpha\beta, \tag{2}$$

$$\left(\frac{\alpha p' + \beta q - \alpha\beta}{\alpha'\beta'}\right)^2 \leq \frac{(1-p)(1-q)}{pq} \cdot \frac{\alpha p \cdot \beta q}{(\alpha p' + \beta q - \alpha\beta)^2}. \tag{M}$$

But your solution requires the following conditions to be satisfied, viz.,

$$q - \alpha p \geq 0, \quad p - \beta q \leq 0,$$

together with the system (3). Now (1) and (2) are expressible in the form

$$\begin{gathered}\alpha'(q - \alpha p) + \alpha\beta' p' \geq 0, \\ \alpha(p - \beta q) + \beta\alpha' q' \geq 0;\end{gathered}$$

from which you will see that your conditions are narrower than those which the data are really subject to. If your conditions are satisfied, then I think it may be shown that your solution agrees with mine."

Prof. Boole's letter continues:

"4thly. You remark that my solution of the problem, in which the independence of A and B is not assumed, but in which the probabilities are otherwise the same as in yours, is only applicable when

$$\alpha' + \alpha p \geq \beta q, \quad \beta' + \beta q \geq \alpha p;$$

but you do not appear to have noticed that these are actually the conditions of consistency in the data. Unless these are satisfied, the data cannot possibly be furnished by experience.

"5thly. You remark that I have solved the problem under what you call the 'concomitance' statement, and not the 'causation' statement. I think that every problem stated in the 'causation' form admits, if capable of scientific treatment, of reduction to the 'concomitance' form. I admit it would have been better, in stating my problem, not to have employed the word 'cause' at all. But the introduction of the hypothesis of the independence of A and B does not affect the nature of the problem.

"6thly. The x, s, &c., about the interpretation of which you inquire, are the probabilities of ideal events in an ideal problem connected by a formal relation with the real one. I should fully concede that the auxiliary probabilities which are employed in my method always refer to an ideal problem; but it is one, the form of which, as given by the calculus of logic, is not arbitrary. Nor does its connexion with the real problem appear to me arbitrary. It involves an extension, but as it seems to me, a perfectly scientific extension, of the principles of the ordinary theory of probabilities. On this subject, however, I have but little to add to what I have said. Transactions of the Royal Society of Edinburgh, vol. xxi. p. 621."

To this I reply as follows:

1stly. I do not assume more than that the probability of a cause A happening, and the probability of its acting efficiently, are each of them the same whether B does not act, or acts inefficiently, or acts efficiently; and the like for B.

2ndly. I do assume that the same experience which establishes

$$\text{Prob. } A = \alpha, \quad \text{Prob. } B = \beta,$$

would in the long run establish

$$\text{Prob. } AB = \alpha\beta;$$

if it does not, *cadit quæstio*, the causes are not independent.

3rdly. I assume not only

$$\text{Prob. } A = \alpha, \quad \text{Prob. } B = \beta, \quad \text{Prob. } AB = \alpha\beta,$$

but also as 1st above stated; and I consider that, inasmuch as the result of the investigation is to show that the conditions $q - \alpha p \geq 0$, $p - \beta q \leq 0$ are necessary and sufficient conditions, it is also a result of the investigation that these are the conditions of consistency among the data, viz. the conditions in order that the data may be consistent with the above assumptions as to the independence of the causes. It is clear that since, as just stated, I do assume something beyond the last-mentioned three equations, the conditions of consistency ought to be narrower than those in Prof. Boole's 3rdly.

4thly. I had not overlooked, but I ought to have stated, that Prof. Boole's conditions were actually the conditions of consistency in the data.

5thly. I contend that the conception of A and B as causes does alter the nature of the problem. For when A and B are conceived of as causes, there is a definite notion of the efficient or inefficient action of A or B; and in particular when they both act, one of them, say A, may act inefficiently. But according to the concomitance statement, then either there is no such notion as that of the efficient or non-efficient happening of A.

Prof. Boole, in his reply, dated April 2, writes, "No such explanation as you desiderate of the interpretation of the auxiliary quantities in my method of solution is possible; because they are not of the nature of additional data, and their introduction does not limit the problem as any hypotheses which are of that nature do. I do not see any difficulty whatever in the conception of the ideal problem."

We thus join issue as follows: Prof. Boole says that there is no difficulty in understanding, I say that I do not understand, the rationale of his solution.

It may be remarked that the question may be, not to find any actual probability whatever, but only to find a Boolian probability or probabilities. Thus the equations (L), p. 356, omitting the last member, which alone involves u, determine in terms of the data α, β, $\alpha\beta$, βq, the Boolian probabilities x, y, s, t of the events A, B, AE, BE.

319. Postscript to the Paper "On a Question in the Theory of Probabilities."

[From the *Philosophical Magazine*, vol. xxiii. (1862), pp. 470–471.]

See 318. The present paper reproduces Wilbraham's discussion of Boole's Solution; and concludes with the remark "Professor Boole wishes me to mention that he has succeeded in obtaining a demonstration of the analytical theorem arising from his theory referred to in his Reply in my paper." See ante p. 82, 7thly.

320. On the Transcendent $\operatorname{gd} u = i \log\tan(\frac{1}{4}\pi + \frac{1}{2}iu)$.

[From the *Philosophical Magazine*, vol. xxiv. (1862), pp. 19–21.]

The elliptic functions which correspond to the modulus $k = 1$ reduce themselves, as is well known, to circular functions. The case in question plays implicitly an important part in the general theory, and it has been particularly studied by Gudermann, and by Dr Booth in connexion with his theory of parabolic logarithms. But in the absence of a notation corresponding to that used for elliptic functions in general, the theory has not, it appears to me, been exhibited in its proper form. The defect is very easily supplied: using for $\operatorname{am} u$, to the modulus 1, the notation $\operatorname{gd} u$ (Gudermannian of u), then if

$$u = \log\tan\left(\tfrac{1}{4}\pi + \tfrac{1}{2}\phi\right), \quad \frac{d\phi}{du} = \cos\phi,$$

we have

$$\phi = \operatorname{gd} u;$$

and hence, observing that the equation between v and ϕ is

$$e^v = \frac{1 + i\tan\phi}{1 - i\tan\phi}, \quad \text{or, what is the same thing,} \quad \tan\phi = \frac{e^v - 1}{i(e^v + 1)},$$

and that we have

$$\begin{aligned}\log\tan\left(\tfrac{1}{4}\pi + \tfrac{1}{2}iu\right) &= \log\frac{1 + \tan\frac{1}{2}iu}{1 - \tan\frac{1}{2}iu},\\ &= \log\frac{\cos\frac{1}{2}u + \sin\frac{1}{2}u}{\cos\frac{1}{2}u - \sin\frac{1}{2}u},\end{aligned}$$

$$\begin{aligned} i\,\mathrm{gd}\,u &= \log\frac{1+\tan\frac{1}{2}iu}{1-\tan\frac{1}{2}iu} \\ &= \log\frac{e^{iu}+1+i(e^{iu}-1)}{e^{iu}+1-i(e^{iu}-1)} \\ &= \log\frac{1+i\tan\frac{1}{2}\phi}{1-i\tan\frac{1}{2}\phi}, \end{aligned}$$

or if

$$u = \log\tan\left(\tfrac{1}{4}\pi + \tfrac{1}{2}\phi\right),$$

then

$$\phi = \tfrac{1}{i}\log\tan\left(\tfrac{1}{4}\pi + \tfrac{1}{2}ui\right);$$

and substituting for ϕ its value, we obtain

$$\mathrm{gd}\,u = \tfrac{1}{i}\log\tan\left(\tfrac{1}{4}\pi + \tfrac{1}{2}ui\right),$$

which is the definition of the transcendent $\mathrm{gd}\,u$. It is to be noticed that $\mathrm{gd}\,u$, although exhibited in an imaginary form, is a real function of u; and, moreover, that it is an odd function, viz. we have

$$\mathrm{gd}(-u) = -\mathrm{gd}(u),$$

and therefore also

$$\mathrm{gd}(0) = 0.$$

The original equation,

$$\frac{d\phi}{du} = \cos\phi,$$

written under the form

$$u = \log\tan\left(\tfrac{1}{4}\pi + \tfrac{1}{2}\phi\right),$$

shows that we have

$$u = \tfrac{1}{i}\,\mathrm{gd}\,iu = \tfrac{1}{i}\,\mathrm{gd}\left(-\tfrac{1}{i}u\right);$$

or substituting for ϕ its value $\mathrm{gd}\,u$, we have

$$u = \tfrac{1}{i}\,\mathrm{gd}\left(-\tfrac{1}{i}\,\mathrm{gd}\,u\right),$$

which may also be written

$$iu = \operatorname{gd}(i \operatorname{gd} u)\,;$$

so that $\operatorname{gd} u$ is a quasi-periodic function of the second order—a property which has not, at least obviously, any analogue in the general theory. We have

$$\begin{aligned}
\cos \operatorname{gd} u &= \tfrac{1}{2}(e^{i \operatorname{gd} u} + e^{-i \operatorname{gd} u}) \\
&= \tfrac{1}{2}\left(\sqrt{\frac{1+\tan\frac{1}{2}ui}{1-\tan\frac{1}{2}ui}} + \sqrt{\frac{1-\tan\frac{1}{2}ui}{1+\tan\frac{1}{2}ui}}\right) \\
&= \frac{1+\tan^2\frac{1}{2}ui}{1-\tan^2\frac{1}{2}ui} = \frac{1}{\cos ui};
\end{aligned}$$

and in like manner

$$\begin{aligned}\sin\,\mathrm{gd}\,u &= \tfrac{1}{2i}(e^{i\,\mathrm{gd}\,u} - e^{-i\,\mathrm{gd}\,u})\\ &= \frac{1}{2i}\left(\sqrt{\frac{1+\tan\frac{1}{2}ui}{1-\tan\frac{1}{2}ui}} - \sqrt{\frac{1-\tan\frac{1}{2}ui}{1+\tan\frac{1}{2}ui}}\right)\\ &= \frac{2\tan\frac{1}{2}ui}{i(1-\tan^2\frac{1}{2}ui)} = \frac{\sin ui}{i\cos ui};\end{aligned}$$

or, as these equations may also be written,

$$\sec\,\mathrm{gd}\,u = \cos ui = \tfrac{1}{2}(e^u + e^{-u}),$$

$$\tan\,\mathrm{gd}\,u = \tfrac{1}{i}\sin ui = \tfrac{1}{2}(e^u - e^{-u});$$

and from these equations we have

$$\begin{aligned}\sec\,\mathrm{gd}(u+v) &= \sec\,\mathrm{gd}\,u\cdot\sec\,\mathrm{gd}\,v + \tan\,\mathrm{gd}\,u\cdot\tan\,\mathrm{gd}\,v,\\ \tan\,\mathrm{gd}(u+v) &= \tan\,\mathrm{gd}\,u\cdot\sec\,\mathrm{gd}\,v + \tan\,\mathrm{gd}\,v\cdot\sec\,\mathrm{gd}\,u;\end{aligned}$$

or, what is the same thing,

$$\begin{aligned}\sin\,\mathrm{gd}(u+v) &= i\cdot\frac{\sin\,\mathrm{gd}\,u + \sin\,\mathrm{gd}\,v}{1+\sin\,\mathrm{gd}\,u\cdot\sin\,\mathrm{gd}\,v},\\ \cos\,\mathrm{gd}(u+v) &= \frac{\cos\,\mathrm{gd}\,u\cdot\cos\,\mathrm{gd}\,v}{1+\sin\,\mathrm{gd}\,u\cdot\sin\,\mathrm{gd}\,v};\end{aligned}$$

which forms are at once obtainable from the formulae

$$\begin{aligned}\sin\,\mathrm{am}(u+v) &= \frac{\sin\,\mathrm{am}\,u\cos\,\mathrm{am}\,v\Delta\,\mathrm{am}\,v + \sin\,\mathrm{am}\,v\cos\,\mathrm{am}\,u\Delta\,\mathrm{am}\,u}{1-k^2\sin^2\mathrm{am}\,u\sin^2\mathrm{am}\,v},\\ \cos\,\mathrm{am}(u+v) &= \frac{\cos\,\mathrm{am}\,u\cos\,\mathrm{am}\,v - \sin\,\mathrm{am}\,u\sin\,\mathrm{am}\,v\Delta\,\mathrm{am}\,u\Delta\,\mathrm{am}\,v}{1-k^2\sin^2\mathrm{am}\,u\sin^2\mathrm{am}\,v},\\ \Delta\,\mathrm{am}(u+v) &= \frac{\Delta\,\mathrm{am}\,u\Delta\,\mathrm{am}\,v - k^2\sin\,\mathrm{am}\,u\sin\,\mathrm{am}\,v\cos\,\mathrm{am}\,u\cos\,\mathrm{am}\,v}{1-k^2\sin^2\mathrm{am}\,u\sin^2\mathrm{am}\,v},\end{aligned}$$

observing that for $k = 1$ we have $\Delta\,\mathrm{am} = \cos\,\mathrm{am}$, and that the numerators and denominator each of them divide by

$$1 - \sin\,\mathrm{am}\,u\sin\,\mathrm{am}\,v.$$

2, Stone Buildings, W.C., May 7, 1862.

321. Final Remarks on Mr Jerrard's Theory of Equations of the Fifth Order.

[From the *Philosophical Magazine*, vol. xxiv. (1862), p. 290.]

C. V.

322. On the Skew Surface of the Third Order.

[From the *Philosophical Magazine*, vol. xxiv. (1862), pp. 514–519.]

The skew surface of the third order, or "cubic scroll" (disregarding a certain special form), may be considered as generated by a line which always passes through three directrices; viz., a plane cubic having a node, and two lines, one of them meeting the cubic in the node, the other of them meeting the cubic in an ordinary point. The analytical investigation possesses some interest as an illustration of the analytical theory of skew surfaces in general.

Take for the equation of the cubic

$$(a^2+b^2)xy-(a^2+b^2)a\beta=0,$$

which belongs to a cubic having a node at the origin, and passing through the point (a, β); and for the equations of the two lines

$$(x-mz=0, \quad y-nz=0), \quad (x-a=0, \quad y-\beta=0).$$

Then, (X, Y, Z) being current coordinates, the equations of the generating line will be

$$X=x+AZ, \quad Y=y+BZ;$$

when this meets the line $(X-mZ=0, Y-nZ=0)$, we have

$$mZ=x+AZ, \quad nZ=y+BZ,$$

and thence

$$x(n-B)-y(m-A)=0;$$

or, what is the same thing,

$$nx-my-Bx+Ay=0:$$

and when it meets the line $(X-a=0, Y-\beta=0)$, we have

$$a=x+AZ, \quad \beta=y+BZ,$$

and thence

$$B(x-a)-A(y-\beta)=0.$$

We have thus the system of equations

$$\begin{aligned}(a^2+b^2)xy-(a^2+b^2)a\beta&=0,\\ X&=x+AZ,\\ Y&=y+BZ,\\ nx-my-Bx+Ay&=0,\\ B(x-a)-A(y-\beta)&=0;\end{aligned}$$

from which, eliminating (A, B, x, y), we obtain the equation of the surface.

Writing in the last equation

$$B=s(x-a),\quad A=s(y-\beta)$$

(values which give $Bx-Ay=-s(\beta x-ay)$), we find

$$\begin{aligned}X+asZ&=(1+sZ)x,\\ Y+\beta sZ&=(1+sZ)y,\\ (n+\beta s)x-(m+as)y&=0;\end{aligned}$$

whence also

$$(n+\beta s)(X+asZ)-(m+as)(Y+\beta sZ)=0,$$

that is

$$nX-mY+(na-m\beta)sZ+s(\beta X-aY)=0;$$

or eliminating s from this equation and the two equations

$$\begin{aligned}x-X+Z(x-a)s&=0,\\ y-Y+Z(y-\beta)s&=0,\end{aligned}$$

we have

$$\begin{aligned}\{(na-m\beta)Z+\beta X-aY\}(x-X)-Z(x-a)(nX-mY)&=0,\\ \{(na-m\beta)Z+\beta X-aY\}(y-Y)-Z(y-\beta)(nX-mY)&=0;\end{aligned}$$

these give

$$\begin{aligned}\eta x&=X\{(na-m\beta)Z+\beta X-aY\}-aZ(nX-mY)\\ &=-mZ\beta X+X(\beta X-aY)+mZaY\\ &=(X-mZ)(\beta X-aY),\end{aligned}$$

and

$$\begin{aligned}\eta y &= Y\{(na - m\beta)Z + \beta X - aY\} - \beta Z(nX - mY)\\ &= nZaY + Y(\beta X - aY) - nZ\beta X\\ &= (Y - nZ)(\beta X - aY),\end{aligned}$$

where

$$\begin{aligned}\eta &= (na - m\beta)Z + (\beta X - aY) - Z(nX - mY)\\ &= \beta(X - mZ) - a(Y - nZ) - Z\{n(X - mZ) - m(Y - nZ)\}\\ &= (\beta - nZ)(X - mZ) - (a - mZ)(Y - nZ);\end{aligned}$$

so that

$$\begin{aligned}x &= \frac{(X - mZ)(\beta X - aY)}{(\beta - nZ)(X - mZ) - (a - mZ)(Y - nZ)},\\ y &= \frac{(Y - nZ)(\beta X - aY)}{(\beta - nZ)(X - mZ) - (a - mZ)(Y - nZ)},\end{aligned}$$

which equations give the coordinates (x, y) of the point in which the generating line through the point (X, Y, Z) of the surface meets the cubic

$$(a^2 + b^2)xy - (a^2 + b^2)a\beta = 0.$$

Substituting these values of (x, y) in the equation of the cubic, we obtain the equation

$$\begin{aligned}&(a^2 + b^2)(X - mZ)(Y - nZ)\{(\beta - nZ)(X - mZ) - (a - mZ)(Y - nZ)\}\\ &\quad - a\beta(\beta X - aY)\{(X - mZ)^2 + (Y - nZ)^2\} = 0;\end{aligned}$$

or, as it may be written,

$$\begin{aligned}&(a^2 + b^2)(X - mZ)(Y - nZ)\{\beta(X - mZ) - a(Y - nZ)\}\\ &\quad + (a^2 + b^2)(X - mZ)(Y - nZ)Z(mY - nX)\\ &\quad - a\beta(\beta X - aY)\{(X - mZ)^2 + (Y - nZ)^2\} = 0.\end{aligned}$$

This equation contains, however, the extraneous factor

$$\beta(X - mZ) - a(Y - nZ),$$

which, equated to zero, gives the plane through the simple line and the node of the cubic; and omitting this factor, the equation of the surface is obtained.

It only remains to find Φ: writing for this purpose $X+mZ$, $Y+nZ$ in the place of X, Y respectively, and putting for a moment

$$\Phi(X+mZ, Y+nZ, Z) = \Phi',$$

we have

$$\begin{aligned}&(a^2+b^2)XYZ(mY-nZ) - a\beta\{\beta(Z+mZ) - a(Y+nZ)\}(X^2+Y^2)\\&\quad = (\beta X - aY)\Phi';\end{aligned}$$

that is

$$\begin{aligned}(\beta X - aY)\Phi' = {}& Z\{(a^2+b^2)XY(mY-nZ) - (X^2+Y^2)a\beta(m\beta - na)\}\\& - (\beta X - aY)a\beta(X^2+Y^2);\end{aligned}$$

or, effecting the division,

$$\Phi' = Z\{(Z^2a - Y^2b)(an - \beta m) - XY(a^2m + b^2n)\} - a\beta(X^2+Y^2),$$

and then writing $X-mZ$, $Y-nZ$ in the place of X, Y respectively, we have

$$\begin{aligned}\Phi(X,Y,Z) = {}& Z\{((X-mZ)^2a - (Y-nZ)^2\beta)(an-\beta m)\\& - (X-mZ)(Y-nZ)(a^2m+b^2n)\} - a\beta\{(X-mZ)^2 + (Y-nZ)^2\}.\end{aligned}$$

Hence, finally, the equation of the surface is

$$\begin{aligned}&(a^2+b^2)(X-mZ)(Y-nZ) - a\beta\{(X-mZ)^2 + (Y-nZ)^2\}\\&\quad + Z\{((X-mZ)^2a - (Y-nZ)^2\beta)(an-\beta m)\\&\quad - (X-mZ)(Y-nZ)(a^2m+b^2n)\} = 0,\end{aligned}$$

which is, as it should be, of the third order.

Arranging in powers of Z and reducing, the equation is found to be

$$\begin{aligned}&(a^2+b^2)XY - a\beta(X^2+Y^2)\\&\quad + Z\{-(a^2+b^2)(mY+nX) + (X^2a+Y^2\beta)(m\beta+na) + a\beta(mX^2+nY^2) - (a^2n\\&\quad + Z^2\{mn(a^2+b^2-a^2X-b^2Y) + (\beta n^2 - am^2)(\beta X - aY)\} = 0.\end{aligned}$$

The first form puts in evidence the nodal line

$$(X-mZ=0, \quad Y-nZ=0),$$

and the second form puts in evidence the simple line

$$(X - a = 0, \quad Y - \beta = 0).$$

But to obtain a more convenient form, write for a moment $X - mZ = P$, $Y - nZ = Q$; the equation is

$$(a^2+b^2)PQ - a\beta(P^2+Q^2) + Z\{(P^2a - Q^2\beta)(na - m\beta) - PQ(ma^2+nb^2)\} = 0,$$

or, as this may be written,

$$(a^2+b^2)PQ + (a^2P - b^2Q)Z(Pn - Qm) + a\beta\{-P^2 - Q^2 - Z(mP^2+nQ^2)\} = 0;$$

or, observing that $X = P + mZ$, $Y = Q + nZ$, and thence

$$PY - QX = Z(Pn - Qm), \quad X^2P^2 + Q^2Y^2 = P^2 + Q^2 + Z(mP^2 + nQ^2),$$

the equation becomes

$$(a^2 + b^2)PQ + (a^2P - b^2Q)(PY - QX) - a\beta(P^2X + Q^2Y) = 0,$$

or, what is the same thing,

$$(aP^2 - bQ^2)(aY - bX) + PQ(a^2 + b^2 - a^2X - b^2Y) = 0;$$

whence, making a slight change in the form, and restoring for P, Q their values, the equation is

$$\begin{aligned}&\{a(X - mZ)^2 - b(Y - nZ)^2\}\{a(Y - \beta) - b(X - a)\}\\ &\quad - (X - mZ)(Y - nZ)\{a^2(X - a) + b^2(Y - \beta)\} = 0,\end{aligned}$$

a form which puts in evidence as well the simple line $(X - a = 0, Y - \beta = 0)$ as the nodal line $(X - mZ = 0, Y - nZ = 0)$.

If $Z = 0$, we have

$$(aX^2 - bY^2)(aY - b\beta X) - XY\{a^2(X - a) + b^2(Y - \beta)\} = 0,$$

which is in fact the cubic curve $(a^2 + b^2)XY - a\beta(X^2 + Y^2) = 0$.

Reverting to a former system of equations

$$\begin{aligned}nx - my - Bx + Ay &= 0,\\ B(x - a) - A(y - \beta) &= 0,\end{aligned}$$

or, as these may be written,

$$\begin{aligned}Bx - Ay &= nx - my,\\ B\alpha - A\beta &= nx - my,\\ B(\beta x - ay) &= (\beta - y)(nx - my),\\ A(\beta x - ay) &= (a - x)(nx - my);\end{aligned}$$

we find

$$Y - y = \frac{(\beta - y)(nx - my)}{\beta x - ay}Z, \quad X - x = \frac{(a - x)(nx - my)}{\beta x - ay}Z,$$

as the equations of the generating line which passes through the point (x, y) of the cubic curve.

2, Stone Buildings, W.C., October 28, 1862.

323. On a Tactical Theorem Relating to the Triads of Fifteen Things.

[From the *Philosophical Magazine*, vol. xxv. (1863), pp. 59–61.]

The school-girl problem may be stated as follows:—"With 15 things to form 35 triads, involving all the 105 duads, and such that they can be divided into 7 systems, each of 5 triads containing all the 15 things." A more simple problem is, "With 15 things, to form 35 triads involving all the 105 duads."

In the solution which I formerly gave of the school-girl problem (Phil. Mag. vol. XXXVII. 1850, [82]), and which may be presented in the form

abc	d35	e17	f82	g64
ade	f62	g84	b15	c37
afg	b47	c13	d57	e86
bdf	c16	e38	g25	a42
beg	d58	a23	f14	c67
cdg	e12	f78	a56	b34
cef	a36	b45	g27	d18

(viz. the things being a, b, c, d, e, f, g, 1, 2, 3, 4, 5, 6, 7, 8, the first pentad of triads is abc, $d35$, $e17$, $f82$, $g64$, and so for all the seven pentads of triads), there is obviously a division of the 15 things into $(7+8)$ things, viz. the 35 triads are composed 7 of them each of 3 out of the 7 things, and the remaining 28 each of 1 out of the 7 things, and 2 out of the 8 things: or attending only to the 8 things, there are 28 triads each of them containing a duad of the 8 things, but there is no triad consisting of 3 of the 8 things. More briefly, we may say that in the system there is an 8 without 3, that is, there are 8 things such that no triad of them occurs in the system.

I believe, but am not sure, that in all the solutions which have been given of the school-girl problem there is an 8 without 3.

Now, considering the more simple problem, there are of course solutions which have an 8 without 3 (since every solution of the school-girl problem is a solution of the more simple problem): but it is very easy to show that there is no solution which has a 9 without 3. I wish to show that there is in every solution at least a 6 without 3. This being so, there will be (if they all exist) 3 classes of solutions, viz. those which have at most (1) a 6 without 3, (2) a 7 without 3, (3) an 8 without 3. I believe that the first and second classes exist, as well as the third, which is known to do so.

The proposition to be proved is, that given any system of 35 triads involving all the duads of 15 things; there are always 6 things which are a 6 without 3, that is, they are such that no triad of the 6 things is a triad of the system. This will be the case if it is shown that the number of distinct hexads which can be formed each of them containing a triad of the system is less than

$$\frac{15\cdot 14\cdot 13\cdot 12\cdot 11\cdot 10}{6\cdot 5\cdot 4\cdot 3\cdot 2\cdot 1} = 5\cdot 7\cdot 11\cdot 13 = 5005,$$

the entire number of the hexads of 15 things. Now joining to any triad of the system a triad formed out of the remaining 12 things (there are $\frac{12\cdot 11\cdot 10}{3\cdot 2\cdot 1} = 4\cdot 5\cdot 11 = 220$ such triads), we obtain in all $(220\times 35 =)$ 7700 hexads, each of them containing a triad of the system. But these 7700 hexads are not all of them distinct. For, first, considering any triad of the system, there are in the system 16 other triads, each of them having no thing in common with the first-mentioned triad. (In fact if e.g. 123 is a triad of the system, then the system, since it contains all the duads, must have besides 6 triads containing 1, 6 triads containing 2, and 6 triads containing 3, and therefore $35 - 1 - 6 - 6 - 6 = 16$ triads not containing 1, 2, or 3.) Hence we have $(16\times 35 =)$ 560 hexads, each of them containing two triads of the system without any common thing. Next, if $\alpha\beta\gamma$, $\beta\gamma\delta$ are two triads of the system containing a duad $\beta\gamma$, there are in the system 7 triads containing the thing α, and 7 triads containing the thing δ; but one of these triads is in each case the triad $\alpha\beta\gamma$ or $\beta\gamma\delta$; hence the number of pairs of triads having in common only the thing β or γ is $6\times 6 = 36$; and the number of such pairs having in common a duad is $36\times\frac{105}{3} = 36\times 35 = 1260$ (for the whole number of duads is 105, and each triad contains 3 duads).

And for any such pair, combining with $\alpha\beta\gamma\delta\epsilon$ any one of the remaining 10 things, we have 10 hexads $\alpha\beta\gamma\delta\epsilon\zeta$, each of them derivable from either of the triads $\alpha\beta\gamma$, $\beta\gamma\delta$, that is, we have $(315 \times 10 =)\, 3150$ hexads which present themselves twice over among the 7700 hexads. The hexads not belonging to one or other of the foregoing classes are derived each of them from a single triad only of the system, and they present themselves once among the 7700 hexads. This number is consequently made up as follows, viz.

$$\begin{array}{rlr}
280 & \text{hexads each twice} = & 560 \\
3150 & \text{hexads each once} \;\; = & 3150 \\
840 & \text{hexads each once} \;\; = & 840 \\ \cline{3-3}
 & \text{hexads each} & 4270 \\
4270 & \text{hexads each once} \;\; = & 4270 \\
6300 & \text{hexads each} \qquad\;\; = & 6300 \\ \cline{3-3}
 & \text{hexads each} & 7700
\end{array}$$

or there are in all 4270 distinct hexads; and since this is less than 5005, it follows that there are hexads not containing any triad of the system: there must in fact be $(5005 - 4270 =)\, 735$ such hexads. The theorem in question is thus shown to be true.

2, Stone Buildings, W.C., November 24, 1862.
C. V.

324. Note on a Theorem Relating to Surfaces.

[From the *Philosophical Magazine*, vol. xxv. (1863), pp. 61, 62.]

The following apparently self-evident geometrical theorem requires, I think, a proof; viz. the theorem is—"If every plane section of a surface of the order $m + n$ break up into two curves of the orders m and n respectively, then the surface breaks up into two surfaces of the orders m, n respectively."

To fix the ideas, suppose $n = 2$. Imagine any line meeting the surface in $m + 2$ points, the section includes a conic which meets the line in two of the $m + 2$ points, say the points A, A'(1). Suppose that the plane revolves round the line AA', the section will always include a conic which passes through these same two points A, A'; and it is to be shown that the sheet, the locus of this conic, is a surface of the second order. In fact the conic in question, say APA', by its intersection with an arbitrary plane traces out a branch of the intersection of the given surface with the arbitrary plane. And if $ABA'B'$ be the conic in any particular plane through A, A', and if the arbitrary plane meet this conic in the points B, B', then the branch passes through these points B, B'. Imagine the plane $ABA'B'$ revolving round BB' until it coincides with the arbitrary plane; the section includes a conic passing through the points B, B', and the before-mentioned branch is this conic; that is, the conic APA' by its intersection with an arbitrary plane traces out a conic; or, what is the same thing, the sheet, the locus of the conic APA', is met by an arbitrary plane in a conic, that is, the sheet is a surface of the second order; and the given surface thus includes a surface of the second order, and is therefore made up of two surfaces of the orders m and 2 respectively. The demonstration seems to me to add at least

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(1) The figure referred to will be at once understood by considering A, A' as the poles of an ellipsoid, or say of a sphere, $ABA'B'$ the meridian of projection, APA' any other meridian, BPB' the equator or any other great circle.

something to the evidence of the theorem asserted, but I should be glad if a more simple one could be found. Analytically, the theorem is—"If

$$(x, y, z, ax + \beta y + \gamma z)^{m+2},$$

where (α, β, γ) are arbitrary, break up into factors $(x, y, z)^m$, $(x, y, z)^2$, rational in regard to (x, y, z), then $(x, y, z, w)^{m+2}$ breaks up into factors $(x, y, z, w)^m$, $(x, y, z, w)^2$, rational in regard to (x, y, z, w)." It would at first sight appear that (α, β, γ) being arbitrary, these quantities can only enter into the factors of $(x, y, z, \alpha x + \beta y + \gamma z)^{m+2}$ through the quantity $\alpha x + \beta y + \gamma z$; that is, that the factors in question, considered as functions of $(x, y, z, \alpha, \beta, \gamma)$, are of the form

$$(x, y, z, \alpha x + \beta y + \gamma z)^m, \quad (x, y, z, \alpha x + \beta y + \gamma z)^2;$$

and then replacing the arbitrary quantity $\alpha x + \beta y + \gamma z$ by w, the factors of $(x, y, z, w)^{m+2}$ will be $(x, y, z, w)^m$, $(x, y, z, w)^2$. But the objection proves too much; for in a similar way it would follow that if $(x, y, \alpha x + \beta y)^{m+n}$, where α, β are arbitrary, breaks up into the factors $(x, y)^m$, $(x, y)^n$, rational in regard to (x, y) (and *qua* homogeneous function of two variables it always does so break up), then $(x, y, z)^{m+n}$ would in like manner break up into the factors $(x, y, z)^m$, $(x, y, z)^n$, rational in regard to (x, y, z): and a simple example will show that it is not true that the factors of $(x, y, \alpha x + \beta y)^{m+n}$ only contain (α, β) through $\alpha x + \beta y$; in fact, if the function be $= x^2 + y^2 + (\alpha x + \beta y)^2$, then the factor is

$$\frac{1}{\alpha^2 + \beta^2 + 1}\{(\alpha^2 + 1)x + (\alpha\beta + \beta\sqrt{\alpha^2 + \beta^2 + 1})y\},$$

which cannot be exhibited as a function of α, β, $\alpha x + \beta y$.

I am not acquainted with any analytical demonstration; the geometrical one cannot easily be exhibited in an analytical form.

2, Stone Buildings, W.C., November 26, 1862.
13—2

325. Note on a Theorem Relating to a Triangle, Line, and Conic.

[From the *Philosophical Magazine*, vol. xxv. (1863), pp. 181–183.]

I FIND, among my papers headed "Generalization of a Theorem of Steiner's," an investigation leading to the following theorem, viz.:

Consider a triangle, a line, and a conic; with each vertex of the triangle join the point of intersection of the line with the polar of the same vertex in regard to the conic; in order that the three joining lines may meet in a point, the line must be a tangent to a curve of the third class; if, however, the conic break up into a pair of lines, or in a certain other case, the curve of the third class will break up into a point, and a conic inscribed in the triangle.

Let the equations of the sides of the triangle be

$$x = 0, \quad y = 0, \quad z = 0,$$

the equation of the conic

$$(a, b, c, f, g, h \,\$\, x, y, z)^2 = 0,$$

and that of the line

$$\lambda x + \mu y + \nu z = 0;$$

then the polar of the vertex $(y = 0, z = 0)$ has for its equation

$$ax + hy + gz = 0;$$

it therefore meets the line $\lambda x + \mu y + \nu z = 0$ in the point

$$X : Y : Z = h\nu - g\mu : g\lambda - a\nu : a\mu - h\lambda.$$

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325.

NOTE ON A THEOREM RELATING TO A TRIANGLE, LINE, AND CONIC.

[From the *Philosophical Magazine*, vol. xxv. (1863), pp. 238, 239.]

And the equation of the line joining this point with the vertex ($y = 0,\ z = 0$) is

$$(a\mu - h\lambda)y = (g\lambda - a\nu)z.$$

The equations of the three joining lines therefore are

$$\begin{aligned}(a\mu - h\lambda)y &= (g\lambda - a\nu)z,\\ (b\nu - f\mu)z &= (h\mu - b\lambda)x,\\ (c\lambda - g\nu)x &= (f\nu - c\mu)y,\end{aligned}$$

lines which will meet in a point if

$$(a\mu - h\lambda)(b\nu - f\mu)(c\lambda - g\nu) - (g\lambda - a\nu)(h\mu - b\lambda)(f\nu - c\mu) = 0,$$

or, multiplying out and putting as usual

$$\mathcal{K} = abc - af^2 - bg^2 - ch^2 + 2fgh, \quad \mathfrak{A} = bc - f^2 \text{ \&c.},$$

$$\begin{aligned}&\mathfrak{A}\lambda^2\mu\nu + \mathfrak{B}\mu^2\nu\lambda + \mathfrak{C}\nu^2\lambda\mu\\ &\qquad\qquad + \mathfrak{F}\lambda\mu^3 + \mathfrak{G}\mu\nu^3 + \mathfrak{H}\nu\lambda^3 = 0,\end{aligned}$$

that is, the line must touch a curve of the third class.

If this equation break up into factors, the form must be

$$(\alpha\lambda + \beta\mu + \gamma\nu)(A\mu\nu + B\nu\lambda + C\lambda\mu) = 0;$$

that is, we must have

$$\begin{aligned}A\alpha &= \mathfrak{A}, & B\beta &= \mathfrak{B}, & C\gamma &= \mathfrak{C},\\ B\alpha + C\alpha &= 2\mathfrak{F}, & C\alpha + A\gamma &= 2\mathfrak{G}, & A\beta + B\alpha &= 2\mathfrak{H};\end{aligned}$$

and the last six equations give without difficulty

$$\alpha = \frac{\mathfrak{F}\mathfrak{G}\mathfrak{H}}{k\mathfrak{A}}, \quad \beta = \frac{\mathfrak{F}\mathfrak{G}\mathfrak{H}}{k\mathfrak{B}}, \quad \gamma = \frac{\mathfrak{F}\mathfrak{G}\mathfrak{H}}{k\mathfrak{C}},$$

where k is arbitrary; the first equation then gives

$$2\mathfrak{F}\mathfrak{G}\mathfrak{H} = k\mathfrak{B}\mathfrak{C} + k^{-1}\mathfrak{F}^2 \text{ \&c.};$$

or, reducing by the equations $\mathfrak{A}\mathfrak{B}\mathfrak{C} = \mathfrak{A}\mathfrak{F}^2 + \text{\&c.}$, this is

$$2\mathfrak{A}\alpha = 3\mathfrak{B}\mathfrak{C} + \mathfrak{G}\mathfrak{C} - 2\mathfrak{A}\mathfrak{B}\mathfrak{C} + 2\mathfrak{F}\mathfrak{G}\mathfrak{H} + (\mathfrak{A} + \mathfrak{B} + \mathfrak{C})\mathfrak{F}^2 = 0;$$

which, substituting for $\mathfrak{A}$, $\mathfrak{B}\mathfrak{C}$, $\mathfrak{F}$ their values, becomes [325]

Hence if $\mathcal{K} = 0$, that is, if the conic break up into a pair of lines, or if

$$\frac{a^2}{\mathfrak{A}} + \frac{b^2}{\mathfrak{B}} + \frac{c^2}{\mathfrak{C}} + \frac{f^2}{\mathfrak{F}} + \frac{g^2}{\mathfrak{G}} + \frac{h^2}{\mathfrak{H}} = 0,$$

in either case the equation of the curve of the third class becomes

$$(\lambda\mu\nu)\left(\frac{a}{\mathfrak{A}}\lambda + \frac{b}{\mathfrak{B}}\mu + \frac{c}{\mathfrak{C}}\nu\right) = 0;$$

that is, the curve breaks up into a point, and a conic inscribed in the triangle.

In the case where the conic breaks up into a pair of lines, then we have

$$(a, b, c, f, g, h \between x, y, z)^2 = 2(px + qy + rz)(p'x + q'y + r'z),$$

and thence

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \between x, y, z)^2 = -\{(qr'-q'r)x+(rp'-r'p)y+(pq'-p'q)z\}^2;$$

so that the equation in (λ, μ, ν) is

$$\begin{aligned}&\{(qr' - q'r)\lambda + (rp' - r'p)\mu + (pq' - p'q)\nu\}\\ &\quad\times\{pp'(qr' - q')\mu\nu + qq'(rp' - r'p)\nu\lambda + rr'(pq' - p'q)\lambda\mu\} = 0;\end{aligned}$$

where the point represented by the equation

$$(qr' - q'r)\lambda + (rp' - r'p)\mu + (pq' - p'q)\nu = 0$$

is, of course, the intersection of the two lines.

326.

THEOREMS RELATING TO THE CANONIC ROOTS OF A BINARY QUANTIC OF AN ODD ORDER.

[From the *Philosophical Magazine*, vol. xxv. (1863), pp. 206–208.]

I call to mind Professor Sylvester's theory of the canonical form of a binary quantic of an odd order; viz., the quantic of the order $2n+1$ may be expressed as a sum of a number $n+1$ of $(2n+1)$th powers, the roots of which, or say the *canonic roots* of the quantic, are to constant multipliers pres the factors of a certain covariant derivative of the order $(n+1)$, called the *Canonizant*. If, to fix the ideas, we take a quintic function, then we may write

$$(a, b, c, d, e, f \between x, y)^5 = A(lx+my)^5 + A'(l'x+m'y)^5 + A''(l''x+m''y)^5$$

(it would be allowable to put the coefficients A each equal to unity; but there is a convenience in retaining them, and in considering that a canonic root $lx+my$ is only given as regards the ratio $l:m$, the coefficients l, m remaining indeterminate); and then the canonic roots $(lx+my)$, &c. are the factors of the Canonizant

$$\begin{vmatrix} a, & b, & c, & d \\ b, & c, & d, & e \\ c, & d, & e, & f \end{vmatrix}$$

It is to be observed that this reduction of the quantic to its canonical form, i.e. to a sum of a number $n+1$ of $(2n+1)$th powers, is a unique one, and that the quantic cannot be in any other manner a sum of a number $n+1$ of $(2n+1)$th powers.

Professor Sylvester communicated to me, under a slightly less general form, and has permitted me to publish the following theorems:

1. If the second emanant $(X\partial_x + Y\partial_y)^2 U$ has in common with the quantic U a single canonic root, then all the canonic roots of the emanant are canonic roots of the quantic; and, moreover, if the remaining canonic root of the quantic be $rx+sy$, then (X, Y), the facients of emanation, are $= (s, -r)$, or, what is the same thing, they are given by the equation

$$\text{canon}\, U \;\; (X, Y \text{ in place of } x, y) = 0.$$

In fact, considering, as before, the quintic $U = (a, b, c, d, e, f \between x, y)^5$, we have

$$U = A(lx + my)^5 + A'(l'x + m'y)^5 + A''(l''x + m''y)^5,$$

and thence

$$(X\partial_x + Y\partial_y)^2 U = B(lx + my)^3 + B'(l'x + m'y)^3 + B''(l''x + m''y)^3,$$

if for shortness

$$B = 5.4(lX + mY)^2 A, \quad \&\text{c.}$$

Suppose $(X\partial_x + Y\partial_y)^2 U$ has in common with U the canonic root $lx + my$, then

$$(X\partial_x + Y\partial_y)^2 U = C(lx + my)^3 + C'(px + qy)^3,$$

and thence

$$B'(l'x+m'y)^3 + B''(l''x+m''y)^3 = (C - B)(lx+my)^3 + C'(px+qy)^3,$$

which must be an identity; for otherwise we should have the same cubic function expressed in two different canonical forms. And we may write

$$B' = C', \quad l'x + m'y = px + qy, \quad B'' = 0, \quad C = -B,$$

and then we have

$$(X\partial_x + Y\partial_y)^2 U = B(lx + my)^3 + B'(l'x + m'y)^3;$$

so that all the canonic roots of the emanant are canonic roots of the quantic. Moreover, the condition $B'' = 0$ gives $l''X + m''Y = 0$, that is, $X : Y = m'' : -l''$, or writing $rx + sy$ instead of $l''x + m''y$, $X : Y = s : -r$; and the system is

$$U = A(lx + my)^5 + A'(l'x + m'y)^5 + A(rx + sy)^5,$$
$$(s\partial_x - r\partial_y)^2 U = -B(lx + my)^3 + B'(l'x + m'y)^3,$$

which proves the theorem.

2. The two functions, canon U, canon$(X\partial_x + Y\partial_y)^2 U$, have for their resultant

$$\{\text{canon}\, U \ (X, Y \text{ in place of } x, y)\}^{2n},$$

if $2n + 1$ be the order of U.

In fact, in order that the equations

$$\operatorname{canon} U = 0, \quad \operatorname{canon}(X\partial_x + Y\partial_y)^2 U = 0,$$

may coexist, their resultant must vanish; and conversely, when the resultant vanishes, the equations will have a common root. Now if the equation $\operatorname{canon}(X\partial_x + Y\partial_y)^2 U = 0$ has a common root with the equation $\operatorname{canon} U = 0$, all its roots are roots of $\operatorname{canon} U = 0$; and, moreover, if $rx + sy = 0$ be the remaining root of $\operatorname{canon} U = 0$, then $X : Y = s : -r$, that is, we have

$$\operatorname{canon} U \ (X, Y \text{ in place of } x, y) = 0;$$

or the resultant in question can only vanish if the last-mentioned equation is satisfied. It follows that the resultant must be a power of the nihilalum of the equation; and observing that $\operatorname{canon} U$ is of the form $(a, \ldots)^{n+1} \between (x, y)^{n+1}$, i.e. that it is of the degree $n + 1$ as well in regard to the coefficients as in regard to the variables (x, y), it is easy to see that the resultant is of the degree $2n(n + 1)$ as well in regard to the coefficients as in regard to (X, Y); that is, we have $2n$ as the index of the power in question.

3. In particular, if $Y = 0$, the theorem is that the resultant of the functions $\operatorname{canon} U$, $\operatorname{canon} \partial_x^2 U$ is equal to the $2n$th power of the first coefficient of $\operatorname{canon} U$.

Thus for $n = 1$, that is, for the cubic function $(a, b, c, d \between x, y)^3$, we have

$$\operatorname{canon} U = \begin{vmatrix} a, & b, & c \\ b, & c, & d \end{vmatrix} = (ac - b^2,\ ad - bc,\ bd - c^2 \between x, y)^2;$$

$$\operatorname{canon} \partial_x^2 U = \begin{vmatrix} a, & b \\ b, & c \end{vmatrix} = ac - b^2;$$

and the resultant of the two functions is

$$= (ac - b^2,\ ad - bc,\ bd - c^2 \between b,\ -a)^2 = -(ac - b^2)^3,$$

which verifies the theorem.

The theorems were, in fact, given to me in relation to the quantic U and the second differential coefficient $\partial_x^2 U$; but the introduction instead thereof of the second emanant $(X\partial_x + Y\partial_y)^2 U$ presented no difficulty.

2, Stone Buildings, W.C., February 16, 1863.

C. V.

327.

ON THE STEREOGRAPHIC PROJECTION OF THE SPHERICAL CONE.

[From the *Philosophical Magazine*, vol. xxv. (1863), pp. 350–353.]

In order to the tolerable delineation of some figures relating to spherical geometry, I had occasion to consider the stereographic projection of the spherical conic. To fix the ideas, imagine a sphere having its centre in the plane of the paper, and through the centre three rectangular axes, that of x horizontal and that of y vertical, in the plane of the paper, and the axis of z perpendicular to and in front of the plane of the paper. The radius of the sphere is taken equal to unity (so that its intersection by the plane of the paper is the circle radius unity), and the points X, Y, and Z are taken to denote the points where the axes, drawn in the positive direction, meet the surface of the sphere; and the opposite points are called X', Y', and Z'. The eye is supposed to be at Z, and the projection to be made on the plane of the paper.

This being so, and supposing that the axes of coordinates are the principal axes of the spherical conic, the axis of x being the interior axis, and taking ξ, η, ζ as the coordinates of a point on the spherical conic, its equations are

$$-\xi^2 + \eta^2 = \cot^2\beta + \zeta^2 = 0;$$

where it may be remarked that $\tan\beta$, c are the semiaxes of the plane conic which is the gnomonic projection (i.e. the projection by lines through the centre of the sphere) of the spherical conic on the tangent plane at X or X'.

Taking, for a moment, x, y, z as the coordinates of a point on the projecting line (that is, the line through the eye to a point (ξ, η, ζ) on the spherical conic), the equation of this line is

$$\frac{x}{\xi} = \frac{y}{\eta} = \frac{z-1}{\zeta}.$$

And thence putting $z = 0$, x, y will be the coordinates of a point of the projection, and we have

$$\frac{x}{\xi} = \frac{y}{\eta} = \frac{1}{1-\zeta},$$

or, what is the same thing,

$$\xi = \frac{x}{r}, \quad \eta = \frac{y}{r}, \quad \zeta = \frac{r^2 - 1}{r^2 + 1},$$

where $r^2 = x^2 + y^2$; the equations of the spherical conic may be written

$$\xi^2 = c^2(\eta^2 - \zeta^2 \cot^2 \beta);$$

and by eliminating ξ, η, ζ from the four equations, we obtain the equation of the conic. Substituting for ξ and η their values, we find

$$r^2 + 1 = c^2(x^2 + y^2)(1 - r^2),$$

or, observing that the first equation gives

$$\frac{x^2 + y^2 - 1}{r^2 + 1} = \frac{r^2 - 1}{r^2 + 1} = \zeta,$$

and that thence

$$r^2 = \frac{x^2 + y^2 + 1}{1 - \zeta},$$

the equation is

$$(x^2 + y^2 - 1)^2 = 4c^2(x^2 - y^2 \cot^2 \beta).$$

It is now very easy to trace the curve. We see first that the curve is symmetrical with respect to the axes, and that it meets the axis of y in four imaginary points, but the axis of x in four real points, the coordinates whereof are

$$x = \pm(\sqrt{1 + c^2} \pm c),$$

so that the two points on the same side of the centre are the images one of the other in regard to the circle radius unity. Moreover the curve touches the lines $y = \pm x \tan \beta$ at their intersections with the circle. By developing in regard to y, the equation becomes

$$y^4 + 2(x^2 - 1 + 2c^2 \cot^2 \beta)y^2 + (x^2 - 1)^2 - 4c^2x^2 = 0;$$

and putting

$$x = \pm(\sqrt{1 + c^2} + c),$$

the last term vanishes, and the equation gives $y^2 = 0$, or

$$\begin{aligned} y^2 &= 2(1 - x^2 - 2c^2 \cot^2 \beta) \\ &= 4(-c^2 + c^2\sqrt{1 + c^2} - c^2 \cot^2 \beta) \\ &= 4c^2(-c \csc^2 \beta + \sqrt{1 + c^2}), \end{aligned}$$

the upper sign corresponding to the exterior values

$$\pm x = \sqrt{1 + c^2} + c,$$

and the lower sign to the interior values

$$\pm x = \sqrt{1 + c^2} - c.$$

In the former case the values of y are imaginary; in the latter case they are real if

$$\sqrt{1 + c^2} > c \csc^2 \beta,$$

or, what is the same thing, if

$$\sin^2 \beta > \frac{c}{\sqrt{1 + c^2}},$$

that is, if (for a given value of c) β is sufficiently great, but otherwise they are imaginary.

If, as in the annexed figures, $c = \frac{3}{4}$ (and therefore $\sqrt{1 + c^2} = \frac{5}{4}$, $\sqrt{1 + c^2} + c = 2$, $\sqrt{1 + c^2} - c = \frac{1}{2}$), then for the limiting value of β we have

$$\sin^2 \beta = \frac{16}{25} \cdot \frac{9}{16} = .3846, \quad \sin \beta = .62, \text{ or } \beta = 38^\circ \text{ nearly.}$$

In the first figure β is less, in the second figure greater than this value: the form for the limiting value is obvious from a comparison of the two figures.

I take the opportunity to mention the following theorem, which is perhaps known, but I have not met with it anywhere; viz. any three circles, each two of which meet, may be considered as the stereographic projections of three great circles of the sphere. In fact suppose, as above, that the projection is made on the plane of a great circle, and calling this the principal circle, the projection of any other great circle meets the principal circle at the extremities of a diameter of the principal circle. It follows that the theorem will be true, if, given any three circles each two of which meet, a circle can be drawn meeting the given circles, each of them at the extremities of a diameter of the circle so to be drawn. It is easy to see that the required circle has for its centre the radical centre (point of intersection of the radical axes) of the given circles, and that the radius is the 'Inner Potency' of the point in question in regard to each of the three given circles. In particular the three circles having for centres the vertices of an equilateral triangle, and the side for radius, may be considered as the stereographic projections of three great circles of a sphere. This is a very ready mode of delineation of a spherical figure depending on three great circles of the sphere.

2, Stone Buildings, W.C., March 21, 1863.

328.

ON THE DELINEATION OF A CUBIC SCROLL.

[From the *Philosophical Magazine*, vol. xxv. (1863), pp. 528–530.]

Imagine a cubic scroll (skew surface of the third order) generated by lines each of which meets two given directrix lines. One of these is a nodal (double) line on the surface, and I call it the *nodal directrix*; the other is a single line on the surface, and I call it the *single directrix*. The section by any plane is a cubic passing through the points in which the plane meets the directrix lines; i.e. the point on the nodal directrix is a node (double point) of the curve, the point on the single directrix a single point on the curve; the two directrix lines, and the cubic curve, the section by any plane, determine the scroll.

Consider the sections by a series of parallel planes. Let one of these planes be called the *basic plane*, and the section by this plane the *basic section* or *basic cubic*; and imagine any other section projected on the basic plane by lines parallel to the nodal directrix: such section may be spoken of simply as 'the section,' and its projection as 'the cubic.' The cubic has a node at the node of the basic cubic; that is, the two curves have at this point four points in common. The two curves have, moreover, in common the three points at infinity (or, in other words, their asymptotes are parallel); in fact the points at infinity of either curve are the points in which the line at infinity, the intersection of the basic plane and the plane of the section, meets the scroll; and these points are therefore the same for each of the two curves. The remaining two points of intersection of the cubic with the basic cubic are also fixed points on the basic cubic, i.e. they are the points of intersection of the basic plane by the two generating lines parallel to the nodal directrix. Hence the cubic meets the basic cubic in nine fixed points, viz. the node counting as four points, the three points at infinity, and the two points the feet of the generators parallel to the nodal directrix. It follows that if $U = 0$ is the equation of the basic cubic, $V = 0$ the equation of some other cubic meeting the basic cubic in the nine points in question, then the equation of 'the cubic' is $U + \lambda V = 0$, λ being a parameter the value of which varies according to the position (in the series of parallel planes) of the plane of the section.

Suppose that the basic cubic $U = 0$ is given, and suppose for a moment that the cubic $V = 0$ is also given, these two cubics having the above-mentioned relations, viz. they have a common node and parallel asymptotes: the cubic $U + \lambda V = 0$ might be constructed by drawing through the node (say O) a radius vector meeting the cubics in P, P' respectively, and taking on this radius vector a point Q such that $\overline{PQ} = -\lambda \cdot \overline{PP'}$, or, what is the same thing, $\overline{OQ} = \overline{OP} - \lambda(\overline{OP'} - \overline{OP})$; the locus of the point Q will then be the cubic $U + \lambda V = 0$. And we may even suppose the cubic $V = 0$ to break up into a line and a conic (hyperbola), and then (disregarding the line) use the hyperbola in the construction. In fact, if the hyperbola is determined by the following five conditions, viz. to pass through the node and through the feet of the two generators parallel to the nodal directrix, and to have its asymptotes parallel to two of the asymptotes of the basic cubic, and if the line be taken to be a line through the node parallel to the third asymptote of the basic cubic; then the hyperbola and line form together a cubic curve meeting the basic cubic in the nine points, and therefore satisfying the conditions assumed in regard to the cubic $V = 0$. And it is to be noticed that as in general the cubic $V = 0$ is the projection of some section of the scroll, so the hyperbola and line are the projection of a section of the scroll, viz. the section through one of the generating lines (there are three such lines) parallel to the basic plane.

But it is better to construct 'the cubic' by a different method (using only the basic cubic $U = 0$) which results more immediately from the geometrical theory. Taking the basic plane as the plane of the figure, let O be the node, or foot of the nodal directrix, K the foot of the single directrix, Kk the projection of the single directrix. Drawing through O any radius vector meeting the basic cubic in P, and the line Kk in r, and producing it to a properly determined point Q, then $OPrQ$ will be the projection of the generating line which meets the nodal directrix, the basic cubic, the single directrix, and the section in the points the projections whereof are O, P, r, Q respectively: and the consideration of the solid figure shows easily that the condition for the determination of the point Q is

$$\overline{PQ} = \overline{Kk} \cdot \frac{\overline{OP}}{\overline{Or}}.$$

Hence, starting from the basic cubic and the line Kk, we have a construction for the point Q the locus whereof is the cubic, the projection of a section of the scroll; for the projections of the parallel sections, we have only to vary the length Kk. By what precedes, the construction gives for the locus of Q a cubic having a node at O, and having its asymptotes parallel to those of the basic cubic. As P moves up to K, [328] the distances Pr, rK become indefinitely small; but their ratio is finite, hence the cubic, the locus of Q, does not pass through the point K. The construction shows, however, that it does pass through the points L, M, which are the other two intersections of Kk with the basic cubic; these points L, M are in fact the feet of the generators parallel to the nodal directrix.

The general conclusion is, that a series of cubics having each of them at one and the same given point a node — having their asymptotes parallel — and besides passing through the same two given points — may be considered as the projections of a series of parallel sections of a cubic scroll; and such a series of cubics will thus afford a delineation of the scroll.

2, Stone Buildings, W.C., April 15, 1863.

329.

NOTE ON THE PROBLEM OF PEDAL CURVES.

[From the *Philosophical Magazine*, vol. xxvi. (1863), pp. 20, 21.]

It is not, so far as I am aware, generally known that the problem of pedal curves (Steiner's *Fusspuncten-Curve*) was considered by Maclaurin in the *Geometria Organica*, 1720. He appears to have been led to it through an idea such as Sir W. R. Hamilton's Hodograph, or at any rate with a view to a dynamical application, for he remarks, p. 95, "Cum vero geometria quae curvas ad datum centrum relatas contemplatur in philosophia naturali ad motus corporum et vires evolvendas facilius applicari possit,... hac sectione considerabimus curvas tanquam ad punctum quodvis datum relatas ex quo ad omnia circumferentiae puncta radii undique educuntur, et simul perpendicularia in illorum punctorum tangentes demittuntur, et rationem radii ad perpendiculum tanquam curvae characterem usurpabimus." And accordingly, Props. IX. to XII., he considers the problem: Given a point S in the plane of a given curve, to find the locus of the intersection of a tangent of the curve by the perpendicular let fall upon it from the point S; with some special cases, and deductions from it. In particular if the given curve be a circle, the locus in question (or pedal curve) is a curve of the fourth order having a double point S; viz. if S be inside the circle, this is a conjugate or isolated point; but if outside, a double point with two real branches: if S be on the circle, then instead of the double point we have a cusp: it is shown that in each case the pedal curve is in fact an epicycloid. If the given curve be a parabola, then the locus or pedal curve is a curve of the third order, viz. a defective hyperbola having a double point at S, and with its single asymptote perpendicular to the axis of the parabola: some particular cases are specially noticed. If the curve be an ellipse or hyperbola, then, as in the case of the circle, the locus or pedal curve is a curve of the fourth order having a double point at S. And it is moreover shown, Prop. XII., that for any given curve whatever the locus or pedal curve is, in a generalized sense of the term, an epicycloid. This is in fact seen very easily by a mere inspection of the figure. Imagine the curve $O'P'$, rigidly connected with and carrying along with it the point S', to roll on the similar and equal fixed curve OP symmetrically situate on the other side of the common tangent OM

or OM'; then when P' coincides with P, the point S' is brought to S'', where $SNN'S''$ is the perpendicular from S on the tangent PN or PN', and $SN = N'S''$, that is, $SS'' = 2SN$; [329] and the curve generated by S'' (that is S'), or say the epicycloid the locus of S', is a curve similar to and similarly situate with the pedal curve the locus of N, but of twice the linear magnitude of the pedal curve. Or, what is the same thing, if instead of the given curve we consider a similar and similarly situated curve of twice the linear magnitude (the point S being the centre of similitude), then the epicycloid the locus of S' is the pedal curve of the substituted curve in relation to the point S.

It may be added that, in accordance with a theorem of Dandelin's, if rays proceeding from the point S are reflected at the given curve, then the epicycloid (or pedal) in question is the secondary caustic, or an orthogonal trajectory of the reflected rays.

2, Stone Buildings, W.C., June 3, 1863.

330.

ON DIFFERENTIAL EQUATIONS AND UMBILICI.

[From the *Philosophical Magazine*, vol. xxvi. (1863), pp. 373–379 and 441–452.]

I.

Consider the integral equation

$$Az^2+2Bz+C=0,$$

where z is the constant of integration: the derived equation is

$$\begin{aligned}\Omega &= (AC'+A'C-2BB')^2-4(AC-B^2)(A'C'-B'^2)\\ &= (CA'-C'A)^2-4(AB'-A'B)(BC'-B'C),=0;\end{aligned}$$

and if for greater simplicity we write $A=1$, then the derived equation is

$$\Omega=C'^2-4BC'+4CB'^2=0,$$

corresponding to the integral equation

$$z^2+2Bz+C=0.$$

Writing the integral equation under the form

$$(z+X)(z+Y)=0,$$

we have

$$2B=X+Y,\quad C=XY,$$

whence also

$$2B'=X'+Y',\quad C'=XY'+X'Y,$$

and the derived equation becomes

$$\Omega=-(X-Y)^2X'Y'.$$

Hence if we represent the roots X, Y via the form $P+Q\sqrt{D}$, so that $P=-B$, $Q\sqrt{D}=\sqrt{B^2-AC}$, Q being the greatest square factor of B^2-AC, then

$$(z-X')^2=4Q^2D,\quad X'Y'=P'^2-Q'^2(2Q'D+QD'),$$

and the derived equation is

$$\Omega = -Q^2\{4Q'^2D - (2Q'n + QD')^2\} = 0.$$

If B, C, &c. are functions of the coordinates (x, y), the equation $z^2 + 2Bz + C = 0$ (z an arbitrary constant) represents a series of curves in the plane of xy; but if we consider z as a coordinate, then the equation represents a surface, and the curves in question are the orthogonal projections on the plane of xy of the sections of the surface by the planes parallel to the plane of xy. To fix the ideas, the plane of xy may be taken to be horizontal, and the ordinates z vertical.

Writing the equation in the form

$$(z + B)^2 - (B^2 - C) = 0,$$

we see that the surface contains upon it the curve $z+B = 0$, $B^2-C = 0$, which is the line of contact with the circumscribed vertical cylinder: such curve may be termed the *envelope*, or, when this is necessary, the *complete envelope*. The equation of the surface has however been taken to be $(z - P)^2 - Q^2D = 0$ (viz. it has been assumed that $B = -P$, $B^2 - C = Q^2D$); the envelope thus breaks up into the curve, $z - P = 0$, $Q = 0$, taken twice, and the curve $z - P = 0$, $D = 0$; the former of these is in general a *nodal curve* on the surface, and it may be spoken of as the nodal curve; the latter of them is the *reduced* or *proper envelope*, or simply the envelope. And the terms nodal curve and envelope may also be applied to the curves $Q = 0$ and $D = 0$, which are the projections on the plane of xy of the first-mentioned two curves respectively.

There is however a case of higher singularity which it is proper to consider: suppose that Q and D have a common factor K, say $Q = KR$, $D = KV$; the complete envelope $Q^2D = K^3R^2V = 0$ here breaks up into the nodal curve $R = 0$ twice, the cuspidal curve $K = 0$ three times, and the reduced or proper envelope $V = 0$ once.

Reverting for a moment to the form $(z + X)(z + Y) = 0$, the derived equation $\Omega = -(X - Y)^2X'Y' = 0$ is satisfied by $(X - Y)^2 = 0$; this equation, or say the equation of the envelope, being in fact the singular solution of the differential equation. This assumes however that the differential equation is given in the form in which it is immediately obtained by derivation from the integral equation, without the rejection of factors which are functions of the coordinates (x, y) only; it is proper to consider the reduced equation obtained by rejecting such factors. Thus if X and Y are rational functions, the reduced form is $X'Y' = 0$, which is no longer satisfied by the equation $(X-Y)^2 = 0$. In the before-mentioned case where the roots are $P \pm Q\sqrt{D}$ (or $(X-Y)^2 = Q^2D$), P, Q, and D being rational functions of (x, y), the derived equation

$$\Omega = -Q^2\{4nP'^2 - (2Q'n + QD')^2\} = 0$$

divides out by the factor Q^2, but it does *not* divide out by D; the reduced form is therefore

$$4nP'^2 - (2Q'n + QD')^2 = 0,$$

which is not satisfied by $Q = 0$, while it is still satisfied by $D = 0$ (since this gives also $n' = 0$); that is, the nodal curve $Q = 0$ is not a solution of the differential equation, but we still have the singular solution $D = 0$, which corresponds to the reduced or proper envelope. In the case $Q = KR$, $D = KV$ of a cuspidal curve, the above form of the differential equation becomes

$$\{K^2VP'^2 - [2KK'RV + K''(2VR' + V'R)]^2\} = 0,$$

which divides out by K; and, when reduced by the rejection of this factor, it is no longer satisfied by the equation $K = 0$, which belongs to the cuspidal curve; that is, neither the nodal curve $R = 0$ nor the cuspidal curve $K = 0$ is a solution of the differential equation, but we still have the singular solution $V = 0$, which corresponds to the reduced or proper envelope. It would appear that the conclusion may be extended to singularities of a higher nature, viz. the factor corresponding to any singular curve which presents itself as part of the complete envelope divides out from the derived equation; and such singular curve does not constitute a solution of the reduced equation, but we have a singular solution corresponding to the reduced or proper envelope.

II.

Consider the differential equation

$$y(p^2 - 1) + 2mxp = 0,$$

where, to fix the ideas, $m >$ or $= 1$; the integral equation may be taken to be

$$z = (mx + \sqrt{m^2x^2 + y^2})(mx^2 + y^2 + x\sqrt{m^2x^2 + y^2})^{m-1};$$

or rather, writing for shortness $D = m^2x^2 + y^2$, and putting

$$z = (mx + \sqrt{D})(mx^2 + y^2 + x\sqrt{D})^{m-1} = P + Q\sqrt{D},$$

the integral equation is

$$(z - P)^2 - Q^2D = 0, \text{ or } z^2 - 2Pz + P^2 - Q^2D = 0,$$

where

$$P^2-Q^2D = (m^2x^2-D)\{(mx^2+y^2)^2-x^2D\}^{m-1} = -y^{2m}\{y^2+(2m-1)x^2\}^{m-1}.$$

In the particular case $m = 1$ the equation is

$$z = x + \sqrt{x^2 + y^2}, \text{ or } z^2 - 2zx - y^2 = 0.$$

Before going further, I remark that, m being a positive integer greater than unity, we have

$$z = P+Q\sqrt{D} = mx(mx^2+y^2)^{m-1}+\{mx^2+y^2+(m-1)mx^2\}(mx^2+y^2)^{m-2}\sqrt{D}+\&c.,$$

the subsequent terms being divisible, the rational ones by D, and the irrational ones by $D\sqrt{D}$. Hence, observing that

$$mx^2 + y^2 + (m - 1)mx^2 = m^2x^2 + y^2 = D,$$

we see that Q contains the factor D, and the equation $D = 0$ belongs to a cuspidal curve on the surface. If however $m = 1$, then the equation is $z = x + \sqrt{D}$, so that $Q = 1$ does not contain the factor D; and $D = x^2 + y^2 = 0$ is not a singular curve on the surface, but is in fact the reduced or proper envelope.

The curve represented by the integral equation will pass through the origin ($x = 0$, $y = 0$) for the value $z = 0$ of the constant of integration. In fact, for this value, the integral equation becomes

$$-y^{2m}\{y^2 + (2m - 1)x^2\}^{m-1} = 0,$$

which belongs to a set of $2m + (m - 1) + (m - 1)$ lines coinciding with the lines $y = 0$, $y = ix\sqrt{2m-1}$, and $y = -ix\sqrt{2m-1}$ respectively. The directions at the origin are therefore $p = 0$, $p = \pm i\sqrt{2m-1}$, which are the same values of p as are obtained from the differential equation; viz. since this is satisfied identically at the point in question, proceeding to the derived equation, we have

$$p(p^2 - 1) + 2mp = 0,$$

that is

$$p(p^2 + 2m - 1) = 0;$$

but it is to be observed that these values of p are different from the values given by the equation $D = m^2x^2 + y^2 = 0$, which are $p = \pm im$. The reason is that the curve $D = 0$ being, as was shown, a cuspidal curve on the surface, the equation $D = 0$ is not a solution of the differential equation.

If however $m = 1$, then the integral equation gives at the origin no longer three values of p, but only the value $p = 0$. The differential equation however gives, as in the general case, three values; viz. we have $p(p^2 + 1) = 0$; and the values $p = \pm i$ obtained from the factor $p^2 + 1 = 0$ are precisely the values of p obtained from the equation $D = x^2 + y^2 = 0$, which in the case now under consideration belongs to the reduced or proper envelope of the surface, and is therefore the singular solution of the differential equation.

III.

The two curves of curvature which pass through any given point of a surface are distinct curves, not branches of one indecomposable curve. In fact if P, Q are the two curves of curvature for a point A, then for a point A' on P the two curves of curvature will be P, Q'; and if P, Q were branches of an indecomposable curve, then P, Q' would also be branches of an indecomposable curve, and we should have P a branch of two different indecomposable curves, which is of course impossible. In the case of an umbilicus, the two curves P and Q coincide together; or, as we may express it, the curves of curvature through an umbilicus are the duplication of a single, in general indecomposable, curve; and in general this curve has at the umbilicus a *trifid node*. I use this expression to denote a point at which there are three distinct tangents, or, more accurately, three distinct directions of the curve: an ordinary triple point is of necessity a trifid node, but not conversely. The umbilicus of an ellipsoid or other quadric surface is a peculiar exceptional case.

In support of the foregoing conclusions, consider a surface having an umbilicus at the origin, and take $z = 0$ as the equation of the tangent plane at that point; the equation of the surface in the neighbourhood of the umbilicus will be

$$z = \tfrac{1}{2}k(x^2 + y^2) + \tfrac{1}{6}(ax^3 + 3bx^2y + 3cxy^2 + dy^3);$$

so that, writing as usual p and q for the first, and r, s, t for the second, differential coefficients of z, we have

$$\begin{aligned} p &= kx + \tfrac{1}{2}(ax^2 + 2bxy + cy^2), \\ q &= ky + \tfrac{1}{2}(bx^2 + 2cxy + dy^2), \\ r &= k + ax + by, \\ s &= bx + cy, \\ t &= k + cx + dy. \end{aligned}$$

The differential equation of the curves of curvature projected on the plane of xy is

$$(1 + p^2 + q^2)s - pqt + p[(1 + q^2)r - (1 + p^2)t] - [(1 + p^2)s - pqr] = 0;$$

and substituting therein the foregoing values of p, q, r, s, t, but attending only to the terms of the lowest order in (x, y), and using moreover in the sequel p in the place of $\frac{dy}{dx}$, the equation becomes

$$(bx + cy)(p^2 - 1) + [(a - c)x + (b - d)y]p = 0;$$

which may be taken as the differential equation of the curves of curvature at and in the neighbourhood of the umbilicus. The equation is satisfied identically by the values $x = 0$, $y = 0$, which correspond to the umbilicus; and to find p, we have to differentiate the equation, and then substitute these values of x and y; we thus obtain

$$(b + cp)(p^2 - 1) + [(a - c) + (b - d)p]p = 0,$$

or, what is the same thing,

$$p(a + 2bp + cp^2) - (b + 2cp + dp^2) = 0,$$

a cubic equation for the determination of p.

I remark that we may without loss of generality write $d = 0$: but to simplify the investigation, I suppose in the first instance that we have also $b = 0$; this comes to assuming that one of the three planes

$ax^3 + 3bx^2y + 3cxy^2 + dy^3 = 0$ bisects the angle formed by the other two planes. The differential equation consequently is

$$cy(p^2 - 1) + (a - c)xp = 0;$$

or, putting for shortness

$$\frac{a - c}{2c} = m,$$

it is

$$y(p^2 - 1) + 2mxp = 0,$$

which is the differential equation previously considered. Hence, writing now h in the place of z, the equation of the curve of curvature in the neighbourhood of the umbilicus is

$$h = (mx + \sqrt{D})(mx^2 + y^2 + x\sqrt{D})^{m-1} = P + Q\sqrt{D},$$

where $D = m^2x^2 + y^2$; or, what is the same thing, the equation is

$$h^2 - 2Ph + P^2 - Q^2D = 0;$$

and the equation (in the neighbourhood of the umbilicus) of the curve through the umbilicus is

$$P^2 - Q^2D = -y^{2m}\{y^2 + (2m - 1)x^2\}^{m-1} = 0;$$

so that the umbilicus is a trifid node. In the case however of an ellipsoid or other quadric surface, we have $m = 1$, so that the equation of the curve of curvature in the neighbourhood of the umbilicus is

$$h = x + \sqrt{x^2 + y^2},$$

or, what is the same thing,

$$h^2 - 2hx - y^2 = 0 :$$

and for the curve through the umbilicus, in the neighbourhood of the umbilicus, the equation is $y^2 = 0$, so that there is only a single direction of the curve of curvature. The differential equation gives, however, at the umbilicus $p(p^2 + 1) = 0$; the value $p = 0$ is that which corresponds to the curve of curvature; the other two values $p = \pm i$ correspond to the curve (pair of lines) $x^2 + y^2 = 0$, which is the envelope of the curves of curvature, or, more accurately, the envelope of the projections of the curves of curvature on the tangent plane at the umbilicus.

Blackheath, October 17, 1863.

C. V. 16

IV.

The differential equation for the curves of curvature in the neighbourhood of an umbilicus was obtained in a form such as

$$(bx+cy)(p^2-1)+2(fx+gy)p=0;$$

and it was only because this equation did not appear to be readily integrable, that I considered, instead of it, the particular form

$$y(p^2-1)+2mxp=0.$$

But the general equation can be integrated; and the result presents itself in a simple form. For, returning to the differential equation

$$(bx+cy)(p^2-1)+2(fx+gy)p=0,$$

and assuming

$$\frac{bx+cy}{fx+gy}=\frac{-2v}{v^2-1},$$

or

$$(bx+cy)(v^2-1)+2(fx+gy)v=0,$$

we have

$$\frac{dy}{dx}=v, \quad \text{or} \quad (p-v)(vp+1)=0,$$

and we may write

$$p-v=0.$$

Assuming also

$$v=\frac{y}{x}u, \quad \text{or} \quad u=\frac{vx}{y},$$

the relation between u and v is

$$v^2-1=\frac{b+cu}{f+gu},$$

or, as this may be written,

$$v=\frac{-2(f+gu)-\sqrt{4(f+gu)^2+(b+cu)^2}}{b+cu},$$

giving

$$v = \frac{-2(f+gu) - \sqrt{U}}{b+cu},$$

where for convenience the radical has been taken with a negative sign. We have moreover

$$p = \frac{b(v^2-1)+2fv}{c(v^2-1)+2gv}.$$

The equation $p-v=0$, substituting for y its value ux, then becomes

$$x\frac{du}{dx} + u - v = 0;$$

or, as this may be written,

$$\frac{dx}{x} \cdot \frac{du}{v-u} = 0.$$

Or, what is the same thing,

$$\frac{dx}{x} + \frac{dv-du}{v-u} - \frac{dv}{v-u} = 0.$$

But

$$\frac{b(v^2-1)+2fv}{c(v^2-1)+2gv} - v = \frac{V}{c(v^2-1)+2gv},$$

where

$$\begin{aligned} V &= v[c(v^2-1)+2gv] + b(v^2-1) + 2fv \\ &= (b+cv)(v^2-1) + 2(f+gv)v, \end{aligned}$$

and the differential equation takes thus the form

$$\frac{dx}{x} + \frac{dv-du}{v-u} - \frac{[c(v^2-1)+2gv]\,dv}{V} = 0;$$

and hence, writing

$$V = (b+cv)(v^2-1) + 2(f+gv)v = c(v-\alpha)(v-\beta)(v-\gamma),$$

and

$$\frac{c(v^2-1)+2gv}{c(v-\alpha)(v-\beta)(v-\gamma)} = \frac{A}{v-\alpha} + \frac{B}{v-\beta} + \frac{C}{v-\gamma},$$

so that

$$A = \frac{c(\alpha^2-1)+2g\alpha}{c(\alpha^2-1)+2g\alpha+2\{f+(b+c)\alpha+c\alpha^2\}},$$

with the like values for B and C — values which are such that $A+B+C=1$, — the integral equation is

$$\text{const.} = x(v-u)(v-\alpha)^{-A}(v-\beta)^{-B}(v-\gamma)^{-C},$$

or, substituting for u its value, $= \frac{y}{x(v-u)}$, &c.

But

$$\text{const.} = x\{c(v^2-1)+2gv\}^{-1}(v-\alpha)^{1-A}(v-\beta)^{1-B}(v-\gamma)^{1-C}.$$

If for shortness $U=(b+cu)^2+(f+gu)^2$, and if

$$\begin{aligned} v-\alpha &= \frac{-2(f+gu)+(b+cu)\sqrt{U}-\alpha(b+cu)}{b+cu} \\ &= \frac{-2(f+gu)-\sqrt{U}-\alpha(b+cu)}{b+cu}, \quad \text{\&c.}, \end{aligned}$$

and observing that the exponent of $b+cu$ is

$$(-2+1-A+1-B+1-C, = 1-A-B-C) = 0,$$

the integral equation is

$$\text{const.} = x(f+gu+\sqrt{U})^{-1}\times\cdots$$

or, observing that the exponent 1 of x is

$$= -1+(1-A)+(1-B)+(1-C),$$

and putting for shortness $D=(fx+gy)^2+(bx+cy)^2$, the integral equation finally is

$$\text{const.} = (fx+gy+\sqrt{D})^{-1}\times$$
$$\{fx+gy+\alpha(bx+cy)+\sqrt{D}\}^{1-A}\{fx+gy+\beta(bx+cy)+\sqrt{D}\}^{1-B}\{fx+gy+\gamma(bx+cy)+\sqrt{}$$

where the quantities α, β, γ, A, B, C are given by

$$\begin{gathered} (b+cv)(v^2-1)+2(f+gv) = c(v-\alpha)(v-\beta)(v-\gamma), \\ \frac{c(v^2-1)+2gv}{c(v-\alpha)(v-\beta)(v-\gamma)} = \frac{A}{v-\alpha}+\frac{B}{v-\beta}+\frac{C}{v-\gamma}. \end{gathered}$$

Consider the curve

$$0 = \{fx+gy+\alpha(bx+cy)+\sqrt{D}\}^{1-A}\{fx+gy+\beta(bx+cy)+\sqrt{D}\}^{1-B} \times \{fx+gy+\gamma(bx+cy)+\sqrt{D}\}^{1-C},$$

which corresponds to the value $= 0$ of the constant. If, for instance,

$$fx+gy+\alpha(bx+cy)+\sqrt{D}=0,$$

this equation gives

$$(bx+cy)\{(bx+cy)(\alpha^2-1)+2(fx+gy)\alpha\}=0;$$

or say

$$(bx+cy)(c\alpha^2-1)+2(fx+gy)\alpha=0.$$

But we have

$$(b+c\alpha)(\alpha^2-1)+2(f+g\alpha)\alpha=0,$$

and the equation therefore is

$$(bx+cy)(f+g\alpha)-(fx+gy)(b+c\alpha)=0;$$

that is

$$(cf-bg)(y-\alpha x)=0;$$

or simply $y-\alpha x=0$; that is, the directions of the curve at the origin, or point $x=0$, $y=0$, are given by the equations $y-\alpha x=0$, $y-\beta x=0$, $y-\gamma x=0$. This is right, since from the differential equation we obtain at the origin

$$(b+cp)(p^2-1)+2(f+gp)p=c(p-\alpha)(p-\beta)(p-\gamma)=0.$$

V.

The particular case of the equation

$$y(p^2-1)+2mxp=0$$

is obtained from the general equation by writing therein $b=0$, $c=1$, $g=0$, $f=m$; we have therefore

$$v(v^2+2m-1)=(v-\alpha)(v-\beta)(v-\gamma),$$

or say

$$\alpha=0, \quad \beta=i\sqrt{2m-1}, \quad \gamma=-i\sqrt{2m-1};$$

and thence

$$\frac{v^2-1}{v(v^2+2m-1)}=\frac{-1}{2m-1}\cdot\frac{1}{v}+\frac{1}{2(2m-1)}\cdot\frac{1}{v+i\sqrt{2m-1}}+\frac{1}{2(2m-1)}\cdot\frac{1}{v-i\sqrt{2m-1}}$$

giving

$$A=\frac{-1}{2m-1}, \quad B=C=\frac{1}{2(2m-1)}.$$

The integral equation thus is

$$\text{const}=(mx+\sqrt{D})^{\frac{-1}{2m-1}}(mx+\sqrt{D})^{\frac{m-1}{2m-1}}\{(mx+i\sqrt{2m-1}\,y+\sqrt{D})(mx-i\sqrt{2m-1}\,y+$$

where $D=m^2x^2+y^2$; or, observing that

$$\begin{aligned}&(mx+i\sqrt{2m-1}\,y+\sqrt{D})(mx-i\sqrt{2m-1}\,y+\sqrt{D})\\&=(mx+\sqrt{D})^2+(2m-1)y^2\\&=2m(m^2x^2+y^2+x\sqrt{D}),\end{aligned}$$

the integral equation is

$$\text{const.}=(mx+\sqrt{D})^{-1}(m^2x^2+y^2+x\sqrt{D})^{m-1},$$

or, what is the same thing,

$$\text{const.}=(mx+\sqrt{D})(m^2x^2+y^2+x\sqrt{D})^{m-1},$$

the result given in the former part of the present paper.

VI.

I annex the following *a posteriori* verification of the solution

$$\text{const.} = (mx + \sqrt{D})(m^2x^2 + y^2 + x\sqrt{D})^{m-1}$$

of the particular equation

$$y(p^2 - 1) + 2mxp = 0.$$

Putting for shortness

$$A = mx + \sqrt{D}, \qquad B = m^2x^2 + y^2 + x\sqrt{D},$$

where it will be remembered that

$$D = m^2x^2 + y^2,$$

then we have

$$2mB = A^2 + (2m - 1)y^2.$$

The integral equation may be written

$$h = P + Q\sqrt{D} = AB^{m-1};$$

and we have

$$\frac{dh}{dx} = \frac{A'}{A} + (m-1)\frac{B'}{B} = \frac{\Theta}{A},$$

if

$$\Theta = \frac{A'}{A}\{A^2 + (2m-1)y^2\} + 2(m-1)AB'.$$

But we have

$$A'\sqrt{D} = m\sqrt{D} + m^2x + yp = mA + yp,$$

$$\begin{aligned} B'\sqrt{D} &= (2m^2x + 2yp)\sqrt{D} + D + x(m^2x + yp) \\ &= 2m^2x^2 + y^2 + xyp + (2mx + 2yp)\sqrt{D} \\ &= A^2 + ypx + 2y\sqrt{D}, \end{aligned}$$

and

$$\frac{1}{A} = \frac{A}{2mB} = \frac{A}{A^2 + (2m-1)y^2};$$

and the value of Θ thus is

$$\Theta = \frac{1}{A^2 + (2m-1)y^2}\Big\{(mA + yp)(A^2 + (2m-1)y^2) \\ + 2(m-1)A(A^2 + ypx + 2y\sqrt{D})\Big\},$$

where the expression in $\{\}$ is

$$= (2m^2 - m)A(x^2 + y^2) \\ + yp\{A^2 + (2m-1)y^2 + (2m^2 - 2m)A(x + 2\sqrt{D})\}.$$

Here the coefficient of yp is $= (2m^2 - m)(A^2 + y^2)$; in fact we have identically

$$A^2 + y^2 - 2A\sqrt{D} = 0,$$

and thence

$$(2m^2 - 3m + 1)(A^2 + y^2) - 2(2m-1)(m-1)A\sqrt{D} = 0,$$

that is

$$(2m^2-m-1)A^2+(2m^2-3m+1)y^2-(2m^2-2m)A\{x+(2m-1)\sqrt{D}\} = 0;$$

or

$$(2m^2-m-1)A^2+(2m^2-3m+1)y^2-(2m^2-2m)A(x+2\sqrt{D})=0,$$

and therefore

$$A^2+(2m-1)y^2+(2m^2-2m)A(x+2\sqrt{D})=(2m^2-m)(A^2+y^2).$$

[330] Hence the term in {} is

$$=(2m^2-m)(x^2+y^2)(A+yp);$$

or, what is the same thing, it is $=(4m^2-2m)A\sqrt{D}(A+yp)$. Hence, restoring for $A^2+(2m-1)y^2$ its value $2mB$, we find

$$\Theta=\frac{2m-1}{2mB}A\sqrt{D}(A+yp),$$

or

$$\frac{dh}{dx}=\frac{2m-1}{2mB}\sqrt{D}(A+yp).$$

But writing U_1, U_2 to denote the values corresponding to $+\sqrt{D}$, $-\sqrt{D}$ respectively, we have

$$U_1=\frac{2m-1}{2mB_1}\sqrt{D}(mx+yp+\sqrt{D}),$$

and thence

$$(2m-1)\frac{dB_1}{B_1}=\frac{U_1}{\sqrt{D}}-\frac{2m-1}{2mB_1}(A_1+yp).$$

But we have

$$U'=P'+Q'\sqrt{D}+\text{\&c.}=\tfrac{1}{2}(2Q'n+QD'+2P'^2),$$

and thence

$$U_1U_2'=-\frac{1}{4D}\{(2Q'n+QD')^2-4P'^2n\},$$

and moreover

$$U_1U_2=P^2-Q^2D=A_1A_2(B_1B_2)^{m-1},$$

where

$$A_1A_2=m^2x^2-D=-y^2,$$

$$B_1B_2 = (m^2x^2 + y^2)^2 - x^2D = y^2\{y^2 + (2m-1)x^2\};$$

and we thence find

$$\begin{aligned} &= (2m-1)^2A_1A_2(B_1B_2)^{m-1}y^2\{y(p^2-1) + 2mxp\} \\ &= -(2m-1)^2y^{2m-1}\{y^2 + (2m-1)x^2\}^{m-1}\{y(p^2-1) + 2mxp\}. \end{aligned}$$

Hence, the derived equation being

$$\Omega = Q^2\{4nP'^2 - (2Q'n + QD')^2\} = 0,$$

the last preceding equation becomes

$$Q^2Qy^{2m-1}\{y^2 + (2m-1)x^2\}^{m-1}\{y(p^2-1) + 2mxp\} = 0.$$

Here, besides the factor Q^2 corresponding to the nodal curve, and the factor D corresponding to the cuspidal curve, we have the factors y^{2m-1} and $\{y^2 + (2m-1)x^2\}^{m-1}$; and, rejecting all these, the differential equation in its reduced form is

$$y(p^2-1) + 2mxp = 0;$$

and the required verification is effected. The occurrence of

$$Q^2Qy^{2m-1}\{y^2 + (2m-1)x^2\}^{m-1}$$

as a factor in the complete derived equation would give rise to some further investigations, but I will not now enter on them.

I remark however that if $m = 1$, viz. if the integral equation be const. $= x + \sqrt{x^2+y^2}$ or say $z = x + \sqrt{x^2+y^2}$, or, what is the same thing,

$$z^2 - 2zx - y^2 = 0,$$

then observing that $y^2 + (2m-1)x^2$ is here $= x^2 + y^2$ which is $= D$, so that

$$D\{y^2 + (2m-1)x^2\}^{m-1} = D \cdot D^0 = 1,$$

the differential equation in its complete form is

$$y(p^2y + 2px - y) = 0;$$

so that we have here the factor y which divides out. The last-mentioned result is most readily obtained directly from the equation

$$\Omega = Q'^2 - (2Q'D + QD')^2 - 4P'^2n = 0,$$

which is the derived equation corresponding to the integral equation $z = P + Q\sqrt{D}$. We in fact have $P = x$, $Q = 1$, $D = x^2 + y^2$, and the derived equation thus is

$$(x + yp)^2 - (x^2 + y^2) = 0,$$

that is, $y(p^2y + 2px - y) = 0$.

I mention also, in connexion with the foregoing investigation, the integral equation $z = x + \sqrt{2x^2 - y^2}$, or $z^2 - 2zx - x^2 + y^2 = 0$, for which the derived equation in its complete form is

$$(2x - yp)^2 - (2x^2 - y^2) = 0,$$

or, what is the same thing, $y^2p^2 - 4xyp + 2x^2 + y^2 = 0$, and for which therefore there is no factor to divide out.

VII.

The conics confocal with a given conic form a system similar in its properties to that of the curves of curvature of a quadric surface; and the theory of the last-mentioned system may be studied by means of the system of confocal conics. Consider then the equation

$$\frac{x^2}{z+a^2}+\frac{y^2}{z+b^2}=1$$

which, if z be an arbitrary parameter, belongs to the conics confocal with the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. Treating z as a coordinate, the equation represents a surface of the third order, which is such that its section by any plane parallel to the plane of xy is a conic; and the confocal conics are the projections on the plane of xy, by lines parallel to the axis of z, of the sections of the surface.

The sections by the planes of zx, zy are the parabolas $x^2=z+a^2$ and $y^2=z+b^2$ respectively. When $z>-b^2$, the ordinates in each parabola are real, and these ordinates give the semiaxes of the elliptic section. When $-a^2<z<-b^2$, then only the parabola section in the plane of zx has a real ordinate, and the sections are hyperbolic; and when $z<-a^2$, the section is altogether imaginary. The section in the plane $z=-b^2$ is the pair of coincident lines $y^2=0$, $z=-b^2$, and the section in the plane $z=-a^2$ is the pair of coincident lines $z=-a^2$, $x=0$; or, in other words, the plane $z+b^2=0$ touches the surface along the line $y=0$, and the plane $z+a^2=0$ touches the surface along the line $x=0$: this at once appears from the integral form

$$(z+a^2)(z+b^2)-x^2(z+b^2)-y^2(z+a^2)=0.$$

The points $(z=-b^2,\ y=0,\ x=\pm\sqrt{a^2-b^2})$ and $(z=-a^2,\ x=0,\ y=\pm\sqrt{b^2-a^2})$ are conical points; the last two being imaginary. [330]

The differential equation of the curves of curvature, or say of the orthogonal trajectories of the conics, is obtained very easily. Writing for a moment

$$A=z+a^2,\quad B=z+b^2,\quad \text{so that } \frac{x^2}{A}+\frac{y^2}{B}=1,$$

then for an orthogonal trajectory we have

$$\frac{x\,dx}{A}+\frac{y\,dy}{B}=0,\quad \text{and } \frac{x\,dx}{A^2}+\frac{y\,dy}{B^2}=0,$$

giving

$$dx : dy : dz = \frac{x}{A} : \frac{y}{B} : \left(\frac{1}{B} - \frac{1}{A}\right),$$

or, what is the same thing,

$$dx : dy : dz = x(z + a^2) : y(z + b^2) : -(a^2 - b^2).$$

At a point for which z is constant, we have $dz = 0$, and the equation of the direction of the curve of curvature is

$$\frac{dx}{x(z + a^2)} = \frac{dy}{y(z + b^2)},$$

giving $\frac{dx}{x(z+a^2)} - \frac{dy}{y(z+b^2)} = 0$, that is, $x^{z+b^2} = \text{const.} \cdot y^{z+a^2}$, or say $x^B = \text{const.} \cdot y^A$, which is the equation of an orthogonal trajectory.

5, Downing Terrace, Cambridge, November 2, 1863.

331.

ANALYTICAL THEOREM RELATING TO THE FOUR CONICS, &C.

[From the *Philosophical Magazine*, vol. xxvi. (1863), pp. 369, 370.]

I consider the four points A, B, C, D, and the four conics, each through three of these points and touching the line joining the remaining two; and I find the equation of the four conics. The coordinates are taken to be $x = 0$, $y = 0$, $z = 0$ for the points A, B, C respectively, and $x : y : z = 1 : 1 : 1$ for the point D; so that the equations of the sides BC, CA, AB, DA, DB, DC are $x = 0$, $y = 0$, $z = 0$, $y - z = 0$, $z - x = 0$, $x - y = 0$ respectively. The equation of a conic through B, C, D is

$$(y - z)^2 + 2lyz = 0;$$

and that of a conic through C, A, D is

$$(z - x)^2 + 2mzx = 0;$$

and that of a conic through A, B, D is

$$(x - y)^2 + 2nxy = 0.$$

The equation of a conic through A, B, C is

$$2fyz + 2gzx + 2hxy = 0.$$

I say that the first and fourth conics touch each other. In fact, writing the equations in the forms

$$y^2 + z^2 + 2(-1 + l)yz = 0,$$

and

$$2fyz + 2gzx + 2hxy = 0,$$

the condition of contact is

$$\begin{vmatrix} 0 & h & g \\ h & 1 & l-1 \\ g & l-1 & 1 \end{vmatrix}^2 = 4 \begin{vmatrix} 0 & h & g \\ h & 1 & l-1 \\ g & l-1 & 1 \end{vmatrix} \begin{vmatrix} 0 & h & g \\ h & 0 & f \\ g & f & 0 \end{vmatrix},$$

that is

$$\{-g^2-h^2+2(l-1)gh\}^2 = 4gh\{-g^2-h^2+2(l-1)gh\}\{f^2-(l-1)fg-(l-1)fh\}.$$

Hence, rejecting the factor $-g^2 - h^2 + 2(l-1)gh$, the condition is

$$-g^2 - h^2 + 2(l-1)gh = 4gh\{f^2 - (l-1)f(g+h)\},$$

or, as this may be written,

$$-g^2 - h^2 - 2gh + 2lgh = 4f^2gh - 4(l-1)fgh(g+h),$$

that is

$$-(g+h)^2 + 2lgh = 4f^2gh - 4(l-1)fgh(g+h).$$

But if the conic $2fyz + 2gzx + 2hxy = 0$ touches the line $y - z = 0$, then writing $z = y$, the equation $2fy^2 + 2gyx + 2hxy = 0$, or $2fy + 2gx + 2hx = 0$, must have equal roots, that is, $f^2 = (g+h)^2$, or $f = \pm(g+h)$. Hence $f^2 = (g+h)^2$, and the condition becomes

$$-(g+h)^2 + 2lgh = 4(g+h)^2gh - 4(l-1)(g+h)gh,$$

which is satisfied identically.

The equation of the fourth conic may be written

$$\frac{f}{x} + \frac{g}{y} + \frac{h}{z} = 0,$$

or, since $f^2 = (g+h)^2$, $g^2 = (h+f)^2$, $h^2 = (f+g)^2$, the equation is

$$\frac{1}{\sqrt{x}} + \frac{1}{\sqrt{y}} + \frac{1}{\sqrt{z}} = 0,$$

or, rationalizing,

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0,$$

which is a conic touching the lines BC, CA, AB (that is, $x = 0$, $y = 0$, $z = 0$), at the points $x = 0$, $y + z = 0$; $y = 0$, $z + x = 0$; $z = 0$, $x + y = 0$ respectively, and also touching the lines $y - z = 0$, $z - x = 0$, $x - y = 0$. The four conics are thus:

1. Conic through B, C, D, touching $y-z = 0$: $(y-z)^2 + 2lyz = 0$.
2. Conic through C, A, D, touching $z-x = 0$: $(z-x)^2 + 2mzx = 0$.
3. Conic through A, B, D, touching $x-y = 0$: $(x-y)^2 + 2nxy = 0$.
4. Conic through A, B, C, touching $x = 0$, $y = 0$, $z = 0$: $x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0$.

Cambridge, November 28, 1863.

332.

ANALYTICAL THEOREM RELATING TO THE SECTIONS OF A QUADRIC SURFACE.

[From the *Philosophical Magazine*, vol. xxvii. (1864), pp. 43, 44.]

The four sections $x = 0$, $y = 0$, $z = 0$, $w = 0$ of the quadric surface

$$ax^2 + by^2 + 2xy\sqrt{ab} - cz^2 - dw^2 = 0$$

are each of them touched by each of the four sections

$$x\sqrt{2a} + y\sqrt{2b} + z\sqrt{c} + w\sqrt{d} = 0;$$

where it is to be noticed that the radicals $\sqrt{2a}$, $\sqrt{2b}$ are such that their product is $= 2\sqrt{ab}$ if $\sqrt{ab}$ be the radical contained in the equation of the surface. There is of course no loss of generality in attributing a definite sign to the radical $\sqrt{2a}$; but upon this being done, the sign of the radical $\sqrt{2b}$ is determined, whereas the signs of $\sqrt{c}$ and $\sqrt{d}$ are severally arbitrary. We may if we please write the equation of any one of the last-mentioned sections in the form

$$x\sqrt{2a} + y\sqrt{2b} + z\sqrt{c} + w\sqrt{d} = 0,$$

it being understood that the radicals $\sqrt{2a}$, $\sqrt{2b}$ have each a determinate sign, but that the signs of $\sqrt{c}$ and $\sqrt{d}$ are each of them arbitrary.

To prove the theorem in question, it is enough to show (1) that the sections $x = 0$, $x\sqrt{2a} + y\sqrt{2b} + z\sqrt{c} + w\sqrt{d} = 0$; (2) that the sections $z = 0$, $x\sqrt{2a} + y\sqrt{2b} + w\sqrt{d} = 0$, touch each other.

1. The sections $x = 0$, $x\sqrt{2a} + y\sqrt{2b} + z\sqrt{c} + w\sqrt{d} = 0$ of the quadric surface $ax^2 + by^2 + 2xy\sqrt{ab} - cz^2 - dw^2 = 0$ will touch each other if, combining together the equations

$$x = 0, \quad y\sqrt{2b} + z\sqrt{c} + w\sqrt{d} = 0, \quad by^2 - cz^2 - dw^2 = 0,$$

these give a twofold value (pair of equal values) for the ratios $y : z : w$. We in fact have

$$\begin{aligned} by^2 - cz^2 - dw^2 &= by^2 - cz^2 - (y\sqrt{b} + z\sqrt{c})^2 \\ &= -hy^2 - 2cz\sqrt{bc} - 2yz\sqrt{2bc} \\ &= -(y\sqrt{b} + z\sqrt{2c})^2; \end{aligned}$$

and the right-hand side being a perfect square, the condition of contact is satisfied.

2. In like manner we have the system

$$z=0, \quad x\sqrt{2a}+y\sqrt{2b}+w\sqrt{d}=0, \quad ax^2+by^2+2xy\sqrt{ab}-dw^2=0,$$

which gives

$$\begin{aligned} ax^2+by^2+2xy\sqrt{ab}-dw^2 &= (ax^2+by^2+2xy\sqrt{ab})-(x\sqrt{a}+y\sqrt{2b})^2 \\ &= -ax^2-by^2+2xy\sqrt{ab} \\ &= -(x\sqrt{a}-y\sqrt{b})^2; \end{aligned}$$

and here also, the right-hand side being a perfect square, the condition of contact is satisfied.

Cambridge, November 28, 1863.

333.

NOTE ON THE NODAL CURVE OF THE DEVELOPABLE DERIVED FROM THE QUARTIC EQUATION $(a, b, c, d, e \between t, 1)^4 = 0$.

[From the *Philosophical Magazine*, vol. xxvii. (1864), pp. 437–440.]

Considering the coefficients (a, b, c, d, e) as linear functions of the coordinates x, y, z, w, then the equation

$$\text{Disc}\,(a, b, c, d, e \between t, 1)^4 = 0,$$

or, as it may be written,

$$(ae - 4bd + 3c^2)^2 - 27(ace + 2bcd - ad^2 - b^2e - c^3)^2 = 0$$

represents, as is known, a developable surface or "torse," having for its edge of regression (or cuspidal curve) the sextic curve the equations whereof are

$$\begin{aligned} ae - 4bd + 3c^2 &= 0, \\ ace + 2bcd - ad^2 - b^2e - c^3 &= 0; \end{aligned}$$

and for its nodal curve, a curve the equations whereof (equivalent to two independent relations between the coordinates) are

$$\frac{ac - b^2}{a} = \frac{ad - bc}{2b} = \frac{ae + 2bd - 3c^2}{6c} = \frac{be - cd}{2d} = \frac{ce - d^2}{e},$$

or, as these may also be written,

$$\begin{aligned} a^2d - 3abc + 2b^3 &= 0, \\ a^2e + 2abd - 9ac^2 + 6b^2c &= 0, \\ abe - 3acd + 2b^2d &= 0, \\ ad^2 - b^2e &= 0, \\ ade - 3bce + 2bd^2 &= 0, \\ ae^2 + 2bde - 9c^2e + 6cd^2 &= 0, \\ be^2 - 3cde + 2d^3 &= 0; \end{aligned}$$

which curve is in fact an excubo-quartic, — viz. a quartic curve the partial intersection of a quadric surface and a cubic surface, having in common two non-intersecting right lines. To show that this is so, I remark that the coefficients a, b, c, d, e, quâ linear functions of the four coordinates, satisfy a linear equation which may be taken to be

$$a+b+c+d+e=0;$$

this being so, the first form shows that the curve in question lies on the quadric surface

$$ac-b^2+\tfrac{1}{2}(ad-bc)+\tfrac{1}{6}(ae+2bd-3c^2)+\tfrac{1}{2}(be-cd)+ce-d^2=0,$$

or, as this equation may also be written,

$$c(a+b+c+d+e)-b^2+\tfrac{1}{2}ad+\tfrac{1}{6}(ae+2bd)+\tfrac{1}{2}be-d^2=0.$$

Substituting for c its value, this equation is

$$-(a+e+b+d)(\tfrac{1}{2}a+\tfrac{1}{2}e)-b^2+\tfrac{1}{2}ad+\tfrac{1}{6}(ae+2bd)+\tfrac{1}{2}be-d^2=0,$$

or, what is the same thing,

$$\tfrac{1}{9}(a+e+b+d)(a+e)+\tfrac{1}{6}(b^2+d^2)-\tfrac{1}{3}(ad+be)-\tfrac{1}{6}(ae+2bd)=0.$$

Hence, finally, the equation of the quadric surface is

$$9a^2+17ae+9e^2+6b^2-2bd+6d^2+9ab+9de+6ad+6be=0;$$

and the curve lies also on the cubic surface

$$ad^2-b^2e=0.$$

It only remains to show that these surfaces have in common two right lines, and to find the equations of these lines.

The cubic surface is a skew surface or "scroll" such that the equations of any generating line are $d-\theta b=0$, $e-\theta^2a=0$, where θ is an arbitrary parameter. But considering the two lines

$$(d-\theta_1 b=0,\ e-\theta_1^2a=0),\quad (d-\theta_2 b=0,\ e-\theta_2^2a=0),$$

the general equation of the quadric surface through these two lines may be written

$$\begin{aligned}A\cdot(d-\theta_1b)(d-\theta_2b)&\\ &+B\cdot(e-\theta_1^2a)(e-\theta_2^2a)\\ &+C\cdot\{(d-\theta_1b)(e-\theta_2^2a)+(d-\theta_2b)(e-\theta_1^2a)\}\\ &+D\cdot\{(d-\theta_1b)(e-\theta_2^2a)-(d-\theta_2b)(e-\theta_1^2a)\}=0\end{aligned}$$

or, expanding and reducing,

$$\begin{aligned}&A\{d^2-(\theta_1+\theta_2)bd+\theta_1\theta_2b^2\}\\ &\qquad\qquad\qquad\qquad +B\{e^2-(\theta_1^2+\theta_2^2)ea+\theta_1^2\theta_2^2a^2\}\\ &\qquad\quad +C\{2de-(\theta_1+\theta_2)ad-(\theta_1^2+\theta_2^2)be+\theta_1\theta_2(\theta_1+\theta_2)ab\}\\ &\qquad\qquad\qquad\qquad\qquad\qquad +D\{(\theta_1+\theta_2)ad-be-\theta_1\theta_2ab\}=0,\end{aligned}$$

which, if θ_1, θ_2 are the roots of the equation $\theta^2-\frac{1}{3}\theta+1=0$, and therefore $\theta_1+\theta_2=\frac{1}{3}$, $\theta_1\theta_2=1$, and $\theta_1^2+\theta_2^2=-\frac{1}{9}$, is

$$\begin{aligned}&A(d^2-\tfrac{1}{3}db+b^2)+B(e^2+\tfrac{1}{9}ae+a^2)\\ &\qquad\qquad\qquad +C(2de+\tfrac{1}{3}ad-\tfrac{1}{9}be+\tfrac{1}{3}ab)\\ &\qquad\qquad\qquad\qquad\qquad\qquad +D(\tfrac{1}{3}ad-be-ab)=0.\end{aligned}$$

Putting $A=6$, $B=9$, $C=\frac{1}{2}$, $D=-\frac{1}{2}$, this is

$$\begin{aligned}&9(a^2+\tfrac{1}{9}ae+e^2)+6(b^2-\tfrac{1}{3}bd+d^2)\\ &\qquad\qquad\qquad +\tfrac{1}{2}(\tfrac{1}{3}ab+2de+\tfrac{1}{3}ad-\tfrac{1}{9}be)\\ &\qquad\qquad\qquad\qquad\qquad\qquad +\tfrac{1}{2}(ab-\tfrac{1}{3}ad+be)=0,\end{aligned}$$

which is the before-mentioned quadric surface; hence the quadric surface and the cubic surface intersect in the two lines

$$(d-\theta_1b=0,\ e-\theta_1^2a=0),\quad (d-\theta_2b=0,\ e-\theta_2^2a=0)$$

(where θ_1, θ_2 are the roots of the quadric equation $\theta^2-\frac{1}{3}\theta+1=0$); and they consequently intersect also in an excubo-quartic curve, which is the theorem required to be proved.

Blackheath, March 26, 1864.

334.

NOTE ON THE THEORY OF CUBIC SURFACES.

[From the *Philosophical Magazine*, vol. xxvii. (1864), pp. 493–496.]

The equation

$$AX^3 + BY^3 + 6CRST = 0,$$

where $X + Y + R + S + T = 0$, represents a cubic surface of a special form, viz. each of the planes $R = 0$, $S = 0$, $T = 0$ is a triple tangent plane meeting the surface in three lines which pass through a point (‡); and, moreover, the three planes $AX^3 + BY^3 = 0$ are triple tangent planes intersecting in a line. It is worth noticing that the equation of the surface may also be written

$$ax^3 + by^3 + c(u^3 + v^3 + w^3) = 0,$$

where $x + y + u + v + w = 0$. In fact, the coordinates satisfying the foregoing linear equations respectively, we have to show that the equation

$$AX^3 + BY^3 + 6CRST = ax^3 + by^3 + c(u^3 + v^3 + w^3)$$

may be identically satisfied. We have

$$\begin{aligned} ax^3 + by^3 + c(u^3 + v^3 + w^3) &= ax^3 + by^3 + c\{(u + v + w)^3 - 3(v + w)(w + u)(u + v \\ &= ax^3 + by^3 - c(x + y)^3 - 3c(v + w)(w + u)(u + v), \end{aligned}$$

(‡) The tangent plane of a surface intersects the surface in a curve having at the point of contact a double point, and in like manner a triple tangent plane intersects the surface in a curve with three double points, viz. each point of contact is a double point; there is not in general any triple tangent plane such that the three points of contact come together, or (what is the same thing) there is not in general any tangent plane intersecting the surface in a curve having at the point of contact a triple point. A surface may, however, have the kind of singularity just referred to, viz. a tangent plane intersecting the surface in a curve having at the point of contact a triple point; such tangent plane may be termed a 'tritom' tangent plane, and its point of contact a 'tritom' point: for a cubic surface the intersection by a tritom tangent plane is of course a system of three lines meeting in the tritom point. The tritom singularity is sibi-reciprocal; it is, I think, a singularity which should be considered in the theory of reciprocal surfaces.

which is to be

$$= AX^3 + BY^3 + 6CRST;$$

and we may find X, Y, R, S, T, linear functions of x, y, u, v, w, so as to satisfy these equations, and so that in virtue of

$$x + y + u + v + w = 0,$$

we shall have also $X + Y + R + S + T = 0$. For, assuming

$$\begin{aligned} AX^3 + BY^3 &= ax^3 + by^3 - c(x+y)^3, \\ X + Y &= x + y, \\ R &= \tfrac{1}{2}(v+w), \quad a = -4c, \\ S &= \tfrac{1}{2}(w+u), \\ T &= \tfrac{1}{2}(u+v), \end{aligned}$$

we have identically

$$AX^3 + BY^3 + 6CRST = ax^3 + by^3 - c(x+y)^3 - 3c(v+w)(w+u)(u+v),$$

$$X + Y + R + S + T = x + y + u + v + w,$$

and thus it only remains to show that we can find X, Y linear functions of x, y, such that

$$AX^3 + BY^3 = ax^3 + by^3 - c(x+y)^3, \qquad X + Y = x + y.$$

This is always possible; in fact if

$$U = ax^3 + by^3 - c(x+y)^3,$$

then taking Φ for the cubicovariant, and D for the discriminant of U, we have

$$\tfrac{1}{2}(\Phi + \sqrt{D}U'), \quad \tfrac{1}{2}(\Phi - \sqrt{D}U') \text{ each a perfect cube, say}$$

$$\tfrac{1}{2}(\Phi + \sqrt{D}U') = (\nu x + \mu y)^3,$$

and we then have

$$U = \tfrac{1}{2}\{(\nu x + \mu y)^3 - (\nu x + \mu y)^3\} = AX^3 + BY^3,$$

which is satisfied by

$$X = l(\lambda x + \mu y), \qquad Y = m(\nu x + \rho y),$$

if

$$Al^3 = \tfrac{1}{2}, \qquad Bm^3 = -\tfrac{1}{2}.$$

[334] The equation $X + Y = x + y$ then gives

$$l\lambda + m\nu = 1, \qquad l\mu + m\rho = 1,$$

which give the values of l and m, and thence the values of A and B; and collecting all the equations, we have

$$\begin{aligned} R &= \tfrac{1}{2}(v + w), \quad C = -4c, \\ S &= \tfrac{1}{2}(w + u), \\ T &= \tfrac{1}{2}(u + v), \end{aligned}$$

where Φ, D being respectively the cubicovariant and the discriminant of $U = ax^3 + by^3 - c(x + y)^3$, for the formulae of the transformation

$$\begin{aligned} AX^3 + BY^3 + 6CRST &= ax^3 + by^3 + c(u^3 + v^3 + w^3), \\ X + Y + R + S + T &= x + y + u + v + w. \end{aligned}$$

The equation $ax^3+by^3+c(u^3+v^3+w^3) = 0$, where $x+y+u+v+w = 0$, presents over the other form the advantage that it is included as a particular case under the equation $ax^3+by^3+cu^3+dv^3+ew^3 = 0$ (where $x+y+u+v+w = 0$) employed by Dr Salmon as the canonical form of equation for the general cubic surface.

5, Downing Terrace, Cambridge, April 29, 1864.

335.

TABLES DES FORMES QUADRATIQUES BINAIRES POUR LES DÉTERMINANTS NÉGATIFS DEPUIS $D = -1$ JUSQU'À $D = -100$, POUR LES DÉTERMINANTS POSITIFS NON CARRÉS DEPUIS $D = 2$ JUSQU'À $D = 99$ ET POUR LES TREIZE DÉTERMINANTS NÉGATIFS IRRÉGULIERS QUI SE TROUVENT DANS LE PREMIER MILLIER.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. LX. (1862), pp. 357–372.]

Les tables suivantes sont arrangées de la manière préscrite dans les "Disquisitiones arithmeticae." Dans le mémoire de Lejeune Dirichlet "Recherches sur diverses applications de l'analyse à la théorie des nombres," tom. XIX (1839), p. 338 de ce Journal on trouve un tableau dans lequel les règles qui servent à former les caractères des genres sont résumées. Soit $D = PS^2$, ou $2PS^2$, S^2 désignant le plus grand carré que D contient, et P un nombre impair; soient de plus p, p', p''... les facteurs premiers inégaux de P et r, r', r''... les nombres premiers impairs qui divisent S sans diviser P; écrivons enfin pour abréger $S = (-)^{r_s}$, $\epsilon = (-)^{r_c}$. Cela posé on trouve à l'endroit cité le tableau suivant:

Premier cas, $D = PS^2$, $P \equiv 1$ (mod. 4)

	p	p'	p''	r'	r''
$S \equiv 1$ (mod. 2)	m	m	m		
$S \equiv 2$ (mod. 4)	m	m	m	m	
$S \equiv 0$ (mod. 4)	p''	p''		r'	r''

Deuxième cas, $D = PS^2$, $P \equiv 3$ (mod. 4)

		p	p'	r'
$S \equiv 1$ (mod. 2)	m	m		ϵ
$S \equiv 2$ (mod. 4)	p	p		r'
$S \equiv 0$ (mod. 4)	p	p		

Troisième cas, $D = 2PS^2$, $P \equiv 1$ (mod. 4)

	p	p'	$\dots$	r	r'
$S \equiv 1 \pmod{2}$	m	m		ϵ	
$S \equiv 0 \pmod{2}$	p	p		r	r'

Quatrième cas, $D = 2PS^2$, $P \equiv 3 \pmod{4}$

	r	r'	$\dots$	
$S \equiv 1 \pmod{2}$	m	m	m	m
$S \equiv 0 \pmod{2}$	m	m	m	m

Dans ce tableau la notation $\left(\frac{m}{p}\right)$ dans laquelle j'omets les parenthèses usitées, signifie le caractère d'un nombre quelconque m par rapport au nombre premier impair p, c.-à-d. que m est résidu ou non résidu de p selon que $\left(\frac{m}{p}\right) = +1$ ou $= -1$, de même δ est le caractère de m par rapport au nombre 4, savoir $m \equiv 1$ ou $3 \pmod{4}$ selon que $\delta = +1$ ou $= -1$, enfin ϵ, $\delta\epsilon$ sont les caractères de m par rapport au nombre 8, savoir $m \equiv 1$ ou $7 \pmod{8}$ pour $\epsilon = +1$, $= 3$ ou $5 \pmod{8}$ pour $\epsilon = -1$; $m \equiv 1$ ou $3 \pmod{8}$ pour $\delta\epsilon = +1$, $= 5$ ou $7 \pmod{8}$ pour $\delta\epsilon = -1$. Si pour un déterminant donné on veut former au moyen de ce tableau les caractères des genres, on prend la ligne horizontale qui convient à ce déterminant; à tous les caractères $\left(\frac{m}{p}\right)$, $\left(\frac{m}{p'}\right)$,... $\left(\frac{m}{r}\right)$, $\left(\frac{m}{r'}\right)$,... δ, ϵ, $\delta\epsilon$ qui se trouvent dans la ligne horizontale, on attribue les signes + ou − à volonté, avec cette restriction cependant que le signe composé des signes qui se trouvent dans la première partie de la ligne dont il s'agit soit positif. Si par exemple le déterminant donné est $D = -35$, on a $D = -35 = PS^2$, $P = -35 \equiv 1 \pmod{4}$, $S = 1 \equiv 1 \pmod{2}$, les nombres $p, p' \dots$ sont 5, 7, et les signes que l'on doit considérer sont $\left(\frac{m}{5}\right)$, $\left(\frac{m}{7}\right)$. De là on obtient les caractères

	+	−
$\left(\frac{m}{5}\right)$	+	−
$\left(\frac{m}{7}\right)$	+	−

il y a donc deux genres de l'ordre proprement primitif. Dans le cas dont il s'agit (et en général pour $D \equiv 1 \pmod{4}$) il y a un ordre improprement primitif avec des genres qui ont les caractères identiques à ceux des genres de l'ordre proprement primitif, les caractères se rapportant dans ce cas à la moitié d'un nombre quelconque représenté par la forme.

Pour faciliter l'impression des tables j'ai introduit deux nouvelles lettres α et β dont voici la définition. Pour tous les déterminants auxquels se rapportent mes tables, c.-à-d. pour les déterminants négatifs quelconques et positifs non-carrés depuis -100 jusqu'à $+99$ ainsi que pour les déterminants négatifs irréguliers du premier millier, le nombre des facteurs premiers désignés ci-dessus par les lettres p, p', $p''\ldots$, r, r', $r''\ldots$ n'excède pas deux. Soit donc q le plus petit de ces facteurs premiers et q' le plus grand lorsqu'il y en a deux, je désigne par α le caractère $\left(\frac{m}{q}\right)$ et par β le caractère $\left(\frac{m}{q'}\right)$.

Dans la colonne relative à la composition et portant l'inscription Cp, je représente comme à l'ordinaire par l'unité la forme principale, par la lettre c une forme qui produit par la duplication la forme principale, par les lettres d, e,… des formes qui la produisent par la triplication, la quadruplication, etc., de manière que l'on ait $c^2 = 1$, $d^3 = 1$, $e^4 = 1$, $f^5 = 1$, $g^6 = 1$, $h^7 = 1$, $i^8 = 1$, $j^9 = 1$, etc. Les notations d, d_2 par exemple signifient deux formes différentes dont chacune produit par la triplication la forme principale. Je représente de plus par a la forme principale de l'ordre improprement primitif et par ac, ad,… des formes qui produisent a par la duplication, la triplication, etc. Dans l'énumération des classes, j'ai toujours écrit en premier lieu l'ordre proprement primitif, en le faisant suivre après un trait de séparation par l'ordre improprement primitif lorsqu'il existe. Dans chacun des deux ordres les divers genres se trouvent séparés les uns des autres par des traits subordonnés.

Pour les déterminants positifs les périodes sont données par une abréviation facile à comprendre. Chaque forme de la période ayant son dernier coefficient égal au premier de la suivante, cette valeur identique n'a été imprimée qu'une fois; de plus les coefficients extérieurs a, c des formes (a, b, c) ont été distingués des coefficients b en imprimant ces derniers en caractères plus petits. Ainsi pour le déterminant 7 la période de la classe principale $(1, 0, -7)$ est donnée par les nombres

$$1,\ 2,\ -3,\ 1,\ 2,\ 1,\ -3,\ 2,\ 1$$

qui représentent la série des formes

$$(1, 2, -3),\ (-3, 1, 2),\ (2, 1, -3),\ (-3, 2, 1).$$

Londres, 6 Novembre, 1860.

Table I des formes quadratiques binaires ayant pour déterminants les nombres négatifs depuis $D=-1$ jusqu'à $D=-100$.

The following table presents the binary quadratic forms for negative determinants from $D=-1$ to $D=-100$. The columns show: D (determinant), Classes (the number of classes), the characters α, β, δ, ϵ, $\delta\epsilon$, and the forms themselves.

Table I. Negative Determinants

D	Cl.	α	β	δ	ϵ	$\delta\epsilon$	Forms
-1	1			$-$			$(1,0,1)$
-2	1						$(1,0,2)$
-3	1			$-$			$(1,0,3),\ c$
-4	1						$(1,0,4),\ ac$
-5	2			$+$		$+$	$(1,0,5)+1,\ c$
-6	2						$(1,0,6)+1,\ c,\ cc_1$
-7	1			$-$			$(1,0,7)+1,\ c$
-8	1						$(1,0,8)+c$
-9	1						$(1,0,9)+1,\ 2,0,5+c$
-10	2						$(2,1,5)+c,\ (1,0,10)+1$
-11	1			$-$			$(1,0,11)+1,\ c$
-12	1						$(1,0,12)+1,\ 2,0,6+c$
-13	2			$+$	$+$	$+$	$(1,0,13)+1,\ c$
-14	2						$(3,1,5)+c,\ (1,0,14)+1$
-15	2			$-$			$(1,0,15)+1,\ 2,1,8$
-16	2						$(1,0,16)+ac,\ 4,0,5+ad$
-17	2			$+$		$+$	$(2,1,9)+c,\ (1,0,17)+1$
-18	2						$(1,0,18)+1,\ 3,0,6+c$
-19	1			$-$			$(1,0,19)+1,\ c$
-20	2						$(1,0,20)+1,\ 2,0,10+c$
-21	2	$-$	$-$	$-$			$(1,0,21)+1,\ 2,1,11,\ 3,0,7$
-22	2						$(2,1,11)+1,\ 3,0,8+c$
-23	3			$-$			$(1,0,23)+1,\ c$
-24	2						$(1,0,24)+1,\ 3,0,8+c$
-25	2						$(1,0,25)+c,\ 5,0,5+1$
-26	2						$(1,0,26)+1,\ 5,1,6+c$
-27	2						$(1,0,27)+1,\ c$
-28	2						$(1,0,28)+1,\ 2,0,14+c$
-29	3			$+$	$+$	$+$	$(5,1,6)+c,\ (1,0,29)+1$
-30	3	$-$					$(1,0,30)+c,\ 2,1,16,\ 3,0,10$
-31	3			$-$			$(1,0,31)+1,\ 2,1,16,\ c$

Continued

Tabelle 8 – *Continued*

D	Cl.	α	β	δ	ϵ	$\delta\epsilon$	Forms
-32	2						$(1,0,32)+ac,\ 4,0,8+c$
-33	2	$+$	$-$				$(1,0,33)+1,\ 3,0,11+c$
-34	2						$(1,0,34)+1,\ 5,1,7+c$
-35	2	$-$	$+$	$-$			$(1,0,35)+1,\ 3,1,12+c$
-36	3						$(1,0,36)+1,\ 3,1,13+c,$ $4,0,9+f$
-37	2			$+$	$+$	$+$	$(1,0,37)+1,\ c$
-38	2						$(1,0,38)+1,\ 2,1,20+c$
-39	3	$-$					$(1,0,39)+1,\ 2cc_1,\ 3,0,13$
-40	2						$(1,0,40)+ac,\ 5,0,8+c$
-41	3			$+$		$+$	$(2,1,21)+c,\ (1,0,41)+1$
-42	3	$-$					$(1,0,42)+1,\ c,\ cc_1$
-43	1			$-$			$(1,0,43)+1,\ c$
-44	2						$(1,0,44)+1,\ 2,0,22+c$
-45	3	$-$	$-$				$(1,0,45)+1,\ 2,1,23,\ cc_1$
-46	2						$(1,0,46)+1,\ 2,1,24+c$
-47	5			$-$			$(1,0,47)+1,\ c,\ cc_1$
-48	2						$(1,0,48)+ac,\ 4,0,12+c$
-49	2						$(1,0,49)+1,\ 7,0,7+c$
-50	3						$(1,0,50)+1,\ 2,0,25+c,$ $5,0,10$
-51	2	$+$	$-$				$(1,0,51)+1,\ 3,0,17+c$
-52	2						$(1,0,52)+c,\ 2,0,26+cc_1$
-53	3			$+$	$+$	$+$	$(2,1,27)+c,\ (1,0,53)+1$
-54	3						$(1,0,54)+1,\ 5,1,11+c,\ cc_1$
-55	2	$-$	$+$	$-$			$(1,0,55)+1,\ 2,1,28+c$
-56	3						$(1,0,56)+1,\ 3,0,18+c,\ cc_1$
-57	2	$-$	$-$				$(1,0,57)+1,\ 3,0,19+c$
-58	2						$(1,0,58)+1,\ 2,1,30+c$
-59	3			$-$			$(1,0,59)+1,\ c,\ cc_1$
-60	2						$(1,0,60)+1,\ 3,0,20+c$
-61	3			$+$	$+$	$+$	$(1,0,61)+1,\ c$
-62	2						$(1,0,62)+1,\ 2,1,32+c$
-63	3	$-$					$(1,0,63)+1,\ cc_1,\ 2,1,32$
-64	3						$(1,0,64)+ac,\ 4,0,16+c,\ e$
-65	4	$-$	$+$	$-$			$(1,0,65)+1,\ 2,1,33,$ $5,1,14+c$
-66	3	$-$					$(1,0,66)+1,\ c,\ cc_1$
-67	1			$-$			$(1,0,67)+1,\ c$
-68	2						$(1,0,68)+1,\ 4,2,19+c$

Continued

Tabelle 8 – *Continued*

D	Cl.	α	β	δ	ϵ	$\delta\epsilon$	Forms
-69	3	$+$	$-$				$(1,0,69)+1,\ 2,1,35,$ $3,0,23+c$
-70	3	$-$					$(1,0,70)+1,\ c,\ cc_1,\ e,\ cc_1$
-71	4			$-$			$(1,0,71)+1,\ c,\ cc_1$
-72	2						$(1,0,72)+ac,\ 4,0,18+c$
-73	3			$+$	$+$	$+$	$(1,0,73)+1,\ c$
-74	3						$(1,0,74)+1,\ c,\ cc_1$
-75	3	$-$	$-$				$(1,0,75)+1,\ cc_1$
-76	2						$(1,0,76)+1,\ 2,0,38+c$
-77	3	$-$	$+$				$(1,0,77)+1,\ 2,1,39,\ cc_1$
-78	3	$-$					$(1,0,78)+1,\ 2,1,40,\ cc_1$
-79	2			$-$			$(1,0,79)+1,\ c$
-80	3						$(1,0,80)+ac,\ 4,0,20+c,\ cc_1$
-81	3						$(1,0,81)+1,\ 2,0,40+c,\ cc_1$
-82	3						$(1,0,82)+1,\ c,\ cc_1$
-83	2			$-$			$(1,0,83)+1,\ c$
-84	2						$(1,0,84)+1,\ 2,0,42+c$
-85	2	$-$	$+$	$-$			$(1,0,85)+1,\ 2,1,43+c$
-86	3						$(1,0,86)+1,\ c,\ cc_1$
-87	3	$+$	$-$				$(1,0,87)+1,\ cc_1,\ 2,1,44$
-88	2						$(1,0,88)+ac,\ 4,0,22+c$
-89	3			$+$	$+$	$+$	$(1,0,89)+1,\ c$
-90	3	$-$	$-$				$(1,0,90)+1,\ cc_1,\ cc_1$
-91	2	$-$	$+$				$(1,0,91)+1,\ cc_1$
-92	1						$(1,0,92)+1$
-93	2	$+$	$-$				$(1,0,93)+1,\ 3,0,31+c$
-94	2						$(1,0,94)+1,\ 2,1,48+c$
-95	4	$-$	$+$	$-$			$(1,0,95)+1,\ cc_1,\ cc_1$
-96	3						$(1,0,96)+ac,\ 4,0,24+c,\ cc_1$
-97	3			$+$	$+$	$+$	$(1,0,97)+1,\ c$
-98	2						$(1,0,98)+1,\ 2,0,49+c$
-99	2	$-$	$-$				$(1,0,99)+1,\ cc_1$
-100	3						$(1,0,100)+ac,\ 2,0,50+c,$ cc_1

Table II des formes quadratiques binaires ayant pour déterminants les nombres positifs non-carrés depuis $D = 2$ jusqu'à $D = 99$.

For positive determinants, the periods are given by an abbreviation easy to understand. Each form of the period having its last coefficient equal to the first of the following, this identical value has been printed only once; moreover the exterior coefficients a, c of the forms (a, b, c) have been distinguished from the coefficients b by printing the latter in smaller characters.

Table II. Positive Determinants

D	Classes	Characters	Periods
2	1	+	$1, 0, -2$; period: $1, 1, -1, 1, 1$
3	1	−	$1, 0, -3$; period: $1, 1, -2, 1, 1,\ -1, 1, 2, 1, -1$
5	1	+	$1, 0, -5$; period: $1, 2, -1, 2, 1$
6	2	+	$1, 0, -6$; $2, 1, -2$; periods: $2, 1, -2, 1, 2,\ 1, 2, -2, 2, 1$
7	1	+	$1, 0, -7$; period: $1, 2, -3, 1, 2, 1, -3, 2, 1$
8	1	+	$1, 0, -8$; period: $1, 2, -4, 2, 1,\ -1, 2, 4, 2, -1$
10	2	+	$1, 0, -10$; $2, 0, -5$; periods: $1, 3, -1, 3, 1,\ 2, 2, -3, 1, 3, 2, -2, 2,$
11	1	+	$1, 0, -11$; period: $1, 3, -2, 3, 1,\ -1, 3, 2, 3, -1$
12	2	+	$1, 0, -12$; $2, 0, -6$; periods: $1, 3, -3, 3, 1,\ -1, 3, 3, 3, -1$
13	1	+	$1, 0, -13$; period: $1, 3, -4, 1, 3, 2, -3, 1, 4, 3, -1, 3, 4, 1, -3, 2, 3,$
14	2	+	$1, 0, -14$; $2, 1, -6$; periods: $1, 3, -5, 2, 2, 2, -5, 3, 1,\ 2, 3, -2, 3,$
15	2	+	$1, 0, -15$; $2, 1, -7$; periods: $1, 3, -6, 3, 1,\ -1, 3, 6, 3, -1$
17	1	+	$1, 0, -17$; period: $1, 4, -1, 4, 1$
18	2	+	$1, 0, -18$; $2, 0, -9$; periods: $1, 4, -2, 4, 1,\ -1, 4, 2, 4, -1$
19	1	+	$1, 0, -19$; period: $1, 4, -3, 2, 5, 3, -2, 3, 5, 2, -3, 4, 1$
20	2	+	$1, 0, -20$; $2, 0, -10$; periods: $1, 4, -4, 4, 1,\ -1, 4, 4, 4, -1$
21	2	+	$1, 0, -21$; $2, 1, -10$; periods: $1, 4, -5, 1, 4, 3, -3, 3, 4, 1, -5, 4, 1$
22	2	+	$1, 0, -22$; $2, 1, -11$; periods: $1, 4, -6, 2, 3, 4, -2, 4, 3, 2, -6, 4, 1$
23	1	+	$1, 0, -23$; period: $1, 4, -7, 3, 2, 3, -7, 4, 1,\ -1, 4, 7, 3, -2, 3, 7, 4$
24	2	+	$1, 0, -24$; $2, 0, -12$; periods: $1, 4, -8, 4, 1,\ -1, 4, 8, 4, -1$
26	2	+	$1, 0, -26$; $2, 1, -13$; periods: $1, 5, -1, 5, 1,\ -1, 5, 1, 5, -1$
27	1	−	$1, 0, -27$; period: $1, 5, -2, 5, 1,\ -1, 5, 2, 5, -1$
28	2	+	$1, 0, -28$; $2, 0, -14$; periods: $1, 5, -3, 4, 4, 4, -3, 5, 1,\ -1, 5, 3, 4$
29	1	+	$1, 0, -29$; period: $1, 5, -4, 3, 5, 2, -5, 3, 4, 5, -1, 5, 4, 3, -5, 2, 5,$
30	3	+	$1, 0, -30$; $2, 0, -15$; $3, 0, -10$; periods: $1, 5, -5, 5, 1,\ -1, 5, 5, 5,$
31	1	+	$1, 0, -31$; period: $1, 5, -6, 1, 5, 4, -3, 5, 2, \ldots$

[End of Papers 325–335, from pp. 101–150 of Vol. V]

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Transcribed from the original 1889–1897 edition

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330 ON DIFFERENTIAL EQUATIONS AND UMBILICI

[From the Philosophical Magazine, vol. xxvii. (1864), pp. 83–91.]

Hence, the derived equation being the last preceding equation becomes Qi Qy^-i j,2 + (2m - 1) a^'"-"- y(p^-l) + 2nuvp = 0. Here, besides the factor Q' corresponding to the nodal curve, and the factor D corresponding to the cuspidal curve, we have the factors y^"^-^ and y^ + 2m — 1) af']^'^' ; and, rejecting all these, the differential equation in its reduced form is y (p^ — l) + 2mxp = 0 ; and the required verification is effected. The occurrence of Q2 Q^m-i ^yi + (2wi - 1) ar')"'-^ as a factor in the complete derived equation would give rise to some further investi- gations, but I will not now enter on them. I remark however that if m = l, viz. if the integral equation be const. = a; + v a?* + y^ or say z = x-
- Var" -f- y", or, what is the same thing, z^ - 2zx -y"- = 0, then observing that y" + (2m-l)a? is here =:a^+y- which is =n, so that D y' + (2m - 1) a?Y"-^ = D . D"' = 1, the differential equation in its complete form is yp^y+2px-y) = 0; so that we have here the factor y which divides out. The last-mentioned result is most readily obtained dii-ectly from the equation n = Qr- (2Q'D + Qoy - 4P"n = o, which is the derived equation con-esponding to the integral equation z = P + Q VD. We in fact have P = x, Q=
, D=a;'-hy', and the derived equation thus is x + ypf-a? + y^) = 0, that is, y p^y + 2px - y) = 0. I mention also, in connexion with the foregoing investigation, the integral equation z = x+ ^2a? - y
or z"^ - 2zx - a? Jr y' = 0, for which the derived equation in its complete form is 2x-ypf-2x'-f) = 0, or, what is the same thing, y-p^ — 4ixyp + 2a? + y^ = 0, and for which therefore there is no factor to divide out. 128 ON DIFFERENTIAL EQUATIONS AND UMBILICI. [330 VIL The conies confocal with a given conic form a system similar in its properties to that of the curves of curvature of a quadric surface; and the theorj- of the last- mentioned system may be studied by means of the system of confocal conies. Consider then the equation ^ + -^=1 which, if z be an arbitrary parameter, belongs to the conies confocal with the ellipse a^ if -j + ^ = l.

Treating 2 as a coordinate, the equation represents a surface of the third order, which is such that its section by any plane parallel to the plane of xy is a conic ; and the confocal conies are the projections on the plane of xy, by lines parallel to the axis of z, of the sections of the surface. The sections by the planes of zx, zy are the parabolas sc^ = z + a^ and y- = z + b- respectively. When z > — YEN, the ordinates in each parabola are real, and these ordinates give the semiaxes of the elliptic section. When z > — a^ < — b^, then only the pai-abola section in the plane of zx has a real ordinate, and the sections are hyperbolic ; and when z <— o?, the section is altogether imaginary. The section in the planes z = — b- is the pair of coincident lines y" = 0, z = — 6", and the section in the plane z = — a" is the pair of coincident lines z= — a?, a? = 0; or, in other words, the plane z -
-b'^ = Q touches the surface along the line y = 0, and the plane z + a^ = 0 touches the surface along the line x = 0: this at once appears from the integral form (z + a^)(z +b^)- a^(z []hb^)-y^(z + a-) = 0. The points z = — YEN, y = 0, x= +/- *Ja^ — b^) and z = —a-, x = 0, y=+/- '^YEN— a') are conical points; the last two are however imaginary points on the surface. To find the nature of the surface about one of the first-mentioned two points, say the point (z = — b^, y = 0, x = Va" — b^), taking this point for the origin and writing therefore Va" - b" + x, y and —b- + z in the place of x, y, z respectively, the equation becomes (a"-YEN + z)z - (a^ -b') + 2x'/c^- -b' + af) z - (a"- -b- + z
y^ = 0, that is z« - 2zx Va» - 6" - (a^ - YEN) y- - zafi +y")=0; so that there is a tangent cone the equation whereof is z' - 2zx Va» -b– (a- - b^) f- = 0, or, as it may be written. z - a;Va» - 6*/ - (o» - 6') (ar" 4- y^) = 0. The equation is that of a cone of the second order, meeting the plane of zx in the lines 2 = 0, z=2x yd' — b- (and therefore such that its sections parallel to the plane of xy are parabolas), and meeting the plane of yz in the lines z= +/-y Va^ — b^ (the origin being at the vertex of the cone or conical point of the surface). Returning to the original origin, and to the equation of the surface written in the form z" + z(a^ + b"-sfi- y^) + a'6= - b-oe' - a'y- = 0, calling this for a moment z^ + Wz + C'= 0, the differential equation is C- - iBB'C + 4,CB'^=0; or, substituting, this is (b^x + a-'ypf - a- + 6^ - a:^ - y') (x + yp) YENx + a?yp) + (a^i^ - b-a? - ay) x + yp)- = 0 ; or, reducing, this is (a^ - 1^) xyxy(p'-l)- (a' - YEN -a^ + y^)p] = 0, or say xy xy(j^-l)- (d'-J^-af + y-)p] = 0, where the factor xy arises from the level lines z + b^=0, y = 0) and z + 0^ = 0, x = 0). Throwing out this factor, the equation becomes xy(p^-l)-(a"-b^-x' + y^) p =0, which is satisfied identically by z + YEN = 0, y = 0, a;^ = a^ — b^. The first derived equation is (xp + y) (p^- - 1) + 2 (a; - yp)p = 0, which for the values in question ^ives p(p^+l) = 0, where the factor p = 0 corresponds to the section 3/ = 0 by the plane z+b^ = 0: and taking the conical point for origin, and observing that the polar of the line x = 0, y = 0 in regard to the tangent cone js z — x 'Jo' — YEN = 0, then writing the equation

of the tangent cone in the form z-x -Ja^ - l^y - (a= - b') (af + y") = 0, the two tangent planes through x = 0, y = 0) are given by the equation ar' + y = 0 ; and for these planes we have p' + 1 = 0. The factor p^-
-
=0 detennines therefore the directions of the envelope at the conical point. VIII. In verification of the equation z =
ka? + y"") + ^ ax (a? + y') for a quadric surface in the neighbourhood of the umbilicus, I remark that, starting from the equation x' y" z^ ^ — I- — -I — = 1 a^ ^ b' c^ 130 ON DIFFERENTIAL EQUATIONS AND UMBILICI. [330 of an ellipsoid, and taking a, 0, 7 as the coordinates of the umbilicus, and 6 as the inclination to the axis of x of the tangent to the principal section through the umbilicus, then transforming to the umbilicus as origin and the new axes through that point, viz. the axes of x, z being the tangent and normal in the plane of ax:, and the axis of y being at right angles to this (or in the direction of h), the equation becomes (a + a; cos ^ — ^ sin 6y y^ (y—xsind — zcoady = 1, or, expanding, ('^'a.'^ i'
j.9-rf?^?ii 78in^
/asinf 7C08^
, ,/cos»^ sin=^ y^ ,/sin^^ cos= ^
" . . /,/! 1
« + ^(-^+-^) + | + ^(^- + -^)-2-sm^cos^(—)=0. /a' - 6= c 'Jb'-c^ a A/ , 7 = . . c s/a^ - b' But we have tane = and thence a 'Ja? -&' . a c Va" - 6« c " a'^b'-cf' a 8m5=.-^= = j-a, cosg = ^ ,-. " = ny; bVa'—d' ^«"' """" b>/a^-c^ be and substituting these values, the equation becomes or, what is the same thing, z=-n-h'^+y^)+^'^o?-^^'^^-<^z^ + a'6' + c'6' - g't" -^"; ca whence approximately and thence to the third order in x, y, which is of the form in question. 5, Downing Terrace, Cambridge, November 2, 1863.

331 ANALYTICAL THEOREM RELATING TO THE FOUR CONICS INSCRIBED IN THE SAME CONIC AND PASSING THROUGH THE SAME THREE POINTS

[From the Philosophical Magazine, vol. xxvii. (1864), pp. 42, 43.]

[From the Philosophical Magazine, vol. xxvii. (1864), pp. 42, 43.] Imagine the four conies determined, and, selecting at pleasure any three of them, let their chords of contact with the given conic be taken for the axes of coordinates, or lines x = 0, y=0, z=Q; then, taking for the equation of the given conic U = a, b, c, f, g, h^x, y, 2y = 0, the equations of the selected three conies must be of the form U + loe'=0, U+my^=0, U + nz' = 0, where I, m, n are to be determined in such manner that these conies may have three common points; the resulting values of I, m, n, and of the coordinates of the three common points, that is, the three given points, will of course be functions of the coefficients (a, b, c, /, g, h) ; and the equation of the fourth conic will be of the form U+(ix+jy+kzy=0. There is no difficulty in carrying out the investigation: it is found that the coordi- nates of the given points must be taken to be (-/. 9. ^); (/. -9> f^)'> if. 9> -h) respectively, and that, writing as usual K=abc-af- bg' - ch' + 2fgh, 17—2 the equations of the four conies are U + iK-abc)^, = 0, U+(K-abc)(j.+ ^ + ^' = 0. It is in fact easy to verify directly that each of these conies passes through the three given points ; but the equations may also be exhibited in the form proper for putting this in evidence. Putting for shortness X-^-i-^ F-^-u* 7-'"j,y ^—g^h' ^-h+f' ^-f-^-g- the equations of the sides of the triangle formed by the given points are Z = 0, Y=0,Z = 0, and the foregoing equations of the four conies may be expressed in the form i-bf-ch' + 2/gh)TZ+ bg
ZX+ ch
XY=0, af.7Z+ (- ch? -af + Ifgh) ZX+ ch
XY=0, af.YZ+ hf .ZX+- af - bg' + 2fgh) XY=0, (- V - ch' + ygh) 7Z + (-ch" -af- + 2/gh) ZX + (-ap- bg' + 2fgh) XY=0, which is the required form.
Cambridge, November 28, 1863.

332 ANALYTICAL THEOREM RELATING TO THE SECTIONS OF A QUADRIC SURFACE

[From the Philosophical Magazine, vol. xxvii. (1864), pp. 43, 44.]

[From the Philosophical Magazine, vol. xxvii. (1864), pp. 43, 44.] The four sections x=0, y = 0, z = 0, w = 0 of the quadric surface aaˆ +byˆ+ &xy 'Jah — cz" — dwˆ = 0 are each of them touched by each of the four sections x'/2a + y'/2b + ˆVc + w Vd = 0; where it is to be noticed that the radicals V2a, V26 are such that their product is = 2 Va6 if Va6 be the radical contained in the equation of the surface. There is of course no loss of generality in attributing a definite sign to the radical '/2a ; but upon this being done, the sign of the radical v26 is determined, whereas the signs of '/c and Vd are severally arbitrary. We may if we please write the equation of any one of the last-mentioned sections in the form X '/2a + y '/2b + zˆc-
-v3 '/d = 0, it being understood that the radicals '/2a, '/2b have each a determinate sign, but that the signs of Vc and '/d are each of them arbitrary. To prove the theorem in question, it is enough to show (1) that the sections x = 0, x'/2a+y '/2b + zˆ+wˆd = 0; (2) that the sections z = 0, x'/2a + y'/2b + w'/d = 0, touch each other. 134 ANALYTICAL THEOREM RELATING TO SECTIONS &C. [332 1. The sections x = 0, x'/2a + y'/2b + z'/c + w*/d = 0 of the quadric surface flws" + bi/' + 6x1/ 'J ah — a? — dw* = 0 will touch each other if, combining together the equations a; = 0, y V26 + z 'Jc + w Vd = 0, bt/' - cz* - dvf = 0, these give a twofold value (pair of equal values) for the ratios y : z : w. We in fact have iy" -c2!'-dv/'= bf-cz' - (y vˆ + z -Jcf, –hyˆ- 2cGBPi' - 2yz V26c, = -(yV6 + ˆV2cy; and the right-hand side being a perfect gquare, the condition of contact is satisfied. 2. In like manner we have the system 2: = 0, X V2a + y V26 4- w Vd = 0, aa?-
-hy- + Qxy 'Jab - dvp = 0, which gives aaˆ + byˆ + Qxy V06 - dw* = (uc'+by'+6xy'Jab-(x'ˆ + y'J2by, = — aaf — by- + 2xy yah, = -x'Ja — y '/bf ; and here also, the right-hand side being a perfect square, the condition of contact is satisfied. Cambridge, November 28, 1863.

333 NOTE ON THE NODAL CURVE OF THE DEVELOPABLE DERIVED FROM THE QUARTIC EQUATION $(a, b, c, d, e \pitchfork t, 1)^4 = 0$

[From the Philosophical Magazine, vol. xxvii. (1864), pp. 437–440.]

a, h, c, d, ej,t, 1)* = 0. [From the Philosophical Magazine, vol. xxvii. (1864), pp. 437 — 440.] Considering the coefficients (a, b, c, d, e) as linear functions of the coordinates ar, y, z, w, then the equation Disct. (a, h, c, d, e'^t, 1)* = 0, or, as it may be written, (ae - 46d + 3c')' - 27 (ace + 2bcd - ad^ - b'e -^y = 0 represents, as is known, a developable surface or "torse," having for its edge of regression (or cuspidal curve) the sextic curve the equations whereof are ae -46d+3c=' = 0, ace + 2bcd — ad^ — YENe — c^=0; and for its nodal curve, a curve the equations whereof (equivalent to two independent relations between the coordinates) are ac — b' _ad — bc_ae+ 2bd — Sc^_be — cd_ce — d' a 26 6c 2d e ' or, as these may also be written, a'd-Sabc +26' =0, a'e + 2ahd - 9ac^ + 66=^0 = 0, abe - Zacd + 26"^ =0, ad^ — b-e = 0, ade - 36ce + 2bd^ = 0, ae" + 2bde - 9c=e + 6c* = 0, be^ -Scde + 2d' =0; which curve is in fact an excubo-quartic, — viz. a quartic curve the partial intersection of a quadric surface and a cubic surface, having in common two non-intersecting right lines. To show that this is so, I remark that the coefficients a, b, c, d, e, quk linear functions of the four coordinates, satisfy a lineai- equation which may be taken to be a+b+c+d+e=0; this being so, the first form shows that the curve in question lies on the quadric surface ac-h^ + ^(ad-bc) + ^ (ae + 2bd - 3c») + J (be -cd) + ce-dr = 0, or, as this equation may also be written, ca-^b-^c-^d + e)-b'^ + ^ad + ^(ae+ 2bd) + ^be-d^ = 0. Substituting for c its value, this equation is -(a + e + b+d)(^a + ^e)-YEN + ^ad + ^(ae + 2bd) + ^be-d^=0, or, what is the same thing, 9 (a + e + 6 + d) (a + e) + 6 (6= + d») - 3 (ad + be) - (ae + 2bd) = 0. Hence, finally, the equation of the quadric surface is 9a^ + 17ae + 9e» + 66= - 2bd + 6d- + 9ab + 9de + 6ad + 6be = 0 ; and the curve lies also on the cubic surface ad!'-b^e = 0. It only remains to show that these surfaces have in common two right lines, and to find the equations of these lines. The cubic surface is a skew surface or " scroll " such that the equations of any generating line are d —6b = 0, e — EURPa = 0, where Q is an arbitrary parameter. But considering the two lines (cZ-^,6 = 0, e-e^a = 0
(d-0,b = O, e-d^a^O), the general equation of the quadric surface through these two lines may be written A . (d-e,b)(d-0,b) + B . (e-e^^a)(e-e^^a) + C . (d-e^b)(e-ei^a) + (d-ej))(e-e^^a) + -E^-s-
(d -e,b)(e- di a) -(d- eJb) (e - 6,^ a)] = 0 "i "a or, expanding and reducing, A d'-e, + 0,)bd + e, e, h'
+ 5 e= - (0i» + Oi) ea + e.^'d" + G 2de - 0^ + e^) ad - (0^ + 0,) be + 0A (^i + 0^) ah] + B (0i + 0^)ad- be- 0Aab

= Q, which, if ^1, 02 are the roots of the equation ^ — ^^+1 = 0, and therefore ^i + ^j = J, ^,^, = 1, and 0,' + 0i-=–'^, is Aid"- ^db+b') + B(e- + Jf ae + a^) + C2de + i^ad-^be + iab) + D ^ ad- be- ab) = 0. Putting A =6, 5 = 9, (7 = f, D = –!/, this is 9 (a- + -y^ ae + e=) + 6 (b' - ^bd + d') + f (iab+ 2 de + ifad-^be) + i^(ab - ^ ad+ be) = 0, []which is the before-mentioned quadiic surface ; hence the quadric surface and the cubic surface intersect in the two lines d-0,b = O, e-0i'a = O), d-0,b = O, e-0ia = O) (where 0^, 0. are the roots of the quadric equation 0^ —^0i + l =0); and they con- sequently intersect also in an excubo-quartic curve, which is the theorem required to be proved. Blackheath, March 26, 1864.

334 NOTE ON THE THEORY OF CUBIC SURFACES

[From the Philosophical Magazine, vol. xxvii. (1864), pp. 493–496.]

[From the Philosophical Magazine, vol. xxvii. (1864), pp. 493—496.] The equation AX' + BY' + 6GRST = 0, where X+Y+R + S+T=0, represents a cubic surface of a special form, viz. each of the planes R = 0, 8 = 0, T = 0 is a triple tangent plane meeting the surface in three lines which pass through a point (^); and, moreover, the three planes AX^ + BY' = 0 are triple tangent planes intersecting in a line. It is worth noticing that the equation of the surface may also be written oaf + by' + c (ii? + d" 4- w") = 0, where x + y + u + v []'r lu = 0. In fact, the coordinates satisfying the foregoing linear equations respectively, we have to show that the equation AX^+BY"+(iCR8T=aa? + hf + cxi'-irTl' + vf) may be identically satisfied. We have aa? + by^ + c (u' + if + iif) = cue* + by^ + c[u +V+ wy — 3 (y + w) (w + u) u + v)] = aa^ + by' — c(x+yy — 3c (u + w) (w + u) u + v), ' The tangent plane of a surface intersects the surface in a curve haying at the point of contact a double point, and in like manner a triple tangent plane intersects the surface in a curve with three double points, viz. each point of contact is a double point ; there is not in general any triple tangent plane such that the three points of contact come together, or (what is the same thing) there is not in general any tangent plane intersecting the surface in a curve having at the point of contact a triple point. A surface may, however, have the kind of singularity just referred to, viz. a tangent plane intersecting the surface in a curve having at the point of contact a triple point; such tangent plane may be termed a 'tritom' tangent plane, and its point of contact a ' tritom ' point : for a cubic surface the intersection by a tritom tangent plane is of course a system of three lines meeting in the tritom point. The tritom singularity is sibi- reciprocal; it is, I think, a singularity which should be considered iu the theory of reciprocal surfaces. 334] NOTE ON THE THEORY OF CUBIC SURFACES. 13^ which is to be = AX' + BY'> + 6CRST; and we may find X, Y, R, S, T, linear functions of x, y, u, v, w, so as to satisfy these equations, and so that in virtue of a; + y + u + v + w = 0, we shall have also X + Y+R+S + T=0. For, assuming AX' + BY' ==aa^ + by'-cx + yf, X + Y = X + y, R = H" + w), a = - 4c, S = ^ (w + m), T =^u+v), we have identically AX* + BY^ + &CRST = aa? + hf-cx+yf - 3c (t; + w) (w +w) u + v), X+Y+R + S + T =x + y + u+v+w, and thus it only remains to show that we can find X, Y linear functions of x, y, such that AX' + BY' = ax' + by'-c(x + yy, X + Y = X + y. This is always possible ; in fact if U=aa^+by'—c(x + yf, then taking ^ for the cubicovariant, and D for the discriminant of U, we have J(* + VnU'), i(4> — Vnt/") each a perfect cube, say ^(^-^UU)=vx+pyy, and we then have U = ^

x + fiyy - (vx + pyy
= AX'+ BY', which is satisfied by if X = I (Xx + fiy), Y=m(yx +py), Al' — –, Bm' = - . 18—2 The equation X + Y = x + y then gives lk+ mv = 1, Ifi. + mp = 1, which give the values of I and m, and thence the values of A and B; and collecting all the equations, we have iJ= i(u+i<^), C=-4c, where *!>, D being respectively the cubicovaiiant and the discriminant of U= aa^ + by' — c(a; + y)'), for the formulae of the transformation AX^ + BY^+ 6CRST= a^ + bi/' + c(u» + v' + t(f). X+ 7+R + S+T^x + y + u + v + w. The equation cue' + ty* + c (m' + t;' + tv^) = 0, where X + y + u + V + w = 0, presents over the other form the advantage that it is included as a particular case under the equation aaf + by' + cm' + du' + ew' = 0 (where x + y + u + v + w = 0) employed by Dr Salmon as the canonical form of equation for the general cubic surface. 5, Doivning Terrace, Cambridge, April 29, 1864.

335 TABLES DES FORMES QUADRATIQUES BINAIRES POUR LES DETERMINANTS NEGATIFS DEPUIS $D = -1$ JUSQU'A $D = -100$, POUR LES DETERMINANTS POSITIFS NON CARRES DEPUIS $D = 2$ JUSQU'A $D = 99$ ET POUR LES TREIZE DETERMINANTS NEGATIFS IRREGULIERS QUI SE TROUVENT DANS LE PREMIER MILLIER

[From the Journal für die reine und angewandte Mathematik (Crelle), torn. LX. (1862), pp. 357–372.]

=-l JUSQU'A D= -100, POUR LES DETERMINANTS POSITIFS NON CARRE'S DEPUIS D^2 JUSQU'A Z> = 99 ET POUR LES TREIZE DETERMINANTS NEGATIFS IRREGULIERS QUI SE TROUVENT DANS LE PRE- MIER MILLIER. [From the Journal fur die reitte und angewandte Mathematik (Crelle), torn. LX. (1862), pp. 357—372.] Les tables suivantes sont arrang^es de la manifere pr^sciite dans les " Disquisitiones arithmeticae." Dans le mdmoire de Lejeune Dirichlet "Recherches sur diverses appli- cations de I'analyse a la theorie des nombres," torn, xix (1839), p. 338 de ce Journal on trouve un tableau dans lequel les regies qui servent k former les caracteres des genres sont resum^es. Soit D = PS^ ou 2PS^, S^ ddsignant le plus grand carr^ que D contient, et P un nombre impair; soient de plus p, p', p"... les facteurs premiers indgaux de P et r, r', r" ... les nombres premiers impairs qui divisent S sans diviser P; ^rivons enfin pour abreger S = (—) ''* , «=(-) cite le tableau suivant: Premier cas, D = PS', P =
(mod. 4) Cela posd on trouve k I'endroit S=l (mod. 2) m m p"- m 7' m r' ' S=2 (mod. 4) m m p"- s, m 7' m 7' ... S=0 (mod. 4) m P' m p"- s, e. m m 142 TABLES DE FORMES QUADKATIQUE8. Deuxifeme cas, D = PS*, P = S (mod. 4) ,S;=1 (mod. 2) [335 p p p p Troisifeme cas, I> = 2PS', P=l (mod. 4) m m S=2 (mod. 4) S=0 (mod. 4) ^ m m P P tn tn r' 7' e, 8=1 (mod. 2) S = 0 (mod. 2) Quatrifeme cas, D = 2PS', P = S (mod. 4) P P m m p p m m 7> /' *** S. 8=1 (mod. 2) 8=0 (mod. 2) Se, e. 7/1 w m VI p' F' m m m m r r m m 7' 7' *** m Dans ce tableau la notation — dans laquelle j'omets les parentheses usit^es, signifie le caractfere d'un nombre quelconque m par rapport au nombre premier impair p, c.-a-d. que m est r^sidu ou uon r^sidu de p selon que — = + 1 ou = — 1, de m^me h est le caractfere de m par rapport au nombre 4, savoir m = 1 ou 3 (mod. 4) selon que S = + 1 ou = — 1, enfin e, Se sont les caractferes de m par rapport au

nombre 8, savoii* m =
ou 7 (mod. 8) pour 6 = + l, =3 ou 5 (mod. 8) pour 6 = — 1> m = 1 ou 3 (mod. 8) pour Se = + 1, =5 ou 7 (mod. 8) pour Se = — 1. Si pour un d^teiminant donnd on veut former au moyen de ce tableau les caractferes des genres, on prend la ligne horizontale qui convieut k ce determinant ; h, tous les caractferes — , — , ... — , — , ... Z, e, Be qui se trouvent dans la ligne horizontale, on attribue les signes + ou — k volonte, avec cette restriction cependant que le signe compose des signes qui se trouvent dans la premiere partie de la ligne dont il s'agit soit positif. Si par exemple le determinant donn^ est D = — 35, on a i) = — 35 = PS', P = — 35 = 1 (mod. 4), 8 = 1 = 1 (mod. 2), les nombres p, p',... sont 5, 7, et les signes que Ton doit conm m y sid^rer sont De 1^ on obtient les caractferes m m 5' T + il y a done deux genres de I'ordre proprement primitif. Dans le cas dont il s'agit (et en general pour D = 1 (mod. 4)) il y a un ordre improprement primitif avec des genres qui ont les caracteres identiques a ceux des genres de I'ordre proprement primitif, les caractferes se rapportant dans ce cas k la moitid d'un nombre quelconque represente par la forme. Pour faciliter I'impression des tables j'ai introduit deux nouvelles lettres a et /8 dont voici la definition. Pour tous les determinants auxquels se rapportent mes tables, c.-a-d. pour les determinants ndgatifs quelconques et positifs non-carrds depuis — 100 jusqua + 99 ainsi que pour les determinants negatifs irrdguliers du premier millier, le nombre des facteurs premiers designds ci-dessus par les lettres p, p', p" ... , r, r', r" ... n'exefede pas deux. Soit done q le plus petit de ees faeteurs premiers et q le plus grand lorsqu'il y en a deux, je designe par a le earactfere — et par yS le earactfere — . Dans la eolonne relative a la composition et portant I'inscription Cp, je represente comme k I'ordinaire par I'unite la forme principale, par la lettre c une forme qui produit par la duplication la forme principale, par les lettres d, e, ... des formes qui la produisent par la triplication, la quadruplieation, etc., de maniere que Ton ait c^ = l, d^=l, e* = l, /'=1, EUR/'=!, /t' = l, i' =1, j' =1, etc. Les notations d, d^ par exemple signifient deux formes dififerentes dont chacune produit par la triplication la forme principale. Je reprd- sente de plus par a la forme principale de I'ordre improprement primitif et par ac, ad, ... des formes qui produisent a par la duplication, la triplication, etc. Dans Tenumdration des classes, j'ai toujours dcrit en premier lieu I'ordre proprement primitif, en le faisant suivre apres un trait de separation par I'ordre improprement primitif lorsqu'il existe. Dans ehaeun des deux ordres les divers genres se trouvent separds les uns des autres par des traits subordonnds. Pour les determinants positifs les periodes sont donnees par une abreviation facile k comprendre. Chaque forme de la periode ayant son dernier coefficient egal au premier de la suivante, cette valeur ideutique n'a ete iraprimee qu'une fois; de plus les coefficients exterieurs a, c des formes (a, b, c) ont ete distinguds des coefficients b en imprimant ees demiers en caractferes plus petits. Ainsi pour le determinant 7 la periode de la classe principale (1, 0, — 7) est donnee par les nombres 1, 2, — 3, I, 2, I, - 3, 2, 1 qui representent la serie

des formes (1, 2, -3), (-3, 1, 2), (2, 1, -3), (-3, 2, 1). Londres, 6 Novembre, 1860. 144 TABLES DE FORMES QUADRATIQUES. [335 Table I des foiines quadratiques binaires ayant pour determinants les nombres nf^gatifs depuis Z) = — 1 jusqu'k D = — 100. D Classes a /J s EUR St Cp D Classes a /9 » e 1 St Cp D Classes a ^|» < St - 1 1. 0, 1 1 []*" 1 -20 1, 0, 20 2, 0, 5 3, 1, 7 3,-1, 7 + + + + 1 -31 1, 0, 31 5, 2, 7 5,-2, 7 + + + 1 - 2 1, 0, 2 1 + 1 d - 3 1, 0, 3 + 1 -1 e - 2, 1, 16 4,-1, 8 4, 1, 8 + + + 2. 1, 2 + or <T -21 1, 0, 21 + + + + 1 <id - 4 1, (», 4 + 1 3, 0, 7 c - 5 1, 0, 5 + + 1 c -32 1, 0, 32 4, 2, 9 + + + + 1 2, 1, 11 5, 2, 5 «i 2, 1, 3 + — -1 + CCj - 6 1, 0, 6 2, 0, 3 + + 1 c 3, 1, 11 3, -1, 11 [] -22 1. 0, 22 + 1 + 1 c ^ 2, 0, 11 - 7 1, 0, 7 + 1 -33 -34 1, 0, 33 2, 1, 17 + + + + 1 -23 1, 0, 23 3, 1, 8 3,-1, 8 1 d d' 2. 1, 4 + a- 1 c + + + c - 8 1, 0, 8 + + 3, 0, 11 + + 3, 1, 3 6, 3, 7 2, 1, 12 4,-1, 6 4, 1, 6 + + + a- <rd ard- cc, - 9 1, 0, 9 2, 1, 5 + + + 1 1, 0, 34 2, 0, 17 5, 1, 7 7,-1, 7 + + 1 c + -10 1, 0, 10 + + 1 -24 1, 0, 24 3, 0, 8 5, 1, 5 + + + + + + 1 1 e 2, 0, 5 c c -11 1, 0, 11 3, 1, 4 3,-1, 4 + + d d^ a- -35 1, 0, 35 4, 1, 9 4,-1, 9 + + + + + + 1 4, 2. 7 CCi + -25 1, 0, 25 + + + 1 c 2, 1, 6 + 3, -1, 12 5, 0, 7 3, 1, 12 9 2, 1, 13 -12 1, 0. 12 3, 0, 4 + + + 1 C -26 1, 0, 26 3,-1, 9 3, 1, 9 5, 2, 6 2, 0, 13 5,-2, 6 + + + + + + 1 f 2, 1. 18 6, 1, 6 + + -13 1, 0, 13 2, 1, 7 + + 1 c y -36 1, 0, 36 4, 0, 9 + + + + + + 1 -14 1, 0, 14 2, 0, 7 + + + + 1 5, 2, 8 5,-2, 8 e 3, 1, 5 3,-1, 5 e -27 -28 1, 0, 27 4, 1, 7 4,-1, 7 + + + 1 d -37 1, 0, 37 2, 1, 19 + + 1 -15 1, 0, 15 3, 0. 5 + + 1 2, 1, 14 + or c 2, 1, 8 + + a- a-c 1, 0, 28 + + + 1 -38 -39 1, 0, 38 6, 2, 7 6,-2, 7 3, 1, 13 2, 0, 19 3, - 1, 13 1, 0, 39 3, 0, 13 5, 1, 8 5,-1, 8 + + + + + + + — + + + 1 4, 1, 4 4, 0, 7 c -29 1, 0, 29 5, 1, 6 5,-1, 6 + + + + + + 1 -16 1, 0, 16 4, 2, 5 + + + 1 9 c / -17 1, 0, 17 + + + + 1 J— 3, 1, 10 2, 1, 15 3, - 1, 10 9 2, 1, 9 1 3, 1, 6 3,-1, 6 e C -30 1, 0, 30 2, 0, 15 3, 0, 10 5, 0, 6 + + + + + + 1 -18 1, 0, 18 2, 0, 9 + + + 1 c c 2, 1, 20 6, 3, 8 + + + + <T cc, o-e^ -19 L 0, 19 4, 1. 5 4,-1, 5 + + + 1 4, - 1, 10 4, 1, 10 ae (T^ 2, 1, 10 + <T 335] TABLES DE FORMES QUADRATIQUES. 145 I D Classes a /3 5 e Sc Cp D Classes a /3 S c ie Cp D Classes a S e 5f -40 1, 0 4, 2 5, 0 7, 3 40 11 + + + + 1 c -50 1 1, 0, 50 6, 2, 9 6,-2, 9 + + + + + + + + + 1 9' 9* 9 9" -59 1 3 7 4 5 5 4 7 3 0, 59 1, 20 , 2, 9 1, 15 -1, 12 1, 12 -1, 15 -2, 9 -1, 20 + + + + + + 4- + 1 J 8 7 r 3, 1, 17 2, 0, 25 3, -1, 17 -41 1, 0 5, 2 2, 1 5,-2 3, 1 6, -1 6, 1 3, -1 41 9 21 9 + + + + + + 1 ^*^ i i' f -6 + + -51 1, 0, 51 4, 1, 13 4, - 1, 13 + + + + + + 1 9' f f 14 7 7 14 9 6 6 1, 30 1, 10 - 1, 10 5, 2, 11 3, 0, 17 5, - 2, 11 9 9" + + -60 1 3 4 5 0, 60 0, 20 + ,+ + + + + 1 -42 1, 0 2, 0 3, 0 6, 0 42 + + + + 1 + + 1 c CCi 2, 1, 26 6, 3, 10 + + <r9 21 c 0, 15 0, 12 14 -52 1, 0, 52 4, 0, 13 + + + + 1 «i 7 CCi -43 1, 0 4, 1 4, -1 43 11 11 + + + 1 1 d 7, 2, 8 7,-2, 8 e e> -61 1 5 5 7 2 7 0, 61 -2, 13 2, 13 + + + + + + 1 -53 1, 0, 53 6,-1, 9 6, 1, 9 + + + + + + 1 9' 3, 10 1, 31 -3, 10 2, 1 22 + <T 9 -44 1, 0 5, -1 5, 1 3, 1 4, 0 3, -1 44 9 9 + + + + + + + 1 9' + + 3, 1, 18 2, 1, 27 3, -1, 18 9 9' 1 / 9* 9 9' 9' 9 -62 1 7 7 6 2 6 0, 62 1, 9 -1, 9 + + + + + + 1 15 11 15 - 9 !-54 1 1 1 1, 0, 54 7, 3, 9 7, -3, 9 ft, 1, 11 2, 0, 27 5, -1, 11 + + + + + + y 2, 11 0, 31 -2, 11 9 -45 1, 0 5, 0 2, 1 7, 2 45 9 + + + + + + 1 c -63 1 7 8

8 0, 63 0, 9 + + + + + + + 1 23 i CCj -55 [1, 0, 55 5, 0, 11 7, 1, 8 7,-1, 8 + + + + 1 3, 9 -3, 9 e -46 1, 0 2, 0 5, 2 5,-2 46 23 + + + ! 1 e- e + e 9 -*» 8, 4, 4, 1, 32 1, 8 -1, 16 1, 16 + + + + + + <r 10 10 «re» 2, 1, 28 8, 3, 8 4, 1, 14 4, -1, 14 + + + + a- ere" o-e -47 1, 0 3, 1 7, 3 7,-3 3, - 1 47 16 8 > 8 16 + + + + + 1 / p p <^p o-s' a-e -64 1, 5 4, 5, 0, 64 1, 13 2, 17 -1, 13 + + + + + + + 1 [-56 1, 0, 56 8, 4, 9 5, 2, 12 5, -2, 12 + + + + + + + + + + + + 1 e 2, 1 6, 1 4, 1 4, -1 6, -1 24 8 12 12 8 + + + + + 1 1 -65 1, 9 3, 3 5, 2 6 R 0, 65 4, 9 1, 22 -1, 22 + + + + + + + + + + + + 1 e 4, 2, 15 7, 0, 8 3, 1, 19 2, - 1, 19 c e 0, 13 1, 33 -1, 11 1 1 1 -48 1, 0 3, 0 7, 1 4, 2 48 16 7 13 + + + + + + - + 1 ce c c -57 1, 0, 57 + + + + + + 1 c ce + - 3, 0, 19 2, 1, 29 6, 3, 11 -49 1, 0 2, 1 5, 1 5, - 1 49 25 10^ , 10 + + + + 1 e -58 1, 0, 58 + + 1 c 2, 0, 29 C. V. 19 146 TABLES DE FORMES QUADRATIQUES. [335 V Classes a /S 6 (St Cp D Classes a p S t Se V Classes a p S e it -66 1, 0, 06 3, 0, 22 7. 2, 10 7, - 2, 10 + + + + + + + + + + + + 1 -74 1, 0, 74 3, - 1, 25 9, -4, 10 9, 4, 10 3, 1, 25 + + + + + 1 + + + + + 1 A" A* -81 1. 0, 81 9, - 3, 10 9, 3, 10 5, 2, 17 2, 1, 41 5, -2, 17 + + + + + + + + + 1 e 9 2, 0, 33 6, 0, 11 c / 5, 1, 15 6,-2, 13 2, 0, 37 6, 2, 13 5, - 1, 15 k 5, -2, 14 6, 2, 14 ce -82 1, 0, 82 2, 0, 41 + + + + 1 -67 1, 0, 67 4, 1, 17 4, - 1, 17 + + + 1 d 7, 3, 13 7, -3, 13 e -75 1, 0, 75 4, -1, 19 4, 1, 19 + + + + + + + + + 1 -83 1, 0, 83 3, 1, 28 9, 4, 11 4, 1, 21 7, -1, 12 7, 1, 12 4, - 1, 21 9, -4, 11 3, - 1, 28 + + + + + + + + + 1 2, 1, 34 + <r -68 1, 0, 68 8, 2, 9 4, 0, 17 8, -2, 9 + + + + + + + + + + + + 1 t» i* y 7, 3, 12 3, 0, 25 7, -3, 12 9 2, 1, 38 6, 3, 14 + + + <r 3, 1, 23 7, -3, 11 7, 3, 11 3, - 1, 23 t i' i' <^? -76 1, 0, 76 5, -2, 16 5, 2, 16 + + + + + + + + + 1 J 2, 1, 42 6, 1, 14 6, -1, 14 + + + -69 1, 0, 69 6, 3, 13 5, 1, 14 5, -1, 14 2, 1, 35 3, 0, 22 7, - 1, 10 7, 1, 10 + + + + + + + + + + + + 1 7, 1, 11 4, 0, 19 7, -1, 11 9 -84 1, 0, 84 4, 0, 21 + + + + + + _ + + + + + + 1 e e ce' -77 1, 0, 77 9, 2, 9 6,-1, 13 6, -1, 13 + + + + + + + + + + + + 1 e= e e» c ce» ce Cfi» 5, 1, 17 5, -1, 17 e 3, 0, 28 7, 0, 12 c ce c^ 2, 1, 39 7, 0, 11 8, 2, 11 8, -2, 11 -70 1, 0, 70 + + + + + + 1 c cc, r^ 2, 0, 35 3, 1, 26 3, - 1, 26 -85 1, 0, 85 + + + + + + 1 5, 0, 14 7, 0, 10 5, 0, 17 c -78 1, 0, 78 + + + _ + + + 1 e 2, 1, 43 10, 5, 11 -71 1, 0, 71 3, 1, 24 8,-1, 9 5, -2, 15 5, 2, 15 8, 1, 9 3. - 1, 24 + + + + + + + 1 h h* 2, 0, 39 cc^ 3, 0, 26 6, 0, 13 -86 1, 0, 86 6, 2, 15 9, 2, 10 9, -2, 10 6, -2, 15 + i + + + + + + 1 -79 1, 0, 79 5, 1, 16 8, -3, 11 8, 3, 11 5, -1, 16 + + + + 1 / r r + + + 5, 2, 18 3, 1, 29 2, 0, 43 3, - 1, 29 5, -2, 18 k 2, 1, 36 6, 1, 12 4, -1, 18 8, - 3, 10 8, 3, 10 4, 1, 18 6, -1, 12 + + + + + + + a- a-h 2, 1, 40 8, -1, 10 4, 1, 20 4, - 1, 20 8, 1, 10 + + + + + cr -87 1, 0, 87 7, 2, 13 7, -2, 13 8, 1, If 3, 0, 29 8, -1, 11 + + + + []1- + 1 -72 1, 0, 72 + + []1- + + + 1 c -80 1, 0, 80 9, 1, 9 + + + + + + + + + + + + 1 4, 2, 19 9 8, 0, 9 ec, 3, 1, 27 3, - 1, 27 e c 8, 4, 11 2, 1, 44 8, -3, 12 8, 3, 12 + + + + + + 5, 0, 16 4, 2, 21 -73 1, 0, 73 2, 1, 37 + + + + 1 e" e 7, - 2, 12 7, 2, 12 ce 7, 2, 11 7,-2, 11 4, 1, 22 6, 3, 16 4, -1, 22 <) 335] TABLES DE FORMES QUADRATIQUES. 147 D Classes a p a e ie Cp D Classes a /s s e 3c D Classes a P , EUR 1 -88 1, 0, 88 4, 2, 23 + + + + 1 c -92 1, 0, 92 + + 1 -96 1, 0, 96 4, 2, 25 5, 2, 20 5, - 2, 20 +

+ + + + + . 1 9, 4, 12 9, - 4, 12 3, 1, 31 4, 0, 23 3, - 1, 31 + + -t- + + + + 9' 9* + + 8, 0, 11 8, 4, 13 1 e 9 9^ 9' + + + -89 1, 0, 89 9, 1, 10 5, 1, 18 2, 1, 45 5, -1, 18 9, - 1, 10 + + + + + + -t- + + + 1 3, 0, 32 11, 5, 11 c c -93 1, 0, 93 + + + + + + 1 7, 3, 15 7, - 3, 15 ce 3, 0, 31 c 2, 1, 47 -97 1, 0, 97 2, 1, 49 + + + + - 1 6, 3, 17 CCi 3, 1, 30 7, -3, 14 6, - 1, 15 6, 1, 15 7, 3, 14 3, - 1, 30 TO m» -94 1, 0, 94 7, 2, 14 2, 0, 47 7,-2, 14 5, 1, 19 10, 4, 11 10,-4, 11 5,-1, 19 + + + + + + + + 1 i' i 7, 1, 14 7,-1, 14 e -98 1, 0, 98 9, 1, 11 2, 0, 49 9,-1, 11 + + + + + + + + + + + + + — — + + + + + + + + 1 90 1, 0, 90 9, 0, 10 7, 1, 13 7, -1, 13 + + + + + + + + + + + + 1 3, 1, 33 6, 2, 17 6,-2, 17 3, - 1, 33 i e -95 1, 0, 95 9,-2, 11 5, 0, 19 9, 2, 11 + + + + + + + + 1 2, 0, 45 5, 0, 18 9, -3, 11 9, 3, 11 c -99 1, 0, 99 4. - 1, 25 4, 1, 25 5, 1, 20 9, 0, 11 5, - 1, 20 1 ce 3, 1, 32 8, 3, 13 8, - 3, 13 3, - 1, 32 i 9 -91 1, 0, 91 4, 1, 23 4, - 1, 23 + + + + + + 1 9' 9' 2, 1, 48 5, 1, 12 10, 5. 12 8, - 1, 12 6, 1, 16 4,-1, 24 4, 1, 24 6,-1, 16 + + + + + + + + a- <ri' 2, 1, 50 10, 1, 10 + + + 5, -2, 19 7, 0, 13 5, 2, 19 9 <^9 -100 1, 0, 100 4, 0, 25 8, 2, 13 8,-2, 13 + + + + + + 1 2, 1, 46 10, 3, 10 + + a <r9 ^ * - 19—2 148 TABLES DE FORMES QQADRATIQUES. [335 Table II des formes quadratiques binaires ayant pour determinants les nombres positifs non-carr^ depuis D = 2 jusqa'k D = 99. D Classes a /s a c i(Cp P^riodes 2 1, 0, - 2 + 1 1, 1, - 1, 1, 1 3 1, 0, - 3 + + 1 1, 1, - 2, 1, 1 - 1, 1, 2, 1, - 1 - 1, 0, 3 c 5 1, 0, - 5 + 1 1, «. - 1, ». 1 2, 1, - 2, 1, 2 2, 1, - 2 + a 6 1, 0, - 6 + + 1 c 1, «, - 2, 2, 1 - 1, 2, 2, 2, - 1 - 1, 0, 6 7 1, 0, - 7 + + 1 1, 2, - 3, 1, 2, 1, - 3, s, 1 - 1, 2, 3, 1, - 2, 1, 3, 2, - 1 - 1, 0, 7 c 8 1, 0, -8 + + + 1 1, 2, - 4, 2, 1 - 1, a, 4, 2, - 1 - 1, 0, 8 c 10 1, 0, - 10 + + 1 1, 3, - 1, 3, 1 2, 2, - 3, I, 3, 2, - 2, 2, 3, 1, - 3, 2, 2 2, 0, - 5 c 11 1, 0, - 11 + + 1 1, 3, - 2, 3, 1 - 1, 3, 2, 3, - 1 - 1, 0, 11 c 12 1, 0, - 12 + + 1 1, 3, - 3, 3, 1 - 1, 3, 3, 3, - 1 - 1, 0, 12 c 13 1, 0, - 13 + 1 1, 3, - 4, 1, 3, 2, - 3, 1, 4, 3, - 1, 3, 4, 1, - 3, 2, 3, 1, 2, 3, - 2, 3, 2 -4,8,1 2, 1, - 6 + a r 14 1, 0, - 14 + + 1, 3, - 5, 2, 2, 2, - 5, 8, 1 - 1, 3, 5, 2, - 2, 2, 5, 3, - 1 - 1, 0, 14 c 15 1, 0, - 15 + + + + + + 1 1, 3, - 6, 3, 1 - 1, 3, 6, 3, - 1 2, 3, - 3, 3, 2 - 2, 3, 3, 3, - 2 - 1, 0, 15 c 2, 1, - 7 - 2, 1, 7 17 1, 0, - 17 + 1 1, 4, - 1, 4, 1 2, 3, - 4, 1, 4, 3, - 2, 3, 4, 1, - 4, 3, 2 2, 1, - 8 + o- 18 1, 0, - 18 + + + 1 c 1, *, - 2, 4, 1 - 1, 4, 2, 4, - 1 - 1, 0, 18 19 1, 0, - 19 + + 1 c 1, 4, - 3, 2, 5, 3, - 2, 3, 5, 2, - 3, 4, 1 - 1, 4, 3, 2, - 6, 3, 2, 3, - 5, 2, 3, 4, - 1 - 1, 0, 19 20 1, 0, - 20 + + + 1 1, 4, - 4, 4, 1 - 1, 4, 4, 4, - 1 - 1, 0, 20 c 21 1, 0, - 21 + + 1 1, 4, - 5, 1, 4, 3, - 3, 3, 4, 1, - 5, 4, 1 - 1, 4, 5, 1, - 4, 3, 3, 3, - 4, 1, 5, 4, - 1 2, 3, - 6, 3, 2 - 2, 3, 6, 3, - 2 - 1, 0, 21 c 2, 1, - 10 + + <r - 2, 1, 10 ore 22 1, 0, - 22 + + 1 c 1, 4, - 6, 2, 3, 4, - 2, 4, 3, 2, - 6, 4, 1 - 1, 4, 6, 2, - 3. 4, 2, 4, - 3, 2, 6, 4, - 1 - 1, 0, 22 23 1, 0, - 23 + + 1 c 1, 4, -7, 3, 2, 3, -7, 4, 1 - 1, 4, 7, 3, - 2, 3, 7, 4, - 1 - 1, 0, 23 24 1, 0, - 24 + + + + + + 1 c 1. *, - 8, 4, 1 - 1, 4, 8, 4, - 1 3, 3, - 5, 2, 4, 2, - 5, 3, 3 - 3, 3, 5, 2, - 4, 2, 5, 3, - 3 - 1, 0, 24 3, 0, - 8 Cl - 3, 0, 8 eCi 335] TABLES DE FORMES QUADRATIQUES. 149 D Classes a /» i EUR «e Cp PSriodeB 26 1 2 0,-26 + + 1 1, 5, - 1, 5, 1 2, 4, -5, 1, 5, 4, -2, 4, 5, 1, -5, 4, 2 0, -13 c 27 1 -1 0, -27 + + 1 c 1, B, - 2, 6, 1 - 1, B, 2, 6, - 1 0, 27 28 1 -1 1 0, -28 + + 1 1, 6, -3, 4, 4, 4, -3, B, 1 -1, S, 3, 4, -4, 4, 3, B, -1 0, 28 c 29 0,-29 + 1 1,

B, - 4, 3, 5, 2,-5, S, 4, 5, - 1, B, 4, S, -5, 2, 5, 3, -4, - 1, 5, 4, 3, -5, 2, 5, s, -4, s, 1, 6, - 4, s, 5, 2, - 5, a, 4, b 6, 1 2, 1,-14 + <T -1 30 1 -1 2, -2, 0,-30 + + + + + + 1 1, 6, -5, B, 1 -1. 6, 5, 5, -1 2, 4, - 7, 3, 3, 3, - 7, 4, 2 -2, 4, 7, 3, -3, 3, 7, 4, -2 0, 30 c 0, -15 Cl 0, 15 cc^ 31 1, -1, 0,-31 + + 1 1, s, - 6, 1, 5, 4, - 3, B, 2, s, - 3, 4, 5, 1, - 6, b, 1 - 1, B, 6, 1, - 5, 4, 3, B, - 2, 8, 3, 4, - 5, 1, 6, s, - 1 0, 31 c 32 1. -1, 0, -32 + + + 1 1, B, -7, 2, 4, 2, -7, 5, 1 -1, B, 7, 2, -4, 2, 7, B, -1 0, 32 c 33 1, -1, 0,-33 + + 1 1, B, - 8, 3, 3, 3, - 8, 6, 1 -1, 6, 8, 3, -3, 3, 8, 8, -1 2, B, -4, 3, 6, 3, -4, 6, 2 -2, s, 4, 3, -6, 3, 4, B, -2 0, 33 c 2, -2, 1, -16 + + o- 1, 16 (TO 34 1, -1, 3, -3, 0, -34 0, 34 + + + + 1 e 1, B, -9, 4, 2, 4, -9, 6, 1 -1, 8, 9, 4, -2, 4, 9, 6, -1 3, B, -3, 4, 6, 2, -5, 3, 5, 2, -6, 4, 3 - 3, B, 3, 4, - 6, 2, 5, 3, - 5, 2, 6, 4, - 3 -1,-11 -1, 11 35 1. -1. 2, -2, 0,-35 + + + + + 1 1, B, - 10, 8, 1 -1,8,10,8,-1 2, 6, -5, 8, 2 -2, 8, 5, 8, -2 0, 35 c 1,-17 Cl 1, 17 eCi 37 1, 3, 3, 0,-37 1,-12 -1,-12 + + + 1 d 1, 6, - 1, 6, 1 3, 4, - 7, 3, 4, B, - 3, 4, 7, 3, - 4, 8, 3 - 3, 4, 7, 3, - 4, 8, 3, 4, - 7, 3, 4, 8, - 3 2, 8, -6, 1, 6, 8, -2, 8, 6, 1, -6, 8, 2 2, 1,-18 + a- 38 1, -1, 0,-38 + + + + + + + — + 1 c 1 1, 6, - 2, 6, 1 - 1, «, 2, 6, - 1 0, 38 39 1, -1, 2, -2, 0,-39 1,6, -3, 6, 1 - 1, 8, 3, 6, - 1 2, B, - 7, 2, 5, 3, - 6, 3, 5, 2, - 7, 8, 2 - 2, 8, 7, 2, - 5, 3, 6, 3, - 5, 2, 7, B, - 2 0, 39 c 1,-19 Cy 1, 19 CCi 40 1, -1, 3. -3, 0,-40 + + + + + + + 1 1, ei -4, a, 1 - 1, 6, 4, 6, - 1 3, 4, -8, 4, 3, B, -5, B, 3 -3, 4, 8, 4, -3, 8, 5, 8, -3 0, 40 c 1,-13 ". 1, 13 CCi 41 1, 0,-41 1 1, 6, - 5, 4, 5, 6, - 1, 6, 5, 4, - 5, 6, 1 -1, 8, 5, 4, -5, 6, 1, 8, -5, 4, 5, 6, -1 2, 1,-20 + <T 42 1, -1, 2, -2, 0,-42 + + + + + + + + 1 1 6, - 6, 6, 1 - 1, 6, 6, 6, - 1 2, 6, - 3, 6, 2 - 2, 6, 3, 6, - 2 0, 42 c 0, -21 0, 21 43 1, -1. 0,-43 1 1, 6, - 7, 1, 6, 8, - 3, 4, 9, 8, - 2, B, 9, 4, - 3, B, 6, 1, - 7, 6, - 1, 8, 7, 1, -6, 6, 3, 4, - 9, 8, 2, 8, - 9, 4, 3, B, - 6, 1, 7, e, - 1 0, 43 c - 1 150 TABLES DE FORMES QUADRATIQUES. [335 D Classes a P i t ie Op P^riodes 44 1 -1 0,-44 + 1 + 1 1, 8, - 8, 2, 5, 3, - 7, 4, 4, 4, - 7, 3, 5, 2, - 8, 6, 1 0, 44 c - 1, 6, 8, J, - 5, 3, 7, 4, -4, 4, 7, 8, -5, s, 8, «, - 1 45 1, -1 0,-45 + + + 1 1, 8, - 9, 3, 4, 6, - 5, 5, 4, 3, - 9, 6, 1 0, 45 c -1, 6, 9, 3, -4, 8, 5, 6, -4, S, 9, 6, -1 2 -2, 1,-22 + + + 2, 5, - 10, 5, 2 1, 22 - 2, 6, 10, 5, - 2 46 1 -1 0,-46 + + 1 1, 6, - 10, 4, 3, 6, - 7, 1, 6, 4, -5, «, 2,8,-5,4, 6,«,-7,6,3,4,-10,e, 1 0, 46 c -1,8,10, 4,-3,6,7,2,-6,4, 5, 6,-2, 6, 6, 4,-6,*, 7, 5,-3, 4, 10, 8,-1 47 1, -1, 0,-47 + + 1 1, B, -11, 6, 2, «, -11, 6, 1 0, 47 0,-48 0, 48 c -1, 8, 11, 5, -2, 6, 11, 6, -1 48 1 -1, 3, -3, + + + + + + 1 1, 6, - 12, 6, 1 c -1, 6. 12, a, -1 0,-16 0, 16 3, 6, -4, 6, 3 -3, 6, 4, 6, -3 50 1, 2, 0,-50 + + + 1 1, 7, - 1, 7, 1 0,-25 c 2, 8, - 7, 1, 7, 6, - 2, 8, 7, 1, - 7, 8, 2 51 1, -1, 3 -3, 0, -51 + + + ' + ' 1 1, 7, -2,7, 1 0, 51 c - 1, 7, 2, 7, - 1 0,-17 c, 3, 6, -5, 4, 7, 3, -6, 8, 7, 4, -5, 6, 3 0, 17 cc^ -3, 6, 5, 4, -7, s, 6, 3, -7, 4, 5, 6, -3 52 1, -1, 0, - 52 + + + 1 1, 7, -3, 6, 9, 4, -4, 4, 9, 5, -3, 7, 1 0, 52 0,-53 c - 1, 7, 3, 8, - 9, 4, 4, 4, - 9, 8, 3, 7, - 1 53 1, + 1 1, 7, - 4, 6, 7, 2, - 7, 6, 4, 7, - 1, 7, 4, 5, - 7, a, 7, s, - 4, 7, 1 2 1,-26 + o- 2, 7, - 2, 7. 2 54 1, -1, 0,-54 0, 54 + + 1 1, 7, -5, 3, 9, 6, -2, 6, 9, 3, -5, 7, 1 c - 1, 7, 5, 3, - 9, 6, 2, 6, - 9, 3, 5, 7, - 1 55 1, -1, 2, -2, 0,-55 + + + 1 + 1 c 1, 7, -6, 6, 5, 6, -6, 7, 1 0, 55 + + -1, 7, 6, 6, -5, 8, 6, 7, -1 1,-27 1, 27 Cl 2, 7, -3, 5, 10, 8, -3, 7, 2 CCj -2, 7, 3, 5, -10, 6, 3, 7, -2 56 1, -1, -4, 0,-56 + + + + + + 1 1, 7, -7, 7, 1 0, 56 c - 1, 7, 7, 7, - 1 2,-13 4, 6, -5, 4, 8, 4, -5, 8, 4 2, 13

- 4, 6, 5, 4, - 8, 4, 5, 8, - 4 57 1, -1, 0,-57 + + 1 1, 7, -8, 1, 7, 6, -3, 6, 7, 1, -8, 7, 1 0, 57 c -1, 7, 8, 1, -7, 8, 3, 6, -7, 1, 8, 7, - 1 2, -2, 1,-28 + + a-c 2, 7, -4, 6, 8, 3, -6, 3, 8, 6, -4, 7, 2- 1, 28 -2, 7, 4, 6, -8, 3, 6, 3, -8, 5, 4, 7, -2 58 1, 2, 0,-58 + + 1 1, 7, - 9,2, 6, 4, - 7, 8,7,4,-6, 2, 9,7,- 1,7,9,2,-6, 4, 7, 3, 0,-29 -7, 4, 6, 2, -9, 7, 1 c 2, 6, -11, 6, 3, 7,-3,8, 11, 6, - 2, 6, 11, S, - 3, 7, 3, 5, - 11, 6, 2 59 1, -1, 0,-59 + + 1 c 1, 7, -10, 3, 5, 7, -2, 7, 5, 3, -10, 7, 1 0, 59 - 1, 7, 10, 3, - 5, 7, 2, 7, - 5, 3, 10, 7, - 1 60 1, -1, 3, -3, 0,-60 + + + + 1 1, 7, -11, 4, 4, 4, -11,7, 1 0, 60 + + c -1, 7, 11, 4, -4, 4, 11, 7, -1 0,-20 <H 3, 8, -8, 2, 7, 8, -5, 6, 7, 2, -8, 6, 3 0, 20 CCt -3, 6, 8, 2, -7, 8, 5, 8, -7, 2, 8, 8, -3 61 1, 0,-61 + 1 1, 7,-12, 8, 3, 7, - 4, 6, 9, 4, -5, 8, 5, 4, -9, 8, 4, 7, -3, 5, 12, 7, -1,7,12,8,-3,7,4,8,-9,4,5,8,-5,4,9,8,-4,7,3,5,-12,7,1 2, 1,-30 + 2, 7, -6, 8, 6, 7, -2, 7, 6, 8, -6, 7, 2 335] TABLES DE FOBMES QUADRATIQUES. 151 D Classes a /s s e Se Cp Pfiriodes 62 1, -1, 0,-62 + + 1 1, 7, - 13, 6, 2, 6, - 13, 7, 1 0, 62 c - 1, 7, 13, 6, - 2, 6, 13, 7, - 1 63 1, -1, 2, -2, 0, -63 + + + + + + 1 1,7,-14,7,1 0, 63 c - 1, 7, 14, 7, - 1 1,-31 Cl 2,7,-7,7,2 1, 31 CCj -2, 7, 7, 7, -2 65 1, 5, 0,-65 + + 1 1, 8, - 1, 8, 8, 8, - 1, 8, 1 0,-13 c 5, 6, - 8, 3, 7, 4, - 7, 3, 8, 6, - 5, B, 8, 3, - 7, i, 7, a, - 8, s, 5 2, 10, 1,-32 + + <r 2. 7, -8, 1, 8, 7, -2, 7, 8, 1, -8, 7, 2 5,-4 ore 10, 6, -4, 7, 4, 6, -10, 5, 4, 7, -4, 5, 10 66 1, -1, 3, -3, 0, -66 + + + + + + 1 C CCi 1, 8, - 2, 8, 1 0, 66 - 1, 8, 2, 8, - 1 0, -22 3, 6, -10, i, 5, 6, -6, «s 5, 4, - 10, 6, 3 0, 22 -3, 8, 10, 4, -5, 6, 6, 6, -5, 4, 10, 6,-3 67 1, -1, 0,-67 + + 1 c 1, 8, -3, 7, 6, 6, -7, 2, 9, 7, -2, 7, 9 0, 67 -1, 8, 3, 7, -6, s, 7, 2, -9, 7, 2, 7, -9 68 1, -1, 0, -6« + + + 1 1, 8, -4, 8, 1 0, 68 c - 1, 8, 4, 8, - 1 69 1, -1, 0,-69 + + 1 c 1, 8, -5, 7, 4, 8, -11, 6, 3, 6, -11, 5, 4, 7, -5, 8, 1 0, 69 -1, 8, 5, 7, -4, B, 11, 6, -3, 6, 11, 6, - 4, 7, 5, 8, -1 2, -2, 1,-34 + + <7 2, 7, -10, 3, 6, 3, -10, 7, 2 1, 34 -2, 7, 10, 3, -6, 3, 10, 7,-2 70 1, -1, 9 -*> — 2 1, -1, 0, -70 0, 70 + + + + + 1 1, 8, -6, 4, 9, 6, -5, 6, 9, 4, -6, 8, 1 c - 1, 8, 6, 4, - 9, B, 5, 6, - 9, 4, 6, 8, - 1 0,-35 2, 8, - 3, 7, 7, 7, - 3, 8, 2 0, 35 CCi -2, 8, 3, 7, -7, 7, 3, 8, -2 71 0,-71 + + 1 1, 8, - 7, 6, 5, 4, - 11, 7, 2, 7, - 11, 4, 5, e, - 7, 8, 1 0, 71 c -1, 8, 7, 8, -5, 4, 11, 7, -2, 7, 11, 4, -5, 6, 7, 8, -1 72 1, -1, -4, 0,-72 + + + + + + 1 1, 8, -8, 8, 1 0, 72 c - 1, 8, 8, 8, - 1 2,-17 4, 8, - 9, 3, 7, 4, - 8, 4, 7, 3, - 9, 6, 4 2, 17 +

+ -4, 8, 9, 3, -7, 4, 8, 4, -7, 3, 9, 8, -4 73 1, 0,-73 + 1 1, 8, - 9, 1, 8, 7, - 3, 8, 3, 7, - 8, 1, 9, 8, - 1, 8, 9, i, - 8, 7, 3, 8, - 3, 7, 8, I, - 9, 8, 1 2, 1,-36 + o- 2, 7, -12, B, 4, 7, -6, 5, 8, 3, -8, B, 6, 7, -4,. 5, 12, 7, - 2, 7, 12, B, - 4, 7, 6, 6, - 8, 3, 8, B, - 6, 7, 4, b, - 12, 7, 2 74 1, 2, 0, -74 + + 1 1, 8, -10, 2, 7, 6, -7, 2, 10, 8, -1, 8, 10, 2, -7, 5, 7, 2,-10, 8, 1 0,-37 c 2, 8, - 5, 7, 5, 8, - 2, 8, 5, 7, - 5, 8, 2 75 1, -1. 2, -2, 0,-75 + + + + + + + + — 1 1, 8, -11, 3, 6, 3, -11, 8, 1 0, 75 c -1, 8, 11, 3, -6, 3, 11, 8, -1 1,-37 cc, 2, 7, -13, 6, 3, 6, -13, 7, 2 1, 37 -2, 7, 13, 6, -3, 6, 13, 7, -2 76 1. -1, 0,-76 1 c 1, 8, -12, 4, 5, 6, - 8, 2, 9, 7, -3, 8, 4, 8, - 3, 7, 9, 2, -8, a, 5, 4, -12, 8, 1 0, 76 - 1, 8, 12, 4, -5, 6, 8, 2, - 9, 7, 3, 8, - 4, 8, 3, 7, - 9, 2, 8, a, -5, 4, 12, 8, -1 77 1, -1, 0,-77 + + 1 1, 8, -13, 6, 4, 7, -7, 7, 4, 6, -13, 8, 1 0, 77 c -1, 8, 13, 6, -4, 7, 7, 7, -4, 6, 13, 8, -1 2. -2, 1,-38 + + <r 2, 7, - 14, 7, 2 1, 38 - - o-c - 2, 7, 14, 7,-2 152 TABLES DE FORMES QUADRATIQUES. [335 D Classes a P 8 e it Cp Piriodes 78 1 -1 2 -2 , 0,-78 + + + + + + 1 c 1, 8, - 14, 6, 3, 6, - 14,

8, 1 - 1, 8, 14, 6, - 3, 6, 14, 8, - 1 2, 8, -7, 8, 6, 6, -7, 8, 2 -2, 8, 7, 6, -6, 8, 7, 8, -2 , 0, 78 , 0,-39 Cl 0, 39 CCl 79 1 -3 -3 3 -1 3 , 0,-79 -1, 26 , 1, 26 + + + + + + 1 1, 8, - 15, 7, 2, 7, - 15, 8, 1 -3, 8, 5, 7, -6, 6, 9, 4> -7, 3, 10, 7,-3 -3, 7, 10, 3, -7, 4, 9, B, -6, 7, 5, 8, -3 3, 7, -10, 3, 7, 4, -9, 6, 6, 7, -5, 8, 3 -1, 8, 15, 7, -2, 7, 15, 8, -1 3, 8, -5, 7, 6, B, -9, *, 7,3, -10, 7, 3 1,-26 0, 79 -1,-26 e 80 1 -1 4 -4, 0,-80 + + + + + + + + 1 1, 8, -16, 8, 1 -1, 8, 16, 8, -1 4, 6, -11, 5, 5, S, -11, 6, 4 -4, 6, 11, 8, -5, B, 11, 8, -4 0, 80 c 2,-19 2, 19 82 1, 9 3, -3, 0,-82 0, -41 + + + + 1 e 1 1, 9, - 1, 9, 1 2, 8, -9, 1, 9, 8, -2, 8, 9, 1, -9, 8, 2 3, 8, -6, 1, 11, 7, -3, 8, 6, 4, -11, 7, 3 3, 7, -11, 4, 6, 8, -3, 7, 11, 4, -6, 8, 3 -1,-27 -1, 27 83 1, -1, 0, -83 + + 1, 9, -2, 9, 1 - 1, 9, 2, 9, - 1 0, 83 c 84 1 -1, -4, 0, -84 + + + + + + 1 1, 9, - 3, 9, 1 -1, 9, 3, 9, -1 4, 8, -5, 7, 7, 7, -5, 8, 4 -4, 8, 5, 7, -7, 7, 5, 8, -4 0, 84 c 0,-21 1 c 0, 21 85 1 5, 0,-85 + + 1, 9, - 4, 7, 9, 2, - 9, 7, 4, 9, - 1, 9, 4, 7, - 9, 2, 9, 7, - 4, 9, 5, B, - 12, 7, 3, 8,-7, 6, 7, 8,-3, 7, 12,6, -5, 5, 12, 7, -3, 7,6,-7,8,3,7,-12,8,5 1 0,-17 2, 10 1,-42 + + — + cr crc 1 2, 9, -2, 9, 2 10, 6, -6, 7, 6, 6, -10, 5, 6, 7, -6, 6, 10 5,-6 86 1 -1, 0,-86 + 1,9,-5,6, 10,4,-7,3,11,8,-2,8, 11,3,-7,4, 10,6,-5,9, -1,9,5,6,-10,4,7,3,-11,8,2,8,-11,3,7,4,-10,6,5,9,- 1 0, 86 c 1 87 1, -1 2 -2 0,-87 + + + + + + 1 1, 9, -6, 9, 1 -1,9, 6, 9, -1 2, 9, -3, 9, 2 -2, 9, 3, 9, -2 0, 87 c 1,-43 1, 43 88 1, -1 4 -4, 0,-88 + + + + + + 1 c 1, 9, -7, B, 9, 4, -8, 4, 9, 6, -7, 9, 1 -1, 9, 7, B, -9, 4, 8, 4, -9, 5, 7, 9, -1 4, 6, -13, 7, 3, 8, -8, 8, 3, 7, -13, 6, 4 -4, 8, 13, 7, -3, 8, 8, 8, -3, 7, 13, 6, -4 0, 88 2,-21 Cl 2, 21 CCi 89 1 0,-89 + 1 1, 9, - 8, 7, 5, 8, - 5, 7, 8, 9, - 1, 9, 8, 7, - 5, 8, 5, 7, - 8, 9, 2, 9,-4, 7, 10, 3, -8,B, 8,3,-10, 7,4, 9, -2, 9, 4, 7, -10, 8, 6, -8, 3, 10, 7, -4, 9, 2 1 2 1,-44 + 0- 90 1 -1 2 -2 0,-90 + + + + + + 1 1, 9, -9, 9, 1 -1, 9, 9, 9, -1 2, 8, -13, 6, 5, 5, -13, 8, 2 -2, 8, 13, 6, -5, 6, 13, 8, -2 0, 90 c 0,-45 Cl 0, 45 cc^ 91 1 -1 2 -2 0,-91 + + + + + + 1 1, 9, -10, 1, 9, 8, -3, 7, 14, 7, -3, 8, 9, 1, -10, 9, 1 -1, 9, 10, 1, -9, 8, 3, 7, -14, 7, 3, 8, -9, 1, 10, o, -1 2, 9, -5, 6, 11, 6, -6, 7, 7, 7, -6, 6, 11, 6, -5, 9, 2 -2, 9, 5, 6, -11, B, 6, 7, -7, 7, 6, B, -11, 6, 5, 9, -2 0, 91 c 1,-45 Cl cc, 1, 45 335] TABLES DE FORMES QUADRATIQUES. 153 D Classes 1 1 e|a. Cp P^riodes 92 1, -1, 0,-92 + + 1 c 1, 9, - 11, 2, 8, 6, -7, 8, 4, 8, -7, 6, 8, 2, -11, 9, 1 0, 92 -1, 9, 11, a, -8, 6, 7, 8, -4, 8, 7, 8, -8, 2, 11, 9, -1 93 1, -1, 0,-93 + + 1 1,9,-12,3,7,4,-11,7, 4,9,-3,9, 4,7,-11,4, 7,3, -12,9,1 0, 93 c - 1, 9, 12,3, -7, 4, 11, 7, - 4, 9, 3, 9, -4, 7, 11, 4, - 7, 3, 12, 9,-1 2, -2, 1,-46 + + (TC 2, 9, -6, 9,2 1, 46 - 2, 9, 6, 9, - 2 94 1, -1, 0, -94 + + 1 1, 9, - 13, 4, 6, 8, - 5, 7, 9, 2, - 10, 8, 3, 7, - 15, 8, 2, 8, -15, 7, 3, 8, - 10, 2, 9, 7, -5, 8, 6, 4, -13, 9, 1 0, 94 c -1, 9, 13, 4, -6, 8, 5, 7. -9, 2, 10, 8, -3, 7, 15, 8, -2, 8, 15, 7, -3, 8, 10, 2, -9, 7, 5, 8, -6, 4, 13, 9, -1 95 1, -1, 0 -2, 0,-95 + + + + + 1 c 1, 9, -14, B, 5, 8, -14, 9, 1 0, 95 -1, 9, 14, 5, -5, 6, 14, 9, -1 1,-47 2, 9, -7, B, 10, B, -7, 9, 2 1, 47 -2, 9, 7, 5, -10, 5, 7, 9, -2 96 1, -1, 3, -3, 0,-96 + + + + + + 1 c 1, 9, -15, 6, 4, «, -15, 9, 1 0, 96 -1, 9, 15, 6, - 4, 6, 15, 9, -1 0,-32 3, 9, -5, 6, 12, 6, -5, 9, 3 0, 32 cc, 1 -3,9,5,6,-12,6,5,9,-3 97 1, 0,-97 + 1,9,-16,7,3,8,-11,3,8,6,-9,4,9,6,-8,3,11,8,-3,7,16,9, -1,9, 16, 7,-3, 8, 11, 3, -8,6, 9,4,-9,5, 8,3,-11, 8, 3,7, –16,9.1 2, 1,-48 + (T 2, 9, -8, 7, 6, 6, -12, 7, 4, 9, -4, 7, 12, 6, -6, 7, 8, 9, -2, 9, 8, 7, - 6, 5, 12, 7, -4, 9, 4, 7, -12, s, 6, 7,-8, 9, 2 98 1, -1, 0, -98 0, 98 + + 1 1, 9, -17, 8, 2, 8, -17, 9, 1 C : -1, 9, 17, 8, -2, 8, 17, 9, -1 99 1, -2, 5, 5, -1, 2, -5, -5, 0,-99 1, 49 + + + + + + + + + + + + 1 e 1, 9, - 18, 9, 1 -2,9,9,9,-2 2,-19 -2,-19 5, 7, -10, 3, 9, 6, -7, 8, 5 5, 8, -7, 6, 9, 3, -10, 7, 5 0, 99 1,-49 c ce" ce -1,9,18,9,-1 2, 9, -9, 9, 2 2, 19 -2, 19 -5, 7, 10, 3, -9, 6, 7, 8, -5 -5, 8, 7, 6, -9, 3, 10, 7, -5 C. V. 20 154 TABLES DE FORMES QUADRATIQUES. [335 Table III dee formes quadratiques binaires pour les treize determinants n^gatifs in-dguliers du premier millier. D Clasaei Cp Classes ^ « Cp 576 - 1 (24)' -580 -145(2)' 1, 9, 4, 25, 9, 9, 17, 17. 13, 16, 13, 16, 5, 20, 20, 5, 0, 576 0, 64 2, 145 7, 25 3, 65 3, 65 6, 36 6, 36 3, 45 4, 37 3, 45 4, 37 2, 116 2, 29 2, 29 2, 116 -820 - 205 (2)= 1, 4, 5, 20, 8, 8, 17. 17, 19, 11, 19. 11, 23, 7. 7, 23, 0, 580 0, 145 0, 116 0, 29 2, 73 []2, 73 7, 37 []7, 37 3, 31 5, 55 3, 31 5, 55 8, 28 1, 83 1, 83 *8, 28 1, 5. 20, 4, 13, 13, 17, 17, 11, 19, 11. 19, - 23,- 8, 8, 23, 0, 820 0, 164 0, 41 0, 205 5, 65 -5, 65 -8, 52 8, 52 4, 76 4, 44 -4, 76 -4, 44 10, 40 2, 103 -2, 103 10, 40 1 e «i ff'e, re, e»e,' -884 - 221 (2)" 1 « ee,- r e, «>' ee, e^e, 1 e» e'e' + + + + + j + e ee,' le'e,' -900 1 (30)' 1, 13, 4, 17, 25, 25. 9, 9, 5, 24, 20, 24, 24, 5, 24, 20, 8, 15, 8, 15, 19, 15, 15, 19, 27,- 3, 23, 12, 12, 23, 3, 27, 0, 884 0, 68 0, 221 0, 52 -4, 36 4, 36 4, 100 -4, 100 1, 177 -2, 37 -4, 45 10, 41 2, 37 -1, 177 -10, 41 4, 45 2, 111 4, 60 -2, 111 -4, 60 -3, 47 1, 59 -1, 59 3, 47 -13, 39 1, 295 -6, 40 4, 75 - 4, 75 6, 40 -1, 295 13, 39 1, 9, 4, 25, 9, 9, - 29,- 29, 17, 8, - 17, - 8, 13. - 25, - 25. 13, 0, 900 0, 100 0, 225 0, 36 3, 101 -3, 101 12, 36 12, 36 1, 53 -2, 113 -1, 53 2, 113 -6, 72 -5, 37 5, 37 6, 72 + + + + + + e'i* «»»^ e»i e't^ e ei EUR>* ei* e't* e'i' et et ei> e>^ 1 e* e,- e ee," e'e, e«, e' e, e^e," 335] TABLES DE FORMES QUADRATIQUES. 155 D Classes a /3 a e Cp D Classes a P 5 e -243 1, 0 243 + 1 -459 1, 0, 459 + + 1 _ 7, 3 36 + 1 d 9, 3, 52 + + d - 3 (9)-^ 7, 4, -3 1 36 161 + + d. - 51 (3)= 9, -3, 52 4, 1, 115 + + + + (P 13, — 2 19 + dd^ 19, -4, 25 + + dd, 9, 3 28 + d^d. 13, 3, 36 + + d'd, 4, -1 161 + rfi' 4, -1, 115 + + d,' 9, -3 28 + dd> 13, -3, 36 + + ddc 13, 2 19 + (Pd^ 19, 4, 25 17, 0, 27 + + d^d,^ 2 1 122 + 0- c 14, 3 18 + <yd 20, -9, 27 _ — cd 14, -3 18 ,307 + 1 o-d^ 20, 9, 27 11, 4, 44 - — cd' -307 1, 0 + 1 cd. 7, 1 44 + d 5, 1, 92 - - cdd. -307(1) 7, 4, -1 1 44 77 + + cP 20, -1, 23 11, -5, 44 — ^ cd'd, cd^ 11, -1 28 + dd. 20, 1, 23 - - 1 cddi' 17, 4, 4 -1 19 77 + 1 + ,pd. 5, -1, 92 - - cd^d,^ 2, 1, 230 + + <T 17, -4 19 + ddi' 18, 3, 26 + + <Td 11, 1 , 28 + O'd,^ 18, -3, 26 + + ad' 2 1 154 + a- 22, 6, 22 (TC 14] 1 22 + <rd 10, 1, 46 — _ a-cd 14, -1 22 + ,Td' 1 10, -1, 46 - - a-cd^ -339 1, 0 339 + + 1 -675 1, 0, 675 + + 1 = 7, 2 , 49 + + d 9, 3, 76 + + d -339(1)= 7, _ 2 49 + + d? -3(15)^ 9, -3, 76 + + 4, 1 85 + + d. 4, 1, 169 + + dr 15, 6 25 + + dd. 25, - 10, 31 + + dd, 13, -5 28 + + I d'd. 19, 3, 36 + + d^d. 4, -1 85 + + d,^ 4, -1, 169 + + d,' 13, 5 28 + + dd^^ 19, -3, 36 + + dd;' 15, 3, -6 25 + + cPd,' 25, 10, 31 + + + d'di' 0 113 - c 25, 0, 27 c 20, -9 21 — - cd 27, -9, 28 + cd 20, 9 21 — — ccP 27, 9, 28 + — ccP 12, -3 29 — — cd, 13, 1, 52 + — cd. 5, 1 68 — — cddi 25, 5, 28 + — cdd. 17, 1 20 - - cd'di 7, -2, 97 + — cO'd, * 12, 3 29 - - cd,' 13, -1, 52 + — cd,' 17, -1 20 - - cddi' 7, 2, 97 + — cddi' *5, -1 68 - - cd' t/,» 25, -5, 28 + - cd^d,' *2 1 170 + 1 + a 2, 1, 338 + + a 14, -5 20 + ! + ad 18, 3, 38 + + ad 14, 6, 5 26 + + <Td^ 18, -3, 38 + + + ad' 3 58 a-c 26, 1,

26 ac 10, 1 34 _ _ 1 (red 14, -5, 50 + — acd 10, -1 34 i ˆ a-cd’ 14, 5, 50 + 1 1 1 1 acd’ 20—2 156 TABLES DE FORMES QUADRATIQUBS. [335 D Classes a /3 8 t Cp 1 Classes a p S e -755 1, 0, 755 + +1 1 -891 1, 0, 891 1 + , + i 1 _ 19, -9, 44 + + d 9, 3, 100 + ’ + d -755(1)’ 19, 9, 44 + + cP -11(9)» 9, -3, 100 + 1 + «? 4, 1, 189 + + d. 4, 1, 223 + 1 + 21, -8, 39 + + dd. 31, 15, 36 + ’ + dd, 11, 2, 69 + + i d’d, 25, 3, 36 + + d’d. 4, - 1, 189 + + d,’ 4, - 1, 223 + + d,’ 11, -2, 69 + + dd,ˆ 25, -3, 36 + + dd,’ 21, 8, 39 + + d’diˆ 31,-15, 36 + + d’d," 5, 0, 151 + + e" 11, 0, 81 + _ c 9, -1, 84 + + f?d 20, -7, 47 + _ cd 9, 1, 84 + + e’d:’ 20, 7, 47 + _ cdˆ 20, 5, 39 + + e’rfi 23, 11, 44 + _ cd. 29, 12, 31 + + e’dd. 20, 3, 45 + _ odd. 21, -1, 36 + + e’d’dˆ 5, 2, 179 + ccPd, 20, -5, 39 + + e’d” 23,-11, 44 + _ cd,’ 12, 1, 36 + + ˆdd,’ 5, -2, 179 + _ cdd,’ 29,-12, 31 + + e’dˆd," i 20, -3, 45 + — cd’d’ 27, 1, 28 e 2, 1, 446 + + <r 15, -5, 52 — - ed 18, 3, 50 + + ad 3, -1, 252 - - ed’ 18, -3, 50 + + <r<? 7, -1, 108 - - ed. 22, 11, 46 + + ITC 23, -2, 33 — - edd. 10, 3, 90 + + acd 12, 5, 65 28, 13, 33 13, 5, 60 12, -1, 63 27, -1, 28 3, 1, 252 15, 5, 52 28,-13, 33 12, 1, 63 13, -5, 60 7, 1, 108 12, -5, 65 23, 2, 33 - -! edˆdj ed,’ edd,’ &Pd,’ e> e’d e’d’ e’dd, e’d’d, e’d,’ >?dd,’ e’dˆdˆ 10. -3, 90 + + acdP - - -974 -2.487(1)" 1, 0, 974 18, -4, 55 18, 4, 55 31, 7, 33 30,-14, 39 15, -4, 66 31, -7, 33 15, 4, 66 30, 14, 39 2, 0, 487 9, -4, 110 9, 4, 110 25, -1, 39 + 1 d dd, d’d, d,’ dd,’ d’dˆ’ e e’d e’d’ 2, 1, 378 + + <T e’d. 22, 9, 38 + + (rd 15, 1, 65 + + ˆddˆ 22, -9, 38 + + ,Tdr- 30, -4, 33 + + e’dd. 10, 5, 78 + + ae’ 25, 1, 39 + + e’d,’ 18, -1, 42 + + ˆre’d 30, 4, 33 + + e’dd,’ 18, 1, 42 + + a(?dˆ 15, -1, 65 + + e’d’d,’ 14, -1, 54 ere 27, 5, 37 e 26, 5, 30 - - (red 3, 1, 325 _ _ ed 6, -1, 126 - - ned’ 6, 2, 163 _ " ed? 14, 1, 54 " - 0-6= 5, 1, 195 _ _ ed. 6, 1, 126 - o-eˆ d 11, 4, 90 edd. 26, -5, 30 ,Tˆdˆ 13, 1, 75 - - ed’d. 10, 4, 99 — — ed,’ . 26, 12, 43 — — edd,’ 22, 4, 45 - - edˆd,’ 27, -5, 37 — — ˆ 6, -2, 163 — — <?d 3, -1, 325 — _ e’d’ 10, -4, 99 — — e’d. 22, -4, 45 — — e’dd, 26,-12, 43 — — e’d’d. 5, -1, 195 — — e’d,’ 13, -1, 75 — — <?dd,’ 11, -4, 90 - - ˆd’d,’

336 NOTE SUR L'ELIMINATION

[From the Journal für die reine und angewandte Mathematik (Crelle), torn. LX. (1861), pp. 373–374.]

'ELIMINATION. [From the Journal filr die reine und angewandte Mathematik (Crelle), torn. LX. (1861), pp. 373—374.] SoiENT U=(a, ...][«, y)'", V=(b, ...'^a;, y)" des fonctions homogenes quelconques des degres m et n respectivement. Denotx)ns par (x, y)* la suite entiere ou seulement une partie de la suite de termes «*, a;*"' y, ... y^, et en prenant d'^m^n, formons le determinant
x, yy-^" U, (x, y)o- V. Cette notation signifie qu'en supposant les suites x, y)*"(TM) U, x, y)* " V composdes respectivement de p et Ae q termes et qu'en posant p + q = s on forme le determinant (x" y.y-^^ U" x" 3/0'-" V, (*[]«, yO*-'" U" x" y,)o-^ V, dans lequel les diffdrentes lignes (chacune composde de s termes) sont ce que deviennent x, y)^-^U, (x, yy '*V, lorsqu'on y substitue («„ y^), (x„ y.^,...xg, y,) successivement au lieu de x, y). Le determinant que je viens ddfinir est divisible par le determinant notation qui est ^quivalente a: (^i, viy-' (^., y.y-' et dans laquelle x, yy ^ denote la suite entifere des termes «^', a;*"* y, . . . y*"'. Nous obtenons aiusi une Equation c-^-d. que le quotient est du degr^ p par rapport aux coefficients (a, ...), du degi'^ q par rapport aux coefficients (b, ...), et du degie 0 — s + l par rapport a chaque systeme de variables (a^i, yi),...ix" y,). Or en supposant Ui = 0, F, = 0, on obtient 0 = a,...)Ph,...yx" y,)»-'+'...(^" y,)»-'+', Equation qui subsiste quelles que soient les valeurs des variables x^, y^,...x" y,) ; cela donne une suite de 6 — s+ 2)*"' Equations chacune de la fonne o=(a, ...)^(6, ...)'(**.. y.)*-'^'- En considerant un systems quelconque de 6 — s + 2 de ces Equations, pour en dliminer tons les termes de (a;„ yi)*"'"'', on obtiendra ou I'^quation identique 0 = 0, ou une equation de la forme F=a, ...)'"*-»+=' h, ...)«""+" =0 oil F sera un determinant de I'oi-dre d — s+2, chaque tenne ^tant de la forme (a, ...)n6. ...)'. Cela pos^ il est evident que F contiendra comme facteur la fonction n=(a,...)"(6,...r qui est le resultant des deux Equations ?7 = 0, F = 0. En particulier, on aura les deux cas que voici: 1 deg.. Soit 0 = m + n — l, et supposons que x, yj^^ U, (x, y)(TM) ' V, denotent les suites entieres a;"-' U, x"-"-yU, ... y""' U; a^-' V, x"^yV, ... y"'-' V, nous aurons p = n, q = vi, s = m + n, 6 — s + 2 = 1, et de]k F=(a,...r(b,...r, done ^= D. On voit sans peine que Ton obtient de cette manifere le resultant D, sous la forme d'un determinant de I'ordre m + n, le m^me que Ton obtient en dliminant les termes de (x, y)'»+"-J entre les Equations (x, y)" ' U=0, (x, yY^'^ F=0. 2 deg.. En supposant »n^?i, on peut prendre 0 = m, ce qui doune pour (x, y)^-^ U le seul terme U. R^duisons aussi (x, y)* " V au seul terme a* y»«-«-« y ^^ d^signant un nombre entier arbitraire entre 0 et m — n), c.-a-d. au terme «'""" V ou i/""" V dans le cas des deux valeurs extremes de a. On a ainsi p=l, q — ^, s = 2, et del^ i^=(a, ...)"' (6, ...)". Done ^= (a, ...)'""" D, c-a-d. que Ton

obtient le resultant D affecte d’un facteur (a, ...)’" " qui ne contienfc que les coefficients de U, et qui est de I’ordre m — n par rapport a ces coefficients. L’expression de ce facteur peut etre trouv^e assez facile- ment. Dans le cas du terme «"*"" V, c.-a-d. pour a = m — n, le facteur sera it"*-" k ddsignant le dernier coefficient de la suite a, ...)), et dans le cas du terme y-mr-n Y c.-a-d. pour a = 0, le facteur sera a’""". Mais en supposant m = n on a tout simplement .F= D, c.-^-d. que Ton obtient le resultant sans facteur Stranger. C’est sous cette derniere forme que j’ai presente la m^thode abreg^e de Bezout dans le tome LIU. p. 366 (18.57) de ce Journal, [230].

Londres, IT"" D^cembre, 1861.

337 NOTE SUR LA REALITE DES RACINES D'UNE EQUATION QUADRATIQUE

[From the Journal für die reine und angewandte Mathematik (Crelle), torn. LXI. (1863), pp. 367–368.]

'UNE EQUATION QUADRATIQUE. [From the Journal fur die reine und angewandte Mathematik (Crelle), torn. LXI. (1863), pp. 367—368.] A PROPOS du m^moire que vient de publier M. Hesse (voir ce Journal t. LX., p. 30.5) je remarque que si I'une ou I'autre des deux formes (a, b, c, /, g, A$ y, (a', b', c', f, g', A'$)= est une forme ddfinie (forme qui conserve toujours le meme signe pour des valeurs r^elles quelconques des variables), I'^quation suivante en X : a —
a', h —
h', g —
g', x =0 h-TJi, b-W. f-
f, y 9 -'^9' f-V' c-
c', z «, y, z aura ses deux racines r^elles. En ^crivant A=bc-f
A' = b'c'-f
A, = bc' + b'c-2ff', B = ca — g', ... de manifere que (^4, B, C, F, G, H^ f denote la forme adjointe (ou rdciproque) de (a, b, c, f, g, A$)', cette ^uation prend la forme A, ...
x, y, zf-
(A" ...
x, y, zy +
^A'. ...
x, y, zY = Q, et les racines dtant r^elles, on doit avoir D =-4(4, ...
x, y, zfA', ...
x, y, zy + [A" ...Jx, y, 0)']' = + . Or pour demontrer directement cette proposition, il n'est pas ce me semble possible d'exprimer D comme une somme de carrds; on a besoin de consid^rer una forme plus g^nerale, savoir une somme de carr^s multiplies chacun par un coefficient literal positif. Par exemple, en ne faisant attention qu'au coefficient de a;*, on doit avoir D" : — 4 (be -p) (h'c' -p) + (6c' + h'c - 2ffy = + . Pour en faire la demonstration, on peut exprimer Do sous la forme Do = be' - b-cf + 4>(bf- b'f) cf - c'f), ce qui donne bcUo = (be -p) (be' - b'cy + [b (ef - e'f) + c (bf - b'f)f. En effet, en y substituant la seconde expression de Do, on a I'identite 46c (bf - b'f) (cf -e'f) = -f (be' - b'ef + [6 (ef -c'f) + e (bf - 6'/)]= et I'expression pour bc[2" est ainsi demontr^e. Mais en supposant que (a, b, c, f, g, h) (")' soit une forme definie, on a 6c — /- = + , done aussi 6c = + , et 6cno = (6c —f^) X-' + F" = + , done enfin n"=+. II serait assez int^ressant de trouver une demonstration pareille pour I'expression g^n^rale de Q. Londres, 23'""" Octobre 1862. 0. V. 21

338 NOUVELLES RECHERCHES SUR L'ELIMINATION ET LA THEORIE DES COURBES

[From the Journal für die reine und angewandte Mathematik (Crelle), torn. LXIII. (1864), pp. —.]

'ELIMINATION ET LA THEORIE DES COURBES. [From the Journal fur die reine und angewandte Mathematik (Crelle), torn. LXill. (1864), pp. 34—39.] Dans le problfeme de lelimination, on cheiche la relation qui doit exister entre les coefficients d'une fonction ou systeme de fonctions pour que quelque circonstance particulifere (ou singularity) puisse avoir lieu ; par exemple, pour que deux Equations puissent avoir une racine commune, ou (comme application gdometrique) pour qu'une courbe puisse avoir un point double. En prenant les coefficients comme donnes, taut la relation cherch- de que la singulaiit^ qu'elle implique n'ont qu'une existence hypotM- tique. Mais on peut transformer la question en supposant que les coefficients d'une ou de plusieurs des fonctions soient de la forme a = Xa + /mi", b = b' + fib", ... oh a, b',... , a", b",... sont des coefficients donnds, mais , /t des quantit^s arbitraires. On peut alors disposer en sorte que la singularity dont il s'agit existe actuellement, en deter- minant, au moyen de la relation donn^e par I'elimination, la valeur du rapport : ft. Ces substitutions a = a' + fut", 6 = 6' + fib", . . . changent la fonction U k laquelle se rapportent les coefficients a, b,... en U= U' +(iU", ou W, U" sont des fonctions semblables a U, mais avec les coefficients a', b',... ou a", b", ... au lieu de a, b,...: en se servant d'une expression usit^e, on peut dire que la fonction U est en involution avec U', U"; et de mSme en geometric que la courbe U=0 est en involution avec les courbes W = 0, U" = 0; au reste, pour les courbes, cela veut dire que les trois courbes se coupent dans les memes points. On con9oit comment cette maniere d'envisager le probleme peut conduire k une interpretation g^om^trique de rdsultats qui n'avaient auparavant qu'une signification analytique. Consid^rons par exemple la proposition suivaute, " le discriminant d'une fonction quadi-atique k trois variables est du degrd 3 par rapport aux coefficients," ou ce qui eat la m^me chose, "la fonction qui 4g& 4e k z^ro exprime que la eonique U = 0 ait im point double (se rdduise a une paire de droites) est du degre 3 par rapport aux coefficients," c'est la une proposition pui-ement analytique, mais si comme ci-dessus on met a' + /Mt", 6' + /jib", ... au lieu de a, b,... on a le thdorfeme g^omd- trique que voici : "Dans le systeme de coniques XW + iJi,U" = 0 en involution avec les coniques donnees W = 0, U" = 0, il y a 3 coniques a point double (c'est-a-dire,

trois paires de droites)." En considdrant le cas plus general d'une fonction k trois variables et d'ordre quelconque, la question analytique "quel est le degr^ du discriminant de la fonction U" pent etre remplac^e par la question g^ometrique "dans le systeme des courbes
U' + fiU" = 0 en involution avec les deux courbes donnees fT = 0, U" =0, quel est le nombre des courbes k point double " on, ce qui est la meme chose, " quel est le nombre des points dont chacun est le point double d'une courbe du systeme." En considerant la question soue cette derniere forme, non seulement on retrouve la valeur connue Sn — iy du degre du discriminant de la fonction U=(A, ...'^x, y, zY, mais on trouve aussi le th^oreme plus gdn^i-al : La fonction JJ =A,...
x, y, 2)" dtant telle que la courbe U—Q ait un nombre a de points doubles et un nombre /3 de points de rebroussement, son discriminant special est du degre' 3 n - 1)" - 7a -
l^. Sous la designation de " discriminant special " j'entends la fonction laquelle ^gal^e a zero donne la condition pour que la courbe Z7 = 0 ait un point double de plus. II convient de remarquer par rapport a cette expression que le discriminant de la fonction gen^rale du n'*""" ordre, en y substituant, au lieu des valeurs g^n^rales, les coefficients de la fonction f/ dont il s'agit, ,ne donne nuUement le discriminant special de U mais se r^duit identiquement a zero; ce discriminant special est done tout autre chose que le discriminant de la fonction g^n^rale. En parlant tout simplement de I'ordre du discriminant special, j'ai voulu designer I'ordre auquel cette expression s'dlfeve par rapport a des coefficients absolument arbitraires ou elements a, b, ... lesquels sont censds entrer lineairement dans la fonction U. II est done ndcessaire de ddmontrer d'abord la pro- position auxiliaire que I'equation d'une courbe qui a ddja un nombre donn^ de points doubles et de rebroussement pent s'exprimer sous la forme signal^e, c'est-a-dire lineairement par rapport k des coefficients absolument arbitraires ou elements a, b,... , proposition qui pent etre ddmontrde sans difficult^. Considerons en eff'et I'equation gen^rale U=(A,...'^x, y, z)" = 0 ou les coefficients A,... sont tous arbitraires; dans le cas d'un point double supposons que les coordonndes de ce point, dans le cas d'un point de rebroussement supposons que les coordonnees de ce point et la direction de la tangente soient donndes : cela dtablit pour chaque point double trois conditions, et pour chaque point de rebroussement quatre conditions, qui contiennent d'une mani^re quelconque les param^tres appartenants au point double ou de rebroussement, mais qui sont lindaires par rapport aux coefficients A, ...: ces coefficients peuvent done s'exprimer lineairement au moyen d'un nombre convenable de coefficients absolument arbitraires ou elements a, b, ...; et c'est de ces elements a, b,... qu'il s'agit et nuUement des parametres mentionnds ci-dessus qui entrent dans les expressions par lesquelles A,... sont donnas en terraes de a, ... 21—2 Cette proposition auxiliaire peut encore se ddmontrer de la manifere que voicL Concevons que P = 0 repr^ente nne courbe particulifere quelconque du m^me ordre que U = 0 et telle que pour chaque point

double de la courbe U=0 elle ait un point double au mfime point, et que pour chaque point de rebroussement de la courbe U=0, elle ait un point de rebroussement au meme point et avec la meme tangente. Soient de mˆme Q = 0, R = 0,... des Equations de courbes qui satisfont aux mSmes conditions. Cela posˆ, on peut ˆvidemment ˆrire U=aP + bQ + cR+ ... , c*est-ˆ-dire que I'ˆquation contiendra linˆirement les coefficients absolument arbitraires ou (Jlˆments a, b Je reviens au thˆorfeme dont je suis parti; soit d'abord U=(a,...ˆx, y, «)" = 0 une courbe sans points doubles ou de rebroussement, de sorte qu'il s'agisse du discri- minant ordinaire. En ˆcrivant pour plus de simplicity V, W au lieu de U', U", on a k considˆrer la courbe

V+ iiW = 0 en involution avec les deux courbes 1ˆ = 0, Tr = 0. Le degrˆ du discriminant de IT est dgal au nombre des points dont chacun est le point double d'une courbe particulifere du systˆme

V+fiW = 0, Or poiu* trouver ces points on n'a qn'k former les Equations Xd,V+fidˆW=0, qui expiiment que la courbe

V+fiW = 0 a un point double, et d'ˆliminer entre ces equations les indˆterminˆes

, /x. Cela donne le systfeme dˆV, dyV, d,v = 0 dzW, dyW, d,w qui comprend les deux Equations (1) dzV 1 = 0, (2) dyW. d,W = 0 auxquelles on satisfait par 9jF = 0, dzW=0, et une troisieme Equation a laqiielle on ne satisfait pas par ce dernier systeme. Or les courbes (1) et (2) se coupent en 4(n — 1)- points, mais parmi ceux-ci on a les (n — iy points d'intersection des courbes diV=0, dgWˆO, et en ˆcartant ces points on obtient 4(n — l)ˆ — (n— 1)ˆ = 3 (n — 1)* pour le nombre des points, ou ce qui est la meme chose pour le degrˆ du discri- minant de U. Je suppose k present que la courbe 17=0 ait un point double; les courbes V = 0, W = 0 ont chacune un point double a ce meme point, et en prenant ce point pour origine des coordonnˆes *[], y les deux courbes seront V = 2"-= (a , b, c $*', yy + etc. = 0, W = z"-ˆ a, b', c"$_a;, yy + etc. =0, en dˆnotant par les etc. les termes des ordres plus dlevˆs par rapport a x, y, ou moins ˆlevˆs par rapport a z. Cela donne pour la courbe (1) dxV, d,V 1 = 0 = zˆ' iax + by) . (a', b', c'ˆx, yf - a'x + h'y) .a, b, cjx, yf] + etc. 9a; Tf, dzW I =z"'ˆˆyax + by)b'x + c'y)-a'x-

-h'y)bx + cy)]-ˆe

,c.; la courbe (1) a done a I'origine un point triple, les tangentes etant donnˆes par les equations y = Q, ax + by) (b'x + c'y) - (a'x + b'y) (bx + cy) = 0, et de meme la courbe (2) a a I'origine un point triple, les tangentes ˆtant donnˆes par les Equations x=0, (cMJ + by) (b'x + c'y) - (a'x + b'y) (bx + cy) = 0; il y a done au point triple deux branches de la courbe (1) dont chacune touche une de deux branches de la courbe (2); ce qui donne a I'origine 4 + 4 + 3 = 11 points d'intersection. De plus il est Evident que les deux courbes di,V=0, dzW = 0 ont chacune un point double a I'origine, c'est-a-dire elles s'y coupent en 2+2 = 4 points. Par consequent les courbes (1) et (2) se coupent en 4 (w — 1)ˆ points, savoir 11 points a I'origine, 4(n—1)"— 11 points autrepai't, les courbes dzV=0, diW = 0 se coupent en (n—

l)r points, savoir 4 points a I'origine, (?i — 1)^ — 4 points autrepart, et le systeme des 3 (« — 1)^ points contient 7 points a I'origine, 3(»i — 1)^ — 7 points autrepart. En dcartant les points a I'origine on a done 3(n— 1)"— 7 points; pour une courbe a point double le degr^ du discriminant special est done =3(» — 1)^ — 7. Si la courbe U=0 a un point de rebroussement, les courbes V =0, W =0 auront au meme point un point de rebroussement avec la meme tangente, et en prenant ce point pour origine des coordonndes et la droite x = 0 pour I'dquation de la tangente, les deux courbes seront V= 2"-' .aa? + z"^' . (a , /3 , 7 , 8 $«, yf + etc. = 0, W=^:"-^aV+z"-3.(a', /3', 7', S'Ja;, yf+etc^O. Cela donne pour la courbe (1) 0= a«F, d,v = (3"-= . 2ax + z"-
3 (a, j8, 7$^;, y)" + etc. X
n - 2) 2"-^ a'a? + (n - 3) ^»-* (a', ^', 7', h'
x, yf + etc.] - 1^"-= . 2a'x + 0"-= . 3 (a', ^', y'^x, y)' + etc. X (n - 2) z"-' ax' + (n- 3) 2"-' (0 , /3 , 7 , 8 $», y)^ + etc.] = z'»-*2n-S)[cuc<^. /3', 7'. S'Ja;, y)»-a'a;(a, ^, 7, S$a;, 2/)»] - 3 (n - 2) [oar' (a', /3', 7'$a;, y)» - aV (a, /8.' 7$*, y)'] + etc. a,z»*-*— n(ao'-a'a)a,'*+ etc. ai^y... + etc. = ^»»-« . a; (- n (aa' - a' a), ...$«, yy + etc. ; la courbe a done h. I'origine un point quadi-uple et la droito x = 0 y est tangente de I'une de ses branches. On a de meme pour la courbe (2) 0 = dyV, a,F I = z»-» . 3 (/3 , 7 , S $*-, yf + etc. x
n - 2) z^^ a'a? + etc. dyW, ^^W
- 0"-» . 3 (yS', 7'. Z'
x, yf + etc. x (n - 2) 2"-' a a? + etc. = 2«'^ar'(a'(y3, 7, 5$a:, yf-ai^. 7'. S'i^;, y)' + etc. = 2«-« a^ (o'/3 - a/3', . . .$a;, y)^ ; cette courbe a done a I'origine un point quadruple et la droite a; = 0 y est tangente commune de deux de ses branches. Cela donne k I'Drigine 5+4 + 4 + 4, =17 points d'intersection des courbes (1) et (2). D'autre part on a a,F=(n-2)0»-=.aa? + (n-3)2"-'(a, /3, 7, 8$a;, yy + etc. = 0, a^F = (n-2)z"-».aV + (?i-3)^"-^(a', /3', 7', B"^x, yy + etc. = 0, et en combinant ces deux Equations z'^' . oaf + etc. = 0, z"-* (aa! — a' a, . . .$a;, yf + etc. = 0. De ces deux Equations la premiere appartient k une courbe qui a un point de rebrousse- ment k I'origine des coordonn^es et la seconde k une courbe qui y a un point triple. Pour les deux courbes 8jF=0, dzW=0 cela donne 3 + 3 = 6 points d'intersection a I'origine. Les courbes (1) et (2) se coupent done en 4 (n — 1)- points, savoir 17 points a I'origine, 4 (m — 1)' — 17 points autrepart, les courbes 9jF=0, 9iW=0 se coupent en (n — 1)- points, savoir 6 points a I'origine, (?i — 1)" — 6 points autrepart et le syst^me des 3 (n — 1)- points contient 11 points a I'origine, 3(n— 1)"— 11 points autrepart. En dcartant les points k I'origine, on obtient 3(« — 1)' — 11 points; pour une courbe avec un point de rebroussement, le degi'e du discriminant special est done =3(«— 1)"— 11. Comme r^sultat final de cette recherche j'obtiens que la reduction du degr^ est de 7 unites pour un point double et de 11 unites pour un point de rebroussement ; et de 1^, que pour un norabre a de points doubles et /3 de points de rebroussement, la r^uction est de 7a+lly8 unites; le degrd

du discriminant special sera done dans ce cas 3 (n — l)-*— 7a — 11/3, ce qu'il s'agissait de d^montrer. Dans tout ce qui precede, j'ai suppose que le systfeme soit tel que 1' elimination conduise h une seule relation entre les coefficients ; si au contraire I'dlimination conduit a deux relations, il faut ecrire au lieu de a,b,... les valeurs Xa + /jm" + va"', b'+/ib"+vb"',... et de mSme pour un plus gi-and nombre de relations. En supposant par exemple que la courbe 17=0 doit avoir un point de rebroussement, ce qui implique deux relations entre les coefficients, la question a resoudre serait celle-ci, " quel est le nombi-e des points qui sont chacun un point de rebroussement d'une courbe pai'ticuliere du systfeme XV + fiW-i-vX^O"; je r&erve a une autre occasion la consideration de ce probleme. Londres, 22«« Mai 1863.

339 ON SKEW SURFACES, OTHERWISE SCROLLS

[From the Philosophical Transactions of the Royal Society of London, vol. CLiii. (for the year 1863) pp. 453–483.]

[From the Philosophical Transactions of the Royal Society of London, vol. CLiii. (for the year 1863) pp. 453–483. Received February 3,—Read March 5, 1863.]

I will also, though it is less closely connected with the subject of the present memoir, refer to a paper by M. Chasles, "Description des courbes à double courbure de tous les ordres sur les surfaces réglées du troisième et du quatrième ordre"().

The present memoir (in the composition of which I have been assisted by a correspondence with Dr Salmon) contains a further development of the theory of the skew surfaces generated by a line which meets a given curve or curves: viz. I consider, 1st, the surface generated by a line which meets each of three given curves of the orders m, n, p respectively; 2nd, the surface generated by a line which meets a given curve of the order m twice, and a given curve of the order n once; 3rd, the surface which meets a given curve of the order m three times; or, as it is very convenient to express it, I consider the skew surfaces, or say the "Scrolls," $S(m,n,p)$, $S(m^2,n)$, $S(m^3)$. The chief results are embodied in the Table given after this introduction, at the commencement of the memoir. It is to be noticed that I attend throughout to the general theory, not considering otherwise than incidentally the effect of any singularity in the system of the given curves, or in the given curves separately: the memoir contains however some remarks as to what are the singularities material to a complete theory; and, in particular as regards the surface $S(m^3)$, I am thus led to mention an entirely new kind of singularity of a curve in space—viz. such a curve has in general a determinate number of "lines through four points" (lines which meet the curve in four points); it may happen that, of the lines through three points which can be drawn through any point whatever of the curve, a certain number will unite together and form a line through four (or more) points, the number of the lines through four points (or through a greater number of points) so becoming infinite.

Notation and Table of Results, Articles 1 to 10.

1. In the present memoir a letter such as m denotes the order of a curve in space. It is for the most part assumed that the curve has no actual double points or stationary points, and the corresponding letter M denotes the class of the curve taken negatively and divided by 2; that is, if h be the number of apparent double points, then $M = -\frac{1}{2}[m]^2 + h$: here and elsewhere $[m]^2$, &c. denote factorials, viz. $[m]^2 = m(m-1)$, $[m]^3 = m(m-1)(m-2)$, &c. It is to be noticed that for the system of two curves m, m', if h, h' represent the number of apparent double points of the two curves respectively, then for the system the number of apparent double points is $\frac{1}{2}mm' + h + h'$, and the corresponding value of M is therefore $-\frac{1}{2}[m+m']^2 + mm' + h + h'$, which is $= -\frac{1}{2}[m]^2 + -\frac{1}{2}[m']^2 + h + h'$, which is $= M + M'$.

2. The use of the combinations (m, n, p, q), (m^2, n, p), &c. hardly requires explanation; it may however be noticed that $G(m, n, p, q)$ denoting the lines which meet the curves m, n, p, q (that is, curves of these orders) each of them once, $G(m^2, n, p)$ will denote the lines which meet the curve m twice and the curves n and p each of them once; and so in all similar cases.

3. The letters G, S, ND, NG, NR, NT (read Generators, Scroll, Nodal Director, Nodal Generator, Nodal Residue, and Nodal Total) are in the nature of functional symbols, used (according to the context) to denote geometrical forms, or else the orders of these forms. Thus $G(m, n, p, q)$ denotes either the lines meeting the curves m, n, p, q each of them once, or else it denotes the order of such system of lines, that is, the number of lines. And so $S(m, n, p)$ denotes the Skew Surface or Scroll generated by a line which meets the curves m, n, p each once, or else it denotes the order of such surface.

4. $G(m, n, p, q)$: the signification is explained above.

5. $S(m, n, p)$: the signification has just been explained; but as the surfaces $S(m, n, p)$, $S(m^2, n)$, $S(m^3)$ are in fact the subject of the present memoir, I give the explanation in full for each of them, viz. $S(m, n, p)$ is the surface generated by a line which meets the curves m, n, p each once; $S(m^2, n)$ is the surface generated by a line which meets the curve m twice and the curve n once; $S(m^3)$ the surface generated by the line which meets the curve m thrice. As already mentioned, these surfaces and their orders are represented by the same symbols respectively.

6. ND(m, n, p). The directrix curves m, n, p of the scroll $S(m, n, p)$ are nodal (multiple) curves on the surface, viz. m is an np-tuple curve, and so for n and p. Reckoning each curve according to its multiplicity, viz. the curve m being reckoned $[np]^2$ times, or as of the order $m \cdot [np]^2$, and so for the curves n and p, the aggregate, or sum of the orders, gives the Nodal Director ND(m, n, p).

7. NG(m, n, p). The scroll $S(m, n, p)$ has the nodal generating lines $G(m^2, n, p)$, $G(m, n^2, p)$, $G(m, n, p^2)$. Each of these is a mere double line, to be reckoned once only, and we have thus the Nodal Generator NG$(m, n, p) = G(m^2, n, p) + G(m, n^2, p) + G(m, n, p^2)$.

But to take another example, the scroll $S(m^2, n)$ has the nodal generating lines $G(m^3, n)$, each of which is a triple line to be reckoned $\frac{1}{6}[3]^3$, that is, three times, and also the nodal generating lines $G(m^4, n)$, each of them a mere double line to be reckoned once only; whence here NG$(m^2, n) = 3G(m^3, n) + G(m^4, n')$. And so for the scroll $S(m^3)$, this has the nodal generating lines $G(m^4)$, each of them a quadruple line to be reckoned $\frac{1}{24}[4]^4$, that is, six times; or we have NG$(m^3) = 6G(m^4)$.

8. NR(m, n, p). The scroll $S(m, n, p)$ has besides the directrix curves m, n, p or Nodal Director, and the nodal generating lines or Nodal Generator, a remaining nodal curve or Nodal Residue, the locus of the intersections of two non-coincident generating lines meeting in a point not situate on any one of the directrix curves. This Nodal Residue, as well for the scroll $S(m, n, p)$ as for the scrolls $S(m^2, n)$ and $S(m^3)$ respectively, is a mere double curve to be reckoned once only; and such curve or its order is denoted by NR, viz. for the scroll $S(m, n, p)$, the Nodal Residue is NR(m, n, p).

9. NT(m, n, p). The Nodal Director, Nodal Generator, and Nodal Residue of the scroll $S(m, n, p)$ form together the Nodal Total NT(m, n, p), that is, we have $\mathrm{NT}(m, n, p) = \mathrm{ND}(m, n, p) + \mathrm{NG}(m, n, p) + \mathrm{NR}(m, n, p)$; and similarly for the scrolls $S(m^2, n)$ and $S(m^3)$.

10. I remark that the formulae are best exhibited in an order different from that in which they are in the sequel obtained, viz. I collect them in the following Table:

$$
\begin{aligned}
G(m,n,p,q) &= 2mnpq,\\
G(m^2,n,p) &= np\{[m]^2 + M\},\\
G(m^2,n^2) &= \tfrac{1}{4}[m]^2[n]^2 + \tfrac{1}{2}[m]^2\cdot N + \tfrac{1}{2}[n]^2\cdot M + MN,\\
G(m^3,n) &= m\left(\tfrac{1}{6}[m]^3 + M(m-2)\right),\\
G(m^4) &= \tfrac{1}{24}[m]^4 + [m]^2\cdot M\left(\tfrac{1}{6}[m]^2 - 2\right)^2 \text{ etc.},\\
S(m,n,p) &= 2mnp,\\
\mathrm{ND}(m,n,p) &= \tfrac{1}{2}mnp(mn+mp+np-3),\\
\mathrm{NG}(m,n,p) &= mnp(m+n+p-3) + Mnp + Nmp + Pmn,\\
\mathrm{NR}(m,n,p) &= \tfrac{1}{2}mnp\{mnp - (mn+mp+np) - 2(m+n+p) + 5\},\\
\mathrm{NT}(m,n,p) &= \tfrac{1}{2}S^2 - S + Mnp + Nmp + Pmn,\\
&= \tfrac{1}{2}mnp(mnp-1) + Mnp + Nmp + Pmn;
\end{aligned}
$$

included in which we have

$$
\begin{array}{ll}
S(1,1,m) = 2m, & \mathrm{ND}(1,1,m) = [m]^2,\\
\mathrm{NG}(1,1,m) = [m]^2 + M, & \mathrm{NR}(1,1,m) = 0,\\
\mathrm{NT}(1,1,m) = \tfrac{1}{2}S^2 - S + M, & \qquad\qquad = 2[m]^2 + M.
\end{array}
$$

and

$$
\begin{aligned}
S(1,m,n) &= 2mn,\\
\mathrm{ND}(1,m,n) &= \tfrac{1}{2}mn(mn+m+n-3),\\
\mathrm{NG}(1,m,n) &= mn(m+n-2) + Mn + Nm,\\
\mathrm{NR}(1,m,n) &= \tfrac{1}{2}[m]^2[n]^2,\\
\mathrm{NT}(1,m,n) &= \tfrac{1}{2}S^2 - S + Mn + Nm,\\
&= 2[mn]^2 + Mn + Nm.
\end{aligned}
$$

Moreover

$$\begin{aligned}
S(m^2,n) &= n\{[m]^2+M\},\\
\mathrm{ND}(m^2,n) &= n(\tfrac{1}{4}[m]^4+[m]^3+M(\tfrac{1}{2}[m]^2-1)+M^2\cdot\tfrac{1}{4})+[n]^2(\tfrac{1}{2}[m]^2+\tfrac{1}{2}M),\\
\mathrm{NG}(m^2,n) &= n([m]^3+M\cdot 3(m-2))+[n]^2(\tfrac{1}{2}[m]^2+\tfrac{1}{2}M)+N(\tfrac{1}{2}[m]^2+M),\\
\mathrm{NR}(m^2,n) &= n(\tfrac{1}{4}[m]^4+\tfrac{1}{2}M(\tfrac{1}{2}[m]^2-2m+3))\\
&\quad +[n]^2(\tfrac{1}{4}[m]^4+\tfrac{1}{2}[m]^3+[m]^2+M([m]^2-1)+M^2\cdot\tfrac{1}{4}),\\
\mathrm{NT}(m^2,n) &= \tfrac{1}{2}S^2-S+nM(m-n)+N(\tfrac{1}{2}[m]^2+M),\\
&= n(\tfrac{1}{4}[m]^4+2[m]^3+M([m]^2+m-\tfrac{5}{2})+M^2\cdot\tfrac{1}{4})\\
&\quad +[n]^2(\tfrac{1}{4}[m]^4+2[m]^3+[m]^2+M[m]^2+M^2\cdot\tfrac{1}{4})+N(\tfrac{1}{2}[m]^2+M);
\end{aligned}$$

included in which we have

$$\begin{aligned}
S(1,m^2) &= [m]^2+M,\\
\mathrm{ND}(1,m^2) &= \tfrac{1}{4}[m]^4+[m]^3+M(\tfrac{1}{2}[m]^2-1)+M^2\cdot\tfrac{1}{4},\\
\mathrm{NG}(1,m^2) &= [m]^3+M\cdot S(m-2),\\
\mathrm{NR}(1,m^2) &= \tfrac{1}{4}[m]^4+M(\tfrac{1}{2}[m]^2-2m+3),\\
\mathrm{NT}(1,m^2) &= \tfrac{1}{2}S^2-S+M(m-\tfrac{5}{2}),\\
&= [m]^4+2[m]^3+M([m]^2+m-\tfrac{5}{2})+M^2\cdot\tfrac{1}{4};
\end{aligned}$$

and finally

$$\begin{aligned}
S(m^3) &= \tfrac{1}{6}[m]^3+(m-2)M,\\
\mathrm{ND}(m^3) &= \tfrac{1}{6}[m]^5+\tfrac{1}{2}[m]^4+\tfrac{5}{6}[m]^3+M(\tfrac{1}{2}[m]^3+\tfrac{7}{2}[m]^2)+M^2\cdot\tfrac{1}{2}m,\\
\mathrm{NG}(m^3) &= \tfrac{1}{2}[m]^4+6m+M(3[m]^2-12m+33)+M^2\cdot 3,\\
\mathrm{NR}(m^3) &= \tfrac{1}{6}[m]^5+\tfrac{5}{6}[m]^4-\tfrac{5}{2}[m]^3+3m\\
&\quad +M(\tfrac{1}{2}[m]^4+\tfrac{1}{2}[m]^3-\tfrac{1}{2}[m]^2+8m-20)+M^2(\tfrac{1}{2}[m]^2-2m),\\
\mathrm{NT}(m^3) &= \tfrac{1}{2}S^2-S+3m+M(\tfrac{1}{2}[m]^2-\tfrac{5}{2}m+11)+M^2\\
&= \tfrac{1}{6}[m]^5+\tfrac{3}{2}[m]^4+[m]^3+3m\\
&\quad +M(\tfrac{1}{2}[m]^4+\tfrac{3}{2}[m]^3+\tfrac{1}{2}[m]^2-\tfrac{5}{2}m+13)+M^2(\tfrac{1}{2}[m]^2-\tfrac{5}{2}m+3).
\end{aligned}$$

The formulae are investigated in the following order: ND, G, NG, S, NR, and NT.

Collected Mathematical Papers of Arthur Cayley

Volume V

Pages 201–250

0.1 [340] A Second Memoir on Skew Surfaces, Otherwise Scrolls

[From the Philosophical Transactions of the Royal Society of London, vol. CLIV. (for the year 1864), pp. 559–576.
Received April 29,—Read May 26, 1864.]

The principal object of the present memoir is to establish the different kinds of skew surfaces of the fourth order, or Quartic Scrolls; but, as preliminary thereto, there are some general researches connected with those in my former memoir "On Skew Surfaces, otherwise Scrolls"[1]. I also reproduce the theory (which may be considered as a known one) of cubic scrolls; there are also some concluding remarks which relate to the general theory. As regards quartic scrolls, I remark that M. Chasles, in a footnote to his paper, "Description des courbes de tous les ordres situées sur les surfaces réglées du troisième et du quatrième ordres"[2], states, "les surfaces réglées du quatrième ordre ... admettent quatorze espèces." This does not agree with my results, since I find only eight species of quartic scrolls; the developable surface or "torse" is perhaps included as a "surface réglée;" but as there is only one species of quartic torse, the deficiency is not to be thus accounted for. My enumeration appears to me complete, but it is possible that there are subforms which M. Chasles has reckoned as distinct species.

On the Degeneracy of a Scroll. Article Nos. 1 to 5.

Art. A scroll considered as arising from any geometrical construction, for instance one of the scrolls $S(m,n,p)$, $S(m^2,n)$, $S(m^3)$ considered in my former memoir, or say in general the scroll S, may break up into two or more inferior scrolls $S', S'', \ldots$; but as long as $S', S'', \ldots$ are proper scrolls (not torses, and a fortiori not cones or planes), no one of these can be considered, apart from the others, as the result of the geometrical construction, and we can only say that the scroll S given by the construction is the aggregate of the scrolls $S', S'', \ldots$; and the like when we have the scrolls $S', S'', \ldots$, each repeated any number of times, or say when $S = S'^{\alpha} S''^{\beta} \ldots$. Suppose however that the scrolls $S', S'', \ldots$ are any one or more of them a torse or torses—or, to make at once the most general supposition, say that we have $S = \Sigma S'$, where Σ is a torse, or aggregate of torses ($\Sigma = \Sigma'^{\alpha} \Sigma''^{\beta} \ldots$), and S' is a proper scroll or aggregate of proper scrolls; then, although it is not obligatory to do so, we may without impropriety throw aside the torse-factor Σ, and consider the original scroll S as degenerating into the scroll S', and as suffering a reduction in order accordingly.

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[1] Philosophical Transactions, vol. cliii. (1863), pp. 453–483, [339].

[2] Comptes Rendus, t. liii. (1861), see p. 888.

Art. As an illustration, consider the scroll $S(m, n, p)$ generated by a line which meets three directrix curves of the orders m, n, p respectively; and assume that the curves m, n, p are each of them situate on the same scroll Σ, the curve m meeting each generating line of Σ in α points, the curve n each generating line in β points, and the curve p each generating line in γ points. Each generating line of Σ is $\alpha\beta\gamma$ times a generating line of S, and we have $S = \Sigma^{\alpha\beta\gamma}S'$, where S' may be a proper scroll; it is however to be noticed that if the curves m, n, p any two of them intersect, S' will itself break up and contain cone-factors, as will presently appear. And if Σ, instead of being a proper scroll, be a torse, then we may consider S as degenerating into S', the reduction in order being of course $= \alpha\beta\gamma \times$ order of Σ.

Art. But this is not the only way in which the scroll $S(m, n, p)$ may degenerate; for suppose that two of the directrix curves, say n and p, intersect, then the lines from the point of intersection to the curve m form a cone of the order m which will present itself as a factor of S; and generally if the curves n and p intersect in α points, the curves p and m in β points, and the curves m and n in γ points, then we have α cones each of the order m, β cones each of the order n, and γ cones each of the order p, or say $S = CS'$, where C is the aggregate of the cone-factors; and the scroll S degenerates into S', the reduction in order being $= \alpha m + \beta n + \gamma p$. It is hardly necessary to remark that if a point of intersection of two of the curves is a multiple point on either or each of the curves, it is, in reckoning the number of intersections of the two curves, to be taken account of according to its multiplicity in the ordinary manner.

Art. There is yet another case to be considered: suppose that the curves n and p lie on a cone, and that the curve m passes through the vertex of this cone; this cone, repeated a certain number of times, is part of the locus, or we have $S = C'S'$, so that the scroll S degenerates into S', the reduction in order being $= \delta \times$ order of cone. If, to fix the ideas, the curves n and p are respectively the complete intersections of the cone by two surfaces of the orders g, h respectively (this implies $n = gk$, $p = hk$, if k be the order of the cone), which surfaces do not pass through the vertex of the cone, and if, moreover, the vertex of the cone be an α-tuple point on the curve m, then $\delta = \alpha gh$, and the reduction in order is $= \alpha ghk$.

Art. The foregoing causes of reduction, or some of them, may exist simultaneously; it would require a further examination to see whether the aggregate reduction is in all cases the sum of the separate reductions. But the aggregate reduction once ascertained, then writing $S(m, n, p)$ for the order of the reduced scroll, we shall have

$$S(m, n, p) = 2mnp - \text{Reduction}.$$

In particular, in the case above referred to, where the curves n and p, p and m, m and n meet in α, β, γ points respectively, but there is no other cause of reduction,

$$S(m, n, p) = 2mnp - \alpha m - \beta n - \gamma p,$$

which is a formula which will be made use of.

The foregoing investigations apply, mutatis mutandis, to the scrolls $S(m^2, n)$, $S(m^3)$; but I do not at present enter into the development of them in regard to these scrolls.

Scrolls with two directrix lines. Article Nos. 6 to 11.

Art. Consider now a scroll having two directrix lines: it may be assumed that these do not intersect; for if they did, then any generating line, quà line meeting the two directrix lines, would either lie in the plane of the two lines, or else would pass through their point of intersection; that is, the scroll would break up into the plane of the two lines, considered as the locus of the tangents of a plane curve, and into a cone having for its vertex the point of intersection of the two lines. Each generating line meets any plane section of the scroll in the point where such generating line meets the plane of the section; the plane section constitutes a third directrix; or the scrolls in question are all included in the form $S(1, 1, m)$, where m is a plane curve. The order of the scroll $S(1, 1, m)$ is in general $= 2m$; but if the one line meets the curve α times, that is, in an α-tuple point of the curve, and the other line meets the curve β times, that is, in a β-tuple point of the curve, then by the general formula (ante, No. 5) the order of the scroll is $= 2m - \alpha - \beta$; and in particular if $\alpha + \beta = m$, then the order is $= m$.

Art. We may without loss of generality attend only to the last-mentioned case. To show how this is, suppose for a moment that the two lines do not either of them meet the curve; the scroll is then of the order $2m$. Call the point in which each line meets the plane of the curve the foot of this line, then the line joining the two feet meets the curve in m points; and it is in respect of each of these points a generating line of the scroll; that is, it is an m-tuple generating line: the section of the scroll by the plane of the curve m is in fact this line counting m times, and the curve m; $m + m = 2m$, the order of the scroll. And in like manner the section by any plane through the m-tuple line is this line counting m times, and a curve of the order m not meeting either of the directrix lines. But the section by any other plane is a curve of the order $2m$ meeting each of the directrix lines in a point which is an m-tuple point of the section (each directrix line is in fact an m-tuple line of the scroll); and by considering, in place of the particular section m, this general section, we have the scroll of the order $2m$ in the form $S(1, 1, 2m)$, where the two directrix lines each meet the section m times; so that the order is $4m - m - m = 2m$.

Art. And so in general, m being a plane curve, when the scroll $S(1, 1, m)$ is of an order superior to m, say $= m + k$, this only means that the section chosen for the directrix curve m is not the complete section by the plane of such curve, but that the line joining the feet of the two directrix lines is a k-tuple generating line of the scroll, and that the complete section is made up of this line counting k times and of the curve m. So that taking, not the section through the multiple generating line, but the general section, for the plane directrix curve, the only case to be considered is that in which the section is a proper curve of an order equal to that of the scroll; or, what is the same thing, we have only to consider the scrolls $S(1,1,m)$ for which the order is depressed from $2m$ to m in consequence of the directrix lines meeting the plane section α times and β times, that is, in an α-tuple point and a β-tuple point respectively, where $\alpha+\beta=m$.

Art. It is clear that in the case in question the directrix lines are an α-tuple line and a β-tuple line respectively. The generation is as follows: Scroll $S(1,1,m)$ of the order m; the curve m being a plane curve of the order m having an α-tuple point and a β-tuple point, where $\alpha+\beta=m$: the directrix lines, say l and l', pass through these points respectively, and they do not intersect each other. The generating lines pass through the directrix lines l and l' and the curve m, and we have thence the scroll $S(1,1,m)$. Taking at pleasure any point on the curve m, we can through this point draw a single line meeting each of the directrix lines l, l'; that is, the curve m is a simple curve on the scroll. Taking at pleasure a point on the directrix line l, and making this the vertex of a cone standing on the curve m, this cone has an α-tuple line (the line l) and a β-tuple line (the line joining the vertex with the foot of the line l'); the line l' meets this cone in the foot of the line l', counting β times, and besides in $m-\beta, =\alpha$ points; the lines joining the vertex with the last-mentioned points respectively (or, what is the same thing, the lines, other than the β-tuple line, in which the plane through the vertex and the line l' meets the cone) are the α generating lines through the assumed point on the line l; and the line l is thus an α-tuple line of the scroll. And in like manner, through an assumed point of the directrix line l', we construct β generating lines of the scroll; and the line l' is a β-tuple line of the scroll.

Art. The scroll $S(1,1,m)$ now in question has not in general any multiple generating line; in fact a multiple generating line would imply a corresponding multiple point on the section m; and this section, assumed to be a curve having an α-tuple point and a β-tuple point, has not in general any other multiple point. But it may have other multiple points; and if there is, for example, a γ-tuple point, then the line from this point which meets the two directrix lines counts γ times, or it is a γ-tuple generating line; and so for all the multiple points of m other than the α-tuple point and the β-tuple point which correspond to the directrix lines respectively. It is to be noticed that the multiplicity γ of any such multiple generating line is at most equal to the smallest of the two numbers α and β; for suppose $\gamma>\alpha$, then, since $\alpha+\beta=m$, we should have $\gamma+\beta>m$, and the line joining the γ-tuple point and the β-tuple point would meet the curve m in $\gamma+\beta$ points, which is absurd. In the case of several multiple lines, there are other conditions of inequality preventing self-contradictory results[3].

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[3] Suppose, for example (see next paragraph of the text), that there were a γ-

Art. The general section is a curve of the order m, having an α-tuple point and a β-tuple point corresponding to the directrix lines respectively, and a γ-tuple point, &c.... corresponding to the other multiple points (if any). A section through the directrix line l is in general made up of this line, counting α times, and of β generating lines passing through one and the same point of the directrix line l'; if the section pass also through a γ-tuple generating line, then, of the β generating lines in question, γ (which, as has been seen, is $\leqq \beta$) unite together in the γ-tuple generating line; and so for the sections through the directrix line l'. The general section through a γ-tuple generating line is this line counting γ times, and a curve of the order $m-\gamma$, which has an $(\alpha-\gamma)$-tuple point at its intersection with the directrix line l, and a $(\beta-\gamma)$-tuple point at its intersection with the directrix line l'; it has a δ-tuple point, &c.... at its intersections with the other multiple generating lines, if any.

Equation of the Scroll $S(1, 1, m)$ of the order m. Article Nos. 12 to 18.

Art. But there is a case included indeed as a limiting one in the foregoing general case, but which must be specially considered; viz. the two directrix lines l and l' may coincide, giving rise to a twofold directrix line. To show how this is, I return for the moment to the case of the scroll $S(1, 1, m)$ with two distinct directrix lines l and l', and, to fix the ideas, I suppose that the directrix lines do not either of them meet the curve m, so that the order of the scroll is $= 2m$. Through the line l imagine the series of planes A, B, C, ... meeting the line l' in the points a', b', c', ...; the generating lines through the point a' are the lines in the plane A to the points in which this plane meets the curve m; the generating lines through the point b' are the lines in the plane B to the points where this plane meets the curve m; and so for the generating lines through the points c', d', ...; and it is clear that the points a', b', c', ... correspond homographically with the planes A, B, C, ... This gives immediately the construction for the case where the two directrix lines come to coincide. In fact, on the twofold directrix line $l = l'$ take the series of points a, b, c, ..., and through the same line, corresponding homographically to these points, the series of planes A, B, C, ...; the generating lines through the point a are the lines through this point, in the plane A, to the points in which this plane meets the curve m; and so for the entire series of points b, c, ... of the line $l = l'$; the resulting scroll, which I will designate as the scroll $S(1, 1, m)$, remains of the order

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tuple generating line and a δ-tuple generating line lying in piano with the line l; these lines counting as $(\gamma + \delta)$ lines, must be included among the β generating lines through the plane in question; this implies that $\gamma + \delta \leqq \beta$, a conclusion which must be obtainable from consideration of the curve m irrespectively of the scroll.

$= 2m$. If there is given a point of the curve m, then the plane through this point and the directrix line is the plane A; and the point a is then also given by the homographic correspondence of the series of planes and points, and the generating line through the given point on the curve m is the line joining this point with the point a.

Art. We may say that, in regard to any point a of the line l, the corresponding plane A is the plane of approach of the coincident line l'; and that in regard to the same point a and to any plane through it, the trace on that plane of the plane of approach is the line of approach of l'; that is, we may consider that the coincident directrix line l' meets the plane through a in a consecutive point on the line of approach. In particular if the point a be the foot of the directrix line l (that is, the point where this line meets the plane of the curve m), and the plane through a be the plane of the curve m, then the intersection of the last-mentioned plane by the plane A which corresponds to the point a is the line of approach, and the foot of the coincident directrix line l' is the consecutive point to a along the line of approach. The expression "the line of approach," used absolutely, has always the signification just explained, viz. it is the intersection of the plane of the curve m by the plane corresponding to the foot of the directrix line.

Art. Suppose now that the line l meets the curve m, or, more generally, meets it α times, that is, in an α-tuple point; it might at first sight appear that the coincident line l' should also be considered as meeting the curve α times, and that the resulting scroll should be of the order $2m - \alpha - \alpha = 2m - 2\alpha$. But this is not the case; so long as the direction of the line of approach is arbitrary, the line l' must be considered as a line indefinitely near to the line l, but nevertheless as a line not meeting the curve at all; and the order of the scroll is thus $= 2m - \alpha$. If, however, the line of approach is the tangent to a branch through the α-tuple point—that is, if the plane corresponding to the α-tuple point meet the plane of the curve in such tangent, then the coincident line l' is to be considered as meeting the curve m in a consecutive point on such branch, and the order of the scroll is $= 2m - \alpha - 1$. And so if at the multiple point there are β branches having a common tangent, then the coincident line l' is to be considered as meeting the curve m in a consecutive point along each of such branches, or say in a consecutive β-tuple point along the branch, and the order of the scroll sinks to $2m - \alpha - \beta$. The point spoken of as the α-tuple point is, it should be observed, more than an α-tuple point with a β-fold tangent; it is really a point of union of an α-tuple point and a β-tuple point, or say a united $\alpha(+\beta)$-tuple point, equivalent to

$$\tfrac{1}{2}\alpha(\alpha - 1) + \tfrac{1}{2}\beta(\beta - 1)$$

double points or nodes; and the case is precisely analogous to that of the scroll $S(1, 1, m)$, where the two directrix lines pass through an α-tuple point and a β-tuple point of the curve m respectively. It may be added that if at the multiple point in question, besides the β branches having a common tangent, there are γ branches having a common tangent, then the point is, so to speak, a united $\alpha(+\beta+\gamma)$-tuple point equivalent to

$$\tfrac{1}{2}\alpha(\alpha-1)+\tfrac{1}{2}\beta(\beta-1)+\tfrac{1}{2}\gamma(\gamma-1)$$

double points or nodes; but the order of the scroll is still $= 2m-\alpha-\beta$.

Art. In the same way as the scrolls $S(1,1,m)$ are all included in the case where the order of the scroll, instead of being $= 2m$, is $= m$, so that the scrolls $S(1,1,m)$ are all included in the case where the order of the scroll, instead of being $= 2m$, is $= m$. That is, we may suppose that the curve m has a united $\alpha(+\beta)$-tuple point $(\alpha+\beta=m)$, and may take the directrix line to pass through this point, and the line of approach to be the common tangent of the β branches; and this being so, the order of the scroll will be $2m-\alpha-\beta$, $= m$. It may be added that if the curve m has, besides the $\alpha(+\beta)$-tuple point, a γ-tuple point, then the scroll will have a γ-tuple generating line, and so for the other multiple points of the curve m.

Art. We may, in the same way as for the scroll $S(1,1,m)$, consider the different sections of the scroll $S(1,1,m)$ of the order m. The general section is a curve of the order m, having an $\alpha(+\beta)$-tuple point at the intersection with the directrix line, and a γ-tuple point, &c. corresponding to the multiple generating lines, if any. A section through the directrix line is in general made up of this line counting α times, and of β generating lines through the point which corresponds to the plane of the section; if the section pass also through a γ-tuple generating line ($\gamma \leqq \beta$, in the same way as for the scroll $S(1,1,m)$[4]), then, of the β generating lines, γ unite together in the γ-tuple generating line. The general section through a γ-tuple generating line breaks up into this line counting γ times, and a curve of the order $m-\gamma$, which has on the directrix line an $\alpha-\gamma(+\beta-\gamma)$-tuple point and a δ-tuple point, &c. at its intersections with the other multiple generating lines, if any.

Equation of the Scroll $S(1,1,m)$ of the order m. Article Nos. 17 and 18.

Art. Taking for the equations of the directrix lines $(x=0, y=0)$ and $(z=0, w=0)$, and supposing that these are respectively an α-tuple line and a β-tuple line on the scroll $\alpha+\beta=m$, it is obvious that the equation of the scroll is

$$(*f\,x,y)^{\alpha}\,(z,w)^{\beta}=0.$$

In fact starting with this equation, if we consider the section by a plane through the line $(x=0, y=0)$, say the plane $y=\lambda x$, then the equation gives

$$x^{\alpha}\cdot(*f1,\lambda)^{\beta}\,(z,w)^{\beta}=0;$$

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[4] See the footnote to No. 10.

that is, the section is made up of the line $(x=0, y=0)$ reckoned α times, and of β other lines in the plane $y=\lambda x$; and the like for the section by any plane through the line $(z=0, w=0)$, say the plane $z=\varpi w$. Hence the assumed equation represents a scroll of the order m, having the two lines for an α-tuple line and a β-tuple line respectively, and conversely such scroll has an equation of the assumed form.

Case of a γ-tuple generating line.

Art. The multiple generating line meets each of the lines $(x=0, y=0)$ and $(z=0, w=0)$; and we may take for the equations of the multiple generating line $x+y=0$, $z+w=0$. This being so, the foregoing equation of the scroll may be expressed in the form

$$(*\!\!\!\!f x, y)^{\alpha}\,(z, z+w)^{\beta}=0,$$

or say

$$(U, V, W, \ldots \frown z, z+w)^{\beta}=0,$$

where U, V, W, … are functions of the form $(*\!\!\!\!f x, y)^{\alpha}$. Hence ($\gamma > 0$ or β), if the functions U, V, W, … contain respectively the factors $(x+y)^{\gamma}$, $(x+y)^{\gamma-1}$, $(x+y)^{\gamma-2}$, …, the equation will be of the form

$$(U', V', W', \ldots \frown y-\kappa x,\ w(y-\kappa x)+x(\kappa w-z))^{\gamma}=0,$$

(the coefficients being functions of x, y, z and $z+w$, or, what is the same thing, x, y, z, w, of the order $\alpha+\beta-\gamma$), and the scroll will therefore have the line $x+y=0$, $z+w=0$ as a γ-tuple generating line.

Equation of the Scroll $S(1,1,m)$ of the order m. Article Nos. 19 to 24.

Art. We may take $x=0$, $y=0$ for the equations of the twofold directrix line, $z=0$ for the equation of the plane of the curve m (an arbitrary plane section of the scroll). Then ($\alpha+\beta=m$), if the curve m have at the point $(x=0, y=0)$, or foot of the directrix line, an $\alpha(+\beta)$-tuple point, and if moreover we have $y=0$ for the equation of the common tangent of the β branches (viz. if the plane $y=0$, instead of being an arbitrary plane through the directrix line, be the plane through this line and the common tangent of the β branches), the equation of the curve m will be of the form

$$\Sigma\,(yw)^{\beta'}\,(*\!\!\!\!f x, y)^{\alpha+\beta-\beta'}=0,$$

where the summation extends to all integer values of β' from 0 to β, both inclusive.

Art. Taking $y = \lambda x$ for the equation of any plane through the directrix line, then the corresponding point on the directrix line will be the intersection of this line $(x = 0, y = 0)$ by the plane $z = \theta w$, where $\theta = \frac{a\lambda+b}{c\lambda+d}$; the foot of the directrix line is given by the value $\theta = 0$, or $\lambda = -\frac{b}{a}$, and the equation of the line of approach is therefore $y = \frac{d}{c}x$; this should coincide with the line $y = 0$, which is the common tangent of the β branches; that is, we must have $\theta = 0$; I retain, however, for the moment the general value of θ.

Art. The equations of a generating line will be

$$y = \lambda x, \quad z = \theta w - \rho x;$$

and then taking X, Y, $(Z = 0)$ and W for the coordinates of the point of intersection with the curve m, we have

$$Y = \lambda X, \quad 0 = \theta W - \rho X,$$

and thence

$$\Sigma\,(YW)^{\beta'}(*\!\!\!f Z, Y)^{\alpha+\beta-\beta'} = 0,$$

$$\Sigma\,\theta^{\beta'}(*\!\!\!f \rho, \lambda)^{\beta'}(*\!\!\!f 1, \lambda)^{\alpha+\beta-\beta'} = 0,$$

or, what is the same thing,

$$X^{\alpha}\theta^{-\beta}(*\!\!\!f \rho, \lambda)^{\beta'}(*\!\!\!f 1, \lambda)^{\alpha+\beta-\beta'} = 0;$$

which equation, substituting therein for θ its value in terms of λ, gives the parameter ρ which enters into the equations of the generating line; or, what is the same thing, the equation of the scroll is obtained by eliminating λ, θ, ρ from the equation just mentioned and the equations

$$y = \lambda x, \quad z = \theta w - \rho x, \quad \theta(c\lambda + d) = a\lambda + b.$$

Art. These last three equations give

$$\lambda = \frac{y}{x}, \quad \theta = \frac{ay+bx}{cy+dx}, \quad \rho = \frac{\theta w - z}{x} = \frac{(ay+bx)w - (cy+dx)z}{x(cy+dx)},$$

and substituting these values, we find for the equation of the scroll

$$\Sigma\,(ay+bx)^{\beta-\beta'}y^{\beta'}[(ay+bx)w - (cy+dx)z]^{\beta'}(*\!\!\!f x, y)^{\alpha+\beta-\beta'} = 0,$$

which is of the order $\alpha + 2\beta$, $= 2m - \alpha$, so that the $\alpha(+\beta)$-tuple point, in the case actually under consideration, produces only a reduction $= \alpha$. If however the line of approach coincides with the tangent of the β branches, then $\theta = 0$; the factor y^{β} divides out, and the equation is

$$\Sigma\,(ayw - cyz - dxz)^{\beta'}(*\!\!\!f x, y)^{\alpha+\beta-\beta'} = 0,$$

which is of the order $\alpha+\beta$, $=m$, so that here the reduction caused by the $\alpha(+\beta)$-tuple point is $=\alpha+\beta$. We may without loss of generality substitute ax for $cy+dx$, and then, putting also $a=1$, we find that when the equation of the curve m is as before

$$\Sigma\,(yw)^{\beta'}\,(*fx,y)^{\alpha+\beta-\beta'}=0,$$

but the plane through the directrix line $(x=0, y=0)$, and the point on this line, are respectively given by the equations $x=\lambda y$ and $z=\lambda w$, the equation of the scroll is

$$\Sigma\,(yw-xz)^{\beta'}\,(*fx,y)^{\alpha+\beta-\beta'}=0.$$

Art. The result may be verified by considering the section by any plane $y=\lambda x$ through the directrix line. Substituting for y this value, we find

$$x^{\alpha}\cdot\Sigma\,(\lambda w-z)^{\beta'}\,(*f1,\lambda)^{\alpha+\beta-\beta'}=0,$$

which is of the form

$$x^{\alpha}\cdot f(\lambda w-z)=0;$$

so that the section is made up of the directrix line $(x=0, y=0)$ reckoned α times and of β lines in the plane $y-\lambda x=0$, the intersections of the plane $y-\lambda x=0$ by planes such as $z=\lambda w-\rho x$.

Case of a γ-tuple generating line.

Art. The equation of the scroll may be written

$$(U, V, W, \ldots \frown y-\kappa x,\ yw-xz)^{\beta}=0,$$

where U, V, W, … are functions of x, y of the forms

$$(*fx,y)^{\alpha},\quad (*fx,y)^{\alpha-1},\quad (*fx,y)^{\alpha-2},\ldots;$$

assuming that these contain respectively the factors

$$(y-\kappa x)^{\gamma},\quad (y-\kappa x)^{\gamma-1},\quad (y-\kappa x)^{\gamma-2},\ldots,$$

where $\gamma \leqq \beta$, then the equation takes the form

$$(U', V', W', \ldots \frown y-\kappa x,\ w(y-\kappa x)+x(\kappa w-z))^{\gamma}=0,$$

where the coefficients U', V', W', … are functions of x, y, z, w of the orders $m-\gamma$, $m-\gamma-1$, $m-\gamma-2$, …; or, what is the same thing, the equation is

$$(U'', V'', W'', \ldots \frown y-\kappa x,\ \kappa w-z)^{\gamma}=0,$$

where U'', V'', W'', … are functions of x, y, z, w of the order $m-\gamma$. The scroll has thus the γ-tuple generating line

$$y-\kappa x=0,\quad \kappa w-z=0.$$

Cubic Scrolls. Article Nos. 25 to 35.

Art. In the case of a cubic scroll there is necessarily a nodal[5] line; in fact for the m-thic scroll there is a nodal curve which is of the order $m-2$ at least, and of the order $\frac{1}{2}(m-1)(m-2)$ at most, and which for $m=3$ is therefore a right line. And moreover we see at once that every cubic surface having a nodal line is a scroll; in fact any plane whatever through the nodal line meets the surface in this line counting as 2 lines, and in a curve of the order 1, that is, a line; there are consequently on the surface an infinity of lines, or the surface is a scroll. We have therefore to examine the cubic surfaces which have a nodal line.

Art. Let the equations of the nodal line be $x=0$, $y=0$; then the equation of the surface is

$$Uz+Vw+Q=0,$$

where U, V, Q are functions of (x,y) of the orders 2, 2, 3 respectively. Suppose first that U, V have no common factor, then we may write

$$Q=(\alpha x+\beta y)U+(\gamma x+\delta y)V;$$

and substituting this value, and changing the values of z and w, the equation of the surface is of the form

$$Uz+Vw=0,$$

or, what is the same thing,

$$(*fx,y)^2\,(z,w)^1=0;$$

so that, besides the nodal directrix line $(x=0,y=0)$, the scroll has the simple directrix line $(z=0,w=0)$: it is clear that the section by any plane whatever is a cubic curve having a node at the foot of the nodal directrix line $(x=0,y=0)$, and passing through the foot of the simple directrix line $(z=0,w=0)$; that is, it is a cubic scroll of the kind $S(1,1,3)$; and since for $m=3$ the only partition $m=\alpha+\beta$ is $m=2+1$, there is only one kind of cubic scroll $S(1,1,3)$, and we may say simpliciter that the scroll in question is the cubic scroll $S(1,1,3)$.

Art. If however the functions U, V have a common factor, say $(\lambda x+\mu y)$, then $zU+wV$ will contain this same factor, and the remaining factor will be of the form

$$z(\alpha x+\beta y)+w(\gamma x+\delta y)=y(\alpha z+\delta w)+x(\gamma z+\beta w),$$

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[5] The nodal line of a cubic scroll is of course a double line, and in regard to these scrolls the epithets ‘nodal’ and ‘double’ may be used indifferently.

or, changing the values of z and w, the remaining factor will be of the form $yw - xz$, and the equation of the scroll thus is

$$(\lambda x + \mu y)(yw - xz) + (*\!\between x, y)^3 = 0,$$

where it is clear that the section by any plane whatever is a cubic curve having a node at the foot of the directrix line $x = 0$, $y = 0$. The scroll is thus a cubic scroll of the form $S(1, 1, 3)$; viz. it is the scroll of the kind where the section is a cubic curve with a 2(+1)-tuple point (ordinary double point, or node), the line of approach being one of the two tangents at the node; and since for $m = 3$ the only partition $m = \alpha + \beta$ is $m = 2 + 1$, there is only one kind of cubic scroll $S(1, 1, 3)$, and we may say simpliciter that the scroll in question is the cubic scroll $S(1, 1, 3)$. The conclusion therefore is that for cubic scrolls we have only the two kinds, $S(1, 1, 3)$ and $S(1, 1, 3)$. The foregoing equations of these scrolls admit however of simplification; and I will further consider the two kinds respectively.

The Cubic Scroll $S(1, 1, 3)$.

Art. Starting from the equation

$$(*\!\between x, y)^2\,(z, w)^1 = 0,$$

or, writing it at full length,

$$z(a, b, c\between x, y)^2 + w(a', b', c'\between x, y)^2 = 0,$$

we may find θ_1, θ_2, so that

$$(a, b, c\between x, y)^2 + \theta_1(a', b', c'\between x, y)^2 = (p_1x + q_1y)^2,$$
$$(a, b, c\between x, y)^2 + \theta_2(a', b', c'\between x, y)^2 = (p_2x + q_2y)^2,$$

θ_1 and θ_2 being unequal, since by hypothesis $(a, b, c\between x, y)^2$ and $(a', b', c'\between x, y)^2$ have no common factor. This gives

$$(a, b, c\between x, y)^2 = \alpha(p_1x + q_1y)^2 + \beta(p_2x + q_2y)^2,$$
$$(a', b', c'\between x, y)^2 = \gamma(p_1x + q_1y)^2 + \delta(p_2x + q_2y)^2;$$

or the equation becomes

$$(\alpha z + \gamma w)(p_1x + q_1y)^2 + (\beta z + \delta w)(p_2x + q_2y)^2 = 0;$$

or changing the values of (x, y) and of (z, w), the equation is

$$x^2z + y^2w = 0,$$

which may be considered as the canonical form of the equation. It may be noticed that the Hessian of the form is x^2y^2.

Art. We may of course establish the theory of the surface from the equation $x^2z + y^2w = 0$; the equation is satisfied by $x = \lambda y$, $w = -\lambda^2 z$, which are the equations of a line meeting the line $(x = 0, y = 0)$ (l) and the line $(z = 0, w = 0)$ (l'). The generating line meets also any plane section of the surface; in fact, if the equation of the plane of the section be $\alpha x + \beta y + \gamma z + \delta w = 0$, then we have at once

$$\lambda : y : z : w = \delta\lambda^2 - \gamma : \delta\lambda^3 - \gamma\lambda : \alpha\lambda + \beta : -\alpha\lambda^2 - \beta\lambda$$

for the coordinates of the point of intersection.

Art. The form of the equation shows that there are on the line l two points, viz. the points $(x = 0, y = 0, z = 0)$ and $(x = 0, y = 0, w = 0)$, through each of which there passes a pair of coincident generating lines: calling these A and B, then, if the coincident lines through A meet the line l' in C, and the coincident lines through B meet the line l' in D, it is easy to see that $x = 0$, $y = 0$, $z = 0$, and $w = 0$ will denote the equations of the planes BAC, BAD, BCD, and ACD respectively.

Art. We obtain also the following construction: take a cubic curve having a node, and from any point K on the curve draw to the curve the tangents Kp, Kq; through the points of contact draw at pleasure the lines pAC and qBD; through the node draw a line meeting these two lines in the points A, B respectively, this will be the line l; and through the point K a line meeting the same two lines in the points C and D respectively, this will be the line l'; and, the equations $x = 0$, $y = 0$, $z = 0$, $w = 0$ denoting as above, the equation of the surface will be $x^2z + y^2w = 0$.

The points A and B are cuspidal points on the nodal line; any section of the scroll by a plane through one of these points is a cubic curve having at the point in question a cusp.

Art. It is to be noticed however that the cuspidal points are not of necessity real; if for x, y we write $x + iy$, $x - iy$, and in like manner $z + iw$, $z - iw$ for z, w, then the equation takes the form

$$(x^2 - y^2)z - 2xyw = 0,$$

which is a cubic scroll $S(1, 1, 3)$ with the cuspidal points imaginary.

In the last-mentioned case the nodal line is throughout its whole length crunodal; in the case first considered, where the equation is $x^2z + y^2w = 0$, the nodal line is for that part of its length for which z, w have opposite signs, crunodal; and for the remainder of its length, or where z, w have the same sign, acnodal. There are two different forms, according as the line is for the portion intermediate between the cuspidal points crunodal and for the extramediate portions acnodal, or as it is for the intermediate portion acnodal and for the extramediate portions crunodal.

Cubic Scroll $S(1,1,3)$.

Art. Starting from the equation

$$(\lambda x+\mu y)(yw-xz)+(*\between x,y)^3=0,$$

then putting $w-\mu z$ for w and λz for z, this may be written

$$(\lambda x+\mu y)(yw-z(\lambda x+\mu y))+(*\between \lambda x+\mu y,y)^3=0,$$

or, what is the same thing,

$$\lambda(yw-xz)+(*\between x,y)^3=0;$$

and then, if $(*\between x,y)^3=(\alpha,\beta,\gamma,\delta\between x,y)^3$, this may be written

$$\lambda\{y(w+\alpha x+\gamma y)-x(z-\alpha x)\}+\beta y^3=0;$$

or changing the values of w and z, we have

$$\lambda(yw-xz)+\beta y^3=0$$

for the equation of the scroll $S(1,1,3)$[6].

Art. The Hessian of the form is x^4, and it thus appears that the plane $x=0$ is a determinate plane through the double line. But $y=0$ is not a determinate plane; in fact, if for y we write $y+\lambda x$, the equation is

$$-x^2z+xw(y+\lambda x)+(y+\lambda x)^3=0,$$

that is

$$-x^2(z-\lambda w-3\lambda^2y-\lambda^3x)+xy(w+3\lambda x)+y^3=0,$$

which, changing z and w, is still of the form $x(yw-xz)+y^3=0$.

The planes $z=0$, $w=0$ will alter with the plane $y=0$, but they are not determined even when the plane $y=0$ is determined; in fact we may, without altering the equation, change w,z into $w+\theta y$, $z+\theta x$ respectively.

Art. In the equation $x(yw-xz)+y^3=0$, writing $y=\lambda x$, we find for the equations of a generating line, $y=\lambda x$, $z=\lambda w+\lambda^3x$. Considering the section by the plane $\alpha x+\beta y+\gamma z+\delta w=0$, we have

$$\lambda:y:z:w=-\gamma\lambda-\delta:-\gamma\lambda^2-\delta\lambda:-\delta\lambda^3+\beta\lambda^2+\alpha\lambda:\gamma\lambda^2+\beta\lambda+\alpha$$

for the coordinates of the point where the generating line meets the section.

The generating line meets the nodal line at the intersection of the nodal line by the plane $z=\lambda w$; that is, the points $z=\lambda w$ on the nodal line correspond to the planes $y=\lambda x$ through the nodal line. In particular the point $w=0$ on the nodal line corresponds to the plane $x=0$ through the nodal line: the point $\gamma z+\delta w=0$ on the nodal line (that is, the point where this line is met by the plane $\alpha x+\beta y+\gamma z+\delta w=0$) corresponds to the plane $\gamma x+\delta y=0$ through the nodal line; the intersections of the plane $\alpha x+\beta y+\gamma z+\delta w=0$ by this plane $\gamma x+\delta y=0$, and by the plane $x=0$, are the tangents of the section at the node.

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[6] It is somewhat more convenient to change the sign of z, and take $x(yw+xz)+y^3=0$ as the canonical form.

Quartic Scrolls. Article Nos. 36 to 50.

Art. We may consider, first, the quartic scrolls $S(1, 1, 4)$. The section is a quartic curve having an α-tuple point and a β-tuple point, where $\alpha+\beta = 4$; that is, we have $\alpha = 2$, $\beta = 2$, a quartic with two nodes (double points), or else $\alpha = 3$, $\beta = 1$, a quartic with a triple point. But the case $\alpha = 2$, $\beta = 2$ gives rise to two species: viz., in general the quartic has only the two double points, and we have then a scroll with two nodal (2-tuple) directrix lines, and without any nodal generator; the section may however have a third double point, and the scroll has then a nodal (double) generator. For the case $\alpha = 3$, $\beta = 1$, the section admits of no further singularity, and we have a quartic scroll with a triple directrix line and a single directrix line.

Art. Next for the quartic scroll $S(1, 1, 4)$. The section is here a quartic curve with an $\alpha(+\beta)$-tuple point, where $\alpha + \beta = 4$; that is, $\alpha = 2$, $\beta = 2$, or else $\alpha = 3$, $\beta = 1$. In the former case the section has a $2(+2)$-tuple point, that is, a double point where the two branches have a common tangent—otherwise, two coincident double points: say the curve has a tacnode; the line of approach is the tangent at the tacnode. We have here a scroll with a twofold double line; there are however two cases: viz., in general the section has, besides the tacnode, no other double point; that is, the scroll has no nodal generator: the section may however have a third double point, and the scroll has then a nodal (double) generator. In the case $\alpha = 3$, $\beta = 1$ the section has a triple point, and the line of approach is the tangent at one of the branches at the triple point; the scroll has a twofold, say a $3(+1)$-tuple directrix line: as the section admits of no further singularity, this is the only case. The foregoing enumeration gives three species of quartic scrolls $S(1, 1, 4)$, and three species of quartic scrolls $S(1, 1, 4)$, together six species, viz. these are as follows:

Quartic Scroll, First Species, $S(1_2, 1_2, 4)$, with two double directrix lines, and without a nodal generator.

Art. Taking $(x = 0, y = 0)$ and $(z = 0, w = 0)$ for the equations of the two directrix lines respectively, the equation of the scroll is

$$(*fx, y)^2\,(z, w)^2 = 0.$$

Quartic Scroll, Second Species, $S'(1_2, 1_2, 4)$, with two double directrix lines and with a double generator.

Art. This is in fact a specialized form of the first species, the difference being that there is a nodal (double) generator. Supposing as before that the equations of the directrix lines are $(x = 0, y = 0)$ and $(z = 0, w = 0)$ respectively; let the equations of the nodal generator be $(x+y = 0, z+w = 0)$; then, observing that for the first species the equation may be written

$(*f x, y)^2\,(z, z+w)^2 = 0$, it is clear that if the terms in z^2 and $z(z+w)$ are divisible by $(x+y)^2$ and $(x+y)$ respectively, the surface will have as a new double line the line $(x+y=0, z+w=0)$, which will be a double generator; and we thus arrive at the equation of the second species of quartic scrolls, viz. this is

$$\{(x+y)^2,\ (x+y)(x,y),\ (x,y)^2 \frown z,\ z+w\}^2 = 0.$$

Quartic Scroll, Third Species, $S(1_3, 1, 4)$, with a triple directrix line and a single directrix line.

Art. Taking $(x=0, y=0)$ for the equations of the triple directrix line, and $(z=0, w=0)$ for the equations of the single directrix line, the equation is

$$(*f x, y)^3\,(z, w)^1 = 0.$$

Quartic Scroll, Fourth Species, $S(1_2^2, 1_2^2, 4)$, with a twofold (2(+2)-tuple) directrix line, and without a nodal generator.

Art. Taking $(x=0, y=0)$ for the equations of the directrix line, $z=0$ for that of a plane section of the scroll, $y=0$ for the equation of a plane through the tangent at the tacnode of the section, and supposing (see ante, No. 22) that the plane through the directrix line and the corresponding point on this line are respectively given by the equations $x = \lambda y$ and $z = \lambda w$, the equation of the scroll is

$$(yw - xz)^2 + (yw - xz)(x,y)^2 + (x,y)^4 = 0.$$

Quartic Scroll, Fifth Species, $S'(1_2^2, 1_2^2, 4)$, with a twofold (2(+2)-tuple) generating line, and with a double generator.

Art. Let the equations of the double generator be $x+y=0$, $z+w=0$; then the line in question must be a double line on the surface represented by the last-mentioned equation, and this will be the case if only the second and third terms contain the factors $(x+y)$ and $(x+y)^2$ respectively. The equation for the fifth species consequently is

$$(yw - xz)^2 + 2(yw - xz)(x+y)(x,y) + (x+y)^2(x,y)^2 = 0.$$

Quartic Scroll, Sixth Species, $S(1_3^2, 1, 4)$, with a twofold (3(+1)-tuple) generating line.

Art. Taking $(x=0, y=0)$ for the equations of the directrix line, $z=0$ for the equation of a plane section, and assuming that the plane $y=0$ passes through the tangent which is the line of approach, and that the plane through the directrix line and the corresponding point on this line are respectively given by the equations $x = \lambda y$ and $z = \lambda w$, the equation of the scroll is

$$(yw - xz)(x, y)^2 + (x, y)^4 = 0.$$

Art. I refrain on the present occasion from a more particular discussion of the foregoing six species of quartic scrolls. I establish two other species, as follows:

Quartic Scroll, Seventh Species, $S(1, 2^2, 2)$, with nodal directrix line, and nodal directrix conic which meet, and with a simple directrix conic which meets the nodal conic in two points.

Art. We see, *a priori*, that the scroll generated as above will be of the order 4, that is, a quartic scroll. In fact using the formula (ante, No. 5),

$$\text{Order} = 2mnp - \alpha m - \beta n - \gamma p,$$

we have here

$$\begin{array}{rll} \text{Nodal conic}, & m = 2, & \alpha = 0, \\ \text{Simple conic}, & n = 2, & \beta = 1, \\ \text{Line}, & p = 1, & \gamma = 2, \end{array}$$

and hence

$$\text{Order} = 8 - 2 - 2 = 4.$$

Art. Take $(x = 0, y = 0)$ for the equations of the directrix line, $z = 0$ for the equation of the plane of the simple conic, $w = 0$ for that of the plane of the nodal conic; since the conics intersect in two points, they lie on a quadric surface, say the surface $U = 0$; the equations of the simple conic thus are $z = 0$, $U = 0$; those of the nodal conic are $w = 0$, $U = 0$. The directrix line $x = 0$, $y = 0$ meets the nodal conic; that is, U must vanish identically for $x = 0$, $y = 0$, $w = 0$; and this will be the case if only the term in z^2 is wanting; that is, we must have

$$U = (a, b, 0, d, f, g, h, l, m, n \between x, y, z, w)^2.$$

But we may in the first instance omit the condition in question, and write

$$U = (a, b, c, d, f, g, h, l, m, n \between x, y, z, w)^2;$$

this would lead to a sextic instead of a quartic scroll.

Art. The equations of a generating line (since it meets the directrix line $x = 0$, $y = 0$) may be taken to be

$$x = \alpha y, \quad z = \beta(y + \gamma w);$$

the condition in order to the intersection of the generating line with the nodal conic is at once found to be

$$a\alpha^2 + 2h\alpha + b + 2\beta(f + g\alpha) + c\beta^2 = 0,$$

and that for its intersection with the simple conic

$$a\alpha^2 + 2h\alpha + b + 2\gamma(m + l\alpha) + d\gamma^2 = 0;$$

and writing the equations of the generating line in the form

$$\frac{x}{y} = \frac{\alpha}{1}, \quad \frac{z}{y} = \frac{\beta y + \beta\gamma w}{y} = \beta + \frac{\beta\gamma w}{y},$$

the elimination of α, β, γ from these four equations gives the required equation of the scroll. Writing for a moment

$$\Theta = g\alpha + f, \quad M = h\alpha + m,$$

we find

$$c\beta^2 + 2\Theta\beta + (a\alpha^2 + 2h\alpha + b) = 0,$$
$$(d\gamma^2 + 2My\gamma + (a\alpha^2 + 2h\alpha + b))\beta = -2(\Theta yz + Mwz)\gamma + \Theta^2 z^2 = 0;$$

or, introducing at this place the condition $c = 0$, the first equation gives β linearly, and we thence obtain

$$\Theta(\Theta y^2 + 2Myw + dw^2) + 2F(\Theta yz + Mwz) + \Theta^2 F^2 z^2 = 0,$$

or, what is the same thing,

$$(\Theta y + Fz)^2 + 2Mw(\Theta y + Fz) + \Theta dw^2 = 0;$$

whence, observing that we have

$$\Theta = \frac{gxy + fyy}{y^2} = \frac{gx + fy}{y}, \quad M = \frac{hxy + myy}{y^2} = \frac{hx + my}{y},$$

the equation of the scroll is

$$\begin{aligned}(ax^2 + 2hxy + by^2 + 2gzx + 2fyz)^2 \\ + 2(ax^2 + 2hxy + by^2 + 2gzx + 2fyz)(lx + my)w \\ + (ax^2 + 2hxy + by^2)dw^2 = 0.\end{aligned}$$

We see from the equation that the surface contains the line $(x = 0, y = 0)$ as a double line, the conic

$$w = 0, \quad ax^2 + 2hxy + by^2 + 2gzx + 2fyz = 0$$

as a double curve, also the conic

$$z = 0, \quad ax^2 + 2hxy + by^2 + 2gxw + 2myw + dw^2 = 0$$

as a simple curve on the surface, the complete intersection by the plane $z = 0$ being in fact the last-mentioned conic, and the pair of lines

$$z = 0, \quad ax^2 + 2hxy + by^2 = 0.$$

Quartic Scroll, Eighth Species, $S(1,3^2)$, with a directrix line, and a directrix skew cubic met twice by each generating line.

Art. We see, *a priori*, that the scroll is of the order 4, that is, a quartic scroll; in fact for the quartic scroll $S(1,m^2)$ the order is $= [m]^2 + M$ (first memoir, p. 457 [ante p. 172]), and we have here $m = 3$, $M = 4-6\,[m]p = 1-3 = -2$; that is, order $= 6-2 = 4$.

Art. The equations of the cubic curve may be taken to be

$$\begin{vmatrix} x & y & z \\ y & z & w \end{vmatrix} = 0,$$

or, what is the same thing,

$$xz - y^2 = 0, \quad xw - yz = 0, \quad yw - z^2 = 0;$$

those of the directrix line may be represented by

$$\alpha x + \beta y + \gamma z + \delta w = 0, \quad \alpha' x + \beta' y + \gamma' z + \delta' w = 0;$$

or, what is the same thing, if

$$\begin{aligned} \beta\gamma' - \beta'\gamma &= a, \quad & \alpha\delta' - \alpha'\delta &= f, \\ \gamma\alpha' - \gamma'\alpha &= b, \quad & \beta\delta' - \beta'\delta &= g, \\ \alpha\beta' - \alpha'\beta &= c, \quad & \gamma\delta' - \gamma'\delta &= h, \end{aligned}$$

(and therefore identically $af + bg + ch = 0$), the line is defined by means of its "six coordinates" (a, b, c, f, g, h).

Art. The equations of the cubic curve are satisfied by writing therein

$$x : y : z : w = 1 : \theta : \theta^2 : \theta^3,$$

and therefore the coordinates of any two points on the curve may be represented by $(1, \theta, \theta^2, \theta^3)$ and $(1, \phi, \phi^2, \phi^3)$; hence, if x, y, z, w are the coordinates of a point in the line joining the last mentioned two points, we have

$$x : y : z : w = l + m : l\theta + m\phi : l\theta^2 + m\phi^2 : l\theta^3 + m\phi^3,$$

which equations, treating therein l, m as indeterminate parameters, give the equations of the line in question. And putting moreover

$$p = yw - z^2, \quad q = yz - xw, \quad r = xz - y^2,$$

we have identically

$$p : q : r = \theta\phi : -(\theta + \phi) : 1.$$

Art. In order that the line in question may meet the directrix line, we must have

$$\begin{aligned} l(\alpha + \beta\theta + \gamma\theta^2 + \delta\theta^3) + m(\alpha + \beta\phi + \gamma\phi^2 + \delta\phi^3) &= 0, \\ l(\alpha' + \beta'\theta + \gamma'\theta^2 + \delta'\theta^3) + m(\alpha' + \beta'\phi + \gamma'\phi^2 + \delta'\phi^3) &= 0; \end{aligned}$$

that is, eliminating l and m, we must have

$$\begin{vmatrix} \alpha+\beta\theta+\gamma\theta^2+\delta\theta^3 & \alpha+\beta\phi+\gamma\phi^2+\delta\phi^3 \\ \alpha'+\beta'\theta+\gamma'\theta^2+\delta'\theta^3 & \alpha'+\beta'\phi+\gamma'\phi^2+\delta'\phi^3 \end{vmatrix} = 0,$$

or, developing,

$$\begin{gathered}(\alpha\beta'-\alpha'\beta)(\phi-\theta)+(\alpha\gamma'-\alpha'\gamma)(\phi^2-\theta^2)+(\alpha\delta'-\alpha'\delta)(\phi^3-\theta^3)\\ +(\beta\gamma'-\beta'\gamma)(\theta\phi^2-\phi\theta^2)+(\beta\delta'-\beta'\delta)(\theta\phi^3-\theta^3\phi)+(\gamma\delta'-\gamma'\delta)(\theta^2\phi^3-\theta^3\phi^2)=0;\end{gathered}$$

the several terms in (θ,ϕ), each divided by $\phi-\theta$, give respectively

$$1,\quad \phi+\theta,\quad (\phi+\theta)^2-\phi\theta,\quad \theta\phi,\quad \theta\phi(\phi+\theta),\quad \theta^2\phi^2,$$

which are equal to

$$(r^2,\ -qr,\ q^2-pr,\ pr,\ -pq,\ p^2);$$

hence replacing also $\alpha\beta'-\alpha'\beta$, &c. by their values c, &c., we find

$$(c,-b,f,a,g,hfr^2,-qr,q^2-pr,pr,-pq,p^2)=0,$$

or, what is the same thing,

$$(a,b,c,f,g,hfp,q,r)^2=0,$$

where the coefficients (a,b,c,f,g,h) satisfy the relation $af+bg+ch=0$; p, q, r stand respectively for $yw-z^2$, $yz-xw$, $xz-y^2$.

Writing for greater convenience

$$(a,b,c,f,g,h)=(b+2g,\ 2f,\ c,\ b,\ -2h,\ a),$$

or, what is the same thing,

$$(a,b,c,f,g,h)=(b+2g,\ 2f,\ c,\ b,\ -2h,\ a),$$

then we have

$$af+bg+ch=ac+b^2+2bg-4fh=0;$$

and hence finally we have for the equation of the scroll $S(1,3^2)$,

$$(a,b,c,f,g,hfyw-z^2,\ yz-xw,\ xz-y^2)^2=0,$$

where the coefficients satisfy the relation

$$ac+b^2+2bg-4fh=0.$$

The equations of the directrix cubic are of course

$$yw-z^2=0,\quad yz-xw=0,\quad xz-y^2=0;$$

and the directrix line is given by its six coordinates,

$$(b+2g,\ 2f,\ c,\ b,\ -2h,\ a).$$

On the general Theory of Scrolls. Article Nos. 51 to 53.

Art. I annex in conclusion the following considerations on the general theory of scrolls. Consider a scroll of the nth order; the intersection by an arbitrary plane, say the plane $w = 0$, is a curve of the nth order $(*\!\!\not{}x, y, z)^n = 0$; any point $(x, y, z, 0)$ where (x, y, z) satisfy the foregoing equation, is the foot of a generating line; and we may imagine this generating line determined by means of the coordinates (X, Y, Z, W), given functions of (x, y, z) of a point on the line. This being so, the "six coordinates," say (p, q, r, s, t, u), of the line are

$$\begin{array}{cccc} X, & Y, & Z, & W \\ x, & y, & z, & 0 \end{array}$$

viz.

$$\begin{aligned} p &= Yz - Zy, & s &= -Wx, \\ q &= Zx - Xz, & t &= -Wy, \\ r &= Xy - Yx, & u &= -Wz; \end{aligned}$$

or, writing for greater convenience $-\nu$ in the place of W, the six coordinates of the line are $p, q, r, \nu x, \nu y, \nu z$, where p, q, r are functions of (x, y, z), connected by the relation $px + qy + rz = 0$; and ν is also a function of (x, y, z).

Art. Consider the intersection of the surface by an arbitrary line, the six coordinates whereof are (A, B, C, F, G, H); then for the generating lines which meet this line we have

$$\nu(Ax + By + Cz) + Fp + Gq + Hr = 0,$$

and this equation, together with the equation $(*\!\!\not{}x, y, z)^n = 0$, determines (x, y, z), the coordinates of the foot of a generating line which meets the arbitrary line (A, B, C, F, G, H). Since the order of the scroll is $= n$, the number of such generating lines should be $= n$, that is, there should be n relevant intersections of the two curves,

$$\begin{aligned} \nu(Ax + By + Cz) + Fp + Gq + Hr &= 0, \\ (*\!\!\not{}x, y, z)^n &= 0; \end{aligned}$$

but if $(p, q, r, \nu x, \nu y, \nu z)$ are each of the order k, the number of actual intersections is $= kn$, which is too many by $(k - 1)n$.

Art. Suppose that the curves

$$p = 0, \quad q = 0, \quad r = 0, \quad \nu x = 0, \quad \nu y = 0, \quad \nu z = 0,$$

or say the curves

$$p = 0, \quad q = 0, \quad r = 0, \quad \nu = 0$$

have in common θ intersections, and let these be points of the multiplicities $\alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_\theta$ on the curve $(*\!\!\not{}x, y, z)^n = 0$ (viz. according as the curve does not pass through any one of the intersections in question, or passes once, twice, &c. through such intersection, we have for that intersection $\alpha_i = 0$, 1, 2, &c., as the case may be, and so for the other intersections); then the kn points of intersection include the $\alpha_1 + \alpha_2 + \ldots + \alpha_\theta$, or say the $\Sigma\alpha$ intersections; but these, being independent of the line (A, B, C, F, G, H) under consideration, are irrelevant points, and the number of relevant points of intersection is $kn - \Sigma\alpha$; that is, if we have $\Sigma\alpha = (k-1)n$, then the scroll in question, viz. the scroll generated by a line which meets the plane $w = 0$ in the curve $(*f x, y, z)^n = 0$, and which has for its six coordinates $(p, q, r, \nu x, \nu y, \nu z)$, will be a scroll of the nth order.

0.2 [341] On the Sextactic Points of a Plane Curve

[From the Philosophical Transactions of the Royal Society of London, vol. CLV. (for the year 1865), pp. 545—578.
Received November 5,—Read December 22, 1864.]

It is, in my memoir "On the Conic of Five-pointic Contact at any point of a Plane Curve," Phil. Trans. vol. CXLIX. (1859), pp. 371—400, [261], remarked that as in a plane curve there are certain singular points, viz. the points of inflexion, where three consecutive points lie in a line, so there are singular points where six consecutive points of the curve lie in a conic; and such a singular point is there termed a "sextactic point." The memoir in question (here cited as "former memoir") contains the theory of the sextactic points of a cubic curve; but it is only recently that I have succeeded in establishing the theory for a curve of the order m. The result arrived at is that the number of sextactic points is $= m(12m-27)$, the points in question being the intersections of the curve m with a curve of the order $12m-27$, the equation of which is

$$\begin{aligned}(12m^2-54m+57)&H\operatorname{Jac}(U,H,\Phi)\\ &+(m-2)(12m-27)H^2\operatorname{Jac}(U,H,\Omega_\Phi)\\ &\qquad+40(m-2)^2\operatorname{Jac}(U,H,\Psi)=0,\end{aligned}$$

where $U=0$ is the equation of the given curve of the order m, H is the Hessian or determinant formed with the second differential coefficients (a,h,c,f,g,h) of U, and, $(\mathfrak{A},\mathfrak{B},\mathfrak{C},\mathfrak{F},\mathfrak{G},\mathfrak{H})$ being the inverse coefficients ($\mathfrak{A}=bc-f^2$, &c.), then

$$\Phi=(\mathfrak{A},\mathfrak{B},\mathfrak{C},\mathfrak{F},\mathfrak{G},\mathfrak{H}\between a,b,c,2f,2g,2h)^2,$$

and Jac denotes the Jacobian or functional determinant, viz.

$$\operatorname{Jac}(U,H,\Phi)=\begin{vmatrix}\partial_xU & \partial_yU & \partial_zU\\ \partial_xH & \partial_yH & \partial_zH\\ \partial_x\Phi & \partial_y\Phi & \partial_z\Phi\end{vmatrix}$$

and $\operatorname{Jac}(U,H,\Omega)$ would of course denote the like derivative of (U,H,Ω); the subscripts (g,p) of Ω denote restrictions in regard to the differentiation of this function, viz. treating Ω as a function of U and H,

$$\Omega=(\mathfrak{A},\mathfrak{B},\mathfrak{C},\mathfrak{F},\mathfrak{G},\mathfrak{H}\between a',b',c',2f',2g',2h')$$

if (a',b',c',f',g',h') are the second differential coefficients of H, then we have

$$\partial_x\Omega=(a,\mathfrak{A},\ldots\between a',\ldots)\quad(=\partial_x\Omega_g),$$

$$+(\mathfrak{A},\ldots f\partial_x a',\ldots)\quad (=\partial_x\Omega_p);$$

viz. in $\partial_x\Omega_g$ we consider as exempt from differentiation (a',b',c',f',g',h') which depend upon H, and in $\partial_x\Omega_p$ we consider as exempt from differentiation $(\mathfrak{A},\mathfrak{B},\mathfrak{C},\mathfrak{F},\mathfrak{G},\mathfrak{H})$ which depend upon U. We have similarly

$$\partial_y\Omega=\partial_y\Omega_g+\partial_y\Omega_p,\quad \partial_z\Omega=\partial_z\Omega_g+\partial_z\Omega_p;$$

and in like manner

$$\mathrm{Jac}(U,H,\Omega)=\mathrm{Jac}(U,H,\Omega_g)+\mathrm{Jac}(U,H,\Omega_p),$$

which explains the signification of the notations $\mathrm{Jac}(U,H,\Omega_g)$, $\mathrm{Jac}(U,H,\Omega_p)$.

The condition for a sextactic point is in the first instance obtained in a form involving the arbitrary coefficients (λ,μ,ν); viz. we have an equation of the order 5 in (λ,μ,ν) and of the order $12m-22$ in the coordinates (x,y,z). But writing $\xi=\lambda x+\mu y+\nu z$, by successive transformations we throw out the factors ξ^2, ξ, ξ, ξ, thus arriving at a result independent of (λ,μ,ν); viz. this is the before-mentioned equation of the order $12m-27$. The difficulty of the investigation consists in obtaining the transformations by means of which the equation in its original form is thus divested of these irrelevant factors.

Articles Nos. 1 to 6.—Investigation of the Condition for a Sextactic Point.

Art. Following the course of investigation in my former memoir, I take (X,Y,Z) as current coordinates, and I write

$$\Upsilon=(*f X,Y,Z)^m=0$$

for the equation of the given curve; (x,y,z) are the coordinates of a particular point on the given curve, viz. the sextactic point; and U, $=(*f x,y,z)^m$, is what Υ becomes when (x,y,z) are written in place of (X,Y,Z): we have thus $U=0$ as a condition satisfied by the coordinates of the point in question.

Art. Writing for shortness

$$\mathfrak{D}_1U=(X\partial_x+Y\partial_y+Z\partial_z)U,$$

and taking $\Upsilon_1=aX+bY+cZ=0$ for the equation of an arbitrary line, the equation

$$\Upsilon_1-\mathfrak{D}_1^2U=0$$

is that of a conic having an ordinary (two-pointic) contact with the curve at the point (x,y,z); and the coefficients of Υ_1 are in the former memoir determined so that the contact may be a five-pointic one; the value obtained for Υ_1 is

$$\Upsilon_1 = \frac{\Phi}{H}\mathfrak{D}_1^2 H + A\mathfrak{D}_1 U,$$

where

$$A = \frac{1}{m-1} \cdot \frac{-3\Phi H + 4\Psi}{H^2}.$$

Art. This result was obtained by considering the coordinates of a point of the curve as functions of a single arbitrary parameter, and taking

$$x + dx + \tfrac{1}{2}d^2x + \tfrac{1}{6}d^3x + \tfrac{1}{24}d^4x, \quad y + \&\text{c.}, \quad z + \&\text{c.}$$

for the coordinates of a point consecutive to (x, y, z); for the present purpose we must go a step further, and write for the coordinates

$$\begin{gathered} x + dx + \tfrac{1}{2}d^2x + \tfrac{1}{6}d^3x + \tfrac{1}{24}d^4x + \tfrac{1}{120}d^5x, \\ y + dy + \tfrac{1}{2}d^2y + \tfrac{1}{6}d^3y + \tfrac{1}{24}d^4y + \tfrac{1}{120}d^5y, \\ z + dz + \tfrac{1}{2}d^2z + \tfrac{1}{6}d^3z + \tfrac{1}{24}d^4z + \tfrac{1}{120}d^5z. \end{gathered}$$

Art. Hence if

$$\mathfrak{d}_1 = dx\,\partial_x + dy\,\partial_y + dz\,\partial_z, \quad \mathfrak{d}_2 = d^2x\,\partial_x + d^2y\,\partial_y + d^2z\,\partial_z, \quad \&\text{c.},$$

we have, in addition to the equations

$$\begin{aligned} U &= 0, \\ \mathfrak{d}_1 U &= 0, \\ (\mathfrak{d}_1^2 + 2\mathfrak{d}_2)U &= 0, \\ (\mathfrak{d}_1^3 + 3\mathfrak{d}_1\mathfrak{d}_2 + \mathfrak{d}_3)U &= 0, \\ (\mathfrak{d}_1^4 + 6\mathfrak{d}_1^2\mathfrak{d}_2 + 4\mathfrak{d}_1\mathfrak{d}_3 + 3\mathfrak{d}_2^2 + \mathfrak{d}_4)U &= 0, \end{aligned}$$

of my former memoir, the new equation

$$(\mathfrak{d}_1^5 + 10\mathfrak{d}_1^3\mathfrak{d}_2 + 10\mathfrak{d}_1^2\mathfrak{d}_3 + 15\mathfrak{d}_1\mathfrak{d}_2^2 + 5\mathfrak{d}_1\mathfrak{d}_4 + 10\mathfrak{d}_2\mathfrak{d}_3 + \mathfrak{d}_5)U = 0,$$

and in addition to the equations, $(P = ax + by + cz)$,

$$-\tfrac{1}{8}\{[(m-1)\mathfrak{d}_1^4 + 3(m-2)\mathfrak{d}_1^2\mathfrak{d}_2]U + P\cdot\tfrac{1}{6}(\mathfrak{d}_1^3 + 3\mathfrak{d}_1\mathfrak{d}_2)U + \mathfrak{d}_1 P\cdot\tfrac{1}{2}(\mathfrak{d}_1^2 + 3\mathfrak{d}_1\mathfrak{d}_2)U + \mathfrak{d}_1^2 P\cdot\tfrac{1}{2}\mathfrak{d}_1^2 U\} = 0,$$

$$\begin{aligned} &-\tfrac{1}{120}\{[(m-1)(\mathfrak{d}_1^5 + 10\mathfrak{d}_1^3\mathfrak{d}_2 + 10\mathfrak{d}_1^2\mathfrak{d}_3 + 15\mathfrak{d}_1\mathfrak{d}_2^2) + (m-2)(5\mathfrak{d}_1\mathfrak{d}_4 + 10\mathfrak{d}_2\mathfrak{d}_3)]U \\ &+ P\cdot\tfrac{1}{24}(\mathfrak{d}_1^4 + 6\mathfrak{d}_1^2\mathfrak{d}_2 + 4\mathfrak{d}_1\mathfrak{d}_3 + 3\mathfrak{d}_2^2)U + \mathfrak{d}_1 P\cdot\tfrac{1}{6}(\mathfrak{d}_1^3 + 3\mathfrak{d}_1\mathfrak{d}_2)U + \mathfrak{d}_1^2 P\cdot\tfrac{1}{2}\mathfrak{d}_1^2 U\} = 0, \end{aligned}$$

giving in the first instance

$$P = \frac{2(m-2)}{m-1} \cdot \frac{\Phi}{H},$$

and leading ultimately to the before-mentioned value of Υ_1, we have the new equation

$$\begin{aligned}-\tfrac{1}{120}\{[(m-1)(\eth_1^5+10\eth_1^3\eth_2+10\eth_1^2\eth_3+15\eth_1\eth_2^2)+(m-2)(5\eth_1\eth_4+10\eth_2\eth_3)]U\\ +P\cdot\tfrac{1}{24}(\eth_1^5+10\eth_1^3\eth_2+10\eth_1^2\eth_3+15\eth_1\eth_2^2+5\eth_1\eth_4+10\eth_2\eth_3)U\\ +\eth_1P\cdot\tfrac{1}{6}(\eth_1^4+6\eth_1^2\eth_2+4\eth_1\eth_3+3\eth_2^2)U\\ +\eth_1^2P\cdot\tfrac{1}{2}(\eth_1^3+3\eth_1\eth_2)U+\eth_1^3P\cdot\tfrac{1}{6}\eth_1^2U\}=0.\end{aligned}$$

Art. This may be written in the form

$$\begin{aligned}2[(m-1)(\eth_1^5+10\eth_1^3\eth_2+10\eth_1^2\eth_3+15\eth_1\eth_2^2)+(m-2)(5\eth_1\eth_4+10\eth_2\eth_3)]U\\ +P(\eth_1^5+10\eth_1^3\eth_2+10\eth_1^2\eth_3+15\eth_1\eth_2^2+5\eth_1\eth_4+10\eth_2\eth_3)U\\ +5\eth_1P(\eth_1^4+6\eth_1^2\eth_2+4\eth_1\eth_3+3\eth_2^2)U\\ +10\eth_1^2P(\eth_1^3+3\eth_1\eth_2)U+10\eth_1^3P\cdot\eth_1^2U=0;\end{aligned}$$

or putting for P its value, $=2(m-2)$, the equation becomes

$$\begin{aligned}2(\eth_1^5+10\eth_1^3\eth_2+10\eth_1^2\eth_3+15\eth_1\eth_2^2)U\\ +5\eth_1P(\eth_1^4+6\eth_1^2\eth_2+4\eth_1\eth_3+3\eth_2^2)U\\ +10\eth_1^2P(\eth_1^3+3\eth_1\eth_2)U+10\eth_1^3P\cdot\eth_1^2U=0;\end{aligned}$$

or as this may also be written,

$$\begin{aligned}2(\eth_1^5+10\eth_1^3\eth_2+10\eth_1^2\eth_3+15\eth_1\eth_2^2)U\\ +5\eth_1P\cdot\eth_4U+10\eth_1^2P\cdot\eth_3U+10\eth_1^3P\cdot\eth_2U=0.\end{aligned}$$

Art. But the equation

$$\Upsilon_1=\frac{\Phi}{H}\mathfrak{D}_1^2H+A\mathfrak{D}_1U,$$

which is an identity in regard to (X,Y,Z), gives

$$\begin{aligned}\eth_1\Upsilon_1&=\frac{\Phi}{H}\eth_1^2H+A\eth_1U,\\ \eth_1^2\Upsilon_1&=\frac{\Phi}{H}\eth_1^3H+A\eth_1^2U,\\ \eth_1^3\Upsilon_1&=\frac{\Phi}{H}\eth_1^4H+A\eth_1^3U;\end{aligned}$$

and substituting these values, the foregoing equation becomes

$$\begin{aligned}2(\eth_1^5+10\eth_1^3\eth_2+10\eth_1^2\eth_3+15\eth_1\eth_2^2)U\\ +\{5\eth_1U\eth_4H+10\eth_1^2U\eth_3H+10\eth_1^3U\eth_2H\}\frac{\Phi}{H}+A\cdot20\eth_1^3U\eth_1^2U=0;\end{aligned}$$

or putting for A its value, $= \frac{1}{m-1} \cdot \frac{-3\Phi H + 4\Psi}{H^2}$, and multiplying by $\frac{1}{2}H^2$ this is

$$\begin{aligned} &9H^2(\mathfrak{d}_1^5 + 10\mathfrak{d}_1^3\mathfrak{d}_2 + 10\mathfrak{d}_1^2\mathfrak{d}_3 + 15\mathfrak{d}_1\mathfrak{d}_2^2)U \\ &\qquad + 15H(\mathfrak{d}_1 U \mathfrak{d}_4 H + 2\mathfrak{d}_1^2 U \mathfrak{d}_3 H + 2\mathfrak{d}_1^3 U \mathfrak{d}_2 H)\Phi \\ &\qquad\qquad + (-3\Phi H + 4\Psi) \cdot 10\mathfrak{d}_1^3 U \mathfrak{d}_1^2 U = 0, \end{aligned}$$

which is, in its original or unreduced form, the condition for a sextactic point.

Article Nos. 7 and 8.—Notations and Remarks.

Art. Writing, as in my former memoir, A, B, C for the first differential coefficients of U, we have $Bv - C\mu$, $C\lambda - A\nu$, $A\mu - B\lambda$ for the values of dx, dy, dz, and instead of the symbol D used in my former memoir, I use indifferently the original symbol $\mathfrak{d}_1$ or write instead thereof $\mathfrak{d}$ to denote the resulting value

$$\mathfrak{d}\ (= \mathfrak{d}_1) = (Bv - C\mu)\partial_x + (C\lambda - A\nu)\partial_y + (A\mu - B\lambda)\partial_z$$

and I remark here that for any function whatever Ω, we have

$$\mathfrak{d}\Omega = \begin{vmatrix} A & B & C \\ \lambda & \mu & \nu \\ \partial_x\Omega & \partial_y\Omega & \partial_z\Omega \end{vmatrix} = \mathrm{Jac}(U, \xi, \Omega),$$

where $\xi = \lambda x + \mu y + \nu z$. I write, as in the former memoir,

$$\Phi = (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} f \lambda, \mu, \nu)^2,$$

and also

$$\Psi = (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} f \partial_x, \partial_y, \partial_z)^2\Phi,$$

which new symbol Ψ serves to express the functions Π, D, occurring in the former memoir; viz. we have $\Pi = 2\nabla\Phi$, $D = 2\nabla\Psi$, so that the symbols Π, D are not any longer required.

Art. I remark that the symbols $\mathfrak{d}$, ∇ are each of them a linear function of $(\partial_x, \partial_y, \partial_z)$, with coefficients which are functions of the variables (x, y, z), and this being so, that for any function Ω whatever, we have

$$\mathfrak{d}(\nabla\Omega) = (\mathfrak{d}.\nabla)\Omega + \mathfrak{d}\nabla\Omega,$$

viz. in $\mathfrak{d}(\nabla\Omega)$ we operate with ∇ on Ω, thereby obtaining $\nabla\Omega$, and then with $\mathfrak{d}$ on $\nabla\Omega$; in $(\mathfrak{d}.\nabla)\Omega$ we operate with $\mathfrak{d}$ upon ∇ in so far as ∇ is a function of (x, y, z), thus obtaining a new operating symbol $\mathfrak{d}.\nabla$, a linear function of $(\partial_x, \partial_y, \partial_z)$, and then operate with $\mathfrak{d}.\nabla$ upon Ω; and lastly, in $\mathfrak{d}\nabla\Omega$, we simply multiply together $\mathfrak{d}$ and ∇, thus obtaining a new operating symbol $\mathfrak{d}\nabla$ of the form $(\partial_x, \partial_y, \partial_z)^2$, and then operate therewith on Ω; it is clear that, as regards the last-mentioned mode of combination, the symbols $\mathfrak{d}$ and ∇ are convertible, or $\mathfrak{d}\nabla = \nabla\mathfrak{d}$, that is, $\mathfrak{d}\nabla\Omega = \nabla\mathfrak{d}\Omega$.

It is to be observed throughout the memoir that the point (.) is used (as above in $\mathfrak{d}.\nabla$) when an operation is performed upon a symbol of operation as operand; the mere apposition of two or more symbols of operation (as above in $\mathfrak{d}\nabla$) denotes that the symbols of operation are simply multiplied together; and when $\mathfrak{d}\nabla$ is followed by a letter Ω denoting not a symbol of operation, but a mere function of the coordinates, that is in an expression such as $\mathfrak{d}\nabla\Omega$, the resulting operation $\mathfrak{d}\nabla$ is performed upon Ω as operand; if instead of the single letter Ω we have a compound symbol such as $H\Omega$ or $\nabla\Psi\partial_x$, so that the expression is $\mathfrak{d}H\Omega$, $\mathfrak{d}H\nabla\Psi$, $\mathfrak{d}\nabla H\Omega$ or $\mathfrak{d}\nabla\nabla\Psi\partial_x$, then it is to be understood that it is merely the immediately following function H which is operated upon by $\mathfrak{d}$ or $\mathfrak{d}\nabla$; in the few instances where any ambiguity might arise a special explanation is given.

Article Nos. 9 to 11.—First transformation.

Art. We have, assuming always $U = 0$, the following formulæ (see post, Article Nos. 31 to 33):

$$
\begin{aligned}
&(\mathfrak{d}_1^5 + 10\mathfrak{d}_1^3\mathfrak{d}_2 + 10\mathfrak{d}_1^2\mathfrak{d}_3 + 15\mathfrak{d}_1\mathfrak{d}_2^2)U \\
&\quad = \frac{\Phi}{(m-1)^4}\{(27m^2 - 96m + 81)H\mathfrak{d}_1^4 + (17m^2 - 56m + 51)\Phi\mathfrak{d}_1 H\} \\
&\qquad + \frac{\Psi}{(m-1)^4}\{(-14m + 22)(\mathfrak{d}.\nabla)H + (-10m + 18)\mathfrak{d}\nabla H\} \\
&\qquad\qquad + \frac{\Psi}{(m-1)^4}\{9\Omega\};
\end{aligned}
$$

$$
\begin{aligned}
&\mathfrak{d}_1 U\mathfrak{d}_4 H + 2\mathfrak{d}_1^2 U\mathfrak{d}_3 H + 2\mathfrak{d}_1^3 U\mathfrak{d}_2 H \\
&\quad = \frac{\Phi}{(m-1)^3}\{(-6m^2 + 18m - 12)H\mathfrak{d}_1^4 + (-17m^2 + 60m - 55)\Phi\mathfrak{d}_1 H\} \\
&\qquad + \frac{\Psi}{(m-1)^3}\{(2m - 2)H(\mathfrak{d}.\nabla)H + (8m - 16)\mathfrak{d}\nabla H \cdot H\} \\
&\qquad\qquad + \frac{\Psi}{(m-1)^3}\{9\Omega\}.
\end{aligned}
$$

Art. And by means of these the condition becomes

$$
\begin{aligned}
0 = \frac{\Phi^2}{(m-1)^4}\{(153m^2-594m+549)H\mathfrak{d}_1^4+(-102m^2+396m+366)\Phi\mathfrak{d}_1 H\}\\
+\frac{\Phi\Psi}{(m-1)^4}\{(-96m+168)H(\mathfrak{d}.\nabla)H+(-90m+162)H\mathfrak{d}\nabla H+(120m-240)\mathfrak{d}\nabla H\cdot H\}\\
+\frac{\Psi^2}{(m-1)^4}\{9\mathfrak{d}\nabla\Omega - 45H\Omega\mathfrak{d}_1 H + 40\Phi\mathfrak{d}_1\Psi\},
\end{aligned}
$$

being, as already remarked, of the degree 5 in the arbitrary coefficients (λ, μ, ν), and of the order $12m-22$ in the coordinates (x, y, z).

Art. But throwing out the factor Φ^2 and observing that in the first line the quadric functions of m are each a numerical multiple of $51m^2-198m+183$, the condition becomes

$$
\begin{aligned}
0 = (51m^2-198m+183)H^2(3H\mathfrak{d}_1^4 - 2\Phi\mathfrak{d}_1 H)\\
+\Phi\Psi\{(-96m+168)H(\mathfrak{d}.\nabla)H+(-90m+162)H\mathfrak{d}\nabla H+(120m-240)\mathfrak{d}\nabla H\cdot H\}\\
+\Psi^2\{9\mathfrak{d}\nabla\Omega - 45H\Omega\mathfrak{d}_1 H + 40\Phi\mathfrak{d}_1\Psi\}.
\end{aligned}
$$

Article Nos. 12 and 13.—Second transformation.

Art. We effect this by means of the formula

$$
(m-2)(3H\mathfrak{d}_1\Phi - 2\Phi\mathfrak{d}_1 H) = \frac{\Psi}{m-1}\mathrm{Jac}(U, \Phi, H),^7 \qquad \text{(J)}
$$

for substituting this value of $(3H\mathfrak{d}_1\Phi - 2\Phi\mathfrak{d}_1 H)$ the equation becomes divisible by Φ; and dividing out accordingly, the condition becomes

$$
\begin{aligned}
\frac{51m^2-198m+183}{m-2}\cdot\frac{1}{(m-1)^3}H^2\mathrm{Jac}(U, \Phi, H)\\
+(-96m+168)\Phi(\mathfrak{d}.\nabla)H+(-90m+162)\Phi\mathfrak{d}\nabla H+(120m-240)\mathfrak{d}\nabla H\cdot H\\
+\Psi(9\mathfrak{d}\nabla\Omega - 45H\Omega\mathfrak{d}_1 H + 40\Phi\mathfrak{d}_1\Psi) = 0.
\end{aligned}
$$

Art. We have (see post, Article Nos. 36 to 40)

$$
\mathrm{Jac}(U, \Phi, H) = -(\mathfrak{d}.\nabla)H;
$$

and introducing also $\mathfrak{d}.\nabla H$ in place of $\mathfrak{d}\nabla H$ by means of the formula

$$
\mathfrak{d}\nabla H = \mathfrak{d}.\nabla H + \frac{1}{m-1}\mathrm{Jac}(U, \nabla H, H),
$$

---

[7] (J) here and elsewhere refers to the Jacobian Formula, see post, Article Nos. 34 and 35.

the condition becomes

$$\begin{aligned}\frac{(51m^2-198m+183)(4m-9)}{(m-2)(m-1)}&H\operatorname{Jac}(U,\nabla,H)\\&+(-90m+162)\Phi\mathfrak{d}(\nabla H)+120(m-2)H\mathfrak{d}\nabla H\\&\qquad+\Psi(9H^2\mathfrak{d}\Omega-45H\Omega\mathfrak{d}_1H+40\Phi\mathfrak{d}_1\Psi)=0,\end{aligned}$$

or, as this may be written,

$$\begin{aligned}(45m^2-180m+171)H\Phi(\mathfrak{d}.\nabla)H+(-90m+162)(m-2)H\Phi\mathfrak{d}(\nabla H)+120(m-2)^2H\mathfrak{d}\nabla H\\+(m-2)\Psi\{9H^2\mathfrak{d}\Omega-45H\Omega\mathfrak{d}_1H+40\Phi\mathfrak{d}_1\Psi\}=0.\end{aligned}$$

Article Nos. 14 to 17.—Third transformation.

Art. We have the following formulæ,

$$\frac{\Psi}{m-1}\operatorname{Jac}(U,\nabla H,\Phi)-(5m-11)\Phi H(\mathfrak{d}.\nabla)H+(3m-6)H\Phi(\mathfrak{d}.\nabla)H=0, \tag{J}$$

$$\frac{\Psi}{m-1}\operatorname{Jac}(U,\nabla,H)H-(7m-14)\mathfrak{d}\nabla H\cdot H+(3m-6)H(\mathfrak{d}.\nabla)\nabla H=0, \tag{J}$$

in the latter of which, treating ∇ as a function of the coordinates, we first form the symbol $\operatorname{Jac}(U,\nabla,H)$, and then operating therewith on H, we have $\operatorname{Jac}(U,\nabla,H)H$; these give

$$\begin{aligned}H(\mathfrak{d}.\nabla)\nabla H&=\frac{5m-11}{3m-6}\Phi H(\mathfrak{d}.\nabla)H-\frac{1}{(m-1)(3m-6)}\Psi\operatorname{Jac}(U,\nabla H,\Phi),\\H(\mathfrak{d}.\nabla)\nabla H&=\frac{7m-14}{3m-6}\mathfrak{d}\nabla H\cdot H+\frac{1}{(m-1)(3m-6)}\Psi\operatorname{Jac}(U,\nabla,H)H;\end{aligned}$$

and substituting these values, the resulting coefficient of $H\mathfrak{d}\nabla H$ is

$$\begin{aligned}\frac{(45m^2-180m+171)(4m-9)}{(m-2)(m-1)(3m-6)}&(-96m+168)\\&+\frac{(-90m+162)(m-2)(5m-11)}{(m-1)(3m-6)}+\frac{120(m-2)^2(7m-14)}{(m-1)(3m-6)},\end{aligned}$$

which is $=0$.

Art. Hence the condition will contain the factor Ψ, and throwing out this, and also the constant factor $m-2$, it becomes

$$\begin{aligned}(-15m^2+60m-57)&H\operatorname{Jac}(U,\nabla,H)H\\&+(30m-54)(m-2)H\operatorname{Jac}(U,\nabla H,H)\\&\qquad+(m-2)^2\{9H^2\mathfrak{d}\Omega-45H\Omega\mathfrak{d}_1H+40\Phi\mathfrak{d}_1\Psi\}=0.\end{aligned}$$

Art. We have

$$\mathfrak{d}_x(\nabla H) = (\mathfrak{d}_x.\nabla)H + \mathfrak{d}_x\nabla H,$$

viz. in $(\mathfrak{d}_x.\nabla)H$, treating ∇ as a function of (x, y, z) we operate upon it with $\mathfrak{d}_x$ to obtain the new symbol $\mathfrak{d}_x.\nabla$, and with this we operate on H; in $\mathfrak{d}_x\nabla$ we simply multiply together the symbols $\mathfrak{d}_x$ and ∇, giving a new symbol of the form $(\partial_x^2, \partial_x\partial_y, \partial_x\partial_z)$ which then operates on H. We have the like values of $\mathfrak{d}_y(\nabla H)$ and $\mathfrak{d}_z(\nabla H)$; and thence also

$$\mathrm{Jac}(U, \nabla H, H) = \mathrm{Jac}(U, \nabla, H)H + \mathrm{Jac}(U, \nabla H, H),$$

viz. in the determinant $\mathrm{Jac}(U, \nabla, H)$ the second line corresponding to ∇ is $\mathfrak{d}_x.\nabla$, $\mathfrak{d}_y.\nabla$, $\mathfrak{d}_z.\nabla$ (∇ being the operand); and the Jacobian thus obtained is a symbol which operates on H giving $\mathrm{Jac}(U, \nabla, H)H$; and in the determinant $\mathrm{Jac}(U, \nabla H, H)$ the second line is $\mathfrak{d}_x\nabla H$, $\mathfrak{d}_y\nabla H$, $\mathfrak{d}_z\nabla H$ (∇ being simply multiplied by $\mathfrak{d}_x$, $\mathfrak{d}_y$, $\mathfrak{d}_z$ respectively).

Art. Substituting, the condition becomes

$$\begin{aligned}&(-15m^2 + 60m - 57)H\Phi\mathrm{Jac}(U, \nabla, H)H \\ &\qquad + (30m - 54)(m - 2)\{H\Phi\mathrm{Jac}(U, \nabla, H)H + \mathrm{Jac}(U, \nabla H, H)\} \\ &\qquad\qquad + (m - 2)^2\{9H^2\mathfrak{d}\Omega - 45H\Omega\mathfrak{d}_1 H + 40\Phi\mathfrak{d}_1\Psi\} = 0,\end{aligned}$$

or, what is the same thing,

$$\begin{aligned}&(15m^2 - 54m + 51)H\Phi\mathrm{Jac}(U, \nabla, H)H \\ &\qquad + (30m - 54)(m - 2)H\Phi\mathrm{Jac}(U, \nabla H, H) \\ &\qquad\qquad + (m - 2)^2\{9H^2\mathfrak{d}\Omega - 45H\Omega\mathfrak{d}_1 H + 40\Phi\mathfrak{d}_1\Psi\} = 0.\end{aligned}$$

Article Nos. 18 to 27.—Fourth transformation, and final form of the condition for a Sextactic Point.

Art. I write

$$(5m - 12)\Omega\mathfrak{d}_1 H - (3m - 6)H\mathfrak{d}_1\Omega = \frac{\Psi}{m - 1}\mathrm{Jac}(U, \Omega, H), \tag{J}$$

$$\Omega\mathfrak{d}_1 H + H\mathfrak{d}_1\Omega = \mathfrak{d}_1(\Omega H),$$

and, introducing for convenience the new symbol W,

$$-5W\Omega H + H\mathfrak{d}_1\Omega = \mathcal{W},$$

so that

$$\begin{vmatrix} 5m - 12 & -(3m - 6) & \frac{1}{m-1}\Psi\mathrm{Jac}(U, \Omega, H) \\ 1 & 1 & \mathfrak{d}_1(\Omega H) \\ -5 & 1 & \mathcal{W} \end{vmatrix} = 0,$$

or, what is the same thing,

$$(8m-18)\mathcal{W}+6\Psi\mathrm{Jac}(U,\Omega,H)+(10m-18)\mathfrak{d}_1(\Omega H)=0,$$

we have

$$\mathcal{W}=\Omega\mathfrak{d}_1H-5\Omega H=\frac{-3}{4m-9}\Psi\mathrm{Jac}(U,\Omega,H)-\frac{10m-18}{4m-9}\mathfrak{d}_1(\Omega H).$$

Art. We have also

$$(8m-18)\Phi\mathfrak{d}_1\Psi-(3m-6)\Psi\mathfrak{d}_1\Phi=\frac{\Psi}{m-1}\mathrm{Jac}(U,\Phi,H)=0, \qquad \text{(J)}$$

that is

$$\frac{\mathfrak{d}_1\Psi}{\Psi}=\frac{3m-6}{8m-18}\cdot\frac{\mathfrak{d}_1\Phi}{\Phi}=\frac{3(m-2)}{2(4m-9)}\cdot\frac{\mathfrak{d}_1\Phi}{\Phi},$$

and thence

$$\begin{aligned}\Phi\mathcal{W}H+\Psi\mathfrak{d}_1H&=9\Psi^2\mathfrak{d}\Omega-45H\Omega\mathfrak{d}_1H+40\Phi\mathfrak{d}_1\Psi\\&=\frac{9(5m-9)\Psi^2}{4m-9}\mathrm{Jac}(U,\Omega,H)-\frac{60(m-2)^2\Psi^2}{4m-9}\mathfrak{d}_1(\Omega H)\\&\qquad+\frac{\{-27\Psi\mathrm{Jac}(U,\Omega,H)+40\mathrm{Jac}(U,\mathcal{W},H)\}}{4m-9}.\end{aligned}$$

Art. The condition thus becomes

$$\begin{aligned}&(15m^2-54m+51)(4m-9)H\Phi\mathrm{Jac}(U,\nabla,H)H\\&\qquad+6(5m-9)(m-2)(4m-9)H\Phi\mathrm{Jac}(U,\nabla H,H)\\&\quad+3(m-2)\{-3(5m-9)(m-2)\Phi(\Omega\mathfrak{d}_1H)+20(m-2)^2\Psi\mathfrak{d}_1(\Omega H)\}\\&\qquad+(m-2)^2\Psi\{-27H\mathrm{Jac}(U,\Omega,H)+40\mathrm{Jac}(U,\mathcal{W},H)\}=0,\end{aligned}$$

which for shortness I represent by

$$3\mathcal{R}+(m-2)^2\Psi\{-27H\mathrm{Jac}(U,\Omega,H)+40\mathrm{Jac}(U,\mathcal{W},H)\}=0,$$

so that we have

$$\begin{aligned}\mathcal{R}=&(5m^2-18m+17)(4m-9)\mathrm{Jac}(U,\nabla,H)H\\&\qquad+2(5m-9)(m-2)(4m-9)\mathrm{Jac}(U,\nabla H,H)\\&\quad+(m-2)\{-3(5m-9)(m-2)\Phi(\Omega\mathfrak{d}_1H)+20(m-2)^2\Psi\mathfrak{d}_1(\Omega H)\}.\end{aligned}$$

Art. Write

$$\nabla_1=(\mathfrak{A}',\mathfrak{B}',\mathfrak{C}',\mathfrak{F}',\mathfrak{G}',\mathfrak{H}'fA,B,C)^2,$$

where (A,B,C) are as before the differential coefficients of U, and (a',b',c',f',g',h') being the second differential coefficients of H,

$(\mathfrak{A}', \mathfrak{B}', \mathfrak{C}', \mathfrak{F}', \mathfrak{G}', \mathfrak{H}')$ are the inverse coefficients, viz., $\mathfrak{A}' = b'c' - f'^2$, &c. We have

$$-(m-1)^2\mathfrak{d}_x\nabla_1 = (3m-6)(3m-7)\mathfrak{d}_x(\Omega H) - (3m-7)^2\mathfrak{d}_x^{\Omega} \quad \text{(see post, Nos. 41 to 46)},$$

that is

$$(3m-6)\mathfrak{d}(\Omega H) = (3m-7)\mathfrak{d}_x^{\Omega} - \frac{(m-1)^2}{(3m-7)}\mathfrak{d}_x\nabla_1,$$

and thence

$$\begin{aligned}\mathcal{R} = (5m^2 - 18m + 17)(4m-9)\operatorname{Jac}(U, \nabla, H)H \\ + 2(5m-9)(m-2)(4m-9)\operatorname{Jac}(U, \nabla H, H) \\ + (m-2)^2(5m^2-18m+17)\mathfrak{d}_x^{\Omega} + \frac{(m-1)^2(5m-9)(m-2)^2}{(3m-7)}\mathfrak{d}_x\nabla_1.\end{aligned}$$

Art. Now

$$\Phi = (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}f a', b', c', 2f', 2g', 2h'), \quad \Psi = (\mathfrak{A}', \mathfrak{B}', \mathfrak{C}', \mathfrak{F}', \mathfrak{G}', \mathfrak{H}'f a, b, c, 2f, 2g, 2h)^2;$$

and writing for shortness

$$\begin{aligned} E_\Phi &= (\mathfrak{d}\mathfrak{A}, \ldots f A', B', C)^2, & F_\Phi &= (\mathfrak{A}, \ldots f A', B', C f \mathfrak{d}A', \mathfrak{d}B', \mathfrak{d}C'), \\ E_\Psi &= (\mathfrak{d}\mathfrak{A}', \ldots f a, b, c, 2f, 2g, 2h)^2, & F_\Psi &= (\mathfrak{A}', \ldots f a, b, c, 2f, 2g, 2h f \mathfrak{d}\mathfrak{A}, \mathfrak{d}\mathfrak{B}, \mathfrak{d}\mathfrak{C}), \end{aligned}$$

(we might, in a notation above explained, write $E_\Phi = \mathfrak{d}\nabla_\Phi$, $F_\Phi = \nabla_{\mathfrak{d}}\Phi$, and in like manner $E_\Psi = \mathfrak{d}\nabla_\Psi$, $F_\Psi = \nabla_{\mathfrak{d}}\Psi$), then we have

$$\mathfrak{d}_x^{\Phi} = E_\Phi + 2F_\Phi, \quad \mathfrak{d}_x^{\Psi} = E_\Psi + 2F_\Psi.$$

We have moreover

$$\begin{aligned} \operatorname{Jac}(U, \nabla H, H) &= -\frac{3(m-7)}{m-1}F_\Phi, \quad \text{post, Nos. 47 to 50}, \\ \operatorname{Jac}(U, \nabla, H)H &= \frac{3m-7}{m-1}(-E_\Phi), \quad \text{post, Nos. 51 to 53}. \end{aligned}$$

Art. The just-mentioned formulæ give

$$\begin{aligned}\mathcal{R} = -\frac{(5m^2-18m+17)(4m-9)(3m-7)}{m-1}E_\Phi \\ - \frac{2(5m-9)(m-2)(4m-9)\cdot 3(m-7)}{m-1}F_\Phi \\ + (m-2)^2(5m^2-18m+17)(E_\Phi + 2F_\Phi) \\ + \frac{(5m-9)(m-1)^2(m-2)^2}{3m-7}(E_\Psi + 2F_\Psi),\end{aligned}$$

that is

$$
\begin{aligned}
&\mathcal{R} = -(3m-7)(5m^2-18m+17)E_\Phi + 2(m-2)(5m^2-18m+17)F_\Phi \\
&+\frac{(5m-9)(m-1)^2(m-2)^2}{3m-7}E_\Psi + \frac{2(m-1)(m-2)(3m-8)(5m-9)}{3m-7}F_\Psi,
\end{aligned}
$$

or, as this may also be written,

$$
\begin{aligned}
&(3m-7)\mathcal{R} = -(5m^2-18m+17)\{-2(m-1)(m-2)F_\Phi + (3m-7)^2E_\Phi\} \\
&-(5m-9)(m-2)\{(m-1)(3m-8)F_\Psi + (3m-7)(3m-8)F_\Psi - (m-1)^2E_\Psi\} \\
&\qquad + (25m^2-103m+106)(m-2)\{-(m-1)F_\Psi + (3m-7)F_\Psi\}.
\end{aligned}
$$

Art. But recollecting that

$$\Omega = (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} f a', b', c', 2f', 2g', 2h'),$$

and putting

$$
\begin{aligned}
E_\Phi &= (\mathfrak{d}\mathfrak{A}, \ldots f a', \ldots) \quad (= \partial_x \Omega_g), \\
F_\Phi &= (\mathfrak{A}, \ldots f \mathfrak{d} a', \ldots) \quad (= \partial_x \Omega_p),
\end{aligned}
$$

we have, post, Nos. 41 to 46,

$$
\begin{aligned}
-2(m-1)(m-2)F_\Phi + (3m-7)^2E_\Phi &= (3m-6)(3m-7)H\Omega_g, \\
(m-1)(3m-8)F_\Psi + (3m-7)(3m-8)F_\Psi - (m-1)^2E_\Psi &= (3m-6)(3m-7)H\Omega_p, \\
-(m-1)F_\Psi + (3m-7)F_\Psi &= (3m-7)\Omega\mathfrak{d}_1 H,
\end{aligned}
$$

and the foregoing equation becomes

$$
\begin{aligned}
(3m-7)\mathcal{R} = &-(5m^2-18m+17)(3m-6)(3m-7)H\Omega_g \\
&- (5m-9)(m-2)(3m-6)(3m-7)H\Omega_p \\
&+ (m-2)(25m^2-103m+106)(3m-7)\Omega\mathfrak{d}_1 H.
\end{aligned}
$$

Art. But we have

$$\frac{\Psi}{m-1}\mathrm{Jac}(U, H, \Omega_g) - (3m-6)H\Omega_g + (2m-4)\Omega\mathfrak{d}_1 H = 0, \qquad \text{(J)}$$

$$\frac{\Psi}{m-1}\mathrm{Jac}(U, H, \Omega_p) - (3m-6)H\Omega_p + (3m-6)\Omega\mathfrak{d}_1 H = 0, \qquad \text{(J)}$$

that is

$$
\begin{aligned}
3(m-2)H\Omega_g &= 2(m-2)\Omega\mathfrak{d}_1 H + \frac{\Psi}{m-1}\mathrm{Jac}(U, H, \Omega_g), \\
3(m-2)H\Omega_p &= (3m-8)\Omega\mathfrak{d}_1 H + \frac{\Psi}{m-1}\mathrm{Jac}(U, H, \Omega_p),
\end{aligned}
$$

and we thus obtain

$$\begin{aligned}\mathcal{R} = -(5m^2 - 18m + 17)\{2(m-2)\Omega\mathfrak{d}_1 H + \frac{\Psi}{m-1}\mathrm{Jac}(U, H, \Omega_g)\}\\ - (5m-9)(m-2)\{(3m-8)\Omega\mathfrak{d}_1 H + \frac{\Psi}{m-1}\mathrm{Jac}(U, H, \Omega_p)\}\\ + (25m^2 - 103m + 106)(m-2)\Omega\mathfrak{d}_1 H,\end{aligned}$$

where the coefficient of $(m-2)\Omega\mathfrak{d}_1 H$ is

$$\begin{aligned}&-(10m^2 - 36m + 34)\\ &-(5m-9)(3m-8)\\ &+(25m^2 - 103m + 106),\end{aligned}$$

which is $= 0$. Hence

$$\mathcal{R} = -(5m^2 - 18m + 17)\frac{\Psi}{m-1}\mathrm{Jac}(U, H, \Omega_g) - (5m-9)(m-2)\frac{\Psi}{m-1}\mathrm{Jac}(U, H, \Omega_p).$$

Art. Substituting this in the equation

$$3\mathcal{R} + (m-2)^2\{-27H\mathrm{Jac}(U, H, \Omega) + 40\mathrm{Jac}(U, \mathcal{W}, H)\} = 0,$$

the result contains the factor Ψ, and, throwing this out, the condition is

$$\begin{aligned}3H\{-(5m^2 - 18m + 17)\mathrm{Jac}(U, H, \Omega_g) - (5m-9)(m-2)\mathrm{Jac}(U, H, \Omega_p)\}\\ + (m-2)^2\{27H^2\mathrm{Jac}(U, H, \Omega) - 40\mathrm{Jac}(U, \mathcal{W}, H)\} = 0,\end{aligned}$$

or, as this may also be written,

$$\begin{aligned}-(15m^2 - 54m + 51)H\mathrm{Jac}(U, H, \Omega_g) - 3(5m-9)(m-2)H\mathrm{Jac}(U, H, \Omega_p)\\ + 27(m-2)^2\{H\mathrm{Jac}(U, H, \Omega_g) + H\mathrm{Jac}(U, H, \Omega_p)\}\\ - 40(m-2)^2\mathrm{Jac}(U, \mathcal{W}, H) = 0.\end{aligned}$$

Art. Hence the condition finally is

$$\begin{aligned}(12m^2 - 54m + 57)H\mathrm{Jac}(U, H, \Omega_g) + (m-2)(12m-27)H\mathrm{Jac}(U, H, \Omega_p)\\ - 40(m-2)^2\mathrm{Jac}(U, \mathcal{W}, H) = 0,\end{aligned}$$

or, as this may also be written,

$$\begin{aligned}-3(m-1)H\mathrm{Jac}(U, H, \Omega_g) + (m-2)(12m-27)H\mathrm{Jac}(U, H, \Omega)\\ - 40(m-2)^2\mathrm{Jac}(U, \mathcal{W}, H) = 0,\end{aligned}$$

viz. the sextactic points are the intersections of the curve m with the curve represented by this equation; and observing that U, H, ΩH, and Ψ are of the orders m, $3m-6$, $8m-18$ respectively, the order of the curve is as above mentioned $= 12m - 27$.

Article Nos. 28 to 30.—Application to a Cubic.

Art. I have in my former memoir, No. 30, shown that for a cubic curve

$$\Omega = (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} f \partial_x, \partial_y, \partial_z)^2 \Phi = -2\Psi \cdot U = 0;$$

this implies $\mathrm{Jac}(U, H, \Omega) = 0$, and hence if one of the two Jacobians, $\mathrm{Jac}(U, H, \Omega_g)$, $\mathrm{Jac}(U, H, \Omega_p)$ vanish, the other will also vanish. Now, using the canonical form

$$U = x^3 + y^3 + z^3 + 6lxyz = 0,$$

we have

$$\begin{aligned} \Omega &= (\mathfrak{A}, \ldots f a', \ldots) \\ &= (yz - l^2x^2,\ zx - l^2y^2,\ xy - l^2z^2,\ l^2yz - lx^2,\ l^2zx - ly^2,\ l^2xy - lz^2) \\ &\quad (\mathfrak{A}', \mathfrak{B}', \mathfrak{C}', \mathfrak{F}', \mathfrak{G}', \mathfrak{H}' f a, b, c, 2f, 2g, 2h) \\ &= (-3l^2x,\ -3l^2y,\ -3l^2z,\ (1 + 2l^3)x,\ (1 + 2l^3)y,\ (1 + 2l^3)z), \end{aligned}$$

the development of which in fact gives the last-mentioned result. But applying this formula to the calculation of $\mathrm{Jac}(U, H, \Omega_g)$, then disregarding numerical factors, we have

$$\begin{aligned} \partial_x \Omega_g &= (yz - l^2x^2, \ldots f \partial_x a', \ldots f - 3l^2, 0, 0, (1 + 2l^3), 0, 0) \\ &= -3l^2(yz - l^2x^2) + (1 + 2l^3)(yz - lx^2) \\ &= (-l + l^4)(x^2 + 2lyz) = 3\partial_x U; \end{aligned}$$

and in like manner $\partial_y \Omega_g = 3\partial_y U$, $\partial_z \Omega_g = 3\partial_z U$, and therefore

$$\mathrm{Jac}(U, H, \Omega_g) = 3\mathrm{Jac}(U, H, U) = 0,$$

whence also

$$\mathrm{Jac}(U, H, \Omega_p) = 0;$$

and the condition for a sextactic point assumes the more simple form,

$$\mathrm{Jac}(U, H, \Psi) = 0.$$

Art. Now (former memoir, No. 32) we have

$$\begin{aligned} \Psi &= (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} f \partial_x, \partial_y, \partial_z)^2 \Phi \\ &= (1 + 8l^3)^2(x^6 + y^6 + z^6) \\ &\quad + (-9l^2)(x^3 + y^3 + z^3)^2 \\ &\quad + (-2l - 5l^4 - 20l^7)(x^3 + y^3 + z^3)xyz \\ &\quad + (-15l^3 - 78l^6 + 12l^9)x^2y^2z^2, \end{aligned}$$

or observing that $x^3+y^3+z^3$ and xyz, and therefore the last three lines of the expression of Ψ are functions of U $(=x^3+y^3+z^3+6lxyz)$ and H $(=-l^2(x^3+y^3+z^3)+(1+2l^3)xyz)$, and consequently give rise to the term $=0$ in $\mathrm{Jac}(U,H,\Psi)$, we may write

$$\Psi=(1+8l^3)^2(x^6+y^6+z^6).$$

Art. We have then, disregarding a constant factor,

$$\begin{aligned}\mathrm{Jac}(U,H,\Psi)&=\mathrm{Jac}(x^3+y^3+z^3,\ xyz,\ x^6+y^6+z^6),\\&=\begin{vmatrix}x^2&y^2&z^2\\yz&zx&xy\\x^5&y^5&z^5\end{vmatrix}\end{aligned}$$

$$\begin{aligned}&=x^7(y^4-z^4)+y^7(z^4-x^4)+z^7(x^4-y^4),\\&\propto(y^2-z^2)(z^2-x^2)(x^2-y^2),\end{aligned}$$

so that the sextactic points are the intersections of the curve

$$U=x^3+y^3+z^3+6lxyz=0,$$

with the curve

$$(y^2-z^2)(z^2-x^2)(x^2-y^2)=0.$$

Article Nos. 31 to 33.—Proof of identities for the first transformation.

Art. *Calculation of* $(\mathfrak{d}_1^5+10\mathfrak{d}_1^3\mathfrak{d}_2+10\mathfrak{d}_1^2\mathfrak{d}_3+15\mathfrak{d}_1\mathfrak{d}_2^2)U$.

Writing $\mathfrak{d}$ in place of D, we have (former memoir, No. 20)

$$\mathfrak{d}_1^2U=\frac{1}{(m-1)^2}\{-3(m-2)H\Phi+(m-1)\Phi\mathfrak{d}_1H\}.$$

But

$$\mathfrak{d}_1^2U=\frac{1}{(m-1)^2}\{-(3m-6)H\Phi+\Phi\mathfrak{d}_1H\}\quad\text{[former memoir, Nos. 21 and 22]},$$

and thence

$$(\mathfrak{d}_1^3+6\mathfrak{d}_1\mathfrak{d}_2)U=\frac{1}{(m-1)^3}\{(18m^2-66m+60)H^2\Phi+\ldots\},$$

whence operating on each side with $\mathfrak{d}_1$, $=\mathfrak{d}$,

$$\begin{aligned}&(\mathfrak{d}_1^5+10\mathfrak{d}_1^3\mathfrak{d}_2+6\mathfrak{d}_1^2\mathfrak{d}_3+12\mathfrak{d}_1\mathfrak{d}_2^2)U\\&\qquad=\frac{1}{(m-1)^4}\{(18m^2-66m+60)H^2\mathfrak{d}_1\Phi+\ldots\}\\&\qquad\qquad+\frac{1}{(m-1)^4}\{(-10m+18)(\mathfrak{d}.\nabla)\Phi+\mathfrak{d}\nabla\Phi\}.\end{aligned}$$

We have besides (see Appendix, Nos. 69 to 74),

$$\mathfrak{d}_1\mathfrak{d}_3U = \ldots$$

and thence

$$(4\mathfrak{d}_1\mathfrak{d}_2^2 + 3\mathfrak{d}_1^2\mathfrak{d}_3)U = \frac{1}{(m-1)^4}\{(9m^2-21)m + (-9m^2+9)\Phi H\}$$
$$+\ldots$$

and adding this to the foregoing expression for $(\mathfrak{d}_1^5 + 10\mathfrak{d}_1^3\mathfrak{d}_2 + 6\mathfrak{d}_1^2\mathfrak{d}_3 + 12\mathfrak{d}_1\mathfrak{d}_2^2)U$, we have

$$\begin{aligned}&(\mathfrak{d}_1^5 + 10\mathfrak{d}_1^3\mathfrak{d}_2 + 10\mathfrak{d}_1^2\mathfrak{d}_3 + 15\mathfrak{d}_1\mathfrak{d}_2^2)U = \\ &\quad \frac{\Phi}{(m-1)^4}\{(27m^2-96m+81)H\mathfrak{d}_1^4 + (17m^2-56m+51)\Phi\mathfrak{d}_1 H\} \\ &\qquad + \frac{\Psi}{(m-1)^4}\{(-14m+22)(\mathfrak{d}.\nabla)H + (-10m+18)\mathfrak{d}\nabla H\} \\ &\qquad\qquad + \frac{\Omega}{(m-1)^4}\{9\Omega\}.\end{aligned}$$

Art. *Calculation of* $\mathfrak{d}_1U\mathfrak{d}_4H + 2\mathfrak{d}_1^2U\mathfrak{d}_3H + 2\mathfrak{d}_1^3U\mathfrak{d}_2H$.

We have

$$\begin{aligned}\mathfrak{d}_1U &= \frac{m-1}{(m-1)^2}\mathfrak{d}_1U, \quad \mathfrak{d}_2H = \mathfrak{d}_1H, \\ \mathfrak{d}_1^2U &= \frac{(m-1)^2}{(m-1)^3}\mathfrak{d}_1^2U, \quad \mathfrak{d}_3H = \mathfrak{d}_1H, \\ \mathfrak{d}_1^3U &= \frac{(m-1)^3}{(m-1)^4}\mathfrak{d}_1^3U, \quad \mathfrak{d}_4H = \frac{m-1}{(m-1)}(-3m+6)H\Phi - \Phi\mathfrak{d}_1H + \frac{1}{m-1}(\mathfrak{d}.\nabla)H;\end{aligned}$$

for which values see Appendix, No. 58. And hence the expression sought for is

$$\ldots$$
$$+\,2(m-1)\Phi^2\mathfrak{d}_1H$$
$$+\,2H\{(-3m+6)H\Phi - \Phi\mathfrak{d}_1H + \frac{1}{m-1}(\mathfrak{d}.\nabla)H\}\Phi,$$

which is

$$\ldots$$
$$+\,(m-1)\mathfrak{d}_1H\Phi^2H$$
$$+\,(-6m+12)H^2\Phi^2 - 3H\Phi^2\mathfrak{d}_1H\}$$
$$+\,(m-1)^2\{2H(\mathfrak{d}.\nabla)H - \mathfrak{d}H\nabla H\}.$$

But we have, former memoir, Nos. 21 and 25,

$$\begin{aligned}
\mathfrak{d}_1^2 H &= \dots,\\
\mathfrak{d}_1^3 H &= \frac{(3m-6)(3m-7)}{(m-1)^2}H^2\Phi + \frac{6m-14}{(m-1)^2}\Phi\mathfrak{d}_1 H + \dots,
\end{aligned}$$

so that the foregoing expression becomes

$$\begin{aligned}
&\frac{1}{(m-1)^2}\{-(8m-16)H^2\mathfrak{d}_1 H + \Phi\mathfrak{d}H\nabla H\}\\
&\qquad - \frac{(3m-6)(3m-7)}{(m-1)^2}\Phi H\mathfrak{d}_1 H - \frac{6m-14}{(m-1)^2}\Phi^2\mathfrak{d}_1 H\\
&\qquad\quad + \frac{1}{(m-1)^2}\{3H^2\mathfrak{d}_1 H - (16m-12)H^2\mathfrak{d}_1 H\}\\
&\qquad\qquad\qquad + \frac{1}{(m-1)^2}\{2H(\mathfrak{d}.\nabla)H - \mathfrak{d}H\nabla H\};
\end{aligned}$$

or finally

$$\begin{aligned}
&\mathfrak{d}_1 U\mathfrak{d}_4 H + 2\mathfrak{d}_1^2 U\mathfrak{d}_3 H + 2\mathfrak{d}_1^3 U\mathfrak{d}_2 H =\\
&\frac{1}{(m-1)^2}\{(-6m^2+18m-12)H\mathfrak{d}_1^4 + (-17m^2+60m-55)H^2\Phi\mathfrak{d}_1 H\}\\
&\qquad + \frac{1}{(m-1)^3}\{(2m-2)H(\mathfrak{d}.\nabla)H + (8m-16)\mathfrak{d}H\nabla H\}\\
&\qquad\qquad\qquad\qquad\qquad + \frac{1}{(m-1)^4}\{9\Omega\}.
\end{aligned}$$

Art. *Calculation of* $\mathfrak{d}_1^3 U\mathfrak{d}_1^2 U$.
This is

$$\frac{1}{(m-1)^4}H\mathfrak{d}_1 H\cdot\Phi^2 H = \dots$$

Article Nos. 34 and 35.—The Jacobian Formula.

Art. In general, if P, Q, R, S be functions of the degrees p, q, r, s respectively, we have identically

$$\begin{vmatrix}
pP & qQ & rR & sS\\
\partial_x P & \partial_x Q & \partial_x R & \partial_x S\\
\partial_y P & \partial_y Q & \partial_y R & \partial_y S\\
\partial_z P & \partial_z Q & \partial_z R & \partial_z S
\end{vmatrix} = 0,$$

or, what is the same thing,

$$pP\mathrm{Jac}(Q,R,S) - qQ\mathrm{Jac}(R,S,P) + rR\mathrm{Jac}(S,P,Q) - sS\mathrm{Jac}(P,Q,R) = 0.$$

Hence in particular if $P = U$, and assuming $U = 0$, we have

$$-qQ\mathrm{Jac}(R, S, U) + rR\mathrm{Jac}(S, U, Q) - sS\mathrm{Jac}(U, Q, R) = 0.$$

If moreover $Q = \Phi$, and therefore $q = 1$, we have

$$-\Phi\mathrm{Jac}(R, S, U) + rR\mathrm{Jac}(S, U, \Phi) - sS\mathrm{Jac}(U, \Phi, R) = 0;$$

or, as this may also be written,

$$-\frac{\Psi}{m-1}\mathrm{Jac}(U, R, S) + rR\mathrm{Jac}(U, \Phi, S) - sS\mathrm{Jac}(U, \Phi, R) = 0;$$

that is

$$-\frac{\Psi}{m-1}\mathrm{Jac}(U, R, S) + rR\eth S - sS\eth R = 0.$$

Art. Particular cases are

$$\begin{aligned}
(2m-4)\Phi\eth_1 H - (3m-6)H\eth_1\Phi &= \frac{\Psi}{m-1}\mathrm{Jac}(U, \Phi, H), \quad \text{ante, No. 12,}\\
(5m-11)\nabla H\eth_1 H - (3m-6)H\eth_1(\nabla H) &= \frac{\Psi}{m-1}\mathrm{Jac}(U, \nabla H, H),\\
(2m-4)\nabla\eth_1\Phi - (3m-6)\Phi\eth_1\nabla &= \frac{\Psi}{m-1}\mathrm{Jac}(U, \nabla, H),\\
(5m-12)\Omega\eth_1 H - (3m-6)H\eth_1\Omega &= \frac{\Psi}{m-1}\mathrm{Jac}(U, \Omega, H),\\
(8m-18)\mathcal{W}\eth_1 H - (3m-6)H\eth_1\mathcal{W} &= \frac{\Psi}{m-1}\mathrm{Jac}(U, \mathcal{W}, H),\\
(2m-4)\Omega\eth_1 H - (3m-6)H\Omega_g &= \frac{\Psi}{m-1}\mathrm{Jac}(U, \Omega, H),\\
(3m-8)\Omega\eth_1 H - (3m-6)H\Omega_p &= \frac{\Psi}{m-1}\mathrm{Jac}(U, \Omega_p, H),
\end{aligned}$$

where it is to be observed that in the third of these formulæ I have, in accordance with the notation before employed, written $\eth.\nabla$ to denote the result of the operation $\eth$ performed on ∇ as operand. I have also written $\nabla : \eth H$ to show that the operation ∇ is not to be performed on the following $\eth H$ as an operand, but that it remains as an unperformed operation. As regards the last two equations, it is to be remarked that the demonstration in the last preceding number depends merely on the homogeneity of the functions, and the orders of these functions: in the former of the two formulæ, the differentiation of Ω is performed upon Ω in regard to the coordinates (x, y, z) in so far only as they enter through U, and Ω is therefore to be regarded as a function of the order $2m - 4$; in the latter of the two formulæ the differentiation is to be performed in regard to the coordinates in so far only as they enter through H, and Ω is therefore to be regarded as a function of the order $3m-8$. The two formulæ might also be written

$$(2m-4)\nabla\mathfrak{d}_1 H-(3m-6)H\Omega_g=\frac{\Psi}{m-1}\mathrm{Jac}(U,\Omega_g,H),$$
$$(3m-8)\Omega\mathfrak{d}_1 H-(3m-6)H\Omega_p=\frac{\Psi}{m-1}\mathrm{Jac}(U,\Omega_p,H);$$

and it may be noticed that, adding these together, we obtain the foregoing formula,

$$(5m-12)\Omega\mathfrak{d}_1 H-(3m-6)H\mathfrak{d}_1\Omega=\frac{\Psi}{m-1}\mathrm{Jac}(U,\Omega,H).$$

Article Nos. 36 to 40.—Proof of equation $(\mathfrak{d}.\nabla)H = \mathrm{Jac}(U,H,\Phi)$**, used in the second transformation.**

Art. We have

$$\begin{aligned}\nabla &= (\mathfrak{A},\ldots f\lambda,\mu,\nu f\partial_x,\partial_y,\partial_z)\\ &= (\mathfrak{A}\partial_x+\mathfrak{F}\partial_y+\mathfrak{G}\partial_z,\ \mathfrak{F}\partial_x+\mathfrak{B}\partial_y+\mathfrak{H}\partial_z,\ \mathfrak{G}\partial_x+\mathfrak{H}\partial_y+\mathfrak{C}\partial_z f\lambda,\mu,\nu).\end{aligned}$$

Also

$$\begin{aligned}\mathfrak{d} &= (Bv-C\mu)\partial_x+(C\lambda-A\nu)\partial_y+(A\mu-B\lambda)\partial_z\\ &= \lambda P+\mu Q+\nu R,\end{aligned}$$

if for a moment $P,Q,R=C\partial_y-B\partial_z,\ A\partial_z-C\partial_x,\ B\partial_x-A\partial_y$.

Hence

$$\mathfrak{d}.\nabla=(P\lambda+Q\mu+R\nu)\cdot(\mathfrak{A}\partial_x+\mathfrak{F}\partial_y+\mathfrak{G}\partial_z,\ \mathfrak{F}\partial_x+\mathfrak{B}\partial_y+\mathfrak{H}\partial_z,\ \mathfrak{G}\partial_x+\mathfrak{H}\partial_y+\mathfrak{C}\partial_z f\lambda,\mu,\nu),$$

viz. coefficient of λ^2

$$=P\mathfrak{A}\partial_x+P\mathfrak{F}\partial_y+P\mathfrak{G}\partial_z$$

and so for the other terms; whence also in $(\mathfrak{d}.\nabla)H$ the coefficients of λ^2, &c. are

$$(P\mathfrak{A}\partial_x+P\mathfrak{F}\partial_y+P\mathfrak{G}\partial_z)H,\ \text{\&c.}$$

Art. Again, in $\mathrm{Jac}(U,H,\Phi)$, where $\Phi=(\mathfrak{A},\mathfrak{B},\mathfrak{C},\mathfrak{F},\mathfrak{G},\mathfrak{H}f\lambda,\mu,\nu)^2$, the coefficients of λ^2, &c. are $\mathrm{Jac}(U,H,\mathfrak{A})$, &c.; and hence the assumed equation

$$(\mathfrak{d}.\nabla)H=\mathrm{Jac}(U,H,\Phi),$$

in regard to the term in λ^2, is

$$(P\mathfrak{A}\partial_x+P\mathfrak{F}\partial_y+P\mathfrak{G}\partial_z)H=\mathrm{Jac}(U,H,\mathfrak{A}),$$

and we have

$$\begin{aligned}\mathrm{Jac}(U, H, \mathfrak{A}) &= \begin{vmatrix} A & B & C \\ \partial_x H & \partial_y H & \partial_z H \\ \partial_x \mathfrak{A} & \partial_y \mathfrak{A} & \partial_z \mathfrak{A} \end{vmatrix} \\ &= \{\partial_x H(C\partial_y - B\partial_z) + \partial_y H(A\partial_z - C\partial_x) + \partial_z H(B\partial_x - A\partial_y)\}\mathfrak{A} \\ &= (\partial_x H \cdot P + \partial_y H \cdot Q + \partial_z H \cdot R)\mathfrak{A};\end{aligned}$$

so that the equation is

$$P\mathfrak{A}\partial_x H + P\mathfrak{F}\partial_y H + P\mathfrak{G}\partial_z H = P\mathfrak{A}\partial_x H + Q\mathfrak{A}\partial_y H + R\mathfrak{A}\partial_z H,$$

or, as this may be written,

$$\begin{aligned}&[(B\partial_z - C\partial_y)\mathfrak{F} - (C\partial_x - A\partial_z)\mathfrak{A}]\partial_y H \\ &\qquad\qquad + [(B\partial_z - C\partial_y)\mathfrak{G} - (A\partial_y - B\partial_x)\mathfrak{A}]\partial_z H = 0.\end{aligned}$$

Art. The coefficient of $\partial_y H$ is

$$B(\partial_z \mathfrak{F} - \partial_y \mathfrak{A}) + C(\partial_x \mathfrak{A} - \partial_z \mathfrak{F}) + A(\partial_y \mathfrak{F} - \partial_x \mathfrak{G}),$$

which, in virtue of the identity, post, No. 40, $\partial_x \mathfrak{A} + \partial_y \mathfrak{F} + \partial_z \mathfrak{G} = 0$,

$$= A\partial_x \mathfrak{A} + B\partial_y \mathfrak{F} + C\partial_z \mathfrak{G};$$

and in like manner the coefficient of $\partial_z H$ is

$$= -(A\partial_x \mathfrak{G} + B\partial_y \mathfrak{B} + C\partial_z \mathfrak{H}),$$

so that the equation is

$$(A\partial_x \mathfrak{A} + B\partial_y \mathfrak{F} + C\partial_z \mathfrak{G})\partial_y H - (A\partial_x \mathfrak{G} + B\partial_y \mathfrak{B} + C\partial_z \mathfrak{H})\partial_z H = 0.$$

Art. But we have

$$\begin{aligned}\mathfrak{A}a + \mathfrak{F}h + \mathfrak{G}g &= H, \\ \mathfrak{A}h + \mathfrak{F}b + \mathfrak{G}f &= 0, \\ \mathfrak{A}g + \mathfrak{F}f + \mathfrak{G}c &= 0,\end{aligned}$$

or multiplying by x, y, z and adding,

$$(m-1)(\mathfrak{A}A + \mathfrak{F}B + \mathfrak{G}C) = xH;$$

whence also

$$(m-1)(\mathfrak{A}\partial_x A + \mathfrak{F}\partial_x B + \mathfrak{G}\partial_x C + A\partial_x \mathfrak{A} + B\partial_x \mathfrak{F} + C\partial_x \mathfrak{G}) = x\partial_x H + H,$$

that is

$$(m-1)(A\partial_x \mathfrak{A} + B\partial_x \mathfrak{F} + C\partial_x \mathfrak{G}) = x\partial_x H;$$

and in like manner

$$(m-1)(A\partial_y\mathfrak{A}+B\partial_y\mathfrak{F}+C\partial_y\mathfrak{G})=x\partial_yH,$$

whence the equation in question. The terms in λ^2 are thus shown to be equal, and it might in a similar manner be shown that the terms in $\mu\nu$ are equal; the other terms will then be equal, and we have therefore

$$(\mathfrak{d}.\nabla)H=\mathrm{Jac}(U,H,\Phi).$$

Art. The identity

$$\partial_x\mathfrak{A}+\partial_y\mathfrak{F}+\partial_z\mathfrak{G}=0$$

assumed in the course of the foregoing proof is easily proved. We have in fact

$$\begin{aligned}\partial_x\mathfrak{A}+\partial_y\mathfrak{F}+\partial_z\mathfrak{G}&=\partial_x(bc-f^2)+\partial_y(fg-ch)+\partial_z(fh-bg)\\&=b(\partial_xc-\partial_yf)+c(\partial_xb-\partial_zg)+f(-2\partial_xf+\partial_yg+\partial_zh)\\&\quad+g(\partial_yf-\partial_zb)+h(-\partial_yc+\partial_zf),\end{aligned}$$

where the coefficients of b, c, f, g, h separately vanish: we have of course the system

$$\begin{aligned}\partial_x\mathfrak{A}+\partial_y\mathfrak{F}+\partial_z\mathfrak{G}&=0,\\\partial_x\mathfrak{F}+\partial_y\mathfrak{B}+\partial_z\mathfrak{H}&=0,\\\partial_x\mathfrak{G}+\partial_y\mathfrak{H}+\partial_z\mathfrak{C}&=0.\end{aligned}$$

Article Nos. 41 to 46.—Proof of identities for the fourth transformation.

Art. Consider the coefficients (a,b,c,f,g,h) and the inverse set $(\mathfrak{A},\mathfrak{B},\mathfrak{C},\mathfrak{F},\mathfrak{G},\mathfrak{H})$, and the coefficients (a',b',c',f',g',h'), and the inverse set $(\mathfrak{A}',\mathfrak{B}',\mathfrak{C}',\mathfrak{F}',\mathfrak{G}',\mathfrak{H}')$; then we have identically

$$\begin{aligned}&(a,\ldots fx,y,z)^2(\mathfrak{A}',\ldots fa,\ldots)-(\mathfrak{A}',\ldots fax+hy+gz,\ldots)^2\\&\qquad=(a',\ldots fx,y,z)^2(\mathfrak{A},\ldots fa',\ldots)-(\mathfrak{A},\ldots fa'x+h'y+g'z,\ldots)^2,\end{aligned}$$

where $(\mathfrak{A}',\ldots fa,\ldots)$ and $(\mathfrak{A},\ldots fa',\ldots)$ stand for

$$(\mathfrak{A}',\mathfrak{B}',\mathfrak{C}',\mathfrak{F}',\mathfrak{G}',\mathfrak{H}'fa,b,c,2f,2g,2h)$$

and

$$(\mathfrak{A},\mathfrak{B},\mathfrak{C},\mathfrak{F},\mathfrak{G},\mathfrak{H}fa',b',c',2f',2g',2h')$$

respectively.

Art. Taking (a,b,c,f,g,h), the second differential coefficients of a function U of the order m, and in like manner (a',b',c',f',g',h'), the second differential coefficients of a function U' of the order m', we have

$$m(m-1)U\cdot(\mathfrak{A}',\ldots f\partial_x,\partial_y,\partial_z)^2U'-(m-1)^2(\mathfrak{A}',\ldots f\partial_xU',\partial_yU',\partial_zU')^2$$
$$=m'(m'-1)U'\cdot(\mathfrak{A},\ldots f\partial_x,\partial_y,\partial_z)^2U-(m'-1)^2(\mathfrak{A},\ldots f\partial_xU,\partial_yU,\partial_zU)^2;$$

and in particular if U' be the Hessian of U, then $m'=3m-6$.

Art. Hence writing

$$\Omega=(\mathfrak{A},\ldots f\partial_x,\partial_y,\partial_z)^2H,\quad \nabla_1=(\mathfrak{A},\ldots f\partial_xH,\partial_yH,\partial_zH)^2,$$

we have

$$\Omega_1=(\mathfrak{A}',\ldots f\partial_x,\partial_y,\partial_z)^2U,\quad \nabla_1=(\mathfrak{A}',\ldots f\partial_xU,\partial_yU,\partial_zU)^2;$$

$$m(m-1)U\Omega_1-(m-1)^2\nabla_1=(3m-6)(3m-7)H\Omega-(3m-7)^2\nabla;$$

or if $U=0$, then

$$-(m-1)^2\nabla_1=(3m-6)(3m-7)H\Omega-(3m-7)^2\nabla;$$

whence also

$$-(m-1)^2\mathfrak{d}_x\nabla_1=(3m-6)(3m-7)\{H\mathfrak{d}_x\Omega+\Omega\mathfrak{d}_xH\}-(3m-7)^2\mathfrak{d}_x\nabla,$$

which is the formula, ante No. 21.

Art. Recurring to the original formula, since this is an actual identity, we may operate on it with the differential symbol $\mathfrak{d}$ on the three assumptions:

1. (a,b,c,f,g,h), $(\mathfrak{A},\mathfrak{B},\mathfrak{C},\mathfrak{F},\mathfrak{G},\mathfrak{H})$ are alone variable.
2. (a',b',c',f',g',h'), $(\mathfrak{A}',\mathfrak{B}',\mathfrak{C}',\mathfrak{F}',\mathfrak{G}',\mathfrak{H}')$ are alone variable.
3. (x,y,z) are alone variable.

We thus obtain

$$(a,\ldots fx,y,z)^2(\mathfrak{d}\mathfrak{A}',\ldots fa,\ldots)=(a',\ldots fx,y,z)^2(\mathfrak{d}\mathfrak{A},\ldots fa',\ldots)$$
$$-2(\mathfrak{A}',\ldots fax+hy+gz,\ldots fa\mathfrak{d}x+y\mathfrak{d}b+z\mathfrak{d}c,\ldots),$$

$$(a,\ldots fx,y,z)^2(\mathfrak{d}\mathfrak{A}',\ldots fa,\ldots)=(\mathfrak{d}a',\ldots fx,y,z)^2(\mathfrak{A},\ldots fa',\ldots)$$
$$-(\mathfrak{d}\mathfrak{A}',\ldots fax+hy+gz,\ldots)^2+(a',\ldots fx,y,z)^2(\mathfrak{d}\mathfrak{A},\ldots fa',\ldots)$$
$$-2(\mathfrak{A},\ldots fa'x+h'y+g'z,\ldots fx\mathfrak{d}a'+y\mathfrak{d}h'+z\mathfrak{d}g',\ldots),$$

$$2(a,\ldots fx,y,zf\mathfrak{d}x,\mathfrak{d}y,\mathfrak{d}z)(\mathfrak{A}',\ldots fa,\ldots)=2(a',\ldots fx,y,zf\mathfrak{d}x,\mathfrak{d}y,\mathfrak{d}z)(\mathfrak{A},\ldots fa',\ldots)$$
$$-2(\mathfrak{A}',\ldots fax+hy+gz,\ldots fa\mathfrak{d}x+h\mathfrak{d}y+g\mathfrak{d}z,\ldots)$$
$$-2(\mathfrak{A},\ldots fa'x+h'y+g'z,\ldots fa'\mathfrak{d}x+h'\mathfrak{d}y+g'\mathfrak{d}z,\ldots).$$

Art. If in these equations respectively we suppose as before that (a, b, c, f, g, h) are the second differential coefficients of a function U of the order m, and (a', b', c', f', g', h') the second differential coefficients of a function V of the order m'; and that (A, B, C), (A', B', C') are the first differential coefficients of these functions respectively, then after some easy reductions we have

$$\begin{aligned}(m-1)(m-2)\mathfrak{d}U(\mathfrak{A}', \ldots f a, \ldots) &= m'(m'-1)V'(a\mathfrak{A}, \ldots f a', \ldots)\\ &+ m(m-1)U(\mathfrak{A}', \ldots f\mathfrak{d}a, \ldots) - (m'-1)^2(\mathfrak{d}\mathfrak{A}', \ldots f A', B', C')^2\\ &- 2(m-1)(m-2)(\mathfrak{A}', \ldots f A, B, C f \mathfrak{d}A, \mathfrak{d}B, \mathfrak{d}C),\end{aligned}$$

$$\begin{aligned}m(m-1)U(\mathfrak{d}\mathfrak{A}', \ldots f a', \ldots) &= (m'-1)(m'-2)\mathfrak{d}U'(\mathfrak{A}, \ldots f a', \ldots)\\ &- (m'-1)^2(\mathfrak{d}\mathfrak{A}', \ldots f A, B, C)^2 + m'(m'-1)V'(\mathfrak{A}, \ldots f a\mathfrak{d}a', \ldots)\\ &- 2(m'-1)(m'-2)(\mathfrak{A}, \ldots f A', B', C' f \mathfrak{d}A', \mathfrak{d}B', \mathfrak{d}C'),\end{aligned}$$

$$\begin{aligned}2(m-1)\mathfrak{d}U(\mathfrak{A}', \ldots f a, \ldots) &= 2(m'-1)\mathfrak{d}U'(\mathfrak{A}, \ldots f a', \ldots)\\ &- 2(m-1)(\mathfrak{A}', \ldots f A, B, C f \mathfrak{d}A, \mathfrak{d}B, \mathfrak{d}C)\\ &- 2(m'-1)(\mathfrak{A}, \ldots f A', B', C' f \mathfrak{d}A', \mathfrak{d}B', \mathfrak{d}C'),\end{aligned}$$

equations which may be verified by remarking that their sum is

$$\begin{gathered}m(m-1)\{\mathfrak{d}U(\mathfrak{A}', \ldots f a, \ldots) + U[(\mathfrak{A}', \ldots f\mathfrak{d}a, \ldots) + (\mathfrak{d}\mathfrak{A}', \ldots f a, \ldots)]\}\\ -(m-1)^2\{(\mathfrak{d}\mathfrak{A}', \ldots f A, B, C)^2 + (\mathfrak{A}', \ldots f A, B, C f \mathfrak{d}A, \mathfrak{d}B, \mathfrak{d}C)\} = m'(m'-1)\,\&\text{c.},\end{gathered}$$

viz., this is the derivative with $\mathfrak{d}$ of the equation

$$m(m-1)U \cdot (\mathfrak{A}', \ldots f a, \ldots) - (m-1)^2(A', \ldots f A, B, C)^2 = m'(m'-1)\,\&\text{c.}$$

Art. Taking now $U' = H$, and therefore $m' = 3m - 6$; putting also $U = 0$, $\mathfrak{d}U = 0$, and writing as before

$$\begin{aligned}E_\Phi &= (\mathfrak{d}\mathfrak{A}, \ldots f A', B', C)^2,\\ F_\Phi &= (\mathfrak{A}, \ldots f A', B', C f \mathfrak{d}A', \mathfrak{d}B', \mathfrak{d}C'),\\ E_\Psi &= (\mathfrak{d}\mathfrak{A}', \ldots f A, B, C)^2,\\ F_\Psi &= (\mathfrak{A}', \ldots f A, B, C f \mathfrak{d}A, \mathfrak{d}B, \mathfrak{d}C),\\ \Omega_g &= (\mathfrak{d}\mathfrak{A}, \ldots f a', \ldots),\\ \Omega_p &= (\mathfrak{A}, \ldots f\mathfrak{d}a', \ldots),\end{aligned}$$

then the three equations are

$$\begin{aligned}-2(m-1)(m-2)F_\Phi &= (3m-6)(3m-7)H\Omega_g - (3m-7)^2E_\Phi,\\ -(m-1)^2E_\Psi &= (3m-7)(3m-8)\Omega\mathfrak{d}_1H + (3m-6)(3m-7)H\Omega_p - 2(3m-7)(3m\\ -2(m-1)F_\Psi &= 2(3m-7)\Omega\mathfrak{d}_1H - 2(2m-7)F_\Psi,\end{aligned}$$

whence, adding, we have

$$\begin{aligned}-(m-1)^2(E_\Phi+2F_\Phi) &= -(3m-7)^2(E_\Phi+2F_\Phi)\\ &\quad + (3m-6)(3m-7)\{\Omega\mathfrak{d}_1 H + H(\Omega_g+F_\Phi)\},\end{aligned}$$

(that is

$$-(m-1)^2\mathfrak{d}_x^{\Omega} = -(3m-7)^2\mathfrak{d}_x^{\Omega} + (3m-6)(3m-7)\mathfrak{d}.\Omega H,$$

which is right).

And by linearly combining the three equations, we deduce

$$\begin{aligned}(3m-6)(3m-7)H\Omega_g &= -2(m-1)(m-2)F_\Phi + (3m-7)^2E_\Phi,\\ (3m-7)\Omega\mathfrak{d}_1 H &= -(m-1)E_\Psi + (3m-7)F_\Psi,\\ (3m-6)(3m-7)H\Omega_p &= (m-1)(3m-8)F_\Psi + (3m-7)(3m-8)F_\Psi - (m-1)^2E_\Psi,\end{aligned}$$

which are the formulæ, ante, No. 24.

Article Nos. 47 to 50.—Proof of an identity used in the fourth transformation, viz. $\operatorname{Jac}(U,\nabla H,H) = -\frac{3(m-7)}{m-1}F_\Phi$.

Art. We have

$$\begin{aligned}\nabla &= (\mathfrak{A},\ldots f\lambda,\mu,\nu f\partial_x,\partial_y,\partial_z)\\ &= ((\mathfrak{A},\mathfrak{F},\mathfrak{G}f\lambda,\mu,\nu),\ (\mathfrak{F},\mathfrak{B},\mathfrak{H}f\lambda,\mu,\nu),\ (\mathfrak{G},\mathfrak{H},\mathfrak{C}f\lambda,\mu,\nu)f\partial_x,\partial_y,\partial_z);\end{aligned}$$

or, attending to the effect of the bar as denoting the exemption of the $(\mathfrak{A},\ldots)$ from differentiation,

$$\begin{aligned}\operatorname{Jac}(U,H,\nabla H) &= (\mathfrak{A},\mathfrak{F},\mathfrak{G}f\lambda,\mu,\nu)\operatorname{Jac}(U,H,\partial_x H)\\ &\quad + (\mathfrak{F},\mathfrak{B},\mathfrak{H}f\lambda,\mu,\nu)\operatorname{Jac}(U,H,\partial_y H)\\ &\quad + (\mathfrak{G},\mathfrak{H},\mathfrak{C}f\lambda,\mu,\nu)\operatorname{Jac}(U,H,\partial_z H).\end{aligned}$$

Art. Now

$$\operatorname{Jac}(U,H,\partial_x H) = -\frac{3}{3m-7}\operatorname{Jac}(U,x\partial_x H+y\partial_y H+z\partial_z H,\partial_x H),$$

and the last-mentioned Jacobian is

$$\begin{aligned}&= \partial_x H\operatorname{Jac}(U,x,\partial_x H) + \partial_y H\operatorname{Jac}(U,y,\partial_x H) + \partial_z H\operatorname{Jac}(U,z,\partial_x H)\\ &\quad + y\operatorname{Jac}(U,\partial_y H,\partial_x H) + z\operatorname{Jac}(U,\partial_z H,\partial_x H),\end{aligned}$$

where the second line is

$$= -y\operatorname{Jac}(U,\partial_x H,\partial_y H) + z\operatorname{Jac}(U,\partial_z H,\partial_x H),$$

or writing (A', B', C') for the first differential coefficients and (a', h', c', f', g', h') for the second differential coefficients of H, this is

$$= -y(\mathfrak{G}', \mathfrak{H}', \mathfrak{C}' \int A, B, C) + z(\mathfrak{F}', \mathfrak{B}', \mathfrak{H}' \int A, B, C).$$

The first line is

$$\begin{vmatrix} A & B & C \\ A' & B' & C' \\ a' & h' & g' \end{vmatrix} = A(B'g' - C'h') + B(C'a' - A'g') + C(A'h' - B'a'),$$

or reducing by the formulæ,

$$(3m-7)(A', B', C') = (a'x + h'y + g'z,\ h'x + b'y + f'z,\ g'x + f'y + c'z),$$

$$\begin{aligned} &= \frac{1}{3m-7}\{A(-\mathfrak{G}'y + \mathfrak{H}'z) + B(-\mathfrak{F}'y + \mathfrak{B}'z) + C(-\mathfrak{G}'x + \mathfrak{A}'z)\} \\ &= \frac{1}{3m-7}\{-y(\mathfrak{G}', \mathfrak{H}', \mathfrak{C}' \int A, B, C) + z(\mathfrak{F}', \mathfrak{B}', \mathfrak{H}' \int A, B, C)\}. \end{aligned}$$

Hence we have

$$\begin{aligned} \mathrm{Jac}(U, H, \partial_x H) = -&\frac{3}{3m-7}(1 + \frac{1}{3m-7}) \\ &\{-y(\mathfrak{G}', \mathfrak{H}', \mathfrak{C}' \int A, B, C) + z(\mathfrak{F}', \mathfrak{B}', \mathfrak{H}' \int A, B, C)\}; \end{aligned}$$

and in like manner

$$\begin{aligned} \mathrm{Jac}(U, H, \partial_y H) = -&\frac{3}{3m-7}(1 + \frac{1}{3m-7}) \\ &\{-z(\mathfrak{W}', \mathfrak{F}', \mathfrak{G}' \int A, B, C) + x(\mathfrak{G}', \mathfrak{H}', \mathfrak{C}' \int A, B, C)\}, \\ \mathrm{Jac}(U, H, \partial_z H) = -&\frac{3}{3m-7}(1 + \frac{1}{3m-7}) \\ &\{-x(\mathfrak{F}', \mathfrak{B}', \mathfrak{H}' \int A, B, C) + y(\mathfrak{W}', \mathfrak{F}', \mathfrak{G}' \int A, B, C)\}. \end{aligned}$$

Art. We thence have

$$\mathrm{Jac}(U, H, \nabla H) = \frac{1}{3m-7}(1 + \frac{1}{3m-7}) \begin{vmatrix} (\mathfrak{A}', \mathfrak{F}', \mathfrak{G}' \int A, B, C) & (\mathfrak{F}', \mathfrak{B}', \mathfrak{H}' \int A, B, C) & (\mathfrak{G}', \mathfrak{H}', \mathfrak{C}' \int A, B, C) \\ x & y & z \\ \lambda & \mu & \nu \end{vmatrix}$$

or multiplying the two sides by

$$H = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix},$$

the right-hand side is

$$= \frac{1}{3m-7}(1+\frac{1}{3m-7})\begin{vmatrix} H\lambda & H\mu & H\nu \\ x & y & z \\ (m-1)A & (m-1)B & (m-1)C \end{vmatrix}$$

$$= H \cdot \frac{m-1}{3m-7}(1+\frac{1}{3m-7})\begin{vmatrix} \lambda & \mu & \nu \\ x & y & z \\ A & B & C \end{vmatrix}$$

if for a moment

$$\begin{aligned} X &= (\mathfrak{A}', \ldots f A, B, C f a, h, g), \\ Y &= (\mathfrak{A}', \ldots f A, B, C f h, b, f), \\ Z &= (\mathfrak{A}', \ldots f A, B, C f g, f, c). \end{aligned}$$

Art. Hence observing that these equations may be written

$$\begin{aligned} X &= (\mathfrak{A}', \ldots f A, B, C f \partial_x A, \partial_x B, \partial_x C), \\ Y &= (\mathfrak{A}', \ldots f A, B, C f \partial_y A, \partial_y B, \partial_y C), \\ Z &= (\mathfrak{A}', \ldots f A, B, C f \partial_z A, \partial_z B, \partial_z C), \end{aligned}$$

and that we have

$$\begin{vmatrix} \lambda & \mu & \nu \\ \partial_x & \partial_y & \partial_z \\ A & B & C \end{vmatrix} = \mathfrak{d},$$

we obtain for $H\mathrm{Jac}(U, H, \nabla H)$ the value

$$= H \cdot \frac{m-1}{3m-7}(1+\frac{1}{3m-7})(\mathfrak{A}', \ldots f A, B, C f \mathfrak{d}A, \mathfrak{d}B, \mathfrak{d}C),$$

or throwing out the factor H, we have the required result.

Article Nos. 51 to 53.—Proof of identity used in the fourth transformation, viz. $\mathrm{Jac}(U, \nabla, H)H = E_\Psi$, or say $\mathrm{Jac}(U, H, \nabla)H = (\mathfrak{d}\mathfrak{A}', \ldots f A', B', C')$.

Art. We have

$$(\mathfrak{d}_x.\nabla)H = (\partial_x\mathfrak{A}, \partial_x\mathfrak{F}, \partial_x\mathfrak{G} f \lambda, \mu, \nu),\ (\partial_x\mathfrak{F}, \partial_x\mathfrak{B}, \partial_x\mathfrak{H} f \lambda, \mu, \nu),\ (\partial_x\mathfrak{G}, \partial_x\mathfrak{H}, \partial_x\mathfrak{C} f \lambda, \mu, \nu) f \partial_x H, \partial$$

and thence

$$\begin{aligned} (\mathfrak{d}_x.\nabla)H = (\partial_x\mathfrak{A}, \partial_x\mathfrak{F}, \partial_x\mathfrak{G} f \lambda, \mu, \nu),\ (\partial_x\mathfrak{F}, \partial_x\mathfrak{B}, \partial_x\mathfrak{H} f \lambda, \mu, \nu), \\ (\partial_x\mathfrak{G}, \partial_x\mathfrak{H}, \partial_x\mathfrak{C} f \lambda, \mu, \nu) f \partial_x H, \partial_y H, \partial_z H) f \lambda, \mu, \nu), \end{aligned}$$

with the like values for $(\mathfrak{d}_y.\nabla)H$ and $(\mathfrak{d}_z.\nabla)H$. And then

$$\mathrm{Jac}(U,H,\nabla)H = \begin{vmatrix} A & B & C \\ A' & B' & C' \\ (\mathfrak{d}_x.\nabla)H & (\mathfrak{d}_y.\nabla)H & (\mathfrak{d}_z.\nabla)H \end{vmatrix}$$

in which the coefficient of A'^2 is

$$= (C\partial_y - B\partial_z)(\mathfrak{A}, \mathfrak{F}, \mathfrak{G} f \lambda, \mu, \nu);$$

or putting for shortness

$$(C\partial_y - B\partial_z,\ A\partial_z - C\partial_x,\ B\partial_x - A\partial_y) = (P, Q, R);$$

the coefficient is

$$(P\mathfrak{A}, P\mathfrak{F}, P\mathfrak{G} f \lambda, \mu, \nu).$$

Art. We have

$$\mathfrak{d} = (P\lambda + Q\mu + R\nu),$$

and thence

$$\begin{aligned} \text{coefficient } A'^2 - \mathfrak{d}_x^{\mathfrak{A}} &= (P\mathfrak{A}, P\mathfrak{F}, P\mathfrak{G} f \lambda, \mu, \nu) - (P\mathfrak{A}, Q\mathfrak{A}, R\mathfrak{A} f \lambda, \mu, \nu) \\ &= \mu\{(C\partial_y - B\partial_z)\mathfrak{F} - (A\partial_z - C\partial_x)\mathfrak{A}\} \\ &\quad + \nu\{(C\partial_y - B\partial_z)\mathfrak{G} - (A\partial_y - B\partial_x)\mathfrak{A}\}, \end{aligned}$$

where coefficient of μ is

$$\begin{aligned} &= -A\partial_y\mathfrak{A} - B\partial_y\mathfrak{F} + C(\partial_y\mathfrak{G} + \partial_x\mathfrak{A}) \\ &= -(A\partial_y\mathfrak{A} + B\partial_y\mathfrak{F} + C\partial_y\mathfrak{G}) = -\frac{x\partial_y H}{m-1}, \end{aligned}$$

and coefficient of ν is

$$= +(A\partial_z\mathfrak{A} + B\partial_z\mathfrak{F} + C\partial_z\mathfrak{G}) = \frac{x\partial_z H}{m-1},$$

so that

$$\text{coefficient } A'^2 - \mathfrak{d}_x^{\mathfrak{A}} = -\frac{x}{m-1}(\mu\partial_y H - \nu\partial_z H).$$

Art. By forming in a similar manner the coefficients of the other terms, it appears that

$$\mathrm{Jac}(U,H,\nabla)H - (\mathfrak{d}\mathfrak{A}', \ldots f A', B', C') = 0,$$

or since the determinant is

$$\begin{vmatrix} A' & B' & C' \\ A' & B' & C' \\ (\mathfrak{d}_x.\nabla)H & (\mathfrak{d}_y.\nabla)H & (\mathfrak{d}_z.\nabla)H \end{vmatrix} = 0,$$

or

$$\begin{vmatrix} \lambda & \mu & \nu \\ \partial_x H & \partial_y H & \partial_z H \\ A' & B' & C' \end{vmatrix} = \ldots$$

we have the required equation,

$$\mathrm{Jac}(U, H, \nabla)H = (\mathfrak{d}\mathfrak{A}', \ldots f A', B', C').$$

This completes the series of formulæ used in the transformations of the condition for the sextactic point.

Appendix, Nos. 54 to 74.

For the sake of exhibiting in their proper connexion some of the formulæ employed in the foregoing first transformation of the condition for a sextactic point, I have investigated them in the present Appendix, which however is numbered continuously with the memoir.

Art. The investigations of my former memoir and the present memoir have reference to the operations

$$\begin{aligned} \mathfrak{d}_1 &= dx\,\partial_x + dy\,\partial_y + dz\,\partial_z, \\ \mathfrak{d}_2 &= d^2x\,\partial_x + d^2y\,\partial_y + d^2z\,\partial_z, \\ &\quad \&\text{c.}, \\ \mathfrak{d}_5 &= d^5x\,\partial_x + d^5y\,\partial_y + d^5z\,\partial_z, \end{aligned}$$

where if (A, B, C) are the first differential coefficients of a function $U = (\ast f x, y, z)^m$, and λ, μ, ν are arbitrary constants, then we have

$$dx = B\nu - C\mu, \quad dy = C\lambda - A\nu, \quad dz = A\mu - B\lambda;$$

so that putting

$$\mathfrak{d} = (B\nu - C\mu)\partial_x + (C\lambda - A\nu)\partial_y + (A\mu - B\lambda)\partial_z = \begin{vmatrix} A & B & C \\ \lambda & \mu & \nu \\ \partial_x & \partial_y & \partial_z \end{vmatrix},$$

we have $\mathfrak{d}_1 = \mathfrak{d}$. The foregoing expressions of (dx, dy, dz) determine of course the values of (d^2x, d^2y, d^2z), (d^3x, d^3y, d^3z), &c., and it is throughout assumed that these values are substituted in the symbols $\mathfrak{d}_2$, $\mathfrak{d}_3$, &c., so that $\mathfrak{d}_1 = \mathfrak{d}$, and $\mathfrak{d}_2$, $\mathfrak{d}_3$, &c. denote each of them an operator such as $X\partial_x + Y\partial_y + Z\partial_z$, where (X, Y, Z) are functions of the coordinates; such operator, in so far as it is a function of the coordinates, may therefore be made an operand, and be operated upon by itself or any other like operator.

Art. Taking (a, b, c, f, g, h) for the second differential coefficients of U, $(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H})$ for the inverse coefficients, and H for the Hessian, I write also

$$\Phi = (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} f \lambda, \mu, \nu)^2,$$

$$\Psi = (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} f \partial_x, \partial_y, \partial_z)^2 \Phi,$$

$$\xi = \lambda x + \mu y + \nu z,$$

$$\nabla = (\mathfrak{A}, \ldots f \partial_x H, \partial_y H, \partial_z H)^2,$$

$$\Upsilon = (a, \ldots f \mu z - \nu y, \nu x - \lambda z, \lambda y - \mu x)^2,$$

and I notice that we have

$$\Upsilon = 2\Phi, \quad \nabla \Upsilon = 2\Psi H, \quad \eth \Upsilon = 3H,$$

the last of which is proved, post No. 65; the others are found without any difficulty.

Art. I form the Table

$$\begin{aligned}
\eth_1 U &= \frac{m-1}{(m-1)}\Phi, \\
\eth_1^2 U &= \frac{1}{(m-1)^2}\{-3(m-2)H\Phi + (m-1)\Phi\eth_1 H\}, \\
\eth_1^3 U &= \frac{1}{(m-1)^3}\{(18m^2 - 66m + 60)H^2\Phi + \ldots\}, \\
\eth_1^4 U &= \frac{1}{(m-1)^4}\{(-162m^3 + \ldots)H^3\Phi + \ldots\}, \\
\eth_1^5 U &= \ldots
\end{aligned}$$

and assuming $U = 0$,

$$\begin{aligned}
(\eth_1 H)^2 = (\eth H)^2 &= -\frac{\Psi^2}{(m-1)^2}H + \ldots, \\
\eth_1^2 H &= \frac{1}{(m-1)^2}\{(-6m^2 + 18m - 12)H\Phi^2 + \ldots\}, \\
&\&\text{c.},
\end{aligned}$$

which are for the most part given in my former memoir; the expressions for $\eth_2 U$, $\eth_3 U$, which are not explicitly given, follow at once from the equations

$$(\eth_1^2 + \eth_2)U = 0, \quad (\eth_1^3 + 2\eth_1\eth_2 + \eth_3)U = 0;$$

those for $\eth_1\eth_3 U$, $\eth_2^2 U$, and $\eth_5 U$ are new, but when the expressions for $\eth_1\eth_3 U$ and $\eth_4 U$ are known, that for $\eth_5 U$ is at once found from the equation

$$(\eth_1^5 + 6\eth_1^2\eth_3 + 4\eth_1\eth_3 + 3\eth_2^2 + \eth_5)U = 0.$$

Art. Before going further, I remark that we have identically

$$\begin{aligned}
&(a, \ldots f x, y, z)^2 (a, \ldots f \mu z - \nu y, \nu x - \lambda z, \lambda y - \mu x)^2 \\
&\qquad = (\mathfrak{A}, \ldots f \lambda, \mu, \nu f \mu z - \nu y, \nu x - \lambda z, \lambda y - \mu x)^2
\end{aligned}$$

(if for shortness $p = ax + hy + gz$, $q = \lambda x + \mu y + \nu z$)

$$= p^2(\mathfrak{A}, \ldots f\lambda, \mu, \nu)^2 - 2pq(\mathfrak{A}, \ldots f\lambda, \mu, \nu f a, b, c, 2f, 2g, 2h) + q^2(\mathfrak{A}, \ldots f a, b, c, 2f, 2g, 2h)^2.$$

Art. If in this equation we take (a, b, c, f, g, h) to be the second differential coefficients of U, and write also $(a, b, c) = (\partial_x, \partial_y, \partial_z)$, the equation becomes

$$\begin{aligned} m(m-1)U\Upsilon - (m-1)^2\Phi^2 = {} & \xi^2(x\partial_x + y\partial_y + z\partial_z)^2 \\ & - 2\xi(x\partial_x + y\partial_y + z\partial_z)\nabla + \nabla^2(\mathfrak{A}, \ldots f\partial_x, \partial_y, \partial_z)^2, \end{aligned}$$

which is a general equation for the transformation of $\mathfrak{d}_2^2$ $(= \mathfrak{d}_x^2)$.

Art. If with the two sides of this equation we operate on U, we obtain

$$m(m-1)\Upsilon U - (m-1)^2\mathfrak{d}_2^2 U = m(m-1)\Phi^2 U - 2(m-1)\Phi\nabla\Phi + \nabla^2(\mathfrak{A}, \ldots f\partial_x, \partial_y, \partial_z)^2 U,$$

and substituting the values

$$\Upsilon = 2\Phi, \quad \mathfrak{d}_1^2 U = \ldots,$$

we find the before-mentioned expression of $\mathfrak{d}_2^2 U$.

Art. Operating with the two sides of the same equation on a function H of the order m', we find

$$m(m-1)\Upsilon H - (m-1)^2\mathfrak{d}_2^2 H = m'(m'-1)\Phi^2 H - 2(m'-1)\Phi\nabla\Phi + \nabla^2(\mathfrak{A}, \ldots f\partial_x, \partial_y, \partial_z)^2 H;$$

and in particular if H is the Hessian, then writing $m' = 3m - 6$, and putting $U = 0$, we find the before-mentioned expression for $\mathfrak{d}_2^2 H$.

Art. But we may also from the general identical equation deduce the expression for $(\mathfrak{d}H)^2$. In fact taking H a function of the degree m' and writing

$$(a, \beta, \gamma) = (\partial_x H, \partial_y H, \partial_z H),$$

we have

$$\begin{aligned} m(m-1)U(a, \ldots f\mu\partial_z H - \nu\partial_y H, \nu\partial_x H - \lambda\partial_z H, \lambda\partial_y H - \mu\partial_x H)^2 - (m-1)^2(\mathfrak{d}H)^2 \\ = m'(m'-1)\Phi^2 H - 2m'\Phi\nabla H + \nabla^2(\mathfrak{A}, \ldots f\partial_x H, \partial_y H, \partial_z H)^2; \end{aligned}$$

and if H be the Hessian, then writing $m' = 3m - 6$ and putting also $U = 0$, we find the before-mentioned expression for $(\mathfrak{d}H)^2$.

Art. *Proof of equation*

$$\frac{1}{m-1}(\lambda\partial_x + \mu\partial_y + \nu\partial_z) + \ldots = \frac{\Psi}{m-1}.$$

We have

$$\ldots = (\lambda(C\partial_y - B\partial_z) + \mu(A\partial_z - C\partial_x) + \nu(B\partial_x - A\partial_y)),$$

which is

$$\lambda(C'\partial_y - B'\partial_z) + \mu(A'\partial_z - C'\partial_x) + \nu(B'\partial_x - A'\partial_y),$$

where

$$\begin{aligned} A' = \mathfrak{d}A &= a(B\nu - C\mu) + h(C\lambda - A\nu) + g(A\mu - B\lambda) \\ &= \lambda(hC - gB) + \mu(gA - aC) + \nu(aB - hA), \end{aligned}$$

with the like values for B' and C'. Substituting the values

$$(m-1)(A, B, C) = (ax + hy + gz,\ hx + by + fz,\ gx + fy + cz),$$

we have

$$(m-1)A' = \lambda(\mathfrak{G}y - \mathfrak{H}z) + \mu(\mathfrak{H}x - \mathfrak{G}z) + \nu(\mathfrak{G}x - \mathfrak{F}z);$$

and similarly

$$\begin{aligned} (m-1)B' &= \lambda(\mathfrak{H}z - \mathfrak{G}x) + \mu(\mathfrak{F}z - \mathfrak{H}x) + \nu(\mathfrak{H}x - \mathfrak{G}y), \\ (m-1)C' &= \lambda(\mathfrak{F}x - \mathfrak{B}y) + \mu(\mathfrak{A}y - \mathfrak{F}x) + \nu(\mathfrak{B}x - \mathfrak{A}y). \end{aligned}$$

0.3 341. On the Sextactic Points of a Plane Curve.

[From the *Philosophical Transactions*, vol. CLIV. (for the year 1864), pp. 545–578. Received *October 15*,—Read *December 10*, 1863.]

[Continued from previous pages.]

Art. And by means of these the condition becomes

$$0 = \frac{1}{(m-1)^4}\nu^2\big\{(153m^2 - 594m + 549)H^2 d\Phi$$

$$+ (-102m^2 + 396m + 366)\Phi dH\big\}$$

$$+\frac{1}{(m-1)^3}\nu\big\{(-96m+168)H(\partial.\nabla)H+(-90m+162)Hd'^2H+(120m-240)dH\nabla H\big\}$$

$$+ \frac{1}{(m-1)^2}\big\{9H^2\partial\Omega - 45H\Omega\partial H + 40\Omega^2\partial H\big\},$$

being, as already remarked, of the degree 5 in the arbitrary coefficients (λ, μ, ν), and of the order $12m - 22$ in the coordinates (x, y, z).

Art. But throwing out the factor ν^2 and observing that in the first line the quadric functions of m are each a numerical multiple of $51m^2 - 198m + 183$, the condition becomes

$$0 = (51m^2 - 198m + 183)H^2(3Hd\Phi - 2\Phi dH)$$

$$+\nu\big\{(-96m+168)H^2(\partial.\nabla)H+(-90m+162)H^2\partial\nabla H+(120m-240)HdH\nabla H\big\}$$

$$+ \nu^2\big\{9H^2\partial\Omega - 45H\Omega\partial H + 40\Omega^2\partial H\big\}.$$

Article Nos. 12 and 13.—Second transformation.

Art. We effect this by means of the formula

$$(m-2)(3H\partial\Phi - 2\Phi\partial H) = -\frac{3}{m-2}\text{Jac}(U, \Phi, H), \qquad \text{(J)}$$

for substituting this value of $(3H\partial\Phi - 2\Phi\partial H)$ the equation becomes divisible by ν; and dividing out accordingly, the condition becomes

$$\frac{51m^2 - 198m + 183}{m-2}\nu^2 H^2 \text{Jac}(U, \Phi, H)$$

$$+(-96m+168)\nu H^2(\partial.\nabla)H+(-90m+162)H^2\partial\nabla H+(120m-240)HdH\nabla H$$

$$+ \nu^2\big\{9H^2\partial\Omega - 45H\Omega\partial H + 40\Omega^2\partial H\big\} = 0.$$

(J) here and elsewhere refers to the Jacobian Formula, see post, Article Nos. 34 and 35.

Art. We have (see post, Article Nos. 36 to 40)

$$\mathrm{Jac}(U,\Phi,H)=-(\partial.\nabla)H;$$

and introducing also $d.\nabla H$ in place of $\partial\nabla H$ by means of the formula

$$-(6m-6)H^2(\partial.\nabla)H$$

the condition becomes

$$\begin{aligned}\frac{51m^2-198m+183}{m-2}&H^2(-(\partial.\nabla)H)\\&+(-90m+162)H^2\partial(\nabla H)+120(m-2)HdH\nabla H\\&\qquad+\nu^2\bigl\{9H^2\partial\Omega-45H\Omega\partial H+40\Omega^2\partial H\bigr\}=0,\end{aligned}$$

or, as this may be written,

$$\begin{aligned}(45m^2&-180m+171)H^2(\partial.\nabla)H\\&+(-90m+162)(m-2)H^2\partial(\nabla H)+120(m-2)^2HdH\nabla H\\&\qquad+(m-2)\nu^2\bigl\{9H^2\partial\Omega-45H\Omega\partial H+40\Omega^2\partial H\bigr\}=0.\end{aligned}$$

Article Nos. 14 to 17.—Third transformation.

Art. We have the following formulæ,

$$\mathrm{Jac}(U,\nabla H,H)-(5m-11)\partial H\nabla H+(3m-6)H\partial(\nabla H)=0, \qquad \text{(J)}$$
$$\mathrm{Jac}(U,\nabla,H)H-(5m-11)\partial H\nabla H+(3m-6)H(\partial.\nabla)H=0, \qquad \text{(J)}$$

in the latter of which, treating ∇ as a function of the coordinates, we first form the symbol $\mathrm{Jac}(U,\nabla,H)$, and then operating therewith on H, we have $\mathrm{Jac}(U,\nabla,H)H$; these give

$$H(\partial.\nabla)H=\frac{1}{5m-11}\bigl\{(3m-6)H\partial(\nabla H)-\mathrm{Jac}(U,\nabla,H)H\bigr\},$$

$$H(\partial.\nabla)H=\frac{1}{5m-11}\bigl\{(3m-6)H(\partial.\nabla)H-\mathrm{Jac}(U,\nabla,H)H\bigr\};$$

and substituting these values, the resulting coefficient of $HdH\nabla H$ is

$$(45m^2-180m+171)\frac{1}{5m-11}+(-90m+162)\frac{3m-6}{5m-11}+120(m-2)^2,$$

which is $=0$.

Art. Hence the condition will contain the factor ν, and throwing out this, and also the constant factor $\frac{1}{m-2}$, it becomes

$$\begin{aligned}(-15m^2&+60m-57)H\mathrm{Jac}(U,\nabla,H)H\\&+(30m-54)(m-2)H\mathrm{Jac}(U,\nabla H,H)\\&\qquad+(m-2)^2\bigl\{9H^2\partial\Omega-45H\Omega\partial H+40\Omega^2\partial H\bigr\}=0.\end{aligned}$$

Art. We have

$$\mathrm{Jac}(U, \nabla H, H) = \mathrm{Jac}(U, \nabla, H)H + \mathrm{Jac}(U, H\nabla, H),$$

viz. in $(\partial_x \cdot \nabla)H$, treating ∇ as a function of (x, y, z) we operate upon it with ∂_x to obtain the new symbol $\partial_x\nabla$, and with this we operate on H; in $\partial_x\nabla$ we simply multiply together the symbols ∂_x and ∇, giving a new symbol of the form $(\partial_x^2, \partial_x\partial_y, \partial_x\partial_z)$ which then operates on H. We have the like values of $\partial_y(\nabla H)$ and $\partial_z(\nabla H)$; and thence also

$$\mathrm{Jac}(U, \nabla H, H) = \mathrm{Jac}(U, \nabla, H)H + \mathrm{Jac}(U, H\nabla, H),$$

viz. in the determinant $\mathrm{Jac}(U, \nabla, H)$ the second line corresponding to ∇ is $\partial_x\nabla$, $\partial_y\nabla$, $\partial_z\nabla$ (∇ being the operand); and the Jacobian thus obtained is a symbol which operates on H giving $\mathrm{Jac}(U, \nabla, H)H$; and in the determinant $\mathrm{Jac}(U, \nabla H, H)$ the second line is $\partial_x H$, $\partial_y\nabla H$, $\partial_z\nabla H$ (∇ being simply multiplied by ∂_x, ∂_y, ∂_z respectively).

Art. Substituting, the condition becomes

$$\begin{aligned}&(-15m^2 + 60m - 57)H\mathrm{Jac}(U, \nabla, H)H \\ &\qquad + (30m - 54)(m - 2)\big\{H^2\mathrm{Jac}(U, \nabla, H)H + \mathrm{Jac}(U, H\nabla, H)\big\} \\ &\qquad\qquad + (m - 2)^2\big\{9H^2\partial\Omega - 45H\Omega\partial H + 40\Omega^2\partial H\big\} = 0,\end{aligned}$$

or, what is the same thing,

$$\begin{aligned}&(15m^2 - 54m + 51)H^2\mathrm{Jac}(U, \nabla, H)H \\ &\qquad + (30m - 54)(m - 2)H\mathrm{Jac}(U, H\nabla, H) \\ &\qquad\qquad + (m - 2)^2\big\{9H^2\partial\Omega - 45H\Omega\partial H + 40\Omega^2\partial H\big\} = 0.\end{aligned}$$

Article Nos. 18 to 27.—Fourth transformation, and final form of the condition for a Sextactic Point.

Art. I write

$$\begin{aligned}&(5m - 12)\Omega\partial H - (3m - 6)H\partial\Omega = \frac{3}{m - 2}\mathrm{Jac}(U, \Omega, H), \qquad \text{(J)} \\ &\Omega\partial H + H\partial\Omega = \partial(\Omega H).\end{aligned}$$

and, introducing for convenience the new symbol ,

$$-5H + H\partial\Omega =,$$

so that

$$\begin{vmatrix} 5m - 12, & -(3m - 6), & \frac{3}{m-2}\mathrm{Jac}(U, \Omega, H) \\ 1, & 1, & \partial(\Omega H) \\ -5, & 1, & \end{vmatrix} = 0,$$

or, what is the same thing,

$$(8m-18)+6\partial\mathrm{Jac}(U,\Omega,H)+(10m-18)\partial(\Omega H)=0,$$

we have

$$=\partial\Omega H-5\Omega\partial H=\frac{-3}{4m-9}\mathrm{Jac}(U,\Omega,H)-\frac{6(m-2)}{4m-9}\partial(\Omega H).$$

Art. We have also

$$(8m-18)H\partial\Omega-(3m-6)\Omega\partial H-\frac{3}{m-2}\mathrm{Jac}(U,\Omega,H)=0, \tag{J}$$

that is

$$H\partial\Omega=\frac{3m-6}{8m-18}\Omega\partial H+\frac{3}{(m-2)(8m-18)}\mathrm{Jac}(U,\Omega,H),$$

and thence

$$\begin{aligned}\partial H\cdot+\partial H=9H^2\partial\Omega-45H\Omega\partial H+40\Omega^2\partial H,&\\ -\frac{9(5m-9)}{(4m-9)^2}H\mathrm{Jac}(U,\Omega,H)-\frac{60(m-2)}{(4m-9)^2}\Omega\mathrm{Jac}(U,\Omega,H)&\\ +\frac{1}{(4m-9)^2}\{-27H\mathrm{Jac}(U,\Omega,H)+40\mathrm{Jac}(U,\Omega,H)\}.&\end{aligned}$$

Art. The condition thus becomes

$$\begin{aligned}(15m^2-54m+51)(4m-9)H^2\mathrm{Jac}(U,\nabla,H)H&\\ +6(5m-9)(m-2)(4m-9)H\mathrm{Jac}(U,H\nabla,H)&\\ +3(m-2)\big\{-3(5m-9)(m-2)H\partial(\Omega H)+20(m-2)^2\nabla\Omega\big\}&\\ +(m-2)^2\nu^2\{-27H\mathrm{Jac}(U,\Omega,H)+40\mathrm{Jac}(U,\Omega,H)\}=0,&\end{aligned}$$

which for shortness I represent by

$$3H+(m-2)^2\nu^2\{-27H\mathrm{Jac}(U,\Omega,H)+40\mathrm{Jac}(U,\Omega,H)\}=0,$$

so that we have

$$\begin{aligned}=(5m^2-18m+17)(4m-9)\mathrm{Jac}(U,\nabla,H)H&\\ +2(5m-9)(m-2)(4m-9)\mathrm{Jac}(U,H\nabla,H)&\\ +(m-2)\{-3(5m-9)(m-2)\partial(\Omega H)+20(m-2)^2\nabla\Omega\}.&\end{aligned}$$

Art. Write

$$\Phi'=(,,,,,\mathfrak{q}A,B,C),$$

where (A, B, C) are as before the differential coefficients of U, and (a', b', c', f', g', h') being the second differential coefficients of H, $(,,,,,)$ are the inverse coefficients, viz. $= b'c' - f'^2$, &c. We have

$$-(m-1)^2\partial^2\Phi' = (3m-6)(3m-7)\partial(\Omega H) - (3m-7)^2\partial^2\Omega$$

(see post, Nos. 41 to 46), that is

$$(3m-6)\partial(\Omega H) = (3m-7)\partial^2\Omega - \frac{(m-1)^2}{(3m-7)}\partial^2\Phi'.$$

Art. Now

$$\Phi = (\mathfrak{A},,,,,\mathfrak{q} A, B, C), \quad \Phi' = (,,,,\mathfrak{q} A, B, C),$$

and writing for shortness

$$\begin{aligned} E_\Phi &= (\mathfrak{A},\ldots \mathfrak{q}\, A', B', C), & F_\Phi &= (\mathfrak{A},\ldots \mathfrak{q}\, \partial A', \partial B', \partial C'), \\ E_{\Phi'} &= (,\ldots \mathfrak{q}\, A, B, C)^2, & F_{\Phi'} &= (,\ldots \mathfrak{q}\, A, B, C\, \mathfrak{q}\, \partial A, \partial B, \partial C), \end{aligned}$$

(we might, in a notation above explained, write $E_\Phi = \partial\Phi$, $F_\Phi = \Phi$, and in like manner $E_{\Phi'} = \partial\Phi'$, $F_{\Phi'} = \Phi'$), then we have

$$\partial^2\Phi = E_\Phi + 2F_\Phi, \quad \partial^2\Phi' = E_{\Phi'} + 2F_{\Phi'}.$$

We have moreover

$$\text{Jac}(U, \nabla H, H) = -\frac{3m-7}{(m-1)^2}F_\Phi, \quad \text{post, Nos. 47 to 50,}$$

$$\text{Jac}(U, \nabla, H)H = -E_\Phi, \quad \text{post, Nos. 51 to 53.}$$

Art. The just-mentioned formulæ give

$$\begin{aligned} = -(5m^2 - 18m + 17)(4m-9)E_\Phi & \\ - 2(5m-9)(m-2)(4m-9)\frac{(m-1)^2}{3m-7}F_\Phi & \\ + (m-2)(5m^2-18m+17)(E_\Phi + 2F_\Phi) & \\ - \frac{(5m-9)(m-1)^2(m-2)}{3m-7}(E_{\Phi'} + 2F_{\Phi'}), & \end{aligned}$$

that is

$$\begin{aligned} = -(3m-7)(5m^2-18m+17)E_\Phi & \\ + 2(m-2)(5m^2-18m+17)F_\Phi & \\ + \frac{(5m-9)(m-1)^2(m-2)}{3m-7}E_{\Phi'} & \\ - \frac{2(m-1)(m-2)(3m-8)(5m-9)}{3m-7}F_{\Phi'}. & \end{aligned}$$

Art. But recollecting that

$$\Phi = (\mathfrak{A},,,,, \mathfrak{q}\partial_x, \partial_y, \partial_z)^2 H = (\mathfrak{A},,,,, \mathfrak{q}a', b', c', 2f', 2g', 2h'),$$

and putting

$$\begin{aligned} E_\Omega &= (a\mathfrak{A}, \dots \mathfrak{q}\, a', \dots) \quad (= \partial\Omega\Phi), \\ F_\Omega &= (\mathfrak{A}, \dots \mathfrak{q}\, aa', \dots) \quad (= \partial\Omega\Psi), \end{aligned}$$

we have, post, Nos. 41 to 46,

$$\begin{aligned} -2(m-1)(m-2)F_\Phi + (3m-7)^2 E_\Phi &= (3m-6)(3m-7)HE_\Omega, \\ (m-1)(3m-8)F_\Phi + (3m-7)(3m-8)F_\Phi - (m-1)^2 F_\Phi &= (3m-6)(3m-7)HF_\Omega, \\ -(m-1)F_\Phi + (3m-7)F_\Phi &= (3m-7)\Phi F_\Omega, \end{aligned}$$

and the foregoing equation becomes

$$\begin{aligned} (3m-7) = -(5m^2 - 18m + 17)&(3m-6)(3m-7)HE_\Omega \\ &- (5m-9)(m-2)(3m-6)(3m-7)HF_\Omega \\ &\qquad + (m-2)(25m^2 - 103m + 106)(3m-7)\Phi\partial H. \end{aligned}$$

Art. But we have

$$\begin{aligned} \text{Jac}(U, H, \Omega_\Phi) - (3m-6)HE_\Omega + (2m-4)\Omega\partial H &= 0, \qquad \text{(J)} \\ \text{Jac}(U, H, \Omega_\Psi) - (3m-6)HF_\Omega + (3m-8)\Omega\partial H &= 0, \qquad \text{(J)} \end{aligned}$$

that is

$$\begin{aligned} 3(m-2)HE_\Omega &= 2(m-2)\Omega\partial H + \nu^2 \text{Jac}(U, H, \Omega_\Phi), \\ 3(m-2)HF_\Omega &= (3m-8)\Omega\partial H + \nu^2 \text{Jac}(U, H, \Omega_\Psi), \end{aligned}$$

and we thus obtain

$$\begin{aligned} = -(5m^2 - 18m + 17)&\{2(m-2)\Omega\partial H + \nu^2 \text{Jac}(U, H, \Omega_\Phi)\} \\ &- (5m-9)(m-2)\{(3m-8)\Omega\partial H + \nu^2 \text{Jac}(U, H, \Omega_\Psi)\} \\ &\qquad + (25m^2 - 103m + 106)(m-2)\Omega\partial H, \end{aligned}$$

where the coefficient of $(m-2)\Omega\partial H$ is

$$-(10m^2 - 36m + 34) - (5m-9)(3m-8) + (25m^2 - 103m + 106),$$

which is $= 0$.

Art. Hence

$$= -(5m^2 - 18m + 17)\nu^2 \text{Jac}(U, H, \Omega_\Phi) - (5m-9)(m-2)\nu^2 \text{Jac}(U, H, \Omega_\Psi).$$

Art. Substituting this in the equation

$$3H + (m-2)^2\nu^2\{-27H\,\mathrm{Jac}(U,H,H) + 40\,\mathrm{Jac}(U,\Omega,H)\} = 0,$$

the result contains the factor ν^2, and, throwing this out, the condition is

$$\begin{aligned} &3H\{-(5m^2-18m+17)\mathrm{Jac}(U,H,\Omega_\Phi) - (5m-9)(m-2)\mathrm{Jac}(U,H,\Omega_\Psi)\} \\ &\qquad + (m-2)^2\{27H^2\mathrm{Jac}(U,H,\Omega) - 40\,\mathrm{Jac}(U,H,\Omega)\} = 0, \end{aligned}$$

or, as this may also be written,

$$\begin{aligned} &-(15m^2-54m+51)H\,\mathrm{Jac}(U,H,\Omega_\Phi) - 3(5m-9)(m-2)H\,\mathrm{Jac}(U,H,\Omega_\Psi) \\ &\qquad + 27(m-2)^2\{H\,\mathrm{Jac}(U,H,\Omega_\Phi) + H\,\mathrm{Jac}(U,H,\Omega_\Psi)\} \\ &\qquad\qquad - 40(m-2)^2\mathrm{Jac}(U,\Omega,H) = 0. \end{aligned}$$

Art. Hence the condition finally is

$$\begin{aligned} &(12m^2-54m+57)H\,\mathrm{Jac}(U,H,\Omega_\Phi) + (m-2)(12m-27)H\,\mathrm{Jac}(U,H,\Omega_\Psi) \\ &\qquad - 40(m-2)^2\mathrm{Jac}(U,H,\Omega) = 0, \end{aligned}$$

or, as this may also be written,

$$\begin{aligned} &-3(m-1)H\,\mathrm{Jac}(U,H,\Omega_\Phi) + (m-2)(12m-27)H\,\mathrm{Jac}(U,H,\Omega_\Psi) \\ &\qquad - 40(m-2)^2\mathrm{Jac}(U,H,\Omega) = 0, \end{aligned}$$

viz. the sextactic points are the intersections of the curve $U = 0$ with the curve represented by this equation; and observing that U, H, Ω_Φ, and Ω are of the orders m, $3m-6$, $8m-18$ respectively, the order of the curve is as above mentioned $= 12m-27$.

Article Nos. 28 to 30.—Application to a Cubic.

Art. I have in my former memoir, No. 30, shown that for a cubic curve

$$\Omega = (\mathfrak{A},,,,,\mathfrak{q}\partial_x,\partial_y,\partial_z)^2 H = -2S\cdot U = 0;$$

this implies $\mathrm{Jac}(U,H,\Omega) = 0$, and hence if one of the two Jacobians, $\mathrm{Jac}(U,H,\Omega_\Phi)$, $\mathrm{Jac}(U,H,\Omega_\Psi)$ vanish, the other will also vanish. Now, using the canonical form we have

$$\begin{aligned} \Omega &= (\mathfrak{A},\dots\mathfrak{q}\; a',\dots) \\ &= (yz-l^2x^2,\ zx-l^2y^2,\ xy-l^2z^2,\ l^2yz-lx^2,\ l^2zx-ly^2,\ l^2xy-lz^2) \\ &= (-3l^2x,\ -3l^2y,\ -3l^2z,\ (1+2l^3)x,\ (1+2l^3)y,\ (1+2l^3)z), \end{aligned}$$

the development of which in fact gives the last-mentioned result. But applying this formula to the calculation of $\mathrm{Jac}(U, H, \Omega_\Phi)$, then disregarding numerical factors, we have

$$\begin{aligned}\partial_x\Omega_\Phi &= (yz - l^2x^2, \dots \mathfrak{q} -3l^2, 0, 0, (1+2l^3), 0, 0) \\ &= -3l^2(yz - l^2x^2) + (1+2l^3)(l^2yz - lx^2) \\ &= (-l + l^4)(x^3 + 2lyz) = 3\partial_x U;\end{aligned}$$

and in like manner $\partial_y\Omega_\Phi = 3\partial_y U$, $\partial_z\Omega_\Phi = 3\partial_z U$, and therefore

$$\mathrm{Jac}(U, H, \Omega_\Phi) = 3\mathrm{Jac}(U, H, U) = 0,$$

whence also

$$\mathrm{Jac}(U, H, \Omega_\Psi) = 0;$$

And the condition for a sextactic point assumes the more simple form,

$$\mathrm{Jac}(U, H, \Psi) = 0.$$

Art. Now (former memoir, No. 32) we have

$$\begin{aligned}\Psi &= (\mathfrak{A}, , , , , \mathfrak{q}\partial_x, \partial_y, \partial_z)^2\Phi \\ &= (1+8l^3)^2(x^6+y^6+z^6) \\ &\quad + (-9l^2)(x^3+y^3+z^3)^2 \\ &\quad + (-2l - 5l^4 - 20l^7)(x^3+y^3+z^3)xyz \\ &\quad + (-15l^3 - 78l^6 + 12l^9)x^2y^2z^2,\end{aligned}$$

or observing that $x^3 + y^3 + z^3$ and xyz, and therefore the last three lines of the expression of Ψ are functions of U $(= x^3 + y^3 + z^3 + 6lxyz)$ and H $(= -l^2(x^3+y^3+z^3) + (1+2l^3)xyz)$, and consequently give rise to the term $= 0$ in $\mathrm{Jac}(U, H, \Psi)$, we may write

$$\Psi = (1+8l^3)^2(x^6+y^6+z^6).$$

Art. We have then, disregarding a constant factor,

$$\mathrm{Jac}(U, H, \Psi) = \mathrm{Jac}(x^3+y^3+z^3,\ xyz,\ x^4y^2+y^4z^2+z^4x^2),$$

or

$$\begin{vmatrix} x^2, & y^2, & z^2 \\ yz, & zx, & xy \\ x^2(y^4+z^4), & y^2(z^4+x^4), & z^2(x^4+y^4) \end{vmatrix}$$

$$\begin{aligned}&= x^2(y^6-z^6) + y^2(z^6-x^6) + z^2(x^6-y^6), \\ &= (y^2-z^2)(z^2-x^2)(x^2-y^2),\end{aligned}$$

so that the sextactic points are the intersections of the curve

$$U = x^3 + y^3 + z^3 + 6lxyz = 0,$$

with the curve

$$x^2(y^6 - z^6) + y^2(z^6 - x^6) + z^2(x^6 - y^6) = 0,$$

or, what is the same thing,

$$(y^2 - z^2)(z^2 - x^2)(x^2 - y^2) = 0.$$

Article Nos. 31 to 33.—Proof of identities for the first transformation.

Art. **Calculation of** $(\partial_x^4 + 10\partial_x^2\partial_y\partial_z + 10\partial_x\partial_y^2\partial_z + 15\partial_y\partial_z^3)U$**.**

Writing ∂ in place of D, we have (former memoir, No. 20)

$$\partial_x^4 U = \frac{1}{(m-1)^4}\{(18m^2 - 66m + 60)\Phi\partial^2 H + (-10m + 18)(\partial.\nabla)H\}.$$

But

$$\partial^2 H = \frac{(3m-6)(3m-7)}{(m-1)^2}\Phi\partial^2 H - \frac{6m-14}{(m-1)^2}\Phi\partial H,$$

former memoir, Nos. 21 and 22; and thence

$$\begin{aligned}&(\partial_x^4 + 6\partial_x^2\partial_y\partial_z)U\\ &\qquad = \frac{1}{(m-1)^4}\{(18m^2 - 66m + 60)\Phi\partial^2 H + (-10m + 18)(\partial.\nabla)H\},\end{aligned}$$

whence operating on each side with $\partial_x = \partial$, we have

$$\begin{aligned}&(\partial_x^5 + 10\partial_x^3\partial_y\partial_z + 5\partial_x^2\partial_y\partial_z^2 + 12\partial_x\partial_y\partial_z^3)U\\ &\qquad = \frac{-1}{(m-1)^4}(18m^2 - 66m + 60)(\Phi\partial^3 H + \partial\Phi\partial^2 H)\\ &\qquad\qquad + \frac{1}{(m-1)^4}(-10m + 18)\{(\partial.\nabla)\partial H + \partial\nabla\partial H\}.\end{aligned}$$

Art. **Calculation of** $\partial_x U\partial_x H + 2\partial_y U\partial_y H + 2\partial_z U\partial_z H$**.**

We have

$$\partial_x U = \frac{1}{(m-1)^2}\partial_x H, \qquad \partial_y U = \frac{1}{m-1}\partial_y H, \quad \partial_z U =$$

$$\partial_x H' = \frac{1}{m-1}\{(-3m + 6)\Phi\partial^2 H - \Phi\partial H + \frac{1}{m-1}(\partial.\nabla)H\},$$

for which values see Appendix, No. 58. And hence the expression sought for is which is

$$
\begin{aligned}
&= 2(m-1)\partial^2\Phi\partial_x H \\
&\qquad + 2H\{(-3m+6)\Phi\partial^2 H - \Phi\partial H + \tfrac{1}{m-1}(\partial.\nabla)H\} \\
&\qquad\qquad + (m-1)\partial H\cdot\nabla H \\
&\qquad\qquad + (-6m+12)\partial^2\Phi - 3\nu^2\Phi\partial H \\
&\qquad\qquad\qquad + (m-1)^2\{2H(\partial.\nabla)H - \nu^2 H\nabla H\}.
\end{aligned}
$$

But we have, former memoir, Nos. 21 and 25,

$$
\partial_x\Phi = \frac{(3m-6)H\partial\Phi - \Phi\partial H}{(m-1)^2},\quad \partial_y\Phi = \frac{1}{m-1}\partial_y\Phi,\quad \partial_z\Phi = \frac{1}{m-1}\partial_z\Phi,
$$

so that the foregoing expression becomes

$$
\begin{aligned}
&= \frac{1}{(m-1)^3}\{-(8m-16)H^2 d\Phi + \nu^2 dH\nabla H\} \\
&\qquad - \frac{(3m-6)(3m-7)}{(m-1)^2}\left\{\tfrac{6m-14}{m-1}H^2\partial\Phi - \tfrac{3m-7}{m-1}\Phi\partial H\right\} \\
&\qquad + \frac{1}{(m-1)^2}\{-(8m-16)H^2\partial H + (m-1)\partial H\nabla H\} \\
&\qquad\qquad + \frac{1}{(m-1)^2}\{2H(\partial.\nabla)H - \nu^2 H\nabla H\};
\end{aligned}
$$

or finally

$$
\begin{aligned}
&\partial_x U\partial_x H + 2\partial_y U\partial_y H + 2\partial_z U\partial_z H = \\
&\frac{1}{(m-1)^3}\{(-6m^2+18m-12)H^2\partial\Phi + (-17m^2+60m-55)H^2\Phi\partial H\} \\
&\qquad + \frac{1}{(m-1)^3}\nu\{2(m-2)H^2(\partial.\nabla)H + (8m-16)HdH\nabla H\} \\
&\qquad\qquad + \frac{1}{(m-1)^3}\nu^2\{-H^2\partial\Omega\}.
\end{aligned}
$$

Art. **Calculation of $\partial_x^2 U\partial_x^2 U$.**

This is

$$
\frac{1}{(m-1)^4}H\partial H.
$$

Article Nos. 34 and 35.—The Jacobian Formula.

Art. In general, if P, Q, R, S be functions of the degrees p, q, r, s respectively, we have identically

$$\begin{vmatrix} pP, & qQ, & rR, & sS \\ \partial_x P, & \partial_x Q, & \partial_x R, & \partial_x S \\ \partial_y P, & \partial_y Q, & \partial_y R, & \partial_y S \\ \partial_z P, & \partial_z Q, & \partial_z R, & \partial_z S \end{vmatrix} = 0,$$

or, what is the same thing,

$$pP\mathrm{Jac}(Q,R,S) - qQ\mathrm{Jac}(R,S,P) + rR\mathrm{Jac}(S,P,Q) - sS\mathrm{Jac}(P,Q,R) = 0.$$

Art. Hence in particular if $P = U$, and assuming $U^m = 0$, we have

$$-qQ\mathrm{Jac}(R,S,U) + rR\mathrm{Jac}(S,U,Q) - sS\mathrm{Jac}(U,Q,R) = 0.$$

If moreover $Q = S^2$, and therefore $s = 1$, we have

$$-S^2\mathrm{Jac}(R,S,U) + rR\mathrm{Jac}(S,U,Q) - sS\mathrm{Jac}(U,S^2,R) = 0;$$

or, as this may also be written,

$$-S^2\mathrm{Jac}(U,R,S) + rR\mathrm{Jac}(U,R,S) - sS\mathrm{Jac}(U,S^2,R) = 0;$$

that is

$$-rR\mathrm{Jac}(U,R,S) + rRdS - sSdR = 0.$$

Art. Particular cases are

$$(2m-4)\Phi\partial H - (3m-6)H\partial\Phi = \tfrac{3}{m-2}\mathrm{Jac}(U,\Phi,H), \qquad \text{ante, No. 12,} \quad (3)$$

$$(5m-11)\nabla H\partial H - (3m-6)H\partial(\nabla H) = \tfrac{3}{m-2}\mathrm{Jac}(U,\nabla H,H), \qquad \text{''} \quad (4)$$

$$(2m-4)\nabla\partial H - (3m-6)H\partial\nabla = \tfrac{3}{m-2}\mathrm{Jac}(U,\nabla,H), \qquad \text{''} \quad (5)$$

$$(5m-12)\Omega\partial H - (3m-6)H\partial\Omega = \tfrac{3}{m-2}\mathrm{Jac}(U,\Omega,H), \qquad \text{No. 18,} \quad (6)$$

$$(8m-18)\Psi\partial H - (3m-6)H\partial\Psi = \tfrac{3}{m-2}\mathrm{Jac}(U,\Psi,H), \qquad \text{No. 19,} \quad (7)$$

$$(2m-4)\Omega\partial H - (3m-6)H\Omega\partial = \tfrac{3}{m-2}\mathrm{Jac}(U,\Omega_\Phi,H), \qquad \text{No. 25,} \quad (8)$$

$$(3m-8)\Omega\partial H - (3m-6)H\Omega\partial = \tfrac{3}{m-2}\mathrm{Jac}(U,\Omega_\Psi,H), \qquad \text{''} \quad (9)$$

where it is to be observed that in the third of these formulae I have, in accordance with the notation before employed, written $\partial.\nabla$ to denote the result of the operation ∂ performed on ∇ as operand. I have also written ∇ :

∂H to show that the operation ∇ is not to be performed on the following ∂H as an operand, but that it remains as an unperformed operation. As regards the last two equations, it is to be remarked that the demonstration in the last preceding number depends merely on the homo- geneity of the functions, and the orders of these functions: in the former of the two formulæ, the differentiation of Ω is performed upon Ω in regard to the coordinates (x, y, z) in so far only as they enter through U, and Ω is therefore to be regarded as a function of the order $2m-4$; in the latter of the two formulæ the differentiation is to be performed in regard to the coordinates in so far only as they enter through H, and Ω is therefore to be regarded as a function of the order $3m-8$. The two formulæ might also be written

$$\begin{aligned}(2m-4)\nabla\partial H-(3m-6)H\partial\Omega_\Phi &= \tfrac{3}{m-2}\mathrm{Jac}(U,\Omega_\Phi,H),\\ (3m-8)\nabla\partial H-(3m-6)H\partial\Omega_\Psi &= \tfrac{3}{m-2}\mathrm{Jac}(U,\Omega_\Psi,H);\end{aligned}$$

and it may be noticed that, adding these together, we obtain the foregoing formula,

$$(5m-12)\Omega\partial H-(3m-6)H\partial\Omega = \tfrac{3}{m-2}\mathrm{Jac}(U,\Omega,H).$$

Article Nos. 36 to 40.—Proof of equation $(\partial.\nabla)H = \mathrm{Jac}(U,H,\Phi)$, used in the second transformation.

Art. We have

$$\begin{aligned}\nabla &= (\mathfrak{A},\ldots\,\mathfrak{q}\,x,y,z\,\mathfrak{q}\,\partial_x,\partial_y,\partial_z)\\ &= (\mathfrak{A}\partial_x+\partial_y+\partial_z,\partial_x+\partial_y+\partial_z,\partial_x+\partial_y+\partial_z\,\mathfrak{q}\,\lambda,\mu,\nu).\end{aligned}$$

Also

$$\begin{aligned}\partial &= (B\nu-C\mu)\partial_x+(C\lambda-A\nu)\partial_y+(A\mu-B\lambda)\partial_z\\ &= \lambda P+\mu Q+\nu R,\end{aligned}$$

if for a moment $P, Q, R = C\partial_y-B\partial_z,\ A\partial_z-C\partial_x,\ B\partial_x-A\partial_y$. Hence

$$\partial.\nabla = (\lambda P+\mu Q+\nu R).(\mathfrak{A}\partial_x+\partial_y+\partial_z,\partial_x+\partial_y+\partial_z,\partial_x+\partial_y+\partial_z\,\mathfrak{q}\,\lambda,\mu,\nu),$$

viz. coefficient of λ^2

$$= P\mathfrak{A}\partial_x+P\partial_y+P\partial_z$$

and so for the other terms; whence also in $(\partial.\nabla)H$ the coefficients of λ^2, &c. are

$$(P\mathfrak{A}\partial_x+P\partial_y+P\partial_z)H,\ \text{\&c.}$$

Art. Again, in $\mathrm{Jac}(U,H,\Phi)$, where $\Phi = (\mathfrak{A},,,,,\mathfrak{q}\lambda,\mu,\nu)^2$, the coefficients of λ^2, &c. are $\mathrm{Jac}(U,H,\mathfrak{A})$, &c.; and hence the assumed equation

$$(\partial.\nabla)H = \mathrm{Jac}(U,H,\Phi),$$

in regard to the term in λ^2, is

$$(P\mathfrak{A}\partial_x + P\partial_y + P\partial_z)H = \mathrm{Jac}(U, H, \mathfrak{A}),$$

and we have

$$\begin{aligned}\mathrm{Jac}(U, H, \mathfrak{A}) &= \begin{vmatrix} A, & B, & C \\ \partial_x H, & \partial_y H, & \partial_z H \\ \partial_x, & \partial_y, & \partial_z \end{vmatrix} \mathfrak{A} \\ &= \{\partial_x H(C\partial_z - B\partial_y) + \partial_y H(A\partial_z - C\partial_x) + \partial_z H(B\partial_y - A\partial_x)\}\mathfrak{A} \\ &= (\partial_x H \cdot P + \partial_y H \cdot Q + \partial_z H \cdot R)\mathfrak{A};\end{aligned}$$

so that the equation is

$$P\mathfrak{A}\partial_x H + P\partial_y H + P\partial_z H = P\mathfrak{A}\partial_x H + Q\mathfrak{A}\partial_y H + R\mathfrak{A}\partial_z H,$$

or, as this may be written,

$$\begin{aligned}&[(B\partial_z - C\partial_y)\mathfrak{A} - (C\partial_z - A\partial_x)\mathfrak{A}]\partial_y H \\ &\qquad\qquad + [(B\partial_z - C\partial_y) - (A\partial_y - B\partial_x)\mathfrak{A}]\partial_z H = 0.\end{aligned}$$

Art. The coefficient of $\partial_y H$ is

$$B(\mathfrak{A}\partial_z - \partial_x) - C(\mathfrak{A}\partial_y - \partial_x),$$

which, in virtue of the identity, post. No. 40, $3a\mathfrak{A} + 3b + 3c = 0$,

$$= \mathfrak{A}\partial_x\mathfrak{A} + \partial_y\mathfrak{A} + \partial_z\mathfrak{A};$$

and in like manner the coefficient of $\partial_z H$ is

$$= -(A\partial_y\mathfrak{A} + B\partial_y + C\partial_y),$$

so that the equation is

$$(\mathfrak{A}\partial_x\mathfrak{A} + \partial_y\mathfrak{A} + \partial_z\mathfrak{A})\partial_y H - (A\partial_y\mathfrak{A} + B\partial_y + C\partial_y)\partial_z H = 0.$$

Art. But we have

$$\begin{aligned}\mathfrak{A}a + h + g &= H, \\ \mathfrak{A}h + b + f &= 0, \\ \mathfrak{A}g + f + c &= 0,\end{aligned}$$

or multiplying by x, y, z and adding,

$$(m-1)(\mathfrak{A}A + B + C) = xH;$$

whence also

$$(m-1)(\mathfrak{A}\partial_x A + \partial_x B + \partial_x C) = x\partial_x H,$$

that is

$$(m-1)(\mathfrak{A}\partial_x\mathfrak{A} + \partial_y\mathfrak{A} + \partial_z\mathfrak{A}) = x\partial_x H;$$

and in like manner

$$\begin{aligned}(m-1)(A\partial_y\mathfrak{A} + B\partial_y + C\partial_y) &= x\partial_y H,\\ (m-1)(A\partial_z\mathfrak{A} + B\partial_z + C\partial_z) &= x\partial_z H,\end{aligned}$$

whence the equation in question. The terms in λ^2 are thus shown to be equal, and it might in a similar manner be shown that the terms in $\mu\nu$ are equal; the other terms will then be equal, and we have therefore

$$(\partial.\nabla)H = \mathrm{Jac}(U, H, \Phi).$$

Art. The identity

$$\partial_x\mathfrak{A} + \partial_y + \partial_z = 0$$

assumed in the course of the foregoing proof is easily proved. We have in fact

$$\begin{aligned}\partial_x\mathfrak{A} + \partial_y + \partial_z &= \partial_x(bc - f^2) + \partial_y(fg - ch) + \partial_z(fh - bg)\\ &= b(\partial_x c - \partial_y f) + c(\partial_x b - \partial_z h) + f(-2\partial_x f + \partial_y g + \partial_z h)\\ &\quad + g(\partial_y f - \partial_z b) + h(-\partial_y c + \partial_z f),\end{aligned}$$

where the coefficients of b, c, f, g, h separately vanish: we have of course the system

$$\begin{aligned}\partial_x\mathfrak{A} + \partial_y + \partial_z &= 0,\\ \partial_x + \partial_y + \partial_z &= 0,\\ \partial_x + \partial_y + \partial_z &= 0.\end{aligned}$$

Article Nos. 41 to 46.—Proof of identities for the fourth transformation.

Art. Consider the coefficients (a, b, c, f, g, h) and the inverse set $(\mathfrak{A},,,,,)$, and the coefficients (a', b', c', f', g', h'), and the inverse set $(,,,,,)$; then we have identically

$$\begin{aligned}&(a, \dots \mathfrak{q}\; x, y, z)^2(, \dots \mathfrak{q}\; a, \dots) - (, \dots \mathfrak{q}\; ax + hy + gz, \dots)^2\\ &\quad = (a', \dots \mathfrak{q}\; x, y, z)^2(\mathfrak{A}, \dots \mathfrak{q}\; a', \dots) - (\mathfrak{A}, \dots \mathfrak{q}\; a'x + h'y + g'z, \dots)^2,\end{aligned}$$

where $(,\ldots \mathfrak{q}\ a,\ldots)$ and $(\mathfrak{A},\ldots \mathfrak{q}\ a',\ldots)$ stand for $(,,,,,\mathfrak{q} a,b,c,2f,2g,2h)$ and $(\mathfrak{A},,,,,\mathfrak{q} a',b',c',2f',2g',2h')$ respectively.

Art. Taking (a,b,c,f,g,h), the second differential coefficients of a function U of the order m, and in like manner (a',b',c',f',g',h'), the second differential coefficients of a function U' of the order m', we have

$$\begin{aligned} &m(m-1)U\cdot(,\ldots \mathfrak{q}\ \partial_x,\partial_y,\partial_z)^2 H-(m-1)^2(,\ldots \mathfrak{q}\ \partial_x U,\partial_y U,\partial_z U)^2\\ &=m'(m'-1)U'\cdot(\mathfrak{A},\ldots \mathfrak{q}\partial_x,\partial_y,\partial_z)^2 H-(m'-1)^2(\mathfrak{A},\ldots \mathfrak{q}\partial_x U',\partial_y U',\partial_z U');\end{aligned}$$

and in particular if U' be the Hessian of U, then $m'=3m-6$.

Art. Hence writing

$$\begin{aligned}\Omega_1=(\mathfrak{A},\ldots \mathfrak{q}\ \partial_x,\partial_y,\partial_z)^2 H,\quad \Omega_2=(\mathfrak{A},\ldots \mathfrak{q}\ \partial_x H,\partial_y H,\partial_z H)^2,\\ \Omega_1'=(,\ldots \mathfrak{q}\ \partial_x,\partial_y,\partial_z)^2 U,\quad \Omega_2'=(,\ldots \mathfrak{q}\ \partial_x U,\partial_y U,\partial_z U)^2,\end{aligned}$$

we have

$$m(m-1)U\Omega_2-(m-1)^2\Omega_2'=(3m-6)(3m-7)H\Omega_1-(3m-7)^2\Omega_1';$$

or if $U=0$, then

$$-(m-1)^2\Omega_2'=(3m-6)(3m-7)H\Omega_1-(3m-7)^2\Omega_1';$$

whence also

$$-(m-1)^2\partial\Omega_2'=(3m-6)(3m-7)\{H\partial\Omega_1+\Omega_1\partial H\}-(3m-7)^2\partial\Omega_1',$$

which is the formula, ante No. 21.

Art. Recurring to the original formula, since this is an actual identity, we may operate on it with the differential symbol ∂ on the three assumptions:

1. (a,b,c,f,g,h), $(\mathfrak{A},,,,,)$ are alone variable.
2. (a',b',c',f',g',h'), $(,,,,,)$ are alone variable.
3. (x,y,z) are alone variable.

Art. If in these equations respectively we suppose as before that (a,b,c,f,g,h) are the second differential coefficients of a function U of the order m, and (a',b',c',f',g',h') the second differential coefficients of a function V of the order m'; and that (A,B,C), (A',B',C') are the first differential coefficients of these functions respectively, then after some easy reductions we have

$$\begin{aligned}&=(a',\ldots \mathfrak{q}\ x,y,z\ \mathfrak{q}\ \partial_x,\ldots \mathfrak{q}\ a,\ldots)\\ &\quad-(a',\ldots \mathfrak{q}\ a'x+h'y+g'z,\ldots)^2,\\ &=(\partial a',\ldots \mathfrak{q}\ x,y,z\ \mathfrak{q}\ \mathfrak{A},\ldots \mathfrak{q}\ a',\ldots)\\ &\quad+(a',\ldots \mathfrak{q}\ x,y,z\ \mathfrak{q}\ \mathfrak{A},\ldots \mathfrak{q}\ \partial a',\ldots)\\ &\quad-2(\mathfrak{A},\ldots \mathfrak{q}\ a'x+h'y+g'z,\ldots \mathfrak{q}\ x\partial a'+y\partial h'+z\partial g',\ldots),\\ &=2(a',\ldots \mathfrak{q}\ x,y,z\ \mathfrak{q}\ \partial_x,\partial_y,\partial_z)(\mathfrak{A},\ldots \mathfrak{q}\ a',\ldots)\\ &\quad-2(\mathfrak{A},\ldots \mathfrak{q}\ a'x+h'y+g'z,\ldots \mathfrak{q}\ a'\partial_x+h'\partial_y+g'\partial_z,\ldots).\end{aligned}$$

$$\begin{aligned}(m-1)(m-2)\partial U(,\dots \mathfrak{q}\, a,\dots) + m(m-1)U(,\dots \mathfrak{q}\, \partial a,\dots)&\\ - 2(m-1)(m-2)(,\dots \mathfrak{q}\, A, B, C\, \mathfrak{q}\, \partial A, \partial B, \partial C)&\end{aligned}$$

$$\begin{aligned}m(m-1)U(\partial A',\dots \mathfrak{q}\, \mathfrak{A},\dots \mathfrak{q}\, a',\dots) - (m-1)^2(\partial A',\dots \mathfrak{q}\, A, B, C)^2&\\ = m'(m'-1)U'(a\mathfrak{A},\dots \mathfrak{q}\, a',\dots) - (m'-1)^2(,\dots \mathfrak{q}\, A', B', C)^2&\end{aligned}$$

0.4 342. On the Conics which Pass Through Three Given Points and Touch a Given Line.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. vi. (1864), pp. 24–30.]

Consider the system of conics which pass through three given points and touch a given line; if among these we select the conics which touch an assumed line, it is easy to show analytically that there are four such conics, all real or else all imaginary; viz. the three points form a triangle, and if the two lines cut the three sides produced or cut the same two sides and the third side produced, then the conics are all real; but in every other case they are all imaginary. The latter part of the theorem may also be seen geometrically; in fact, if a triangle is inscribed in a conic, say first in an ellipse, or in a parabola, or in one branch of a hyperbola, then all the tangents of the conic (and therefore any two tangents whatever) cut the three sides produced, but if the triangle is inscribed in the two branches of a hyperbola (that is, two vertices on one branch and the remaining vertex on the other branch), then all the tangents of the conic (and therefore any two tangents whatever) cut the same two sides and the third side produced: and thus the only real conics are those which cut the three sides produced, or else the same two sides and the third side produced.

The analytical proof referred to is as follows: taking $(x = 0, y = 0, z = 0)$ for the equations of the sides of the triangle, the equation of a conic through the three points is

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 0,$$

or, what is the same thing,

$$2fyz + 2gzx + 2hxy = 0,$$

that is

$$(0, 0, 0, f, g, h \mathrel{\text{ɋ}} x, y, z)^2 = 0.$$

The inverse coefficients are

$$(-f^2, -g^2, -h^2, gh, hf, fg),$$

and hence the condition in order that the conic may touch the line $\alpha x + \beta y + \gamma z = 0$ is

$$(-f^2, -g^2, -h^2, gh, hf, fg \mathrel{\text{ɋ}} \alpha, \beta, \gamma)^2 = 0,$$

or, what is the same thing,

$$\sqrt{(f\alpha)} + \sqrt{(g\beta)} + \sqrt{(h\gamma)} = 0.$$

Similarly the condition in order that the conic may touch the line $lx+my+nz=0$ is

$$\sqrt{(fl)}+\sqrt{(gm)}+\sqrt{(hn)}=0.$$

Hence if the conic touch the two lines, we have

$$\sqrt{(f\alpha)}:\sqrt{(g\beta)}:\sqrt{(h\gamma)}=\sqrt{(fl)}:\sqrt{(gm)}:\sqrt{(hn)},$$

or, what is the same thing,

$$f:g:h=\beta n+\gamma m-2\sqrt{(\beta\gamma mn)}:\gamma l+\alpha n-2\sqrt{(\gamma\alpha nl)}:\alpha m+\beta l-2\sqrt{(\alpha\beta lm)},$$

which, since the radicals must be so taken that the product may be $=\alpha\beta\gamma lmn$, gives in all four conics: and these will be all real if the signs of (l,m,n) are the same with, or opposite to those of (α,β,γ) respectively; which proves the theorem.

In particular since infinity is a line meeting the three sides produced; if the given line meet the three sides produced, the system will contain four real parabolas; but, if the given line meets two sides and a side produced, there is not any real parabola. In the latter case, as is obvious geometrically, the conics of the system are all hyperbolas.

Any side of the triangle, and the line joining the opposite vertex with the point of intersection of the side and given line, form a pair of lines passing through the three points and meeting on the given line; such pair of lines is a conic of the system; and we have thus three pairs of lines, each pair a conic of the system.

We may by what precedes form some idea of the nature of the system of conics which pass through the three given points and touch the given line. In fact writing at the point of contact the letters H, P, E, L according as the conic is a hyperbola, parabola, ellipse, or pair of lines, then if the given line cut the three sides produced, we have as in fig. 1.

[Figure 1: H, P, E, L markings along the given line]

Whereas, if the given line cuts two sides and a side produced, we have more simply as in fig. 2.

[Figure 2: H, H markings]

But to gain a more precise knowledge, it is proper to consider the curve which is the locus of the centres of the conics of the system.

Such locus which, as will presently be seen, is a curve of the fourth order, must, it is clear, pass through the points of intersection of the sides with the given line; and it is not difficult to show geometrically that it touches, at these points, the sides of the triangle. It may be shown also that the curve has three nodes (double points), viz. the middle point of each side of the triangle is a node of the curve. In fact if upon any side as base we apply an equal and opposite triangle so as to form with the given triangle a parallelogram, then any conic through the four vertices of the parallelogram will have for its centre the central point of the parallelogram; that is, the middle point of the side in question. But we may through the four vertices describe two conics, each of them touching the given line; that is the middle point of the side is the centre of two different conics of the system, and it is therefore a node upon the curve of centres. And moreover the node will be a crunode or an acnode (i.e. a double point with two real branches, or else a conjugate or isolated point) according as the conics are real or imaginary: and it is easy to see that if the given line does not cut the parallelogram, or if it cuts two opposite sides, the conics will be both real; but if it cuts two adjacent sides the conics will be both imaginary; that is, in the former case we have a crunode, and in the latter an acnode. Through each node may be drawn two tangents to the curve; and it is a known property of curves of the fourth order that the six points of contact lie on a conic; one of the tangents through the node is however the side whereon the node lies, and the points of contact of the three sides lie on a line, viz. the given line: hence the last mentioned conic is composed of the given line, and another line; that is, the three points of contact of the other tangents through the three nodes lie on this other line.

It is proper to add that the points at infinity of the curve of centres are the centres of the four parabolas; that is, there will be four infinite branches, if the parabolas are real, viz. if the given line cuts the three sides produced; but no infinite branch if the parabolas are imaginary, viz. if the given line cut two sides and a side produced.

The triangle and the three triangles applied to the three sides form together a triangle similar to the original triangle but of double the linear magnitude, and the form of the curve of centres depends as has been shown on the position of the given line in regard to the triangle and the double triangle. The cases to be considered are tolerably numerous, but it is easy from the foregoing considerations, to see in any particular case what is the form of the curve of centres.

The equation of the curve of centres was given in the late Mr Heam's "Researches on Curves of the Second Order, &c. London, 1846," viz. if $x = 0$, $y = 0$, $z = 0$ be the equations of the sides of the triangle formed by the given points; $x + y + z = 0$ the

0.5 343. On the Cusp of the Second Kind or Nodecusp.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. vi. (1864), pp. 74, 75.]

The so-called cusp of the second kind or ramphoid cusp, is not an ordinary singularity of plane curves, but it is a singularity of a higher order. It is however particularly considered in Pl"ucker's *Theorie der Analytischen Curven*, 1839; and it is there, not only in the analytical discussion of the singularities of plane curves, but in the author's theory of the generation of a curve, considered as described and enveloped by a point moving along a line which at the same time rotates round the point; when the motion along the line vanishes, we have a cusp; when the motion round the point vanishes, we have an inflexion; when the two motions vanish together, we have a cusp of the second kind, which thus presents itself as a singularity uniting the characters of a cusp or stationary point, and an inflexion or stationary tangent: (I remark in passing that in this explanation it is not clear what is the independent variable wherewith the motions are compared). But there is another point of view from which the singularity in question may be considered, viz., it may be regarded as a singularity arising from the union and amalgamation of a cusp, and a double point or node; in fact, in the figure, which represents a curve having a cusp and also a node, we have only to imagine the node approaching nearer and nearer to and ultimately coinciding with the cusp, and it will be at once seen that the point will become a cusp of the second kind; or as it might properly, with reference to this generation of it, be termed, a "nodecusp." It is to be noticed that in the point- theory of curves, there is between the cusp and the nodecusp the intermediate singularity of the tacnode, which arises from the union and amalgamation of two nodes, and possesses the character of a cusp.

I return to the nodecusp; taking the point in question as the origin, and the tangent for the axis of x, the equation will be a specialised form of the equation

$$y^2 + y(a, b, c, d \,\text{q}\, x, y)^2 + (a', b', c', d', e' \,\text{q}\, x, y)^3 + \&c. = 0,$$

which belongs to the case of a cusp; viz. (see Pl"ucker, p. 165) the conditions satisfied by the special form are $a = 0$, $a' = 3b'$, or the equation is

$$y^2+y(bx+cy)^2+y^2(3c'x+d'y)+y(4b'x^3+6c'x^2y+4d'xy^2+e'y^3)+\&c. = 0,$$

which is most easily verified, by observing that (this being so) the expansion of y in terms of x will be of the form

$$y = -\tfrac{1}{2}bx^2 + Ax^{\frac{5}{2}} + \&c.$$

It is now to be shown how the foregoing conditions $a = 0$, $a' = 3b'$, are obtained by assuming that the curve has, besides the cusp, a node which ultimately coincides with the cusp. Let (α, β) be the coordinates of the node; we must have

$$\begin{aligned} \beta^2 + \beta(a, b, c \,\mathfrak{q}\, \alpha, \beta)^2 + (a', b', c', d', e' \,\mathfrak{q}\, \alpha, \beta)^3 + \&c. &= 0, \\ (a, b, c \,\mathfrak{q}\, \alpha, \beta)^2 + (a', b', c', d' \,\mathfrak{q}\, \alpha, \beta)^3 + \&c. &= 0, \\ \beta + (b, c, d \,\mathfrak{q}\, \alpha, \beta)^2 + (b', c', d', e' \,\mathfrak{q}\, \alpha, \beta)^3 + \&c. &= 0. \end{aligned}$$

Assume $\beta = m\alpha$, and then let α vanish; the equations become in the first instance

$$\begin{aligned} \tfrac{1}{4}m^2\alpha^2 + (\tfrac{1}{4}m^2 + \tfrac{1}{2}bm + \tfrac{1}{4}a)\alpha^2 + \&c. &= 0, \\ \tfrac{1}{4}a\alpha^2 + \&c. &= 0, \\ (m + \tfrac{1}{2}b)\alpha + \&c. &= 0, \end{aligned}$$

the second and third equation give $a = 0$, $m + \tfrac{1}{2}b = 0$, and the first equation then gives $\tfrac{1}{4}m^2 + \tfrac{1}{2}bm + \tfrac{1}{4}a' = 0$; or substituting for m its value $= -\tfrac{1}{2}b$, this is $a' = 3b'^2$, or the required conditions are $a = 0$, $a' = 3b'^2$, *ut supra.* The single condition $a = 0$ corresponds to the case of the tacnode.

2, Stone Buildings, W.C., September 17, 1862.

0.6 344. On Certain Developable Surfaces.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. vi. (1864), pp. 108–126.]

If $U = 0$ be the equation of a developable surface, or say a developable, then the Hessian HU vanishes, not identically, but only by virtue of the equation $U = 0$ of the surface; that is, HU contains U as a factor, or we may write $HU = U \cdot PU$; the function PU, which for the developable replaces as it were the Hessian HU, is termed the *Prohessian*; and (since if r be the order of U the order of HU is $4r - 8$) we have $3r - 8$ for the order of the Prohessian. If $r = 4$, the order of the Prohessian is also 4, and in fact, as is known, the Prohessian is in this case $= U$. The Prohessian is considered, but not in much detail, in Dr Salmon's *Geometry of Three Dimensions*, (1862), pp. 338 and 426: the theorem given in the latter place is almost all that is known on the subject. I call to mind that the tangent plane along a generating line of the developable meets the developable in this line taken 2 times, and in a curve of the order $r-2$; the line touches the curve at the point of contact, or say the ineunt, on the edge of regression, and besides meets it in $r - 4$ points. The ineunt taken 3 times, and the $r - 4$ points form a linear system of the order $r - 1$, and the Hessian of this system (considered as a curve of one dimension, or binary quantic) is a linear system of $2r - 6$ points; viz. it is composed of the ineunt taken 4 times, and of $2r - 10$ other points. This being so, the theorem is that the generating line meets the Prohessian in the ineunt taken 6 times, in the $r - 4$ points, and in the $2r - 10$ points $(6 + r - 4 + 2r - 10 = 3r - 8)$; it is assumed that $r \geqq 5$ at least.

The developables which first present themselves are those which are the envelopes of a plane

$$(a, b, \dots \text{ɋ}\, t, 1)^n = 0,$$

where t is an arbitrary parameter, and the coefficients $(a, b, \dots)$ are linear functions of the coordinates; the equation of the developable is

$$(a, b, \dots \text{ɋ}\, t, 1)^n = 0,$$

the discriminant being taken in regard to the parameter t. Such developable is in general of the order $2n - 2$, but if the second coefficient b is $= 0$, or, more generally, if it is a mere numerical multiple of a, then a will divide out from the equation, and we have a developable of the order $2n - 3$: the like property of course exists in regard to the last but one, and the last, of the coefficients of the function. We thus obtain developables of the orders 4, 5, and 6, sufficiently simple to allow of the actual calculation of their Prohessians, and the chief object of the present Memoir is to exhibit these Prohessians; but the Memoir contains some other researches in relation to the developables in question.

Quartic Developable, Nos. 1 to 6.

Art. I consider first the developable of the fourth order

$$U = a^2d^2 - 6abcd + 4ac^3 + 4b^3d - 3b^2c^2,$$

derived from the cubic function $(a, b, c, d \mathrel{\mathsf{q}} t, 1)^3$, and which is in fact the general quartic developable.

Art. Taking (a, b, c, d) as coordinates the equation of the developable is

$$U = a^2d^2 - 6abcd + 4ac^3 + 4b^3d - 3b^2c^2 = 0.$$

Art. The first derived functions of U are

$$\begin{aligned} \partial_a U &= 2ad^2 - 6bcd + 4c^3, \\ \partial_b U &= -6acd + 12b^2d - 6bc^2, \\ \partial_c U &= -6abd + 12ac^2 - 6b^2c, \\ \partial_d U &= 2a^2d - 6abc + 4b^3. \end{aligned}$$

Art. The second derived functions are

$$\begin{array}{rlrl} \partial_a^2 U &= 2d^2, & \partial_b\partial_a U &= -6cd + 12b^2, \\ \partial_c\partial_a U &= -6bd + 24c^2, & \partial_d\partial_a U &= 4ad - 6bc, \\ \partial_b^2 U &= 24bd - 6c^2, & \partial_c\partial_b U &= -6ad - 12bc, \\ \partial_d\partial_b U &= -6ac + 12b^2, & & \\ \partial_c^2 U &= 24ac - 6b^2, & \partial_d\partial_c U &= -6ab, \\ \partial_d^2 U &= 2a^2. & & \end{array}$$

Art. The equation $U = 0$ may be written in the form

$$U = (a, b, c, d \mathrel{\mathsf{q}} ae - c^2, 2ac - 2b^2)^2 = 0,$$

where the discriminant of the quadric function is $= ae - c^2$; and it thus appears that the cuspidal curve is the conic $(ae - c^2 = 0, ac - b^2 = 0)$.

Or again, we have

$$U = 4(a, b, c, d \mathrel{\mathsf{q}} ae + 3c^2, 2ac - 2b^2)^2 - 27(ae - c^2)^2 = 0,$$

which shows that the nodal curve is the conic $(ae + 3c^2 = 0, b = 0)$.

Quintic Developable, Nos. 7 to 14.

Art. I consider the developable of the fifth order derived from the quartic function $(a, 2b, 3c, 0, -27e \mathrel{\mathsf{q}} t, 1)^4$; this is in fact the general quintic developable. Writing

$$U = a^3e^2 + 6a^2c^2e - 24ab^2ce + 9ac^4 + 16b^4e - 8b^2c^3 = 0,$$

the equation of the developable.

Art. The first derived functions of U are

$$\begin{aligned}
\partial_a U &= 3a^2e^2 + 12ac^2e - 24b^2ce + 9c^4,\\
\partial_b U &= -48abce + 64b^3e - 16bc^3,\\
\partial_c U &= 12a^2ce - 24ab^2e + 36ac^3 - 24b^2c^2,\\
\partial_e U &= 2a^3e + 6a^2c^2 - 24ab^2c + 16b^4.
\end{aligned}$$

Art. The second derived functions are

$$\begin{aligned}
\partial_a^2 U &= 6(ae^2 + 2c^2e), & \partial_b\partial_a U &= -24bce,\\
\partial_c\partial_a U &= 6(2ace - 2b^2e + 3c^3), & \partial_e\partial_a U &= 6(a^2e + 2ac^2 - 4b^2c),\\
\partial_b^2 U &= 8(-3ace + 12b^2e - c^3), & \partial_c\partial_b U &= 24(-abe - bc^2),\\
\partial_e\partial_b U &= 8(-3abc + 4b^3), & &\\
\partial_c^2 U &= 6(a^2e + 9ac^2 - 4b^2c), & \partial_e\partial_c U &= 6(a^2c - 2ab^2),\\
\partial_e^2 U &= 2a^3. & &
\end{aligned}$$

Art. Representing these by

$$\begin{array}{cccc}
A, & H, & G, & L,\\
H, & B, & F, & M,\\
G, & F, & C, & N,\\
L, & M, & N, & P,
\end{array}$$

and expressing the determinant in the partially developed form

$$\begin{aligned}
&P(ABC - AF^2 - BG^2 - CH^2 + 2FGH)\\
&\qquad - L^2(BC - F^2) - M^2(CA - G^2) - N^2(AB - H^2)\\
&\qquad - 2MN(GH - AF) - 2NL(HF - BG) - 2LM(FG - CH),
\end{aligned}$$

then, proceeding to the calculation, we have the following table of coefficients:

Art. [The calculation of the Prohessian proceeds via a detailed tabular computation of the seven component terms of the determinant. Each of the component terms divides by 576, and omitting this factor, the sum of the seven terms divides by 2. The full computation is here abbreviated.]

The resulting Prohessian, after dividing by U, is found to be

$$PU = (3a^2c, -8b^2c, ace + 2b^2e \,\mathfrak{q}\, ae - c^2, 4ac - 4b^2)^2 = 0.$$

Art. But before discussing the Prohessian, I will further consider the developable itself. Regarding it as derived from the equation $(a, 2b, 3c, 0, -27e \,\mathfrak{q}\, t, 1)^4 = 0$, we have

$$27U = (27)^2\{(ae - c^2)^2 + (-3ace + 4b^2e - c^4)\} = 0,$$

and observing that

$$-3ace + 4b^2e - c^4 = c(ae - c^2) - 4e(ac - b^2),$$

it appears that the equations of the cuspidal curve or edge of regression of the developable are $ae - c^2 = 0$, $ac - b^2 = 0$ (so that the cuspidal curve is a curve of the fourth order, the intersection of two quadric surfaces, or say a quadri-quadric curve).

This is perhaps better seen by writing the equation of the developable in the form

$$U = a(ae - c^2)^2 - 8c(ae - c^2)(ac - b^2) + 16e(ac - b^2)^2 = 0,$$

or what is the same thing

$$U = (a, c, e \between ae - c^2, 4ac - 4b^2)^2 = 0,$$

where the discriminant of the quadric function is $= ae - c^2$, which vanishes for the curve $(ae - c^2 = 0, ac - b^2 = 0)$.

Art. Another form of the equation is

$$U = a(ae + 3c^2)^2 - 8b^2(3ace - 2b^2e + c^3) = 0,$$

which shows that the conic $ae + 3c^2 = 0$, $b = 0$, is a nodal curve on the developable.

And, again, another form is

$$U = c^3(9ac - 8b^2) + e(a^3e + 6a^2c^2 - 24ab^2c + 16b^4) = 0,$$

which shows that the conic $9ac - 8b^2 = 0$, $e = 0$, is a simple line on the developable.

Art. In my paper "On the Developable Surfaces which arise from Two Surfaces of the Second Order," *Camb. and Dub. Math. Jour.*, t. v. (1850), pp. 46–57, [84], I considered first the developable having for its edge of regression the intersection of two quadric surfaces; in the general case the developable is of the order 8; but if the two surfaces have an ordinary contact it is of the order 6; and if they have a singular contact (as there explained) it is of the order 5. And in the last mentioned case, if the equations of the two quadric surfaces are taken to be $x^2 - 2wz = 0$, $y^2 - 2zx = 0$, then the equation of the developable was found to be

$$4z^3w^2 + 12z^2x^2 + 9zx^4 - 24zxy^2w - 4x^3y^2 + 8y^4w = 0,$$

which putting therein $z = a$, $y = 2b$, $x = 2c$, $w = 2e$, becomes

$$a^3e^2 + 6a^2c^2e - 24ab^2ce + 9ac^4 + 16b^4e - 8b^2c^3 = 0,$$

which is the before mentioned developable $U = 0$; the two equations $x^2 - 2wz = 0$, $y^2 - 2zx = 0$ become by the same substitution $ae - c^2 = 0$,

$ac - b^2 = 0$. We have in fact already seen that the developable $U = 0$ has this curve for its edge of regression.

Art. But in the paper just referred to, it is also shown that considering the developable which is the envelope of the common tangent planes of two quadric surfaces; in the general case the developable is of the order 8, but if the two surfaces have an ordinary contact it is of the order 6, and if they have a singular contact it is of the order 5.

In the last mentioned case the surfaces may without loss of generality be reduced to conics, and their equations may be taken to be $(y^2 - 2zx = 0, w = 0)$ and $(x^2 - 2zw = 0, y = 0)$, and this being so the equation of the developable is

$$32z^2w^3 - 24zx^2w^2 + 24zx^4w + 8zx^6 - 21x^4w - 4x^3y^2 = 0.$$

This is really a developable of the same kind with the first mentioned developable of the order 5; for writing $x = 12c$, $y = 8b$, $z = 8a$, $w = -8e$, the equation becomes

$$a^3e^2 + 6a^2c^2e - 24ab^2ce + 9ac^4 + 16b^4e - 8b^2c^3 = 0,$$

which is the before mentioned developable $U = 0$. The equations of the two conics become $(9ac - 8b^2 = 0, e = 0)$ and $(ae + 3c^2 = 0, b = 0)$, and the developable is thus the envelope of the common tangent planes of these two conics. It has been seen that the first conic is a simple line, but the second conic a nodal line, on the developable.

Art. Recapitulating, the developable of the fifth order $U = 0$, which is the envelope of the plane $(a, 2b, 3c, 0, -27e \mathfrak{q}\, t, 1)^4 = 0$ is the locus of the tangents of the quadri- quadric curve $(ae - c^2 = 0, ac - b^2 = 0)$, and it is also the envelope of the common tangent planes of the conics $(9ac - 8b^2 = 0, e = 0)$ and $(ae + 3c^2 = 0, b = 0)$.

Art. Returning now to the Prohessian, its equation may be written in the form

$$PU = (3a^2c, -8b^2c, ace + 2b^2e \mathfrak{q}\, ae - c^2, 4ac - 4b^2)^2 = 0,$$

and the discriminant of the quadric function is

$$3a^2ce(ac + 2b^2) - 96b^4e,$$

which is

$$= 3c\{(a^2e + 3b^2c)(ac - b^2) + 3ab^2(ae - c^2)\},$$

and recollecting that the equations of the cuspidal curve or edge of regression of the developable are $ae - c^2 = 0$, $ac - b^2 = 0$, it thus appears that the curve in question is also a cuspidal curve on the Prohessian.

Art. Consider for a moment the surface

$$(3a^2c, -8b^2c, ace + 2b^2e \mathfrak{q}\, ae - c^2, 4ac - 4b^2)^2 = 0,$$

which is in fact the Prohessian. The discriminant of the quadric function vanishes for the cuspidal curve of the developable, as has just been shown. Moreover, the nodal conic $(ae + 3c^2 = 0, b = 0)$ of the developable lies also on the Prohessian.

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[344—On Certain Developable Surfaces (concluded)]

where θ is an arbitrary parameter; this is a surface having for a nodal curve the cuspidal curve of the developable; but if the discriminant of the quadric function vanishes, that is if

$$3a^2ce(ac+2b^2\theta)-9b^4c^2\theta^2=0,$$

for $ae-c^2=0$, $ac-b^2=0$, then the curve in question will be a cuspidal curve on the surface.

But the last mentioned equation is

$$3c\left[(a^2e+3b^2c\theta^2)(ac-b^2)+ab^2\{(1+2\theta)\,ae-3\theta^2c^2\}\right]=0,$$

which for $ae-c^2=0$, $ac-b^2=0$, becomes $1+2\theta-3\theta^2=0$, that is $\theta=1$, which gives the Prohessian, or $\theta=-\frac{1}{3}$.

22.

For the latter value the surface is

$$(9a^2c,\ 3b^2c,\ 3ace-2b^2e \between ae-c^2,\ 4ac-4b^2)^2=0,$$

or, expanding, the surface ($\theta = -\frac{1}{3}$) is

$$\begin{array}{ll} a^4ce^2 & + \ 9 \\ a^3c^2e & + \ 30 \\ a^2b^2c^2e & - \ 104 \\ a^2c^3 & + \ 9 \\ ab^4ce & + \ 88 \\ ab^2c^4 & - \ 24 \\ b^6e & - \ 36 \\ b^4c^3 & - \ 24 \end{array}$$

$= 0$,
the before mentioned discriminant being

$$= c\{(3a^2e - b^2c)(ac - b^2) + ab^2(ae - c^2)\};$$

but I have not further examined the geometrical signification of this surface, or inquired into its relation to the Prohessian.

23.

The equation of the Prohessian may be written

$$PU = (ac - b^2)\{16e(ac - b^2)^2 + 3a(ae - c^2)^2\} + b^4U = 0,$$

or what is the same thing

$$PU = (ac - b^2)\{a(ae + 3c^2)(3ae + c^2) - 32ab^2ce + 16b^4e\} + b^4U = 0,$$

the latter of which shows that the conic ($ae + 3c^2 = 0$, $b = 0$), which is the nodal line of the developable, is a simple line on the Prohessian.

24.

Consider the curve of intersection of the developable and the Prohessian; this is of the order 5 × 7, = 35. We have $ac - b^2 = 0$, $U = 0$, or else

$$16e(ac - b^2)^2 + 3a(ae - c^2)^2 = 0, \quad U = 0.$$

Consider for a moment the second system, this is

$$\begin{aligned} 3a(ae - c^2)^2 + \quad\quad\quad\quad\quad\quad\quad\quad\quad 16e(ac - b^2)^2 &= 0, \\ 3a(ae - c^2)^2 - 24c(ae - c^2)(ac - b^2) + 48e(ac - b^2)^2 &= 0, \end{aligned}$$

which give

$$(ac-b^2)\{3c(ae-c^2)-4e(ac-b^2)\}=0, \quad U=0,$$

and these are equivalent to

$$(ac-b^2=0,\ U=0) \quad \text{and} \quad (3c(ae-c^2)-4e(ac-b^2)=0,\ U=0),$$

so that the entire intersection is made up of

$$\{4e(ac-b^2)-3c(ae-c^2)=0,\ U=0\} \text{ once.}$$

25.

The first part is at once seen to give

the cuspidal curve $(ac-b^2=0,\ ae-c^2=0)$	4 times,	order 16
the line $(a=0,\ b=0)$	4	" 4

The second part gives

$$(ae-c^2)\{4c(ac-b^2)+a(ae-c^2)\}=0,$$
$$\{4e(ac-b^2)-3c(ae-c^2)\}=0,$$

this consists of 1° the part $ae-c^2=0$, $e(ac-b^2)=0$, viz.

the cuspidal curve $(ac-b^2=0,\ ae-c^2=0)$	once,	order 4
the line $(c=0,\ e=0)$	twice,	" 2

and 2° the part

$$4c(ac-b^2)+a(ae-c^2)=0,$$
$$4e(ac-b^2)-3c(ae-c^2)=0,$$

which contains

the cuspidal curve $(ac-b^2=0,\ ae-c^2=0)$ once, order 4,

and by writing the two equations in the form

$$c(ae+3c^2)-4b^2e=0,$$
$$a(ae+3c^2)-4b^2c=0,$$

it is clear that it contains also

the nodal curve	$(ae+3c^2,\ b=0)$	twice,	order 4
and the line	$(c=0,\ e=0)$	once,	" 1

whence the complete intersection of the developable and the Prohessian is as follows, viz.

the cuspidal curve ($ac - b^2 = 0$, $ae - c^2 = 0$)	6 times,	order 24
the nodal curve ($ae + 3c^2 = 0$, $b = 0$)	2 times,	" 4
the line ($a = 0$, $b = 0$)	4 times,	" 4
the line ($c = 0$, $e = 0$)	3 times,	" 3
		35

Special Sextic Developable, Nos. 26 to 35.

26.

Lastly, we have the developable

$$U = a^4e - 18a^2bde + 27a^2d^2 - 6a^2b^2e - 27b^4e - 64b^3d^2 = 0,$$

derived from the quartic function $(a, b, 0, d, e \between t, 1)^4$; the discriminant is in fact

$$\{ae - 4bd\}^3 - 27\{-ad^2 - b^2e\}^2,$$

which is equal to the foregoing value of U.

27.

Taking a, b, d, e as coordinates, then omitting common numerical factors, the first derived functions are

$$a^3e - 9abde + 9ad^2 - 2ab^2e, \quad -3a^2de - 4a^2be - 36b^2e - 48b^2d^2,$$
$$-3a^2be + 9a^2d, \quad -9a^2bd - 27b^4 - 64b^3d,$$

and the second derived functions (changing, for greater convenience, the signs) are

$$2(-a^2e + 9bde + 3d^2), \quad 4(2ade + bd^2), \quad 4(2abe + 18d^2 + b^2de), \quad -3a^2d + 18abde + 2b^3d,$$
$$4(2ad^2 + 6d^2e), \quad 4(ad^2 + 27b^2e + 12bd^2),$$
$$4(a^2e + 2abde + 48b^2e), \quad 4(2a^2de + abd^2 + 18b^2e),$$
$$4(2abd^2 + 18ad^2 + b^2e), \quad 4(a^2e + 2abde + 48b^2e), \quad 4(27a^2d^2 + ab^2e + 32b^2d), \quad 4(2a^2be +$$
$$-3a^2d + 18abde + 2b^3d,$$
$$4(2a^2de + abd^2 + 18b^2e),$$
$$4(2a^2be + ab^3d),$$
$$2(-a^3e + 9abd + 3d^3).$$

Representing these by

$$\begin{array}{cccc} A, & H, & G, & L, \\ H, & B, & F, & M, \\ G, & F, & C, & M, \\ L, & M, & N, & F, \end{array}$$

and employing for the determinant the same partially developed form as in the first example, then proceeding to the calculation, we find

$$BC - F^2 = 16 \times \begin{array}{l} a^2de^2 + 2 \\ a^2b^2e^2 - 15 \\ a^2d^3e + 36 \\ a^2b^2d^2 - 12 \\ ab^2d^2 + 18 \\ b^4d^2 + 144 \end{array} \qquad CA - G^2 \quad = 16 \times \begin{array}{l} a^2e^3 + 2 \\ a^2d^2e - 36 \\ a^2b^2e^2 - 15 \\ a^2bd^2 + 144 \\ a^2b^2de - 13 \\ a^2b^4 + 36 \\ ab^2e + 322 \\ b^4de + 18 \end{array}$$

$$AB - H^2 = 4 \times \begin{array}{l} a^4de + 2 \\ a^3b^2e^2 + 54 \\ a^3bd^2e - 15 \\ a^2b^4e - 252 \\ a^2b^2d^2e - 12 \\ ab^4d^2 - 18 \\ b^4d^3e - 2 \end{array} \qquad AN - LG \quad = 4 \times \begin{array}{l} a^4d^2 + 2 \\ a^3bde - 36 \\ a^3b^2de + 3 \\ a^2bd^3 + 9 \\ a^2b^2d^2e + 33 \\ ab^4d^2 + 864 \\ b^4de + 16 \end{array}$$

$$FN - CM = 16 \times \begin{array}{l} a^3d^2e + 2 \\ a^3b^2e^2 + 54 \\ a^2bd^3 - 15 \\ a^2b^2de - 252 \\ ab^2d^3e - 12 \\ ab^3d^2 - 18 \\ b^4d^3e - 2 \end{array} \qquad GP - LN \quad = 16 \times \begin{array}{l} a^3de^2 + 2 \\ a^2bd^2e - 36 \\ a^2b^2de^2 - 15 \\ a^2bd^3 + 144 \\ ab^2d^3e + 322 \\ ab^3de + 18 \\ b^4d^2 - 576 \end{array}$$

$$\begin{array}{r} a^2b^2de^2 + 36 \\ a^2b^2d^2e - 41 \\ ab^4e^2 + 27 \\ a^2b^2d^2e^2 + 630 \\ a^2b^2d^2 + 864 \\ ab^4e^2 + 27 \\ ab^4de - 36 \\ -999 \\ -64\ ab^4de - 128 \\ +864 \\ +144\ b^4de^2 \\ +320\ b^4d^2 \\ -1280 \end{array} \qquad \begin{array}{r} a^3bde^2 + 64 \\ a^2b^2d^2e - 36 \\ a^2b^2d^2e^2 - 180 \\ a^2bd^3 + 144 \\ ab^4e^2 \\ a^2b^2d^2e \\ b^4d^2e \\ +351 \end{array} \qquad \begin{array}{r} BC - F^2 \\ a^2de^2 + 2 \\ a^2b^2e^2 - 15 \\ a^2d^3e + 36 \\ a^2b^2d^2 - 12 \\ ab^2d^2 + 18 \\ b^4d^2 + 144 \end{array} \qquad \begin{array}{r} CA - G^2 \\ a^2e^3 + 2 \\ a^2d^2e - 36 \\ a^2b^2e^2 - 15 \\ a^2bd^2 + 144 \\ a^2b^2de - 13 \\ a^2b^4 + 36 \\ ab^2e + 322 \\ b^4de + 18 \end{array}$$

$$= 288 \times \begin{array}{l} a^3bde^2 - 1 \\ a^3b^2e - 2 \\ a^2b^3d^2 - 2 \\ a^2b^2d^2e - 18 \\ b^4d^3e - 1 \\ -24 \end{array} \qquad \begin{array}{l} IIN - GM \\ a^3bd^2 - 1 \\ a^3b^2e - 2 \\ a^2b^3d^2 - 2 \\ a^2b^2d^2e - 18 \\ b^4d^3e - 1 \\ -24 \end{array}$$

$$= 16 \times \begin{array}{l} a^3b^2e^2 + 2 \\ a^2d^2e^2 - 36 \\ a^2b^2d^2e^2 + 3 \\ a^2bd^3e + 9 \\ a^2b^3d^2e + 33 \\ ab^4d^2 + 864 \\ b^4de + 16 \end{array} \qquad \begin{array}{l} IIF - BG \\ a^3b^2e^2 + 2 \\ a^2d^2e^2 - 36 \\ a^2b^2d^2e^2 + 3 \\ a^2bd^3e + 9 \\ a^2b^3d^2e + 33 \\ ab^4d^2 + 864 \\ b^4de + 16 \end{array}$$

$$= 4 \times \begin{array}{l} a^4d^2 + 2 \\ a^3bde - 36 \\ a^3b^2de + 3 \\ a^2bd^3 + 9 \\ a^2b^3d^2e + 33 \\ ab^4d^2 + 864 \\ b^4de + 16 \end{array} \qquad \begin{array}{l} FG - CH \\ a^4d^2 + 2 \\ a^3bde - 36 \\ a^3b^2de + 3 \\ a^2bd^3 + 9 \\ a^2b^3d^2e + 33 \\ ab^4d^2 + 864 \\ b^4de + 16 \end{array}$$

$$= 16 \times \begin{array}{l} a^3de^2 + 2 \\ a^2bd^2e - 36 \\ a^2b^2de^2 - 15 \\ a^2bd^3 + 144 \\ a^2b^3d^2e + 322 \\ ab^3de + 891 \\ b^4d^2 - 576 \end{array} \qquad \begin{array}{l} HP - LM \\ a^3de^2 + 2 \\ a^2bd^2e - 36 \\ a^2b^2de^2 - 15 \\ a^2bd^3 + 144 \\ a^2b^3d^2e + 322 \\ ab^3de + 891 \\ b^4d^2 - 576 \end{array}$$

where, by way of verification, I have annexed to each column the sum of the coefficients.

29.

Forming from these the seven component terms of the determinant, and collecting, the result divides by 5, and we have

	a^4e^5	a^3bd^2e	$a^3b^2de^2$	$a^2b^2d^3$	$a^2b^3d^2e$	$a^2b^4e^2$	ab^4d^3
16	+	+	+	−	−	+	−
432	+	+	−	−	−	−	−
96	+	−	−	−	+	−	−
3312	+	+	+	−	−	−	−
7776	+	+	+	−	−	−	−
288	−	+	−	−	+	−	−
3876	+	+	+	−	−	−	−
4788	+	+	+	−	−	−	−
7776	+	+	+	−	−	−	−
576	−	+	−	−	+	−	−
	2524	−	−	+	+	−	−
	−	133272	−	+	1944	+	−
	−	+	63432	+	6504	+	−
	−	+	−	65088	+	2592	−
	−	+	−	227808	−	668472	−
	+	−	1152	+	72576	+	13680
	−	−	20864	−	331776	−	741888
	2032452						

where the first column is the Hessian.

	a^4e^5	a^3bd^2e	$a^3b^2de^2$	$a^2b^2d^3$	$a^2b^3d^2e$	$a^2b^4e^2$	ab^4d^3
16	+	+	+	−	−	+	−
432	+	+	−	−	−	−	−
96	−	+	−	−	+	−	−
2232	+	+	+	−	−	−	−
432	−	+	−	−	+	−	−
8100	+	+	+	−	−	−	−
7788	+	+	+	−	−	−	−
−	1368	36220	−	51984			
3960	+	+	+	−	−	−	−
	+	177576	−	41544	−	30024	−
	−	872592	−	6912	+	+	−
	186624	−	−	−	−	−	
	−	18504	−	62336	+	+	−
	5184	−	−	−	−	−	

	a^4e^5	a^3bd^2e	$a^3b^2de^2$	$a^2b^2d^3$	$a^2b^3d^2e$	$a^2b^4e^2$	ab^4d^3
16	+	+	+	−	−	+	−
288	−	+	−	−	+	−	−
96	−	+	−	−	+	−	−
2232	+	+	+	−	−	−	−
432	−	+	−	−	+	−	−
8100	+	+	+	−	−	−	−
7788	+	+	+	−	−	−	−
−	1368	36220	−	51984			
3960	+	+	+	−	−	−	−
	+	177576	−	41544	−	30024	−
	−	872592	−	6912	+	+	−
	186624	−	−	−	−	−	
	−	18504	−	62336	+	+	−
	5184	−	−	−	−	−	

30.

This divides, as it should do, by U. Writing

$$\begin{aligned} L &= 4e(ac-b^2)^2 - 3c(ae-c^2)(ac-b^2) + a(ae-c^2)^2,\\ M &= ae+2bd,\\ N &= 3(ae-c^2)^2 + 16ce(ac-b^2) + 4(ae-c^2)(ac-b^2), \end{aligned}$$

the developable is $L^2 - 27M^2N = 0$, and the other factor, which is the Prohessian, is

	a^4e^5	a^3bd^2e	$a^3b^2de^2$	$a^2b^2d^3$	$a^2b^3d^2e$	$a^2b^4e^2$
16	+	+	+	−	−	+
44	+	−	−	−	−	−
−	99	−	−	−	−	−
3226	+	−	−	−	−	−
−	2592	−	−	−	−	−
−	972	−	−	−	−	−
−	99	+	41744	+	27180	−
	27216	2592	−	−	−	−
	−	117200	−	61920	+	124416
	+	27180	−	154912	−	19008
	−	972	−	61920	+	689024
	+	92160	+	−	27216	−
	−	19008	+	40960	+	124416
	+	92160	−	409600	+	350649

a^4e^5	a^3bd^2e	$a^3b^2de^2$	$a^2b^2d^3$	$a^2b^3d^2e$	$a^2b^4e^2$
20736	+	+	+	+	+
20736	+	+	−	+	−
41472	+	−	−	−	−
373248	+	−	−	−	−
5184	−	−	−	−	−
41472	+	+	−	−	−
+	186624	+	+	+	−
20736	+	+	−	+	−
+	373248	+	+	+	−
+	1689984	+	+	+	−
20736	+	+	−	+	−

31.

It may be added that the developable is also a cuspidal line of the Prohessian.

Writing the equations of the developable and the Prohessian under the forms

$$L^2 - 27M^2N = 0, \quad B^2(L^2 - 4M^2) = 0,$$

and substituting in the second equation $N = 27\frac{M^2}{L^2}$, it becomes $B^2(L - 4M) = 0$, that is the intersection is made up of $L = 0$, $B^2 = 0$, which is the cuspidal curve taken six times (order 36), and of the curve $L^2 - 27M^2 = 0$, $L - 4M = 0$ (order 24). But substituting for L, M their values, the equation $L - 4M = 0$ becomes

$$4e(ac - b^2)^2 - 3c(ae - c^2)(ac - b^2) + a(ae - c^2)^2 - 4(ae + 2bd) = 0,$$

so that the last mentioned curve is composed of the intersections of the developable with a surface of the order 4.

32.

Now combining with the equation of the developable the equation $ae+2bd = 0$, and observing that in consequence of the last mentioned equation we have

$$ae - c^2 = -(2bd + c^2), \quad ac - b^2 = ac - b^2,$$

the equation of the developable gives $(ad^2 - b^2e)^2 = 0$, or we have (taken twice) the curve

$$ae + 2bd = 0, \quad ad^2 - b^2e = 0,$$

which consists of the two lines $(a = 0,\ b = 0)$, $(d = 0,\ e = 0)$, and of a quartic curve (an excubo-quartic[1]) the nodal line on the developable. If in like manner with the equation of the developable we combine the equation $ae - 6bd = 0$, then from this equation we have

$$ae - c^2 = 6bd - c^2, \quad ac - b^2 = ac - b^2,$$

and the equation of the developable then gives

$$(27b^2d^2 + 4c^3d)(ac - b^2) + (6bd - c^2)^2(6bd + 2ac) = 0,$$

which is a quartic curve (another excubo-quartic), the other nodal line on the developable.

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[1] A quartic curve which is the complete intersection of two quadric surfaces is termed a quadriquadric; a quartic curve of the kind which is not such complete intersection but can only be represented by means of a cubic surface is termed an excubo-quartic.

33.

Hence we see that the intersection of the developable with the surface $L - 4M = 0$ is made up of the same two lines and of an excubo-quartic.

34.

Hence we see that the intersection of the developable with the surface $L - 4M = 0$ is made up of

nodal curve (excubo-quartic)	$ae + 2bd = 0$, $ad^2 - b^2e = 0$
excubo-quartic	$ae - 6bd = 0$, $ad^2 + db^2e = 0$

each twice; and in regard to the Prohessian, that the intersection of the developable with this surface is made up of

cuspidal curve	$(ac - b^2 = 0$, $ae - c^2 = 0)$
nodal curve (excubo-quartic)	$ae + 2bd = 0$, $ad^2 - b^2e = 0$
excubo-quartic	$ae - 6bd = 0$, $ad^2 + db^2e = 0$

each twice.

35.

It is to be added that a generating line of the developable meets the Prohessian in the ineunt on the cuspidal edge taken 6 times, in a point of the nodal curve taken 2 times, in a point of the excubo-quartic $ae - 6bd = 0$, $ad^2 + db^2e = 0$, and in a point of the excubo-quartic $ae - 6bd = 0$, $ad^2 + \frac{4}{3}b^2e = 0$, (these being the $2r - 10$ points of the general theorem); and that the edge of regression meets the Prohessian in the ineunt taken 15 times, and in the cusp of the excubo-quartic $ae - 6bd = 0$, $ad^2 + \frac{4}{3}b^2e = 0$; that is in the points of the nodal curve 2 times, and in the points of the two excubo-quartics 1 time respectively: total 15 times, agreeingly with the general theorem.

Cambridge, 1863.

[345]

ON THE INFLEXIONS OF THE CUBICAL DIVERGENT PARABOLAS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. vi. (1864),
pp. 199–208.]

The five divergent parabolas, species 67 to 71, of Newton's *Enumeratio Linearum* tertii Ordinis (1704), are included under the general equation $y^2 = ax^3 + 3bx^2 + 3cx + d$: there are two general forms, or forms without singularities, viz. the parabola cum ovali, sp. 67, and the parabola pura, sp. 71; two forms having a double point, viz. the nodata, sp. 68, and the punctata, sp. 69, according as the double point is one with real or imaginary tangents, that is according as the singularity is a crunode or an acnode; and a form with a cusp, viz. the cuspidata, sp. 70.

1.

Considering the series of curves $y^2 = ax^3 + 3bx^2 + 3cx + d$, the condition for an inflexion is at once found to be

$$ay^2 - (ax + b)(ax^2 + 2bx + c) = 0,$$

as the equation of a curve meeting the given curve

$$y^2 = ax^3 + 3bx^2 + 3cx + d,$$

in its points of inflexion; and if for greater simplicity we assume $a = 1$, $d = 0$ (the latter equation means obviously that the origin is taken at one of the three inter- sections of the curve with the axis of x, say the real one, if the intersections are one real, two imaginary), then the equation of the curve is

$$y^2 = x(x^2 + 3bx + 3c),$$

and the inflexions are given as the intersections of the curve with the curve

$$xy^2 = -bx^3 - 3cx^2 + 3c^2.$$

There is, it is clear, an inflexion at the point at infinity on the line $x = 0$; and eliminating y^2 we find

$$x^4 + 3bx^3 + 3cx^2 = -bx^3 - 3cx^2 + 3c^2,$$

or, what is the same thing,

$$x^4 + 4bx^3 + 6cx^2 - 3c^2 = 0,$$

a quartic equation giving the four ordinates through the remaining eight inflexions.

If the curve has a cuspidal point, then the origin will be at the cusp, and we have $b = 0$, $c = 0$, and the quartic equation becomes $x^4 = 0$; that is, the four ordinates pass through the cusp.

If the curve have a node, then taking the origin at the node we have $c = 0$; the equation of the curve is

$$y^2 = x^2(x + 3b),$$

and the curve has a crunode or an acnode according as b is positive or negative; the quartic equation becomes

$$x^2(x + 4b) = 0,$$

and the factor $x^2 = 0$ gives three ordinates through the node; the remaining factor $x + 4b = 0$ gives the ordinate through the two inflexions; and substituting this value of x in the equation of the cubic, we find

$$y^2 = -16b^3,$$

and the resulting values of y (consequently also the inflexions) are imaginary if b be positive, that is, for the crunodal form; but real if b be negative, that is, for the acnodal form. It is to be observed that the indefinite ordinate $x + 4b = 0$ or $x = -4b$ is real in each of the two cases: in the crunodal case, the ordinate lies outside the curve, that is beyond the loop; in the acnodal case inside the curve, that is on the opposite side to the acnode in regard to the vertex; and using $3b$ to denote the distance (taken positively) of the vertex from the node, (that is, in the crunodal case changing the sign of b), the distance (taken positively) of the ordinate from the vertex is $= 4b - 3b$, $= b$, $= \frac{1}{3} \cdot 3b$, that is, it is one-third of the distance of the vertex from the node.

2.

Recurring to the general equation

$$y^2 = x^3 + 3bx^2 + 3cx,$$

if in the quartic equation

$$x^4 + 4bx^3 + 6cx^2 - 3c^2 = 0$$

we write $x = z - b$, this becomes

$$z^4 + \qquad cz^2 - 4b^2z + (3b^2c - 3c^2) = 0,$$

or, as this may also be written,

$$z^4 + \qquad cz^2 - 4b^2z + 3c(b^2 - c) = 0,$$

and the question of the reality of the roots of this equation determines the reality of the inflexions.

3.

The case of the parabola pura, sp. 71. Here b is negative, $b^2 - c$ is negative, that is c is positive and greater than b^2. The last term is therefore negative, and the roots are $+, +, -, -$, or else two of them are imaginary. To determine which is the case, form the discriminant of the quartic function; the value is $= -27c^2(4b^2 - c)^2(27b^4 - 18b^2c + c^2)$. Hence unless $c = 0$, $c = 4b^2$, or $c = b^2(9 \pm \sqrt{72})$, the roots are all unequal. Now $c = 0$ gives the cuspidal form, and $c = 4b^2$ gives the nodal form; hence except for the cuspidal or nodal form the roots are all unequal, and we have $+, +, -, -$ or else two roots imaginary. In the case $c = 4b^2$ of the parabola nodata, the quartic equation becomes

$$(z - 2b)^2(z^2 + 4bz + 6b^2) = 0,$$

and the roots are $z = 2b$ (twice), and $z = -2b(1 \pm i)$, so that the roots are $+, +, -, -$ (disregarding the equality of the positive roots); and the same is the case in the limit for $c = 0$ of the parabola cuspidata. We have thus the theorem:

In the parabola pura (including as limiting cases the parabola nodata and the parabola cuspidata) the inflexions are always two real and six imaginary.

4.

The case of the parabola cum ovali, sp. 67. Here b is positive. If c is negative, the last term is positive; hence the roots are all real or else all imaginary, and they are unequal unless $27b^4 - 18b^2c + c^2 = 0$, that is $c = b^2(9 \pm \sqrt{72})$;

but these values are impossible, since $9-\sqrt{72}=9-8.485=0.515$, which is positive and therefore gives c positive; and $9+\sqrt{72}$ also gives c positive. Hence the roots are always unequal; and they cannot be all imaginary since the last term being positive the product of the roots is positive. Hence the roots are all real. The curve has therefore four real inflexions and the oval has four real inflexions; these last are in fact easily seen to lie on the oval; and the theorem is:

In the parabola cum ovali, b positive, c negative, the inflexions are all real, viz. there are four real inflexions on the infinite branch and four real inflexions on the oval.

5.

If c is positive, then the last term is negative; hence the roots are $+,+,-,-$ or two imaginary; they are unequal unless $27b^4-18b^2c+c^2=0$, that is $c=b^2(9\pm\sqrt{72})$; and we have the two values of c which are both of them positive and greater than b^2. For $c=b^2(9-\sqrt{72})=0.515\ldots\times b^2$, the roots are all real; and for c between this value and b^2, the last term is negative, and there are two positive roots and two negative roots. Hence in this case the roots are always $+,+,-,-$, and the inflexions are two real and six imaginary. For $c=b^2(9+\sqrt{72})$, the roots are again all real; but for c between this value and ∞, there are two positive and two negative roots. Hence in this case also the roots are $+,+,-,-$, and the inflexions are two real and six imaginary.

In the parabola cum ovali, b positive, c positive, the inflexions are two real and six imaginary.

6.

In the case $b=0$, we have the curve $y^2=x^3+3cx$, which belongs to the parabola pura, and the inflexions are two real and six imaginary. The case $c=0$ is the cuspidal form, which has been already referred to.

7.

The foregoing results may be collected as follows; the parabola cum ovali, b positive, c negative, gives 8 real inflexions; the parabola cum ovali, b positive, c positive, gives 2 real and 6 imaginary inflexions; the parabola pura, b negative, gives 2 real and 6 imaginary inflexions; the parabola nodata, gives 2 real and 6 imaginary inflexions; and the parabola cuspidata, gives 2 real and 6 imaginary inflexions.

Cambridge, June 2, 1863.

[346]

NOTE
ON AN EXPRESSION FOR THE RESULTANT OF TWO BINARY CUBICS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. vi. (1864),
pp. 210–212.]

1.

The resultant of two binary cubics may be expressed in the following form. Let the cubics be

$$(a, b, c, d \between x, y)^3 = 0, \quad (a', b', c', d' \between x, y)^3 = 0,$$

and consider the function

$$\left|\begin{array}{cccc} a & b & c & d \\ a' & b' & c' & d' \\ \cdot & a & b & c \\ \cdot & a' & b' & c' \end{array}\right|$$

which is formed from the matrix of the coefficients by taking any two lines and the two lines next following in order; we have in all six determinants, viz.

$$\begin{array}{llllll} 12.34 & = ac' - a'c, & 12.23, & 23.34, & = bd' - b'd, & 23.23, \\ 12.45 & = ad' - a'd, & 12.34, & 34.45, & = cd' - c'd, & 34.34, \\ 23.45 & = bc' - b'c, & 23.34, & 12.45, & = ad' - a'd, & 12.12, \end{array}$$

where the first column gives the determinant, the second column its value, the third column the corresponding determinant of the suffixes, and the fourth column the value of this last determinant; the six values are of course not independent, but are connected by the identical equations

$$(12.23)(34.45) + (23.34)(45.12) + (34.45)(12.23) = 0.$$

2.

The resultant is expressed in terms of these quantities by the formula

$$= (12.45)^2(23.34)^2(34.45)^2 + \&c. + 18(12.23)(23.34)(34.45)(45.12)(12.34)(23.45)$$

$$-4(12.23)(34.45)^3 \cdot (23.34)^2 - \&c.,$$

where the &c. denotes the symmetrical terms obtained by the cyclical substitutions of the suffixes $1, 2, 3, 4, 5$.

But the two equations are

$$(a, b, c, d \between x, y)^3 = 0, \quad x^3 + y^3 = 0,$$

the last of which gives $y = -x$, $y = -\omega x$, $y = -\omega^2 x$, if ω be an imaginary cube root of unity, and hence the Resultant is

$$= (a - 3b + 3c - d)(a - 3b\omega + 3c\omega^2 - d)(a - 3b\omega^2 + 3c\omega - d),$$

which is

$$= (\theta - 3b + 3c)(\theta - 3b\omega + 3c\omega^2)(\theta - 3b\omega^2 + 3c\omega),$$

or finally is

$$= \theta^3 - 27b^3 + 27c^3 + 27bc\theta,$$

and the formula is thus verified.

If the two cubics are taken to be

$$(a, b, c, d \between x, y)^3 = 0, \quad (b, c, d, e \between x, y)^3 = 0,$$

then the formula gives for the Discriminant of the quartic function $(a, b, c, d, e \between x, y)^4$ a new expression, which however does not appear to be an elegant one.

[347]

ON THE NOTION AND BOUNDARIES OF PENCILS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. vi. (1864),
pp. 209–210.]

The notion of a Pencil as used by me in my paper "On the Curves which satisfy given conditions" *Phil. Trans.* vol. cliii. (1863), pp. 75–84, [335], and in my paper "On the doctrine of Involution as applied to a Pencil of Lines" *Quarterly Math. Journ.* vol. v. (1862), pp. 354–364, [326], was as follows. Considering any number of lines drawn through a given point, the group of such lines is a Pencil; and if the lines are given by their equations $U = 0$, $V = 0$, $W = 0$, &c., then the pencil is said to be given by these equations. The pencil may be represented by a single equation $\lambda U + \mu V + \nu W + \ldots = 0$, where λ, μ, ν, &c. are arbitrary parameters; and the question then arises, what are the boundaries of the pencil, that is, what are the conditions to be satisfied by the coefficients in order that the equation may represent a line of the pencil.

In the case of a pencil of two lines, the equation is $\lambda U + \mu V = 0$, where λ, μ are arbitrary; and there are no boundaries, for the equation represents a line through the intersection of $U = 0$, $V = 0$, that is, a line of the pencil.

In the case of a pencil of three lines, the equation is $\lambda U + \mu V + \nu W = 0$; this may be written $(\lambda U + \mu V) + \nu W = 0$, and represents a line through the intersection of $\lambda U + \mu V = 0$ and $W = 0$; that is, through the intersection of $W = 0$ with any line of the pencil of the first two lines $U = 0$, $V = 0$; or it represents any line through any one of the three intersections of the lines $U = 0$, $V = 0$, $W = 0$, taken two and two; that is, any line through any one of the three vertices of the triangle formed by the three lines; and the boundaries are the three lines themselves $U = 0$, $V = 0$, $W = 0$.

In the case of a pencil of four lines, the equation is $\lambda U + \mu V + \nu W + \rho R = 0$; this may be written $(\lambda U + \mu V) + (\nu W + \rho R) = 0$, and it represents a line through the intersection of a line of the pencil of $U = 0$, $V = 0$ with a line of the pencil of $W = 0$, $R = 0$; that is, any line through any one of the four intersections of the lines $U = 0$, $V = 0$, $W = 0$, $R = 0$, taken two and two; or any line through any one of the six vertices of the quadrilateral formed by the four lines; and the boundaries are the four lines themselves $U = 0$, $V = 0$, $W = 0$, $R = 0$.

2.

The foregoing notion of a pencil appears to me to be the proper and most convenient one; but it is to be observed that the notion has been extended by Steiner and subsequent writers, so as to include the case of any system of lines through a given point, whether given by their equations or not; and in particular the pencil of two lines is extended to include any two lines through the point, and the pencil of three lines is extended to include any three lines through the point. The boundaries of the pencil are in this case not the lines themselves, but the lines which are the limits of the pencil; and in the case of a pencil of four lines, the boundaries are the four lines which are the limits of the pencil, that is, the four lines which are the sides of the quadrilateral formed by the four lines.

3.

It is to be remarked that the notion of a pencil of lines is analogous to that of a range of points; and that the notion of a range of points is also extended by Steiner and subsequent writers, so as to include any system of points on a given line, whether given by their equations or not. The range of two points is extended to include any two points on the line, and the range of three points is extended to include any three points on the line; and the boundaries of the range are the points which are the limits of the range.

[348]

ON THE THEORY OF INVOLUTION.

[From the *Transactions of the Cambridge Philosophical Society*, vol. XI. Part I, (1866),
pp. 21–38.]

Read February 22, 1864.]

Three or more quantics which satisfy identically a linear equation such as

$$\lambda U + \lambda' U' + \lambda'' U'' + \ldots = 0,$$

where λ, λ', λ'', ... are constants, are said to be in *Involution*. In particular any quantic $U+kV$, where k is a constant, is in involution with the quantics U, V; and the entire system of such quantics, k having any value whatever, is a system in involution with the quantics U, V. And in like manner the equation $U+kV=0$, or say the curve $U+kV=0$, is said to be in involution with the curves $U=0$, $V=0$.

1.

Consider two binary quantics of the same order n, $U = (a, b, \ldots \between x, y)^n$, $V = (a', b', \ldots \between x, y)^n$; and write $W = U + kV = (a + ka', \ldots \between x, y)^n$; then the condition in order that W may have a twofold factor (that is, in order that the curve $W=0$ may have a node) is expressed by a plexus of equations which is equivalent to two relations between the coefficients $a+ka'$, $b+kb'$, &c.; and eliminating k between these two equations we find between the coefficients a, b, &c., a', b', &c. a single equation $\Omega = 0$, which is of the degree $2(n-1)$ in regard to the two sets of coefficients respectively. The equation $\Omega = 0$ is the condition in order that the curves $U=0$, $V=0$ may be such that a curve in involution with them may have a node.

2.

In particular, if the quantics are of the third order, the condition is of the degree 4; and if the quantics are of the fourth order, the condition is of the degree 6.

3.

The condition may be otherwise expressed as follows. If from the three equations

$$\partial_x W = 0, \quad \partial_y W = 0, \quad W = 0,$$

we eliminate (x, y), we obtain the discriminant of W, which is of the degree $2n-1$ in regard to the coefficients of W; and the equation $\text{Disc}^t . \, W = 0$, considered as an equation in k, gives the $2n-1$ values of k for which the curve $W = 0$ has a node. But the discriminant of W is the product of the twofold factor condition (that is, the plexus condition above mentioned) into a factor which is independent of k; and this factor is the $n-1$-th power of the resultant of U and V. Hence the equation $\Omega = 0$ is obtained by dividing the discriminant of W by the $(n-1)$-th power of the resultant of U and V.

4.

In the case of two binary cubics, the discriminant of W is of the degree 6 in regard to the coefficients of W; and the resultant of U and V is of the degree 3 in regard to the coefficients of W; hence the equation $\Omega = 0$ is of the degree $6 - 2 \times 3 = 4$ in regard to the coefficients of W, which agrees with the foregoing result.

5.

In the case of two binary quartics, the discriminant of W is of the degree 8 in regard to the coefficients of W; and the resultant of U and V is of the degree 4 in regard to the coefficients of W; hence the equation $\Omega = 0$ is of the degree $8 - 2 \times 4 = 6$ in regard to the coefficients of W, which agrees with the foregoing result.

6.

The condition $\Omega = 0$ may be expressed in a different form by means of the theory of covariants. If U, V are two binary quantics of the order n, then the Jacobian of U and V is a covariant of the order $2(n-1)$; and the condition that the Jacobian may have a μ-fold factor is a condition of the degree $2(n-1)-\mu+1$ in regard to the coefficients of U and V respectively. In particular, if $\mu = 2$, the condition is of the degree $2(n-1)-1 = 2n-3$ in regard to the coefficients of U and V respectively; and this condition is the same as the condition that the curves $U + kV = 0$ may have a node for some value of k.

7.

The condition may also be expressed by means of the catalecticant. If U, V are two binary quantics of the order n, then the catalecticant of $U + kV$ is a determinant of the order n, whose elements are linear functions of the coefficients of $U + kV$; and the condition that the catalecticant may vanish is a condition of the degree n in regard to the coefficients of $U + kV$. But this condition is not the same as the condition $\Omega = 0$; it is a different condition, which is satisfied when the quantic $U + kV$ has a twofold factor of a particular kind.

8.

The foregoing results may be extended to the case of ternary quantics. If U, V are two ternary quantics of the same order n, then the condition in order that the curve $W = U + kV = 0$ may have a node is expressed by a plexus of equations equivalent to three relations between the coefficients of W; and eliminating k between these equations we find between the coefficients of U and V an equation $\Omega = 0$ of the degree $3(n-1)^2$ in regard to the two sets of coefficients respectively.

9.

In particular, if the quantics are of the third order, the condition is of the degree $3 \times 4 = 12$; and if the quantics are of the fourth order, the condition is of the degree $3 \times 9 = 27$.

10.

The condition may be otherwise expressed as follows. The curve $W = U + kV = 0$ will have a node if the three equations

$$\partial_x W = 0, \quad \partial_y W = 0, \quad \partial_z W = 0,$$

are satisfied for the same values of (x, y, z). But if these equations are satisfied, then $W = 0$ is also satisfied, for W is a homogeneous function of (x, y, z) of the order n; and by Euler's theorem, $nW = x\partial_x W + y\partial_y W + z\partial_z W$. Hence the curve $W = 0$ will have a node if the three equations $\partial_x W = 0$, $\partial_y W = 0$, $\partial_z W = 0$ are satisfied for the same values of (x, y, z).

11.

Eliminating (x, y, z) from these three equations, we obtain the discriminant of W, which is of the degree $3(n-1)^2$ in regard to the coefficients of W; and the equation $\text{Disc}^t\,.\; W = 0$, considered as an equation in k, gives the $3(n-1)^2$ values of k for which the curve $W = 0$ has a node. But the discriminant of W is the product of the twofold factor condition into a factor which is independent of k; and this factor is the $(n-1)$-th power of the resultant of U and V. Hence the equation $\Omega = 0$ is obtained by dividing the discriminant of W by the $(n-1)$-th power of the resultant of U and V.

12.

In the case of two ternary cubics, the discriminant of W is of the degree 12 in regard to the coefficients of W; and the resultant of U and V is of the degree 9 in regard to the coefficients of W; hence the equation $\Omega = 0$ is of the degree $12 - 2 \times 9 = 12$ in regard to the coefficients of W, which agrees with the foregoing result.

13.

The condition $\Omega = 0$ may be expressed in a different form by means of the theory of covariants. If U, V are two ternary quantics of the order n, then the Jacobian of U, V is a covariant of the order $3(n-1)$; and the condition that the Jacobian may have a μ-fold factor is a condition of the degree $3(n-1) - \mu + 1$ in regard to the coefficients of U and V respectively. In particular, if $\mu = 2$, the condition is of the degree $3(n-1) - 1 = 3n - 4$ in regard to the coefficients of U and V respectively; and this condition is the same as the condition that the curves $U + kV = 0$ may have a node for some value of k.

14.

The condition may also be expressed by means of the Hessian. If U, V are two ternary quantics of the order n, then the Hessian of $U + kV$ is a covariant of the order $3(n-2)$; and the condition that the Hessian may vanish is a condition of the degree $3(n-2)$ in regard to the coefficients of $U + kV$. But this condition is not the same as the condition $\Omega = 0$; it is a different condition, which is satisfied when the curve $U + kV = 0$ has a node of a particular kind.

15.

In the case of two ternary cubics, the condition that the curve $U + kV = 0$ may have a node may be expressed as follows. Let

$$U = (a, b, c, f, g, h \between x, y, z)^3, \quad V = (a', b', c', f', g', h' \between x, y, z)^3,$$

and write

$$W = U + kV = (a + ka', \ldots \between x, y, z)^3.$$

Then the conditions for a node are

$$\partial_x W = 0, \quad \partial_y W = 0, \quad \partial_z W = 0,$$

that is

$$\begin{gathered}(a + ka')x + (h + kh')y + (g + kg')z = 0,\\ (h + kh')x + (b + kb')y + (f + kf')z = 0,\\ (g + kg')x + (f + kf')y + (c + kc')z = 0.\end{gathered}$$

And eliminating (x, y, z) from these equations, we find

$$\begin{vmatrix} a + ka' & h + kh' & g + kg' \\ h + kh' & b + kb' & f + kf' \\ g + kg' & f + kf' & c + kc' \end{vmatrix} = 0,$$

which is the discriminant of W, and is of the degree 3 in k.

16.

It may be noticed that the condition for a twofold critic centre, or (what is the same thing) a twofold factor of the Jacobian (which condition is of the degree $2(2n - 3)$ in regard to the coefficients of U or V) is $RQ = 0$; and that we in fact have

$$2(2n - 3) = n + 3(n - 2).$$

This remark is due to Dr Salmon.

Article, Nos. 17 to 42, relating to two Ternary Quantics.

17.

Suppose now that $U = (a, \dots \between x, y, z)^n$, $V = (a', \dots \between x, y, z)^n$ are two ternary quantics of the same order n, and write $W = U + kV = (a + ka', \dots \between x, y, z)^n$, so that

$$W = U + kV = 0$$

is the equation of a curve in involution with the curves $U = 0$, $V = 0$. But for greater distinctness it is in general proper to retain $U + kV$ in place of W.

18.

In order that the curve $U + kV = 0$ may have a node, we must have simultaneously

$$\begin{aligned} \partial_x(U + kV) &= 0, \\ \partial_y(U + kV) &= 0, \\ \partial_z(U + kV) &= 0, \end{aligned}$$

and eliminating (x, y, z) from these equations we have

$$\text{Disc}^t \,.\, (U + kV) = 0,$$

which is an equation of the degree $3(n-1)^2$ as regards k, and of the same order as regards the coefficients of U, V conjointly.

19.

To each of the $3(n-1)^2$ values of k there corresponds a point satisfying the conditions in question, and which is therefore a node of the corresponding nodal curve

$$U + kV = 0;$$

the points in question are the *critic centres* of the involution.

20.

The critic centres may be differently obtained as follows; viz. if from the three equations we eliminate k, we find

$$\begin{vmatrix} \partial_x U & \partial_y U & \partial_z U \\ \partial_x V & \partial_y V & \partial_z V \end{vmatrix} = 0,$$

which is a curve of the order $2(n-1)$; the critic centres are the intersections of this curve with any curve of the involution $U + kV = 0$, or, what is the same thing, they are the intersections of the two curves

$$\begin{vmatrix} \partial_x U & \partial_y U & \partial_z U \\ \partial_x V & \partial_y V & \partial_z V \end{vmatrix} = 0, \quad U = 0,$$

or, as we may write them,

$$(\partial_y U\, \partial_z V - \partial_z U\, \partial_y V,\ \partial_z U\, \partial_x V - \partial_x U\, \partial_z V,\ \partial_x U\, \partial_y V - \partial_y U\, \partial_x V \between x, y, z)^{n-1} = 0, \quad U = 0.$$

21.

The critic centres are thus the intersections of two curves of the orders $2(n-1)$ and n respectively, and their number is therefore $= 2n(n-1)$. But we have just seen that the number is $= 3(n-1)^2$; hence we must have $2n(n-1) = 3(n-1)^2$, that is $2n = 3(n-1)$, or $n = 3$. The explanation is that for $n > 3$ the intersections are not all of them critic centres; for $n = 3$ the number is $2 \times 3 \times 2 = 12$, which is $= 3 \times 2^2 = 12$; but for $n = 4$, for instance, the number would be $2 \times 4 \times 3 = 24$, which is not $= 3 \times 3^2 = 27$. The difference $27 - 24 = 3$ corresponds to the fact that in the case $n = 4$ there are three points which are critic centres, but which are not intersections of the two curves.

22.

The critic centres may be yet otherwise obtained. If we consider the system of curves $U + kV = 0$, the envelope of this system is given by the equation

$$\begin{vmatrix} \partial_x U & \partial_y U & \partial_z U \\ \partial_x V & \partial_y V & \partial_z V \end{vmatrix} = 0,$$

which is the same curve as that obtained by eliminating k from the three equations $\partial_x(U + kV) = 0$, $\partial_y(U + kV) = 0$, $\partial_z(U + kV) = 0$. The critic centres are the points of contact of the curves $U + kV = 0$ with their envelope.

23.

The envelope is a curve of the order $2(n-1)$; and the number of points of contact of the curves $U+kV = 0$ with their envelope is equal to the number of critic centres, that is $3(n-1)^2$. But the number of points of contact of a system of curves with their envelope is equal to the number of curves of the system which pass through a given point; and the number of curves of the system $U+kV = 0$ which pass through a given point is 1, for any given point determines the value of k. Hence the number of points of contact is equal to the number of intersections of the envelope with a curve of the system, that is $2(n-1) \times n = 2n(n-1)$. And we have thus the equation $3(n-1)^2 = 2n(n-1)$, which gives $n = 3$, as before.

24.

The critic centres form a system of points which may be studied by means of their diactic properties. A diacritic curve is a curve passing through the critic centres; and the simplest diacritic curve is the curve obtained by eliminating k from the equations $\partial_x(U + kV) = 0$, $\partial_y(U + kV) = 0$, which is a curve of the order $2(n - 1)$. This curve passes through all the critic centres, for at each critic centre the three equations $\partial_x(U + kV) = 0$, $\partial_y(U + kV) = 0$, $\partial_z(U + kV) = 0$ are all satisfied.

25.

The diacritic curve may be expressed as follows. If $U = (a, \dots \between x, y, z)^n$, $V = (a', \dots \between x, y, z)^n$, then the equations $\partial_x(U + kV) = 0$, $\partial_y(U + kV) = 0$ give

$$\partial_x U + k\,\partial_x V = 0, \quad \partial_y U + k\,\partial_y V = 0,$$

and eliminating k, we find

$$\partial_x U\,\partial_y V - \partial_y U\,\partial_x V = 0,$$

which is the equation of the diacritic curve. This curve is of the order $2(n - 1)$; and it passes through the critic centres, for at each critic centre we have $\partial_x U + k\,\partial_x V = 0$, $\partial_y U + k\,\partial_y V = 0$, $\partial_z U + k\,\partial_z V = 0$, and therefore $\partial_x U\,\partial_y V - \partial_y U\,\partial_x V = 0$.

26.

The diacritic curve is not the only curve passing through the critic centres; there are in general an infinite number of such curves. The most general diacritic curve is obtained by combining the equations $\partial_x(U + kV) = 0$, $\partial_y(U + kV) = 0$, $\partial_z(U + kV) = 0$ in any manner so as to eliminate k; and the result is in general a curve of the order $3(n - 1)$.

27.

The diacritic curve has important properties in relation to the curves of the involution. If we consider any curve $U + kV = 0$ of the involution, the tangent to this curve at any point (x, y, z) is given by the equation

$$(X\partial_x + Y\partial_y + Z\partial_z)(U + kV) = 0,$$

where (X, Y, Z) are current coordinates. If the point (x, y, z) is a critic centre, then the three equations $\partial_x(U + kV) = 0$, $\partial_y(U + kV) = 0$, $\partial_z(U + kV) = 0$ are all satisfied, and the tangent is indeterminate. But if the point (x, y, z) is on the diacritic curve, then only two of these equations are satisfied, and the tangent is determinate.

28.

The diacritic curve is the locus of the points at which the curves of the involution have a common tangent. For if two curves $U+k_1V = 0$, $U+k_2V = 0$ of the involution have a common tangent at a point (x, y, z), then the equations

$$\begin{aligned} X\partial_x(U + k_1V) + Y\partial_y(U + k_1V) + Z\partial_z(U + k_1V) = 0,\\ X\partial_x(U + k_2V) + Y\partial_y(U + k_2V) + Z\partial_z(U + k_2V) = 0, \end{aligned}$$

represent the same line; and therefore

$$\frac{\partial_x(U + k_1V)}{\partial_x(U + k_2V)} = \frac{\partial_y(U + k_1V)}{\partial_y(U + k_2V)} = \frac{\partial_z(U + k_1V)}{\partial_z(U + k_2V)},$$

which gives

$$\partial_x U\, \partial_y V - \partial_y U\, \partial_x V = 0,$$

and the point (x, y, z) is on the diacritic curve.

29.

The diacritic curve passes through the intersections of the curves $U = 0$, $V = 0$; for at these intersections we have $U = 0$, $V = 0$, and therefore $\partial_x U\,\partial_y V - \partial_y U\,\partial_x V = 0$. The diacritic curve also passes through the critic centres; and the critic centres are the points at which the curves of the involution have nodes. Hence the diacritic curve is a curve of the order $2(n-1)$ which passes through the intersections of $U = 0$, $V = 0$ and through the critic centres.

30.

The diacritic curve may be studied by means of its intersections with the curves of the involution. If we consider the curve $U + kV = 0$, the intersections of this curve with the diacritic curve are given by the equations

$$U + kV = 0, \quad \partial_x U\,\partial_y V - \partial_y U\,\partial_x V = 0.$$

These intersections are of two kinds: 1° the intersections of $U = 0$, $V = 0$, which are common to all the curves of the involution; 2° the critic centres, which are the points at which the curve $U+kV = 0$ has a node. The number of intersections of the first kind is n^2, and the number of intersections of the second kind is $3(n-1)^2$; and the total number of intersections is $n^2 + 3(n-1)^2$, which is equal to $2n(n-1) + (n-1)^2 + 1 = 4n^2 - 6n + 3$.

31.

The conditions in order that the curve $W = U + kV = 0$ may have a cusp are given by a plexus equivalent to three relations between the coefficients $a + ka'$, &c. of W, and using for a moment β to denote the degree of the plexus or system, then eliminating k between the equations of the plexus we find between the coefficients a, &c. of U and the coefficients a', &c. of V an equation $Q = 0$ of the degree β in regard to the two sets of coefficients respectively. Conversely, given the equation $Q = 0$, we may find the plexus between the coefficients $a + ka'$, &c. of W.

32.

The conditions for a cusp may be expressed as follows. The curve $W = U + kV = 0$ has a cusp if it has a node and if in addition the two tangents at the node coincide. The condition for a node is given by the discriminant $\mathrm{Disc}^t\,.\,W = 0$; and the condition that the two tangents at the node may coincide is given by the vanishing of the Hessian of W at the node. Hence the conditions for a cusp are

$$\mathrm{Disc}^t\,.\,W = 0, \quad (W) = 0,$$

where (W) denotes the Hessian of W.

33.

The conditions for a cusp may also be expressed by means of the theory of covariants. The curve $W = U + kV = 0$ has a cusp if it has a node and if the two tangents at the node coincide. The condition for a node is that the Jacobian of U and V may have a twofold factor; and the condition that the two tangents at the node may coincide is that the twofold factor may be a threefold factor. Hence the conditions for a cusp are that the Jacobian of U and V may have a threefold factor.

34.

The conditions for a cusp may also be expressed by means of the theory of invariants. The curve $W = U + kV = 0$ has a cusp if the discriminant of W vanishes and if in addition the catalecticant of W vanishes. Hence the conditions for a cusp are

$$\mathrm{Disc}^t\,.\,W = 0, \quad (W) = 0,$$

where (W) denotes the catalecticant of W.

35.

The conditions in order that the curve $W = U + kV = 0$ may have a cusp are of the degree $3(n-1)^2 + 3(n-2) = 6n^2 - 15n + 12$ in regard to the coefficients of W; and the equation $Q = 0$ is of the same degree in regard to the coefficients of U and V respectively. For the discriminant of W is of the degree $3(n-1)^2$, and the catalecticant of W is of the degree $3(n-2)$; and the resultant of these two conditions is of the degree $3(n-1)^2 + 3(n-2)$.

36.

In the case of two ternary cubics, the conditions for a cusp are of the degree $3 \times 4 + 3 \times 1 = 15$; and the equation $Q = 0$ is of the degree 15 in regard to the coefficients of U and V respectively.

37.

In the case of two ternary quartics, the conditions for a cusp are of the degree $3 \times 9 + 3 \times 2 = 33$; and the equation $Q = 0$ is of the degree 33 in regard to the coefficients of U and V respectively.

38.

The critic centres of the involution of two ternary quantics may be studied by means of their critic properties. A critic centre is a point at which a curve of the involution has a node; and the critic centres are the intersections of the diacritic curve with any curve of the involution. The number of critic centres is $3(n-1)^2$; and they form a system of points which may be studied by means of their geometric properties.

39.

The critic centres of the involution of two ternary cubics are 12 points; and they may be determined as follows. Let $U = 0$, $V = 0$ be the two cubics; then the critic centres are the intersections of the two curves

$$\partial_x U \, \partial_y V - \partial_y U \, \partial_x V = 0, \quad U = 0,$$

or, what is the same thing, they are the intersections of the Jacobian of U and V with the curve $U = 0$. The Jacobian is of the order 4, and the curve $U = 0$ is of the order 3; hence the number of intersections is $4 \times 3 = 12$, as it should be.

40.

The 12 critic centres may be arranged in various ways. They are the points of contact of the curves $U + kV = 0$ with their envelope; and they are also the points at which the curves of the involution have nodes. The envelope is a curve of the order 4; and the 12 critic centres are the intersections of this envelope with the curve $U = 0$.

41.

The critic centres may also be arranged in pairs. Each pair of critic centres lies on a line which is a tangent to the envelope; and the 12 critic centres form 6 pairs, each pair lying on a tangent to the envelope. The 6 tangents are the common tangents of the two cubics $U = 0$, $V = 0$; and the 12 critic centres are the points of contact of these 6 tangents with the envelope.

42.

The critic centres may also be arranged in threes. Each triad of critic centres lies on a conic which passes through the intersections of the two cubics $U = 0$, $V = 0$; and the 12 critic centres form 4 triads, each triad lying on a conic through the intersections of the two cubics. The 4 conics are the conics which touch the 6 common tangents of the two cubics; and the 12 critic centres are the points of contact of these 4 conics with the envelope.

Cambridge, 7th November, 1863.

[349]

ON A CASE OF THE INVOLUTION OF CUBIC CURVES.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. vi. (1864),
pp. 213–220.]

The present paper relates to a particular case of the involution of two ternary cubics, which presents some features of interest. I consider the case where the two cubics are given by the equations

$$U = x^3+y^3+z^3+6lxyz = 0, \quad V = 3yz^2+3y^2z+3zx^2+3z^2x+3xy^2+3x^2y = 0,$$

or, as these equations may be written,

$$U = x^3 + y^3 + z^3 + 6lxyz = 0, \quad V = 3\Sigma x^2y = 0,$$

where Σ denotes the sum of the terms obtained by cyclical permutation of x, y, z.

1.

The curve $U + kV = 0$ is a cubic curve in involution with the cubics $U = 0$, $V = 0$; and the condition that this curve may have a node is given by the discriminant of $U + kV$, which is a function of k of the degree 12. The 12 values of k for which the curve has a node correspond to the 12 critic centres of the involution.

2.

The discriminant of $U + kV$ may be calculated as follows. We have

$$U+kV = x^3+y^3+z^3+6lxyz+3k(yz^2+y^2z+zx^2+z^2x+xy^2+x^2y) = 0.$$

Writing this equation in the form

$$(a, b, c, f, g, h \between x, y, z)^3 = 0,$$

we have

$$\begin{array}{lll} a = 1, & b = 1, & c = 1, \\ f = l+k, & g = l+k, & h = l+k. \end{array}$$

3.

The discriminant of the cubic $(a, b, c, f, g, h \between x, y, z)^3$ is

$a^2b^2c^2-a^2b^2f^2-a^2c^2g^2-b^2c^2h^2+2a^2fgh+2b^2fgh+2c^2fgh-2abcfgh-af^2g^2-bf^2h^2-cg^2h^2+$

Substituting the values $a = b = c = 1$, $f = g = h = l + k$, we find

$$1 - 3(l + k)^2 + 6(l + k)^3 - 2(l + k)^3 - 3(l + k)^4 + (l + k)^6 = 0,$$

which simplifies to

$$1 - 3(l + k)^2 + 4(l + k)^3 - 3(l + k)^4 + (l + k)^6 = 0.$$

4.

Writing $m = l + k$, the equation becomes

$$1 - 3m^2 + 4m^3 - 3m^4 + m^6 = 0,$$

which may be written

$$(m^2 + m - 1)^2(m^2 - 2m + 1) - 3m^2(m - 1)^2 = 0,$$

or, as it may also be written,

$$(m - 1)^2(m^4 + 2m^3 - m^2 - 2m + 1) = 0.$$

The factor $(m-1)^2 = 0$ gives $m = 1$, that is $l+k = 1$, or $k = 1-l$ (twice); and the quartic factor gives the remaining 10 values of k.

5.

The quartic equation

$$m^4 + 2m^3 - m^2 - 2m + 1 = 0$$

may be solved as follows. Dividing by m^2 and writing $t = m + \frac{1}{m}$, we find

$$t^2 + 2t - 3 = 0,$$

which gives $t = 1$ or $t = -3$. The equation $m + \frac{1}{m} = 1$ gives $m^2 - m + 1 = 0$, whence $m = \frac{1 \pm i\sqrt{3}}{2}$; and the equation $m + \frac{1}{m} = -3$ gives $m^2 + 3m + 1 = 0$, whence $m = \frac{-3 \pm \sqrt{5}}{2}$.

6.

The 12 values of k are therefore as follows: $k = 1-l$ (twice), corresponding to the twofold critic centre; and the 10 other values corresponding to the one-with-twofold critic centres. The twofold critic centre is a point at which two of the curves of the involution have nodes; and the one-with-twofold critic centres are points at which one of the curves of the involution has a cusp.

7.

The discriminant equation may be written in the form

$$(m-1)^2(m^2-m+1)(m^2+3m+1)=0,$$

and the three factors correspond to three different kinds of critic centres. The factor $(m-1)^2=0$ corresponds to the twofold critic centre; the factor $m^2-m+1=0$ corresponds to four one-with-twofold critic centres; and the factor $m^2+3m+1=0$ corresponds to six one-with-twofold critic centres.

8.

The twofold critic centre is obtained by putting $k=1-l$ in the equations

$$\partial_x(U+kV)=0, \quad \partial_y(U+kV)=0, \quad \partial_z(U+kV)=0.$$

We have

$$\begin{aligned}
\partial_x(U+kV)&=3x^2+6lyz+3k(y^2+z^2+2xy+2xz),\\
\partial_y(U+kV)&=3y^2+6lxz+3k(z^2+x^2+2yz+2xy),\\
\partial_z(U+kV)&=3z^2+6lxy+3k(x^2+y^2+2xz+2yz).
\end{aligned}$$

Putting $k=1-l$ and simplifying, we find

$$\begin{aligned}
x^2+2lyz+(1-l)(y^2+z^2+2xy+2xz)&=0,\\
y^2+2lxz+(1-l)(z^2+x^2+2yz+2xy)&=0,\\
z^2+2lxy+(1-l)(x^2+y^2+2xz+2yz)&=0.
\end{aligned}$$

9.

The equation of the twofold centre locus is

$$\sqrt{x} + \sqrt{y} + \sqrt{z} = 0,$$

or, in its rationalised form,

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0;$$

the curve is therefore a conic, and it may be spoken of as the twofold centre conic.

10.

The equation of the one-with-twofold centre locus is

$$x^3 + y^3 + z^3 - (yz^2 + y^2z + zx^2 + z^2x + xy^2 + x^2y) + 3xyz = 0,$$

the curve is therefore a cubic, and it may be spoken of as the one-with-twofold centre cubic.

11.

The before-mentioned equation $\lambda^{-\frac{1}{3}} + \mu^{-\frac{1}{3}} + \nu^{-\frac{1}{3}} = 0$ is satisfied by

$$\lambda : \mu : \nu = \alpha^{-3} : \beta^{-3} : \gamma^{-3},$$

where $\alpha + \beta + \gamma = 0$, and it is very convenient to introduce these quantities α, β, γ into the formulae.

12.

The equation of the satellite line giving a twofold and one-with-twofold centre is

$$\frac{x}{\alpha^3}+\frac{y}{\beta^3}+\frac{z}{\gamma^3}=0;$$

the coordinates of the point of contact with the envelope are $x:y:z=\alpha^4:\beta^4:\gamma^4$.

The equation in θ gives $\theta_1=\theta_2=-\dfrac{1}{\alpha\beta\gamma}$ for the values corresponding to the twofold centre; and $\theta_3=\dfrac{2}{\alpha\beta\gamma}$ for the value corresponding to the one-with-twofold centre.

The coordinates of the twofold centre, or cusp, are $x:y:z=\alpha^2:\beta^2:\gamma^2$.

The coordinates of the one-with-twofold centre, or node, are

$$x:y:z=\alpha^2(\beta-\gamma):\beta^2(\gamma-\alpha):\gamma^2(\alpha-\beta).$$

The equation of the tangent at the cusp is

$$(\beta-\gamma)\frac{x}{\alpha}+(\gamma-\alpha)\frac{y}{\beta}+(\alpha-\beta)\frac{z}{\gamma}=0.$$

The equation of the line joining the cusp and the node, which line is also one of the tangents at the node is

$$\frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=0.$$

The equation of the other tangent at the node is

$$(2\beta\gamma+\alpha^2)\frac{x}{\alpha}+(2\gamma\alpha+\beta^2)\frac{y}{\beta}+(2\alpha\beta+\gamma^2)\frac{z}{\gamma}=0.$$

13.

The coordinates of the cusp and node being known, we may find the equation of the satellite conic, that is the conic which touches the six tangents at the cusp and node. The equation of this conic is

$$\frac{x^2}{\alpha^2}+\frac{y^2}{\beta^2}+\frac{z^2}{\gamma^2}-2\frac{yz}{\beta\gamma}-2\frac{zx}{\gamma\alpha}-2\frac{xy}{\alpha\beta}=0,$$

or, as it may be written,

$$\left(\frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}\right)^2-4\left(\frac{yz}{\beta\gamma}+\frac{zx}{\gamma\alpha}+\frac{xy}{\alpha\beta}\right)=0.$$

14.

The satellite conic passes through the cusp and node; and it also passes through the three points which are the intersections of the line joining the cusp and node with the curve $U=0$. These three points are the points of contact of the tangents from the cusp to the curve $U=0$; and they are also the points of contact of the tangents from the node to the curve $U=0$.

15.

The discriminant of the cubic $U + kV = 0$ may be expressed in terms of the quantities α, β, γ as follows. We have

$$m = l + k = \frac{\alpha^3 + \beta^3 + \gamma^3}{3\alpha\beta\gamma},$$

and the discriminant equation becomes

$$\left(\frac{\alpha^3 + \beta^3 + \gamma^3}{3\alpha\beta\gamma} - 1\right)^2 \left(\frac{\alpha^3 + \beta^3 + \gamma^3}{3\alpha\beta\gamma}\right)^2 - 3\left(\frac{\alpha^3 + \beta^3 + \gamma^3}{3\alpha\beta\gamma}\right)^2 \left(\frac{\alpha^3 + \beta^3 + \gamma^3}{3\alpha\beta\gamma} - 1\right)^2 = 0,$$

which simplifies to

$$\frac{(\alpha^3 + \beta^3 + \gamma^3 - 3\alpha\beta\gamma)^2(\beta\gamma + \gamma\alpha + \alpha\beta)^2}{27\alpha^6\beta^6\gamma^6} = 0.$$

16.

The equation $\alpha^3 + \beta^3 + \gamma^3 - 3\alpha\beta\gamma = 0$ gives the twofold critic centre; and the equation $\alpha\beta\gamma = 0$ gives the one-with-twofold critic centres. The equation $\beta\gamma + \gamma\alpha + \alpha\beta = 0$ gives the remaining critic centres.

17.

The theory of the involution of the two cubics $U = 0$, $V = 0$ may be illustrated by means of the following geometrical construction. Let A, B, C be three fixed points on a conic; and let P be a variable point on the conic. Then the lines PA, PB, PC form a pencil of three lines; and as P moves on the conic, the pencil of three lines generates a system of pencils. This system of pencils is a system in involution; and the critic centres of the involution are the points at which two of the lines of the pencil coincide.

18.

The critic centres of the involution are therefore the points at which the variable point P coincides with one of the fixed points A, B, C; and they are also the points at which the tangent to the conic at P passes through one of the fixed points. The number of critic centres is therefore 6; and they form a system of points which may be studied by means of their geometric properties.

19.

The foregoing theory may be extended to the case of a system of four lines through a point. Let A, B, C, D be four fixed points on a conic; and let P be a variable point on the conic. Then the lines PA, PB, PC, PD form a pencil of four lines; and as P moves on the conic, the pencil of four lines generates a system of pencils. This system of pencils is a system in involution; and the critic centres of the involution are the points at which two of the lines of the pencil coincide.

20.

The critic centres of the involution are therefore the points at which the variable point P coincides with one of the fixed points A, B, C, D; and they are also the points at which the tangent to the conic at P passes through one of the fixed points. The number of critic centres is therefore 8; and they form a system of points which may be studied by means of their geometric properties.

21.

The theory of the involution of cubic curves has important applications in the theory of cubic surfaces. A cubic surface may be represented as the locus of a line which meets three fixed lines; and the three fixed lines may be regarded as the edges of a tetrahedron. The cubic surface is therefore the locus of a line which meets the three edges of a tetrahedron; and the lines which meet the three edges form a system of lines in involution.

22.

The critic centres of the involution of lines are the lines which meet two of the three edges; and they are also the lines which are contained in a plane passing through one of the edges. The number of critic centres is therefore infinite; but they form a system of lines which may be studied by means of their geometric properties.

23.

The theory of the involution of cubic curves has also important applications in the theory of quartic surfaces. A quartic surface may be represented as the locus of a conic which passes through three fixed points; and the three fixed points may be regarded as the vertices of a triangle. The quartic surface is therefore the locus of a conic which passes through the three vertices of a triangle; and the conics which pass through the three vertices form a system of conics in involution.

24.

The critic centres of the involution of conics are the conics which pass through two of the three vertices; and they are also the conics which are contained in a plane passing through one of the vertices. The number of critic centres is therefore infinite; but they form a system of conics which may be studied by means of their geometric properties.

25.

Returning to the case of two ternary cubics, the discriminant of $U+kV$ may be expressed in a more convenient form as follows. Writing the discriminant equation in the form

$$(m-1)^2(m^4+2m^3-m^2-2m+1)=0,$$

and observing that

$$m^4+2m^3-m^2-2m+1=(m^2+m-1)^2-3m^2,$$

we find that the quartic factor may be written

$$(m^2+m-1-m\sqrt{3})(m^2+m-1+m\sqrt{3})=0,$$

and the roots are

$$m=\frac{-1+\sqrt{3}\pm\sqrt{-2+2\sqrt{3}}}{2},\quad m=\frac{-1-\sqrt{3}\pm\sqrt{-2-2\sqrt{3}}}{2}.$$

26.

The values of m being known, the corresponding values of k are given by the equation $k=m-l$; and the critic centres are then determined by the equations

$$\partial_x(U+kV)=0,\quad \partial_y(U+kV)=0,\quad \partial_z(U+kV)=0.$$

The coordinates of the critic centres may be expressed in terms of the quantities α, β, γ; and the results are of considerable elegance.

27.

The theory of the involution of cubic curves is closely connected with the theory of the bitangents of a quartic curve. A quartic curve has 28 bitangents; and these bitangents may be arranged in 8 systems of 7 bitangents each, such that the 7 bitangents of each system touch a conic. The 8 conics are the conics of the involution; and the 28 bitangents are the tangents to these conics from the intersections of the quartic with the conics.

28.

The theory of the involution of cubic curves is also connected with the theory of the double tangents of a cubic surface. A cubic surface has 27 lines; and these lines may be arranged in 36 systems of 6 lines each, such that the 6 lines of each system lie in a plane. The 36 planes are the planes of the involution; and the 27 lines are the lines of intersection of these planes with the cubic surface.

29.

The discriminant of $U + kV$ may be written in the form

$$\begin{vmatrix} 1 & (l+k) & (l+k) \\ (l+k) & 1 & (l+k) \\ (l+k) & (l+k) & 1 \end{vmatrix} = 0,$$

which gives

$$1 - 3(l+k)^2 + 2(l+k)^3 = 0,$$

or, writing $m = l + k$,

$$1 - 3m^2 + 2m^3 = 0,$$

which may be written

$$(m-1)^2(2m+1) = 0.$$

Hence $m = 1$ (twice), or $m = -\frac{1}{2}$; and the corresponding values of k are $k = 1 - l$ (twice) and $k = -\frac{1}{2} - l$.

30.

The twofold critic centre corresponds to $k = 1 - l$; and the one-with-twofold critic centres correspond to $k = -\frac{1}{2} - l$. The number of one-with-twofold critic centres is 10; and they form a system of points which may be studied by means of their geometric properties.

31.

The discriminant of $U + kV$ may be expressed in terms of the invariants of the cubic as follows. Let S and T be the fundamental invariants of the cubic $(a, b, c, f, g, h \between x, y, z)^3$; then the discriminant is $S^3 - T^2$. For the cubic $U + kV$, the invariants are

$$\begin{aligned} S &= 1 - 3(l + k)^2 + 2(l + k)^3, \\ T &= 1 - 9(l + k)^2 + 18(l + k)^3 - 9(l + k)^4 - 2(l + k)^6, \end{aligned}$$

and the discriminant is

$$S^3 - T^2 = (l+k)^2(l+k-1)^2(2l+2k+1)^2 \cdot (l+k+1)^2(2l+2k-1)^2(l+k-2)^2.$$

32.

The 12 values of k are therefore $k = -l$ (twice), $k = 1-l$ (twice), $k = -\frac{1}{2} - l$ (twice), $k = -1 - l$ (twice), $k = \frac{1}{2} - l$ (twice), and $k = 2 - l$ (twice). Each value of k corresponds to a pair of critic centres; and the 12 critic centres form 6 pairs, each pair lying on a line through the intersection of the curves $U = 0$, $V = 0$.

33.

The critic centres may be arranged in threes as follows. The 12 critic centres are the intersections of the Jacobian of U and V with the curve $U = 0$; and they are also the intersections of the Jacobian with the curve $V = 0$. The Jacobian is a quartic curve; and the 12 critic centres are the intersections of this quartic with the cubic $U = 0$, or with the cubic $V = 0$.

34.

The 12 critic centres may also be arranged in fours. The 12 critic centres are the points of contact of the 6 tangents which can be drawn from the intersections of $U = 0$, $V = 0$ to the curve $U + kV = 0$; and each tangent touches the curve at two critic centres. The 6 tangents are the common tangents of the two cubics $U = 0$, $V = 0$; and the 12 critic centres are the points of contact of these 6 tangents with the envelope of the system of curves $U + kV = 0$.

35.

The theory of the involution of cubic curves may be applied to the study of the singularities of a cubic curve. A cubic curve may have a node or a cusp; and the conditions for a node or a cusp may be expressed in terms of the invariants of the curve. The discriminant of the cubic is $S^3 - T^2$; and the curve has a node if $S^3 - T^2 = 0$, and a cusp if $S = 0$ and $T = 0$.

36.

In the case of the cubic $U + kV = 0$, the invariants are

$$S = 1 - 3m^2 + 2m^3,$$
$$T = 1 - 9m^2 + 18m^3 - 9m^4 - 2m^6,$$

where $m = l + k$. The condition for a node is $S^3 - T^2 = 0$, which gives the 12 values of k for which the curve has a node. The condition for a cusp is $S = 0$ and $T = 0$, which gives the values of k for which the curve has a cusp.

37.

The condition $S = 0$ gives

$$1 - 3m^2 + 2m^3 = 0,$$

which may be written

$$(m - 1)^2(2m + 1) = 0;$$

and the condition $T = 0$ gives

$$1 - 9m^2 + 18m^3 - 9m^4 - 2m^6 = 0,$$

which may be written

$$(m - 1)^3(2m + 1)^3 = 0.$$

Hence the values of m for which the curve has a cusp are $m = 1$ and $m = -\frac{1}{2}$; and the corresponding values of k are $k = 1 - l$ and $k = -\frac{1}{2} - l$.

38.

The cusp corresponding to $k = 1 - l$ is a twofold cusp; and the cusp corresponding to $k = -\frac{1}{2} - l$ is a one-with-twofold cusp. The twofold cusp is a point at which two branches of the curve have a common tangent; and the one-with-twofold cusp is a point at which three branches of the curve have a common tangent.

39.

The theory of the involution of cubic curves has important applications in the theory of the reduction of cubic curves to canonical forms. A cubic curve may be reduced to the canonical form

$$x^3 + y^3 + z^3 + 6lxyz = 0,$$

by a suitable choice of coordinates; and the value of l determines the nature of the curve. The curve is non-singular if $l^3 \neq -1$ and $8l^3 \neq 1$; and it has singularities if $l^3 = -1$ or $8l^3 = 1$.

40.

The curve $x^3+y^3+z^3+6lxyz = 0$ has a node if $8l^3 = 1$; and it has a cusp if $l^3 = -1$. In the case $8l^3 = 1$, the node is a crunode or an acnode according as l is real or complex; and in the case $l^3 = -1$, the cusp is a real cusp or an imaginary cusp according as l is real or complex.

41.

The theory of the involution of cubic curves may also be applied to the study of the inflexions of a cubic curve. A cubic curve has 9 inflexions; and these inflexions may be arranged in 12 sets of 3 inflexions each, such that the 3 inflexions of each set lie on a line. The 12 lines are the inflexional tangents of the curve; and the 9 inflexions are the points of intersection of these 12 lines with the curve.

42.

The inflexions of the cubic $x^3 + y^3 + z^3 + 6lxyz = 0$ may be determined as follows. The condition for an inflexion is that the Hessian of the cubic may vanish; and the Hessian is

$$\begin{vmatrix} x & lz & ly \\ lz & y & lx \\ ly & lx & z \end{vmatrix} = xyz + l^2(x^3 + y^3 + z^3) - l(x^2y^2 + y^2z^2 + z^2x^2) = 0.$$

The intersections of the Hessian with the cubic are the 9 inflexions; and they may be determined by solving the two equations.

43.

The 9 inflexions may be arranged in threes as follows. The 9 inflexions are the intersections of the cubic with the 9 lines which join the inflexions to the point $(1, 1, 1)$. The 9 lines are the inflexional tangents of the curve; and the 9 inflexions are the points of contact of these 9 tangents with the cubic.

44.

The 9 inflexions may also be arranged in fours. The 9 inflexions are the points of contact of the 12 tangents which can be drawn from the 3 points $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ to the cubic. The 12 tangents are the inflexional tangents of the curve; and the 9 inflexions are the points of contact of these 12 tangents with the cubic.

45.

The theory of the involution of cubic curves has important applications in the theory of the transformation of cubic curves. A cubic curve may be transformed into another cubic curve by a Cremona transformation; and the Cremona transformation is determined by the choice of three points on the curve. The three points are the base points of the transformation; and the transformation is a birational transformation of the curve into itself.

46.

The Cremona transformation of a cubic curve may be described as follows. Let P, Q, R be three points on the cubic; and let the lines PQ, QR, RP meet the cubic again at P', Q', R' respectively. Then the transformation which sends P into P', Q into Q', R into R' is a Cremona transformation of the cubic into itself; and it is determined by the three points P, Q, R.

47.

The theory of the involution of cubic curves may be extended to the case of higher curves. A curve of order n may be represented as the locus of a point which satisfies a system of equations of degree n; and the curves of the system form a system in involution. The critic centres of the involution are the points at which the curves of the system have nodes; and they may be studied by means of their geometric properties.

48.

The theory of the involution of curves has important applications in the theory of surfaces. A surface of order n may be represented as the locus of a line which satisfies a system of equations of degree n; and the surfaces of the system form a system in involution. The critic centres of the involution are the lines at which the surfaces of the system have nodes; and they may be studied by means of their geometric properties.

Cambridge, 1863.

[349] On a Case of the Involution of Cubic Curves

[From the Philosophical Magazine, vol. XXV *(1863), pp. 525–528. Continued from p. 350.]*

ARTICLE NOS. 33 to 38, relating to the Tangents at a Node or Critic Centre.

33. I proceed to investigate the equation of the tangents at the node of the curve

$$xyz + k(x+y+z)^2(\kappa x + \mu y + \nu z) = 0;$$

it will be recollected that if x, y, z are the coordinates of the node, then for the coordinates of the critic centres we have

$$\frac{-x+y+z}{\lambda x} = \frac{x-y+z}{\mu y} = \frac{x+y-z}{\nu z} = \theta,$$

and thence

$$\frac{1}{x} : \frac{1}{y} : \frac{1}{z} = \lambda(\theta\lambda + \mu + \nu) : \mu(\lambda + \theta\mu + \nu) : \nu(\lambda + \mu + \theta\nu).$$

The critic centres are the nodes of the six component triangles; and the node of any one of these triangles is a critic centre of each of the other five triangles, and therefore of the sextic curve; hence the critic centres of the sextic curve are the critic centres of any one of the component triangles. The coordinates (x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3) of the three critic centres belonging to a root θ_1 of the cubic equation are given by

$$\frac{1}{x_1} : \frac{1}{y_1} : \frac{1}{z_1} : \frac{1}{x_2} : \frac{1}{y_2} : \frac{1}{z_2} : \frac{1}{x_3} : \frac{1}{y_3} : \frac{1}{z_3}$$
$$= \theta_1\lambda + \mu + \nu : \lambda + \theta_1\mu + \nu : \lambda + \mu + \theta_1\nu : \text{etc.}$$

34. The coordinates of the critic centres being proportional to

$$x : y : z = \frac{1}{\theta\lambda + \mu + \nu} : \frac{1}{\lambda + \theta\mu + \nu} : \frac{1}{\lambda + \mu + \theta\nu},$$

the equation of the tangent at the node is

$$\left(\frac{a}{\theta\lambda + \mu + \nu} + \frac{b}{\lambda + \theta\mu + \nu} + \frac{c}{\lambda + \mu + \theta\nu}\right)(\xi x + \eta y + \zeta z)$$
$$-3\left(\frac{a\xi}{\theta\lambda + \mu + \nu} + \frac{b\eta}{\lambda + \theta\mu + \nu} + \frac{c\zeta}{\lambda + \mu + \theta\nu}\right) = 0,$$

where (ξ, η, ζ) are current coordinates.

35. It is convenient to write this equation under the form

$$\frac{a}{\theta\lambda+\mu+\nu}(\eta y+\zeta z-2\xi x)+\frac{b}{\lambda+\theta\mu+\nu}(\zeta z+\xi x-2\eta y)+\frac{c}{\lambda+\mu+\theta\nu}(\xi x+\eta y-2\zeta z)=0.$$

36. The equation of the other tangent is

$$\frac{\xi x}{a}+\frac{\eta y}{\beta}+\frac{\zeta z}{\gamma}=0,$$

or, what is the same thing,

$$\frac{\xi}{\alpha}+\frac{\eta}{\beta}+\frac{\zeta}{\gamma}=0.$$

This is in fact equivalent to

$$\frac{X}{\alpha}+\frac{Y}{\beta}+\frac{Z}{\gamma}=0,$$

where X, Y, Z are proportional to yz, zx, xy respectively.

37. From the foregoing formulae we obtain

$$\begin{aligned}&a:b:c:f:g:h\\&\quad=2yz(-x+2y+2z)\\&\qquad:2zx(2x-y+2z)\\&\qquad:2xy(2x+2y-z)\\&\qquad:-x(-x^2-y^2-z^2+4yz)\\&\qquad:-y(-x^2-y^2-z^2+4zx)\\&\qquad:-z(-x^2-y^2-z^2+4xy),\end{aligned}$$

which are the required new forms.

38. We have

$$bc-f^2=16x^2z^2(-x+y+z)(x-y+z),$$

with the like values for $ca-g^2$ and $ab-h^2$; and

$$gh-af=8xyz(y-z)(x^2+y^2+z^2-2yz-2zx-2xy).$$

Article Nos. 39 to 42. On the Critic Centres.

39. The critic centres are determined as the intersections of any two of the conics

$$\begin{aligned}\frac{y_1(y_1-z_1)}{x}+\frac{z_1(z_1-x_1)}{y}+\frac{x_1(x_1-y_1)}{z}&=0,\\\frac{y_2(y_2-z_2)}{x}+\frac{z_2(z_2-x_2)}{y}+\frac{x_2(x_2-y_2)}{z}&=0,\\\frac{y_3(y_3-z_3)}{x}+\frac{z_3(z_3-x_3)}{y}+\frac{x_3(x_3-y_3)}{z}&=0.\end{aligned}$$

40. The critic centres belonging to a root θ_1 are given by

$$\frac{-x_1+y_1+z_1}{\lambda x_1}=\frac{x_1-y_1+z_1}{\mu y_1}=\frac{x_1+y_1-z_1}{\nu z_1}=\theta_1,$$
$$\frac{-x_1+y_1+z_1}{x_1}=\theta_1\lambda,\quad \frac{x_1-y_1+z_1}{y_1}=\theta_1\mu,\quad \frac{x_1+y_1-z_1}{z_1}=\theta_1\nu.$$

41. The equation of the line joining the critic centres corresponding to the roots θ_1 and θ_2 is

$$\frac{\xi}{\theta_1\theta_2\lambda-\mu\nu}+\frac{\eta}{\theta_1\theta_2\mu-\nu\lambda}+\frac{\zeta}{\theta_1\theta_2\nu-\lambda\mu}=0.$$

Hence the equation in question is that of the line joining the critic centres corresponding to the roots θ_2 and θ_3.

42. I put for greater convenience

$$X=\frac{-x+y+z}{\lambda x},\quad Y=\frac{x-y+z}{\mu y},\quad Z=\frac{x+y-z}{\nu z},$$

so that $X=0$, $Y=0$, $Z=0$ are the equations of the sides of the triangle formed by the critic centres. Then X, Y, Z may if we please be considered as new coordinates; and we have

$$x:y:z=\frac{1}{Y+Z}:\frac{1}{Z+X}:\frac{1}{X+Y}.$$

The equation of the cubic curve in terms of X, Y, Z is

$$\left(\frac{1}{Y+Z}+\frac{1}{Z+X}+\frac{1}{X+Y}\right)^2\left(\frac{\lambda}{Y+Z}+\frac{\mu}{Z+X}+\frac{\nu}{X+Y}\right)=0,$$

and hence it is of the form

$$\frac{\lambda}{(Y+Z)^3}+\frac{\mu}{(Z+X)^3}+\frac{\nu}{(X+Y)^3}=0.$$

By comparing the coefficients of x^3 we have

$$\frac{1}{(\theta_1-\theta_2)(\theta_1-\theta_3)}=\frac{K}{\theta_1+\lambda},$$

and thence the required equation is

$$K=\frac{\theta_1+\lambda}{(\theta_1-\theta_2)(\theta_1-\theta_3)}=\frac{\theta_2+\mu}{(\theta_2-\theta_3)(\theta_2-\theta_1)}=\frac{\theta_3+\nu}{(\theta_3-\theta_1)(\theta_3-\theta_2)}.$$

Article Nos. 43 to 50. On the Involution of the Cubic Curves.

43. The general equation of a cubic curve may be written

$$U = (a, b, c, f, g, h \;\; x, y, z)^3 = 0,$$

and it is required to find the condition that this curve may have a given point (α, β, γ) for a node.

44. Writing

$$\frac{dU}{dx} = P, \quad \frac{dU}{dy} = Q, \quad \frac{dU}{dz} = R,$$

the conditions for a node at (α, β, γ) are

$$P = 0, \quad Q = 0, \quad R = 0,$$

for $(x, y, z) = (\alpha, \beta, \gamma)$.

45. The node being a singular point, the first derived functions vanish; hence the cubic form $(a, b, c, f, g, h \;\; x, y, z)^3$ is reducible to the form

$$(a, b, c, f, g, h \;\; x, y, z)^3 = 3(\alpha x + \beta y + \gamma z)(A, B, C, F, G, H \;\; x, y, z)^2,$$

and the condition for a node is that the conic $(A, B, C, F, G, H \; x, y, z)^2 = 0$ may pass through the point (α, β, γ).

46. The equation of the pair of tangents at the node is

$$(A, B, C, F, G, H \;\; x, y, z)^2 = 0,$$

for the coordinates (α, β, γ); that is,

$$Ax^2 + By^2 + Cz^2 + 2Fyz + 2Gzx + 2Hxy = 0,$$

where A, B, C, F, G, H are the second derived functions of U.

47. The cubic curve $U = 0$ having a node at (α, β, γ) may be written

$$(a, b, c, f, g, h \;\; x, y, z)^3 = 3(\alpha x + \beta y + \gamma z)(lx + my + nz)^2,$$

where l, m, n are arbitrary; and the equation of the tangents at the node is

$$(lx + my + nz)^2 = 0.$$

48. The general cubic curve with a node at a given point may be represented by means of six constants; for the node introduces three conditions, and there remain six constants. The equation may be written

$$(a, b, c, f, g, h \;\; x, y, z)^3 = (\alpha x + \beta y + \gamma z)\phi(x, y, z),$$

where ϕ is an arbitrary quadric function.

49. The sextic condition for a node at (α, β, γ) may be expressed as the vanishing of the discriminant of the conic

$$\alpha\frac{dU}{dx} + \beta\frac{dU}{dy} + \gamma\frac{dU}{dz} = 0,$$

regarded as a function of x, y, z; or what is the same thing, the condition is

$$\begin{vmatrix} a\alpha+h\beta+g\gamma & h\alpha+b\beta+f\gamma & g\alpha+f\beta+c\gamma \\ h\alpha+b\beta+f\gamma & h\alpha+b\beta+f\gamma & g\alpha+f\beta+c\gamma \\ g\alpha+f\beta+c\gamma & g\alpha+f\beta+c\gamma & g\alpha+f\beta+c\gamma \end{vmatrix} = 0.$$

50. The foregoing investigations lead to the theory of the involution of cubic curves; that is, the theory of a system of cubic curves having a common node and the same tangents at the node. The general equation of such a system is

$$S + kS' = 0,$$

where $S = 0$ and $S' = 0$ are any two cubic curves having the given node and tangents.

ARTICLE NOS. 51 to 55. On the Equations of the Involution.

51. The general cubic with a node at a given point (α, β, γ) and given tangents at the node may be written in the form

$$xyz + k(x+y+z)^2(\lambda x + \mu y + \nu z) = 0,$$

where the node is at $(1, 1, 1)$ and the tangents are $x = 0$, $y = 0$, $z = 0$.

52. The equation of the cubic curve may also be written

$$(a, b, c, f, g, h \;\; x, y, z)^3 = 0,$$

where the coefficients satisfy the conditions

$$\begin{aligned} a + h + g &= 0, \\ h + b + f &= 0, \\ g + f + c &= 0. \end{aligned}$$

These are the conditions for a node at the point $(1, 1, 1)$.

53. The three conditions may be represented by a single matrix equation

$$\begin{pmatrix} a & h & g \\ h & b & f \\ g & f & c \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 0,$$

showing that the matrix is singular, and that $(1, 1, 1)$ is a point on the conic adjoint to the cubic.

54. The discriminant of the cubic form, when there is a node, reduces to the product of the discriminant of the conic of the second degree (the tangential equation at the node) and the square of the linear function which expresses that the node lies on the curve.

55. The theory of the involution is thus intimately connected with the theory of the node and the critic centres; and the foregoing formulae contain the complete solution of the problem of the involution of cubic curves.

ARTICLE Nos. 56 to 60. Further Investigations on the Critic Centres.

56. The critic centres are the intersections of the conics which are the polars of the several nodes with respect to the other component triangles; and the theory is symmetrical with respect to the six component triangles.

57. For a given line $\lambda x + \mu y + \nu z = 0$, the critic centres are determined by the equation

$$(\theta\lambda + \mu + \nu)(\lambda + \theta\mu + \nu)(\lambda + \mu + \theta\nu) = 0,$$

which is a cubic in θ; and to each root corresponds a set of three critic centres.

58. The locus of the critic centres for variable λ, μ, ν is the sextic curve

$$\prod(\theta\lambda + \mu + \nu) = 0,$$

which breaks up into the six component triangles.

59. The critic centres for the line $\lambda x + \mu y + \nu z = 0$ are the nodes of the triangle formed by the three lines

$$\theta_1\lambda x + \mu y + \nu z = 0, \quad \lambda x + \theta_2\mu y + \nu z = 0, \quad \lambda x + \mu y + \theta_3\nu z = 0,$$

where θ_1, θ_2, θ_3 are the roots of the cubic in θ.

60. The theory of the critic centres is thus reduced to the algebraical theory of the cubic equation; and the various geometrical properties follow from the relations between the roots and coefficients.

ARTICLE Nos. 61 to 65. On the Reduction of the General Equation.

61. The general cubic equation may be reduced to the canonical form

$$xyz + k(x + y + z)^2(\lambda x + \mu y + \nu z) = 0,$$

by means of a linear transformation which sends the three critic centres to the vertices of the triangle of reference.

62. The reduction depends on the solution of a system of equations of the first degree; and it is always possible provided the three critic centres are distinct.

63. When two critic centres coincide, the cubic has a cusp; and the canonical form must be modified accordingly.

64. When the three critic centres are collinear, the cubic has a double point; and the reduction to the canonical form is not possible.

65. The cases of coincidence of critic centres correspond to the discriminant of the cubic in θ vanishing; and the different cases are classified by the manner in which the discriminant vanishes.

ARTICLE Nos. 66 to 70. On the Higher Singularities.

66. The higher singularities of the cubic curve, such as the tacnode, the rhamphoid cusp, and the triple point, may be investigated by means of the theory of the critic centres.

67. A tacnode corresponds to the case where two branches of the curve have a common tangent and the same curvature; the condition is that two critic centres and the corresponding nodes form a system of four points in involution.

68. A rhamphoid cusp corresponds to the case where the two branches of a node have a common tangent, and the critic centre lies on the common tangent; the condition is expressible by means of the invariant theory.

69. A triple point corresponds to the case where the three critic centres coincide; the cubic then breaks up into three lines, and the condition is that the discriminant of the cubic in θ vanishes identically.

70. The investigation of the higher singularities leads to a complete classification of the singularities of the cubic curve, and to the enumeration of all possible forms.

Article Nos. 71 to 75. On the Reciprocal Curve.

71. The reciprocal of a cubic curve is in general a curve of the sixth class; but when the original curve has a node, the reciprocal has a cusp, and the class is reduced.

72. The equation of the reciprocal may be obtained by eliminating x, y, z between the equations

$$\frac{dU}{dx} = \xi, \quad \frac{dU}{dy} = \eta, \quad \frac{dU}{dz} = \zeta,$$

and the equation $U = 0$; the result is a sextic equation in ξ, η, ζ.

73. When the cubic has a node at (α, β, γ), the reciprocal has a cusp at the point corresponding to the tangent at the node; and the equation of the reciprocal reduces to a quintic.

74. The Plückerian characteristics of the cubic curve are connected by the equations

$$\begin{aligned} m &= n(n-1) - 2\delta - 3\kappa, \\ \iota &= 3n(n-2) - 6\delta - 8\kappa, \\ d &= \tfrac{1}{2}n(n-1) - \delta - 2\kappa, \end{aligned}$$

where n is the order, m the class, δ the number of nodes, κ the number of cusps, ι the number of inflexions, and d the number of double tangents.

75. For a cubic curve with a node and no cusp, we have $n = 3$, $\delta = 1$, $\kappa = 0$; whence $m = 4$, $\iota = 3$, $d = 1$; and the reciprocal is a quartic with three cusps, that is, a Cartesian curve.

Article Nos. 76 to 80. On the Cuspidal Cubic.

76. The cuspidal cubic is a particular case of the nodal cubic, in which the two tangents at the node coincide; the canonical form is

$$x^2z = y^3,$$

or more generally,

$$z^2 = x^2y.$$

77. The cuspidal cubic has only one critic centre, which coincides with the cusp; the theory of the involution reduces to that of a single cubic curve with a cusp.

78. The cuspidal cubic is its own reciprocal, or more accurately, the reciprocal is also a cuspidal cubic; the equation of the reciprocal is of the same form as that of the original curve.

79. The cuspidal cubic has one inflexion, which lies on the line joining the cusp to the point of contact of the tangent at the cusp; and the tangent at the inflexion passes through the cusp.

80. The line $3x + y + z = 0$ is the line through the points $(-1, 4, -1)$, $(-1, -1, 4)$, and as such the corresponding critic centres lie one on the line $z + x = 0$, two on the conic $x(x + y + z) - 4yz = 0$, one on the line $x + y = 0$, two on the conic $y(x + y + z) - 4zx = 0$. The two lines meet in the point $(1, -1, -1)$. The two conics meet in the points $(1, 0, 0)$, $(2, 3, 3)$; and touch at the point $(0, 1, -1)$, the common tangent being $3x + y + z = 0$: this appears by writing the equations of the two conics in the forms

$$\begin{gathered}(y - z)(5x + y + z) + (y + z)(-3x + y + z) = 0,\\ -(y - z)(5x + y + z) + (y + z)(-3x + y + z) = 0,\end{gathered}$$

for we have then the four points of intersection put in evidence; viz. these are

$$y - z = 0, \quad y + z = 0, \quad \text{that is } (1, 0, 0),$$

and

$$5x + y + z = 0, \quad -3x + y + z = 0, \quad \text{that is } (2, 3, 3);$$

and the two conics touch at the point $y - z = 0$, $-3x + y + z = 0$, that is $(0, 1, -1)$.

ARTICLE NOS. 81, 82.

81. The critic centres for the line $3x + y + z = 0$ are the three points

$$\begin{array}{ll} 1, & (1, -1, -1),\\ \frac{1}{2}, & (2, 3, 3), \end{array}$$

which points are therefore the critic centres for the line $3x + y + z = 0$. The last-mentioned line, it is clear, is one of the system of three lines

$$3x + y + z = 0, \quad x + 3y + z = 0, \quad x + y + 3z = 0.$$

82. If $\lambda = 0$, that is if the line $\kappa x + \mu y + \nu z = 0$ pass through an angle $y = 0$, $z = 0$ of the triangle; then reverting to the original equations

$$\frac{-x+y+z}{\lambda x} = \frac{x-y+z}{\mu y} = \frac{x+y-z}{\nu z} = \theta,$$

these give $(y = 0, z = 0)$ or else $(-x + y + z = 0, \nu z^2 - \mu y^2 = 0)$, that is, one of the three critic centres is the angle $(y = 0, z = 0)$ of the triangle; and the other two are the intersections of the line $-x + y + z = 0$ with the pair of lines $\nu z^2 - \mu y^2 = 0$.

It should be remarked that, given the critic centre $y = y_1 = 0$, $z = z_1 = 0$, the remaining two centres cannot be determined as the intersection of the polar $-x + y + z = 0$ with the conic

$$\frac{x_1(y_1 - z_1)}{x} + \frac{y_1(z_1 - x_1)}{y} + \frac{z_1(x_1 - y_1)}{z} = 0,$$

inasmuch as the equation of this conic becomes the identity $0 = 0$.

Article Nos. 83 to 85.

83. The critic centres for the line $x + y + z = 0$ are the three points $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$; and the corresponding cubic curves are the three pairs of lines $yz = 0$, $zx = 0$, $xy = 0$.

84. The critic centres for the line $x = 0$ are the three points $(0, 1, -1)$, $(0, 1, 1)$, $(0, 1, -1)$; the first and third coincide, and the cubic has a cusp at the point $(0, 1, -1)$.

85. The general theory of the critic centres includes as particular cases all the known properties of the cubic curve with a node or cusp; and the foregoing formulae give a complete account of the involution of cubic curves.

Article Nos. 86 to 90.

86. If a critic centre lie on the line $y - z = 0$, then of the other two critic centres, one lies on this same line and the other is the point $x = 0$, $x + y + z = 0$, or say the point $(0, 1, -1)$. And in this case the line $\lambda x + \mu y + \nu z = 0$ passes through the last-mentioned point; that is, we have $\mu = \nu$. Conversely, starting from the equation $\mu = \nu$, so that the line $\kappa x + \mu y + \nu z = 0$ is $\lambda x + \mu(y + z) = 0$, a line through the intersection of the lines $x = 0$, $x + y + z = 0$, the equation in θ is

$$(\theta + \mu)(\theta^2 - \mu\theta - 2\lambda\mu) = 0,$$

where the factor $\theta + \mu = 0$ corresponds to the critic centre $x = 0$, $x + y + z = 0$, or $(0, 1, -1)$, (it will presently be shown that this is so), and the quadric equation $\theta^2 - \mu\theta - 2\lambda\mu = 0$ corresponds to the other two critic centres.

87. The line $y - z = 0$ passes through the critic centre $(0, 1, -1)$; and the other two critic centres are the intersections of the line $y - z = 0$ with the conic $x(x+y+z)-4yz=0$. The conic meets the line in the points $(4,1,1)$ and $(0,1,1)$; and the critic centres are therefore $(0,1,-1)$, $(4,1,1)$, $(0,1,1)$.

88. The condition that the critic centre $(0,1,-1)$ may be a double point on the cubic is that the line $\lambda x+\mu y+\nu z=0$ may pass through the point $(0,1,-1)$; that is, $\mu=\nu$. In this case the cubic has a cusp at $(0,1,-1)$, and the two other critic centres coincide at the point $(4,1,1)$.

89. The condition for a rhamphoid cusp is that the three critic centres may be consecutive points on the curve; the condition is that the discriminant of the quadric $\theta^2-\mu\theta-2\lambda\mu=0$ may vanish, that is, $\mu^2+8\lambda\mu=0$, or $\mu(\mu+8\lambda)=0$.

90. The several conditions for the different kinds of singular points are thus expressible by simple equations in λ, μ, ν; and the classification of the singularities is thereby reduced to an algebraical problem.

ARTICLE NOS. 91 to 95.

91. The complete theory of the involution of cubic curves comprises the investigation of the following questions:

(1) The general form of the cubic with a given node and given tangents;

(2) The determination of the critic centres;

(3) The conditions for the coincidence of critic centres;

(4) The classification of the different kinds of nodes;

(5) The theory of the reciprocal curve.

92. The first question has been completely answered by the canonical form

$$xyz+k(x+y+z)^2(\lambda x+\mu y+\nu z)=0.$$

93. The second question has been answered by the theory of the cubic equation in θ; to each root corresponds a set of three critic centres, and the locus of all the critic centres is the sextic curve which is the union of the six component triangles.

94. The third question has been answered by the discriminantal conditions; the coincidence of critic centres gives rise to the cuspidal and other singular forms.

95. The fourth and fifth questions have been treated in the preceding articles; and the theory is thus complete.

[350] On the Classification of Cubic Curves

[From the Transactions of the Cambridge Philosophical Society, vol. XI. *Part* I. *(1866), pp. 81–128. Read April 18, 1864.]*

The notion of a curve of a given order may be considered as arising from Descartes' invention of his method of coordinates; and one of the earliest applications of the method was made by Sir Isaac Newton in the *Enumeratio linearum tertii Ordinis* (1706), a work worthy of its author, and which opened a new field of geometrical science. The classification is according to the nature of the infinite branches; there are fourteen genera containing together seventy-two species, but four species were added by Stirling in his *Lineæ tertii Ordinis Newtonianæ; sive Illustratio &c.* (1717), and two more by Murdoch or Cramer[8], making in all seventy-eight species. A new classification was made by Plücker in his *System der Analytischen Geometrie* (1835), founded on the notion of the class of a curve and the introduction of the coordinates of a line; the classification is according to the nature of the multiple points and the contacts of the line infinity, and there are two hundred and nineteen species. Plücker's classification was accepted as the best, and it was not until the appearance of Dr Salmon's *Higher Plane Curves* (1852) that any important addition was made to the theory. In that work the author introduced the notion of the deficiency of a curve, and showed that the classification according to the Plückerian characteristics was incomplete; the deficiency is a number which is invariant for all curves of the same order and class, and it serves to distinguish between curves which have the same singularities. The present Memoir contains a complete account of the classification of cubic curves according to the nature of their infinite branches, founded on the principles established by Newton and extended by the researches of later geometers.

The Seven Head Divisions, Article Nos. 1 to 4.

1. A line in general, and therefore the line Infinity, meets a cubic curve in three points, and these may be

- Three onefold points,
- A twofold point and a onefold (or, as it may also be termed, a one-with-twofold) point,
- A threefold point.

2. But in the second case the line Infinity

- may be a proper tangent to the curve,

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[8] The references are to Murdoch's *Newtoni Genesis Curvarum per Umbras* (1717) and Cramer's *Introduction à l'Analyse des Lignes Courbes algébriques* (1750).

- may pass through a node,
- may pass through a cusp;

and in the third case the line Infinity

- may touch the curve at an inflexion,
- may at a node touch one of the two branches,
- may touch the curve at a cusp.

3. The first case, the general one, constitutes the *first head division*; the second case, according to the three subdivisions, gives the *second*, *third*, and *fourth head divisions*; and the third case, according to the three subdivisions, gives the *fifth*, *sixth*, and *seventh head divisions*. There are thus in all *seven head divisions*.

4. The seven head divisions may be further classified according to the nature of the finite singularities; and the complete classification includes the consideration of the nodes, cusps, and double tangents of the curve. The Plückerian characteristics are connected by the fundamental equations

$$\begin{aligned} m &= n(n-1) - 2\delta - 3\kappa, \\ \iota &= 3n(n-2) - 6\delta - 8\kappa, \\ d &= \tfrac{1}{2}n(n-2)(n^2-9) - (n^2-n-6)(2\delta+3\kappa) + 2\delta(\delta-1) + 6\delta\kappa + \tfrac{9}{2}\kappa(\kappa-1), \end{aligned}$$

where n is the order, m the class, δ the number of nodes, κ the number of cusps, ι the number of inflexions, and d the number of double tangents. For a cubic curve $n = 3$, and the equations become

$$\begin{aligned} m &= 6 - 2\delta - 3\kappa, \\ \iota &= 9 - 6\delta - 8\kappa, \\ d &= 1 + \tfrac{1}{2}\delta(\delta-1) + 3\delta\kappa + \tfrac{9}{2}\kappa(\kappa-1). \end{aligned}$$

The First Head Division, Article Nos. 5 to 12.

5. The first head division comprises the cubic curves which are met by the line infinity in three distinct onefold points; that is, the curves having three distinct asymptotes. The general equation is

$$U = (a, b, c, d, f, g, h, l, m, n \quad x, y, z)^3 = 0,$$

where the triangle of reference is formed by the three asymptotes.

6. The three asymptotes may be concurrent or non-concurrent; in the first case the curve has a centre, and in the second case it has not. The condition for concurrency is

$$l : m : n = f : g : h,$$

and the point of concurrence is the centre of the curve.

7. When the asymptotes are concurrent, the cubic may be written in the form

$$xyz + k(x + y + z)^3 = 0,$$

and the centre is at the point $(1, 1, 1)$. The curve is then symmetrical with respect to the centre, and is called a *central cubic.*

8. When the asymptotes are non-concurrent, the cubic may be written in the form

$$xyz + (lx + my + nz)(px + qy + rz)^2 = 0,$$

where $lx + my + nz = 0$ is the line of the asymptotes and $px + qy + rz = 0$ is a transversal.

9. The central cubic has nine inflexions, which lie by threes on twelve lines; the configuration is that of the nine points of inflexion of a cubic curve, which is well known in the theory of elliptic functions.

10. The non-central cubic has three inflexions, which lie on the line $lx + my + nz = 0$; the line of the three inflexions is called the *inflexion line* of the curve.

11. The first head division contains the following species, classified according to the nature of the finite singularities:

(1) The general cubic, with no singularities; $m = 6$, $\iota = 9$, $d = 1$.

(2) The nodal cubic, with one node; $m = 4$, $\iota = 3$, $d = 0$.

(3) The cuspidal cubic, with one cusp; $m = 3$, $\iota = 1$, $d = 0$.

12. The three species may be further subdivided according to the nature of the points at infinity; and the complete enumeration gives twenty-two species in the first head division.

The Second Head Division, Article Nos. 13 to 20.

13. The second head division comprises the cubic curves which are met by the line infinity in a twofold point and a onefold point, where the twofold point is a proper point of contact; that is, the curves having a parabolic asymptote and a rectilinear asymptote. The general equation is

$$U = (a, b, c, d, f, g, h, l, m, n \;\; x, y, z)^3 = 0,$$

where the line $z = 0$ is the line infinity, and the point $(1, 0, 0)$ is the point of contact.

14. The parabolic asymptote is the line $z = 0$, which touches the curve at the point $(1, 0, 0)$; and the rectilinear asymptote is the line $y = 0$, which meets the curve at the point $(0, 1, 0)$. The third point of intersection of the line infinity with the curve is at the point $(1, 0, 0)$, and is a twofold point.

15. The cubic curves of the second head division may be divided into two classes, according as the parabolic asymptote is or is not parallel to the rectilinear asymptote; the two cases correspond to different arrangements of the branches at infinity.

16. The equation of the curve may be written in the form

$$yz^2 = ax^3 + 3bx^2y + 3cxy^2 + dy^3,$$

where $z = 0$ is the line infinity; and the parabolic asymptote is $z^2 = 0$.

17. The curve has a single inflexion at infinity, which lies on the rectilinear asymptote; and the tangent at the inflexion is the line infinity itself.

18. The second head division contains the following species:

(1) The general cubic of the division, with no finite singularities; $m = 4$, $\iota = 3$, $d = 0$.

(2) The nodal cubic, with one node; $m = 2$, $\iota = 1$, $d = 0$.

(3) The cuspidal cubic, with one cusp; $m = 1$, $\iota = 0$, $d = 0$.

19. The species may be further subdivided according to the nature of the branches at infinity; and the complete enumeration gives eighteen species in the second head division.

20. The classification of the species in this division depends on the discriminant of the cubic form $ax^3 + 3bx^2y + 3cxy^2 + dy^3$; and the different cases correspond to the different ways in which the discriminant may vanish.

The Third Head Division, Article Nos. 21 to 28.

21. The third head division comprises the cubic curves which are met by the line infinity in a twofold point and a onefold point, where the twofold point is a node; that is, the curves having a node at infinity and a rectilinear asymptote. The general equation is

$$U = (a, b, c, d, f, g, h, l, m, n \;\; x, y, z)^3 = 0,$$

where the line $z = 0$ is the line infinity, and the node is at the point $(1, 0, 0)$.

22. The node at infinity is formed by the intersection of two branches of the curve, which have distinct tangents; the tangents at the node are the lines $y = 0$ and $z = 0$, and the rectilinear asymptote is the line $x = 0$.

23. The cubic curves of this division may be divided into two classes, according as the node at infinity is a crunode or an acnode; the two cases correspond to different arrangements of the branches at infinity.

24. The equation of the curve may be written in the form

$$xyz = ax^3 + 3bx^2y + 3cxy^2 + dy^3,$$

where $z = 0$ is the line infinity, and the node is at $(1, 0, 0)$ with tangents $y = 0$ and $z = 0$.

25. The curve has no inflexions at infinity; but it may have finite inflexions, the number of which depends on the nature of the cubic form.

26. The third head division contains the following species:

(1) The general cubic of the division, with no finite singularities; $m = 4$, $\iota = 3$, $d = 0$.

(2) The nodal cubic, with one finite node; $m = 2$, $\iota = 1$, $d = 0$.

(3) The cuspidal cubic, with one finite cusp; $m = 1$, $\iota = 0$, $d = 0$.

27. The species may be further subdivided according as the node at infinity is a crunode or an acnode; and the complete enumeration gives sixteen species in the third head division.

28. The classification depends on the discriminant of the cubic form, and on the condition that the node at infinity may be a crunode or an acnode; the different cases are classified by the simultaneous vanishing of certain invariants.

The Fourth Head Division, Article Nos. 29 to 36.

29. The fourth head division comprises the cubic curves which are met by the line infinity in a twofold point and a onefold point, where the twofold point is a cusp; that is, the curves having a cusp at infinity and a rectilinear asymptote. The general equation is

$$U = (a, b, c, d, f, g, h, l, m, n \;\; x, y, z)^3 = 0,$$

where the line $z = 0$ is the line infinity, and the cusp is at the point $(1, 0, 0)$.

30. The cusp at infinity is formed by the contact of two branches of the curve, which have a common tangent; the tangent at the cusp is the line $y = 0$, and the rectilinear asymptote is the line $x = 0$.

31. The cubic curves of this division have a peculiar character, arising from the cusp at infinity; the branch at infinity is a single branch with a point of return, and the curve has no finite singularities in the general case.

32. The equation of the curve may be written in the form

$$y^2 z = ax^3 + 3bx^2 y + 3cxy^2 + dy^3,$$

where $z = 0$ is the line infinity, and the cusp is at $(1, 0, 0)$ with tangent $y = 0$.

33. The curve has one inflexion at infinity, which lies on the rectilinear asymptote; and the tangent at the inflexion is the line infinity itself.

34. The fourth head division contains the following species:

(1) The general cubic of the division, with no finite singularities; $m = 3$, $\iota = 1$, $d = 0$.

(2) The nodal cubic, with one finite node; $m = 1$, $\iota = 0$, $d = 0$.

35. The species may be further subdivided according to the nature of the branch at infinity; and the complete enumeration gives twelve species in the fourth head division.

36. The classification depends on the discriminant of the cubic form $ax^3 + 3bx^2y + 3cxy^2 + dy^3$, and on the condition that the cusp at infinity may give rise to different arrangements of the finite branches.

The Fifth Head Division, Article Nos. 37 to 44.

37. The fifth head division comprises the cubic curves which are met by the line infinity in a threefold point, where the threefold point is an inflexion; that is, the curves having an inflexion at infinity, the line infinity being the tangent at the inflexion. The general equation is

$$U = (a, b, c, d, f, g, h, l, m, n \ \ x, y, z)^3 = 0,$$

where the line $z = 0$ is the line infinity, and the inflexion is at the point $(1, 0, 0)$.

38. The inflexion at infinity is a point where the curve has three-point contact with the line infinity; the tangent at the inflexion is the line infinity itself, and the curve has no asymptote in the ordinary sense.

39. The cubic curves of this division may be divided into two classes, according as the inflexion at infinity is of the first or second kind; the distinction depends on the nature of the branches at the inflexion.

40. The equation of the curve may be written in the form

$$z^3 = ax^3 + 3bx^2y + 3cxy^2 + dy^3,$$

where $z = 0$ is the line infinity, and the inflexion is at $(1, 0, 0)$ with tangent $z = 0$.

41. The curve has no asymptotes, but it has three inflexions, which lie on a line called the *inflexion line* of the curve.

42. The fifth head division contains the following species:

(1) The general cubic of the division, with no finite singularities; $m = 3$, $\iota = 3$, $d = 0$.

(2) The nodal cubic, with one finite node; $m = 1$, $\iota = 1$, $d = 0$.

43. The species may be further subdivided according to the nature of the inflexion at infinity; and the complete enumeration gives fourteen species in the fifth head division.

44. The classification depends on the discriminant of the cubic form, and on the condition that the inflexion at infinity may be of the first or second kind; the different cases are classified by the vanishing of certain invariants.

The Sixth Head Division, Article Nos. 45 to 52.

45. The sixth head division comprises the cubic curves which are met by the line infinity in a threefold point, where the threefold point is a node;

that is, the curves having a node at infinity, the line infinity touching one of the branches at the node. The general equation is

$$U = (a, b, c, d, f, g, h, l, m, n \;\; x, y, z)^3 = 0,$$

where the line $z = 0$ is the line infinity, and the node is at the point $(1, 0, 0)$.

46. The node at infinity is a point where two branches of the curve intersect, and the line infinity touches one of the branches; the tangent to the other branch is a finite line, and the curve has a parabolic branch at infinity.

47. The cubic curves of this division have a peculiar character, arising from the node at infinity with the line infinity as a tangent; the curve has no asymptote in the ordinary sense, but it has a parabolic branch at infinity.

48. The equation of the curve may be written in the form

$$yz^2 = ax^3 + 3bx^2y + 3cxy^2 + dy^3,$$

where $z = 0$ is the line infinity, the node is at $(1, 0, 0)$, and the tangent to one branch is $z = 0$.

49. The curve has one finite inflexion, which lies on the line joining the node to the point of contact of the tangent at the node; and the tangent at the inflexion passes through the node.

50. The sixth head division contains the following species:

(1) The general cubic of the division, with no finite singularities; $m = 2$, $\iota = 1$, $d = 0$.

(2) The nodal cubic, with one finite node; $m = 0$, $\iota = 0$, $d = 0$.

51. The species may be further subdivided according to the nature of the node at infinity; and the complete enumeration gives ten species in the sixth head division.

52. The classification depends on the discriminant of the cubic form, and on the condition that the node at infinity may be a crunode or an acnode; the different cases are classified by the vanishing of certain invariants.

The Seventh Head Division, Article Nos. 53 to 60.

53. The seventh head division comprises the cubic curves which are met by the line infinity in a threefold point, where the threefold point is a cusp; that is, the curves having a cusp at infinity, the line infinity being the tangent at the cusp. The general equation is

$$U = (a, b, c, d, f, g, h, l, m, n \;\; x, y, z)^3 = 0,$$

where the line $z = 0$ is the line infinity, and the cusp is at the point $(1, 0, 0)$.

54. The cusp at infinity is a point where two branches of the curve have contact of the second order with the line infinity; the curve has no asymptote in the ordinary sense, but it has a cuspidal branch at infinity.

55. The cubic curves of this division are the simplest of all cubic curves, having no finite singularities in the general case; the cusp at infinity is the only singularity, and the curve is rational.

56. The equation of the curve may be written in the form

$$z^3 = x^2(ax + by),$$

where $z = 0$ is the line infinity, and the cusp is at $(1, 0, 0)$ with tangent $z = 0$.

57. The curve has one finite inflexion, which lies on the line joining the cusp to the point of contact of the tangent at the cusp; and the tangent at the inflexion passes through the cusp.

58. The seventh head division contains the following species:

(1) The general cubic of the division, with no finite singularities; $m = 1$, $\iota = 0$, $d = 0$.

(2) The nodal cubic, with one finite node; this case is impossible, since the cusp at infinity already exhausts the singularities.

59. The species may be further subdivided according to the nature of the cusp at infinity; and the complete enumeration gives eight species in the seventh head division.

60. The classification of the seventh head division is the simplest of all, since the only variable element is the nature of the cusp at infinity; and the theory is complete.

Summary of the Seven Head Divisions, Article Nos. 61 to 65.

61. The seven head divisions may be tabulated as follows:

Division	**Nature of contact with line Infinity**	m	ι
I	Three onefold points	6	9
II	Twofold (proper tangent) + onefold	4	3
III	Twofold (node) + onefold	4	3
IV	Twofold (cusp) + onefold	3	1
V	Threefold (inflexion)	3	3
VI	Threefold (node)	2	1
VII	Threefold (cusp)	1	0

where m is the class and ι the number of inflexions for the general cubic of each division.

62. The total number of species in the seven head divisions is 219, agreeing with Plücker's enumeration; the distribution among the divisions is as follows:

Division	Species
I	22
II	18
III	16
IV	12
V	14
VI	10
VII	8
Total	100

which is the number of species having no finite singularities.

63. Taking into account the finite singularities, the total number of species is 219; and the enumeration is complete. The finite singularities give rise to additional species in each division, according to the number of nodes and cusps.

64. The complete table of the 219 species is given in the *Transactions of the Cambridge Philosophical Society*, vol. XI. Part I., and is reproduced in the present Memoir.

65. The classification is founded on the principles established by Newton, and extended by the researches of Plücker, Dr Salmon, and others; and it is believed to be the most complete classification of cubic curves which has yet been given.

The Table of Species, Article Nos. 66 to 75.

66. The table of species is arranged according to the seven head divisions, and in each division according to the nature of the finite singularities; the Plückerian characteristics are given for each species, and the corresponding Newtonian genus and species are indicated.

67. The table is as follows:

First Head Division. Three onefold points at Infinity.

No.	Description	m	ι	d
1	General cubic	6	9	1
2	Nodal cubic	4	3	0
3	Cuspidal cubic	3	1	0

68. The first head division contains twenty-two species, classified according to the nature of the finite singularities; the species are numbered from 1 to 22, and the Plückerian characteristics are given for each.

69. The twenty-two species of the first head division are as follows:

Species	δ	κ	m	ι
1	0	0	6	9
2	1	0	4	3
3	0	1	3	1
4	2	0	2	0
5	1	1	1	0
6	0	2	0	0
7	3	0	0	0
8	2	1	0	0
9	1	2	0	0
10	0	3	0	0
11	1	0	4	3
12	0	0	6	9
13	2	0	2	0
14	1	1	1	0
15	0	2	0	0
16	1	0	4	3
17	0	1	3	1
18	1	0	4	3
19	0	0	6	9
20	0	1	3	1
21	1	0	4	3
22	0	0	6	9

where δ is the number of nodes and κ the number of cusps.

70. The second head division contains eighteen species, classified according to the nature of the finite singularities and the arrangement of the branches at infinity; the species are numbered from 23 to 40.

71. The third head division contains sixteen species, classified according to the nature of the finite singularities and the nature of the node at infinity; the species are numbered from 41 to 56.

72. The fourth head division contains twelve species, classified according to the nature of the finite singularities and the nature of the cusp at infinity; the species are numbered from 57 to 68.

73. The fifth head division contains fourteen species, classified according to the nature of the finite singularities and the nature of the inflexion at infinity; the species are numbered from 69 to 82.

74. The sixth head division contains ten species, classified according to the nature of the finite singularities and the nature of the node at infinity; the species are numbered from 83 to 92.

75. The seventh head division contains eight species, classified according to the nature of the cusp at infinity; the species are numbered from 93 to 100.

Newton's Genera and Plücker's Species, Article Nos. 76 to 85.

76. Newton's classification of cubic curves is founded on the nature of the infinite branches; there are fourteen genera, containing together seventy-two species, which were afterwards increased to seventy-eight by Stirling, Murdoch, and Cramer.

77. The fourteen genera of Newton correspond to different arrangements of the infinite branches; and each genus contains a certain number of species, classified according to the nature of the finite singularities.

78. The correspondence between Newton's genera and Plücker's species is given in the following table:

Newton's Genus 1, contains 5 Species, viz.

No.	1	2	3	4	5
corresponding to Plücker's Species	I.	II.	III.	IV.	V.
Species	1	2	3	4	5

79. The correspondence is as follows:

Newton	1	2	3	4	5	6
Plücker	I.	II.	III.	IV.	V.	VI.

and similarly for the remaining genera.

80. Newton's Genus 2, contains 14 Species, viz.

10, 10′, 11, 12, 13, 13′, 14, 15, 16, 17, 18, 19, 20, 21

corresponding to Plücker's Species
IX. 38, 39, 40, ...

81. Newton's Genus 3, contains 7 Species, viz.

22, 23, 24, 25, 26, 27, 28

corresponding to Plücker's Species
XIV. 47, 48, 49, 50, 51, 52, 53

82. Newton's Genus 4, contains 7 Species, viz.

29, 30, 31, 32, 33, 34, 35

corresponding to Plücker's Species
XXI. 54, 55, 56, 57, 58, 59, 60

83. Newton's Genus 5, contains 4 Species, viz.

36, 37, 38, 39

corresponding to Plücker's Species
XXVIII. 61, 62, 63, 64

84. The remaining genera are classified in a similar manner; and the complete correspondence between Newton's seventy-eight species and Plücker's two hundred and nineteen species is given in the table.

85. The table of correspondence shows that Plücker's classification is much more complete than Newton's; and that the two hundred and nineteen species of Plücker include all the seventy-eight species of Newton, with additional species arising from the more minute classification of the singularities.

The Complete Enumeration, Article Nos. 86 to 95.

86. The complete enumeration of the 219 species of cubic curves is given in the following table, which shows the number of species in each head division, classified according to the nature of the finite singularities:

	Head Division						
δ	I	II	III	IV	V	VI	VII
0	22	18	16	12	14	10	8
1	18	14	12	8	10	6	4
2	12	8	6	4	6	2	0
3	6	2	0	0	2	0	0
Total	58	42	34	24	32	18	12

where δ is the number of nodes.

87. The total number of species is thus

$$58 + 42 + 34 + 24 + 32 + 18 + 12 = 220,$$

but one of these is impossible (the case $\delta = 3$, $\kappa = 0$ in the seventh division), so that the actual number is 219.

88. The classification is verified by the Plückerian equations; and the number of species in each division is in accordance with the theory of the singularities of cubic curves.

89. The present Memoir contains the complete theory of the classification of cubic curves, founded on the principles established by Newton and extended by the researches of Plücker, Dr Salmon, and others.

90. The classification is the most complete which has yet been given; and it is believed that the 219 species include all possible forms of cubic curves.

91. The theory of the classification of cubic curves is intimately connected with the theory of the singularities of curves; and the investigation of the singularities leads to a complete understanding of the different forms of cubic curves.

92. The present Memoir contains also the theory of the involution of cubic curves, which is a particular case of the general theory of the classification; and the two theories are combined in the complete enumeration of the 219 species.

93. The classification of cubic curves is a subject which has occupied the attention of geometers for more than two centuries; and the present Memoir contains the most complete account of the theory which has yet been given.

94. The author hopes that the present Memoir will be found useful to those who are engaged in the study of the higher geometry; and that it will contribute to the advancement of the science.

95. The Memoir concludes with a table of the 219 species, arranged according to the seven head divisions, and showing the Plückerian characteristics and the corresponding Newtonian genus and species for each.

Cambridge, February 8, 1864.

Plate III

Note.—The plate contains figures representing the different forms of cubic curves arising in the several head divisions. The points marked C_3, D_3, E_3, A_3 *are critic centres and singular points of the curves. The diagrams illustrate the arrangement of the branches and the nature of the singularities for representative species in each division.*

[Plate III — Figure of Cubic Curves]

The original plate contains diagrams illustrating the various species of cubic curves classified in the Memoir.

Abbildung 1: *
Plate III. Diagrams of Cubic Curves (see Article [350]).

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[350] On the Classification of Cubic Curves (concluded)

72. We may fix the absolute magnitudes of the coordinates by writing $x+z=1$; and the equation then becomes

$$(x^2+y^2)(1-x)+4x\{(\lambda-\nu)x+\mu y+\nu\}=0.$$

The origin is at the asymptote-point, or intersection of the imaginary asymptotes; the equation of the real asymptote is $x=1$; that of the imaginary asymptotes is $x^2+y^2=0$.

If x and y are ordinary rectangular coordinates then the pair of lines represented by this equation will be an indefinitely small circle, and conversely, if the Asymptote-Point be an indefinitely small circle, then x and y will be rectangular coordinates; and we may without loss of generality assume that this is so.

73. The equation of the envelope is

$$\sqrt[3]{x(x+yi)}+\sqrt[3]{x(x-yi)}+\sqrt[3]{z}=0;$$

this gives successively

$$\begin{aligned}\sqrt[3]{x(x+yi)}+\sqrt[3]{x(x-yi)}-\sqrt[3]{z}&=-\sqrt[3]{3}\sqrt[6]{x^2+y^2},\\ x+z-\sqrt{x^2+y^2}&=2\sqrt[3]{z}\{\sqrt[3]{x(x+iy)}+\sqrt[3]{x(x-iy)}\},\\ (x+z)^2+x^2+y^2-2(x+z)\sqrt{x^2+y^2}&=4z(x+\sqrt{x^2+y^2}),\\ (x-z)^2+x^2+y^2&=2(x+z)\sqrt{x^2+y^2},\\ \{(x-z)^2+x^2+y^2\}^2&=4(x+z)^2(x^2+y^2);\end{aligned}$$

that is

$$(x-z)^2+2(x^2+y^2)\{(x-z)^2-2(x+z)^2\}+(x^2+y^2)^2=0.$$

Putting $z=1-x$, and therefore $x-z=2x-1$, and $x+z=1$, this becomes

$$(2x-1)^2+2(x^2+y^2)\{(2x-1)^2-2\}+(x^2+y^2)^2=0,$$

that is

$$(x^2+y^2)^2-2(x^2+y^2)(3x^2-20x+17)+(2x-1)^2=0,$$

which may also be written

$$y^4-2y^2(3x^2-20x+17)+9x^4+8x^3-40x^2-8x+1=0,$$

or, what is the same thing,

$$y^4 - 2y^2(x-1)(3x-17) + (x-1)(9x-1)(x+1)^2 = 0,$$

for the equation of the envelope. The solution of the equation in y gives

$$y^2 = (x-1)(3x-17) + (2x-3)\sqrt{-32(x-1)}.$$

74. The original curve $\sqrt[3]{x} + \sqrt[3]{y} + \sqrt[3]{z} = 0$ had the three nodes

$$(2,-1,-1), \quad (-1,2,-1), \quad (-1,-1,2);$$

and thence writing

$$\tfrac{1}{2}(x+yi), \quad \tfrac{1}{2}(x-yi), \quad z, \text{ for } x, y, z,$$

we find the nodes

$$(x=\tfrac{1}{2},\ y=\tfrac{1}{2}i), \quad (x=\tfrac{1}{2},\ y=-\tfrac{1}{2}i), \quad (x=-1,\ y=0);$$

the first and second of these are acnodes, or the curve has a pair of imaginary acnodes; the third is a crunode; and to find the directions at this point, if in the equation of the curve we write $x-\frac{1}{2}$ for x, the equation becomes

$$y^4 - 2y^2(x-2)(3x-20) + (x-2)(9x-10)x^2 = 0;$$

the lowest terms therefore are $20(-4y^2+x^2)=0$; and we have $y=\pm\frac{1}{2}(x+1)$ for the equation of the tangents at the crunode.

$y=0$ gives $x=\frac{1}{2}$, $x=\frac{1}{2}$ and (as a twofold value) $x=-1$, which belongs to the crunode.

$x=1$ gives $y^2=0$, or the line $x=1$ is a tangent of four-pointic intersection.

$x=0$ gives $y^4-34y^2+1=0$, that is $y^2=17\pm12\sqrt{2}$, or $y=\pm(3\pm2\sqrt{2})$.

75. The curve has a pair of asymptotic parabolas, and taking for the equation of one of them

$$(y-x\sqrt{3}+\beta)^2 = -4f(x+\alpha),$$

this gives

$$y = x\sqrt{3} - \beta + \sqrt{-4f(x+\alpha)},$$

and thence

$$2y' = 3x^2 - 2\beta x\sqrt{3} + \beta^2 - 4f(x+\alpha) + 2(x-\sqrt{3})\sqrt{-32(x+\alpha)},$$

which agrees with the value of y' in the envelope as to the terms x^2, x, and by properly determining α, β, it may be made to agree as to the terms x and x^0.

We have in the parabola

$$y_1 = 3x^2 - 2\beta x\sqrt{3} + \beta^2 + \sqrt{-32}\{2\alpha\sqrt{x} + (\alpha-\beta)\sqrt{x}\},$$

and in the curve

$$y' = 3x^2 - 20x + 11 + \sqrt{-32}\{2x\sqrt{x} - 4\sqrt{x}\}.$$

We have therefore

$$-2\beta\sqrt{3} = -20, \quad \alpha - \beta = -4,$$

giving

$$\beta = \frac{10}{\sqrt{3}}, \quad \alpha = \frac{10}{\sqrt{3}} - 4 = \frac{10-4\sqrt{3}}{\sqrt{3}},$$

so that the equation of the parabola is

$$\left(y - x\sqrt{3} + \tfrac{10}{\sqrt{3}}\right)^2 = -\tfrac{16}{\sqrt{3}}\left(x + \tfrac{10}{\sqrt{3}} - 4\right);$$

that of the other parabola is of course obtained by merely changing the sign of $\sqrt{3}$.

76. From the foregoing results we may trace the curve, but this may be done somewhat more easily by means of polar coordinates, viz. writing $x = r\cos\theta$, $y = r\sin\theta$, and therefore $z = 1 - r\cos\theta$, we have

$$\sqrt[3]{r^2\cos\tfrac{1}{2}\theta} + \sqrt[3]{r^2\cos\tfrac{1}{2}\theta} + \sqrt[3]{1 - r\cos\theta} = 0,$$

that is

$$r(\cos\theta + 8\cos^3\tfrac{1}{2}\theta) = 1;$$

or since

$$\cos\theta = 1 - 8\sin^2\tfrac{1}{4}\theta\cos^2\tfrac{1}{4}\theta = 1 - 8\cos^4\tfrac{1}{2}\theta + 8\cos^4\tfrac{1}{2}\theta,$$

we have

$$\cos\theta + 8\cos^3\tfrac{1}{2}\theta = (1 - 4\cos^2\tfrac{1}{2}\theta)^2 = (1 + 2\cos\theta)^2,$$

and the equation is

$$r = \frac{1}{(1+2\cos\theta)^2}.$$

$\theta = 0°$ gives $r = 1$. $\theta = 180°$ gives $r = 1$. $\theta = 240°$ gives $r = \infty$. $\theta = 360°$ gives $r = 1$,

values which agree with the results obtained by rectangular coordinates.

77. The form is shown in the figure; we see that the curve consists of a lower branch without any singularity; and of an upper branch which cuts itself in the crunode.

78. The equation of the twofold centre locus (making in the form $\sqrt[3]{x} + \sqrt[3]{y} + \sqrt[3]{z} = 0$, the foregoing transformation) is

$$\sqrt[3]{x(x+yi)} + \sqrt[3]{x(x-yi)} + \sqrt[3]{1-x} = 0,$$

that is

$$x + \sqrt{x^2+y^2} = 1-x, \quad \text{or} \quad \sqrt{x^2+y^2} = 1-2x,$$

which is

$$3x^2 - 4x - y^2 = -1,$$

or

$$9(x-\tfrac{2}{3})^2 - 3y^2 = 1.$$

This is a hyperbola having its centre at the harmonic point $x = \frac{2}{3}$, $y = 0$; having $x = \frac{1}{3}$, $x = 1$ for the extremities of the transverse axis, and such that the asymptotes are inclined to the axis of x at an angle $= 60°$; this curve is also shown in the figure.

79. Similarly making the transformation in the equation of the one-with-twofold centre locus written under the form $-(-x+y+z)(x-y+z)(x+y-z) + xyz = 0$, this becomes

$$-(-yi+z)(yi+z)(x-z) + \tfrac{1}{4}(x+yi)(x-yi)z = 0,$$

that is

$$-\tfrac{1}{4}(y^2+z^2)(x-z) + \tfrac{1}{4}(x^2+y^2)z = 0,$$

which is

$$-\tfrac{1}{4}z^2(x-z) + x^2z + y^2(xz-4x) = 0,$$

or, what is the same thing,

$$z(x-2z)^2 + y^2(5z-4x) = 0,$$

or, putting for z its value $= 1-x$, this is

$$(1-x)\{3x-2+y^2(5-9x)\} = 0,$$

that is

$$y^2 = \frac{(x-1)(3x-2)}{9x-5},$$

which is the equation of the one-with-twofold centre locus.

80. The curve is symmetrical in regard to the axis of x. And moreover

$x < \frac{2}{3}$, y is impossible,
$x = \frac{2}{3}$, $y^2 = \infty$, or the line $x = \frac{2}{3}$ is an asymptote,
$x = \frac{1}{2}$, $y^2 = 0$, which is a crunode,
$x = 1$, $y^2 = 0$,
$x > 1$, y is impossible.

The equation of the tangents at the crunode are $y^2 = 3(x - \frac{1}{2})$, or the tangents are inclined to the axis of x at angles $= 60°$. The curve consists, as shown in the figure, of a single branch cutting itself in the crunode, and tending on each side towards the asymptote.

81. The equation of the harmonic conic, making the foregoing transformation in the equation

$$\frac{\mu-\nu}{x} + \frac{\nu-\lambda}{y} + \frac{\lambda-\mu}{z} = 0,$$

becomes

$$\frac{\lambda-\nu+\mu i}{x+yi} + \frac{\lambda-\nu-\mu i}{x-yi} + \frac{\mu}{z} = 0,$$

that is

$$2(\lambda-\nu)x - 2\mu yi + 2\mu ix - \frac{\mu(x^2+y^2)}{z} = 0,$$

which is

$$\mu(x^2+y^2) - 2z\{(\lambda-\nu)y + \mu x\} = 0,$$

or, what is the same thing,

$$\mu(x^2+y^2-2zx) + 2(\lambda-\nu)yz = 0,$$

which, putting for z its value $= 1 - x$, is

$$\mu(3x^2+y^2-2x) + 2(\lambda-\nu)y(1-x) = 0,$$

or developing,

$$\mu(3x^2-2x) - 2(\lambda-\nu)xy + \mu y^2 + 2(\lambda-\nu)y = 0,$$

either of which is the equation of the harmonic conic corresponding to the satellite line $(\lambda-\nu)x + \mu y + \nu = 0$, or, since the direction is alone material, to the satellite line $(\lambda-\nu)x + \mu y = 0$. The second form shows that the conic is

$$\begin{array}{rl} \text{an ellipse} & \text{for } 3\mu^2 > (\lambda-\nu)^2, \\ \text{a parabola} & \text{for } 3\mu^2 = (\lambda-\nu)^2, \\ \text{a hyperbola} & \text{for } 3\mu^2 < (\lambda-\nu)^2. \end{array}$$

82. The first form shows that the conic passes through the four points which are the intersection of the ellipse $3x^2+y^2-2x = 0$, (or, as the equation may also be written, $9(x-\frac{1}{3})^2 + 3y^2 = 1$), with the pair of lines $(x-1)y = 0$: this is right, for the points in question are the three points $(x = 0,\ y = 0)$, $(x = 1,\ y = i)$, $(x = 1,\ y = -i)$, which are the vertices of the triangle formed by the asymptotes $(x^2+y^2)(x-1) = 0$; and the point $x = \frac{1}{3}$, $y = 0$, which is the harmonic point.

83. Putting for shortness $\lambda - \nu = \kappa$, so that the equation of the satellite line is $\kappa x + \mu y + \nu = 0$, and that of the corresponding harmonic conic is

$$\mu(3x^2 + y^2 - 2x) + 2\kappa y(1 - x) = 0,$$

the coordinates of the centre are found from the formulae

$$\tfrac{1}{2}(x + yi) : \tfrac{1}{2}(x - yi) : z = (\kappa + \mu i)^2 : (\kappa - \mu i)^2 : -\tfrac{1}{4}\mu^2,$$

whence, since $x + z = 1$, we have for the coordinates of the centre

$$x = \frac{\kappa^2 - \mu^2}{\kappa^2 + 3\mu^2}, \quad y = \frac{-2\mu\kappa}{\kappa^2 + 3\mu^2},$$

and it is easy to verify that these belong to a point on the twofold centre conic $3x^2 - 4x - y^2 + 1 = 0$.

84. The asymptotes of the harmonic conic meet at the centre; and they again cut the twofold centre conic in two points, the intersections of the last-mentioned conic with the line

$$(\kappa + \mu i)\sqrt[3]{x(x + yi)} + (-\kappa + \mu i)\sqrt[3]{x(x - yi)} - 2\mu z = 0,$$

that is

$$\mu x + \kappa y - 2\mu z = 0,$$

or, what is the same thing,

$$3\mu x + \kappa y - 2\mu = 0.$$

85. I remark that the equation of the asymptotes of the harmonic conic is

$$(3\mu^2 - \kappa^2)\{\mu(3x^2 + y^2 - 2x) + 2\kappa y(1 - x)\} + \mu(\mu^2 + \kappa^2) = 0,$$

and that the theorem for the construction of the asymptotes depends on the identical equation

$$\begin{aligned}(3\mu^2-\kappa^2)\{\mu(3x^2+y^2-2x)+2\kappa y(1-x)\}+\mu(y+\kappa)+3\mu(\mu^2+\kappa^2)(3x^2-4x+1)\\ = -2\{-(3\mu^2 - \kappa^2)x + 2\mu\kappa y + \mu^2 + \kappa^2\}(3\mu x + \kappa y - 2\mu),\end{aligned}$$

which is easily verified; and where

$$-(3\mu^2 - \kappa^2)x + 2\mu\kappa y + \mu^2 + \kappa^2 = 0$$

is the equation of the tangent of the twofold centre conic $3x^2-y^2-4x+1 = 0$ at the centre of the harmonic conic.

86. On account of the symmetry in regard to the axis of x, it will be sufficient to attend to the series of curves corresponding to a direction $y = \xi x$ of the satellite line, for which the ratio $\dfrac{\mu}{\lambda - \nu}$ has a given sign; and the inclination of the satellite line to the asymptote will pass from 90° to 0° according as the value of the ratio $\frac{\mu}{\kappa}$ passes from 0 to ∞.

87. First, if $\frac{\mu}{\kappa} = 0$, that is, if the satellite line be perpendicular to the asymptote, then the harmonic conic is the ellipse

$$3x^2 + y^2 - 2x = 0,$$

or, as it may also be written,

$$9(x - \tfrac{1}{3})^2 + 3y^2 = 1.$$

As $\frac{\mu}{\kappa}$ increases from 0 to $\sqrt{3}$, that is, as the inclination of the satellite line diminishes from 90° to 30°, the harmonic conic becomes an ellipse of greater and greater excentricity, and ultimately a parabola.

88. I notice the particular case $\frac{\mu}{\kappa} = \frac{1}{2}$, which corresponds to the direction parallel to one of the nodal tangents of the envelope: the harmonic conic is in this case the ellipse

$$3x^2 + xy + y^2 - 2x - y = 0,$$

which, it will be observed, passes through the point $(x = 0,\ y = \frac{1}{2})$ which is one of the points of intersection of the twofold centre locus or hyperbola $3x^2 - 4x - y^2 = -1$ by the line $x = 0$.

89. For the value $\frac{\mu}{\kappa} = \sqrt{3}$ corresponding to a direction inclined at an angle = 30° to the asymptote, the harmonic conic becomes the parabola $(x\sqrt{3}-y)^2-2(x-y\sqrt{3}) = 0$: this equation may also be written $(x-\frac{1}{3})^2+y^2 = (x+y\sqrt{3}-1)^2$, a form which puts in evidence the focus and directrix of the parabola.

90. For a value $\frac{\mu}{\kappa} > \sqrt{3}$, that is, when the satellite line is inclined to the asymptote at an angle $< 30°$, the harmonic conic is a hyperbola, and ultimately when $\frac{\mu}{\kappa} = \infty$, or the satellite line is parallel to the asymptote, the harmonic conic becomes the pair of lines $y(1 - x) = 0$.

91. I have in the figure shown the following forms of the harmonic conic; viz.

- hyperbola, corresponding to inclination $< 30°$ of satellite line to asymptote.
- parabola, to inclination $= 30°$.
- ellipse $\frac{\mu}{\kappa} = \frac{1}{2}$, to inclination $(= \tan^{-1} 2)$ of a nodal tangent of the envelope.
- ellipse $\frac{\mu}{\kappa} > \frac{1}{2}$;
- ellipse $\frac{\mu}{\kappa} < \frac{1}{2}$;

and for these forms respectively the successive positions of the satellite line are indicated as follows:

92. For the inclination $< 30^\circ$, the positions are A, C, M, C', D, E, A', viz.

A, at infinity,
C, touching lower branch of envelope,
M, between C and C',
C', touching upper branch of envelope,
D, passing through asymptote point D_1,
E, passing through crunode of envelope,
A', at infinity.

The corresponding positions of the critic centres on the hyperbola are

A_1, A_2, each at infinity,
C_2, a twofold centre,
M_2, a real centre,
C_2', a one-with-twofold centre,
D_2, (Asymptote-Point), D_3,
E_2, and E_3,
A_1', A_2', at infinity, harmonic point.

93. For inclination $= 30^\circ$ the positions are (AC), M, C', D, E, $(A'C)$, viz.

AC, at infinity, touches envelope at infinity,
M, between (AC) and C',
C', touching upper branch of envelope,
D, passing through asymptote point D_1,
E, passing through crunode of envelope,
$A'C$, at infinity touches envelope at infinity.

The corresponding positions of the critic centres on the parabola are

A_2, (harmonic point) a one-with-twofold centre; $A_3 = A_3'$ at infinity, a twofold centre,
M_1, real centre, the other two centres being imaginary,
C_3', a twofold centre, C_3', a one-with-twofold centre,
D_1, (asymptote point), D_2; D_3,
E_1, E_3; E_2.

94. For inclination $= \tan^{-1} 2$, the positions are A, M, C', D, (EC''), N, A', viz.

A, at infinity,
M, between A and C,
C', touching upper branch of envelope,
D, through asymptote point D_1,
EC'', touching upper branch of envelope at crunode N between EC'' and A',
A', at infinity.

95. And for the inclinations $<$ and $>$ $\tan^{-1} 2$, the only difference is that the positions are A, C', D, E, C'', N, A', viz.

A, at infinity,
M, between A and C'',

C', touching upper branch of envelope,
D, through asymptote point D_1,
E, through crunode,
C'', touching upper branch of envelope,
N, between C'' and A',
A', at infinity,

and that instead of the points $C'E_3$ and $C''E_2$ we have separately the points C_3'', C_3' and E_2, E_3, E_1 as shown in the figure.

96. For the better understanding of the figure it is to be observed that the points D_1 and D_3 lie on the line $x = \frac{1}{2}$: this depends on the theorem No. 81 of the Memoir on Involution, viz. of the critic centres which belong to a satellite line through the vertex $(0, 0, 1)$, one is the vertex itself, the other two lie on the line $x + y - 2 = 0$; or making the transformation $\frac{1}{2}(x + iy)$, $\frac{1}{2}(x - iy)$, z for x, y, z and writing $x + z = 1$, of the critic centres which pass through the vertex $(0, 0, 1)$ (the asymptote point), one is this point itself, the other two lie on the line $x - z = 0$, that is $x = \frac{1}{2}$.

97. Again it is to be observed that the centre E_2 lies on the line $x = 0$, and the centres E_3, E_1 on the circle (indicated by a dotted line in the figure) $(x + \frac{1}{2})^2 + y^2 = \frac{1}{4}$: this depends on the theorem Nos. 73 and 74 of the Memoir on Involution, viz. of the critic centres for satellite lines through the node (acnode) $(1, 1, -4)$, one lies on the line $x + y = 0$, and the other two lie on the conic $z(x + y + z) - 4xy = 0$; making the substitution

$$\tfrac{1}{2}(x + yi), \quad \tfrac{1}{2}(x - yi), \quad z \quad \text{for } x, y, z,$$

and writing also $x + z = 1$, we find that, for the satellite lines which pass through the crunode $(1, 0, 2)$, the critic centres lie one on the line $x = 0$, the other two on the conic

$$z(x + z) - x^2 - y^2 = 0,$$

that is, on the circle $x^2 + y^2 + x - 1 = 0$, or $(x + \frac{1}{2})^2 + y^2 = \frac{1}{4}$.

98. The circle in question cuts the twofold centre conic $3x^2 - 4x - y^2 = -1$ at its intersections with the line $x = 0$, viz. in the points $x = 0$, $y = \pm 1$; and it moreover touches the one-with-twofold centre locus $y^2 = \dfrac{(x - 1)(3x - 2)}{9x - 5}$ at the point where this same circle meets the ellipse $3x^2 + xy + y^2 - 2x - y = 0$, which is the harmonic conic corresponding to the inclination $\tan^{-1} 2$. In fact, writing down the three equations,

$$\begin{aligned} x^2 + y^2 + x - 1 &= 0, \\ 3x^2 + xy + y^2 - 2x - y &= 0, \\ y^2 &= \frac{(x - 1)(3x - 2)}{9x - 5}, \end{aligned}$$

the first and third equations give

$$-(x - 1)(3x - 2) + (9x - 5)(x^2 + x - 1) = 0,$$

or, reducing,

$$(5x-3)^2 = 0,$$

that is $x = \frac{3}{5}$, and then from the first or third equation $y^2 = \frac{1}{25}$, or $y = \pm\frac{1}{5}$; hence the circle touches the one-with-twofold centre locus at the points $(\frac{3}{5}, \pm\frac{1}{5})$ and by means of the second equation we see that the first of these points, viz. the point $x = \frac{3}{5}$, $y = -\frac{1}{5}$, is a point of the ellipse or harmonic conic $3x^2 + xy + y^2 - 2x - y = 0$.

99. I consider the analytical theory of the case where the satellite line is parallel to the asymptote; this is in fact similar to the theory ante Nos. 67–69; writing $\frac{1}{2}(x+yi)$, $\frac{1}{2}(x-yi)$, z, $\lambda+\mu i$, $\lambda-\mu i$, ν in the place of $x, y, z, \lambda, \mu, \nu$, and putting afterwards $\mu = 0$, that is, in the transformed equation $\lambda x + \mu y + \nu z = 0$ writing $\mu = 0$, we find for the satellite line $\lambda x + \nu(1-x) = 0$; the equation in θ (the factor $\theta + \lambda = 0$ being disregarded) is

$$\theta^2 - \lambda\theta - 2\lambda\nu = 0,$$

and the corresponding critic centres lie on the line $y = 0$, at the distances given by the equation

$$\frac{x}{z} = \frac{\theta - (\lambda - \nu)}{\nu},$$

and the values of z are given by the equation

$$\lambda(1-z)^2 - \lambda z(1-z) - 2\nu z^2 = 0,$$

that is

$$2(\lambda - \nu)z^2 - 3\lambda z + \lambda = 0,$$

which, putting $\dfrac{\nu}{\lambda - \nu} = 1 - \alpha$, or $\alpha = \dfrac{\nu}{\lambda - \nu}$ (α is the distance of the satellite line from the asymptote $z = 0$), becomes

$$2Z^2 - 3\alpha Z + \alpha = 0,$$

or we have

$$Z = \tfrac{1}{4}\{3\alpha \pm \sqrt{\alpha(9\alpha - 8)}\}.$$

The condition for a twofold centre is ($\alpha = 0$ which may be disregarded, or else) $9\alpha - 8 = 0$; or, what is the same thing, $\lambda + 8\nu = 0$.

100. If z_1, z_2 are the coordinates of the two critic centres, then we have

$$2z_1^2 = \tfrac{1}{\nu}\{(\lambda - \nu)z_1 - \lambda\}, \quad 2z_2^2 = \tfrac{1}{\nu}\{(\lambda - \nu)z_2 - \lambda\},$$

and thence

$$\frac{z_1^2}{z_2^2} = \frac{3z_1 - 1}{3z_2 - 1},$$

or, reducing,

$$3z_1z_2 - z_1 - z_2 = 0,$$

or in terms of the x-coordinates

$$3x_1x_2 - 2(x_1 + x_2) + 1 = 0,$$

which equation however merely expresses that the two centres are harmonics to each other in regard to the twofold centre conic $3x^2 - 4x - y^2 + 1 = 0$. It is right to remark that the formulae, although referring to a different system of coordinates, are absolutely identical with those given Nos. 67–69, writing therein ν for n, and z for x.

101. An inspection of the form of the envelope shows what are the positions of the satellite line which gives rise to Pluecker's Groups for the Hyperbolas A Defective. We have in fact.

HYPERBOLAS A DEFECTIVE, ASYMPTOTE NOT OSCULATING.

The satellite line is not parallel to the asymptote, and the different positions give the following six of Pluecker's Groups, viz.

XVIII. Satellite line cuts upper, cuts lower, branch of envelope.
XIX. Satellite line touches upper, cuts lower, branch.
XX. Satellite line cuts upper, cuts lower, branch and passes through asymptote point.
XXI. Satellite line does not cut upper, cuts lower, branch.
XXII. Satellite line cuts upper, cuts lower, branch, touches lower branch.
XXIII. Satellite line cuts upper, cuts lower, branch, does not cut lower branch.

HYPERBOLAS A DEFECTIVE, ASYMPTOTE OSCULATING.

The satellite line is parallel to the asymptote, and we have the six groups,

XXVIII. Satellite line cuts upper branch, cuts lower branch of envelope, viz. it lies above the asymptote point.
XXIX. Do., Do., but it passes through the asymptote point.
XXX. Do., Do., but it lies below the asymptote point.
XXXI. Satellite line touches upper branch, cuts lower branch.
XXXII. Satellite line does not cut upper branch, cuts lower branch.
XXXIII. Satellite line does not cut upper branch, does not cut lower branch, viz. it lies below the asymptote.

And finally,

HYPERBOLAS A DEFECTIVE, THREE OSCULATING ASYMPTOTES.

Satellite line at infinity, giving the single group

XXXV.

But the division gives rise to a remark such as is made ante No. 71.

As to the Groups of the Hyperbolas c. Article Nos. 102 to 104.

102. Taking $z = 0$ as the equation of the line infinity, and $x = 0$, $y = 0$ as the equation of any two lines through the point of intersection of the asymptotes, or 'asymptote point,' then the equation of the cubic may be taken to be

$$(a, b, c, d\,,\ x, y)^3 + kz^2(\lambda x + \mu y + \nu z) = 0.$$

103. To determine the critic centres we have

$$\begin{aligned}(a, b, c\,,\ x, y)^2 + kz^2\lambda &= 0,\\ (b, c, d\,,\ x, y)^2 + kz^2\mu &= 0,\\ 2z(\lambda x + \mu y + \nu z) + z^2k &= 0,\end{aligned}$$

and thence

$$\mu(a, b, c\,,\ x, y)^2 - \lambda(b, c, d\,,\ x, y)^2 = 0,$$

or as it may also be written

$$(\mu a - \lambda b, \mu b - \lambda c, \mu c - \lambda d\,,\ x, y)^2 = 0,$$

and also

$$2(\lambda x + \mu y) - 3\nu z = 0,$$

which two equations determine the critic centres for a given position of the satellite line; the first of them gives a pair of lines through the asymptote point; the latter is a line parallel to the satellite line: there are thus two critic centres.

104. The condition for a twofold centre is

$$(ac - b^2, bc - ad, bd - c^2\,,\ \lambda, \mu)^2 = 0,$$

so that there are a pair of twofold centres which will be real if

$$(bc - ad)^2 - 4(ac - b^2)(bd - c^2) = +,$$

imaginary if

$$(bc - ad)^2 - 4(ac - b^2)(bd - c^2) = -,$$

that is, the twofold centres will be real or imaginary, according as the equation $(a, b, c, d\,,\ x, y)^3 = 0$ has its roots one real and two imaginary, or all three real; viz. the twofold centres are real for the Hyperbolas c Defective; imaginary for the Hyperbolas c Redundant. And we see also that for the Hyperbolas c Redundant the critic centres are always real; but that for the Hyperbolas c Defective, they may be both real, or both imaginary, or may coincide together, giving a twofold centre. But the two cases are best studied by assuming different special forms for the equation.

The Hyperbolas c Redundant. Article Nos. 105 to 107.

105. The equation may be taken to be

$$xy(x-y)+kz^2(\lambda x+\mu y+\nu z)=0,$$

or writing $z=1$, then the equation is

$$xy(x-y)+k(\lambda x+\mu y+\nu)=0.$$

We may, to fix the ideas, consider the case where the three asymptotes are parallel to the sides of an equilateral triangle; x, y, and $x-y$ will then denote the perpendicular distances of the point from the three asymptotes respectively.

106. The critic centres are given by the equations

$$\begin{gathered} 2(\lambda x+\mu y)-3\nu=0, \\ \lambda x^2-2(\lambda+\mu)xy+\mu y^2=0; \end{gathered}$$

or, what is the same thing, they are the intersections of the line

$$2(\lambda x+\mu y)-3\nu=0$$

by the two real lines

$$\lambda x=\{(\lambda+\mu)\pm\sqrt{\lambda^2+\lambda\mu+\mu^2}\}y.$$

107. The groups are

HYPERBOLAS C REDUNDANT.

No osculating asymptote.

The satellite line not parallel to any asymptote, that is $\lambda=0$, $\mu=0$, $\lambda+\mu=0$. We have the two groups

VII. Satellite line does not pass through asymptote point ($\nu\neq 0$).

VIII. Satellite line passes through asymptote point ($\nu=0$).

HYPERBOLAS C REDUNDANT. One osculating asymptote. Satellite line is parallel to an asymptote, suppose to the asymptote $x=0$; that is, $\mu=0$, or the satellite line is $\lambda x+\nu=0$. We have only the group

XV.

HYPERBOLAS C REDUNDANT. Three osculating asymptotes. Satellite line lies at infinity, that is, $\lambda=0$, $\mu=0$. We have only the group

XVII.

The Hyperbolas c Defective. Article Nos. 108 to 110.

108. The equation may be taken to be

$$\tfrac{1}{4}x(x^2+y^2)+kz^2(\lambda x+\mu y+\nu z)=0,$$

or writing $z = 1$, then it is

$$\tfrac{1}{4}x(x^2+y^2)+k(\lambda x+\mu y+\nu)=0,$$

and if to fix the ideas we take the case where the two imaginary asymptotes are the asymptotes of a circle, then x, y will be ordinary rectangular coordinates.

109. The critic centres are given by the equations

$$\begin{aligned} 2(\lambda x+\mu y)-3\nu&=0,\\ 3\mu x^2-2\lambda xy+\mu y^2&=0, \end{aligned}$$

that is they are the intersections of the line $2(\lambda x+\mu y)-3\nu=0$ (which is a line parallel to the satellite line, on the other side of the asymptote point and at a distance from it $= \frac{3}{2}$ distance of satellite line) by the pair of lines

$$3\mu x=(\lambda\pm\sqrt{\lambda^2-3\mu^2})y.$$

Hence the critic centres are real if $\lambda^2 > 3\mu^2$, that is, if the satellite line is inclined to the asymptote at an angle $> 60°$; imaginary if $\lambda^2 < 3\mu^2$, that is, if the satellite line is inclined to the asymptote at an angle $< 60°$: and there is a twofold centre if $\lambda^2 = 3\mu^2$, that is, if the inclination is $= 60°$. This assumes however that $\nu \neq 0$, that is that the satellite line does not pass through the asymptote point; when it does the distinction of the cases disappears.

Hence the groups are

110. HYPERBOLAS C DEFECTIVE. No osculating asymptote. The Satellite line is not parallel to the asymptote, and the groups are,

Satellite line not passing through asymptote point.

XXIV. Satellite line inclined to asymptote at angle $> 60°$.
XXV. Satellite line inclined to asymptote at angle $= 60°$.
XXVI. Satellite line inclined to asymptote at angle $< 60°$.

Satellite line passes through asymptote point, the single group

XXVII.

HYPERBOLAS C DEFECTIVE. Real osculating asymptote. The satellite line is parallel to the asymptote, and we have the single group

XXXIV.

HYPERBOLAS C DEFECTIVE. Three osculating asymptotes. Satellite line is at infinity and we have the single group

XXXVI.

The foregoing theory of the hyperbolas A and c completes the enumeration of the groups I. to XXXVI.

As to the Groups of the parabolic hyperbolas. Article Nos. 111 to 115.

111. I consider the equation in the form

$$x(by^2 + cz^2 + 2gzx) + kz^2(\mu y + \nu z) = 0,$$

viz. the cubic $x(by^2+cz^2+2gzx) = 0$ is made up of a conic $by^2+cz^2+2gzx = 0$, and a line $x = 0$; the other cubic $z^2(\mu y+\nu z) = 0$ is made up of a tangent of the conic, regarded as a twofold line, $z^2 = 0$, and of a line $\mu y + \nu z = 0$ through the point of contact of such tangent.

112. To determine the critic centres we have

$$\begin{aligned} x \cdot gz + k(by^2 + cz^2 + tgzx) &= 0, \\ x \cdot by + kz \cdot \mu z &= 0, \\ x(cz + gz) + kz(2\mu y + 3\nu z) &= 0; \end{aligned}$$

eliminating k from the second and third equations

$$\mu z(cz + gx) - by(2\mu y + 3\nu z) = 0,$$

that is

$$-2b\mu y^2 + c\mu z^2 + g\mu zx - 3b\nu yz = 0;$$

or reducing by means of the first equation written under the form

$$by^2 + cz^2 + gzx = 0,$$

we find

$$3c\mu z^2 + 9g\mu zx - 3b\nu yz = 0,$$

that is $z = 0$, which may be rejected, or else

$$c\mu z + 3g\mu x - b\nu y = 0,$$

or, as it may be written,

$$\mu(3gx + cz) - \nu by = 0.$$

Hence the entire series of critic centres lie on the conic

$$by^2 + cz^2 + 4gzx = 0,$$

and corresponding to the satellite line $\mu y + \nu z = 0$, we have the two critic centres which are the intersections of the conic by the line

$$\mu(3gx + cz) - \nu by = 0,$$

the lines pass through the fixed point $3gx + cz = 0$, $y = 0$, and form a pencil homographic with the satellite lines $\mu y + \nu z = 0$.

113. We have a twofold centre if the line touches the conic, that is the condition for this is

$$(bc, -4g^2, 0, 0, -2bg, 0\,,\ 3g\mu, -b\nu, c\mu)^2 = 0,$$

that is $3bg(c\mu^2 + 4b\nu^2) = 0$, or simply,

$$3c\mu^2 + 4b\nu^2 = 0,$$

and from this and the equation $\mu y + \nu z = 0$, eliminating μ and ν we find

$$4by^2 + 3cz^2 = 0,$$

for the equation of the satellite lines which respectively give rise to a twofold centre; the lines in question are real or imaginary according as the lines $by^2 + cz^2 = 0$ are real or imaginary, that is, according as the line $x = 0$ cuts the conic $by^2 + cz^2 + 2gzx = 0$ in two real, or in two imaginary, points.

114. Writing now $b = 1$, $c = -mn$, $2g = n$ and $x + mz$ in the place of z, the equation is

$$(x + mz)(y^2 + nzx) + 2kz^2(\mu y + \nu z) = 0,$$

and we may consider $z = 0$ as the equation of the line infinity: writing in the formulae $z = 1$, the critic centres are given as the intersection of the parabola

$$y^2 + 2nx + mn = 0,$$

with the line

$$\nu y = \mu m(3x + m);$$

and the condition for a two-fold centre is

$$4\nu^2 = -3mn\mu^2 = 0;$$

the equation of the satellite lines corresponding respectively to a two-fold centre is

$$4y^2 = 3mn = 0;$$

the lines are real or imaginary according as mn is positive or negative, or (observing that the equations $x + m = 0$, $y^2 + nx = 0$ give $y^2 - mn = 0$), according as the line $x+m = 0$ cuts or does not cut the parabola $y^2+nx = 0$. Suppose for a moment that the line does cut the parabola and that y_1^2 is the corresponding value of y, then $y_1^2 = mn$; and the equation $4y^2 - 3mn = 0$ of the satellite lines which correspond respectively to the case of a two-fold centre is $y^2 = \frac{3}{4}y_1^2$. We have thus $y = \pm y_1$ and $y = \pm\frac{\sqrt{3}}{2}y_1$ as special positions of the satellite line $\mu y + \nu = 0$. In the case where the line $x+m = 0$ touches the parabola $y^2 + nx = 0$, the value of y_1^2 is $= 0$, and we have only the special position $y = 0$; finally, when the line does not cut the parabola there is no special position.

115. Pluecker's groups are consequently as follows:

PARABOLIC HYPERBOLAS; ordinary asymptote and five-pointic asymptotic parabola, that is the line $\mu y + \nu = 0$ is not at infinity.

Asymptote cuts parabola, $mn = +$.

XXXVII. Satellite line lies outside the lines $y = \pm y_1$.

XXXVIII. Satellite line passes through a point of intersection, that is, coincides with one of the lines $y = \pm y_1$.

XXXIX. Satellite line lies between the lines $y = \pm y_1$ and $y = \pm\frac{\sqrt{3}}{2}y_1$.

XL. Satellite line coincides with one of the lines $y = \pm\frac{\sqrt{3}}{2}y_1$, which give respectively a two-fold centre.

XLI. Satellite line lies between the lines $y = \pm\frac{\sqrt{3}}{2}y_1$.

Asymptote touches parabola, viz. $m = 0$.

XLIII. Satellite line does not pass through the point of contact.

XLIV. Satellite line passes through point of contact or its equation is $y = 0$.

Asymptote does not cut parabola, viz. $mn = -$. This gives the single group

XLII.

PARABOLIC HYPERBOLAS. Osculating asymptote and six-pointic asymptotic parabola. The satellite line is here at infinity, and there is no new distinction of groups. The groups therefore are

Asymptote cuts parabola,

XLV.

Asymptote does not cut parabola,

XLVI.

Asymptote touches parabola,

XLVII.

As to the Groups of the Central and Parabolic Hyperbolisms. Article No. 116.

116. For the Hyperbolisms, Central and Parabolic, since these have a node or a cusp at infinity, they cannot acquire a new node, and the theory of critic centres does not arise. There is, however, as regards the Hyperbolisms of the Hyperbola a distinction in the position of the satellite line, viz. this may lie outside, or between, the parallel asymptotes.

The groups are

HYPERBOLISMS OF THE HYPERBOLA. Ordinary asymptote. The satellite line is not at infinity, and it may lie in either of the positions just mentioned. We have therefore

XLVIII. Satellite line lies between the parallel asymptotes.

XLIX. Satellite line lies outside the parallel asymptotes.

Osculating asymptote; the satellite line is at infinity.

L.

We have

HYPERBOLISMS OF THE ELLIPSE. Ordinary asymptote. The satellite line is not at infinity, and we have

LI.

Osculating asymptote.

LII. Satellite line is at infinity,

HYPERBOLISMS OF THE PARABOLA. Ordinary asymptote. Satellite line is not at infinity we have

LIII.

Osculating asymptote: satellite line is at infinity: we have

LIV.

As to the Groups of the Divergent Parabolas. Article Nos. 117 and 118.

117. Taking the equation under the form

$$ax^3 + by^2z + kz^2(\lambda x + \nu z) = 0,$$

we find for a critic centre

$$\begin{aligned} 3ax^2z + kz \cdot \lambda z &= 0, \\ 2byz &= 0, \\ by^2 + kz(2\lambda x + 3\nu z) &= 0; \end{aligned}$$

hence there is a single critic centre $y = 0$, $2\lambda x + 3\nu z^2 = 0$, the critic value of k is $k = -\dfrac{2\lambda\nu^2}{27a}$, and with this value of k the equation in fact is

$$4\lambda^2(ax^3 + by^2z) - 27a\nu^2(\lambda x + \nu z)z^2 = 0,$$

that is

$$a(4\lambda^3x^3 - 27\lambda\nu^2xz^2 - 27\nu^3z^3) + 4\lambda^2by^2z = 0,$$

or, as it may be written,

$$a(2\lambda x + 3\nu z)^2(\lambda x - 3\nu z) + 4\lambda^2by^2z = 0,$$

which puts in evidence the critic centre or node of the curve.

But, as there is here only a single critic centre, there is of course no further theory of the two-fold centre, &c.

118. The groups are as given a priori by the relation of the satellite line $\lambda x + \nu z = 0$, to the semicubical parabola $ax^3 + by^2z = 0$, viz. writing $z = 1$ and changing the constants,

Divergent Parabolas, the semicubical parabola $y^2 = x^3$, which is

LV.

Divergent Parabolas, Asymptotic Semicubical Parabola of seven-pointic contact, viz. equation of the asymptotic parabola being $y^2 - x^3 = 0$, and writing for convenience $k(\lambda x + \nu) = -3ax + 2b = 0$ for the equation of the satellite line, the equation of the curve is $y^2 = x^3 - 3ax + 2b$. And the groups are

LVI. Satellite line cuts asymptotic parabola.
LVII. Satellite line does not cut asymptotic parabola.
LVIII. Satellite line passes through vertex of parabola.

And further

Divergent Parabolas. Asymptotic Semicubical Parabola of nine-pointic intersection. The satellite line is at infinity, and the equation is $y^2 = x^3 + 2b$. This is group

LIX.

As to the Trident Curve and the Cubical Parabola. Article Nos. 119 and 120.

119. For the Trident Curve, equation is $x(x^2 + 4y) + 16 = 0$, or the satellite line is at infinity, and there is no distinction of groups; we have only group

LX.

120. For the Cubical Parabola this is $x^3 + 16y = 0$, there is no distinction of groups, and the curve is group

LXI.

As to the Division into Species: Comparison of Newton and Pluecker. Article Nos. 121 and 122.

121. The division into species is obtained without difficulty when the groups are once established; in fact it only remains to trace for each given form of V and s the series of curves $V + \mu s = 0$, as μ passes from ∞ to $-\infty$ through the value 0 and the critic values which correspond to nodal curves: I have nothing to add to what has been done by Pluecker, and it is unnecessary to reproduce the investigation. It may be remarked that the mere inspection of Pluecker's figures is sufficient to show which of his species correspond to the same Newtonian species; the species which do so belong to the same Newtonian species in some instances closely resemble each other in form, although in others the difference of form is apparent enough: but the Plueckerian species which correspond to the same Newtonian species belong for the most part to different groups and are thus distinguished from each other by the characters which distinguish the groups to which they respectively belong: thus for instance Newton's Species 1 (a hyperbola A Redundant) is characterised as consisting of three hyperbolic branches, one inscribed, one circumscribed, and one ambigeue, with an oval. Such a curve may exist with three different positions of the satellite line in regard to the asymptotes, viz. the satellite line may cut the three sides produced, or it may pass through a vertex, cutting the opposite side produced, or it may cut two sides and the third side produced, not cutting the envelope — which are the characters of Pluecker's groups I, II, IV, respectively, and there belongs to the Newtonian species 1, a species out of each of these groups, viz. they are I. 1; II. 9, and IV. 18.

122. The correspondence of the Plueckerian Species with those of Newton is shown in the following Table.

NEWTON'S GENUS 1, CONTAINS 9 SPECIES, VIZ.

I.	1	corresponding to Pluecker's Species	1
II.	9	"	11
III.	18	"	4
IV.	3	"	8
V.	7	"	7
VI.	2	"	3
	12	"	12
	10	"	10
	21	"	21

PART OF NEWTON'S GENUS 4, CONTAINS 3 SPECIES, VIZ.

VII.	27	corresponding to Pluecker's Species	27
VIII.	19	"	19
	17	"	17

NEWTON'S GENUS 2, CONTAINS 14 SPECIES, VIZ.

IX.	5, 6		corresponding to Pluecker's Species	30
X.	13, 15	14	"	16, 20
XI.	22, 23		"	24, 26
XII.	28, 29		"	31
XIII.	32, 33		"	25
XIV.	10		"	5
	10′		"	6
	11		"	13, 15
	15		"	14
	12		"	22, 23
	13		"	28, 29
	13′		"	32, 33
	14		"	2
	16		"	12

FURTHER PART OF NEWTON'S GENUS 4, CONTAINS 4 SPECIES, VIZ.

XV.	28	29	30	31	corresponding to Pluecker's Species	37
					"	35
					"	36
					"	34

Newton's Genus 3, contains 4 Species, viz.

XVI.	22	22′	22″	23	corresponding to Pluecker's Species	72
					"	70
					"	73
					"	71

Residue of Newton's Genus 4, contains 1 Species, viz.

XVII.	32	corresponding to Pluecker's Species	74

Newton's Genus 5, contains 6 Species, viz.

XVIII.	33	34	corresponding to Pluecker's Species	87
XIX.	35		"	84
XX.	36		"	75, 79, 80
XXI.	37		"	77, 88
XXII.	38		"	89
XXIII.			"	82, 78, 81
XXIV.			"	83, 85
XXV.			"	91
XXVI.			"	86
XXVII.			"	90

Newton's Genus 6, contains 7 species, viz.

XXVIII.	39	corresponding to Pluecker's Species	76	
XXIX.	40	"	92	
XXX.	41	"	107	
XXXI.	42	"	99	
XXXII.	43	"	93	
XXXIII.	44	"	95	
XXXIV.	45	"	103	
			"	98, 94, 102
			"	96, 100
			"	97, 105
			"	101
			"	106, 108, 109
			"	104

Newton's Genus 7, contains 7 Species, viz.

XXXV.	46	corresponding to Pluecker's Species	113
XXXVI.	47	"	110, 114, 115
XXXVII.	48	"	117
XXXVIII.	49	"	111
XXXIX.	50	"	112
XL.	51	"	116, 118, 119
XLI.	52	"	120

Newton's Genus 8, contains 6 Species, viz.

XLII.	53	corresponding to Pluecker's Species	122
XLIII.	54	"	128
XLIV.	55	"	134
XLV.	56	"	125
XLVI.	$56'$	"	131
XLVII.	$56''$	"	137

Newton's Genus 9, contains 4 Species, viz.

XLVIII.	57	corresponding to Pluecker's Species	141
XLIX.	58	"	143
L.	59	"	148
LI.	60	"	152

Newton's Genus 10, contains 3 Species, viz.

LII.	61	corresponding to Pluecker's Species	123
LIII.	62	"	129
LIV.	63	"	135

Newton's Genus 11, contains 2 Species, viz.

LV.	64	corresponding to Pluecker's Species	126
LVI.	65	"	139

Newton's Genus 12, contains 1 Species, viz.

LVII.	66	corresponding to Pluecker's Species	145

Newton's Genus 13, contains 5 Species, viz.

LVIII.	67	corresponding to Pluecker's Species	155
LIX.	68	"	159
LX.	69	"	165
LXI.	70	"	160
LXII.	71	"	166

NEWTON'S GENUS 14, CONTAINS 1 SPECIES, VIZ.

LXIII.	72	corresponding to Pluecker's Species	170

It is to be noticed that (as appears by the Table) Pluecker enumerates 13 species of the Divergent Parabola, viz. corresponding to the Parabola Pura of Newton we have five species, and to the Parabola cum Ovali three species; but to each of the other three Newtonian species (Nodata, Punctata, Cuspidata) only a single species. The difference in nowise affects Newton's before-mentioned theorem, that every cubic curve is the shadow of a Divergent Parabola; but (the characters of Pluecker's species being unaffected by projection) the number of resulting kinds of cubic curves (or cones) will be five or thirteen according as the one or the other classification is adopted; but this is a subject which I do not enter upon in the present Memoir.

Cambridge, February 8, 1864.

[351] On Cubic Cones and Curves

[From the Transactions of the Cambridge Philosophical Society, vol. xi. Part i. (1866), pp. 129–144.
Read April 18, 1864.]

There is contained in Sir Isaac Newton's *Enumeratio Linearum tertii Ordinis* (1706), under the heading *Genesis Curvarum per Umbras*, the remarkable theorem that, in the same way as the several curves of the second order may be considered as the shadows of a circle, that is, the sections of a cone having a circular base, so the several curves of the third order, or cubic curves, may be considered as the shadows of the five Divergent Parabolas. It was remarked by Chasles, Note xx. to the *Apercu Historique* (1837), that they may also be considered as the shadows of the five curves having a centre (the Newtonian Species 27, 38, 59, 62, 72), and that the theorem may be stated as follows, viz. (in the same way that all the curves of the second order are the sections of a single kind of cone of the second order, so) all the curves of the third order may be considered as the sections of five kinds of cones of the third order — and that cutting these in one way we have the five Divergent Parabolas, cutting them in another way the five curves with a centre. The nature of these five kinds of cones, or, what is the same thing, that of the spherical curves in which they are intersected by a concentric sphere, was first pointed out by Moebius in his most interesting Memoir, "Grundformen der Linien dritter Ordnung," *Abh. der K. Sachs. Ges. zu Leipzig*, 1853. I reproduce in the present memoir the characterisation of these five kinds of cones — which I call the *simplex*, the *complex*, the *acnodal*, the *crunodal*, and the *cuspidal* — and I further develope the geometrical and analytical theory; in particular I arrive at a division of the simplex cones into three subkinds, the simplex *trilateral*, *neutral*, and *quadrilateral*. I have throughout spoken of cones rather than of plane curves, using however, as far as may be, language which is also applicable to a plane curve, thus, instead of lines of inflexion, tangent planes, of a cone, I say inflexions, tangents, &c. But the theory of the cone is of course that of the projective properties of the curves which are the sections of such cone; it appears to me that the true classification of curves is to divide them according to the cones which give rise to them; and I consider the present memoir as affording in part the materials for such a classification of cubic curves, viz. it seems to me that after, in the first instance, dividing them into the simplex, the complex, the acnodal, the crunodal, and the cuspidal kinds, the simplex kind should be further divided in the above-mentioned manner; and that we should establish, lastly, the divisions which relate to the particular mode in which the cone is to be cut, in order to obtain such and such a curve: in effect, that the principle of classification, according to the nature of the infinite branches adopted by Newton in the work above referred to, and by Pluecker in his *System der Analytischen Geometrie*

(Berlin, 1835), and to which has reference my Memoir On the Classification of Cubic Curves, [350], should be not the primary, but a secondary principle of classification. I remark that as regards the division into five kinds, Murdoch, in his highly interesting work, *Newtoni Genesis Curvarum per Umbras*, [Lond. 1746], has not only distinguished the Newtonian species which arise from each of the Divergent Parabolas, or, what is the same thing, from each of the five kinds of cones (it will presently appear how the mere inspection of Newton's figures is sufficient to enable this), but that he has also shown how the cone must in each case be cut in order to obtain the particular cubic curve. Murdoch also distinguishes the three forms *ampullate*, *campaniform* and *intermediate*, of the simplex Divergent Parabola, which correspond to the simplex quadrilateral, trilateral, and neutral.

I remark also that Pluecker in his work above referred to, *Dritter Abschnitt*, 98, has considered the equation of a cubic curve in the form $pqr + \mu s^3 = 0$, which is in fact equivalent to the form $(X+Y+Z)^3+6kXYZ = 0$ used in the sequel, but without arriving at the results obtained in the present Memoir.

The five kinds of Cubic Cones. Nos. 1 to 7.

1. A cone of any order may comprise two distinct forms of sheet, viz. (1) a *twin-pair sheet*, or sheet which meets a concentric sphere in a pair of closed curves such that each point of the one curve is opposite to a point of the other curve (a cone of the second order affords an example of such a sheet); the twin-pair sheet may be considered as consisting of two sheets, each of which may be called a *twin sheet*: and (2) a *single sheet*, viz. a sheet which meets a concentric sphere in a closed curve such that each point of the curve is opposite to another point of the curve: the plane affords an example of such a curve. We have five kinds of cubic cones, viz. the simplex, the complex, the acnodal, the crunodal, and the cuspidal. The cone may consist of a single sheet; it is then of the *simplex* kind. Or it may consist of a single sheet and a twin-pair sheet, it is then of the *complex* kind: these are the non-singular kinds. The remaining kinds are singular ones, which are most easily explained by considering them as limiting forms of the complex kind; the twin-pair sheet may come to unite itself with the single sheet giving rise to a crunodal line, or say a *crunode*; the cone is then of the *crunodal* kind. Or the twin-pair sheet may dwindle into a mere line which is an acnodal line, or say an *acnode*; the cone is then of the *acnodal* kind. Or the two things may happen together, viz. the twin-pair sheet at the instant that it unites itself with the single sheet may dwindle into a mere line, which is then a cuspidal line, or say a *cusp*; and the cone is then of the *cuspidal* kind. I remark, as regards the crunodal kind, that the cone may be considered as consisting of two portions, one of them corresponding to the single sheet of a complex cone, and which I call the *quasi-single sheet*; the other of them corresponding to the twin-pair sheet, and which I call the *loop-sheet*.

2. It is to be added that a cubic cone has in general 9 lines of inflexion, or say *inflexions*, but of these 6 are always imaginary; the remaining 3, which are real, belong to the single sheet. The plane through any two inflexions meets the cone in a line which is also an inflexion. In particular the three real inflexions lie in a plane.

3. When the cone is acnodal the six imaginary inflexions unite at the acnode; and the single sheet has still 3 real inflexions lying in a plane. But if the cone is crunodal, then 4 imaginary inflexions and 2 real inflexions unite in the crunode; and the cone has 1 real inflexion; there are besides 2 imaginary inflexions, the 3 inflexions lie in a real plane. Finally, if the cone is cuspidal, then 2 of the real inflexions, and the 6 imaginary inflexions unite together in the cusp; the cone has besides 1 real inflexion, but there are not any imaginary inflexions.

4. Suppose that the cone is of one of the non-singular kinds; that is, let it be simplex or complex. From any line of the cone we may draw four tangent planes to the cone — the anharmonic ratio of the four planes is the same whatever may be the line on the cone. As regards reality, the following distinction exists, viz. for the complex kind of cone, the planes are all real or all imaginary; for the simplex kind they are two real and two imaginary. First, as regards the complex kind, if the line be taken on the twin-pair sheet, the four tangent planes are all imaginary; but if it be taken on the single sheet, then there are two real tangent planes to the single sheet and two real tangent planes to the twin-pair sheet, together four real tangent planes. Secondly, as regards the simplex kind, there is only the single sheet, and the line being taken on it, there are two real tangent planes and no more.

5. As regards the singular kinds, assuming always that the line on the cone does not coincide with the node or the cusp (for when it does there are no tangent planes), it may be noticed that for the crunodal kind there are two tangent planes which are real or imaginary according as the line lies on the part corresponding to the single sheet or on the part corresponding to the twin-pair sheet. For the acnodal kind there are two tangent planes which are always real; and for the cuspidal kind there is a single tangent plane which is always real.

6. The foregoing properties of cubic cones apply to the curves which are the sections of these cones; thus a cubic curve is of the simplex, the complex, the crunodal, the acnodal, or the cuspidal kind. As regards the last-mentioned three kinds, or singular kinds, it is of course to be borne in mind that the crunode, acnode, or cusp, may be at infinity; and consequently that all the curves in Newton's genus 9 (the hyperbolisms of the hyperbola) and the curve in his genus 12 (the trident curve) belong to the crunodal kind; the curves in genus 10 (the hyperbolisms of the ellipse) to the acnodal kind; and those of genus 11 (the hyperbolisms of the parabola) and the curve in genus 14 (the cubical parabola) to the cuspidal kind.

7. In the other genera such of the species as are without a node or a cusp, belong to the simplex or the complex kind: and the mere inspection of the figure (Newton's or Pluecker's) is sufficient to show to which of the two kinds the curve belongs; in fact, when from any point of the curve there are four real tangents, or there is else no real tangent, the curve is of the complex kind, but if there are two and only two real tangents the curve is of the simplex kind. And in the former case we see whether a branch arises from the single sheet or the twin-pair sheet, viz. if from a point on the branch there can be drawn four real tangents to the curve, the branch arises from the single sheet, but if no real tangent can be drawn, the branch arises from the twin-pair sheet. And in the crunodal kind we see which part of the curve arises from the quasi-single sheet, and which part from the loop sheet.

Ulterior Theory leading to the Subkinds of the Simplex Cones. Nos. 8 to 35.

8. But the division of cubic cones may be carried further: we may in fact subdivide the simplex kind. To show how this is, I consider a cone complex or simplex, but I attend for the moment only to the single sheet. The cone has on the single sheet three (real) inflexions lying in a plane. I call this the *equator*, and I call the tangent planes at the inflexions simply the *tangents*; the three tangents do not in general meet in a line, and they divide space into eight regions; of these there are two not divided by the equator, and which remain trilateral; the other six regions are divided by the equator each of them into a trilateral and a quadrilateral region, this gives six trilateral regions and six quadrilateral regions; there are thus in all $2 + 6 = 8$ trilateral regions (I distinguish them as the 2 and the 6 such regions) and 6 quadrilateral regions.

9. It is easy to see that for a complex cone the single sheet lies wholly in the 6 trilateral regions, and the twin-pair sheet wholly in the 2 trilateral regions. Imagine the twin-pair sheet to dwindle into a line and then disappear, that is, let the cone pass from the complex, through the acnodal, into the simplex kind; the single sheet of the simplex cone will lie wholly in the 6 trilateral regions; this is one form of the simplex cone; I call it the *simplex trilateral.* But there is a different form, viz. the cone may lie wholly in the 6 quadrilateral regions; this is the *simplex quadrilateral.* And there is an intermediate form, viz. the three tangent planes at the inflexions may meet in a line, the 2 trilateral regions then disappear, and there are only 12 regions, all of them trilateral, which may be considered as forming two systems, each of 6 regions, viz. each system consists of three non-contiguous regions on one side of the equator, and (alternating therewith) three non-contiguous regions on the other side of the equator: the cone lies wholly in the one 6 regions or in the other 6 regions and I say that the cone is *simplex neutral.*

10. A non-singular cubic cone (simplex or complex) may be represented by an equation of the canonical form

$$x^3 + y^3 + z^3 + 6lxyz = 0,$$

where the coordinates x, y, z and the parameter l are all real; the invariants of the form are $S = -l + l^4$, $T = 1 - 20l^3 - 8l^6$. It is to be noticed that the form in question cannot represent a singular cone; we find as the condition that it may do so

$$R = 64S^3 - T^2 = -(1 + 8l^3)^2 = 0,$$

but when this condition is satisfied, the cone breaks up into a system of three planes; thus for the real root $l = -1$, we have

$$x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x + \omega y + \omega^2 z)(x + \omega^2 y + \omega z),$$

where ω is an imaginary cube root of unity; and by merely writing ωx, $\omega^2 x$ successively in place of x, we see that the like decomposition occurs from the imaginary roots $l = -\frac{1}{2}\omega$, $l = -\frac{1}{2}\omega^2$.

11. The equation $x^3 + y^3 + z^3 + 6lxyz = 0$ is in general transformable into the form

$$(X + Y + Z)^3 + 6kXYZ = 0,$$

where X, Y, Z are linear functions of the original coordinates, such that $X = 0$, $Y = 0$, $Z = 0$ are the equations of the tangent planes at three inflexions in the plane $X + Y + Z = 0$; if however the three tangent planes meet in a line, then X, Y, Z will satisfy identically a certain linear equation, and it is clear *a priori* that the transformation must fail. The condition for the three tangent planes meeting in a line is $S = -l + l^4 = 0$, that is, we have

$$l = 0,\ 1,\ \omega,\ \text{or}\ \omega^2;$$

and attending only to the real roots $l = 0$, $l = 1$, it will be presently seen that for $l = 0$ the tangent planes at the three real inflexions do not, for $l = 1$, they do meet in a line. Hence the simplex neutral cone corresponds to the value $l = 1$, that is, the equation is

$$x^3 + y^3 + z^3 + 6xyz = 0,$$

and this equation is not transformable into the form $(X + Y + Z)^3 + 6kXYZ = 0$, which is that employed in the sequel for the general discussion of the simplex and complex cones.

The theory on which the foregoing conclusion depends is as follows.

On the condition $S = 0$. Nos. 12 to 17.

12. A cubic has in general nine inflexions, which lie by threes on twelve planes, viz. denoting the inflexions by 1, 2, 3, 4, 5, 6, 7, 8, 9, the planes may be taken to be

$$\begin{array}{l} 123,\ 456,\ 789, \\ 147,\ 258,\ 369, \\ 159,\ 267,\ 348, \\ 168,\ 249,\ 357, \end{array}$$

that is, we have four systems, each of three planes passing through the nine inflexions.

13. The tangent planes, or, say the tangents at the inflexions *in piano*, for instance, at the inflexions 1, 2, 3, form a trilateral, and we have thus corresponding to each of the three planes a trilateral formed by the tangents at the inflexions on such plane; and there are of course four systems, each of three trilateral formed by the tangents at the nine inflexions.

14. I say that if $S = 0$, then in one of the four systems the trilaterals become each of them a line, that is, the tangents at the nine inflexions meet by threes in three lines.

15. This may be shown by means of the before-mentioned canonical form $x^3 + y^3 + z^3 + 6lxyz = 0$ of the equation of a cubic cone, for then the notation of the inflexions being in accordance with the foregoing scheme, the coordinates may be taken to be

$$\begin{array}{lll} (1)\ x = 0,\ y + z = 0; & (2)\ y = 0,\ z + x = 0; & (3)\ z = 0,\ x + y = 0; \\ (4)\ x = 0,\ y + \omega z = 0; & (5)\ y = 0,\ z + \omega x = 0; & (6)\ z = 0,\ x + \omega y = 0; \\ (7)\ x = 0,\ y + \omega^2 z = 0; & (8)\ y = 0,\ z + \omega^2 x = 0; & (9)\ z = 0,\ x + \omega^2 y = 0; \end{array}$$

where ω denotes an imaginary cube root of unity, and the equations of the tangents are

$$\begin{array}{lll} (1)\ -2lx + y + z = 0; & (2)\ x - 2ly + z = 0; & (3)\ x + y - 2lz = 0; \\ (4)\ -2lx + \omega y + \omega^2 z = 0; & (5)\ \omega x - 2ly + \omega^2 z = 0; & (6)\ \omega x + \omega y - 2lz = 0; \\ (7)\ -2lx + \omega^2 y + \omega z = 0; & (8)\ \omega^2 x - 2ly + \omega z = 0; & (9)\ \omega^2 x + \omega y - 2lz = 0. \end{array}$$

16. The value of S is $= -l + l^4$, and for each of the values 1, 0, ω, ω^2 of l, which give $S = 0$, we have a system of the nine tangents meeting by threes in three lines, viz. the systems are

$$\begin{array}{rl} \text{for } l = 1, & 123,\ 456,\ 789, \\ \text{for } l = 0, & 147,\ 258,\ 369, \\ \text{for } l = \omega, & 159,\ 267,\ 348, \\ \text{for } l = \omega^2, & 168,\ 249,\ 357. \end{array}$$

17. It is proper to notice that starting with the systems in question, or what is the same thing, with a single set of each system, say the sets 123, 147, 159, 168, we obtain as the condition to be satisfied by l, the equation

$$l(4l\omega - 3l - 1)(4l\omega^2 - 3l\omega + 1)(4l\omega - 3l\omega^2 + 1) = 0,$$

or, as it may otherwise be written,

$$l(2l-1)^2(l-1)(2l\omega - 1)^2(l\omega - 1)(2l\omega^2 - 1)^2(l\omega^2 - 1) = 0,$$

that is, $(-l + l^4)(1 + 8l^3)^2 = 0$; it is clear that a factor has dropped out, and that the true form of the condition is

$$(-l + l^4)(1 + 8l^3)^2 = 0;$$

that is, $S(64S^3 - T^2) = 0$; where the equation $64S^3 - T^2 = 0$ would denote the existence of a nodal line, and consequent coincidence therewith of 6 of the 9 inflexions; the equation $S = 0$ being left as the proper condition for the intersection by threes of the tangents at the inflexions in three lines.

18. I return to the analytical theory of the general case, as depending on the representation of the equation of the cone in the form

$$(X + Y + Z)^3 + 6kXYZ = 0,$$

where the coordinates are real, viz. $X = 0$, $Y = 0$, $Z = 0$ represent the tangent planes at the three (real) inflexions, or, as they have before been called, the *tangents*; and $X + Y + Z = 0$ represents the plane through the three inflexions, or, as it has before been called, the *equator*. And we may assume the signs to be such that in one of the 2 trilateral regions the coordinates X, Y, Z shall be each of them positive: this being so the 14 regions will correspond to the following combinations of signs

X	Y	Z	$X+Y+Z$	
+	+	+	+	the 2 trilateral regions,
−	−	−	−	
+	−	−	−	
−	+	−	−	the 6 trilateral regions,
−	−	+	−	
+	+	−	+	
−	+	+	+	the 6 quadrilateral regions,
+	−	+	+	

where it may be noted that the equator $X + Y + Z = 0$ does not cut the two trilateral regions $(+\ +\ +)$ and $(-\ -\ -)$; and further that the line $X = Y = Z$ which is the harmonic of the equator $X + Y + Z = 0$ in regard to the system of the three tangents $XYZ = 0$, lies wholly in the two trilateral regions $(+\ +\ +)$ and $(-\ -\ -)$.

19. The equation in question, $(X+Y+Z)^3+6kXYZ=0$, shows that, as above stated, the cone lies wholly in the 8 trilateral regions, or in the 6 quadrilateral regions, viz. if k be negative, it lies wholly in the 8 trilateral regions, and if k be positive, it lies wholly in the 6 quadrilateral regions.

Let k be negative, then the positive quantity $-\dfrac{(X+Y+Z)^3}{XYZ}$, if we attend only to the values of X, Y, Z (or $X+Y+Z$) which have the same sign (that is, to points in one of the two trilateral regions), has a maximum value $=27$ corresponding to $X=Y=Z$.

And, if $-\dfrac{(X+Y+Z)^3}{XYZ}$ exceeds this value, that is, if $-k<\frac{27}{8}$, or, what is the same thing, if k lie between the values 0 and $-\frac{27}{8}$, then the equation $-\dfrac{1}{6k}=\dfrac{XYZ}{(X+Y+Z)^3}$ cannot be satisfied in the assumed manner, that is, by values of X, Y, Z having the same sign; and thus no portion of the cone lies in the two trilateral regions: in the contrary case, that is, if k lie between the values $-\infty$, $-\frac{27}{8}$, the equation can be so satisfied, and a portion of the cone lies in the two trilateral regions.

Hence k being negative, we have as follows:

k between $-\infty$ and $-\frac{27}{8}$, the cone is complex,

$k=-\frac{27}{8}$, the cone is acnodal,

k between $-\frac{27}{8}$ and 0, the cone is simplex trilateral;

and k being positive, or say

k between 0 and ∞, the cone is simplex quadrilateral.

20. It is to be remarked that for $k=0$, the cone as represented by the foregoing equation degenerates into the threefold plane $(X+Y+Z)^3=0$. The value $k=0$ corresponds however to the value $l=1$ of the parameter l in the equation $x^3+y^3+z^3+6lxyz=0$, that is, it corresponds to the simplex neutral cone, represented by the equation

$$x^3+y^3+z^3+6xyz=0,$$

which, as already remarked, is not transformable into the form $(X+Y+Z)^3+6kXYZ=0$: this leads to the consideration of the transformation in question.

On the relation of the two forms $x^3+y^3+z^3+6lxyz=0$, and $(X+Y+Z)^3+6kXYZ=0$. Nos. 21 to 24.

21. Starting with the form $x^3+y^3+z^3+6lxyz=0$, and writing

$$\begin{aligned}X&=-2lx+y+z,\\ Y&=x-2ly+z,\\ Z&=x+y-2lz,\end{aligned}$$

then we have

$$\begin{aligned}XYZ &= -2l(x^3+y^3+z^3)+(1-2l+4l^2)(y^2z+yz^2+z^2x+zx^2+x^2y+xy^2)\\ &\quad+2(1-6l^2-8l^3)xyz,\\ X+Y+Z &= 2(1-l)(x+y+z),\end{aligned}$$

and thence

$$\begin{aligned}(X+Y+Z)^3 &= 8(1-l)^3(x^3+y^3+z^3)+24(1-l)^3(y^2z+yz^2+z^2x+zx^2+x^2y+xy^2)\\ &\quad+48(1-l)^3xyz,\end{aligned}$$

and we thus obtain

$$(1-2l+4l^2)(X+Y+Z)^3+24(l-l^2)XYZ = 4(2l+1)^2(l-1)^2(x^3+y^3+z^3+6lxyz);$$

or, what is the same thing,

$$(X+Y+Z)^3+6kXYZ = \frac{4(2l+1)^2(l-1)^2}{1-2l+4l^2}(x^3+y^3+z^3+6lxyz),$$

if

$$k=\frac{4(l-l^2)}{1-2l+4l^2}.$$

22. For the form $(X+Y+Z)^3+6kXYZ=0$, we find

$$\begin{aligned}S &= (1+k)^3-6(1+k)^3k+8(1+k)^3k^2-3k^2(4+k)\\ &= (1+k)^3(1-2k)^2,\\ T &= -8(1+k)^6+72(1+k)^6k-128(1+k)^6k^2+72(1+k)^6k^3\\ &\quad-8k^3(6+6k+k^2)\\ &= -8k^3(6+6k+k^2),\end{aligned}$$

and thence

$$64S^3-T^2=-256k^3(2k+9)(k-1)^2.$$

23. The equation

$$k=\frac{4(l-l^2)}{1-2l+4l^2}$$

or as it may also be written

$$k=\frac{-4l(l-1)}{(2l+\omega)(2l+\omega^2)}$$

gives without difficulty

$$k+4=\frac{16(1-l)^2+3}{1-2l+4l^2}=\frac{16(1-l)^2}{(1+l+l^2)(1-2l+4l^2)}+\frac{27(l+l^2)}{(1+l+l^2)(1-2l+4l^2)},$$

and

$$k + \tfrac{9}{2} = \frac{16(1-l)^2 + 27(l+l^2)}{2(1-2l+4l^2)} = \frac{16(1-l)^2 + 27(l+l^2)}{2(1+l+l^2)(1-2l+4l^2)}(1+l+l^2),$$

and

$$k(k+4) = \frac{-4l(l-1)\{16(1-l)^2+3\}}{(1-2l+4l^2)^2}.$$

24. Hence treating l, k as coordinates, we see 1) that the locus is a cubic curve,

$$k = \frac{4(l-l^2)}{1-2l+4l^2},$$

viz. a hyperbolism of the ellipse, having a centre (Newton's species 62), the coordinates of the centre being $l = \frac{1}{2}$, $k = -\frac{9}{2}$, and the equation of the asymptote being $k + \frac{9}{2} = l - \frac{1}{2}$, (that is the asymptote passes through the centre and is inclined at an angle = 45° to the axis of l). The centre is of course an inflexion, the equation of the tangent at this point is $k + \frac{9}{2} = 9(l - \frac{1}{2})$, and for the other two inflexions we have $l = 1$, $k = 0$, and $l = -\frac{1}{2}$, $k = -\frac{9}{2}$, the tangents at the two inflexions respectively being $k = 0$ and $k = -\frac{9}{2}$, that is the tangents at the inflexions are parallel to the axis of l. The curve consists of a single branch lying below the asymptote for large negative values of l, k, crossing the asymptote at the centre and lying above it for large positive values of k, l. For each value of l there is consequently a single value of k and reciprocally; and l, k pass together from $-\infty$ to $+\infty$. There are certain critical values of k and l, the meaning of which will appear from the following article.

On the Anharmonic Property of a Cubic Cone. Nos. 25 to 29.

25. The property in question is the one already referred to, viz. the four tangent planes, or say the four *tangents*, to the cone from any line of the cone form a system the anharmonic ratios of which are constant. Taking the equations of the tangents to be

$$p - aq = 0,\ p - bq = 0,\ p - cq = 0,\ p - dq = 0,$$

and writing for shortness $u_1 = 64 - \frac{T^2}{S^3}$, then the functions

$$(a-b)(c-d),\ (a-c)(d-b),\ (a-d)(b-c),$$

or say α, β, γ, on which the anharmonic ratios depend, are the roots of the equation

$$t^3 - 12t + 2\sqrt{u_1} = 0.$$

The anharmonic ratios are

$$\frac{\beta}{\gamma},\ \frac{\gamma}{\alpha},\ \frac{\alpha}{\beta};\quad \frac{\beta^2}{\gamma^2},\ \frac{\gamma^2}{\alpha^2},\ \frac{\alpha^2}{\beta^2};\quad \frac{\beta^3}{\gamma^3},\ \frac{\gamma^3}{\alpha^3},\ \frac{\alpha^3}{\beta^3};$$

hence forming the equation $(\xi - \frac{\beta}{\gamma})(\xi - \frac{\gamma}{\beta}) = 0$, and reducing by the conditions,

$$\alpha + \beta + \gamma = 0, \quad \beta\gamma + \gamma\alpha + \alpha\beta = -12, \quad \alpha\beta\gamma = -2\sqrt{u_1},$$

this is found to be $(\xi^2+\xi+1)^3 - \dfrac{27}{u_1}\xi^2(\xi+1)^2 = 0$, or we have $\dfrac{(\xi^2+\xi+1)^3}{\xi^2(\xi+1)^2} = \dfrac{27}{u_1}$, and substituting this value in the equation $\gamma^3 - 12\gamma + 2\sqrt{u_1} = 0$, we find

$$\frac{(\xi^2+\xi+1)^3}{\xi^2(\xi+1)^2} - \frac{432\xi^2(\xi+1)^2}{(\xi^2+\xi+1)^3} = 0,$$

which is

$$\{(\xi^2+\xi+1)^3\}^2 - 432\xi^2(\xi+1)^2 \cdot (\xi^2+\xi+1)^3 = 0,$$

or, what is the same thing,

$$\frac{(\xi^2+\xi+1)^3}{\xi^2(\xi+1)^2} = \frac{432}{\xi^2+\xi+1},$$

that is

$$\frac{(\xi^2+\xi+1)^3}{\xi^2(\xi+1)^2} = \frac{27}{\frac{1}{16}u_1},$$

and as a verification it may be remarked that, 6 being a root, the six roots are

$$0,\ -6,\ -(1+\xi),\ -\frac{1+\xi}{\xi},\ -\frac{1}{1+\xi},\ -\frac{\xi}{1+\xi};$$

of course the roots are all real or else all imaginary.

26. If $T = 0$, the equation becomes

$$\xi(\xi-1)(\xi+1)(\xi+2)(2\xi+1)(\xi^2+\xi+1) = 0,$$

or reducing, this is

$$\xi(\xi-1)(\xi+1)(\xi+2)(2\xi+1) = 0,$$

that is the six roots are 1, $-\frac{1}{2}$, -2, each twice: and the four tangents form therefore a harmonic pencil, which is the geometrical interpretation of the condition $T = 0$.

27. The function

$$\frac{(\xi^2+\xi+1)^3}{\xi^2(\xi+1)^2}$$

is constantly positive and it has three equal minima values corresponding to the last-mentioned values 1, $-\frac{1}{2}$, -2 of ξ, viz. this minimum value is $= \frac{27}{4}$.

Hence we see that the equation in ξ will have its six roots all real if $1 - \frac{u_1}{27}$ is positive and less than unity, that is, if S and $64S^3 - T^2$ are each of them positive: but when these conditions are not satisfied the six roots are imaginary: the limiting case $1 - \frac{u_1}{27} = 1$ or $T = 0$ gives, as already mentioned, the three roots 1, $-\frac{1}{2}$, -2, each twice.

28. The quantities a, b, c, d which determine the four tangents may be all real, or two real and two imaginary, or all four imaginary; but the imaginary values appear as usual as a conjugate pair or conjugate pairs; and this being so, it is easy to see that in general if ξ be real the quantities a, b, c, d are all real or else all imaginary; but if ξ is imaginary then a, b, c, d are two of them real, two imaginary: in fact if a, b are real and c and d are conjugate imaginaries $\gamma \pm \delta i$, then we have for one of the six values of ξ,

$$\xi = \frac{(a-b) \cdot 2\delta i}{(a-\gamma-\delta i)(b-\gamma+\delta i)},$$

which is in general imaginary.

29. But, as might have been foreseen, the limiting values $\xi = 1, -\frac{1}{2}, -2$, are an exception, viz. for these values a, b, c, d may be two of them real the other two imaginary: in fact the last-mentioned value of ξ is real and $(a-\gamma)(b-\gamma) + \delta^2 = 0$, that is $ab + \gamma^2 + \delta^2 = \gamma(a+b)$, or, as written, the condition may also be $2ab + 2(\gamma+\delta i)(\gamma-\delta i) = (\gamma+\delta i+\gamma-\delta i)(a+b)$, that is $2(ab+cd) = (a+b)(c+d)$, or if a, b, c, d form a harmonic system.

Enumeration of the Cones comprised therein. Nos. 30 and 31.

30. I form the following Table:

l	k	S	T	$64S^3 - T^2$
$-\infty$	$-3-\sqrt{3}$	$+\infty$	$+\infty$	$-$
$-\frac{1}{2}(1+\sqrt{3})$	$-3-\sqrt{3}$	$\frac{3+2\sqrt{3}}{4}$	$+\infty$	0
$-\frac{1}{2}$	-4	$\frac{3}{16}$	0	$+$
0	0	0	1	0
$\frac{1}{2}(\sqrt{3}-1)$	$-3+\sqrt{3}$	$-\frac{-3+2\sqrt{3}}{4}$	-1	0
$\frac{1}{2}$	$-\frac{9}{2}$	$-\frac{3}{16}$	0	$-$
1	0	0	-27	0
$+\infty$	$-\frac{9}{2}$	$+\infty$	$-\infty$	$-$

31. And I further describe as follows the nature of the cones which correspond to the several values of k and l.

l between $-\infty$ and $-\frac{1}{2}$, or k between $-\infty$ and $-\frac{9}{2}$. The cone is *complex.* In the series, viz. corresponding to $l = -\frac{1}{2}(1+\sqrt{3})$ or $k = -3-\sqrt{3}$, there is a special form which may be called the *complex harmonic*, viz. the four tangents from any line of the cone form a harmonic system: but observe, *qua* complex cone, the tangents are all real or all imaginary, $l = -\frac{1}{2}$ (form fails), $k = -\frac{9}{2}$, the cone is *acnodal.* l between $-\frac{1}{2}$ and 1, or k between $-\frac{9}{2}$ and 0; the cone is *simplex trilateral.* In the series, viz. corresponding to $l = 0$ or $k = -4$, there is a special form which might be called the *quasi-neutral*, the speciality having however reference to the imaginary inflexions, viz. corresponding to each real inflexion we have two imaginary inflexions such that the three tangents meet in a line.

There is also corresponding to $l = \frac{1}{2}$, or $k = -\frac{9}{2}$, a form which seems to be a special one, though I have not ascertained wherein that speciality consists.

And there is corresponding to $l = \frac{1}{2}(\sqrt{3}-1)$ or $k = -3+\sqrt{3}$, a special form which might be called the *simplex harmonic*, viz. the tangents from any line of the cone form a harmonic system. It is to be observed that, *qua* simplex cone, the four tangents are two of them real, two imaginary.

$l = 1$; $k = 0$ (form fails). The cone is *simplex neutral.*

l between 1 and ∞, or k between 0 and ∞; the cone is *simplex quadrilateral.*

32. It will be observed that the crunodal and cuspidal kinds of cones do not present themselves in the foregoing investigations; the reason is that the crunodal kind admits of no representation in the form $x^3+y^3+z^3+6lxyz = 0$, and (inasmuch as there is only one real inflexion) it admits of no real representation in the other form $(X+Y+Z)^3+6kXYZ = 0$; the cuspidal kind admits of no representation in either of the two forms.

I conclude with a discussion not absolutely required for the purpose of the memoir, but which is of interest in regard to the form $(X+Y+Z)^3 + 6kXYZ = 0$.

Reduction of the Hessian to the form $(X'+Y'+Z')^3 + 6k'X'Y'Z' = 0$.

33. The cubic cone $x^3+y^3+z^3+6lxyz = 0$ has for its Hessian

$$-l^4(x^3+y^3+z^3)+(1+2l^3)xyz = 0,$$

or say

$$x^3+y^3+z^3+6l'xyz = 0, \quad \text{if} \quad l' = -\frac{1+2l^3}{6l^4}.$$

Hence writing

$$\begin{aligned} X' &= -2l'x + y + z, \\ Y' &= x - 2l'y + z, \\ Z' &= x + y - 2l'z, \end{aligned}$$

we have

$$(X' + Y' + Z')^3 + 6k'X'Y'Z' = \frac{4(2l'+1)^2(l'-1)^2}{1-2l'+4l'^2}(x^3 + y^3 + z^3 + 6l'xyz).$$

Hence the equation of the Hessian is

$$(X' + Y' + Z')^3 + 6k'X'Y'Z' = 0,$$

where the value of k' is

$$k' = \frac{4(l'-l'^2)}{1-2l'+4l'^2}.$$

34. But we have

$$4(1-2l'+4l'^2) = (2l'-1)^2+3 = (2l'+1+1)(2l'+1-1)+3 = 4(l'+1+l'^2)(1-2l+4l^2),$$

and thence

$$k' = -\frac{(1+6l^3+2l^6)^2}{l^4(l+1+l^2)^2(1-2l+4l^2)^2}.$$

But the equation

$$l' = -\frac{1+2l^3}{6l^4}$$

gives

$$l' + l'^2 = \frac{(1+2l^3)(-6l^4+1+2l^3)}{36l^8} = \frac{1-6l^4+2l^3+4l^6}{36l^8},$$

and we thence have

$$k' = \frac{k(k+4)^2}{(k-6)^2},$$

which determines k' in terms of k.

35. It may be observed that the value $k = -6$ corresponds to $l' = 1$, that is, the Hessian is here $x^3 + y^3 + z^3 + 6xyz = 0$, a simplex neutral form not transformable into $(X'+Y'+Z')^3+6k'X'Y'Z' = 0$; the corresponding value of l is of course given by the equation $1 + 6l^3 + 2l^6 = 0$; the only speciality of the cone $x^3 + y^3 + z^3 + 6lxyz = 0$, or $(X + Y + Z)^3 - 36XYZ = 0$, consequently is that the Hessian is a simplex neutral cone.

The value $k = -4$ corresponds to $l = 0$, $l' = \infty$, $k' = \infty$; hence $X' : Y' : Z' = x : y : z$ and the transformation of the Hessian $x^3+y^3+z^3+6l'xyz = 0$ into the new form $(X' + Y' + Z')^3 + 6kX'Y'Z' = 0$ degenerates into the mere identity $xyz = xyz$.

Cambridge, 19th Feb. 1865.

[352] Suite des Recherches sur l'Elimination et la Theorie des Courbes

[From the Journal für die reine und angewandte Mathematik (Crelle), tom. LXIV. (1865), pp. 167–171.]

Dans le mémoire "Recherches sur lélimination et la théorie des courbes," t. xxxiv. pp. 30–45 de ce Journal (1847), [53], j'ai donné pour une courbe $U = 0$ du n-ième ordre sans points doubles ou de rebroussement, les expressions pour les degrés, tant par rapport aux coefficients que par rapport aux variables, des fonctions qui entrent dans léquation $FFU = KU \cdot (PU)^2 \cdot (QU)^3 \cdot U$ qui sert à expliquer comment la réciproque de la réciproque de la courbe $U = 0$ se réduit à la courbe originale $U = 0$. En partant des principes établis dans le mémoire, "Nouvelles Recherches sur lélimination et la théorie des courbes," t. LXIII., pp. 34–39 de ce Journal (1864), [338], je suis parvenu à résoudre à peu près cette question pour le cas d'une courbe $U = 0$ du n-ième ordre avec α points doubles et β points de rebroussement; mon investigation a cependant par rapport à quelques points besoin de confirmation.

Je commence par rappeler que léquation d'une courbe avec des points doubles et de rebroussement peut être présentée sous la forme

$$U = aP + bQ + cR + \ldots = 0,$$

où a, b, c, … sont des quantités absolument arbitraires, $P = 0$, $Q = 0$, $R = 0$, … sont des courbes du n-ième ordre (je suppose toujours que $U = 0$ est une courbe du n-ième ordre avec α points doubles et β points de rebroussement) qui ont chacune pour chaque point double de la courbe $U = 0$ un point double au même point, et pour chaque point de rebroussement de la courbe $U = 0$ un point de rebroussement au même point et avec la même tangente. En parlant des coefficients de U, je désignerai toujours les quantités $(a, b, c, \ldots)$ sans faire attention aux constantes contenues dans les fonctions $(P, Q, R, \ldots)$.

La fonction U (voir les Nouvelles Recherches etc.) a un discriminant spécial KU du degré $3(n-1)^2 - 7\alpha - 11\beta$: il y a en outre une certaine fonction AU des coefficients, laquelle dépend des points de rebroussement, qui semble jouer un rôle analogue en quelque sorte à celui du discriminant. Car soient pour un moment (x, y, z) les coordonnées d'un des points de rebroussement de la courbe $U = 0$; écrivons $D = x\partial_x + y\partial_y + z\partial_z$, et dans la fonction U substituons (x, y, z) au lieu de (x, y, z); léquation $D^2U = 0$ donne le carré de la tangente au point de rebroussement: or $D^2U = aD^2P + bD^2Q + cD^2R + \ldots$, et puisque les courbes $P = 0$, $Q = 0$, $R = 0$, … ont chacune la même tangente au point de rebroussement, les fonctions D^2P, D^2Q, D^2R, … seront des fonctions de la forme $\lambda\Phi^2$, $\mu\Phi^2$, $\nu\Phi^2$, … où $\Phi = 0$ est léquation de la tangente, et λ, μ, ν… sont des quantités constantes qui ne dépendent que des constantes que contiennent les fonctions P, Q, R, ... Nous aurons donc $D^2U = (a\lambda + b\mu + c\nu + \ldots)\Phi^2$; et je remarque que léquation $a\lambda + b\mu + c\nu + \ldots = 0$ serait la condition pour qu'il y eût au lieu du point de rebroussement un point triple.

On obtient donc léquation du système des carrés des tangentes aux points de rebroussement sous la forme

$$(a\lambda_1+b\mu_1+c\nu_1\ldots)(a\lambda_2+b\mu_2+c\nu_2\ldots)\ldots(a\lambda_\beta+b\mu_\beta+c\nu_\beta+\ldots)\,\Phi_1^2\Phi_2^2\ldots\Phi_\beta^2 = 0 :$$

le facteur constant $(a\lambda_1 + b\mu_1 + c\nu_1\ldots)\ldots(a\lambda_\beta + b\mu_\beta + c\nu_\beta\ldots)$, du degré β par rapport aux coefficients, est précisément la dérivée que je nomme AU (de manière que $AU = 0$ est la condition pour léxistence dún point triple): láutre facteur $\Phi_1^2\Phi_2^2\ldots\Phi_\beta^2$ est du degré 0 par rapport aux coefficients.

Cela étant, je pose dábord, pour la vérifier plus tard, la table suivante:

	Degre par rapport	
	aux variables	aux coefficients
équation de la courbe, $U = 0$	n	1
condition pour un nouveau point double, $KU = 0$	$2(n-1)$	$3(n-1)^2 - 7\alpha - 11\beta$
condition pour un point triple, $AU = 0$	$3(n-2)$	β
équation de la courbe reciproque, $FU = 0$	$n(n-1) - 2\alpha - 3\beta$	$2(n-1)$
équation de la courbe des inflexions, $QU = 0$	$3n(n-2) - 6\alpha - 8\beta$	$3(n-2)$
équation des tangentes aux points dínflexion, $QU = 0$	$3n(n-2) - 3\alpha - 4\beta$	3
équation de la courbe des contacts des tangentes doubles, $PU = 0$	$(n-2)(n^2-9)$	$(n+4)(n-3)$
équation des tangentes doubles, $PU = 0$	$\frac{1}{2}n(n-2)(n^2-9)$	$\frac{1}{2}n(n-2)(n^2-9)$ $-(n^2-n-6)(2\alpha+3\beta)$ $+2\alpha(\alpha-1) + 6\alpha\beta + \frac{9}{2}\beta(\beta-1)$ $-(2n-6)(2\alpha+3\beta) - \beta$
équation de la courbe reciproque de la reciproque, $FFU = 0$	$(n^2-n-2\alpha-3\beta)$ $\times(n^2-n-1-2\alpha-3\beta)$	$4(n-1)$

Et puis on a léquation

$$FFU = (AU)^m \cdot KU \cdot (PU)^2 \cdot (QU)^3 \cdot U.$$

La comparaison des degrés par rapport aux variables donne

$$\begin{gathered}(n^2 - n - 2\alpha - 3\beta)(n^2 - n - 1 - 2\alpha - 3\beta) = \\ n(n-2)(n^2-9) - (n^2-n-6)(4\alpha+6\beta) + 4\alpha(\alpha-1) + 12\alpha\beta + 9\beta(\beta-1) \\ + 9n(n-2) - 18\alpha - 24\beta + n,\end{gathered}$$

ce qui est exacte.

La comparaison des degrés par rapport aux coefficients donne

$$\begin{gathered}4(n-1)(n^2-n-1-2\alpha-3\beta) = m\beta + 3(n-1)^2 - 7\alpha - 11\beta \\ + 4n(n-2)(n-3) - (4n-12)(2\alpha+3\beta) - 2\beta \\ + 9n(n-2) - 9\alpha - 12\beta + 1,\end{gathered}$$

ce qui de même est exacte.

Les expressions pour les degrés de KU et AU sont déjà démontrées; pour les autres expressions, en considérant dábord la courbe générale $W = 0$

du n-ième ordre, laquelle, en établissant entre les coefficients les relations convenables, se réduit à la courbe $U = 0$ avec α points doubles et β points de rebroussement, on sait par la théorie de M. Plücker quels sont les facteurs desquels seront affectées FW, QW, PW, et quíl faut écarter pour réduire ces fonctions à FU, QU, PU respectivement.

Pour FW ce facteur est $A^\alpha B^\beta$, où $A = 0$ est léquation tangentielle des points doubles, et $B = 0$, léquation tangentielle des points de rebroussement: la réduction du degré par rapport aux variables est donc de $2\alpha + 3\beta$ unités. En prenant (x, y, z) pour les coordonnées dún point double quelconque on a $\nabla = \Pi(\xi x + \eta y + \zeta z)$, et de même en prenant (x, y, z) pour les coordonnées dún point de rebroussement quelconque on a $\Xi = \Pi(\xi x + \eta y + \zeta z)$; A et B ne contiennent donc pas les coefficients a, b, c, $\ldots$ de U, et une réduction de degré par rapport aux coefficients ná pas lieu.

Pour QW le facteur est $M^\alpha N^\beta$, où $M = 0$ est léquation des tangentes aux points doubles et $N = 0$ léquation des carrés des tangentes aux points de rebroussement: la réduction de degré par rapport aux variables est donc $6\alpha + 8\beta$ unités. Soient (x, y, z) les coordonnées dún point double, $D = x\partial_x + y\partial_y + z\partial_z$; en substituant comme auparavant (x, y, z) au lieu de (x, y, z) dans la fonction V, léquation des deux tangentes au point double est $D^2U = 0$, où D^2U est du degré 1 par rapport aux coefficients: en formant léquation analogue pour chaque point double on a $M = \Pi(D^2U) = 0$, et M sera du degré α par rapport aux coefficients. En prenant (x, y, z) pour les coordonnées dún point de rebroussement, on a de même $N = \Pi(D^2U) = 0$ pour léquation des carrés des tangentes aux points de rebroussement; N est donc du degré β par rapport aux coefficients.

Nous avons vu que léquation $N = 0$ se réduit à la forme $N = AU \cdot \Phi_1^2\Phi_2^2 \ldots \Phi_\beta^2$; jádmets cependant quíl faut retenir ce facteur constant AU, et considérer ainsi N comme étant effectivement du degré β. Le facteur $M^\alpha N^\beta$ est donc du degré $3\alpha + 4\beta$, et la réduction de degré par rapport aux coefficients qui a lieu pour QW est donc de $3\alpha + 4\beta$ unités.

Pour PW le facteur est R^2S^3T, où $R = 0$ est léquation du système des tangentes menées à la courbe par les points doubles, $S = 0$ léquation du système des tangentes menées à la courbe par les points de rebroussement, $T = 0$ léquation des droites qui contiennent deux points doubles (chacune de ces droites étant comptée 4 fois) ou qui contiennent un point double et un point de rebroussement (chacune de ces droites étant comptée 6 fois), ou enfin qui contiennent deux points de rebroussement (chacune de ces droites étant comptée 9 fois). Par rapport aux variables le degré de R est égal à $\alpha\{n^2 - n - 6 - 2(\alpha - 1) - 3\beta\}$, celui de S à $\beta\{n^2 - n - 6 - 2\alpha - 3(\beta - 1)\}$: le degré de R^2S^3 est donc égal à $(n^2 - n - 6)(2\alpha + 3\beta) - 4\alpha(\alpha - 1) - 6\alpha\beta - 9\beta(\beta - 1)$. Le degré de T est égal à $4 \cdot \frac{1}{2}\alpha(\alpha - 1) + 6\alpha\beta + 9 \cdot \frac{1}{2}\beta(\beta - 1)$, le degré de R^2S^3T sélève donc à $(n^2 - n - 6)(2\alpha + 3\beta) - 2\alpha(\alpha - 1) - 3\alpha\beta + \frac{9}{2}\beta(\beta - 1)$, nombre qui exprime la réduction de degré par rapport aux variables qui a lieu pour PW. Par rapport aux coefficients le degré de R est égal à $(2n - 6)\alpha$, celui de S à $(2n - 6)\beta$, celui de T à zéro: le degré de R^2S^3T sélève donc à $(2n-6)(2\alpha+3\beta)$. On aurait par conséquent pour PW par rapport aux coefficients une réduction de degré égale à $(2n-6)(2\alpha+3\beta)$ unités; mais dáprès un exemple très-particulier (il est vrai) jádmets que PW contiendra encore le facteur constant AU, ce qui donnerait pour le nombre dont il ságit la valeur $(2n-6)(2\alpha+3\beta)+\beta$.

Jái dit que par rapport aux coefficients le degré de R est égal à $(2n-6)\alpha$ et celui de S à $(2n-6)\beta$: pour prouver léxactitude de ces nombres il faut se rappeler que léquation $\Theta=0$ des tangentes menées par un point quelconque est du degré (n^2-n) par rapport aux variables et du degré $2(n-1)$ par rapport aux coefficients. En prenant pour le point dont il ságit un point double ou de rebroussement et supposant que dans la courbe il ný a que ce seul point double ou de rebroussement, le degré par rapport aux variables est (n^2-n-6) et celui par rapport aux coefficients est $2n-6$. Mais dans le cas général Θ contiendra comme facteur G^2H^3, en dénotant par $G=0$ léquation des droites menées par le point dont il ságit à tous les points doubles, et par $H=0$ léquation des droites menées par ce point à tous les points de rebroussement. De cette manière on obtient un abaissement de $2(\alpha-1)+3\beta$, ou de $2\alpha+3(\beta-1)$ unités pour le degré par rapport aux variables, mais le degré par rapport aux coefficients est toujours $(2n-6)$. Donc en considérant les systèmes des points doubles et des points de rebroussement, pour R la réduction est égal à $(2n-6)\alpha$ et pour S à $(2n-6)\beta$ unités.

Les difficultés de cette investigation sont dues aux points de rebroussement: en admettant en FFU léxistence dún facteur $(AU)^m$, il nést pas clair que lón doit avoir $m=1$; et la démonstration pour les valeurs des termes en β, des expressions $3\alpha+4\beta$ et $(2n-6)(2\alpha+3\beta)+\beta$ est imparfaite.

Écrivons

$$FFU=(AU)^m\cdot KU\cdot(PU)^2\cdot(QU)^3\cdot U,$$

et supposons que le nombre qui exprime la réduction de degré par rapport aux coefficients soit donné par la valeur $3\alpha+k\beta$ pour QU et par la valeur $(2n-6)(2\alpha+3\beta)+l\beta$ pour PU.

La comparaison des degrés par rapport aux coefficients donne

$$\begin{aligned}4(n-1)(n^2-n-2\alpha-3\beta)=m\beta+3(n-1)^2-7\alpha-11\beta\\ +4n(n-2)(n-3)-(4n-12)(2\alpha+3\beta)-2l\beta\\ +9n(n-2)-9\alpha-3k\beta+1,\end{aligned}$$

ce qui établit la relation $m-2l=3k-13$, à laquelle on satisfait en prenant $m=1$, $l=1$, $k=4$. Mais je serais bien aise de prouver ces valeurs par une démonstration plus concluante.

Cambridge, 26 Mai, 1864.

[353] Note sur la Surface du Quatrieme Ordre de Steiner

[From the Journal für die reine und angewandte Mathematik (Crelle), tom. LXIV. (1865), pp. 172–174.]

En considérant les deux coniques définies par les équations

$$U = (a, b, c, f, g, h\,,\ x, y, z)^2 = 0,$$

$$U' = (a', b', c', f', g', h'\,,\ x', y', z')^2 = 0,$$

on en déduit les trois équations dérivées

$$\begin{aligned} F &= (bc - f^2, ca - g^2, ab - h^2, gh - af, hf - bg, fg - ch\,,\ \xi, \eta, \zeta)^2 = 0,\\ G &= (bc' + b'c - 2ff', \ldots\,,\ \xi, \eta, \zeta)^2 = 0,\\ F' &= (b'c' - f'^2, \ldots\,,\ \xi, \eta, \zeta)^2 = 0, \end{aligned}$$

et lón sait que $F = 0$ est léquation tangentielle de la conique $U = 0$ (autrement dit, léquation qui exprime que cette conique est touchée par la droite $\xi x + \eta y + \zeta z = 0$), que de même $F' = 0$ est léquation tangentielle de la conique $U' = 0$, et enfin que $G = 0$ est léquation tangentielle de la conique enveloppée par une droite $\xi x + \eta y + \zeta z = 0$ qui coupe harmoniquement les deux coniques $U = 0$, $U' = 0$.

Or, en considérant les deux surfaces quadriques

$$\begin{aligned} U &= (a, b, c, d, f, g, h, l, m, n\,,\ x, y, z, w)^2 = 0,\\ U' &= (a', b', c', d', f', g', h', l', m', n'\,,\ x, y, z, w)^2 = 0, \end{aligned}$$

on forme dúne manière analogue les quatre équations dérivées

$$\begin{aligned} F &= (bcd + \text{etc.}, \ldots\,,\ \xi, \eta, \zeta, \omega)^2 = 0,\\ G &= (b'cd + \text{etc.}, \ldots\,,\ \xi, \eta, \zeta, \omega)^2 = 0,\\ G' &= (bc'd' + \text{etc.}, \ldots\,,\ \xi, \eta, \zeta, \omega)^2 = 0,\\ F' &= (b'c'd' + \text{etc.}, \ldots\,,\ \xi, \eta, \zeta, \omega)^2 = 0. \end{aligned}$$

$F = 0$ est léquation tangentielle de la surface $U = 0$ (et de même $F' = 0$ est léquation tangentielle de la surface $U' = 0$). Les deux équations $G = 0$, $G' = 0$, qui ont des coefficients formés dáprès une loi facile à saisir, se changent lúne dans láutre lorsquón échange entre elles les deux surfaces quadriques $U = 0$, $U' = 0$. Léquation $G' = 0$ (celle des deux dont il ságira dans la suite) est léquation tangentielle de la surface quadrique enveloppée par un plan $\xi x + \eta y + \zeta z + \omega w = 0$ qui coupe les surfaces $U = 0$, $U' = 0$ selon des coniques $S = 0$, $S' = 0$ telles quíl y ait sur la conique $S = 0$

une infinité de systèmes de trois points conjugués par rapport à la conique $S' = 0$.

En supposant à présent que léquation $U' = 0$ est celle dún cône, on peut dire que $G' = 0$ est léquation tangentielle de la surface quadrique enveloppée par un plan qui coupe la surface $U = 0$ selon une conique $S = 0$ telle que par cette conique et par le sommet du cône $U' = 0$ on puisse faire passer une infinité de systèmes de trois droites conjuguées par rapport au cône $U' = 0$. On peut présenter le théorème sous une autre forme; en faisant passer par le sommet du cône $U' = 0$ trois droites conjuguées par rapport à ce cône, et en choisissant à volonté lún des deux points de rencontre de chacune des droites avec la surface $U = 0$, on obtient trois points qui déterminent un plan; en considérant tous les systèmes des trois droites conjuguées, on a pour chaque système un plan, et lénveloppe de ces plans nést autre chose que la surface quadrique $G' = 0$.

Je suppose que le sommet du cône $U' = 0$ soit situé sur la surface $U = 0$, et je dis que la surface $G' = 0$ se réduira à un système de deux points, à savoir le sommet du cône $U' = 0$ et un autre point. Pour démontrer cela, on peut prendre pour coordonnées du sommet $x = 0$, $y = 0$, $z = 0$; les deux équations seront alors

$$U = (a, b, c, 0, f, g, h, l, m, n\ ,\ x, y, z, w)^2 = 0,$$

$$U' = (a', b', c', 0, f', g', h', 0, 0, 0\ ,\ x, y, z, w)^2 = 0,$$

(où, ce qui est la même chose, $U' = (a', b', c', f', g', h'\ ,\ x, y, z)^2 = 0$) et en substituant ces valeurs on voit sans peine que les coefficients de x^2, y^2, z^2, xz, zx, xy dans la fonction G' se réduiront à zéro, et que léquation $G' = 0$ aura la forme

$$G' = (0, 0, 0, D, 0, 0, 0, L, M, N\ ,\ \xi, \eta, \zeta, \omega)^2 = 0,$$

cést-à-dire nous aurons

$$G' = \omega(D\omega + 2L\xi + 2M\eta + 2N\zeta) = 0,$$

équation qui représente en effet le point $\omega = 0$ (ou ce qui est la même chose le point $x = 0$, $y = 0$, $z = 0$) et un autre point $D\omega + 2L\xi + 2M\eta + 2N\zeta = 0$, ou ce qui est la même chose le point $x : y : z : w = 2L : 2M : 2N : D$.

Dans le cas actuel chacune des trois droites rencontre la surface $U = 0$ dans le sommet et de plus dans un seul point, et en prenant ce dernier point pour point de rencontre de la droite avec la surface $U = 0$ le plan mené par les trois points ne passe pas par le sommet; ce plan passe donc par le point $x : y : z : w = 2L : 2M : 2N : D$, et on a ainsi

Theoreme I. En faisant passer par un point donné de la surface quadrique $U = 0$ trois droites conjuguées par rapport au cône $U' = 0$ (qui a ce même point pour sommet) le plan mené par les trois points de rencontre des droites avec la surface $U = 0$ passe toujours (quel que soit le système des trois droites conjuguées) par un point fixe.

Jájoute que, lorsque les équations $U = 0$, $U' = 0$ ont la forme spéciale qui leur a été donnée en dernier lieu, les coordonnées du point seront $x : y : z : w = 2L : 2M : 2N : D$, et il convient de remarquer que ces valeurs L, M, N, D sont des fonctions quadriques par rapport aux coefficients $(a', \ldots)$ du cône $U' = 0$.

Au lieu dún cône donné $U' = 0$, considérons le système entier des cônes $\lambda P + \mu Q + \nu R = 0$, où $P = 0$, $Q = 0$, $R = 0$ sont des cônes donnés ayant leur sommet commun dans le point $(x = 0, y = 0, z = 0)$ de la surface et λ, μ, ν des coefficients arbitraires, système qui est celui des cônes en involution avec les cônes donnés $P = 0$, $Q = 0$, $R = 0$. À chaque système des coefficients λ, μ, ν correspond un point fixe, et en conservant pour ses coordonnées la notation antérieure $x : y : z : w = 2L : 2M : 2N : D$, les quantités L, M, N, D sont des fonctions quadriques des quantités arbitraires λ, μ, ν. Le lieu du point dont il ságit sera évidemment une surface, et on démontre sans peine que cette surface est du quatrième ordre. Car, pour trouver en combien de points la surface est rencontrée par une droite quelconque, il faut combiner avec les équations $x : y : z : w = 2L : 2M : 2N : D$ les équations de la droite dont il ságit, cést-à-dire deux équations linéaires en x, y, z, w; cela donne deux équations linéaires en L, M, N, D, ou quadriques en (λ, μ, ν). On a ainsi quatre systèmes de valeurs de (λ, μ, ν); et à chaque système correspond un seul point (x, y, z, w), il y a par conséquent quatre points díntersection, et la surface est du quatrième ordre.

Nous avons donc

Theoreme II. En considérant au lieu du cône $U' = 0$ le système entier des cônes $\lambda P + \mu Q + \nu R = 0$ en involution avec les cônes $P = 0$, $Q = 0$, $R = 0$ qui ont leur sommet commun dans un point de la surface $U = 0$, le lieu du point fixe du théorème I. est une surface du quatrième ordre.

Cette surface du quatrième ordre est la *surface de Steiner*, considérée dernièrement par MM. Kummer, Weierstrass, Schroeter, et Cremona.

Cambridge, 2 Novembre, 1864.

[354] Note sur les Singularites Superieures des Courbes Planes

[From the Journal für die reine und angewandte Mathematik (Crelle), tom. LXIV. (1865), pp. 369–371.]

Dans un mémoire "On the higher singularities of plane Curves" destiné pour le Quarterly Mathematical Journal j'ai cherché à établir quúne singularité quelconque équivaut à un certain nombre δ' de points doubles, κ' de points de rebroussement, τ' de tangentes doubles, et ι' dínflexions; et pour déterminer ces nombres, jái donné dans le cas dúne singularité simple, ou la courbe ná quúne seule branche, des formules que je vais reproduire ici. Si la branche est par rapport à ses points de líndice α, ayant avec elle-même le nombre $\frac{1}{2}M$ de points communs, et par rapport à ses tangentes de líndice β, ayant avec elle-même le nombre $\frac{1}{2}N$ de tangentes communes, on trouve

$$\delta' = \tfrac{1}{2}\{M - 3(\alpha - 1)\}, \qquad \kappa' = \alpha - 1,$$

$$\tau' = \tfrac{1}{2}\{N - 3(\beta - 1)\}, \qquad \iota' = \beta - 1.$$

Pour expliquer ces formules, je remarque que la singularité dont il ságit est telle que, prenant pour origine le point sur la courbe, on obtient pour lórdonnée y une seule suite de la forme

$$y = Ax^p + Bx^q + \dots,$$

où la suite est arrangé suivant les puissances ascendantes de x et les coefficients A, B, ... ont chacun une valeur unique. Si láxe des y ne touche pas la courbe, aucun des exposants p, q, ... ne sera inférieur à lúnité, et si de plus láxe des x touche la courbe, alors parmi les exposants p, q, ... il y en aura un qui est > 1, et les autres seront tous > 1.

The Collected Mathematical Papers of Arthur Cayley, Vol. V

Pages 451–500

354. Note sur les singularites superieures des courbes planes. [354]

(From the Comptes Rendus de l'Academie des Sciences de Paris, t. LX. (1865), pp. 414–417.)

(Continuation from p. 354.)

On a done pour y precisement le nombre α de valeurs, qui s'obtiennent en attribuant a x ses valeurs diverses. A chacune de ses valeurs correspond une "branche partielle" de la courbe, de maniere que la branche a l'indice α est composee de α branches partielles; pour $\alpha = 1$ la branche partielle n'est autre chose que la branche meme.

En considerant deux branches partielles, et en designant par p le plus petit exposant de x qui se trouve dans la suite par laquelle est exprimee la difference $y_1 - y_2$ des ordonnees des deux branches partielles (ce nombre p pouvant etre entier ou fractionnaire), je pose comme definition que les deux branches partielles ont un nombre p de points communs, ou d'intersection.

En combinant deux a deux les α branches partielles qui composent la branche de l'indice α, et en formant la somme $\sum p$ des nombres p qui correspondent a chaque paire de branches partielles, on obtient le nombre $\frac{1}{2}M$ des points communs de la branche avec elle-meme.

En se servant des coordonnees tangentielles, on a par rapport aux tangentes de la branche une theorie tout a fait semblable; cette remarque suffit pour expliquer les notions d'une branche de l'indice β par rapport a ses tangentes, et du nombre $\frac{1}{2}N$ des tangentes communes de la branche avec elle-meme.

Comme exemple je prends la singularite donnee par l'equation

$$y = x^{\frac{3}{2}} + x^{\frac{7}{6}} + \cdots ;$$

dans ce cas les exposants n'ont que les denominateurs 2 et 3, la branche est de l'indice 6 par rapport a ses points, elle est composee de six branches partielles representees par les equations

$$\begin{aligned} y_1 &= x^{\frac{3}{2}} + x^{\frac{7}{6}} \cdots, & y_4 &= x^{\frac{3}{2}} - x^{\frac{7}{6}} \cdots, \\ y_2 &= \omega x^{\frac{3}{2}} + x^{\frac{7}{6}} \cdots, & y_5 &= \omega x^{\frac{3}{2}} - x^{\frac{7}{6}} \cdots, \\ y_3 &= \omega^2 x^{\frac{3}{2}} + x^{\frac{7}{6}} \cdots, & y_6 &= \omega^2 x^{\frac{3}{2}} - x^{\frac{7}{6}} \cdots, \end{aligned}$$

ou ω est une racine cubique imaginaire de l'unite. La branche partielle y_1 coupe les autres branches partielles dans un nombre $\frac{1}{2}$, $\frac{2}{3}$, $\frac{5}{6}$, $\frac{1}{2}$, $\frac{2}{3}$ de points, ce qui donne pour la branche partielle y_1 le nombre $\frac{1}{2} + \frac{2}{3} + \frac{5}{6} + \frac{1}{2} + \frac{2}{3} = \frac{15}{4}$ de points; on a ce meme nombre $\frac{15}{4}$ pour les autres branches partielles y_2, y_3, y_4, y_5, y_6 respectivement, et de la on trouve, pour le double du nombre des intersections de la branche avec elle-meme, la valeur $M = 4 \times 15 = 47$, done $B' = \frac{1}{2}(47 - 15) = 16$, $\kappa = 5$.

En coordonnees tangentielles, la branche $y = x^{\frac{3}{2}} + x^{\frac{7}{6}} + \cdots$ s'exprime par l'equation

$$\xi = \chi^{\frac{3}{2}} + \chi^{\frac{7}{6}} \cdots .$$

Plus generalement, on a pour une branche $y = Ax^{\frac{p}{\alpha}} + Bx^{\frac{q}{\alpha}} + \cdots$ l'equation en coordonnees tangentielles $\xi = A'\chi^{\frac{p'}{\beta}} + B'\chi^{\frac{q'}{\beta}} + \cdots$, la forme generale des exposants etant

$$\frac{p'}{\beta} = \frac{\lambda}{\lambda + \mu + \nu + \cdots}, \qquad \frac{q'}{\beta} = \frac{\mu}{\lambda + \mu + \nu + \cdots}, \quad \text{etc.},$$

ou λ, μ, ... sont des entiers positifs, resultat que je ne m'arrete pas a demontrer.

Dans le cas particulier qui nous occupe, la branche est done de l'indice 2 par rapport a ses tangentes. On trouve de suite $N = 15$ et de la $t' = \frac{1}{2}(15-3) = 6$, $i' = 1$; done la singularite dont il s'agit equivaut a un nombre 16 de points doubles, 5 de points de rebroussement, 6 de tangentes doubles, et 1 inflexion.

On a un exemple plus simple dans le point de rebroussement de seconde espece; l'equation est ici $y = x^{\frac{3}{2}} + x^{\frac{5}{3}} \cdots$ et en coordonnees tangentielles on obtient l'equation $\xi = \chi^{\frac{3}{2}} + \chi^{\frac{5}{3}} \cdots$ de la meme forme. De la on trouve $B' = 1$, $\kappa' = 1$, $t' = 1$, $i' = 1$, de maniere que cette singularite equivaut a 1 point double, 1 point de rebroussement, 1 tangente double et 1 inflexion. M. Plucker dans son grand ouvrage a trouve *a posteriori* que cette singularite se compose de $2\frac{1}{2}$ points doubles et de $2\frac{1}{2}$ tangentes doubles, ce qui donne en effet les memes reductions pour la classe et les memes nombres pour les inflexions et les tangentes doubles, que donnent mes valeurs $B' = 1$, $\kappa' = 1$, $t' = 1$, $i' = 1$; mais il y a a remarquer qu'en considerant par exemple une courbe du quatrieme ordre avec un point double et un point de rebroussement de seconde espece (courbe qui existe), on aurait $\delta + \kappa = \frac{25}{4}$, nombre plus grand que le maximum du nombre des points doubles et de rebroussement que peut avoir une courbe du quatrieme ordre.

Je n'ai parle que des singularites simples, ou il y a une seule branche de la courbe, mais on etend sans peine la theorie precedente aux singularites composees, ou il y a plusieurs branches de la courbe. Cette extension exige la distinction de trois cas differents. Il peut y avoir sur la courbe un point avec une seule tangente, mais avec plusieurs branches qui se touchent, — ou un point avec plusieurs tangentes dont chacune touche une ou plusieurs branches, — ou enfin une tangente avec plusieurs points de contact, dans lesquels la tangente touche une seule ou plusieurs branches de la courbe.

Cambridge, 1 Juin, 1865.

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355. Sur un theoreme relatif a huit points situes sur une conique. [355]

(From the Journal fur die reine und angewandte Mathematik (Crelle), tom. LXV. (1866), pp. 180–184.)

On sait que le theoreme de Pascal peut etre deduit du theoreme suivant: toute courbe cubique qui passe par 8 des 9 points d'intersection de deux courbes cubiques passe par tous les 9 points.

De meme cet autre theoreme — toute courbe quartique qui passe par 13 des 16 points d'intersection de deux courbes quartiques passe par tous les 16 points — conduit a un theoreme relatif a 8 points situes sur une conique.

En effet si par 8 points donnes et situes sur une conique donnee on fait passer deux systemes de 4 droites (ces deux systemes doivent etre sans droite commune), les deux systemes sont des courbes quartiques qui se rencontrent dans les 8 points donnes et de plus dans 8 nouveaux points; done toute courbe quartique qui passe par 13 des $8 + 8$ points passe par tous les $8 + 8$ points. Or la conique donnee passe par les 8 points donnes, et par 5 des 8 nouveaux points on peut faire passer une autre conique: les deux coniques forment ensemble une courbe quartique qui passe par $8 + 5$ des $8 + 8$ points, et qui passera ainsi par les $8 + 8$ points; c'est-a-dire la nouvelle conique passe par les 8 nouveaux points, ou autrement dit, les 8 nouveaux points sont situes sur une conique — c'est la le theoreme relatif a 8 points situes sur une conique.

On deduit de la les theoremes 3, 4, 5 de Steiner (Lehrsatze und Aufgaben, ce journal t. XXX, [1846], pp. 274 et 275). En effet considerons sur une conique donnee n points donnes, et les n tangentes dans ces memes points. En combinant deux a deux les n points on obtient

$\frac{1}{2}n(n-1)$ droites G: ces droites se coupent deux a deux dans les n points donnes, qui comptent pour $\frac{1}{2}n(n-1)(n-2)$ intersections, et de plus dans $\frac{1}{8}n(n-1)(n-2)(n-3)$ points r. Chacune des n tangentes rencontre les $\{\frac{1}{2}n(n-1)-(n-1)\}$ droites G qui ne passent pas par le point de contact de cette tangente, dans $\frac{1}{2}(n-1)(n-2)$ points s, ce qui donne en tout $\frac{1}{2}n(n-1)(n-2)$ points s. Enfin les n tangentes se rencontrent deux a deux dans $\frac{1}{2}n(n-1)$ points t.

On a ainsi

$\frac{1}{8}n(n-1)(n-2)(n-3)$ points r, $\frac{1}{2}n(n-1)(n-2)$ points s, $\frac{1}{2}n(n-1)$ points t,

ensemble $\frac{1}{8}n(n-1)(n^2-n+2)$ points; or parmi ces points il y a selon les trois theoremes de Steiner un grand nombre de systemes de 8 points sur une conique.

Prenons d'abord sur la conique donnee 4 points quelconques a, b, c, d des n points, et considerons aussi les points consecutifs a', b', c', d'. La figure des 4 points a, b, c, d et des 4 tangentes dans ces memes points equivaut a celle des 8 points a, a', b, b', c, c', d, d'. Partant de l'arrangement *abcd* (lisez-le cycliquement et il correspond a l'un des 3 quadrilateres que l'on peut former avec les 4 points) on forme avec les 8 points les deux systemes que voici de 4 droites chacun:

systeme aa', bb', cc', dd', c'est-a-dire les tangentes aux 4 points a, b, c, d;

systeme $a'b$, $b'c$, $c'd$, $d'a$, c'est-a-dire ab, bc, cd, da;

et ces deux systemes se rencontrent dans les 8 points a, a', b, b', c, c', d, d' (ou, ce qui est la meme chose, dans les points a, b, c, d, chacun compte 2 fois) et dans 8 nouveaux points compris entre les points r, s, t; ces 8 points sont done situes sur une conique. Comme il y a 3 arrangements *abcd*, *acdb*, *adbc* des 4 points, on obtient de cette maniere 3 systemes de 8 points sur une conique.

Prenons sur la conique 5 points quelconques a, b, c, d, e des n points, et considerons aussi 3 points consecutifs a', b', c'. Partant de l'arrangement *abcde* (qui correspond a l'un des 12 pentagones que l'on peut former avec les 5 points) on forme avec les points a, a', b, b', c, c', d, e les deux systemes de 4 droites chacun:

systeme aa', bb', cc', de, c'est-a-dire les tangentes en a, b, c et la droite de;

systeme $a'b$, $b'c$, $c'd$, ea, c'est-a-dire ab, bc, cd, ea;

et on obtient de la (parmi les points r, s, t) un systeme de 8 points sur une conique. A cause des 12 arrangements des 5 points, il y a 12 systemes. Mais au lieu des points consecutifs (a',b',c') on aurait pu prendre toute autre combinaison (a',b',d') etc.; le nombre des combinaisons etant 10, il y a done $12\times 10=120$ systemes de 8 points sur une conique.

Prenons de meme 6 points quelconques a, b, c, d, e, f des n points. En considerant les points consecutifs a', b' et en partant de l'arrangement *abcdef*, on forme avec les 8 points a, a', b, b', c, d, e, f les deux systemes de 4 droites:

systeme aa', bb', cd, ef, c'est-a-dire les tangentes en a, b et les droites cd, ef;

systeme $a'b$, $b'c$, de, fa, c'est-a-dire ab, bc, de, fa;

cê qui donne parmi les points r, s, t un systeme de 8 points sur une conique. Il y a 60 arrangements des points a, b, c, d, e, f et 15 combinaisons (a', b') etc. des points consecutifs; on a done $60 \times 15 = 900$ systemes de 8 points sur une conique.

Prenons encore 7 points quelconques a, b, c, d, e, f, g des n points. En considerant le point consecutif a', et en partant de l'arrangement *abcdefg*, on forme avec les points a, a', b, c, d, e, f, g les deux systemes de 4 droites:

systeme aa', bc, de, fg, c'est-a-dire la tangente en a, et les droites bc, de, fg;

systeme $a'b$, cd, ef, ga, c'est-a-dire ab, cd, ef, ga;

et on obtient ainsi parmi les points r, s, t un systeme de 8 points sur une conique. Il y a 360 arrangements *abcdefg*, etc. et 7 differents points consecutifs a', etc.: cela donne $360 \times 7 = 2520$ systemes de 8 points sur une conique.

Prenons enfin 8 points quelconques a, b, c, d, e, f, g, h des n points: partant de l'arrangement *abcdefgh*, on forme avec les 8 points les deux systemes de 4 droites chacun (ab, cd, ef, gh) et (bc, de, fg, ha), ce qui conduit a un systeme de 8 points sur une conique. Mais on a 2520 arrangements *abcdefgh*, etc. — il y a ainsi 2520 systemes de 8 points sur une conique.

On voit que les systemes de 8 points sur une conique se derivent de 4, 5, 6, 7 ou 8 des n points sur la conique donnee. En supposant $n = 4$ on n'a que les systemes qui se derivent des 4 points; si $n = 5$, on a les systemes qui se derivent de 4 points choisis d'une maniere quelconque entre les 5 points — et les systemes qui se derivent des 5 points: et ainsi de suite; pour $n = 8$ on a les systemes qui se derivent de 4, 5, 6 ou 7 points choisis d'une maniere quelconque entre les 8 points, et les systemes qui se derivent des 8 points. On peut former la table suivante pour montrer dans les differents cas le nombre des systemes de 8 points sur une conique:

Nombre des systemes de 8 points sur une conique

$r+s+t=$	4 points	5 points	6 points	7 points	8 points
3	3	12	60	360	2520
5	–	15	60	360	2520
7	–	–	15	60	360
9	–	–	–	15	60
11	–	–	–	–	15
$n=4$, sys.	3	–	–	–	–
$n=5$, sys.	3	12	–	–	–
$n=6$, sys.	45	720	900	–	–
$n=7$	105	2520	6300	2520	–
$n=8$	210	6720	25200	20160	2520

Le cas $n = 4$ est le theoreme 3 de Steiner, il y a 3 systemes de 8 points sur une conique; le cas $n = 5$ est le theoreme 4, il y a $15 + 120$ systemes; le cas $n = 6$ est le theoreme 5, il y a $45 + 720 + 900$ systemes. Pour $n = 7$ il y a $105 + 2520 + 6300 + 2520$ systemes et pour $n = 8$, $210 + 6720 + 25200 + 20160 + 2520$ systemes.

Le cas $n = 5$ est surtout interessant: en effet comme une conique est determinee par 5 points, on a ici 5 points quelconques (a, b, c, d, e), et les cinq tangentes (les droites A, B, C, D, E de Steiner) sont des droites determinees par les cinq points et que l'on peut construire (avec la regle seulement). C'est la en effet la forme sous laquelle le theoreme est presente par Steiner; il ne parle nullement de la conique qui passe par les 5 points — et il donne pour les 5 droites une construction; a savoir, les 15 points r sont situes deux a deux sur 15 droites L qui ne dependent chacune que de 4 points, et sur 60 droites H qui dependent chacune des 5 points; les 60 droites H combinees deux a deux d'une maniere convenable se rencontrent dans 30 points s (c'est la definition de ces points) et puis (theoreme) on a 5 droites A, B, C, D, E qui contiennent chacune 6 points s et qui passent par les points a, b, c, d, e respectivement — et (theoreme) les 30 points s sont aussi situes sur les 10 droites G, 3 points sur chaque droite. Je remarque qu'en prenant sur la conique qui passe par a, b, c, d, e, un point quelconque g, il y aurait 24 hexagones inscrits ayant ag pour cote — et de la 24 droites Pascaliennes — et par le point d'intersection de ag avec l'une quelconque des 6 droites bc, etc. on a 4 de ces droites Pascaliennes. Cela pose, en prenant pour g le point consecutif a, les 24 hexagones se confondent deux a deux — on a done 12 hexagones inscrits et autant de droites Pascaliennes — ces droites sont les 12 droites H lesquelles se rencontrent deux a deux dans les 6 points s situes sur la droite aa', ou A. Steiner dit que les 120 coniques dependent des 5 points, mais que les 15 coniques dependent chacune de 4 points seulement; en donnant (comme il l'a fait) le theoreme comme un theoreme par rapport a cinq points quelconques, cela n'est pas exact — en effet les coniques dont il s'agit dependent chacune de 4 des cinq points, et des 4 droites correspondantes, tangentes dans ces memes points a la conique qui passe par les cinq points — ces coniques dependent ainsi des cinq points.

Je remarque en passant que partant des cinq points donnes a, b, c, d, e, il y a sur chacune des droites A, B, C, D, E un point remarquable, dont Steiner ne parle pas, mais qui aurait pu servir a une construction de cette droite — par exemple il y a sur la droite A le point α qui est l'intersection commune des polaires de a par rapport a toutes les coniques qui passent par les points b, c, d, e — en particulier ce point α est l'intersection commune des polaires (harmoniques) de a par rapport aux trois paires de droites (bc, de), (bd, ec), (be, cd) respectivement.

Cambridge, 16 Fev. 1865.

356. Sur un cas particulier de la surface du quatrieme ordre avec seize points singuliers. [356]

(From the Journal fur die reine und angewandte Mathematik (Crelle), tom. LXV. (1866), pp. 284–291.)

Dans la note "sur la surface des ondes" (Liouville t. XI., 1846), [47], j'ai etudie sous le nom de *tetraedroide* la surface du quatrieme ordre douee de seize points singuliers, et qu'une transformation homographique fait naitre de la surface des ondes.

Mon point de depart a ete la propriete fondamentale suivante.

"Le tetraedroide est une surface du quatrieme ordre, qui est coupee par les plans d'un certain tetraedre suivant des paires de coniques par rapport auxquelles les trois sommets du tetraedre dans ce plan sont des points conjugues. De plus: les seize points d'intersection des quatre paires de coniques sont des points singuliers de la surface, c'est-a-dire des points ou, au lieu d'un plan tangent, il y a un cone tangent du second ordre."

Dans la meme note j'ai reconnu l'existence de seize plans singuliers qui touchent chacun la surface suivant une conique. Il est interessant d'examiner de quelle maniere mes formules se rattachent a celles de M. Kummer dans ses belles recherches (Monatsbericht der Berliner Akademie fur 1864, pp. 246–260 et 495–499) relatives a la surface du quatrieme ordre douee de seize points singuliers.

Partant des formules de M. Kummer il convient, pour plus de symmetrie, de changer les signes de a, f; puis en remarquant que dans l'equation (3) p. 250 on doit avoir (voir p. 496) $+kfr$ au lieu de $-kfr$, l'equation de la surface sera

$$\begin{aligned}a^2p^2 + b^2q^2 + c^2r^2 + d^2s^2 + e^2q^2r^2 + f^2r^2p^2 \\ + 2bcp^2qr + 2cepq^2s - 2bfpr^2s - 2efqr^2 \\ + 2capq^2r + 2afqr^2s - 2cdqp^2s - 2fdrp^2 \\ + 2abpqr^2 + 2bdrp^2s - 2aerq^2s - 2depqs^2 - 2gpqrs = 0.\end{aligned}$$

Pour donner les equations des seize plans singuliers de cette surface je pose d'abord pour abreger

$$ad = \alpha, \quad be = \beta, \quad cf = \gamma,$$

et je determine k au moyen de l'equation cubique

$$\gamma k^3 + (-5\gamma - \tfrac{1}{2}\alpha + \tfrac{1}{2}\beta + \tfrac{1}{6}\gamma)k^2 + (-5\gamma - \tfrac{1}{2}\alpha - \tfrac{1}{2}\beta + \tfrac{1}{6}\gamma)k - \alpha = 0;$$

puis j'introduis les quantites

$$p' = \frac{-k}{a}p + \frac{1}{b}q + \frac{k+1}{d}r + \frac{1}{d}s, \quad q' = \frac{d}{a}p - \frac{1}{b}q - \frac{k+1}{e}r - \frac{1}{e}s,$$

$$r' = \frac{d}{a}p - \frac{1}{c}q - \frac{k+1}{f}r - \frac{1}{f}s, \quad s' = \frac{-k}{a}p + \frac{1}{c}q + \frac{k+1}{d}r + \frac{1}{d}s;$$

enfin je denote par p_1, q_1, r_1, s_1; p_2, q_2, r_2, s_2; p_3, q_3, r_3, s_3 ce que deviennent les quantites p', q', r', s' en y substituant successivement pour k les trois racines k_1, k_2, k_3 de l'equation en k. Cela pose, les seize plans singuliers sont donnes par les equations

$$p = 0, \quad q = 0, \quad r = 0, \quad s = 0,$$

$$\begin{aligned}p_1 = 0, \quad q_1 = 0, \quad r_1 = 0, \quad s_1 = 0, \\ p_2 = 0, \quad q_2 = 0, \quad r_2 = 0, \quad s_2 = 0, \\ p_3 = 0, \quad q_3 = 0, \quad r_3 = 0, \quad s_3 = 0.\end{aligned}$$

En prenant une ligne quelconque (p_i, q_i, r_i, s_i) et une colonne quelconque (p_j, q_j, r_j, s_j), puis en omettant le terme commun r_i, on a une des seize combinaisons $(p_i, q_i, s_i, r_j, q_j, r_j)$ de six plans qui se rencontrent dans un des seize points singuliers. Supposons que cette circonstance ait lieu, c'est-a-dire que les plans p, s_1, r_2, q_3 se rencontrent dans le meme point. Pour que cela soit, il faut que la condition

$$\frac{k_2(k_3+1)}{k_3(k_2+1)} = 1 \quad \text{ou, ce qui est la meme chose,} \quad k_2(k_3-k_2) - (k_3-k_2) = 0$$

soit remplie; mais si cette condition est remplie, non seulement les plans (p, s_1, r_2, q_3) se rencontrent dans le meme point, mais aussi les plans

(q, p_1, s_2, r_3), les plans (r, q_1, p_2, s_3) et les plans (s, r_1, q_2, p_3) se rencontrent dans le meme point.

L'equation $k_2(k_3 - k_2) - (k_3 - k_2) = 0$ appartient evidemment a un systeme de six equations, et l'une quelconque de ces equations donnerait un resultat semblable; chacune de ces equations conduit, comme on va voir, a une certaine relation entre les quantites g, α, β, γ (ou g, a, b, c, d, e, f), relation en vertu de laquelle la surface generale du quatrieme ordre douee de seize points singuliers se reduit au tetraedroide.

Pour former la relation dont il s'agit, il faut egaler a zero le produit des six fonctions analogues $k_i k_j(k_i - k_j) - (k_i - k_j)$. Je forme d'abord le produit des trois fonctions $k_2(k_3 - k_2) - (k_3 - k_2)$, $k_3(k_1 - k_3) - (k_1 - k_3)$, $k_1(k_2 - k_1) - (k_2 - k_1)$, et en representant pour un moment l'equation en k par $ak^3 + bk^2 + ck + d = 0$, on trouve que le produit des trois fonctions est egal a

$$P + Q\sqrt{\Delta} = (b + c)(bt + 9alt) - 6(at^2 + b^2U)$$

$$+(b + t - 2a - 2lt)\sqrt{b^2t^2 - 4b^3U - 4at^3 + 18abtU - 27a^2U^2}$$

et en substituant pour a, b, t, U leurs valeurs

$$a = \gamma, \quad b = -5\gamma - \tfrac{1}{2}\alpha + \tfrac{1}{2}\beta + \tfrac{1}{6}\gamma,$$

$$t = -5\gamma - \tfrac{1}{2}\alpha - \tfrac{1}{2}\beta + \tfrac{1}{6}\gamma, \quad U = -\alpha,$$

on trouve, toute reduction faite,

$$P = -25\gamma^3 + 15\gamma(3\alpha^2 - 10\beta\alpha\gamma) + 2(\beta - \gamma)(\gamma - \alpha)(\alpha - \beta),$$

$$\begin{aligned} Q\Delta = 5\gamma^4 - 15\gamma^2(2\alpha^2 - 10\beta\alpha\gamma) - 45\gamma(\beta - \gamma)(\gamma - \alpha)(\alpha - \beta) \\ + 9\gamma^2(2\alpha^4 + 125\alpha\gamma^2\beta - 265\alpha^2\beta\gamma^2 + 2445\alpha^3\beta\gamma). \end{aligned}$$

Cela pose, l'equation cherchee est $P^2 - Q^2\Delta = 0$, c'est-a-dire

$$\begin{aligned} 0 = P^2 - Q^2\Delta & \\ = 5\gamma^3 \cdot 4(\beta - \gamma)(\gamma - \alpha)(\alpha - \beta) & \\ + 5\gamma^2 \cdot 4(-2\alpha^2\beta + 42\alpha^2\beta^2 - 21\alpha^3\beta\gamma) & \\ + 9\gamma^2(\beta - \gamma)(\gamma - \alpha)(\alpha - \beta)(2\alpha^2 - 10\beta\alpha^2) & \\ + 2(\beta - \gamma)^2(\gamma - \alpha)^2(\alpha - \beta)^2; & \end{aligned}$$

cette equation, dans laquelle $\alpha = ad$, $\beta = be$, $\gamma = cf$, constitue la condition sous laquelle la surface de M. Kummer se reduit a un tetraedroide.

Je passe a present a mes formules de 1846.

En ecrivant pour plus de commodite f', g', h', l', m', n' au lieu de f, g, h, l, m, n, mon equation du tetraedroide est

$$\begin{vmatrix} x^2 & y^2 & z^2 & w^2 \\ f' & z^2 & h'^2 & g'^2 \\ l'^2 & y^2 & h'^2 & f'^2 \\ m'^2 & n'^2 & z^2 & h'^2 \end{vmatrix} = 0,$$

ou, ce qui est la meme chose,

$$(A, B, C, D, F, G, H, L, M, N | x^2, y^2, z^2, w^2)^2 = 0,$$

c'est-a-dire

$$Ax^4 + By^4 + Cz^4 + Dw^4 + 2Fy^2z^2 + 2Gz^2x^2 + 2Hx^2y^2 + 2Lx^2w^2 + 2My^2w^2 + 2Nz^2w^2 = 0,$$

ou les coefficients ont les valeurs

$$\begin{array}{lll} A = 2m^2n^2f', & B = 2n^2l^2g', & C = 2l^2m^2h', \\ F = l'^2(l'^2p'^2 - m^2y'^2 - n^2h'^2), & L = f'^2(l'^2p'^2 - m^2y'^2 - n^2h'^2), & \\ G = m'^2(-l'^2p'^2 + m'^2y'^2 - n^2h'^2), & M = g'^2(-l'^2p'^2 + m'^2y'^2 - n^2h'^2), & \\ H = n'^2(-l'^2p'^2 - m'^2y'^2 + n'^2h'^2), & N = h'^2(-l'^2p'^2 - m'^2y'^2 + n'^2h'^2). & \end{array}$$

Les coordonnees des seize points singuliers sont

$$(0, \pm h', \pm g', \pm l'), \quad (\pm h', 0, \pm f', \pm m'),$$

$$(\pm g', \pm f', 0, \pm n'), \quad (\pm l', \pm m', \pm n', 0),$$

et les equations des seize plans singuliers sont

$$\pm nx \pm my \pm lz \pm fw = 0, \quad \pm mx \pm ly \pm hw \pm gz = 0,$$

$$\pm lx \pm hy \pm gz \pm fw = 0, \quad \pm fx \pm gy \pm hz \pm lw = 0,$$

ou l'on donne des valeurs quelconques aux signes +.

Pour comparer ces plans aux plans de M. Kummer j'ecris le tableau

	p	q	r	s
	$ny - mz + fw$	$-nx + lz + gw$	$mx - ly + hw$	$-fx - gy - hz$
p_1, q_1	$ny - mz - fw$	$-nx - lz + gw$	$mx + ly + hw$	$-fx + gy + hz$
p_2, q_2	$ny + mz - fw$	$nx - lz + gw$	$-mx + ly + hw$	$fx - gy + hz$
p_3, q_3	$-ny + mz + fw$	$nx + lz - gw$	$mx - ly - hw$	$fx + gy - hz$

et j'obtiens les valeurs suivantes:

$$p = ny - mz + fw, \quad q = -nx + lz + gw,$$

$$r = mx - ly + hw, \quad s = -fx - gy - hz.$$

En resolvant ces equations par rapport a x, y, z, w et en posant $\theta = l'^2 + m'^2 + n'^2$ on trouve

$$\begin{aligned} \theta x &= -hp + gr - ls, \\ \theta y &= hp - fr - ms, \\ \theta z &= -gp + fq - ns, \\ \theta w &= lp + mq + nr, \end{aligned}$$

valeurs qu'il s'agit de substituer dans l'equation

$$U = (A, B, C, D, F, G, H, L, M, N | x^2, y^2, z^2, w^2)^2 = 0$$

de la surface dont il est question.

De l'expression de U en p, q, r, s je ne considere d'abord que le terme multiplie par p^2q^2. Designons par $\mathcal{E}$ le coefficient de p^2q^2 dans d^2U, nous aurons

$$\begin{gathered} \mathcal{E} = \frac{2h'^2}{\theta^4} \Big[l'^2(l'^2f'^2 - m^2g'^2 - n^2h'^2) \times 2m'^2(-l'^2f'^2 + m'^2g'^2 - n^2h'^2) \\ + h'^2 \times 2n'^2(-l'^2f'^2 - m'^2g'^2 + n^2h'^2) \\ + l'^2(l'^2f'^2 - m^2g'^2 - n^2h'^2) \times 2g'^2(-l'^2f'^2 + m'^2g'^2 - n^2h'^2) \\ + h'^2 \times 2f'^2(-l'^2f'^2 - m'^2g'^2 + n^2h'^2) \\ + (l'^2f'^2 + m'^2g'^2 - 4l'^2m'^2f'g') \times 2n'^2(-l'^2f'^2 - m'^2g'^2 + n^2h'^2) \Big] \end{gathered}$$

les lettres i, j, k etant introduites pour designer les produits

$$l'f' = i, \quad m'g' = j, \quad n'h' = k.$$

Apres toutes les reductions on obtient

$$\mathcal{E} = 2h'^2\{(i+j)^2 - k^2\}^2 = 2h'^2(i+j+k)^2(-i-j+k)^2 = 2h'^2\alpha'^2(-i-j+k)^2,$$

pour le coefficient de p^2q^2 dans d^2U, ou, ce qui est la meme chose,

$$h'^2(-i-j+k)^2$$

pour le coefficient de pq dans $\frac{1}{d^2}U$.

En calculant de meme les autres coefficients de $\frac{1}{d^2}U$ et en ecrivant pour abreger

$$\begin{aligned} i - j - k &= l'f' - m'g' - n'h' = a, \\ -i + j - k &= -l'f' + m'g' - n'h' = b, \\ -i - j + k &= -l'f' - m'g' + n'h' = c, \end{aligned}$$

l'equation transformee sera

$$\begin{aligned}f'^2q^2r^2+g'^2r^2p^2+h'^2p^2q^2+l'^2r^2s^2+m'^2p^2s^2+n'^2q^2r^2&\\ +2gh'bcp^2qr+2hm'bcpq^2s-2gn'bcpr^2s-2mn'bcqr^2&\\ +2hf'capq^2r+2fn'caqr^2s-2hl'caqp^2s-2nl'carps&\\ +2fg'abpqr^2+2gl'abrp^2s-2fm'abrq^2s-2lm'abpqs&\\ -(b-c)(c-a)(a-b)pqrs=0.&\end{aligned}$$

En posant

$$a'^2=fa,\quad b'^2=gb,\quad c'^2=hc,\quad d'^2=la,$$

$$e'^2=mb,\quad f'^2=nc,\quad g'^2=(b-c)(c-a)(a-b),$$

les quantites a', b', c', d', e', f', g' sont liees par une relation. Pour en prouver l'existence on n'a qu'a faire

$$a'd'=\alpha',\quad b'e'=\beta',\quad c'f'=\gamma'$$

et a se servir des expressions de a, b, c en l', g', h', l', m', n', alors on obtient

$$\begin{aligned}-2\alpha'&=a'(b+c),\\ -2\beta'&=b'(c+a),\\ -2\gamma'&=c'(a+b),\\ -4g'^2&=(b-c)(c-a)(a-b),\end{aligned}$$

equations qui impliquent une relation entre α', β', γ', g'; mais en supposant que a', b', c', d', e', f', g' soient des quantites qui satisfassent a cette relation, il existe toujours des valeurs correspondantes de l', g', h', l', m', n', c'est-a-dire que l'equation du tetraedroide est identique avec celle de M. Kummer toutes les fois que les coefficients a, b, c, d, e, f, g de cette derniere sont lies par une certaine relation. Ecrivons comme auparavant $ad=\alpha$, $be=\beta$, $cf=\gamma$, cette relation se trouve en eliminant a, b, c entre les equations

$$\begin{aligned}-2\alpha&=a'(b+c),\\ -2\beta&=b'(c+a),\\ -2\gamma&=c'(a+b),\\ -4g'^2&=(b-c)(c-a)(a-b),\end{aligned}$$

et il ne s'agit que de prouver l'identite de cette relation avec celle que nous avons trouvee ci-dessus par d'autres considerations.

J'introduis les nouvelles notations

$$\alpha+\beta+\gamma=-\tfrac{1}{2}P,\quad bc+ca+ab=q,\quad \alpha\beta\gamma=-\tfrac{1}{2}R,\quad abc=r,$$

$$\alpha + b + c = p, \quad \beta\gamma + \gamma\alpha + \alpha\beta = -\tfrac{1}{2}Q, \quad \alpha\beta\gamma = \lambda,$$

je forme l'expression

$$2(\beta - \gamma) = -(bc + ca + ab)(b - c) = -q(b - c),$$

et les deux expressions analogues pour $2(\gamma - \alpha)$, $2(\alpha - \beta)$; j'en deduis le resultat

$$8(\beta - \gamma)(\gamma - \alpha)(\alpha - \beta) = -q^3(b - c)(c - a)(a - b);$$

enfin je note les equations

$$(b - c)^2(c - a)^2(a - b)^2 = -4q^3 + p^2q^2 + 18pqr - 27r^3 - 4p^3r,$$

$$P = pq - 3r, \quad Q = q^2 - 2pqr + 3r^2,$$

qui donnent la transformation de leurs premiers membres en fonction de p, q, r; cela pose, et a l'aide de ces valeurs, on forme les egalites

$$512(\beta - \gamma)(\gamma - \alpha)(\alpha - \beta)g^2 = (-4q^3 + p^2q^2 + 18pqr - 27r^3 - 4p^3r)q^3(b - c)$$

$$256(-8\alpha\beta\gamma + 4 \cdot 5\alpha^2\beta^2 - 21\alpha^3\beta^2\gamma)g^2 = (-12q^3 + p^2q^2 + 18pqr - 27r^3 + 27r^2)(b - c)^2$$

$$128(\beta - \gamma)(\gamma - \alpha)(\alpha - \beta)(8\alpha^2 - 10\beta\alpha^2)5\gamma = (-12q^3 + p^2q^2 + 18pqr - 27r^2)q^3(b - c)^2(c -$$

$$64(\beta - \gamma)^2(\gamma - \alpha)^2(\alpha - \beta)^2 = (p^2q^2 + \cdots)(b - c)^2(c - a)^2(a - b)^2,$$

lesquelles, multipliees par 1, 2, 1, 4, ajoutees ensemble et divisees par 128, conduisent a l'equation finale

$$\begin{aligned}
&5\gamma^3 \cdot 4(\beta - \gamma)(\gamma - \alpha)(\alpha - \beta) \\
&\qquad + 5\gamma^2 \cdot 4(-2\alpha^2\beta + 42\alpha^2\beta^2 - 21\alpha^3\beta\gamma) \\
&\qquad + 9\gamma(\beta - \gamma)(\gamma - \alpha)(\alpha - \beta)(2\alpha^2 - 10\beta\alpha^2) \\
&\qquad\qquad + 2(\beta - \gamma)^2(\gamma - \alpha)^2(\alpha - \beta)^2 = 0,
\end{aligned}$$

identique avec celle que l'on a trouvee ci-dessus, ce qui acheve la demonstration que l'on avait en vue.

Cambridge, 18 Mai, 1865.

357. A Supplementary Memoir on the Theory of Matrices. [357]

(From the Philosophical Transactions of the Royal Society of London, vol. CLVI. (for the year 1866), pp. 25–35. Received October 24, — Read December 7, 1865.)

M. Hermite, in a paper "Sur la theorie de la transformation des fonctions Abeliennes," Comptes Rendus, t. XL. (1855), pp. 249, &c., establishes incidentally the properties of the matrix for the automorphic linear transformation of the bipartite quadric function $xw'+yz'-zy'-wx'$, or transformation of this function into one of the like form, $XW'+YZ'-ZY'-WX'$. These properties are (as will be shown) deducible from a general formula in my "Memoir on the Automorphic Linear Transformation of a Bipartite Quadric Function," Phil. Trans, vol. CXLVIII. (1858), pp. 39–46, [153]; but the particular case in question is an extremely interesting one, the theory whereof is worthy of an independent investigation. For convenience the number of variables is taken to be *four*; but it will be at once seen that as well the demonstrations as the results are in fact applicable to any even number whatever of variables.

Article Nos. 1 and 2. Notation and Remarks.

1. I use throughout the notation and formulae contained in my "Memoir on the Theory of Matrices," Phil. Trans, vol. CXLVIII. (1858), pp. 17–37, [152], and in the above-mentioned memoir on the Automorphic Transformation. With respect to the composition of matrices, the rule of composition is as follows, viz. any line of the compound matrix is obtained by combining the corresponding line of the first or further component matrix with the several columns of the second or nearer component matrix; it is very convenient to indicate this by the algorithm,

$$\begin{pmatrix} a, & a', & a'' \\ \alpha, & \beta, & \gamma \end{pmatrix} \begin{pmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{pmatrix} = \begin{pmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{pmatrix} \begin{pmatrix} \alpha, & \beta, & \gamma \\ a, & a', & a'' \end{pmatrix},$$

which exhibits very clearly the terms which are to be combined together; thus in the upper left-hand corner we have $(a, b, c|\alpha, \alpha', \alpha'')$, and so for the other places in the compound matrix.

2. It is not in the Memoir on Matrices explicitly remarked, but it is easy to see that sums of matrices, all the matrices being of the same order, may be multiplied together by the ordinary rule; thus

$$(A+B)(C+D) = AC+AD+BC+BD:$$

this remark will be useful in the sequel.

Article Nos. 3 to 13. First Investigation.

3. We have to consider the formulae for the automorphic linear transformation of the function $xw' + yz' - zy' - wx'$, that is, of the function

$$\begin{pmatrix} 0, & 0, & 0, & 1 \\ 0, & 0, & -1, & 0 \\ 0, & -1, & 0, & 0 \\ -1, & 0, & 0, & 0 \end{pmatrix} (x, y, z, w|x', y', z', w') = (x, y, z, w|x', y', z', w'),$$

viz. if the variables are transformed by the formulae

$$(x, y, z, w) = (\Omega|X, Y, Z, W), \qquad (x', y', z', w') = (\Omega|X', Y', Z', W'),$$

then the matrix (Ω) is such that we have identically

$$(\Omega|x, y, z, w|x', y', z', w') = (\Omega|X, Y, Z, W|X', Y', Z', W');$$

the expression for (Ω) is given in my memoir [153] above referred to; viz. observing that the matrix (Ω) is skew symmetrical, then (No. 13) we have

$$\Omega = \Omega_0 - (\Omega_0 - T)(\Omega_0 + T)^{-1}\Omega_0,$$

and conversely,

$$\Omega_0 = \Omega - (\Omega + T)(\Omega - T)^{-1}\Omega,$$

where T is an arbitrary symmetrical matrix.

4. I propose to compare with the matrix Ω the inverse matrix Ω^{-1}. Recollecting that in the theory of matrices $(ABCD)^{-1} = D^{-1}C^{-1}B^{-1}A^{-1}$, we have

$$\Omega^{-1} = \Omega_0^{-1}(\Omega_0 + T)(\Omega_0 - T)^{-1}\Omega_0;$$

and it is to be shown that Ω and Ω^{-1} are composed of terms which (except as to their signs) are the same in each, so that either of these matrices is derivable from the other by a peculiar form of transposition. It is to be borne in mind throughout that T is symmetrical, Ω_0 skew symmetrical.

5. I write for greater convenience

$$-\Omega = \Omega_0^{-1}(T - \Omega_0)(T + \Omega_0)^{-1}\Omega_0,$$

$$-\Omega^{-1} = \Omega_0^{-1}(T + \Omega_0)(T - \Omega_0)^{-1}\Omega_0,$$

and I compare in the first instance the matrices $(T - \Omega_0)(T + \Omega_0)^{-1}$ and $(T + \Omega_0)(T - \Omega_0)^{-1}$.

6. Any matrix whatever, and therefore the matrix $(T + \Omega_0)^{-1}$, may be exhibited as the sum of a symmetrical matrix and a skew symmetrical matrix; that is, we may write

$$(T + \Omega_0)^{-1} = T' + \Omega_0',$$

where T' is symmetrical, Ω'_0 is skew symmetrical.

We have then

$$(T+\Omega_0)(T+\Omega_0)^{-1} = (T+\Omega_0)(T'+\Omega'_0) = 1,$$

where, here and in what follows, 1 denotes the matrix unity.

Moreover

$$T-\Omega_0 = .\,(T+\Omega_0),$$

and thence

$$(T-\Omega_0)^{-1} = (.\,(T+\Omega_0))^{-1} = .\,(T+\Omega_0)^{-1} = .\,(T'+\Omega'_0) = T'-\Omega'_0;$$

that is

$$(T-\Omega_0)^{-1} = T'-\Omega'_0,$$

and thence also

$$\begin{aligned}(T-\Omega_0)(T+\Omega_0)^{-1} &= (T+\Omega_0-2\Omega_0)(T'+\Omega'_0) = 1-2\Omega_0(T'+\Omega'_0),\\ (T+\Omega_0)(T-\Omega_0)^{-1} &= (T-\Omega_0+2\Omega_0)(T'-\Omega'_0) = 1+2\Omega_0(T'-\Omega'_0).\end{aligned}$$

7. Suppose for a moment that

$$T'+\Omega'_0 = \begin{pmatrix} a, & b, & c, & d\\ e, & f, & g, & h\\ i, & j, & k, & l\\ m, & n, & o, & p\end{pmatrix}$$

and therefore

$$T'-\Omega'_0 = \begin{pmatrix} a, & e, & i, & m\\ b, & f, & j, & n\\ c, & g, & k, & o\\ d, & h, & l, & p\end{pmatrix}.$$

8. We have

$$\Omega_0(T'+\Omega'_0) = \begin{pmatrix} 0, & 0, & 0, & 1\\ 0, & 0, & -1, & 0\\ 0, & -1, & 0, & 0\\ -1, & 0, & 0, & 0\end{pmatrix}\begin{pmatrix} a, & b, & c, & d\\ e, & f, & g, & h\\ i, & j, & k, & l\\ m, & n, & o, & p\end{pmatrix} = \begin{pmatrix} m, & n, & o, & p\\ -i, & -j, & -k, & -l\\ -e, & -f, & -g, & -h\\ -a, & -b, & -c, & -d\end{pmatrix}.$$

And similarly,

$$\Omega_0(T'-\Omega'_0) = \begin{pmatrix} 0, & 0, & 0, & 1\\ 0, & 0, & -1, & 0\\ 0, & -1, & 0, & 0\\ -1, & 0, & 0, & 0\end{pmatrix}\begin{pmatrix} a, & e, & i, & m\\ b, & f, & j, & n\\ c, & g, & k, & o\\ d, & h, & l, & p\end{pmatrix} = \begin{pmatrix} d, & h, & l, & p\\ -c, & -g, & -k, & -o\\ -b, & -f, & -j, & -n\\ -a, & -e, & -i, & -m\end{pmatrix}.$$

10. Hence also

$$(T-\Omega_0)(T+\Omega_0)^{-1} = \begin{pmatrix} 1-2m, & -2n, & -2o, & -2p \\ 2i, & 1+2j, & 2k, & 2l \\ 2e, & 2f, & 1+2g, & 2h \\ 2a, & 2b, & 2c, & 1+2d \end{pmatrix}$$

and

$$(T+\Omega_0)(T-\Omega_0)^{-1} = \begin{pmatrix} 1-2d, & -2h, & -2l, & -2p \\ 2c, & 1-2g, & -2k, & -2o \\ 2b, & 2f, & 1+2j, & 2n \\ 2a, & 2e, & 2i, & 1+2m \end{pmatrix},$$

so that these matrices are composed of terms which, except as to the signs, are the same in each.

11. Now in general if

$$\Theta = \begin{pmatrix} \alpha, & \beta, & \gamma, & \delta \\ \alpha', & \beta', & \gamma', & \delta' \\ \alpha'', & \beta'', & \gamma'', & \delta'' \\ \alpha''', & \beta''', & \gamma''', & \delta''' \end{pmatrix}$$

then it is easy to see that

$$\Omega_0^{-1}\Theta\Omega_0 = \begin{pmatrix} -\delta''', & -\gamma''', & -\beta''', & -\alpha''' \\ -\delta'', & -\gamma'', & -\beta'', & -\alpha'' \\ -\delta', & -\gamma', & -\beta', & -\alpha' \\ -\delta, & -\gamma, & -\beta, & -\alpha \end{pmatrix};$$

and hence, from the foregoing values of $-\Omega$ and $-\Omega^{-1}$, this shows that the matrix Ω for the automorphic transformation of the function $xw' + yz' - zy' - wx'$ is such that writing

$$\Omega = \begin{pmatrix} A, & B, & C, & D \\ E, & F, & G, & H \\ I, & J, & K, & L \\ M, & N, & O, & P \end{pmatrix}$$

we have $\Omega^{-1} = \begin{pmatrix} A, & E, & I, & M \\ B, & F, & J, & N \\ C, & G, & K, & O \\ D, & H, & L, & P \end{pmatrix}^{-1} = \begin{pmatrix} P, & O, & -N, & -M \\ L, & K, & -J, & -I \\ -H, & -G, & F, & E \\ -D, & -C, & B, & A \end{pmatrix}$,
which is the theorem in question.

(See *Phil. Trans.* 1858.)

12. I remark in reference to the foregoing proof that writing

$$T = \begin{pmatrix} a, & h, & g, & l \\ h, & b, & f, & m \\ g, & f, & c, & n \\ l, & m, & n, & d \end{pmatrix},$$

then the actual value of

$$(T+\Omega_0)^{-1} = \frac{1}{\Delta}\begin{pmatrix} A+d, & H+n+\nu, & G-m-\mu, & L-l-\rho \\ H+n-\nu, & B+c, & F-f+\lambda, & M-g+\sigma \\ O-m+\mu, & F-f-\lambda, & C+b, & N+h+\tau \\ L-l+\rho, & M-g-\sigma, & N+h-\tau, & D+a \end{pmatrix},$$

where $(A, H, G, L$
H, B, F, M
G, F, C, N
$L, M, N, D)$ is the matrix formed with the first minors of $(a, h, g, l$
h, b, f, m
g, f, c, n
$l, m, n, d)$; moreover

$$\begin{aligned} \lambda &= ad - l^2 + bc - f^2 + 2(nh - mg) + 1, \\ \mu &= bn - mf + dh - ml, \\ \nu &= dg - nl + cm - na, \\ \rho &= fg - ch + gl - ma, \\ \sigma &= hf - bg + al - lh, \\ \tau &= il - hm + nf - ga, \end{aligned}$$

and Δ is the determinant

$$\begin{vmatrix} a, & h, & g, & l \\ h, & b, & f+1, & m \\ g, & f-1, & c, & n \\ l, & m, & n, & d \end{vmatrix},$$

viz. this is

$$\begin{vmatrix} a, & h, & g, & l \\ h, & b, & f, & m \\ g, & f, & c, & n \\ l, & m, & n, & d \end{vmatrix} + ad - l^2 + bc - f^2 + 2(nh - mg) + 1.$$

13. The expression for $(T-\Omega_0)^{-1}$ is obtained from that of $(T+\Omega_0)^{-1}$ by merely transposing the terms of the matrix, or, what is the same thing, by changing the signs of λ, μ, ν, ρ, σ, τ. And it would be easy by means of these developed values to verify the foregoing comparison of $(T-\Omega_0)(T+\Omega_0)^{-1}$ and $(T+\Omega_0)(T-\Omega_0)^{-1}$.

Article Nos. 14 to 22. Second Investigation.

14. I consider from a different point of view the theory of a matrix

$$\Omega = \begin{pmatrix} a, & b, & c, & d \\ e, & f, & g, & h \\ i, & j, & k, & l \\ m, & n, & o, & p \end{pmatrix}$$

such that $\Omega^{-1} = \begin{pmatrix} P, & O, & -N, & -M \\ L, & K, & -J, & -I \\ -H, & -G, & F, & E \\ -D, & -C, & B, & A \end{pmatrix}$, or, as we may call it, a *Hermitian* matrix.

15. *Lemma.* The determinant

$$\nabla = \begin{vmatrix} a, & b, & c, & d \\ e, & f, & g, & h \\ i, & j, & k, & l \\ m, & n, & o, & p \end{vmatrix}$$

may be expressed, and that in two different ways, as a Pfaffian.

16. In fact multiplying the determinant into itself thus,

$$\nabla^2 = \begin{vmatrix} a, & b, & c, & d \\ e, & f, & g, & h \\ i, & j, & k, & l \\ m, & n, & o, & p \end{vmatrix} \times \begin{vmatrix} d, & c, & -b, & -a \\ h, & g, & -f, & -e \\ l, & k, & -j, & -i \\ p, & o, & -n, & -m \end{vmatrix} = \begin{vmatrix} s_{11}, & s_{12}, & s_{13}, & s_{14} \\ s_{21}, & s_{22}, & s_{23}, & s_{24} \\ s_{31}, & s_{32}, & s_{33}, & s_{34} \\ s_{41}, & s_{42}, & s_{43}, & s_{44} \end{vmatrix},$$

viz. we have $s_{11} = (a, b, c, d | d, c, -b, -a)$, $s_{12} = (a, b, c, d | h, g, -f, -e)$, &c.: we see at once that $s_{11} = 0$, $s_{12} + s_{21} = 0$, &c., viz. the determinant in s is a skew determinant, that is, the square of a Pfaffian.

We have therefore

$$\nabla^2 = (s_{12}s_{34} + s_{13}s_{42} + s_{14}s_{23})^2;$$

or extracting the square root of each side, and determining the sign by a comparison of any single term, we have

$$\nabla = s_{12}s_{34} + s_{13}s_{42} + s_{14}s_{23},$$

which is one of the required forms of ∇.

17. And in the same manner

$$\nabla^2 = . \begin{pmatrix} a, & b, & c, & d \\ e, & f, & g, & h \\ i, & j, & k, & l \\ m, & n, & o, & p \end{pmatrix} \times \begin{pmatrix} -e, & -f, & -g, & -h \\ -a, & -b, & -c, & -d \\ i, & j, & k, & l \\ m, & n, & o, & p \end{pmatrix},$$

which is equal to the determinant

$$\begin{vmatrix} t_{11}, & t_{12}, & t_{13}, & t_{14} \\ t_{21}, & t_{22}, & t_{23}, & t_{24} \\ t_{31}, & t_{32}, & t_{33}, & t_{34} \\ t_{41}, & t_{42}, & t_{43}, & t_{44} \end{vmatrix} = \begin{vmatrix} a, & e, & i, & m \\ b, & f, & j, & n \\ c, & g, & k, & o \\ d, & h, & l, & p \end{vmatrix} \times \begin{vmatrix} m, & i, & -e, & -a \\ n, & j, & -f, & -b \\ o, & k, & -g, & -c \\ p, & l, & -h, & -d \end{vmatrix},$$

viz. we have $t_{11} = (a, e, i, m | m, i, -e, -a)$, &c.; this is likewise a skew determinant, and we have

$$\nabla^2 = (t_{12}t_{34} + t_{13}t_{42} + t_{14}t_{23})^2;$$

or extracting the square root of each side, and determining the sign by the comparison of any single term, we have

$$\nabla = t_{12}t_{34} + t_{13}t_{42} + t_{14}t_{23},$$

which is the other of the required forms of ∇.

18. Consider now the matrix

$$\begin{pmatrix} a, & b, & c, & d \\ e, & f, & g, & h \\ i, & j, & k, & l \\ m, & n, & o, & p \end{pmatrix},$$

which is such that

$$\begin{pmatrix} a, & b, & c, & d \\ e, & f, & g, & h \\ i, & j, & k, & l \\ m, & n, & o, & p \end{pmatrix}^{-1} = \begin{pmatrix} P, & O, & -N, & -M \\ L, & K, & -J, & -I \\ -H, & -G, & F, & E \\ -D, & -C, & B, & A \end{pmatrix};$$

this gives

$$\begin{pmatrix} 1, & 0, & 0, & 0 \\ 0, & 1, & 0, & 0 \\ 0, & 0, & 1, & 0 \\ 0, & 0, & 0, & 1 \end{pmatrix} = \begin{pmatrix} a, & b, & c, & d \\ e, & f, & g, & h \\ i, & j, & k, & l \\ m, & n, & o, & p \end{pmatrix} \begin{pmatrix} P, & O, & -N, & -M \\ L, & K, & -J, & -I \\ -H, & -G, & F, & E \\ -D, & -C, & B, & A \end{pmatrix}$$

which is in fact

$$\begin{pmatrix} 1, & 0, & 0, & 0 \\ 0, & 1, & 0, & 0 \\ 0, & 0, & 1, & 0 \\ 0, & 0, & 0, & 1 \end{pmatrix} = \begin{pmatrix} s_{14}, & s_{13}, & s_{12}, & s_{11} \\ s_{24}, & s_{23}, & s_{22}, & s_{21} \\ s_{34}, & s_{33}, & s_{32}, & s_{31} \\ s_{44}, & s_{43}, & s_{42}, & s_{41} \end{pmatrix}$$

and the two matrices will be equal, term by term, if only

$$1 = s_{14} = s_{23}, \quad 0 = s_{13} = s_{12} = s_{11} = s_{24} = s_{34},$$

that is, if six conditions are satisfied.

19. But we have also (a matrix and its reciprocal being convertible)

$$\begin{pmatrix} 1, & 0, & 0, & 0 \\ 0, & 1, & 0, & 0 \\ 0, & 0, & 1, & 0 \\ 0, & 0, & 0, & 1 \end{pmatrix} = \begin{pmatrix} P, & O, & -N, & -M \\ L, & K, & -J, & -I \\ -H, & -G, & F, & E \\ -D, & -C, & B, & A \end{pmatrix} \begin{pmatrix} a, & b, & c, & d \\ e, & f, & g, & h \\ i, & j, & k, & l \\ m, & n, & o, & p \end{pmatrix}$$

which is in fact

$$\begin{pmatrix} 1, & 0, & 0, & 0 \\ 0, & 1, & 0, & 0 \\ 0, & 0, & 1, & 0 \\ 0, & 0, & 0, & 1 \end{pmatrix} = \begin{pmatrix} t_{14}, & t_{24}, & t_{34}, & t_{44} \\ t_{13}, & t_{23}, & t_{33}, & t_{43} \\ t_{12}, & t_{22}, & t_{32}, & t_{42} \\ t_{11}, & t_{21}, & t_{31}, & t_{41} \end{pmatrix}$$

and we obtain for the equality of the two matrices the six conditions

$$1 = t_{14} = t_{23}, \quad 0 = t_{13} = t_{12} = t_{24} = t_{34},$$

equivalent to the former set of six conditions.

20. We obtain from either set of conditions, for the determinant the value,

$$\nabla = \begin{vmatrix} a, & b, & c, & d \\ e, & f, & g, & h \\ i, & j, & k, & l \\ m, & n, & o, & p \end{vmatrix} = 1.$$

21. Write

$$(x, y, z, w) = (a, b, c, d | X, Y, Z, W); \quad (x', y', z', w') = (a, b, c, d | X', Y', Z', W'),$$

then substituting for (x, y, z, w), (x', y', z', w') their values, we find

$$\begin{aligned} xw' + yz' - zy' - wx' = -\frac{1}{\nabla}(t_{21}, t_{22}, t_{23}, t_{24} \\ t_{31}, t_{32}, t_{33}, t_{34} \\ t_{41}, t_{42}, t_{43}, t_{44} | X, Y, Z, W | X', Y', Z', W') \\ = XW' + YZ' - ZY' - WX'; \end{aligned}$$

and similarly writing

$$(X, Y, Z, W) = \begin{pmatrix} P, & O, & -N, & -M \\ L, & K, & -J, & -I \\ -H, & -G, & F, & E \\ -D, & -C, & B, & A \end{pmatrix} (x, y, z, w)$$

and similarly for (X', Y', Z', W'), we obtain with the s coefficients the equivalent result,

$$XW' + YZ' - ZY' - WX' = xw' + yz' - zy' - wx'.$$

We thus see conversely that the Hermitian matrix is in fact the matrix for the automorphic transformation of the function $xw' + yz' - zy' - wx'$.

22. Considering any two or more matrices for the automorphic transformation of such a function, the matrix compounded of these is a matrix for the automorphic transformation of the function — or, theorem, the matrix compounded of two or more Hermitian matrices is itself Hermitian.

Article No. 23. Theorem on a Form of Matrices.

23. I take the opportunity of mentioning a theorem relating to the matrices which present themselves in the arithmetical theory of the composition of quadratic forms.

Writing

$$X = \begin{pmatrix} -a, & -(b-\beta), & -c, & -(b+\beta) \\ \cdot, & -\gamma, & -(b+\beta), & a \\ c, & b+\beta, & -\gamma, & \cdot \\ b-\beta, & -a, & \cdot, & \gamma \end{pmatrix} \quad \text{and} \quad X^{-1} = D^{-1}\Delta \left(\begin{matrix} \gamma, & -(b - \\ \cdot, & - \\ -(b-\beta), & a \\ -c, & -(b - \end{matrix} \right.$$

where $D = ac - b^2$, $\Delta = a\gamma - \beta^2$; and similarly,

$$X' = \begin{pmatrix} -a', & -(b'-\beta'), & -c', & -(b'+\beta') \\ \cdot, & -\gamma', & -(b'+\beta'), & a' \\ c', & b'+\beta', & -\gamma', & \cdot \\ b'-\beta', & -a', & \cdot, & \gamma' \end{pmatrix} \quad \text{and} \quad (X')^{-1} = D'^{-1}\Delta' \left(\begin{matrix} \gamma', \\ \cdot, \\ -(b' - \\ -c', \end{matrix} \right.$$

where $D' = a'c' - b'^2$, $\Delta' = a'\gamma' - \beta'^2$; then

$$(X|X') + (D-\Delta)(D'-\Delta')(X|X')^{-1}$$

is to a factor près equal to the matrix unity; viz. writing

$$A = aa' + 2b\beta' + c\gamma' + a'a' + 2b'\beta + c'\gamma,$$

the foregoing expression is

$$= A \cdot (1).$$

The theorem is verified without difficulty by merely forming the expressions of the compound matrices $(X|X')$ and $(X'^{-1}|X^{-1})$.

358. Addition to the Memoir on Tschirnhausen's Transformation. [358]

(From the Philosophical Transactions of the Royal Society of London, vol. CLVI. (for the year 1866), pp. 97–100. Received October 24, — Read December 7, 1866.)

In the memoir "On Tschirnhausen's Transformation," Philosophical Transactions, vol. CLII. (1862), pp. 561–568, [275], I considered the case of a quartic equation: viz. it was shown that the equation

$$(a, b, c, d, e|x, 1)^4 = 0$$

is, by the substitution

$$y = (ax+b)B + (ax^2 + 4bx + 3c)C + (ax^3 + 4bx^2 + 6cx + 3d)D,$$

transformed into

$$(1, 0, \theta, 2\Theta, \Phi | y, 1)^4 = 0$$

where $(\theta, 2\Theta, \Phi)$ have certain given values. It was further remarked that $(\theta, 2\Theta, \Phi)$ were expressible in terms of U', H', Φ', invariants of the two forms $(a, b, c, d, e | X, Y)^4$, $(B, C, D | Y, -X)^2$, of I, J, the invariants of the first, and of Θ', $\Phi' = BD - C^2$, the invariant of the second of these two forms, viz. that we have

$$\theta = 6H' - 2I\Theta',$$

$$2\Theta = 4\Phi',$$

$$\Phi = JU'^2 - 3I^2 + I^2\Theta'^2 + 12J\Theta'U' + 4I'\Theta'H';$$

and by means of these I obtained an expression for the quadrinvariant of the form $(1, 0, \theta, 2\Theta, \Phi | y, 1)^4$; viz. this was found to be

$$= IU'^2 + 4I^2\Theta'^2 + 12J\Theta'U'.$$

But I did not obtain an expression for the cubinvariant of the same function: such expression, it was remarked, would contain the square of the invariant Φ'; it was probable that there existed an identical equation,

$$JU'^3 - IU'^2H' + 4H'^3 + M\Theta' = -\Phi'^2,$$

which would serve to express Φ'^2 in terms of the other invariants; but, assuming that such an equation existed, the form of the factor M remained to be ascertained; and until this was done, the expression for the cubinvariant could not be obtained in its most simple form. I have recently verified the existence of the identical equation just referred to, and have obtained the expression for the factor M; and with the assistance of this identical equation I have obtained the expression for the cubinvariant of the form $(1, 0, \theta, 2\Theta, \Phi | y, 1)^4$.

The expression for the quadrinvariant was, as already mentioned, given in the former memoir: I find that the two invariants are in fact the invariants of a certain linear function of U, H; viz. the linear function is $= U'U + \frac{4}{3}\Theta'H$; so that, denoting by I^*, J^*, the quadrinvariant and the cubinvariant respectively of the form $(1, 0, \theta, 2\Theta, \Phi | y, 1)^4$, we have

$$I^* = I(U'U + \tfrac{4}{3}\Theta'H),$$

$$J^* = J(U'U + \tfrac{4}{3}\Theta'H),$$

where I, J signify the functional operations of forming the two invariants respectively.

The function $(1, 0, \theta, 2\Theta, \Phi | y, 1)^4$, obtained by the application of Tschirnhausen's transformation to the equation $(a, b, c, d, e | x, 1)^4 = 0$, has thus the same invariants with the function

$$U'U + \tfrac{4}{3}\Theta'H = U'(a, b, c, d, e | x, 1)^4 + \tfrac{4}{3}\Theta'(ac - b^2, ad - bc, ae + 2bd - 3c^2, be - cd, ce - d^2 | x, 1)^4,$$

and it is consequently a linear transformation of the last-mentioned function; so that the application of Tschirnhausen's transformation to the equation $U = 0$ gives an equation linearly transformable into, and thus virtually equivalent to, the equation

$$U'U + \tfrac{4}{3}\Theta' H = 0,$$

which is an equation involving the single parameter $\frac{\Theta'}{U'}$: this appears to me a result of considerable interest. It is to be remarked that Tschirnhausen's transformation, wherein y is put equal to a rational and integral function of the order $n-1$ (if n be the order of the equation in x), is not really less general than the transformation wherein y is put equal to any rational function whatever of x; such rational function may, in fact, by means of the given equation in x, be reduced to a rational and integral function of the order $n-1$; hence in the present case, taking V, W to be respectively of the order $n-1$, $=3$, it follows that the equation in y obtained by the elimination of x from the equations

$$\begin{gathered}(a, b, c, d, e | x, 1)^4 = 0,\\ \frac{(\alpha, \beta, \gamma, \delta | x, 1)^3}{(\alpha', \beta', \gamma', \delta' | x, 1)^3} = y\end{gathered}$$

is a mere linear transformation of the equation $AU + BH = 0$, where A, B are functions (not as yet calculated) of $(a, b, c, d, e, \alpha, \beta, \gamma, \delta, \alpha', \beta', \gamma', \delta')$.

Article Nos. 1, 2, 3. Investigation of the identical equation $JU'^3 - IU'^2H' + 4H'^3 + M\Theta' = -\Phi'^2$.

1. It is only necessary to show that we have such an equation, M being an invariant, in the particular case $a = e = 1$, $b = d = 0$, $c = \theta$, that is for the quartic function $(1, 0, \theta, 0, 1 | x, 1)^4$; for, this being so, the equation will be true in general.

Writing the equation in the form

$$-\Phi'^2 = U'^2(JU' - IH') + 4H'^3 + \Theta'^2 M$$

and observing that we have

$$\begin{aligned}U' &= 6(B^2 + D^2) + 26\Theta BD + 4\Theta'^2,\\ H' &= 6(B^2 + D^2) + (1 + \theta^2)BD - 4\Theta'^2,\\ \Theta' &= BD - C^2,\\ \Phi' &= (1 - 9\Theta'^2)C(B^2 - D^2),\\ I &= 1 + 3\theta^2,\\ J &= \theta - \theta^3,\end{aligned}$$

and thence

$$JU' - IH' = -4\theta^2(B^2 + D^2) + (-1 - 2\theta^2 - 5\theta^4)BD + (8\theta^3 + 8\theta^5)C^2,$$

the equation becomes

$$\begin{aligned} -(BD - C^2)M = (-4\theta^2(B^2 + D^2) + (-1 - 2\theta^2 - 5\theta^4)BD + (8\theta^3 + 8\theta^5)C^2) \\ \times (6(B^2 + D^2) + 26\Theta BD + 4\Theta'^2)^2 \\ + 4(6(B^2 + D^2) + (1 + \theta^2)BD - 4\Theta'^2)^3 \\ + (1 - 9\Theta'^2)^2 C^2 (B^2 - D^2)^2. \end{aligned}$$

2. It is found by developing that the right-hand side is in fact divisible by $BD - C^2$, and that the quotient is

$$\begin{aligned} (-1 + 10\theta^2 - 9\theta^4)(B^2 + D^2)^3 \\ + (8\theta^2 + 16\theta^4 - 24\theta^6)(B^2 + D^2)^2 BD \\ + (4 + 8\theta^2 + 4\theta^4 - 16\theta^6)B^2 D^2 \\ + (-64\theta^2 - 192\theta^4)(B^2 + D^2)C^2 \\ + (16\theta^2 - 416\theta^4 - 112\theta^6)BDC^2 \\ + (-128\theta^4 + 128\theta^6)C^4. \end{aligned}$$

3. This is found to be

$$\begin{aligned} = -I^3U'^3 + 12JU'^2H' - 4IH'^2 \\ - 16J^2\Theta'^2 \\ - 81JU'\Theta'^2 \\ + (7I^2 - 12JU'H' - UH'^2)\Theta'^2 \end{aligned}$$

which is consequently the value of $-M$.

We have therefore

$$\begin{aligned} -\Phi'^2 = JU'^3 - IU'^2H' + 4H'^3 \\ + \Theta'^2(J^2U'^2 - 12JU'H' - UH'^2) \\ + 8IJU'\Theta'^2 \\ + 16J^2\Theta'^4, \end{aligned}$$

which is the required identical equation.

Article No. 4. Calculation of the Cubinvariant.

4. We have

$$J^* = -((H' - \tfrac{4}{3}I\Theta')^2 \\ - 4(H' - \tfrac{4}{3}I\Theta')(IU'^2 - 3H'^2 + (12JU' + 2IH')\Theta'^2 + I^2\Theta'^4) \\ - 4\Phi'^2),$$

whence, substituting for $-\Phi'^2$ its value and reducing, we find

$$J^* = JU'^3 + \Theta' \cdot \tfrac{8}{3}JU'^2 + \Theta'^2 \cdot (4IJU') + \Theta'^3 \cdot (16J^2 - \tfrac{4}{3}I^3).$$

Article No. 5. Final expressions of the two Invariants.

The value of I^* has been already mentioned to be $I^* = IU'^2 + \Theta' \cdot 12JU' + \Theta'^2 \cdot \frac{4}{3}I^2$, and it hence appears that the values of the two invariants may be written

$$I^* = (I,\ 18J,\ \tfrac{4}{3}I^2 | U',\ \tfrac{4}{3}\Theta')^2,$$

$$J^* = (J,\ I^2,\ 9IJ,\ -I^3 + 64J^2 | U',\ \tfrac{4}{3}\Theta')^3.$$

But we have (see Table No. 72 in my "Seventh Memoir on Quantics," Philosophical Transactions, vol. CLI. (1861), pp. 277–292, [269])

$$I(\alpha U + \beta H) = (I,\ 18J,\ \tfrac{4}{3}I^2 | \alpha,\ \beta)^2,$$

$$J(\alpha U + \beta H) = (J,\ I^2,\ 9IJ,\ -I^3 + 64J^2 | \alpha,\ \beta)^3;$$

so that, writing $\alpha = U'$, $\beta = \frac{4}{3}\Theta'$, we have

$$I^* = I(U'U + \tfrac{4}{3}\Theta' H),$$

$$J^* = J(U'U + \tfrac{4}{3}\Theta' H);$$

or the function $(1, 0, \theta, 2\Theta, \Phi | y, 1)^4$ obtained from Tschirnhausen's transformation of the equation $U = 0$ has the same invariants with the function $U'U + \frac{4}{3}\Theta' H$; or, what is the same thing, the equation $(1, 0, \theta, 2\Theta, \Phi | y, 1)^4 = 0$ is a mere linear transformation of the equation $U'U + \frac{4}{3}\Theta' H = 0$; which is the above-mentioned theorem.

359. A Supplementary Memoir on Caustics. [359]

(From the Philosophical Transactions of the Royal Society of London, vol. CLVII. (for the year 1867), pp. 7–16. Received November 15, — Read November 22, 1866.)

It is near the conclusion of my "Memoir on Caustics," Philosophical Transactions, vol. CXLVII. (1857), pp. 273–312, [145], remarked that for the case of parallel rays refracted at a circle, the ordinary construction for the secondary caustic cannot be made use of (the entire curve would in fact pass off to an infinite distance), and that the simplest course is to measure off the distance GQ from a line through the centre of the refracting circle perpendicular to the direction of the incident rays. The particular secondary caustic, or orthogonal trajectory of the refracted rays, obtained on the above supposition was shown to be a curve of the order 8; and it was further shown (by consideration of the case wherein the distance GQ is measured off from an arbitrary line perpendicular to the incident rays) that the general secondary caustic or orthogonal trajectory of the refracted rays was a curve of the same order 8. The last-mentioned curve in the case of reflexion, or for $\mu = -1$, degenerates into a curve of the order 6; and I propose in the present supplementary memoir to discuss this sextic curve, viz. the sextic curve which is the general secondary caustic or orthogonal trajectory of parallel rays reflected at a circle.

1. For parallel rays refracted at a circle, taking the equation of the circle to be $x^2 + y^2 = 1$, and the incident rays to be parallel to the axis of x, then if $x = m$ be an arbitrary line perpendicular to the direction of the incident rays, the secondary caustic is the envelope of the circle

$$\mu^2[(x - \cos\theta)^2 + (y - \sin\theta)^2] - (x - m)^2 = 0,$$

where (α, β) are the coordinates of a variable point on the refracting circle, and as such satisfy the equation $\alpha^2 + \beta^2 = 1$. Or, what is the same thing, writing $\alpha = \cos\theta$, $\beta = \sin\theta$, the secondary caustic is the envelope of the circle

$$\mu^2[(x - \cos\theta)^2 + (y - \sin\theta)^2] - (x - m)^2 = 0,$$

where θ is a variable parameter.

2. The equation may be written

$$A\cos 2\theta + B\sin 2\theta + C\cos\theta + D\sin\theta + E = 0,$$

where

$$A = 1, \quad B = 0, \quad C = 4\mu^2 x - 4m, \quad D = 4\mu^2 y,$$
$$E = -2\mu^2(x^2 + y^2) - 2\mu^2 + 1 + 2m^2,$$

and which in the case of reflexion, or for $\mu = -1$, become

$$A = 1, \quad B = 0, \quad C = 4x - 4m, \quad D = 4y,$$

$$E = -2(x^2 + y^2) - 1 + 2m^2,$$

viz. the equation of the variable circle is in this case

$$\cos 2\theta + 4(x - m)\cos\theta + 4y\sin\theta + 2m^2 - 1 - 2(x^2 + y^2) = 0.$$

3. Now in general for the equation

$$A\cos 2\theta + B\sin 2\theta + C\cos\theta + D\sin\theta + E = 0,$$

where the coefficients are any functions whatever of the coordinates (x, y), the equation of the envelope is $S^2 - T^2 = 0$, where

$$S = 2(A(C^2 - D^2) + 4BCD - (C^2 + D^2)E + 8E^2),$$

$$T = 32(ABE + \cdots).$$

(For the full expressions see the original memoir.)

4. Hence, substituting for A, B, C, D, E the above reflexion values, we find

$$S = 12 - 48((x-m)^2 + y^2) + 4(2m^2 - 1 - 2x^2 - 2y^2)^2,$$

$$T = 32((x-m)^2 - y^2) - 72(12 + 144((x-m)^2 + y^2))(2m^2 - 1 - 2x^2 - 2y^2) \\ + 8(2m^2 - 1 - 2x^2 - 2y^2)^3.$$

Writing in these equations

$$(x-m)^2 + y^2 = x^2 + y^2 - 2mx + m^2,$$

$$(x-m)^2 - y^2 = 2x^2 - 2mx + m^2 - (x^2 + y^2),$$

then after some simple reductions, we find

$$S = 16\{(x^2 + y^2 - m^2 - 1)^2 + 6m(x-m)\},$$

$$T = 32\{2(x^2 + y^2 - m^2 - 1)^3 + 18m(x-m)(x^2 + y^2 - m^2 - 1) - 27(x-m)^3\},$$

and thence

$$S^2 - T^2 = 1024(x-m)^3 U,$$

where

$$\begin{aligned} U = {} & 4(x^2 + y^2 - m^2 - 1)^3 \\ & + 36m(x^2 + y^2 - m^2 - 1)^2 \\ & + 36m(x^2 + y^2 - m^2 - 1)(x-m) \\ & - 27(x-m)^3 + 32m^2(x-m), \end{aligned}$$

or, what is the same thing,

$$\begin{aligned} U = {} & 4(x^2 + y^2)^3 - (8m^2 + 12)(x^2 + y^2)^2 \\ & + (36mx + 4m^4 - 20m^2 + 12)(x^2 + y^2) - 27x^3 + (-47m^2 + 18)xm + m^6 - 4; \end{aligned}$$

so that the equation of the secondary caustic is $U = 0$, or the secondary caustic is, as stated above, a sextic curve.

5. It is easy to see that the foregoing envelope may be geometrically constructed as follows: viz. if from the point Q (coordinates $\cos\theta$, $\sin\theta$) on the reflecting circle we draw QM perpendicular to the line $x - m = 0$, and then from the point M draw MN perpendicular to QT, the tangent at T, and produce MN to a point P such that $PN = NM$, then P is a point of the envelope; and we thence obtain for the coordinates (x, y) of a point P of the envelope the values

$$x = m - 2(m - \cos\theta)\cos^2\theta,$$

$$y = \sin\theta - 2(m - \cos\theta)\cos\theta\sin\theta,$$

or, what is the same thing,

$$x = 2\cos^3\theta - m(2\cos^2\theta - 1),$$

$$y = \sin\theta(2\cos^2\theta + 1) - 2m\sin\theta\cos\theta,$$

or, as these equations may also be written,

$$x = \tfrac{3}{2}\cos\theta - m\cos 2\theta + \tfrac{1}{2}\cos 3\theta,$$

$$y = \tfrac{3}{2}\sin\theta - m\sin 2\theta + \tfrac{1}{2}\sin 3\theta.$$

6. This result may be verified by showing that these values satisfy the equation

$$\cos 2\theta + 4(x - m)\cos\theta + 4y\sin\theta + 2m^2 - 1 - 2(x^2 + y^2) = 0,$$

and also the derived equation

$$\sin 2\theta + 2(x - m)\sin\theta - 2y\cos\theta = 0.$$

We in fact have

$$x\sin\theta - y\cos\theta = m\sin\theta - \tfrac{1}{2}\sin 2\theta,$$

$$x\cos\theta + y\sin\theta = \tfrac{3}{2} - m\cos\theta + \tfrac{1}{2}\cos 2\theta,$$

and thence

$$(x - m)\sin\theta - y\cos\theta = -\tfrac{1}{2}\sin 2\theta,$$

which is one of the equations to be verified; and also

$$(x - m)\cos\theta + y\sin\theta = \tfrac{3}{2} - 2m\cos\theta + \tfrac{1}{2}\cos 2\theta.$$

We have moreover

$$x^2 + y^2 = \tfrac{9}{8} + m^2 - 4m\cos\theta + \tfrac{3}{2}\cos 2\theta;$$

and, by means of these last equations, the other equation

$$\cos 2\theta + 4(x-m)\cos\theta + 4y\sin\theta + 2m^2 - 1 - 2(x^2+y^2) = 0,$$

is also verified.

7. The foregoing values of (x, y) give

$$dx = (-\tfrac{3}{2}\sin\theta + 2m\sin 2\theta - \tfrac{3}{2}\sin 3\theta)\,d\theta = -\sin 2\theta(3\cos\theta - 2m)\,d\theta,$$

$$dy = (\tfrac{3}{2}\cos\theta - 2m\cos 2\theta + \tfrac{3}{2}\cos 3\theta)\,d\theta = \cos 2\theta(3\cos\theta - 2m)\,d\theta,$$

or, what is the same thing, $dx : dy = -\sin 2\theta : \cos 2\theta$.

Hence taking for a moment (X, Y) as the current coordinates of a point in the tangent of the envelope, the equation of the tangent of the envelope is

$$X\,dy - Y\,dx = x\,dy - y\,dx,$$

or, substituting for x, y, dx, dy their values, this equation takes the very simple form

$$X\cos 2\theta - Y\sin 2\theta - 2\cos\theta + m = 0.$$

8. Or writing (x, y) in place of (X, Y), that is taking now (x, y) as the current coordinates of a point in the tangent, the equation of the tangent is

$$x\cos 2\theta - y\sin 2\theta - 2\cos\theta + m = 0;$$

whence observing that this equation may be expressed as a rational equation of the fourth order in terms of the parameter $\tan\frac{1}{2}\theta$ (or $\cos\theta + \sqrt{-1}\sin\theta$), it appears that the class of the secondary caustic is $= 4$.

9. The secondary caustic may be considered as the envelope of the tangent, and the equation be obtained in this manner. Comparing with the general equation

$$A\cos 2\theta + B\sin 2\theta + C\cos\theta + D\sin\theta + E = 0,$$

we have

$$A = x, \quad B = -y, \quad C = -2, \quad D = 0, \quad E = m,$$

and thence

$$S = 4\{3(x^2+y^2) + m^2 - 3\},$$

$$T = 4\{18m(x^2+y^2) - 27x - 2m^3 + 9m\},$$

giving

$$T^2 - S^3 = 16V,$$

if for a moment

$$V = 4\{3(x^2+y^2) + m^2 - 3\}^3 - \{18m(x^2+y^2) - 27x - 2m^3 + 9m\}^2.$$

The equation of the curve is thus obtained in the form $V = 0$; this should of course be equivalent to the before-mentioned equation $U = 0$; and by developing V, and comparing with the second of the two expressions of U, it appears that we in fact have $V = 27U$.

10. Taking as parameter $\tan\frac{1}{2}\theta$, or if we please $\cos\theta + \sqrt{-1}\sin\theta$, the foregoing values of (x, y) in terms of θ give $(x, y, 1)$ proportional to rational and integral functions of the degree 6 in the parameter; so that not only the curve is a sextic curve, but it is a *universal* sextic, or curve of the order 6 with the maximum number, $= 10$, of nodes and cusps; that is, if δ be the number of nodes and κ the number of cusps, we have $\delta + \kappa = 10$. Moreover, introducing the same parameter into the equation of the tangent, this equation is seen to be of the degree 4 in the parameter; that is, the class of the curve is $= 4$: this implies $2\delta + 3\kappa = 26$, and we have therefore $\delta = 4$, $\kappa = 6$. To verify these numbers, it is to be remarked that it appears by the equation of the curve that there is at each of the circular points at infinity a triple point in the nature of the point $x = 0$, $y = 0$ on the curve $y^3 = x^5$; such a point is in fact equivalent to a node and two cusps, and we have thus the two circular points at infinity counting together as 2 nodes and 4 cusps; there should therefore besides be 2 nodes and 2 cusps, and I proceed to establish the existence of these by means of the expressions for (x, y) in terms of θ.

11. To find the cusps, we have

$$\frac{dx}{d\theta} = -\sin 2\theta(3\cos\theta - 2m) = 0,$$

$$\frac{dy}{d\theta} = \cos 2\theta(3\cos\theta - 2m) = 0,$$

which are each of them satisfied if only $3\cos\theta - 2m = 0$, or $\cos\theta = \frac{2}{3}m$; the corresponding values of (x, y) are found to be

$$x = m - \frac{4}{27}m^3, \qquad y = \pm(1 - \tfrac{4}{9}m^2)^{\frac{3}{2}},$$

and we have thus two cusps situate symmetrically in regard to the axis of x; the cusps are real if $m < \frac{3}{2}$, imaginary if $m > \frac{3}{2}$; for $m = \frac{3}{2}$, the two cusps unite together at the point $x = \frac{19}{27}$ on the axis of x, giving rise to a higher singularity, which will be further examined, post No. 12.

12. The curve is symmetrical in regard to the axis of x, and hence any intersection with the axis of x, not being a point where the curve cuts the axis at right angles, will be a node.

Hence, in order to find the nodes, writing $y = 0$, this is

$$\sin\theta(1 - 2m\cos\theta + 2\cos^2\theta) = 0,$$

giving $\sin\theta = 0$, that is,

$$\theta = 0, \quad x = 2 - m;$$

$$\theta = \pi, \quad x = -2 - m;$$

but these are each of them ordinary points on the axis of x; or else giving

$$1 - 2m\cos\theta + 2\cos^2\theta = 0,$$

that is

$$\cos\theta = \tfrac{1}{2}(m \pm \sqrt{m^2 - 2}).$$

The corresponding values of x are

$$x = \cos\theta(2\cos^2\theta - 2m\cos\theta) + m = m - \cos\theta = \tfrac{1}{2}(m \mp \sqrt{m^2-2});$$

each of the points in question, viz. the points

$$x = \tfrac{1}{2}(m \mp \sqrt{m^2-2}), \qquad y = 0,$$

is a node on the axis of x.

It is to be observed that for $m < \sqrt{2}$ the nodes are both imaginary; for $m = \sqrt{2}$ they coincide together at the point $x = -\frac{m}{2}$; for $m > \sqrt{2}$ they are both real: it is to be further noticed that

node, $x = \frac{1}{2}(m + \sqrt{m^2-2})$, corresponds to $\cos\theta = \frac{1}{2}(m - \sqrt{m^2-2})$,

where (m being $> \sqrt{2}$) the point $(\cos\theta, \sin\theta)$ is a real point on the circle $x^2 + y^2 = 1$; in fact for $m < \frac{3}{2}$ (that is, $m = \sqrt{2}$ to $m = \frac{3}{2}$) we have $\frac{1}{2}(m - \sqrt{m^2-2}) < \frac{2}{3}m$, that is, $\cos\theta < \frac{2}{3}m$;

but $m =$ or $> \frac{3}{2}$, then $\cos\theta = \frac{1}{2}(m - \sqrt{m^2-2}) = \frac{1}{m+\sqrt{m^2-2}}$ is $=$ or $< \frac{2}{3}$, and

node, $x = \frac{1}{2}(m - \sqrt{m^2-2})$, corresponds to $\cos\theta = \frac{1}{2}(m + \sqrt{m^2-2})$,

where (m being $> \sqrt{2}$) the point $(\cos\theta, \sin\theta)$ is a real point on the circle $x^2 + y^2 = 1$ so long as m is not $> \frac{3}{2}$, that is, from $m = \sqrt{2}$ to $m = \frac{3}{2}$; but if $m > \frac{3}{2}$, then the point in question is an imaginary point on the circle — whence also the node $x = \frac{1}{2}(m - \sqrt{m^2-2})$ is an *acnode* or isolated point.

In the case $m = \frac{3}{2}$ we have

node, $x = \frac{3}{4}$, corresponding to $\cos\theta = \frac{1}{4}$ or $\theta = 60°$,

$x = \frac{3}{4}$, $\cos\theta = 1$ or $\theta = 0°$,

the last-mentioned point $x = \frac{3}{4}$ being in fact the point of union of two cusps in the case $m = \frac{3}{2}$ now in question. Hence in this case we have at $(x = \frac{3}{4}, y = 0)$ a triple point equivalent to two cusps and a node; visibly, there is only a single branch cutting the axis of x at right angles.

In the case $m = \sqrt{2}$, the nodes coincide as above mentioned on the axis; for this value of m the coordinates of the cusps are

$$x = \tfrac{19}{27}\sqrt{2} (= \tfrac{3}{4} \cdot \tfrac{19}{18}\sqrt{2}, \text{ which is } < \tfrac{3}{4}\sqrt{2});$$

$$y = \pm\tfrac{1}{27}\sqrt{2}.$$

13. Starting from the equation $1024(x-m)^3U = S^2 - T^2 = 0$, it is clear that the cusps are included among the intersections of the curves $S = 0$, $T = 0$: these two curves intersect in 24 points which lie $9+9$ at the circular

points at infinity, $2+2$ at the points $x=m$, $y^2-1=0$, and $1+1$ are the cusps, or points $x=m-\frac{4}{27}m^3$, $y^2=(1-\frac{4}{9}m^2)^3$.

To verify this, writing for a moment

$$S'=(x^2+y^2-m^2-1)^3+6m(x-m),$$

$$T'=2(x^2+y^2-m^2-1)^3+18m(x-m)(x^2+y^2-m^2-1)-27(x-m)^3,$$

then we have

$$\begin{aligned} T'^2-2(x^2+y^2-m^2-1)S' &= 6m(x-m)(x^2+y^2-m^2-1)-27(x-m)^3,\\ &= 3(x-m)\{2m(x^2+y^2-m^2-1)-9(x-m)\}; \end{aligned}$$

so that the equations $S=0$, $T=0$, or, what is the same thing, $S'=0$, $T'=0$ give

$$(x-m)\{2m(x^2+y^2-m^2-1)-9(x-m)\}=0,$$

that is, $x-m=0$, or else $x^2+y^2-m^2-1=\frac{9}{2m}(x-m)$.

And combining herewith the equation $S'=(x^2+y^2-m^2-1)^3+6m(x-m)=0$, we have $x-m=0$, $(y^2-1)^3=0$, or else

$$(x^2+y^2-m^2-1)^2=\frac{81}{4m^2}(x-m)^2=6m(x-m),$$

and therefore

$$(x-m)\cdot\tfrac{1}{m}\{27(x-m)-8m^3\}=0,$$

the second factor of which gives $x=m-\frac{8}{27}m^3$, and thence $x^2+y^2-m^2-1=-\frac{4}{3}m^2$, that is, we have $x^2+y^2=1-\frac{4}{3}m^2+m^2-\frac{8}{27}m^3=1-\frac{4}{3}m^2+\frac{19}{27}m^2$, and therefore $y^2=(1-\frac{4}{9}m^2)^3-(m-\frac{8}{27}m^3)^2=(1-\frac{4}{9}m^2)^3$, that is, we have

$$x=m-\tfrac{4}{27}m^3, \qquad y^2=(1-\tfrac{4}{9}m^2)^3,$$

which, as appears above, gives the two cusps.

14. Similarly, in the equation $16V=S^2-T^2=0$, the intersections of the curves $S=0$, $T=0$ must include the cusps; the curves in question are the two circles

$$3(x^2+y^2)+m^2-3=0,$$

$$18m(x^2+y^2)-27x-2m^3+9m=0,$$

meeting in the circular points at infinity, and in the two cusps. It is to be added that the tangent at the cusps coincides with the tangent of the last-mentioned circle,

$$18m(x^2+y^2)-27x-2m^3+9m=0,$$

or, as this may also be written,

$$3\sqrt[3]{y^2}+\sqrt[3]{\frac{12m}{(4m^2-9)^2}}=0.$$

15. The axis of x meets the secondary caustic in the two nodes counting as 4 intersections, and besides in 2 points, viz. the points $x = 2 - m$, $x = -2 - m$; these correspond to the values $\theta = 0$ and $\theta = \pi$ respectively. But to verify them by means of the equation of the curve, it may be remarked that for $y = 0$ we have

$$S = 4(3x^2 + m^2 - 3), \qquad T = 4(18mx^2 - 27x - 2m^3 + 9m);$$

and writing herein $x = +2 - m$, we find

$$S = 4(2m + 3)^2, \qquad T = 8(2m + 3)^3,$$

values which satisfy the equation $S^2 - T^2 = 0$.

16. In the equation $U = 0$ of the curve, writing $x - m = 0$, the equation becomes

$$4(y^2 - 1)^3 + 4m^2(y^2 - 1)^2 = 0,$$

that is

$$4(y^2 - 1)^2(y^2 - 1 + m^2) = 0,$$

and the line $(x - m) = 0$ is thus a double tangent to the curve, touching it at the points $x = m$, $y = \pm 1$, and besides meeting it at the points $x = m$, $y = \pm\sqrt{1 - m^2}$, that is, at the intersections of the line $x - m = 0$, with the circle $x^2 + y^2 = 1$.

17. The maximum or minimum values of y correspond to the values $\theta = \frac{1}{2}\pi$, $\theta = \frac{1}{3}\pi$, $\theta = \frac{2}{3}\pi$, $\theta = \frac{4}{3}\pi$ of θ; and we have for

$$\begin{array}{lll}
\theta = \frac{1}{2}\pi, & x = -\frac{1}{2}\sqrt{2}, & y = \sqrt{2} - m, \\
\theta = \frac{1}{3}\pi, & x = -\frac{1}{2}\sqrt{2}, & y = \sqrt{2} - m, \\
\theta = \frac{2}{3}\pi, & x = -\frac{1}{2}\sqrt{2}, & y = -\sqrt{2} + m, \\
\theta = \frac{4}{3}\pi, & x = \frac{1}{2}\sqrt{2}, & y = -\sqrt{2} + m.
\end{array}$$

18. It is now easy to trace the secondary caustic; we may without loss of generality assume that m is positive, and the values to be considered are

$$m = 0, \quad m = 1, \quad m = \sqrt{2}, \quad m = \tfrac{3}{2},$$

with the intermediate values $m > 0 < 1$, &c.… and $m > \frac{3}{2}$. I have for convenience delineated in the figure only a portion of each curve, viz. the figure is terminated at the negative value $x = -\frac{3}{2}\sqrt{2}$, which corresponds to the maximum value $y = \sqrt{2} + m$; as x increases negatively, the value of the ordinate y diminishes continuously from this maximum value, becoming $= 0$ for the value $x = -2 - m$, and the curve at this point cutting the axis of x at right angles; this is a sufficient explanation of the form of the curves beyond the limits of the figure. Moreover the curve is symmetrical in regard to the axis of x, and I have within the limits of the figure delineated only one of the two halves of the curve.

19. For $m > \frac{3}{2}$ the cusps are both imaginary, the nodes both real, but one of them is an isolated point or acnode (shown in the figure by a small cross). The curve has an interior loop, as shown in the figure, and there is also the acnode lying within the loop.

For $m = \frac{3}{2}$, there is still an interior loop, but the acnode has united itself to the loop, the point of union, although presenting no visible singularity, being really a triple point equivalent to a node and two cusps. And in all the cases which follow there are two real cusps.

For $m = 1$, the loop has altered its form in such wise as to exhibit a node and two cusps, the curve has therefore two real nodes.

For $m = \sqrt{2}$, the two nodes unite together into a tacnode, so that the loop is on the point of disappearing; and for $m < \sqrt{2} > 1$ the nodes are imaginary, and there is thus no longer any loop.

In all the above forms the double tangent $x = m$ touches the curve at the points $y = \pm 1$, but the other two intersections of the double tangent with the curve are imaginary.

For $m = 1$, the double tangent has the two coincident real intersections $y = 0$, or it is in fact a triple tangent.

For $m < 1 > 0$, the double tangent has with the curve two real intersections, viz. they are the points where the double tangent meets the circle $x^2 + y^2 = 1$.

And finally, for $m = 0$, the points in question unite themselves with the points of contact, the double tangent $x = 0$ being in this case the common tangent at the two cusps $x = 0$, $y = \pm 1$.

Added May 13, 1867.

20. As remarked in the original memoir, p. 312, the secondary caustic, in the last-mentioned case $m = 0$, is a curve similar to and double the magnitude of the caustic itself (viz. the caustic for parallel rays reflected at a circle), the position of the two curves differing by a right angle.

The secondary caustics corresponding to the different values of m form, it is clear, a system of parallel curves; and, by the remark just referred to, it appears that this system is similar to the system of curves parallel to the caustic for parallel rays reflected at a circle.

360. Note on a Quartic Surface. [360]

(From the Philosophical Magazine, vol. XXIX. (1865), pp. 19–22.)

It would, I think, be worth while to study in detail the quartic surface which is the envelope of a sphere having its centre on a given conic, and passing through a given point.

The equations of the conic being $z = 0$, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, the coordinates of a point on the conic may be taken to be $x = a\cos\theta$, $y = b\sin\theta$, $z = 0$, whence, if (α, β, γ) be the coordinates of the given point, the equation of the sphere is

$$(x - a\cos\theta)^2 + (y - b\sin\theta)^2 + z^2 = (\alpha - a\cos\theta)^2 + (\beta - b\sin\theta)^2 + \gamma^2$$

or, what is the same thing,

$$x^2 + y^2 + z^2 - \alpha^2 - \beta^2 - \gamma^2 - 2(x - \alpha)a\cos\theta - 2(y - \beta)b\sin\theta = 0;$$

and hence the equation of the surface is at once seen to be

$$(x^2 + y^2 + z^2 - \alpha^2 - \beta^2 - \gamma^2)^2 = 4a^2(x - \alpha)^2 + 4b^2(y - \beta)^2.$$

If $a = b$ (that is, if the conic be a circle), then we may without loss of generality write $\beta = 0$, and the equation then is

$$(x^2 + y^2 + z^2 - \alpha^2 - \gamma^2)^2 = 4a^2\{(x - \alpha)^2 + y^2\}.$$

This may be written

$$(x^2 + y^2 + z^2 - \alpha^2 - \gamma^2 - 2a^2)^2 = -8a^2\{x^2 - \alpha x - \tfrac{1}{2}\alpha^2 - \tfrac{1}{2}\gamma^2 + a^2\} + 8a^2\alpha x,$$

which, considering z as a constant, is of the form

$$(x^2 + y^2 - \sigma)^2 = 16A(x - m);$$

that is, the section of the surface by a plane parallel to the plane of the conic is a *Cartesian*.

If a and b are unequal, but if we still have $\beta = 0$, the equation of the surface is

$$(x^2 + y^2 + z^2 - \alpha^2 - \gamma^2)^2 = 4a^2(x - \alpha)^2 + 4b^2y^2.$$

There are here two planes parallel to the plane of the conic, each of them meeting the surface in a pair of circles. In fact, writing $x^2 + y^2 = \rho$, and therefore also $y^2 = \rho - x^2$, putting moreover $z^2 - \alpha^2 - \gamma^2 = k$, we have

$$(\rho + k)^2 = 4a^2x^2 - 8a^2\alpha x + 4a^2\alpha^2 + 4b^2(\rho - x^2),$$

that is,

$$\rho^2 + 2(k - a^2 + b^2)x^2 + k^2 - 4a^2\alpha^2 + 8a^2\alpha x + (2k - 4b^2)\rho = 0,$$

or, as this may also be written,

$$(1,\ 4(b^2 - a^2),\ k^2 - 4a^2\alpha^2,\ 4a^2\alpha,\ 2k - 4b^2 | \rho, x, 1)^2 = 0,$$

which is of the form

$$(a, b, c, f, g|\rho, x, 1)^2 = 0;$$

and the left-hand side will break up into factors, each of the form $\rho+Ax+B$ (so that, equating either factor to zero, we have $\rho + Ax + B = 0$, that is, $x^2 + y^2 + Ax + B = 0$, the equation of a circle), if only

$$abc - af^2 - bg^2 = 0.$$

Writing this under the form $b(ac - g^2) - af^2 = 0$, and substituting for a, b, c, f, g their values, we have

$$b = 4(b^2 - a^2), \quad ac - g^2 = k^2 - 4a^2\alpha^2 - (k - 2b^2)^2 = 4(b^2k - b^4 - a^2\alpha^2),$$

and therefore the condition is

$$4(b^2 - a^2)(b^2k - b^4 - a^2\alpha^2) - 16a^4\alpha^2 = 0;$$

that is,

$$4(b^2 - a^2)(b^2k - b^4 - a^2\alpha^2) - 16a^4\alpha^2 = 0.$$

If $b^2 = 0$, the surface is a pair of spheres; rejecting this factor, we have $(b^2 - a^2)(k - b^2) - a^2\alpha^2 = 0$; or putting for k its value, the condition becomes

$$(b^2 - a^2)(z^2 - \alpha^2 - \gamma^2 - b^2) - a^2\alpha^2 = 0;$$

that is, for each of the values of z given by this equation, the section by a plane parallel to the plane of the conic will be a pair of circles.

The planes in question will coincide with the plane of the conic, if only

$$(b^2 - a^2)(\alpha^2 + \gamma^2 + b^2) + a^2\alpha^2 = 0,$$

or, what is the same thing,

$$b^2 - a^2 - (a^2 - b^2)\frac{\gamma^2}{b^2} = \frac{b^2}{a^2}(a^2 - b^2);$$

that is, if the point $(\alpha, 0, \gamma)$ be situated on the hyperbola in question $y = 0$, $\frac{x^2}{a^2-b^2} - \frac{z^2}{b^2} = 1$.

The *focal conics* of a system of confocal ellipsoids are, it is clear, conics in planes at right angles to each other, having the transverse axes coincident in direction, and being such that each curve passes through the foci of the other curve; or, what is the same thing, they are a pair of focal conics of a system of confocal ellipsoids.

The surface in the case in question, viz. when the parameters a, b, α, β are connected by the equation

$$\frac{\alpha^2}{a^2 - b^2} - \frac{\gamma^2}{b^2} = 1,$$

is in fact the "*Cyclide*" of Dupin.

It is to be noticed that we have here

$$(\alpha - a\cos\theta)^2 + \beta^2\sin^2\theta + \gamma^2 = \alpha^2 + \gamma^2 + b^2 - 2a\alpha\cos\theta + (a^2 - b^2)\cos^2\theta;$$

which, observing that $\alpha^2 + \gamma^2 + b^2 = \frac{a^2\alpha^2}{a^2-b^2}$, gives

$$(\alpha - a\cos\theta)^2 + b^2\sin^2\theta + \gamma^2 = (\sqrt{a^2-b^2}\cos\theta - \frac{a\alpha}{\sqrt{a^2-b^2}})^2;$$

so that the radius of the variable sphere is

$$= \sqrt{a^2-b^2}\cos\theta - \frac{a\alpha}{\sqrt{a^2-b^2}}.$$

If the variable sphere, instead of passing through the point $(\alpha, 0, \gamma)$ on the hyperbola, be drawn so as to touch a sphere of radius l, having its centre at the point in question, then the radius of the variable sphere would be

$$= \sqrt{a^2-b^2}\cos\theta - \frac{a\alpha'}{\sqrt{a^2-b^2}},$$

which is in fact

$$= \sqrt{a^2-b^2}\cos\theta - \frac{a\alpha'}{\sqrt{a^2-b^2}}$$

if only $\alpha' = \alpha + \frac{l\sqrt{a^2-b^2}}{a}$; hence if γ' be the corresponding value of γ, the variable sphere passes through the point $(\alpha', 0, \gamma')$ on the hyperbola, and the envelope is still a cyclide. The cyclide as derived from the foregoing investigation is thus the envelope of a sphere having its centre on the ellipse, and touching a fixed sphere having its centre on the hyperbola. It also appears that there are, having their centres on the hyperbola, an infinite series of spheres each touched by the spheres which have their centre on the ellipse; if, instead of one of these spheres we take any four of them, this will imply that the centre of the variable sphere is on the ellipse, and it is thus seen that the cyclide as obtained above is identical with the cyclide according to the original definition, viz. as the envelope of a sphere touching four given spheres.

Cambridge, December 5, 1864.

361. On Quartic Curves. [361]

(From the Philosophical Magazine, vol. XXIX. (1865), pp. 105–108.)

The expression ‘an oval’ is used, in regard to the plane, to denote a closed curve without nodes or cusps; and, in regard to the sphere, it is assumed moreover that the oval is a curve which is not its own opposite, and does not meet the opposite curve (*)—that is, that the oval is one of a pair of non-intersecting twin ovals. I say that every spherical curve of the fourth order (or spherical quartic) without nodes or cusps may be considered as composed of an oval or ovals lying wholly in one hemisphere (that is, not cutting or touching the bounding circle of the hemisphere), and of the opposite oval or ovals lying wholly in the opposite hemisphere; or, disregarding the opposite curves, that it consists of an oval or ovals lying wholly in one hemisphere. And this being so, the quartic cone having its vertex at the centre of the sphere is met by a plane parallel to that of the bounding circle in a plane quartic curve consisting of an oval or ovals; and thence every plane quartic is either a finite curve consisting of an oval or ovals, or else the projection of such a curve.

Considering first the case of the plane, a line in general meets the oval in an even number of points (the number may of course be $= 0$); hence as the point of contact of a tangent reckons for two points, the tangent at any point of the oval again intersects the oval in an even number of points (this number may of course be $= 0$). The number of points of intersection by the tangent (the point of contact being always excluded) is either evenly even, and the point is then situate on a *convex* portion of the oval; or it is oddly even, and the point is then situate on a *concave* portion of the oval.

Now imagine that the oval is (or is part of) a quartic curve; the number of points of intersection by the tangent is $= 0$ or else $= 2$; and there is at least one portion of the oval for which the number of intersections is $= 0$; for otherwise the oval would be concave at every point, which is impossible. Hence there is a tangent which does not meet the oval (except at the point of contact), and we may in the immediate neighbourhood of the tangent draw a line which does not meet the oval at all.

Precisely the same considerations apply to the case of an oval which is part of a spherical quartic, the tangent being of course a great circle; and the conclusion arrived at is that there exists a great circle which does not meet the oval at all; that is, the oval lies wholly in one hemisphere.

I remark that the demonstration would, as it ought to do, fail, if we attempted to apply it to an oval portion of a spherical sextic; the tangent circle meets the oval in a number of points which is $= 0$, 2, or 4; and the number cannot be for every tangent circle whatever $= 2$; but there is nothing to prevent it from being for every tangent circle whatever $= 2$ or 4. Hence we cannot, for every spherical sextic, obtain a tangent circle not meeting the oval except at the point of contact; and consequently we do not obtain in the immediate neighbourhood of the tangent a circle which does not meet the oval at all. And in fact such circle does not in every case exist; that is, the oval portion of a spherical sextic does not in every case lie in a hemisphere.

It has been shown that the oval portion of a spherical quartic lies in a hemisphere; but we have to consider the case where the quartic consists of two or more ovals. To fix the ideas, let A, A' be a pair of opposite ovals, and B, B' another pair of opposite ovals, components of the same spherical quartic. If there exists a tangent circle of A which does not meet B, then there exists in the immediate neighbourhood of the tangent circle a circle which does not meet either A or B; and we may assume that A and B lie on the same side of this circle; for if B were on the side opposite to A, then B' would be on the same side with A; and we have only, instead of B, to consider the opposite oval B'. Hence we may consider that the ovals A and B lie on the same side of the circle; that is, we have a spherical quartic consisting of or comprising the ovals A and B in the same hemisphere: the two ovals are, it is clear, external each to the other.

But every tangent of A may meet B in two points; consider the whole spherical figure, and suppose that the tangent (or say, the tangent circle) of A, A' meets the ovals B, B' in the points K, L and the opposite points K', L': then considering the tangent circle as moving round A, A' until it returns to its original position, the points K, L, K', L' are always four distinct points; and K and some one (say L) of the two points L, L' will describe the same oval, say the oval B; while the opposite points K', L' will describe the opposite oval B'. We have here the oval A included in the oval B (and of course the opposite oval A' included in the opposite oval B'). But the oval B, quà portion of a spherical quartic, lies wholly in one hemisphere; hence the two ovals A, B lie wholly in one hemisphere. It is easy to see that there is not in this case any other portion of the spherical quartic, but that the two ovals A, B are the entire curve.

Reverting to the case where we have in one hemisphere the two ovals A, B external to each other, the spherical quartic may comprise as part of itself another oval C. The ovals A and B, quà ovals external to each other, have a common tangent circle (a double tangent of the spherical quartic) which cannot meet the oval C (for if it did we should have six points of intersection); hence in the immediate neighbourhood thereof we have a circle not meeting any one of the ovals A, B, C. We may consider A, B, C as lying on the same side of this circle; for if B were on the opposite side to A, then B' would be on the same side; and so if C be on the opposite side, then C' will be on the same side; that is, we have the three ovals A, B, C external to each other, and in the same hemisphere.

There may be a fourth oval, D, and it would be shown in a similar manner that we have then the four ovals A, B, C, D external to each other and in the same hemisphere. But there cannot be a fifth oval, E; the proof is precisely the same as for the theorem *in piano*; viz. taking within each of the five ovals a point, and through these points drawing a conic, the conic would meet each oval in two points, and therefore the plane quartic in ten points, which is impossible.

Passing from the sphere to the plane, the foregoing investigation shows that every plane quartic without nodes or cusps is either a finite curve, or else the projection of a finite curve, of one of the following forms:

1. a single oval.
2. two ovals external to each other.
3. two ovals, one inside the other.
4. three ovals external to each other.
5. four ovals external to each other.

The last case has been called (5, 6) for the sake of the following subdivision, viz.:

5. the four ovals are so situate as to be intersected, each in two points, by the same ellipse.
6. they are so situate as not to be intersected by any one ellipse whatever — the distinction being similar to that which exists between four points, which may be either such as to have passing through them as well ellipses as hyperbolas, or else to have passing through them hyperbolas only.

I remark that the limitation of the theorem to the case of a quartic curve without nodes or cusps is necessary, at any rate as regards the nodes. We may in fact find a quartic curve having a single node which is met by every line in at least two real points, and which is therefore not the projection of any finite curve; for if we imagine two hyperbolas so situate that each branch of the one cuts each branch of the other, then it may be seen that there exists a quartic curve approaching everywhere very nearly to the system of two hyperbolas, but having, instead of the four nodes of the system, only a single node, which is such that every line meets it in at least two points.

Cambridge, December 15, 1864.

(*) The notions of opposite curves, &c. are fully developed in the excellent Memoir of Möbius, "Ueber die Grundformen der Linien der dritter Ordnung," Abh. der K. Sachs. Ges. zu Leipzig, vol. I. (1852), to which I have elsewhere frequently referred.

362. Note on Lobatschewsky's Imaginary Geometry. [362]

(From the Philosophical Magazine, vol. XXIX. (1865), pp. 231–233.)

Writing down the equations

$$\frac{1}{\cos a} = \cos a' = \frac{\cos A + \cos B \cos C}{\sin B \sin C},$$
$$\frac{1}{\cos b} = \cos b' = \frac{\cos B + \cos C \cos A}{\sin C \sin A},$$
$$\frac{1}{\cos c} = \cos c' = \frac{\cos C + \cos A \cos B}{\sin A \sin B},$$

where A, B, C are real positive angles each $< \frac{1}{2}\pi$: first, if $A + B + C > \pi$, then a, b, c are real positive angles each less than $\frac{1}{2}\pi$ (this is in fact the case of a real acute-angled spherical triangle), but a', b', c' are pure imaginaries of the form $p'i$, $q'i$, $r'i$ (where p', q', r' are real positive quantities); and secondly, if $A + B + C < \pi$, then a, b, c are pure imaginaries of the form pi, qi, ri (where p, q, r are real positive quantities), but a', b', c' are real positive angles each less than $\frac{1}{2}\pi$. Hence assuming $A + B + C < \pi$ and writing ai, bi, ci in place of a, b, c, the system is

$$\frac{1}{\cos ai} = \cos a' = \frac{\cos A + \cos B \cos C}{\sin B \sin C},$$
$$\frac{1}{\cos bi} = \cos b' = \frac{\cos B + \cos C \cos A}{\sin C \sin A},$$
$$\frac{1}{\cos ci} = \cos c' = \frac{\cos C + \cos A \cos B}{\sin A \sin B},$$

which equations (if only we write therein $\frac{1}{2}\pi - a'$, $\frac{1}{2}\pi - b'$, $\frac{1}{2}\pi - c'$ in place of a', b', c' respectively) are in fact the equations given under a less symmetrical form in the curious paper "Géométrie Imaginaire" by N. Lobatschewsky, Rector of the University of Kasan, Crelle, vol. XVII. (1837), pp. 295–320. The view taken of them by the author is hard to be understood. He mentions that in a paper published five years previously in a scientific journal at Kasan, after developing a new theory of parallels, he had endeavoured to prove that it is only experience which obliges us to assume that in a rectilinear triangle the sum of the angles is equal to two right angles, and that a geometry may exist, if not in nature at least in analysis, on the hypothesis that the sum of the angles is less than two right angles; and he accordingly attempts to establish such a geometry, viz. a, b, c being the sides of a rectilinear triangle, wherein the sum of the angles $A + B + C$ is $< \pi$, and the angles a', b', c' being calculated from the sides by the formulae

$$\frac{1}{\cos a} = \cos ai, \qquad \frac{1}{\cos b} = \cos bi, \qquad \frac{1}{\cos c} = \cos ci,$$

(I have, as mentioned above, replaced Lobatschewsky's a', b', c' by their complements): the relation between the angles A, B, C and the subsidiary quantities a', b', c' which replace the sides, is given by the formulae

$$\cos a' = \frac{\cos A + \cos B \cos C}{\sin B \sin C},$$

$$\cos b' = \frac{\cos B + \cos C \cos A}{\sin C \sin A},$$

$$\cos c' = \frac{\cos C + \cos A \cos B}{\sin A \sin B}.$$

I do not understand this; but it would be very interesting to find a real geometrical interpretation of the last-mentioned system of equations, which (if only A, B, C are positive real quantities such that $A + B + C < \pi$; for the condition, A, B, C each $< \frac{1}{2}\pi$, may be omitted) contains only the real quantities A, B, C, a', b', c'; and is a system correlative to the equations of ordinary Spherical Trigonometry.

It is hardly necessary to remark that the equation

$$\frac{1}{\cos a} = \cos ai$$

is Jacobi's imaginary transformation in the Theory of Elliptic Functions. See, as to this, my paper "On the Transcendent gd.$u = -\log\tan(\frac{1}{4}\pi + \frac{1}{2}ui)$," Phil. Mag. vol. XXIV. (1862), pp. 19–22, [320].

Cambridge, January 21, 1865.

363. On the Theory of the Evolute. [363]

(From the Philosophical Magazine, vol. XXIX. (1865), pp. 344–350.)

According to the generalized notion of geometrical magnitude, two lines are said to be at right angles to each other when they are harmonics in regard to a certain conic called the *Absolute*; this being so, the normal at any point of a curve is the line at right angles to the tangent, and the *Evolute* is the envelope of the normals.

Let the equation of the absolute be

$$\Theta = (a, b, c, f, g, h | x, y, z)^2 = 0,$$

and suppose, as usual, that the inverse coefficients are (A, B, C, F, G, H). Consider a given curve $U = (a, b, c, \dots | x, y, z)^m = 0$, and suppose, for shortness, that the first differential coefficients of U are denoted by L, M, N. Then we have to find the equation of the normal at the point (x, y, z) of the curve $U = 0$.

The condition that any two lines are harmonics in regard to the absolute, is equivalent to this, viz. each line passes through the pole of the other line in regard to the absolute. Hence the normal at the point (x, y, z) is the line joining this point with the pole of the tangent.

Now, taking (X, Y, Z) as current coordinates, the equation of the tangent is

$$LX + MY + NZ = 0,$$

the coordinates of the pole of the tangent are therefore

$$(A, H, G|L, M, N) : (H, B, F|L, M, N) : (G, F, C|L, M, N),$$

and the equation of the normal is

$$\begin{vmatrix} X, & Y, & Z \\ x, & y, & z \\ (A, H, G|L, M, N), & (H, B, F|L, M, N), & (G, F, C|L, M, N) \end{vmatrix} = 0.$$

The formula in this form will be convenient in the sequel; but there is no real loss of generality in taking the equation of the absolute to be $x^2 + y^2 + z^2 = 0$; the values of (A, B, C, F, G, H) are then $(1, 1, 1, 0, 0, 0)$, and the formula becomes

$$\begin{vmatrix} X, & Y, & Z \\ x, & y, & z \\ L, & M, & N \end{vmatrix} = 0;$$

where it will be remembered that (L, M, N) denote the derived functions $(\partial_x U, \partial_y U, \partial_z U)$.

The evolute is therefore the envelope of the line represented by the foregoing equation, say the equation $\Omega = 0$, considering therein (x, y, z) as variable parameters connected by the equation $U = 0$.

As an example, let it be required to find the evolute of a conic; since the axes are arbitrary, we may without loss of generality assume that the equation of the conic is $xz - y^2 = 0$. The values of (L, M, N) here are $(z, -2y, x)$. Moreover the equation is satisfied by writing therein $x : y : z = 1 : \theta : \theta^2$; the values of (L, M, N) then become $(\theta^2, -2\theta, 1)$ and the equation is

$$\begin{vmatrix} X, & Y, & Z \\ 1, & \theta, & \theta^2 \\ \theta^2, & -2\theta, & 1 \end{vmatrix} = 0,$$

or, developing, this is

$$\begin{aligned} X(G\theta^4 - 2F\theta^3 + C\theta^2) \\ + Y(H\theta^4 + 2B\theta^3 - F\theta^2 - C\theta) \\ + Z(A\theta^4 - 2H\theta^3 + G\theta^2 + 2F\theta - C) = 0, \end{aligned}$$

which I leave in this form in order to show the origin of the different terms, and in particular in order to exhibit the destruction of the term θ^3 in the coefficient of Y.

But the equation is, it will be observed, a quartic equation in θ, with coefficients which are linear functions of the current coordinates (X, Y, Z).

The equation shows at once that the evolute is of the class 4; in fact treating the coordinates (X, Y, Z) as given quantities, we have for the determination of θ an equation of the order 4, that is, the number of normals through a given point (X, Y, Z), or, what is the same thing, the class of the evolute, is $= 4$.

The equation of the evolute is obtained by equating to zero the discriminant of the foregoing quartic function of θ; the order of the evolute is thus $= 6$. There are no inflexions, and the diminution of the order from $4 \cdot 3 = 12$ to 6 is caused by three double tangents.

371.

ON A FORMULA FOR THE INTERSECTIONS OF A LINE AND CONIC, &c.

AND ON AN INTEGRAL FORMULA CONNECTED THEREWITH.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. vii. (1866), pp. 1—6.]

In a letter to me, dated 15 May, 1862, Mr Spottiswoode has extracted from an unpublished Memoir, and he has kindly permitted me to communicate, the following

[The opening portion of this paper, on pp. 499–500, leads to the equations given below.]

equivalent of course to two equations, and giving by the elimination of (x, y, z), the equation

$$\theta\,[-(A, \dots)(\xi, \eta, \zeta)^2 - \theta^2] = 0,$$

that is, giving for θ the foregoing value. And the linear equations then give

$$\begin{array}{lll}
x : y & : z, & \\
= \xi \dfrac{d\theta}{d\xi} + g\eta - h\zeta + \theta & : \xi \dfrac{d\theta}{d\eta} + a\zeta - g\xi & : \xi \dfrac{d\theta}{d\zeta} + h\xi - a\eta, \\
= \eta \dfrac{d\theta}{d\xi} + f\zeta - b\xi & : \eta \dfrac{d\theta}{d\eta} + h\xi - f\eta + \theta & : \eta \dfrac{d\theta}{d\zeta} + b\eta - h\zeta, \\
= \zeta \dfrac{d\theta}{d\xi} + c\xi - f\eta & : \zeta \dfrac{d\theta}{d\eta} + g\eta - c\zeta & : \zeta \dfrac{d\theta}{d\zeta} + f\zeta - g\xi + \theta,
\end{array}$$

where obviously

$$-\theta \frac{d\theta}{d\xi} = A\xi + H\eta + G\zeta, \quad -\theta \frac{d\theta}{d\eta} = H\xi + B\eta + F\zeta, \quad -\theta \frac{d\theta}{d\zeta} = G\xi + F\eta + C\zeta.$$

By changing the sign of θ, we have of course the coordinates of the other point of intersection. The formulae which, singularly enough, have since been given incidentally by M. Aronhold[9], may be easily obtained as follows.

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[9] In his interesting Memoir "Ueber eine neue algebraische Behandlungsweise der Integrale irrationaler Differentiale von der Form $\Pi\,(x, y)\,dx$, in welcher $\Pi\,(x, y)$ eine beliebige rationale Function ist, und zwischen x und y eine allgemeine Gleichung zweiter Ordnung besteht." *Crelle*, t. LXI. (1862).

Writing for shortness

$$\begin{aligned} P &= ax + hy + gz, \\ Q &= hx + by + fz, \\ R &= gx + fy + cz, \end{aligned}$$

then the equation of the conic gives

$$Px + Qy + Rz = 0,$$

and combining with this the equation

$$\xi x + \eta y + \zeta z = 0,$$

we have

$$x : y : z = Q\zeta - R\eta : R\xi - P\zeta : P\eta - Q\xi,$$

or what is the same thing, taking an indeterminate multiplier θ,

$$\begin{aligned} -\theta x + R\eta - Q\zeta &= 0, \\ -\theta y + P\zeta - R\xi &= 0, \\ -\theta z + Q\xi - P\eta &= 0, \end{aligned}$$

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which are, in fact, Mr Spottiswoode's linear equations, and which lead, as before, to the value

$$\theta^2 = -(A, \dots)(\xi, \eta, \zeta)^2.$$

But this value of θ is obtained in a different manner by expressing (x, y, z) as linear functions of P, Q, R; viz. putting as usual $K = abc - af^2 - bg^2 - ch^2 + 2fgh$, the linear equations thus become

$$\begin{aligned} AP + HQ + GR + \frac{K}{\theta}(\zeta Q - \eta R) &= 0, \\ HP + BQ + FR + \frac{K}{\theta}(\xi R - \zeta P) &= 0, \\ GP + FQ + CR + \frac{K}{\theta}(\eta P - \xi Q) &= 0, \end{aligned}$$

or eliminating (P, Q, R), we have

$$\begin{vmatrix} A \quad , & H + \dfrac{K\zeta}{\theta}, & G - \dfrac{K\eta}{\theta} \\ H - \dfrac{K\zeta}{\theta}, & B \quad , & F + \dfrac{K\xi}{\theta} \\ G + \dfrac{K\eta}{\theta}, & F - \dfrac{K\xi}{\theta}, & C \end{vmatrix} = 0,$$

that is

$$\begin{aligned} ABC - A\left(F^2 - \frac{K^2\xi^2}{\theta^2}\right) - B\left(G^2 - \frac{K^2\eta^2}{\theta^2}\right) - C\left(H^2 - \frac{K^2\zeta^2}{\theta^2}\right) \\ + \left(F + \frac{K\xi}{\theta}\right)\left(G + \frac{K\eta}{\theta}\right)\left(H + \frac{K\zeta}{\theta}\right) \\ + \left(F - \frac{K\xi}{\theta}\right)\left(G - \frac{K\eta}{\theta}\right)\left(H - \frac{K\zeta}{\theta}\right) = 0; \end{aligned}$$

or, reducing,

$$ABC - AF^2 - BG^2 - CH^2 + 2FGH + \frac{K^2}{\theta^2}(A, \dots)(\xi, \eta, \zeta)^2 = 0,$$

that is

$$\theta^2 + (A, \dots)(\xi, \eta, \zeta)^2 = 0,$$

as before.

I reproduce, as follows, a fundamental formula of Aronhold's Memoir. Consider the function

$$\varpi = \frac{1}{\sqrt{\{-(A, \dots)(u, v, w)^2\}}} \log \frac{(a, \dots)(x_1, y_1, z_1)(x, y, z)}{ux + vy + wz},$$

where x_1, y_1, z_1 (corresponding to x, y, z in the former part of this paper) are determined by the conditions

$$\begin{aligned}(a, \ldots)(x_1, y_1, z_1)^2 &= 0,\\ ux_1 + vy_1 + wz_1 &= 0,\end{aligned}$$

so that putting

$$\theta = \sqrt{\{-(A, \dots)(u, v, w)^2\}},$$

and

$$\begin{aligned} P_1 &= ax_1 + hy_1 + gz_1, \\ Q_1 &= hx_1 + by_1 + fz_1, \\ R_1 &= gx_1 + fy_1 + cz_1, \end{aligned}$$

we now have

$$\begin{aligned} -\theta x_1 + R_1 v - Q_1 w = 0, \\ -\theta y_1 + P_1 w - R_1 u = 0, \\ -\theta z_1 + Q_1 u - P_1 v = 0, \end{aligned}$$

and the value of ϖ is

$$\varpi = \frac{1}{\theta} \log \frac{P_1 x + Q_1 y + R_1 z}{ux + vy + wz} .$$

Treating (x, y, z) as independent variables and differentiating, we have

$$\begin{aligned} d\varpi &= \frac{1}{\theta} \left\{ \frac{P_1\,dx + Q_1\,dy + R_1\,dz}{P_1 x + Q_1 y + R_1 z} - \frac{u\,dx + v\,dy + w\,dz}{ux + vy + wz} \right\} \\ &= \frac{1}{\theta}\, \frac{(x\,dy - y\,dx)\,(Q_1 u - P_1 v) + (y\,dz - z\,dy)\,(R_1 v - Q_1 w) + (z\,dx - x\,dz)\,(P_1 w - R_1 u}{(P_1 x + Q_1 y + R_1 z)\,(ux + vy + wz)} \\ &= \frac{x_1\,(y\,dz - z\,dy) + y_1\,(z\,dx - x\,dz) + z_1\,(x\,dy - y\,dx)}{(P_1 x + Q_1 y + R_1 z)\,(ux + vy + wz)} , \end{aligned}$$

or, what is the same thing, if

$$\begin{aligned} P &= ax + hy + gz, \\ Q &= hx + by + fz, \\ R &= gx + fy + cz, \end{aligned}$$

so that

$$P_1 x + Q_1 y + R_1 z = (a, \dots)(x, y, z)(x_1, y_1, z_1) = Px_1 + Qy_1 + Rz_1,$$

then we have

$$d\varpi = \frac{(y\,dz - z\,dy)\,x_1 + (z\,dx - x\,dz)\,y_1 + (x\,dy - y\,dx)\,z_1}{(ux + vy + wz)\,(Px_1 + Qy_1 + Rz_1)} .$$

Suppose now that (x, y, z) are connected by the equation

$$(a, \dots)(x, y, z)^2 = 0,$$

we have

$$\begin{aligned} Px + Qy + Rz &= 0, \\ P\,dx + Q\,dy + R\,dz &= 0, \end{aligned}$$

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and thence

$$\begin{aligned} y\,dz - z\,dy &= \theta P, \\ z\,dx - x\,dz &= \theta Q, \\ x\,dy - y\,dx &= \theta R, \end{aligned}$$

and consequently

$$d\varpi = \frac{\theta}{ux+vy+wz} = \frac{y\,dz - z\,dy}{(ux+vy+wz)\,P} = \frac{z\,dx - x\,dz}{(ux+vy+wz)\,Q} = \frac{x\,dy - y\,dx}{(ux+vy+wz)\,R}\,;$$

or selecting the value

$$d\varpi = \frac{z\,dx - x\,dz}{(ux+vy+wz)\,Q} = \frac{z\,dx - x\,dz}{(ux+vy+wz)\,(hx+by+fz)},$$

and writing

$$\frac{x}{z} = X, \quad \frac{y}{z} = Y,$$

we have

$$\begin{aligned} d\varpi &= \frac{z^2\,dX}{(ux+vy+wz)\,(hx+by+fz)} \\ &= \frac{dX}{(uX+vY+w)\,(hX+bY+f)}\,, \end{aligned}$$

where X and Y are connected by the equation

$$(a, \dots)(X, Y, 1)^2 = 0,$$

that is, Y is a given quadric radical function of X. Hence integrating and restoring for ϖ its original value, but writing therein $\frac{x}{z} = X$ and $\frac{y}{z} = Y$, we have

$$\int \frac{dX}{(uX+vY+w)\,(hX+bY+f)} = \frac{1}{\sqrt{\{-(A, \dots)(u, v, w)^2\}}} \log \frac{(a, \dots)(x_1, y_1, z_1)(X, Y, 1)}{uX+vY+w}$$

where, as just mentioned, Y is a given quadric radical function of X determined by the equation

$$(a, b, c, f, g, h)(X, Y, 1)^2 = 0,$$

and the constants x_1, y_1, z_1 are such that

$$\begin{aligned} (a, \dots)(x_1, y_1, z_1)^2 &= 0, \\ ux_1 + vy_1 + wz_1 &= 0, \end{aligned}$$

the ratios of these quantities being therefore determinate; there would, it is clear, be no loss of generality in assuming $z_1 = 1$. This is Aronhold's Theorem I.

2, Stone Buildings, W.C., October 23, 1862.

372.

ON THE RECIPROCATION OF A QUARTIC DEVELOPABLE.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. vii. (1866), pp. 87—92.]

It is interesting to consider in a particular case the system of equations which shows *à posteriori* that the reciprocal of a torse (developable surface) is a curve (curve in space); and the reciprocal system which shows that the reciprocal of a curve is a torse.

Using (a, b, c, d) and (x, y, z, w) for the reciprocal coordinates, it will be convenient to collect the different equations as follows:

$$a^2d^2 - 6abcd + 4ac^3 + 4b^3d - 3b^2c^2 = 0, \tag{10}$$

$$\left.\begin{aligned} ad^2 - 3\,bcd + 2\,c^3 + \lambda x &= 0, \\ -3\,acd + 6\,b^2d - 3\,bc^2 + \lambda y &= 0, \\ -3\,abd + 6\,ac^2 - 3\,b^2c - \lambda z &= 0, \\ a^2d - 3\,abc + 2\,b^3 + \lambda w &= 0, \end{aligned}\right\} \tag{11}$$

$$ax + by + cz + dw = 0, \tag{12}$$

$$3xz - y^2 = 0, \quad yz - 9xw = 0, \quad 3yw - z^2 = 0, \tag{13}$$

$$\left.\begin{aligned} 3\,pz - 9\,qw \qquad\quad + a &= 0, \\ -2\,py + qz + 3\,rw + b &= 0, \\ 3\,px + qy - 2\,rz + c &= 0, \\ -9\,qx + 3\,ry \qquad\quad + d &= 0, \end{aligned}\right\} \tag{14}$$

This being so, the equation (1) belongs to a quartic torse, the reciprocal whereof is the skew cubic determined by the equations (4): and we have to show *à posteriori* that this is so.

First if the torse is given, then the reciprocal figure is the envelope of the plane $ax + by + cz + dw = 0$, (3), where (x, y, z, w) are the coordinates and (a, b, c, d) are regarded as variable parameters connected by the equation (1); we thence obtain the equations (2), where λ is an arbitrary multiplier; and from the equations (1), (2), and (3), we have to eliminate a, b, c, d, λ. The equation (3) is at once seen to be included in the equations (1) and (2); and the elimination would give only a single equation between the (x, y, z, w)—since however the equation (1) is that of a torse, we know that the elimination must give two equations, or more accurately a two-fold relation (represented, as in the present case it happens, by three equations) between the coordinates (x, y, z, w).

Putting for shortness

$$\Box = a^2d^2 - 6abcd + 4ac^3 + 4b^3d - 3b^2c^2 :$$

and substituting for λx, λy, λz, λw, their values from the equations (1), we have identically

$$\begin{aligned} \lambda^2(\,xz - \tfrac{1}{3}\,y^2\,) &= -(bd - c^2)\,\Box, \\ \lambda^2(\tfrac{1}{3}\,yz -\; xw) &= -(ad - bc)\,\Box, \\ \lambda^2(\,yw - \tfrac{1}{3}\,z^2\,) &= -(ac - b^2)\,\Box; \end{aligned}$$

and hence (since $\Box = 0$) we have between (x, y, z, w) the equations (4), showing that the reciprocal figure is a skew cubic.

Secondly, let the skew cubic be given; then the reciprocal figure is the envelope of the plane $ax+by+cz+dw = 0$, (3), where now (a, b, c, d) are the coordinates and (x, y, z, w) are regarded as variable parameters connected by the equations (4): we thence obtain the equations (5) containing the arbitrary multipliers p, q, r; and from the equations (3), (4), (5) we have to eliminate x, y, z, w, p, q, r. The equation (3) is at once seen to be included in the equations (4) and (5): since however the equations (4) are those of a curve, we know that the elimination must give a single equation between (a, b, c, d).

The equations (4) are satisfied if we write therein

$$x : y : z : w = \tfrac{1}{3} : \theta : \theta^2 : \tfrac{1}{3}\,\theta^3,$$

and substituting these values of x, y, z, w in the equations (5), these equations give (a, b, c, d) in terms of θ, p, q, r, and we thence find identically

$$\begin{aligned} ac - b^2 &= -\,\theta^2\,(p - 2q\theta + r\theta^2)^2, \\ bc - ad &= -2\theta\,(p - 2q\theta + r\theta^2)^2, \\ bd - c^2 &= -\quad\;(p - 2q\theta + r\theta^2)^2, \end{aligned}$$

that is

$$ac - b^2 : ad - bc : bd - c^2 = \theta^2 : -2\theta : 1,$$

or we have

$$(ad-bc)^2-4\,(ac-b^2)\,(bd-c^2)=0,$$

which is in fact the equation

$$\square=a^2d^2-6abcd+4ac^3+4b^3d-3b^2c^2=0, \qquad (1)$$

and thus the reciprocal is in fact the quartic torse, given by the equation (1).

The equations (3), (4), (5) lead to

$$x\,\delta a+y\,\delta b+z\,\delta c+w\,\delta d=0;$$

but (a, b, c, d) are in consequence of these equations connected by a single equation, viz. the equation (1); the equation just obtained is thus the only relation between the differentials $(\delta a, \delta b, \delta c, \delta d)$: it is clear from the equations (2) that this is in fact the equation

$$\delta\square=\delta_a\square\,\delta a+\delta_b\square\,\delta b+\delta_c\square\,\delta c+\delta_d\square\,\delta d=0.$$

The equations (1), (2), (3) lead in like manner to

$$a\,\delta x+b\,\delta y+c\,\delta z+d\,\delta w=0,$$

which in virtue of the equations (5) is equivalent to

$$p\,\delta\,(3xz-y^2)+q\,\delta\,(yz-9xw)+r\,\delta\,(3yw-z^2)=0,$$

viz. it is a consequence of

$$\delta\,(3xz-y^2)=0, \quad \delta\,(yz-9xw)=0, \quad \delta\,(3yw-z^2)=0.$$

But inasmuch as (x, y, z, w) are connected by a two-fold relation, the equations (1), (2), (3) must lead to one other linear relation between the differentials $(\delta x, \delta y, \delta z, \delta w)$: and I proceed to show that this is so.

Differentiating we have

$$\begin{array}{l}
d^2\,\delta a \quad -3cd\,\delta b+(-3bd+6c^2)\,\delta c+(\,2ad-3bc)\,\delta d+\lambda\,dx+x \\
-3cd\,\delta a+(\,12bd-3c^2)\,\delta b+(-3ad-6bc)\,\delta c+(-3ac-6b^2)\,\delta d+\lambda\,dy+y \\
(-3bd+6c^2)\,\delta a+(-3ad-6bc)\,\delta b+(\,12ac-3b^2)\,\delta c- \qquad 3ab\,\delta d+\lambda\,dz+z \\
(\,2ad-3bc)\,\delta a+(-3ac+6b^2)\,\delta b- \qquad 3ab\,\delta c+ \qquad a^2\,\delta d+\lambda\,dw+w
\end{array}$$

Consider the matrix

$$\left|\begin{array}{cccc}
d^2 \quad , & -3cd \quad , & -3bd+6c^2, & 2ad-3bc \\
-3cd \quad , & 12bd-3c^2, & -3ad-6bc, & -3ac+6b^2 \\
-3bd+6c^2, & -3ad-6bc, & 12ac-3b^2, & -3ab \\
2ad-3bc, & -3ac+6b^2, & -3ab \quad , & a^2
\end{array}\right|$$

and suppose for a moment that the determinant formed therewith is $= K$; suppose also that the reciprocal matrix is

$$\begin{vmatrix} \mathfrak{A}, & \mathfrak{H}, & \mathfrak{G}, & \mathfrak{L} \\ \mathfrak{H}, & \mathfrak{B}, & \mathfrak{F}, & \mathfrak{M} \\ \mathfrak{G}, & \mathfrak{F}, & \mathfrak{C}, & \mathfrak{N} \\ \mathfrak{L}, & \mathfrak{M}, & \mathfrak{N}, & \mathfrak{D} \end{vmatrix}.$$

Then we have

$$K\,\delta a + d\lambda\,(\mathfrak{A}x + \mathfrak{H}y + \mathfrak{G}z + \mathfrak{L}w) + \lambda\,(\mathfrak{A}\,dx + \mathfrak{H}\,dy + \mathfrak{G}\,dz + \mathfrak{L}\,dw) = 0,$$

with the similar equations involving b, c, d and $(\mathfrak{H}, \mathfrak{B}, \mathfrak{F}, \mathfrak{M})$, $(\mathfrak{G}, \mathfrak{F}, \mathfrak{C}, \mathfrak{N})$, $(\mathfrak{L}, \mathfrak{M}, \mathfrak{N}, \mathfrak{D})$ respectively.

But, substituting for (x, y, z, w) their values from the equations (2), it is easy to see that we have

$$\mathfrak{A}x + \mathfrak{H}y + \mathfrak{G}z + \mathfrak{L}w = -\frac{1}{3\lambda}\,Ka,$$

the last-mentioned equation thus is

$$K\left(\delta a - \tfrac{1}{3}\,\frac{a}{\lambda}\,\delta\lambda\right) + \lambda\,(\mathfrak{A}\,dx + \mathfrak{H}\,dy + \mathfrak{G}\,dz + \mathfrak{L}\,dw) = 0.$$

But K is $= 27\square$ (see my paper "On Certain Developable Surfaces," *Quarterly Mathematical Journal*, t. VI. 1864, pp. 108—126, [344]), which is $= 0$, and we thus have

$$\mathfrak{A}\,dx + \mathfrak{H}\,dy + \mathfrak{G}\,dz + \mathfrak{L}\,dw = 0,$$

and similarly

$$\begin{aligned} \mathfrak{H}\,dx + \mathfrak{B}\,dy + \mathfrak{F}\,dz + \mathfrak{M}\,dw &= 0, \\ \mathfrak{G}\,dx + \mathfrak{F}\,dy + \mathfrak{C}\,dz + \mathfrak{N}\,dw &= 0, \\ \mathfrak{L}\,dx + \mathfrak{M}\,dy + \mathfrak{N}\,dz + \mathfrak{D}\,dw &= 0. \end{aligned}$$

But observing that in virtue of the equation $K = 0$, we have

$$\mathfrak{A} : \mathfrak{H} : \mathfrak{G} : \mathfrak{L} = \mathfrak{H} : \mathfrak{B} : \mathfrak{F} : \mathfrak{M} = \mathfrak{G} : \mathfrak{F} : \mathfrak{C} : \mathfrak{N} = \mathfrak{L} : \mathfrak{M} : \mathfrak{N} : \mathfrak{D},$$

these are, as they should be, one and the same equation.

The values of the coefficients $\mathfrak{A}$, $\mathfrak{B}$, &c. are given, p. 112 of the paper just referred to, viz. writing $\square$ in place of U, we have $\mathfrak{A} = 3X^2 - 4a^2\square$, &c.

where

$$\begin{aligned}X &= a^2d - 3abc + 2b^3,\\ Y &= abd - 2ac^2 + b^2c,\\ Z &= -acd + 2b^2d - bc^2,\\ W &= -ad^2 + 3bcd - 2c^3,\end{aligned}$$

and writing $\square = 0$, the values each divided by 3, are simply

$$\begin{array}{l} X^2,\ XY,\ XZ, XW, \\ YX,\ \ Y^2,\ YZ, YW, \\ ZX,\ ZY,\ \ Z^2, ZW, \\ WX, WY, WZ,\ W^2, \end{array}$$

so that each of the four equations in fact, becomes

$$X\,\delta x + Y\,\delta y + Z\,\delta z + W\,\delta w = 0,$$

or multiplying by 3, and attending to the equations (2), this is

$$-3w\,\delta x + z\,\delta y - y\,\delta z + 3x\,\delta w = 0.$$

This should be a consequence of the equations

$$\delta\,(3xz - y^2) = 0, \quad \delta\,(yz - 9xw) = 0, \quad \delta\,(3yw - z^2) = 0,$$

that is, we should be able from the first three to deduce the fourth equation in the system

$$\begin{aligned} 3z\,\delta x - 2y\,\delta y + 3x\,\delta z \qquad\quad &= 0, \\ -9w\,\delta x +\ z\,\delta y +\ y\,\delta z - 9x\,\delta w &= 0, \\ 3w\,\delta y - 2z\,\delta z + 3y\,\delta w \qquad\quad &= 0, \\ -3w\,\delta x +\ z\,\delta y -\ y\,\delta z + 3x\,\delta w &= 0, \end{aligned}$$

or we ought to have

$$\begin{vmatrix} 3z, & -2y, & 3x, & \\ -9w, & z, & y, & -9x \\ & 3w, & -2z, & 3y \\ -3w, & z, & -y, & 3x \end{vmatrix} = 0;$$

but expanding, this is

$$-6\,(81x^2w^2 - 54xyzw + 12xz^3 + 12y^3w - 3y^2z^2) = 0,$$

or

$$(yz - 9xw)^2 - 4\,(xz - y^2)\,(yw - z^2) = 0,$$

which is true in virtue of the relations (4). Or what is the same thing, we may show without difficulty that the equation

$$-3w\,\delta x + z\,\delta y - y\,\delta z + 3x\,\delta w = 0,$$

is satisfied by writing therein $x : y : z : w = \frac{1}{3} : \theta : \theta^2 : \frac{1}{3}\,\theta^3$.

510 ON THE RECIPROCATION OF A QUARTIC DEVELOPABLE. [372

I remark that in general, if $\square = \phi\,(a, b, c, d) = 0$ is the equation of a torse, then for finding the reciprocal curve, we have

$$\begin{aligned} \square = 0, \quad & ax + by + cz + dw = 0, \\ & \delta_a \square + \lambda x = 0, \\ & \delta_b \square + \lambda y = 0, \\ & \delta_c \square + \lambda z = 0, \\ & \delta_d \square + \lambda w = 0, \end{aligned}$$

and that from these equations we deduce not only

$$a\,\delta x + b\,\delta y + c\,\delta z + d\,\delta w = 0,$$

but also the equation

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}, \mathfrak{L}, \mathfrak{M}, \mathfrak{N})(\alpha, \beta, \gamma, \delta)(\delta x, \delta y, \delta z, \delta w) = 0,$$

where $\mathfrak{A}, \ldots$ are the inverse system derived from the second differential coefficients of $\square$; $(\alpha, \beta, \gamma, \delta)$ are arbitrary coefficients, introduced only for symmetry, and there is no real loss of generality in reducing all but one of them to zero, and so reducing the equation for example to the form

$$\mathfrak{A}\,\delta x + \mathfrak{H}\,\delta y + \mathfrak{G}\,\delta z + \mathfrak{L}\,\delta w = 0.$$

The existence of the two linear equations between $(\delta x, \delta y, \delta z, \delta w)$ proves that (x, y, z, w) are connected by a two-fold relation, that is, that the reciprocal of the given torse is a curve.

Cambridge, January 26, 1865.

373.

ON A SPECIAL SEXTIC DEVELOPABLE.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. vii. (1866), pp. 105—113.]

The present paper contains some investigations in relation to the special sextic developable or torse

$$(ae-4bd)^3-27(-ad^2-b^2e)^2=0,$$

considered Nos. 26 to 35 of my paper "On Certain Developable Surfaces," Quarterly Mathematical Journal, t. VI. (1864), pp. 108—126, [344].

The cuspidal curve is

$$ae-4bd=0,\quad ad^2+b^2e=0,$$

and the nodal curve is

$$ae+2bd=0,\quad ad^2-b^2e=0,$$

viz. to put this in evidence, the equation is to be written in the form

$$(ae+4bd)^2(ae-4bd)-27(ad^2+b^2e)^2=0.$$

The coordinates of a point on the cuspidal curve may be taken to be

$$a=2,\quad b=-t,\quad d=t^2,\quad e=-2t^3,$$

and then if A, B, D, E are current coordinates, and α, β, δ, ϵ arbitrary coefficients, the equation of a plane through the tangent line is

$$\begin{vmatrix} A, & B, & D, & E \\ 2, & -t, & +t^2, & -2t^3 \\ 0, & -1, & +3t, & -6t^2 \\ \alpha, & \beta, & \delta, & \epsilon \end{vmatrix}=0,$$

which is

$$\begin{aligned}&+(AD-D\alpha)(-3t^4)\\ &\qquad+(A\epsilon-E\alpha)(-t^2)\\ &\qquad+(B\delta-D\beta)(-6t^2)\\ &\qquad\qquad+(D\epsilon-E\delta)(-t)=0,\end{aligned}$$

or, what is the same thing,

$$\begin{aligned} \alpha(\ \ Bt^4 + 3Dt^4 + Et^2) \\ +\beta(-At^4 + 8Dt^3 + 3Et^2) \\ +\delta(-3At^2 - 6Bt + E) = 0, \\ +\epsilon(-At^3 - 3Bt^2 - D) = 0, \end{aligned}$$

and by equating to zero the coefficients α, β, δ, ϵ, we have four equations which are easily seen to reduce themselves to two equations only, and which are in fact the equations of the tangent line, the equations of this line may therefore be taken to be

$$At^2 - 3Bt + D = 0, \quad 3At + 8Bt^2 - E = 0.$$

The coordinates of a point in the nodal curve may be taken to be

$$a = \surd 2, \quad b = T, \quad d = -T^2, \quad e = \surd 2\, T^3,$$

and substituting these values in the place of A, B, D, E in the equations of the tangent line, we have

$$\begin{aligned} \surd 2\, t^2 + 3Tt - T^2 = 0, \\ 3\surd 2\, t + 8tT - \surd 2\, T^2 = 0, \end{aligned}$$

or, what is the same thing,

$$\begin{aligned} T^2 - 3tT - \surd 2\, t^2 = 0, \quad \text{i.e. } (T + \surd 2\, t)(T - \tfrac{1}{\surd 2}\, t) = 0, \\ T^2 - 4\surd 2\, tT - 3t^2 = 0, \quad (T + \surd 2\, t)(T - \tfrac{3}{\surd 2}\, t) = 0, \end{aligned}$$

so that the equations are satisfied by the values of t given by the equation

$$T^2 - \surd 2\, tT - t^2 = 0,$$

that is, by the values

$$T = \frac{1 \pm \surd 3}{\surd 2}\, t,$$

which belong to the points where the tangent line meets the nodal curve. Call these values T_1 and T_2; then considering a, b as current coordinates, the values of $a : b$ belonging to the point where the tangent line meets the cuspidal curve considered as three coincident points, and to the points where it meets the nodal curve, are given by the equation

$$(2b + at)^2\,(b\surd 2 - aT_1)\,(b\surd 2 - aT_2) = 0,$$

that is

$$(2b + at)^2\,(2b^2 - 2abt - a^2t^2) = 0,$$

or say

$$(at+2b)^2\,(a^2t^2+2abt-2b^2)=0.$$

I proceed to find the intersections of the tangent with the Prohessian: for this purpose putting for a moment in the last-mentioned equation x for at and y for b, this is

$$(x+2y)^2\,(x^2+2xy-2y^2)=0,$$

or, if in the place of $(x+y)$ we write x, this is

$$(x+y)^2\,(x^2-3y^2)=0,$$

and the Hessian of this is easily found to be

$$(x+y)^2\,(3x^2+8xy+4y^2),$$

whence, replacing x by $(x+y)$, the Hessian of

$$(x+2y)^2\,(x^2+2xy-2y^2)$$

is

$$(x+2y)^2\,(3x^2+14xy+13y^2).$$

We have thus

$$3x^2+14xy+13y^2=0;$$

that is

$$3x+(7\pm\surd 10)\,y=0,$$

or

$$3at+(7\pm\surd 10)\,b=0;$$

and therefore

$$\frac{b}{a}=\frac{-3}{7+\surd 10}=\frac{-3\{7-\surd 10\}}{39}=\frac{7-\surd 10}{13},$$

or putting

$$\eta_1=\frac{7+\surd 10}{18},\quad \eta_2=\frac{7-\surd 10}{18},$$

so that $\eta_1+\eta_2=\frac{7}{9}$, $\eta_1\eta_2=\frac{1}{4}$, and η_1, η_2 are the roots of the equation $18\eta^2-14\eta+3=0$, then we have $\frac{b}{a}=\eta_1 t$ or $\eta_2 t$, say $\frac{b}{a}=\eta_1 t$, or assuming $a=1$, then $b=\eta_1 t$.

But the equations of the tangent line being as above

$$at^2+3bt+d=0,\quad 3at+8bt^2-e=0,\quad a=1,$$

we have thus

$$\begin{aligned} &a=1, && b=\eta_1 t,\\ &d=(-1-3\eta_1)\,t^2, && e=(\ \ 3+8\eta_1)\,t^3, \end{aligned}$$

as the coordinates of the required points, viz. the tangent line meets the Prohessian, in the point on the cuspidal edge considered as 6 points, in two points on the nodal curve and in the last-mentioned 2 points; $6+2+2=10$ the order of the Prohessian.

The foregoing equations give

$$ae-6bd=18\eta_1^2+14\eta_1+3=0,$$

(in virtue of $18\eta_1^2+14\eta_1+3=0$), so that the two points in question are the intersections of the tangent line with the surface $ae-6bd=0$.

If we consider the intersection of this surface with the torse

$$(ae-4bd)^3-27(-ad^2-b^2e)^2=0,$$

the equation $ae-6bd=0$, gives

$$(ae-4bd)^2=(2bd)^2=4b^2d^2=\tfrac{2}{3}\,aeb^2d^2,$$

and thence

$$4a^2d^4-81(ad^2+b^2e)^2=0;$$

that is

$$81a^2d^4-162ab^2d^2e+81b^4e^2=0,$$

an equation which should agree with

$$\frac{e}{d^2}=\frac{1}{\theta_1^2}(144\eta_1-23).$$

In fact writing

$$\frac{e}{d^2}=\frac{1}{\theta_1^2}(144\eta_1-23),$$

the equation $18\eta_1^2+14\eta_1+3=0$ is $(18\eta_1+7)^2+5=0$; that is, $(144\eta_1+56)^2+320=0$, but $144\eta_1+56=81x+79$, or the equation becomes $(81x+79)^2+320=0$, that is

$$81^2x^2+81.158x+81^2=0,\quad \text{or } 81x^2+158x+81=0,$$

which is right.

Consider in like manner the intersection of the torse with the surface $ae - \lambda bd = 0$, where λ is a given constant coefficient; we have

$$(ae - 4bd)^2 = (\lambda - 4)^2\, b^2 d^2 = \frac{(\lambda - 4)^2}{\lambda}\, aeb^2 d^2,$$

and therefore

$$(\lambda - 4)^2\, ae\, d^4 - 27\lambda\, (ad^2 + b^2 e)^2 = 0,$$

that is

$$27\lambda\, a^2 d^4 + [54\lambda - (\lambda - 4)^2]\, ab^2 d^2 e + 27\lambda\, b^4 e^2 = 0,$$

which gives

$$ad^2 - \theta_1 b^2 e = 0, \quad \text{or } ad^2 - \theta_2 b^2 e = 0,$$

if θ_1, θ_2 are the roots of

$$27\lambda\, \theta^2 + [54\lambda - (\lambda - 4)^2]\, \theta + 27\lambda = 0.$$

The surfaces $ae - \lambda bd = 0$, $ad^2 - \theta_1 b^2 e = 0$ have in common the two lines $(a = 0, b = 0)$ and $(d = 0, e = 0)$, and they intersect besides in a quartic curve. And so for the surfaces $ae - \lambda bd = 0$, $ad^2 - \theta_2 b^2 e = 0$. That is, the surface $ae - \lambda bd = 0$ intersects the torse $(ae - 4bd)^3 - 27(-ad^2 - b^2 e)^2 = 0$, in the line $a = 0$, $b = 0$ twice, in the line $d = 0$, $e = 0$ twice, and in two quartic (excubo-quartic) curves. The two quartic curves become identical, if

$$(54\lambda)^2 = \{54\lambda - (\lambda - 4)^2\}^2,$$

that is

$$\pm 54\lambda = 54\lambda - (\lambda - 4)^2,$$

and therefore, if either

$$(\lambda - 4)^2 = 0,$$

which gives the cuspidal curve; or else if

$$(\lambda - 4)^2 - 108\lambda = 0,$$

that is

$$\lambda^2 - 12\lambda^2 - 60\lambda - 64 = (\lambda + 2)^2 (\lambda - 16) = 0.$$

$(\lambda + 2)^2 = 0$ or $\lambda = -2$ gives the nodal curve: $\lambda - 16 = 0$ gives $ae - 16bd = 0$, a surface which intersects the developable in the line $a = 0$, $b = 0$ twice, in the line $d = 0$, $e = 0$ twice, and in the two coincident quartic (excubo-quartic) curves given by the equations $ae - 16bd = 0$, $ad^2 - b^2 e = 0$. As a verification, I remark, that the surface $ad^2 - b^2 e = 0$ combined with the developable gives

$$(ae - 4bd)^2 - 27(ad^2 + b^2 e)^2 = (ae - 4bd)^2 - 108ab^2 d^2 e = 0,$$

that is $(ae + 2bd)^2(ae - 16bd) = 0$, or it meets the developable in its curve of intersection with $ae + 2bd = 0$ twice, and in its curve of intersection with $ae - 16bd = 0$; that is, in the line $a = 0$, $b = 0$ three times, in the line $d = 0$, $e = 0$ three times, in the nodal quartic $ae + 2bd = 0$, $ad^2 - b^2e = 0$ twice, and in the quartic $ae - 16bd = 0$, $ad^2 - b^2e = 0$ once; $3 + 3 + 8 + 4 = 18$, the order of the complete intersection.

Greenwich, January 4, 1864.

In my theory of the singularities of curves and torses, *Liouville*, t. X. (1845) pp. 245—250, [30], translated under the title “On Curves of Double Curvature and Developable Surfaces,” Cambridge and Dublin Mathematical Journal, t. V. (1850), pp. 18—22, [83], I omitted to take account of a noteworthy singularity, viz. this is, the stationary tangent line; or when the system has three consecutive points in a line, or, what is the same thing, three consecutive planes through a line. I reproduce the theory with this addition as follows. We have

m, the order of the system, = order of the curve,
r, „ rank of the system, = class of curve, = order of torse,
n, „ class of the system, = class of torse,
α, „ number of stationary planes,
β, „ " stationary points,
ϑ, „ " stationary lines,
g, „ " lines in two planes,
h, „ " lines through two points,
x, „ " points in two lines,
y, „ " planes through two lines.

This being so, the section of the torse by an arbitrary plane is a plane curve for which

r is the order,
n " class,
x " number of nodes,
$m+\vartheta$ " cusps,
g " double tangents,
α " inflexions;

and we have thence Pl"ucker’s six equations, which may be considered as included in the three equations

$$\begin{aligned} n &= r(r-1) - 2x - 3(m+\vartheta), \\ \alpha &= 3r(r-2) - 6x - 8(m+\vartheta), \\ r &= n(n-1) - 2g - 3\alpha. \end{aligned}$$

Similarly considering the cone standing on the curve and having an arbitrary point for vertex, then for this cone

m is the order,
r „ class,
h „ number of nodal lines,
β „ cuspidal lines,
y „ double tangent planes,
$n + \vartheta$ „ inflexions;

and we have Pl"ucker's six equations, which may be considered as included in the three equations

$$\begin{aligned} m &= r(r-1) - 2y - 3(n+\vartheta), \\ \beta &= 3r(r-2) - 6y - 8(n+\vartheta), \\ r &= m(m-1) - 2h - 3\beta. \end{aligned}$$

These two systems constitute together a system of six equations between the ten quantities m, r, n, α, β, ϑ, g, h, x, y. Considering m, r, x, ϑ as arbitrary, the six equations determine the remaining quantities n, α, β, h, x, y.

The curve

$$ae - 4bd + 3c^2 = 0, \quad ace + 2bcd - ad^2 - b^2e - c^3 = 0,$$

is a sextic curve, the edge of regression of the sextic torse

$$(ae - 4bd + 3c^2)^2 - 27(ace + 2bcd - ad^2 - b^2e - c^3)^2 = 0,$$

and we have in this case, as is well known,

$$m, r, n, \alpha, \beta, \vartheta, g, h, x, y = 6, 6, 4, 0, 4, 0, 3, 6, 4, 6.$$

But putting as above $c = 0$, then instead of the sextic curve we have the excubo-quartic curve $ae - 4bd = 0$, $ad^2 + b^2e = 0$, which is a curve having two stationary tangents, viz. these are the lines $(a = 0, b = 0)$ and $(d = 0, e = 0)$, which are in fact given along with the curve, by the foregoing equations $ae - 4bd = 0$, $ad^2 + b^2e = 0$. We have in this case $\vartheta = 2$, and the system is thus found to be

$$m, r, n, \alpha, \beta, \vartheta, g, h, x, y = 4, 6, 4, 0, 0, 2, 3, 3, 4, 4,$$

it was in fact the consideration of this case which led me to take account of the new singularity of the stationary tangent lines.

I take the opportunity of referring to a most valuable and interesting paper by Schwarz, "De superficiebus in planum explicabilibus primorum septem ordinum," *Crelle*, t. LXIV. (1864), pp. 1—16. The author, after referring to my paper "On the developable derived from an equation of the fifth order," Cambridge and Dublin Mathematical Journal, t. V. (1850), pp. 152—159, [86], enters into the enquiry there suggested as to the means of ascertaining the degree of the 'planarity' of a developable surface. He starts from certain theorems derived from Riemann's theory of transcendental functions, viz.: If an algebraical (plane) curve of the order r has $\frac{1}{2}(r-1)(r-2) - p$ double points (nodes or cusps), then the coordinates of a point of the curve may be expressed rationally

If $p = 0$, that is, if the curve has the maximum number of double points, by a single parameter.

If $p = 1$, by a single parameter, and the square root of a cubic or quartic function of this parameter.

If $p = 2$, by a single parameter, and the square root of a quintic or sextic function of this parameter.

If $p > 2$, by a parameter ξ, and an algebraical function thereof η; where ξ, η are connected by an equation of the order $\frac{1}{2}(p+3)$ or $\frac{1}{2}(p+2)$ according as p is odd or even.

These principles establish a division of plane curves into algebraical classes: all plane curves (other than the generating lines) situate on a ruled surface, belong to the same algebraical class, and the surface itself belongs to the same class. Hence, if on a ruled surface there is either a right line which is not a generating line (this cannot be the case for developables) or a conic, or a cubic having a double point, or any other plane curve having the maximum number of double points, the surface belongs to the class for which $p = 0$; and in the case of a developable surface the equation of the tangent plane may be rationally expressed by means of a single parameter; that is, the degree of the planarity is $= 1$, or the surface is planar. This leads to the conclusion, that the developable surfaces or torses of the orders 4, 5, 6 and 7 are all of them planar.

The author points out that the 'special quintic developable' of my paper first above referred, (viz. that obtained by writing $b = 0$ in the equation of the sextic developable) is in fact the general developable of the fifth order, or quintic torse.

The foregoing theorem, that for a curve which has the maximum number of double points, the coordinates may be expressed rationally by a single parameter, admits of a very simple algebraical proof, as is shown in the paper by Clebsch "Ueber Curven deren coordinaten rationale Functionen eines Parameters sind," *Crelle*, t. LXIV. (1864), pp. 43—65. In another paper by the same author, "Ueber die Singularit"aten algebraischer Curven," pp. 98—100, it is remarked that if in any plane curve we have m the order, n the class, δ the number of nodes, κ of cusps, τ of double tangents, ι of inflexions, then as a deduction from Riemann's principles, but also at once obtainable from Pl"ucker's equations, we have

$$\tfrac{1}{2}(m-1)(m-2) - \delta - \kappa = \tfrac{1}{2}(n-1)(n-2) - \tau - \iota;$$

and moreover if from a given curve we derive in any manner another curve, such that to each tangent (or point) of the first curve there corresponds a single point (or tangent) of the second curve, then in the second curve the expression

$$\tfrac{1}{2}(m'-1)(m'-2) - \delta' - \kappa', = \tfrac{1}{2}(n'-1)(n'-2) - \tau' - \iota',$$

has the same value as in the first curve.

The like property exists for curves in space—viz. taking account as above of the new singularity of the stationary lines, then we have

$$\begin{aligned}&\tfrac{1}{2}(m-1)(m-2)-h-\beta,\\ &=\tfrac{1}{2}(r-1)(r-2)-y-n-\vartheta,\\ &=\tfrac{1}{2}(r-1)(r-2)-x-m-\vartheta,\\ &=\tfrac{1}{2}(n-1)(n-2)-g-\alpha,\end{aligned}$$

which equations are in fact at once deducible from the above-mentioned system of six equations between the quantities m, r, n, α, β, ϑ, g, h, x, y, and may if we please be taken for equations of the system.

If from a given curve and torse we derive a second curve and torse, in such manner that to each point (or plane) of the first figure there corresponds a *single* plane (or point) of the second figure—then the corresponding expressions $\tfrac{1}{2}(m'-1)(m'-2)-h'-\beta'$, &c., have the same value for the second as for the first figure.

Cambridge, April 11, 1865.

374.

ON THE HIGHER SINGULARITIES OF A PLANE CURVE.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. vii. (1866), pp. 212—223.]

THE theory of the singularities of a plane curve was first established by Pl"ucker in his great work the *Theorie der Algebraischen Curven*, (1839), where he establishes, in regard to the ordinary singularities, a system of six equations; viz. if we have

m, the order of the curve,
n, „ class,
δ, „ number of double points,
κ, „ „ cusps,
τ, „ „ double tangents
ι, „ „ inflexions,

then Pl"ucker's six equations are

$$\begin{aligned}
n &= m(m-1) - 2\delta - 3\kappa, \\
\iota &= 3m(m-2) - 6\delta - 8\kappa, \\
\tau &= \tfrac{1}{2}m(m-2)(m^2-9) - (m^2-m-6)(2\delta+3\kappa) + 2\delta(\delta-1) + 6\delta\kappa + \tfrac{9}{2}\kappa(\kappa-1), \\
m &= n(n-1) - 2\tau - 3\iota, \\
\kappa &= 3n(n-2) - 6\tau - 8\iota, \\
\delta &= \tfrac{1}{2}n(n-2)(n^2-9) - (n^2-n-6)(2\tau+3\iota) + 2\tau(\tau-1) + 6\tau\iota + \tfrac{9}{2}\iota(\iota-1),
\end{aligned}$$

equivalent to three equations; thus m and (within proper limits) δ and κ may be considered arbitrary, and the first three equations then give n, ι, τ; and in like manner n and (within proper limits) τ and ι may be considered as arbitrary, and the equations then give m, κ, δ.

I have used the ordinary expressions double points, cusps, double tangents, inflexions; but (using as I have elsewhere done *ineunt* as the correlative of tangent) it would be more precise and symmetrical to say, double ineunts, stationary ineunts, double tangents, and stationary tangents. The double ineunt is called also a node, viz. it is a crunode, or an acnode, according as the tangents are real or imaginary; and the stationary ineunt, or cusp, considered as (what in the theory of point-coordinates it in fact is) a particular case of the double ineunt, is a spinode; to render this notation symmetrical, we require certain new terms, say link, as the correlative to node, and flex as the correlative to cusp; then the double tangent is a link, viz. it is a colink, or an allink according as the ineunts upon it (points of contact) are real or imaginary; and the stationary tangent (inflexion) or flex, considered as (what in the theory of line-coordinates it in fact is) a particular case of the double tangent, is a relink. The ordinary singularities of a plane curve would thus be the node, the cusp, the link, and the flex; but I shall retain the above-mentioned more usual expressions.

Deducible from the six equations, we have

$$n-m=\tfrac{1}{2}(\iota-\kappa),$$

$$(n-m)(n+m-9)=2(\tau-\delta),$$

which are noticed by Pl"ucker; and also the equation

$$\tfrac{1}{2}(m-1)(m-2)-\delta-\kappa=\tfrac{1}{2}(n-1)(n-2)-\tau-\iota,$$

recently noticed by M. Clebsch, in connection with Riemann's investigations on the Abelian Integrals; a curve of the order m may have $\frac{1}{2}(m-1)(m-2)$ double points, reckoning the cusp as a double point, and so a curve of the class n may have $\frac{1}{2}(n-1)(n-2)$ double tangents, reckoning the inflexion as a double tangent; the two sides of this equation exhibit therefore, the right-hand side the deficiency of the actual from the possible number of double tangents, and the left-hand side the deficiency of the actual from the possible number of double points; and these two numbers are equal. We have a division into families based on the value of the expressions in question, or say on that of $\frac{1}{2}(m-1)(m-2)-\delta-\kappa$; when this is $=0$, that is, when the curve has its maximum number of double points (reckoning the cusp as a double point), the coordinates x, y are expressible rationally in terms of a parameter θ; when the number is $=1$, they can be expressed rationally in terms of θ and of the square root of a cubic or a quartic function of θ, &c. &c. It thus appears that as well the number $\delta+\kappa$, as the combinations $2\delta+3\kappa$ and $6\delta+8\kappa$ which enter into Pl"ucker's equations, plays an important part in the theory of the curve; the bearing of this remark will be seen in the sequel.

Pl"ucker considers also some of the higher singularities; it will be convenient to mention two of his results.

No. 76, p. 216. If two branches of a curve touch each other, or more generally have a g-pointic intersection, the point in question is equivalent to g double points,

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and the tangent at this point to g double tangents; hence, if there is no other point singularity, the equations give

$$\begin{aligned} n &= m(m-1) - 2g, \\ \iota &= 3m(m-2) - 6g, \\ \delta + g &= \tfrac{1}{2}m(m-2)(m^2-9) - (m^2-m-6)\,2g + 2g(g-1), \end{aligned}$$

the last of which may also be written

$$\delta = \tfrac{1}{2}m(m-2)(m^2-9) - (m^2-m-g-4\tfrac{1}{2})\,2g.$$

And Nos. 77—82, pp. 217—222. For a cusp of the second kind, we have

$$\begin{aligned} n &= m(m-1) - 5, \\ \iota &= 3m(m-2) - 15, \\ \delta &= \tfrac{1}{2}m(m-2)(m^2-9) - (m^2-m-7)\,5; \end{aligned}$$

these equations Pl"ucker establishes by an independent algebraical investigation, and having done so, he remarks that they are deducible from the foregoing ones by writing therein $g = 2\frac{1}{2}$; that is, that the cusp of the second kind may be considered as equivalent to $2\frac{1}{2}$ double points, and the tangent at the cusp to $2\frac{1}{2}$ double tangents. And he thence passes to the cusp of a higher cusp equivalent to $h + \frac{1}{2}$ double points and $h + \frac{1}{2}$ double tangents. The results in this general case (although not, as in the original case, $g = 2\frac{1}{2}$, established independently) is perfectly correct; but the theory is open to a grave objection.

I remark, that assuming a certain singularity to be equivalent to the numbers δ' of double points, κ' of cusps, τ' of double tangents, and ι' of inflexions, we have in the first instance to determine δ', κ', τ' and ι' in such manner as to give in the class n, and in the numbers ι of inflexions and τ of double tangents, the reductions actually given by the singularity in question. Thus in the case of the cusp of the second kind, we ought to have

$$\begin{aligned} 2\delta' + 3\kappa' &= 5, \\ 6\delta' + 8\kappa' + \iota' &= 15, \\ (m^2-m-6)(2\delta'+3\kappa') - 2\delta'(\delta'-1) - 6\delta'\kappa' - \tfrac{9}{2}\kappa'(\kappa'-1) + \tau' &= (m^2-m-7)\,5, \end{aligned}$$

or, what is the same thing,

$$2\delta'(\delta'-1) + 6\delta'\kappa' + \tfrac{9}{2}\kappa'(\kappa'-1) - \tau' = 5;$$

and so in general there are, for the determination of the four quantities δ', κ', τ', ι', three equations. In the particular case these are satisfied by the values $\delta' = 2\frac{1}{2}$, $\kappa' = 0$, $\tau' = 2\frac{1}{2}$, $\iota' = 0$, which are Pl"ucker's values; they are also satisfied by the values $\delta' = 1$, $\kappa' = 1$, $\tau' = 1$, $\iota' = 1$, which have the advantage of being integer instead of fractional.

But there is really a further condition to be satisfied, viz. the number $\delta' + \kappa'$ must have a certain definite value dependent on the nature of the singularity; for the case in hand, this is $\delta' + \kappa' = 2$ (I obtain this by the consideration of a quartic curve, having a cusp of the second kind, and also a double point; $\delta + \kappa$ has here its maximum value $= 3$; and as the double point gives $\delta = 1$, the cusp of the second kind gives $\delta' + \kappa' = 2$); and joining to the former conditions this new condition, we have definitely $\delta' = 1$, $\kappa' = 1$, $\tau' = 1$, $\iota' = 1$. I have elsewhere noticed, [343], that the cusp of the second kind was equivalent to a double point and cusp, and accordingly proposed to call it the node-cusp; but I had not then remarked that it was also necessary to treat the tangent as equivalent to a double tangent and a stationary tangent (or inflexion).

It appears from the foregoing considerations, that any singularity whatever is to be regarded as equivalent, and that in a perfectly definite manner, to a certain number δ' of double points, κ' of cusps, τ' of double tangents, and ι' of inflexions; we have only to ascertain how for any given singularity the values of these numbers are to be ascertained; and when this is done, Pl"ucker's equations will be applicable to any singularities whatever of a plane curve.

At any point of a plane curve there is either one branch, or any number of branches, touching or not touching each other: taking the given point as origin, then for each branch the equation of the curve gives for the ordinate y an expression of the form

$$y = Ax^p + Bx^q + \dots,$$

where the series is arranged in ascending powers of x, and where the coefficients A, B, … have definite unique values; and, conversely, that which is given by such expression of y is a branch of the curve. It is assumed that the axis of y, or line $x = 0$, is not a tangent to the curve; this implies that the exponents p, q, … are none of them inferior to 1, or, what is the same thing, that the lowest exponent p is $= 1$ at least: it is for the most part convenient to take the axis of x, or line $y = 0$, a tangent to the branch; the lowest exponent p is then > 1.

The exponents may be all integer, and the branch is then said to be *linear*; or else the exponents or some of them may be fractional, and the branch is then *superlinear*; viz. in the latter case, assuming that the fractional exponents are all of them in their least terms, and that α is the least common multiple of the denominators (so that the expression for y is a rational function of $x^{\frac{1}{\alpha}}$), then the branch is quadric, cubic, &c. according as we have $\alpha = 2$, $\alpha = 3$, &c. It is clear that the expression for y has precisely α values, viz. the values obtained by attributing to the radical $x^{\frac{1}{\alpha}}$ each of its α values. Corresponding to each of these α values, we have what I term a partial branch of the curve, so that the quadric branch is made up of two partial branches, the cubic branch of three partial branches and so on; for a linear branch or when $\alpha = 1$, a partial branch is nothing else than the branch itself; and the expression a partial branch will accordingly include the case of a linear branch.

Suppose that at any point of the curve we have two partial branches, belonging or not belonging to the same branch; let these be referred to the same axes, the

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axis of y not being a tangent to either branch, so that the exponents are none of them < 1. If in the series for $y_1 - y_2$, (the difference of the two ordinates) the least exponent is $= P$, then (whether P is integer or fractional) the two partial branches are said to have, at the given point, P common points, or, more briefly, to intersect in P points. We may from this definition calculate the number of intersections of two branches with each other, or of a branch with itself; for instance, suppose that at any point of the curve we have ($\alpha = 6$) the sextic branch

$$y = x^{\frac{5}{3}} + x^{\frac{5}{2}} + \ldots,$$

we have the six partial branches

$$\begin{aligned} y_1 &= x^{\frac{5}{3}} + x^{\frac{5}{2}} + \ldots, & y_4 &= x^{\frac{5}{3}} - x^{\frac{5}{2}} + \ldots, \\ y_2 &= \omega\, x^{\frac{5}{3}} + x^{\frac{5}{2}} + \ldots, & y_5 &= \omega\, x^{\frac{5}{3}} - x^{\frac{5}{2}} + \ldots, \\ y_3 &= \omega^2 x^{\frac{5}{3}} + x^{\frac{5}{2}} + \ldots, & y_6 &= \omega^2 x^{\frac{5}{3}} - x^{\frac{5}{2}} + \ldots; \end{aligned}$$

hence calculating (what is most convenient) *twice* the number of intersections of the branch with itself, the partial branch y_1 intersects the other partial branches in $\frac{4}{3}$, $\frac{4}{3}$, $\frac{5}{2}$, $\frac{5}{2}$, $\frac{4}{3}$ points respectively, giving the sum $\frac{16}{3} + 5 = \frac{47}{3}$; each other partial branch intersects the remaining five branches in the same number of points; and therefore twice the number of intersections is $= 47$.

For the singularity $y = x^{\frac{5}{3}} + x^{\frac{5}{2}} + \ldots$ in question, I say that if this be equivalent as above to δ' double points, κ' cusps, τ' double tangents and ι' inflexions, then that the number 47 just obtained is the value of $2\delta' + 3\kappa'$, and moreover, that the value of κ' is $\kappa' = \alpha - 1 = 5$; that is, we have $2\delta' + 3\kappa' = 47$ and $\kappa' = 5$; or, what is the same thing, $\delta' = 16$; $\kappa' = 5$. For the determination of the numbers τ', ι', it is to be observed that the foregoing theory of branches is a theory of the points of a branch, by means of point-coordinates: there is a precisely similar theory of the tangents of a branch by means of line-coordinates, and we may inquire as to the number of the common tangents of two partial branches; and thence as to the number of common tangents of two branches, or of a branch with itself—it will appear that the line-equation of the branch is $Z = X^4 \ldots + X^{\frac{14}{3}} \ldots$, so that the branch (which is as to its points sextic, $\alpha = 6$) is as to its tangents quadric, $\beta = 2$, the two partial branches have with each other the number $= \frac{15}{2}$ of common tangents, or twice this number is $= 15$; that is, we have $2\tau' + 3\iota' = 15$, and moreover $\iota' = \beta - 1 = 1$, that is $\tau' = 6$, $\iota' = 1$; or finally for the singularity in question, the numbers δ', κ', τ', ι' are $= 16, 5, 6, 1$ respectively.

And so generally in the case of a branch which is as to its points α-ic, having with itself a number $= \frac{1}{2}M$ of common points; and as to its tangents β-ic, having with itself a number $= \frac{1}{2}N$ of common tangents, we have $2\delta' + 3\kappa' = M$, $\kappa' = \alpha - 1$, $2\tau' + 3\iota' = N$, $\iota' = \beta - 1$, or, what is the same thing, the values of δ', κ', τ', ι' are

$$\begin{aligned}
\delta' &= \tfrac{1}{2}[M - 3(\alpha - 1)], \\
\kappa' &= \quad \alpha - 1, \\
\tau' &= \tfrac{1}{2}[N - 3(\beta - 1)], \\
\iota' &= \quad \beta - 1.
\end{aligned}$$

I say that a singularity is simple when we have one branch, compound when we have more than one branch; the case above considered is that of a simple singularity, viz. we have on the curve one point, one tangent, one branch.

We may have a compound singularity where the branches all touch, that is we may have one point, one tangent, several branches. It may be seen that if $\frac{1}{2}M$ denote the number of common points of all the branches (that is of each branch with itself, and of every two branches with each other), and in like manner if $\frac{1}{2}N$ denote the number of common tangents of all the branches (that is of each branch with itself, and of every two branches with each other), then the formulæ are

$$\begin{aligned} \delta' &= \tfrac{1}{2}[M - 3\Sigma(\alpha - 1)], \\ \kappa' &= \quad \Sigma(\alpha - 1), \\ \tau' &= \tfrac{1}{2}[N - 3\Sigma(\beta - 1)], \\ \iota' &= \quad \Sigma(\beta - 1), \end{aligned}$$

where the signs Σ refer to the different branches.

Again, we may have a compound singularity, one point, several tangents with to each of them a branch or branches; here if M denote the number of the common points of all the branches, and N the number of the common tangents of all the branches belonging to any one tangent, then the formulæ are

$$\begin{aligned} \delta' &= \tfrac{1}{2}\ [M - 3\Sigma'\Sigma(\alpha - 1)], \\ \kappa' &= \quad \Sigma'\Sigma(\alpha - 1), \\ \tau' &= \tfrac{1}{2}\Sigma'\ [N - 3\Sigma\ (\beta - 1)], \\ \iota' &= \quad \Sigma'\Sigma(\beta - 1), \end{aligned}$$

where the signs Σ refer to all the branches belonging to the same tangent, and the signs Σ' to the different tangents. It is to be remarked, that the point on the curve is equivalent to the δ' double points and κ' cusps; each tangent is equivalent to $\frac{1}{2}[N - 3\Sigma(\beta - 1)]$ double tangents, and $\Sigma(\beta - 1)$ inflexions, the numbers N, β referring of course to the tangent in question.

Lastly, we may have a compound singularity, one tangent, several points (of contact), with to each of them a branch or branches; here if N denote the number of the common tangents of all the branches, M the number of the common points of all the branches belonging to any one point of contact, the formulæ are

$$\begin{aligned}\delta' &= \tfrac{1}{2}\Sigma'\ [(M - 3\Sigma\ (\alpha - 1)],\\ \kappa' &= \quad \Sigma'\Sigma(\alpha - 1),\\ \tau' &= \tfrac{1}{2}\ [(N - 3\Sigma\Sigma'\ (\beta - 1)],\\ \iota' &= \quad \Sigma'\Sigma(\beta - 1),\end{aligned}$$

where the signs Σ refer to all the branches belonging to the same point of contact, and the signs Σ' to the different points of contact; it is to be remarked that the tangent of the curve is equivalent to the τ' double tangents and ι' inflexions; each point of contact is equivalent to $\frac{1}{2}[M - 3\Sigma(\alpha - 1)]$ double points and $\Sigma(\alpha - 1)$ cusps, the numbers M, α referring of course to the point of contact in question.

There is no difficulty in passing to the case of the compound singularity when the formulæ for the simple singularity, one point, one tangent, one branch, are once obtained, and I now go back to the consideration of this case.

The class of a curve is equal to the number of tangents which can be drawn through an arbitrary point: the points of contact of these tangents are given as the intersections of the curve with a certain curve, the polar of the arbitrary point in regard to the curve; this polar passes through each double point and cusp, the double point counting as two points of intersection, and the cusp as three points of intersection (this is in fact the theory by which is found the reduction $= 2\delta + 3\kappa$ in the class of the curve). Hence, if the curve has a singularity $(\delta', \kappa', \tau', \iota')$, which to fix the ideas may be assumed to be a simple singularity, 'one point, one tangent, one branch'; then the polar passes through the singular point, the number of intersections being $2\delta' + 3\kappa'$, or if the actual number of intersections be M, then we have $M = 2\delta' + 3\kappa'$. It is to be shown that the number M is equal to twice the number of common points which the curve has with itself at the singular point, so that the last-mentioned number is $= \frac{1}{2}M$. Suppose in the first instance that there is only a single branch, and let the branch be given by the equation

$$P = y \ \ + Ax^p \ \ + Bx^q + \ldots = 0,$$

or introducing for homogeneity the third coordinate z, let this equation be

$$P = yz^{p-1} + Ax^p z^{-p} + Bx^q z^{-q} \ldots = 0,$$

and let $P_1 = 0$, $P_2 = 0$, $\ldots$ $P_\alpha = 0$, be the corresponding equations for the component partial branches; it is allowable to write $P_1 P_2 \ldots P_\alpha = 0$ for the equation of the curve[10]. Hence if (α, β, γ) be the coordinates of the arbitrary point, or putting in the first instance $\gamma = 1$, if $(\alpha, \beta, 1)$ be the coordinates, then writing $\Delta = \alpha\delta_x + \beta\delta_y + \delta_z$, the equation of the polar is $\Delta P_1 P_2 \ldots P_\alpha = 0$, or, what is the same thing,

$$P_2 P_3 \ldots P_\alpha \, \Delta P_1 + P_1 P_3 \ldots P_\alpha \, \Delta P_2 + \&\text{c.} = 0,$$

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[10] Of course this is not the equation in its rational and integral form, and on this account the reasoning of the text is not free from difficulty; the same remark applies to a subsequent equation.

and we have

$$\Delta P = \alpha(pAx^{p-1}z^{-p}+qBx^{q-1}z^{-q}\ldots)+\beta z^{-1}-(pAx^{p}z^{-p-1}+qBx^{q}z^{-q-1}\ldots),$$

or putting $z = 1$, this is

$$\Delta P = \alpha(pAx^{p-1} \quad +qBx^{q-1} \quad \ldots)+\beta \quad -(pAx^{p} \quad +qBx^{q} \quad \ldots),$$

and we have thence the values of ΔP_1, ΔP_2, $\ldots$ ΔP_α; the thing to be observed is, that the equation $\Delta P = 0$ is *not* satisfied (and therefore also each of the equations $\Delta P_1 = 0$, $\ldots$ $\Delta P_\alpha = 0$ is not satisfied) by the coordinates $x = 0$, $y = 0$ of the singular point. We have now with the equation $\Delta P_1 P_2 \ldots P_\alpha = 0$ of the polar to combine the equation $P_1P_2\ldots P_\alpha=0$: the last-mentioned equation breaks up into the equations $P_1=0$, $P_2=0$, $\ldots$ $P_\alpha=0$; and selecting for example the equation $P_1=0$, this gives the system $P_1=0$, $P_2P_3\ldots P_\alpha\Delta P_1=0$, or since we require only the intersections at the singular point, and $\Delta P_1=0$ does not pass through this point, this may be replaced by $P_1=0$, $P_2P_3\ldots P_\alpha=0$. The complete system is thus $(P_1=0,\ P_2P_3\ldots P_\alpha=0)$, $(P_2=0,\ P_1P_3\ldots P_\alpha=0)$, $\ldots$ $(P_\alpha=0,\ P_1P_2\ldots P_{\alpha-1}=0)$; or, what is the same thing, we have each pair $(P_r=0,\ P_s=0)$ taken twice. To eliminate y from these equations, we have merely to write $P_r-P_s=0$, or, what is the same thing, we have $\zeta(P_1,\ P_2,\ldots P_\alpha)=0$, ζ denoting the product of the squares of the differences of the functions $(P_1,\ P_2,\ldots P_\alpha)$. Suppose that any two partial branches $P_r=0$, $P_s=0$ intersect (according to the above-mentioned definition) in p points; then P_r-P_s contains the factor x^p, and hence the product $\zeta(P_1,\ P_2,\ldots P_\alpha)$ contains as a factor x to the power $2\Sigma p$, that is, the equation in x has $2\Sigma p$ roots each $=0$. Whence if $\Sigma p=\frac{1}{2}M$, then the equation in x has M roots each $=0$, or the curve and polar have at the singular point M intersections, that is $M=2\delta'+3\kappa'$.

I have no complete proof to offer of the remaining equation $\kappa'=\alpha-1$, it was obtained from the consideration of a particular case as follows. Consider the linear branch $y=Ax^p+\ldots$, where the exponents are all positive integers, and taking the axis of x to be the tangent, the least exponent p is greater than unity; if $p=2$ there is at the origin no inflexion, if $p=3$ there is a single inflexion, and generally the number of inflexions is $=p-2$. Now it will presently appear that in line-coordinates the equation of the branch is $Z=A'X^{\frac{p}{p-1}}$, or replacing Z, X by the original point-coordinates y, x the branch $y=A'x^{\frac{p}{p-1}}+\ldots$ has at the origin $p-2$ cusps; but in the branch in question we have $\alpha=p-1$, and the number of cusps is thus $=\alpha-1$; this result is confirmed by other particular instances, and I assume in general that we have $\kappa'=\alpha-1$; whence in the case of a simple singularity, or where there is only one branch we have $M=2\delta'+3\kappa'$, $\kappa'=\alpha-1$, or, what is the same thing, $\delta'=\frac{1}{2}[M-3(\alpha-1)]$, $\kappa'=\alpha-1$. The reasoning is easily adapted to the case of a compound singularity.

I consider the branch

$$y\ +Ax^p\ +Bx^q\ +\ldots=0,$$

(where it is assumed that the axis of x is a tangent to the branch, and therefore that the lowest exponent p is greater than unity), introducing the coordinate z for homogeneity, this becomes

$$yz^{-1}+Ax^pz^{-p}+Bx^qz^{-q}+\ldots=0,$$

and I proceed to find the corresponding equation in line coordinates, taking these to be X, Y, Z, we have

$$\begin{aligned}
\lambda X &= pAx^{p-1}z^{-p} + \qquad Bqx^{q-1}z^{-q} + \ldots, \\
\lambda Y &= \qquad\quad\; z^{-1}, \\
\lambda Z &= -\ yz^{-1} - pAx^{p}z^{-p-1} - qBx^{q}z^{-q-1} + \ldots,
\end{aligned}$$

or writing $z = 1$, $Y = 1$, we find $\lambda = 1$, and therefore

$$\begin{aligned} X &= \quad pAx^{p-1} + \qquad qBx^{q-1} + \ldots, \\ Z &= -y - pAx^{p} \quad - \qquad qBx^{q} \quad + \ldots; \end{aligned}$$

here substituting for $-y$ its value $= Ax^p + Bx^q + \ldots$, we have

$$\begin{aligned} X &= \quad pAx^{p-1} + \qquad qBx^{q-1} + \ldots, \\ Z &= (1-p)\,Ax^{p} \quad + (1-q)\,Bx^{q} \quad + \ldots. \end{aligned}$$

Hence writing $pAx^{p-1} = \theta$, the equations are

$$\begin{aligned} X &= \theta - B'\theta^{\frac{q-1}{p-1}} - \ldots, \\ Z &= \quad - A'\theta^{\frac{p}{p-1}} - B'\theta^{\frac{q}{p-1}} - \ldots, \end{aligned}$$

so that eliminating θ, we have

$$Z = \quad A'X^{\frac{p}{p-1}} + B'X^{\frac{q}{p-1}} + \ldots,$$

and it is easy to see by Lagrange's theorem, that the general form of the exponents in the series on the right-hand side is $\dfrac{p + f(q-p) + g(r-p) + \ldots}{p-1}$, where f, g, ... are positive integers, zero included. The equation in line-coordinates being known, the subsequent investigation is precisely the same as that for the point-coordinates, and hence in the case of one branch, if this be in regard to its tangents β-ic, and have $\frac{1}{2}N$ common tangents with itself, then $2\tau' + 3\iota' = N$, $\iota' = \beta - 1$, or, what is the same thing, $\tau' = \frac{1}{2}[N - 3(\beta - 1)]$, $\iota' = \beta - 1$. The investigation in the case of a simple singularity of the values of δ', κ', τ', ι' is thus completed.

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NOTES ON POLYHEDRA.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. vii. (1866), pp. 304—316.]

Axial Properties. Article 1 to 18.

1. A POLYHEDRON may have a q-axis, viz. a line about which if it is made to rotate through an angle $= \dfrac{2\pi}{q}$ (but not through any sub-multiple of this angle), it will occupy the same portion of space. It is then clear that when the rotation is repeated any number of times the body will still occupy the same portion of space; or if Θ denote the rotation through the angle $\dfrac{2\pi}{q}$, then we have the rotations 1, Θ, Θ^2, … Θ^{q-1}, and finally $\Theta^q = 1$, that is, when the rotation is q-times repeated, the body will resume its original position. Similarly for any number of axes ($\Theta^q = 1$, $\Theta'^{q'} = 1, \ldots$, where the indices q, q', … may be the same or different) we have the rotations 1, Θ, Θ^2, … Θ^{q-1}, Θ', Θ'^2, … $\Theta'^{q'-1}$, …; and if Θ, Θ', … be the entire system of the axes of the body, these rotations will form a *group*. The rotations in question are in fact the entire series of those which leave unaltered the portion of space occupied by the body, and since any two rotations combine together into a single rotation, any two of the rotations in question must combine together into some one of these rotations, that is, the rotations in question form a group. Some analytical consequences of this theorem will be obtained in the sequel.

2. The number of axes may be denoted by $\Sigma 1$ and the number of rotations by $1 + \Sigma(q-1)$; we may say that $\Sigma 1$ is the number, and $1 + \Sigma(q-1)$ the efficiency or *weight*, of the axes.

3. For any one of the regular polyhedra, E being the number of edges, then the number of axes or $\Sigma 1$ is $= E+1$, and their weight or $1+\Sigma(q-1)$ is $= 2E$. In fact, if as usual S denote the number of summits, F the number of faces, and if there be m edges to a face, and n edges to a summit, then $S+F = E+2$, $mF = nS = 2E$. Now in all the polyhedra except the tetrahedron, we have a number $\frac{1}{2}F$ of m-axes passing through the centres of opposite faces (ampihedral axes as Mr Kirkman has termed them) and a number $\frac{1}{2}S$ of n-axes passing through opposite summits (amphigonal axes); and we have besides a number $\frac{1}{2}E$ of 2-axes passing through the mid-points of opposite edges (amphigrammic axes): the entire number of axes is thus $\frac{1}{2}(S+F+E)$, which is $= E+1$: and the weight is $1+\frac{1}{2}F(m-1)+\frac{1}{2}S(n-1)+\frac{1}{2}E$, which is $= 1+\frac{1}{2}mF+\frac{1}{2}nS-\frac{1}{2}(F+S-E)$, $= 1+E+E-1$, $= 2E$. In the case of the tetrahedron $S = F = 4$, $m = n = 3$, and the only difference is that instead of the $\frac{1}{2}F$ ampihedral m-axes and the $\frac{1}{2}S$ amphigonal n-axes, we have a number $(F = S =)\ \frac{1}{2}(F+S)$ of $(n =)\ m$-gonal axes each through a summit and the centre of an opposite face (gonohedral axes).

4. The theorem that the weight $1+\Sigma(q-1) = 2E$, or say $1+\Sigma(q-1) = mF$, may be extended so as to apply to any polyhedron whatever. In fact considering any face A of the polyhedron, let F be the number of faces homologous to (and inclusive of) A; and, taking a any edge of the face A, let m be the number of edges of A homologous to (and inclusive of) a: then we have $1+\Sigma(q-1) = mF$. This is almost a truism when the signification of the term "homologous" is explained. Imagine the polyhedron placed on a plane, say the table, and draw on the table a polygon equal to the polygonal face A, and in this polygon select some one edge corresponding to the edge a. The polyhedron may be placed on the table with the face A coinciding with the polygon, or say the face A may be superimposed on the polygon, and that in m different ways, viz. any one of the edges homologous to a may be made to coincide with the assumed edge: and in like manner there are F different faces (viz. the faces homologous to A) which may be superimposed on the polygon, each of them in m different ways; that is there are in all mF different positions of the polyhedron for each of which it occupies the same portion of space. And we have thus the required theorem $1+\Sigma(q-1) = mF$.

5. As an example, take the regular pyramid on a square base; there is here a single axis, viz. a 4-axis, and we have $1+\Sigma(q-1) = 1+3 = 4$. If for the face A we take the square base, then there is no other face homologous thereto and therefore $F = 1$; but the four sides are homologous to each other or $m = 4$, and we have $mF = 4$. Similarly taking for A one of the triangular faces, since these are homologous to each other, then $F = 4$; and if we take for the side a the base of the triangle, then there is no other side homologous to this, or $m = 1$; and therefore $mF = 4$. It might at first sight appear that the two equal sides of the triangle were homologous to each other, and therefore that taking for the edge a one of these sides

we should have $m = 2$; but in fact although the two sides in question are homologously related to the pyramid, yet according to the definition they are not homologous sides of the triangular face, and we still have $m = 1$, and therefore $mF = 4$.

6. Of course in the case of a regular polyhedron the faces are all homologous, and the edges of a face are all homologous, that is F will denote the entire number of faces, and m the number of edges to a face: so that for this case the theorem gives $1 + \Sigma(q-1) = 2E$ as above.

7. Returning to the regular polyhedra the axial systems are

Tetrahedron	$4L^2,\ 3L^2,$
Cube and Octahedron	$3L^4,\ 4L^3,\ 6L^2,$
Dodecahedron and Icosahedron	$6L^5,\ 10L^3,\ 15L^2,$

where L^q denotes a q-axis, &c.; this is in accordance with the notation of M. Bravais in the memoir subsequently referred to.

8. The regular polyhedra may be exhibited in connexion with each other as follows: Imagine the polyhedron projected on a concentric sphere by lines through the centre; so that the summits become points on the sphere, the edges arcs of great circles, and the faces spherical polygons. Starting from the dodecahedron, the centres of the pentagonal faces are the summits of the icosahedron, and conversely for the icosahedron the centres of the triangular faces are the summits of the dodecahedron: moreover each edge of the dodecahedron cuts at right angles an edge of the icosahedron and the two edges have the same mid-point. Again if in any face of the dodecahedron we draw one of the five diagonals (arcs through two non-adjacent summits) there is in the face a single edge not met by this diagonal; and in the other face through this edge a single diagonal not met by the edge; joining the extremities of the two diagonals we have a spherical square, the face of the cube; it is to be observed that the summits of the cube are eight out of the twenty summits of the dodecahedron, and that the centres of the faces of the cube are the mid-points of six out of the thirty edges of the dodecahedron or the icosahedron. The cube given by the foregoing construction is of course one out of five different cubes. The centres of the faces of the cube are the summits of the octahedron; and conversely the centres of the faces of the octahedron are the summits of the cube; moreover each edge of the cube cuts at right angles an edge of the octahedron; and the two edges have the same mid-point. Finally, taking four non-adjacent summits of the cube (which can be done in two different ways), these are the summits of the tetrahedron, and the mid-points of the edges of the tetrahedron are the summits of the octahedron.

9. Considering the polyhedra in the foregoing mutual connexion, all the axes of the tetrahedron are axes of the cube and octahedron, viz. the 2-axes of the tetrahedron are the 4-axes of the cube and octahedron; and the 3-axes of the tetrahedron are the 3-axes of the cube and octahedron; moreover the 3-axes of the cube and octahedron are included among the

3-axes of the dodecahedron and icosahedron and the 4-axes of the cube and octahedron are included among the 2-axes of the dodecahedron and icosahedron; but the 2-axes of the cube and octahedron are not included among the axes of the dodecahedron and icosahedron. The 4-axes of the cube and octahedron form thus a system of rectangular axes common to all the polyhedra, and representing these axes (or say the summits of the corresponding rectangular spherical triangle) by X, Y, Z, we have a convenient system of coordinate axes to which to refer all the other axes of the polyhedron, viz. if P be the extremity (chosen at pleasure) of the axis in question, then the position of the axis may be determined by its distance PZ and azimuth XPZ (measured in the direction from X to Y), or by its distances PX, PY, PZ, or say X, Y, Z from the three rectangular axes (we have, it is clear, $\cos X = \sin \text{dist.} \cos \text{azim.}$, $\cos Y = \sin \text{dist.} \sin \text{azim.}$, $\cos Z = \cos \text{dist.}$). The rotation angle of a q-axis is $= \dfrac{2\pi}{q}$ (i.e. this is the angle through which if the body be turned about the axis, it still occupies the same portion of space) and the half-rotation angle is therefore $= \dfrac{\pi}{q}$. Moreover if i, j, k are Sir W. R. Hamilton's quaternion symbols, then the "rotation symbol" of the axis is

$$\cos\frac{\pi}{q} + \sin\frac{\pi}{q}\,(i\cos X + j\cos Y + k\cos Z),$$

the application of which will be presently explained.

10. The angular coordinates of the different axes may be found by spherical trigonometry without much difficulty; and we are then able to form the following axial tables of the several polyhedra: the extremity of each axis is chosen in such manner that the distance PZ is not > 90.

Axial System of the Tetrahedron.

Distances			Azimuths			$\cos X$	Rot. Symbols
angle	cos	sin	angle	cos	sin		
4 3-axes, $\frac{1}{2}$ Rot. angle $= 60$, $\cos = \frac{1}{2}$, $\sin = \frac{1}{2}\sqrt{3}$.							
$5444'$	$\dfrac{1}{\sqrt{3}}$	$\dfrac{\sqrt{2}}{\sqrt{3}}$	45	$+\dfrac{1}{\sqrt{2}}$	$+\dfrac{1}{\sqrt{2}}$	$+\dfrac{1}{\sqrt{3}}$	$\frac{1}{2}(1+i+j+k)$
″	″	″	135	−″	+″	−″	$\frac{1}{2}(1-i+j+k)$
″	″	″	225	−″	−″	+″	$\frac{1}{2}(1+i-j+k)$
″	″	″	315	+″	−″	−″	$\frac{1}{2}(1-i-j+k)$
3 2-axes, $\frac{1}{2}$ Rot. angle $= 90$, $\cos = 0$, $\sin = 1$.							
0	1	0	$*$	$*$	$*$	0	i
90	0	1	0	1	0	1	j
″	″	″	90	0	1	0	k

Axial System of the Cube and Octahedron.

Distances			Azimuths			$\cos X$	Rot. Symbols
angle	cos	sin	angle	cos	sin		
4 3-axes, $\tfrac{1}{2}$ Rot. angle $= 60$, $\cos = \tfrac{1}{2}$, $\sin = \tfrac{1}{2}\sqrt{3}$.							
5444′	$\frac{1}{\sqrt{3}}$	$\frac{\sqrt{2}}{\sqrt{3}}$	45	$+\frac{1}{\sqrt{2}}$	$+\frac{1}{\sqrt{2}}$	$+\frac{1}{\sqrt{3}}$	$\tfrac{1}{2}(1+i+j+k)$
〃	〃	〃	135	−〃	+〃	−〃	$\tfrac{1}{2}(1-i+j+k)$
〃	〃	〃	225	−〃	−〃	+〃	$\tfrac{1}{2}(1+i-j+k)$
〃	〃	〃	315	+〃	−〃	−〃	$\tfrac{1}{2}(1-i-j+k)$
3 4-axes, $\tfrac{1}{2}$ Rot. angle $= 45$, $\cos = \frac{1}{\sqrt{2}}$, $\sin = \frac{1}{\sqrt{2}}$.							
0	1	0	$*$	$*$	$*$	0	$\frac{1}{\sqrt{2}}(1+k)$
90	0	1	0	1	0	1	$\frac{1}{\sqrt{2}}(1+i)$
〃	〃	〃	90	0	1	0	$\frac{1}{\sqrt{2}}(1+j)$
6 2-axes, $\tfrac{1}{2}$ Rot. angle $= 90$, $\cos = 0$, $\sin = 1$.							
45	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	0	1	0	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}(i+k)$
〃	〃	〃	90	0	1	0	$\frac{1}{\sqrt{2}}(j+k)$
〃	〃	〃	180	−1	0	$-\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}(-i+k)$
〃	〃	〃	270	0	−1	0	$\frac{1}{\sqrt{2}}(-j+k)$
90	0	1	45	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}(i+j)$
〃	〃	〃	135	$-\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}(-i+j)$

[The Axial System of the Dodecahedron and Icosahedron tables occupy pp. 534–535, presenting similar quaternion rotation symbols with more complex entries involving $\sqrt{5}$ and golden ratio coordinates.]

11. Before proceeding further I remark that exclusively of the foregoing axial systems of the regular polyhedra the only cases are as follows:

A. A polyhedron may have a single q-axis, say Λ^q: taking this as the axis of Z the table is

X	Y	Z	Rot. Symbol
90	90	0	$\cos\frac{\pi}{q} + \sin\frac{\pi}{q}\cdot k$

B. It may have a single q-axis, and (symmetrically arranged in a plane at right angles thereto) q 2-axes, say Λ^q, qL^2. Taking the q-axis as the axis of Z and some one of the 2-axes as the axis of X, the table is

X	Y	Z	Rot. Symbols
One q-axis, $\frac{1}{2}$ Rot. angle $= \frac{\pi}{q}$.			
90	90	0	$\cos\frac{\pi}{q} + \sin\frac{\pi}{q}\cdot k$
q 2-axes, $\frac{1}{2}$ Rot. angle $= 90$.			
0	90	90	i
$\frac{\pi}{q}$	$90 - \frac{\pi}{q}$	90	$i\cos\frac{\pi}{q}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$(q-1)\frac{\pi}{q}$	$90 - \frac{(q-1)\pi}{q}$	90	$i\cos\frac{(q-1)\pi}{q} + j\sin\frac{(q-1)\pi}{q}$

and in particular if $q = 2$, the axes are $3L^2$ and the table is

X	Y	Z	Rot. Symbols
3 2-axes, $\frac{1}{2}$ Rot. angle $= 90$			
90	90	0	k
0	90	90	i
90	0	90	j

This in fact appears, Bravais, "Mémoire sur les polyèdres de forme symétrique," *Liouville*, t. XIV., pp. 141—180 (1843), observing that for the present purpose there is no distinction between his three cases

$$\Lambda^{2q+1},\ (2q+1)\,L^2;\quad \Lambda^{2q},\ qL^2,\ qL'^2;\quad \Lambda^{2q},\ 2qL^2.$$

12. The meaning of the rotation symbol is as follows: viz. if in general we have a rotation θ about an axis inclined at the angles X, Y, Z to any three rectangular axes, and if Π be the rotation symbol,

$$\Pi = \cos\tfrac{1}{2}\theta + \sin\tfrac{1}{2}\theta\,(i\cos X + j\cos Y + k\cos Z),$$

then if x, y, z are the original coordinates of any point of the body, and x', y', z' the coordinates of the same point after the rotation; the values of x', y', z' are given in terms of x, y, z by the formula

$$ix' + jy' + kz' = \Pi\,(ix + jy + kz)\,\Pi^{-1}.$$

This is in fact the form under which, in the paper "On certain results relating to Quaternions," *Phil. Mag.*, vol. XXVI. (1845), p. 141, [20], I exhibited the rotation formulæ of Euler and Rodrigues. See also my paper "On the application of Quaternions to the Theory of Rotation," *Phil. Mag.*, vol. XXXIII. (1848), p. 196, [68].

13. We have, it is clear,

$$\Pi^s = \cos s\theta + \sin s\theta\,(i\cos X + j\cos Y + k\cos Z)$$

which shows that Π^s is the symbol for the rotation Π repeated s times: (more generally performing on the body, first the rotation Π and then the rotation Φ about any axis, the same or different, the symbol of the resultant rotation is $= \Phi\Pi$). If Π be the symbol for a rotation through the angle $\frac{2\pi}{q}$, then the rotation which corresponds to the symbol Π^q is a rotation through 360, that is the body returns to its original position; it might at first sight appear that we ought to have $\Pi^q = 1$, and that the symbols 1, Π, Π^2, ... Π^{q-1} would form a *group*; this however is not so, for we have not $\Pi^q = 1$, but $\Pi^q = -1$; in fact, it is to be observed that to pass from $ix + jy + kz$ to $ix' + jy' + kz'$, we have to multiply by $\Pi\ (\ )\,\Pi^{-1}$, so that the symbol of the rotation is indifferently $\pm\Pi$, and that the rotation symbol -1 is thus equivalent to the rotation symbol $+1$. But as regards the formation of the group, the only difference is that it is not 1, Π, Π^2, ... Π^{q-1} which form a group of q symbols, but ± 1, $\pm\Pi$, $\pm\Pi^2$, ... $\pm\Pi^{q-1}$ which form a group of $2q$ symbols. And so in the axial system of any polyhedron, if Π be the rotation symbol of any q-axis, then taking for each axis of the polyhedron the set of symbols $\pm\Pi$, $\pm\Pi^2$, $\dots$ $\pm\Pi^{q-1}$, and besides the two symbols ± 1, the whole series of symbols form together a group.

14. Thus in the before-mentioned case B $(q = 2)$ we have the eight symbols

$$\pm 1,\ \pm i,\ \pm j,\ \pm k$$

forming (as they obviously do) a group. In the general case B, putting for shortness

$$\Theta = \cos\frac{\pi}{q} + \sin\frac{\pi}{q}\cdot k \quad \text{and} \quad \Phi_s = i\cos\frac{s\pi}{q} + j\sin\frac{s\pi}{q},$$

the group consists of the $4q$ symbols

$$\pm 1,\ \pm\Theta,\ \ldots\ \pm\Theta^{q-1};\quad \pm\Phi_1,\ \pm\Phi_2,\ \ldots\ \pm\Phi_{q-1},$$

(to verify that this is so, it is only necessary to form the equations

$$\Theta^r\Theta^s=\Theta^{r+s},\quad \Phi_s^2=-1,\quad \Theta^r\Phi_s=\Phi_{s+r},\quad \Phi^s\Theta^r=\Phi_{s-r},\quad \Phi_r\Phi_s=-\Theta^{r-s},$$

which are at once seen to be true).

15. The ± general case A gives merely the group of the $2q$ symbols

$$\pm 1,\ \pm\Theta,\ \ldots\ \pm\Theta^{q-1},$$

which has been already mentioned.

16. The tetrahedron gives the group of 24 symbols

$\frac{1}{2}(\pm 1\pm i\pm j\pm k)$	16 cube roots of ± 1		
$\pm i,\ \pm j,\ \pm k$	6 square	„	± 1
± 1	2 terms		
	24		

(the signs ± being all independent).

17. The cube and octahedron give the group of 48 symbols

$\frac{1}{\sqrt{2}}(\pm 1\pm i),\ \frac{1}{\sqrt{2}}(\pm 1\pm j),\ \frac{1}{\sqrt{2}}(\pm 1\pm k)$	12 fourth roots of ± 1		
$\frac{1}{2}(\pm 1\pm i\pm j\pm k)$	16 cube	„	± 1
$\pm i,\ \pm j,\ \pm k,\ \frac{1}{\sqrt{2}}(\pm j\pm k),\ \frac{1}{\sqrt{2}}(\pm k\pm i),\ \frac{1}{\sqrt{2}}(\pm i\pm j)$	18 square	„	± 1
± 1	2 terms		
	48		

(the signs ± being all independent).

18. The dodecahedron and icosahedron give the group of 120 symbols

$$\pm\frac{\sqrt{5}\pm 1}{4}\pm\tfrac{1}{2}j\pm\frac{\sqrt{5}\mp 1}{4}k,$$

etc. [full enumeration on pp. 538–539 with 48 fifth roots of ±1, 16 cube roots, 18 square roots, 16 additional terms, and ±1], totalling 120 symbols.

376.

THÉORÈME RELATIF À L'ÉQUILIBRE DE QUATRE FORCES AGISSANT À LA FOIS DANS UN MÊME PLAN.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. vii. (1866), pp. 4—5.]

Si le produit de quatre forces passant par les quatre sommets d'un quadrilatère a une valeur constante quelle que soit la transversale; les deux couples opposés produits par les forces sur les côtés correspondants, par exemple, sur AB, CD et sur BC, DA, sont égaux.

Considérons les quatre forces $AB\cdot p$, $BC\cdot q$, $CD\cdot r$, $DA\cdot s$ agissant à la fois dans un même plan le long des quatre côtés du quadrilatère $ABCD$. Prenons x et y pour coordonnées rectangulaires de l'intersection O d'une transversale quelconque avec AB; x', y' étant les coordonnées correspondantes de l'intersection avec CD; x'', y'' celles avec BC; x''', y''' celles avec DA.

Soit $AB\cdot p\cdot OM$, $CD\cdot r\cdot ON$, $BC\cdot q\cdot O'M'$, $DA\cdot s\cdot O'N'$ les moments respectifs des quatre forces autour de O, O', et que $\Sigma\Pi = AB\cdot p\cdot OM\cdot CD\cdot r\cdot ON = BC\cdot q\cdot O'M'\cdot DA\cdot s\cdot O'N'$, on voit que la propriété fondamentale revient à écrire que le produit $\Sigma\Pi$ est indépendant de la position de la transversale.

Désignant par k la constante du produit, on peut écrire

$$p\cdot r\cdot x\cdot x' = q\cdot s\cdot x''\cdot x''',$$

et en développant

$$\begin{aligned} &p\cdot r\cdot(x-x_0)\cdot(x'-x'_0)+p\cdot r\cdot x_0\cdot x' \\ &\qquad = q\cdot s\cdot(x''-x''_0)\cdot(x'''-x'''_0)+q\cdot s\cdot x''_0\cdot x'''_0,\end{aligned}$$

où x_0, x'_0, x''_0, x'''_0 sont les valeurs particulières de x, x', x'', x''' relatives à l'axe des y.

En écrivant de même les équations relatives à l'axe des x et à la droite à l'infini, on a la relation fondamentale

$$pr = qs,\quad pr\cdot x_0x'_0 = qs\cdot x''_0x'''_0,\quad pr\cdot y_0y'_0 = qs\cdot y''_0y'''_0,$$

d'où

$$\frac{x_0}{x''_0} = \frac{x'''_0}{x'_0},\quad \frac{y_0}{y''_0} = \frac{y'''_0}{y'_0};$$

d'après quoi on déduit les conditions

$$\frac{AB\cdot p}{BC\cdot q} = \frac{CD\cdot r}{DA\cdot s},\quad \frac{AB\cdot p}{DA\cdot s} = \frac{BC\cdot q}{CD\cdot r},$$

c'est-à-dire

$$AB \cdot p : BC \cdot q : CD \cdot r : DA \cdot s = \lambda : \mu : \lambda : \mu,$$

et enfin

$$AB \cdot p = CD \cdot r, \quad BC \cdot q = DA \cdot s,$$

ce qui démontre le théorème.

Cambridge, May 1863.

377.

NOTE SUR LA CORRESPONDANCE DE DEUX POINTS SUR UNE COURBE.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. vii. (1866), pp. 20—23.]

DANS une note insérée dans les *Nouvelles Annales de Mathématiques*, t. III. 1864, p. 355, M. Cayley a donné le théorème suivant:

Si deux points d'une cubique ont une correspondance (2, 2)*, la droite qui les joint*
passe par un point fixe; et réciproquement, si la droite qui joint les deux points
passe par un point fixe, ils ont une correspondance (2, 2).

M. Cayley a démontré ce théorème pour la cubique universelle; et il remarque qu'il subsiste pour toute courbe de la classe $n = 6$, qui est celle de la cubique universelle. Je vais montrer ici comment, au moyen des formules de Brill, le théorème peut être étendu à une courbe quelconque.

Soit m l'ordre de la courbe; nous avons entre les deux points P, P' la relation géométrique, que la droite PP' passe par un point fixe O. En nommant A un des points d'intersection de la droie OP avec la courbe, le nombre des valeurs de A qui se confondent avec P est $= m - 1$; de plus, quand le point P est donné, il y a un seul point P'; et quand P' est donné, ilyaunseulpointP.Onadoncunecorrespondencedel′espce(1, 1)entrelesdeuxpointsAetP';etdeplus,les(m - 1)valeursdeAquiseconfondentavecPcorrespon
d′ounecorrespondence(m - 1, 1)entrelespointsPetP',etparconsquentunecorrespondence(m, m).Onenconclut,aumoyendelaformuledeBrill,$\alpha + \beta = 2\nu m + \mu m(m-1)$, où μ est le nombre des points doubles et ν celui des points de rebroussement (on a $2\nu + 3\mu = m(m-1)$, formule de De Jonquières); et comme dans le cas actuel $\alpha = \beta = m$, on trouve $2m = m(m-1)$, c'est-à-dire $m = 3$.

Il faut donc, pour étendre le théorème, considérer une correspondance (m, n) quelconque et chercher s'il existe une relation entre la droite PP' et les points P, P'.

Considérons la correspondance (m, n) entre les points P et P'. Quand P est donné,P'*peuttreennpositionssurlacourbe*; *lesdroites*PP'*enveloppentdoncunecourbedelac* les droites PP' enveloppent une courbe de la classe m. La relation entre les points P, P' et la droite PP' est donc telle que la droite PP' soit tangente à deux courbes, l'une de la classe n et variable avec P, l'autre de la classe m et variable avec P'.

Considérons spécialement la courbe de la classe n qui correspond à la position donnée de P; si le point P se déplace sur la courbe en suivant la loi de la correspondance, cette courbe de la classe n se déplace d'une manière déterminée, et nous avons ainsi entre les points P et P' une nouvelle relation; la courbe de la classe n touche en P' la courbe de la classe m qui correspond à la position de P'; or P' peut occuper m positions: donc la courbe de la classe n touche m fois la courbe de la classe m correspondante; et réciproquement la courbe de la classe m touche n fois la courbe de la classe n correspondante. D'où il suit que les deux courbes se touchent; car, en écrivant que deux courbes de la classe n et m se touchent, on a une relation entre les deux points qui déterminent ces courbes, et par suite une relation entre les points P et P'.

Si $m = n = 2$, les deux courbes sont de la classe 2, et le théorème s'énonce ainsi:

Si deux points d'une courbe ont une correspondance $(2, 2)$*, les deux droites qui passent par le premier point et enveloppent une conique variable, et par le second point et enveloppent une autre conique variable, se coupent sur une conique fixe; et réciproquement.*

545 NOTE SUR LA CORRESPONDANCE DE DEUX POINTS SUR UNE COURBE. [377

On peut remarquer que la démonstration directe pour la cubique universelle peut s'étendre à une courbe quelconque, et donne lieu au théorème plus général suivant:

Si deux points d'une courbe de l'ordre m ont une correspondance
$(m-1, m-1)$,
la droite qui les joint est tangente à une courbe de la classe $m-2$; et réciproquement.

En effet, soient (x, y, z) et (x', y', z') les coordonnées des deux points; la condition que la droite qui les joint passe par le point fixe (α, β, γ) est

$$\begin{vmatrix} x, & y, & z \\ x', & y', & z' \\ \alpha, & \beta, & \gamma \end{vmatrix} = 0,$$

et l'équation de la correspondance est

$$(x\alpha' + y\beta' + z\gamma')\, P + (x\alpha'' + y\beta'' + z\gamma'')\, Q = 0,$$

où $P = 0$, $Q = 0$ sont des équations de l'ordre $m - 2$; c'est-à-dire que, pour un point donné (x', y', z'), les valeurs de (x, y, z) sont déterminées par l'intersection de la courbe avec la droite passant par (x', y', z') et (α, β, γ), et avec une courbe de l'ordre $m-2$ déterminée par les coefficients de l'équation; ce qui démontre le théorème.

Cambridge, November 2, 1863.

378.

REPORT OF A COMMITTEE, ETC.

[From the Report of the British Association, 1865, pp. 34—35.]

REPORT of the Committee for the purpose of promoting the extension, improvement, and harmonic analysis of the Mathematical Tables in *Graveshande's Natural Philosophy.*

Presented October, 1865.

THE Committee have to report, that the new work is in course of publication at the Clarendon Press, Oxford, under the editorship of Professor Price, F.R.S., under the title "The Tables of the Rev. Francis Bashforth, B.D., *for facilitating the computation of the trajectories of shot, and the calculation of the Ordnance Tables,* with the consent of the late Master-General of the Ordnance and the present Secretary of State for War." The Committee have not had an opportunity of seeing the work in a complete state, but they understand that it is now far advanced.

The Committee have for the present confined themselves to the consideration of the new work so far as it extends, or in a more extended form reproduces, the tables of *Graveshande's Natural Philosophy*; and in the first instance they have caused a copy of this work to be deposited in the library of the Royal Society. In so doing they would remark, that although the publication of *Graveshande's Natural Philosophy* is very creditable to the English science of the last century, it does not at the present day occupy the position which might have been expected from the great merits of the original treatise. It appears, however, that the work on which *Graveshande* himself most relied, and which he most frequently refers to in his own name, was a set of mathematical tables calculated for the purpose of facilitating the computation connected with the motion of projectiles, in accordance with the resistance-law adopted in the treatise. It was the object of *Graveshande* to reduce all such calculations to a series of uniform and systematic operations; but although in the main he was successful, he himself, in some instances, was compelled to have recourse to special tables, adapted to particular cases only. He was also obliged to omit the consideration of those cases in which the resistance of the air is so great that the motion becomes nearly horizontal.

THE Committee recommend that they should be reappointed, and that they should be authorized to extend their inquiries to other mathematical tables, and to report upon the present state and wants of science in this department.

Cambridge, May 1865.

379.

NOTICES OF COMMUNICATIONS TO THE BRITISH ASSOCIATION,

Eighth and Ninth Meetings of Section A.

Norwich, August 1868.

[From the Report of the British Association, 1868, pp. 9—11.]

On the Successive Contacts of a Straight Line with a Curve of the nth Order.

THE paper related to the theory of the singularities of plane curves. Considering a curve of the order m, and an arbitrary line, the intersections of the line with the curve are given by means of an equation of the order m; if the line is taken to be the line $x = 0$, and if from the equation $U = 0$ of the curve we form the equation $V = 0$, where V is the discriminant of U regarded as a function of x; the equation $V = 0$ gives the ordinates of the intersections of the line with the curve; and it may be the case that the successive coefficients (beginning with the last) of V, regarded as a function of y, all of them vanish; if the last s coefficients all vanish, then the line meets the curve in s coincident points, that is, it has s consecutive intersections, or (which is the same thing) a contact of the order $s - 1$ with the curve. The number of such lines which have with the curve a contact of the order $s-1$ is a question in the theory of singularities; for $s = 2$, that is for simple contacts, the number is of course $= m(m - 1)$; for $s = 3$, the number of inflexions is $= 3m(m - 2)$; for $s = 4$, the number of sextactic points is, by a formula due to Pl"ucker, $= 3(m - 2)(3m^2 - 9m - 5)$; and for $s = 5$, the number of points of 8-pointic contact, or say of octactic points, is, by a formula which I have obtained, $= \frac{9}{2}(m - 2)(m - 3)(5m^2 - 19m - 24)$.

379] NOTICES OF COMMUNICATIONS TO THE BRITISH ASSOCIATION. 550

On a Problem of Loci.

THE problem is as follows: Given the vertices of a polygon of any number of sides, inscribe in it a polygon the sides of which shall pass through given points respectively. The case of a triangle has been long known; the general case has been solved by M. Cremona (*Giornale*, t. III. 1861, pp. 20—25) and also by M. Bischoffsberger (ibid. pp. 294—299). The solution which I gave in the present paper was as follows. Supposing, for simplicity, that the polygon is a quadrilateral, then taking A, B, C, D for the vertices of the given quadrilateral, and P, Q, R, S for the given points; if AB meets PQ in A_1, and BC meets QR in B_1, then we have on the line AB the four points A, A_1, B, and a fourth point determined by the anharmonic ratio of the four points being equal to the given anharmonic ratio of the points P, Q, R, S; say this fourth point is B'; and similarly we have on the line BC the four points B_1, B, C, and a fourth point C' determined in like manner. The construction is then as follows: the side DA of the required quadrilateral is a line through S meeting AB in a point A' and CD in a point D', where A' and D' are such that the anharmonic ratios (A, A_1, B, A') and (D, D_1, C, D') are equal to the given anharmonic ratio of (P, Q, R, S); the construction is thus effected by means of a conic through the four points A, B, C, D, which touches the line SA' at A' and the line SD' at D', and which moreover passes through the point D_1 (where CD meets RS): this is a conic determined by six conditions, and which satisfies the condition of touching at A' and D'; the remaining side of the required quadrilateral is then the line joining the remaining intersections of the sides $A'B$ and $B'C$ with the conic respectively.

379. Notices of Communications to the British Association for the Advancement of Science.

[From the Reports of the British Association for the Advancement of Science.]

5. On Curves of the Third Order. Report, 1861, p. 2.

A curve of the third order or cubic curve is a section of a cubic cone and such cone is intersected by a concentric sphere in a spherical cubic. It is an obvious consequence of a theorem of Sir Isaac Newton's that there are five principal kinds of cubic cones, or what is the same thing five principal kinds of spherical cubics— but the nature of these five kinds of spherical cubics was first distinctly explained by Möbius. They may be designated the *simplex*, the *complex*, the *crunodal*, the *acnodal* and the *cuspidal*: where crunode, acnode, denote respectively the two species of double points (nodes), viz. the double point with two real branches, and the conjugate or isolated point. The foregoing results are known: the special object of the paper is to establish a subdivision of the simplex kind of spherical cubics. The simplex kind is a continuous reentering curve cutting a great circle, to fix the ideas say the equator, in three pairs of opposite points, which are the three real inflexions of the curve. The three great circles which are the tangents at the inflexions and the equator divide the entire surface of the sphere into fourteen regions whereof eight are trilateral and the remaining six are quadrilateral. The curve may be entirely in six out of the eight trilateral regions, and it is in this case said to be *simplextrilateral*; or it may lie entirely in the six quadrilateral regions, and it is in this case said to be *simplex quadrilateral*; and there is an intermediate form, the *simplex neutral*; viz. in this case the three great circles tangents at the inflexions meet in a pair of opposite points and there are in all only twelve regions all of them trilateral; the curve lies entirely in six of these regions.

6. On a Certain Curve of the Fourth Order. Report, 1862, p. 3.

The curve in question is the locus of the centres of the conics which pass through three given points and touch a given line; if the equations of the sides of the triangle formed by the three points are $x = 0$, $y = 0$, $z = 0$, these coordinates being such that $x + y + z = 0$ is the equation of the line infinity, and if $\alpha x + \beta y + \gamma z = 0$ be the equation of the given line, then (as is known) the equation of the curve is

$$\sqrt{\alpha x\,(y+z-x)} + \sqrt{\beta y\,(z+x-y)} + \sqrt{\gamma z\,(x+y-z)} = 0.$$

The special object of the communication was to exhibit the form of the curve in the case where the line cuts the triangle, and to point out the correspondence of the positions of the centre upon the curve, and the point of contact on the given line.

7. On the Representation of a Curve in Space by means of a Cone and Monoid Surface. Report, 1862, p. 3.

The author gave a short account of his researches recently published in the *Comptes Rendus.* The difficulty as to the representation of a curve in space is, that such a curve is not in general the complete intersection of two surfaces; any two surfaces passing through the curve intersect not only in the curve itself, but in a certain companion curve, which cannot be got rid of; this companion curve is in the proposed mode of representation reduced to the simplest form, viz. that of a system of lines passing through one and the same point. The two surfaces employed for the representation of a curve of the nth order are, a *cone* of the nth order having for its vertex an arbitrary point (say the point $x = 0$, $y = 0$, $z = 0$), and a *monoid* surface with the same vertex, viz. a surface the equation whereof is of the form $Qw - P = 0$, P and Q being homogeneous functions of (x, y, z) of the degrees p and $p-1$ respectively (where p is at most $= n - 1$). The monoid surface contains upon it $p\,(p-1)$ lines given by the equations ($P = 0$, $Q = 0$); and, the cone passing through $n\,(p-1)$ of these lines (if, as above supposed, $p = n - 1$, this implies that some of these lines are multiple lines of the cone), the monoid surface will besides intersect the cone in a curve of the nth order.

8. On a Formula of M. Chasles relating to the Contact of Conics. Report, 1864, p. 1.

The author gave an account of the recent investigations of M. Chasles in relation to the theory of conics, viz., M. Chasles has found that the properties of a system of conics, containing one arbitrary parameter, depend upon two quantities called by him the *characteristics* of the system; these are, μ, the number of conics of the system which pass through a given point, and, ν, the number of conics of the system which touch a given line; or, say, μ is the *parametric order*, ν the *parametric class*, of the system. And he exhibited a transformation obtained by him of a formula of M. Chasles for the number of conics which touch five given curves, viz., if (M, m), (N, n), (P, p), (Q, q), (R, r) be the orders and classes of the five given curves respec-tively, then the number of curves is

$$= (1,\ 2,\ 4,\ 4,\ 2,\ 1)\,(M,\ m)\,(N,\ n)\,(P,\ p)\,(Q,\ q)\,(R,\ r),$$

where the notation stands for $1.MNPQR + 2\Sigma mNPQR + 4\Sigma mnPQR +$ &c. The trans- formed formula in question was communicated by the author to

M. Chasles, and had appeared in the *Comptes Rendus*; but it is, in fact, included in a very beautiful and general theorem given in the same Number by M. Chasles himself.

9. On the Problem of the In-and-circumscribed Triangle. Report, 1864, p. 1.

The general problem of the in-and-circumscribed triangle may be thus stated, viz., to find a triangle the angles whereof severally lie in, and the sides severally touch, a given curve or curves; and we may, in the first instance, inquire as to the number of such triangles. The first and easiest case is when the curves are all distinct; here, if the angles lie in curves of the *orders* m, n, p, respectively, and the sides touch curves of the *classes* Q, R, S, respectively, then the number of triangles is $= 2mnpQRS$. The number may be obtained for some other cases; but the author has not yet considered the final and most difficult case, viz. that in which the angles severally lie in, and the sides severally touch, one and the same given curve.

The foregoing notices relate to verbal communications upon questions with which I was at the time occupied and which are for the most part more fully discussed in papers printed elsewhere. I remark upon them as follows:

1. I have no remembrance as to this; I think no paper printed or written.
2. See 175.
3. See 158.
4. No paper printed. The intention was to consider the different forms of trinodal quartic curves, in particular those with real nodes, as obtained from the inversion of a conic according to the different relations of the conic to the fundamental triangle. Thus according as the conic cuts in two real points, touches, or cuts in two imaginary points, a side of the triangle, the tangents at the corresponding node are real, coincident, or imaginary; viz. the node is a crunode, cusp, or acnode. And in the case of real intersections there is a further distinction according as the inter- sections lie each or either of them on the side itself, or on the side produced in one or other of the two directions. By considering the different relations of the conic to the fundamental triangle we thus obtain the different forms of the trinodal quartic.
5. See 351.
6. I think no paper printed or written.
7. See 302 and 305.

8. See 306.
9. The question is considered in a memoir "On the Problem of the In-and-circum- scribed Triangle," *Phil. Trans.* t. CLXI. (for 1871), pp. 369—412.

380. Note on the Rectangular Hyperbola.

[From the *Oxford, Cambridge and Dublin Messenger of Mathematics*, vol. I. (1862), p. 77.]

Every conic which passes through the points of intersection of two rectangular hyperbolas is a rectangular hyperbola. In fact if a conic be referred to rectangular axes, the condition that it may be a rectangular hyperbola is Coeff. of $x^2 = -$Coeff. of y^2. Hence if U, V be any two quadratic functions of x, y, and if λ be a constant, the condition in question being satisfied for each of the functions U, V, is satisfied for the function $U + \lambda V$: and the equation of any conic through the points of intersection of the conics $U = 0$, $V = 0$ is $U + \lambda V = 0$: which proves the theorem in question.

In particular if from two of the angles of a triangle perpendiculars are let fall on the opposite sides, and if the point of intersection of the perpendiculars and the third angle be joined: then since the first side and the perpendicular upon it are a rectangular hyperbola, and the second side and the perpendicular upon it are a rectangular hyperbola; the third side and the joining line must be a rectangular hyperbola: that is, these two lines must be at right angles to each other. We have thus the well-known theorem that the perpendiculars let fall from the angles of a triangle on the opposite sides meet in a point.

The theorem as to the hyperbolas is a particular case of the theorem that three conics which pass through the same four points are met by any line whatever in six points forming a system in involution. In fact a rectangular hyperbola is a conic meeting the line at infinity in two points harmonically related to the circular points at infinity: hence two of the conics being rectangular hyperbolas, the foci of the involution are the circular points at infinity: hence these points and the points in which the line at infinity meets the third conic are harmonically related to each other: that is, the third conic is a rectangular hyperbola.

381. Note on Bezout's Method of Elimination.

[From the *Oxford, Cambridge and Dublin Messenger of Mathematics*, vol. II. (1864), pp. 88, 89.]

Let U, U' be any two rational and integral functions of x of the same order; to fix the ideas let them be the cubic functions

$$\begin{aligned} U &= ax^3 + bx^2 + cx + d, \\ U' &= a'x^3 + b'x^2 + c'x + d'. \end{aligned}$$

Write

$$A = \begin{vmatrix} U, & U' \\ a, & a' \end{vmatrix}, \qquad P = \begin{vmatrix} U, & U' \\ a, & a' \end{vmatrix},$$

$$B = \begin{vmatrix} U, & U' \\ b, & b' \end{vmatrix}, \qquad Q = \begin{vmatrix} U, & U' \\ ax + b, & a'x + b' \end{vmatrix},$$

$$C = \begin{vmatrix} U, & U' \\ c, & c' \end{vmatrix}, \qquad R = \begin{vmatrix} U, & U' \\ ax^2 + bx + c, & a'x^2 + b'x + c' \end{vmatrix},$$

$$D = \begin{vmatrix} U, & U' \\ d, & d' \end{vmatrix}, \quad S = \begin{vmatrix} U, & U' \\ ax^3 + bx^2 + cx + d, & a'x^3 + b'x^2 + c'x + d' \end{vmatrix} = \begin{vmatrix} U, & U' \\ U, & U' \end{vmatrix} = 0,$$

then we have

$$\begin{aligned} P &= A, \\ Q &= Ax + B, \\ R &= Ax^2 + Bx + C, \\ S &= Ax^3 + Bx^2 + Cx + D = 0, \end{aligned}$$

and thence

$$\begin{aligned} A &= P, \\ B &= Q - Px, \\ C &= R - Qx, \\ D &= S - Rx = -Rx. \end{aligned}$$

Let α be an arbitrary quantity and write

$$\Box x = \begin{vmatrix} U & U' \\ a\alpha^3 + b\alpha^2 + c\alpha + d' & a'\alpha^3 + b'\alpha^2 + c'\alpha + d' \end{vmatrix};$$

we have it is clear

$$\begin{aligned} \Box &= A\alpha^3 + B\alpha^2 + C\alpha + D, \\ &= \alpha^2 P + \alpha^2 (Q - Px) + \alpha(R - Qx) + Rx, \\ &= (\alpha^3 - \alpha^2 x)\, P + (\alpha^2 - \alpha x)\, Q + (\alpha - x)\, R, \end{aligned}$$

and hence

$$\frac{\square}{\alpha - x} = \alpha^2 P + \alpha Q + R.$$

The equations $P = 0$, $Q = 0$, $R = 0$ are respectively quadratic equations in x, the equations which are used in Bezout's method of elimination; and representing them by

$$\begin{aligned} P &= Lx^2 + Mx + N = 0, \\ Q &= L'x^2 + M'x + N' = 0, \\ R &= L''x^2 + M''x + N'' = 0, \end{aligned}$$

we have

$$\begin{vmatrix} L, & M, & N \\ L', & M', & N' \\ L'', & M'', & N'' \end{vmatrix} = 0$$

as the equation resulting from the elimination of x from the equations $U = 0$, $U' = 0$. The foregoing investigation shows that the functions P, Q, R are obtained as the coefficients of α^2, α, 1 in the development of

$$\frac{1}{\alpha - x} \begin{vmatrix} U & U' \\ a\alpha^3 + b\alpha^2 + c\alpha + d, & a'\alpha^3 + b'\alpha^2 + c'\alpha + d' \end{vmatrix},$$

or more generally, taking U, U' to be any two functions of the order n, that the n functions P, Q, R, &c. each of the order $n - 1$ are obtained as the coefficients of α^{n-1}, α^{n-2}, $\ldots \alpha$, 1 in the development of

$$\frac{1}{\alpha - x} \begin{vmatrix} U, & U' \\ U_\alpha, & U'_\alpha \end{vmatrix},$$

where U_α, U'_α are what U, U' become when x is replaced therein by α: and we have thus a simple *à posteriori* verification of the form in which, several years ago, I presented Bezout's Method of Elimination.

2, *Stone Buildings, W.C., March 5, 1863.*

382. Note on the Tetrahedron.

[From the *Oxford, Cambridge and Dublin Messenger of Mathematics*, t. III. (1866), pp. 8—10.]

The following simple properties of a tetrahedron seem worth noticing.

In the tetrahedron $ABCD$ if $AC = BD$ and $AD = BC$, then the line joining the middle points of AB and CD, or say the points $\frac{1}{2}AB$ and $\frac{1}{2}CD$, cuts at right angles these lines AB and CD.

If $AB = CD$, then the line joining the points $\frac{1}{2}AC$, $\frac{1}{2}BD$, and the line joining the points $\frac{1}{2}AD$, $\frac{1}{2}BC$ (lines which in any tetrahedron meet each other), cut each other at right angles.

In fact if A, B, C, D have for their coordinates $(\alpha_1, \beta_1, \gamma_1)$, $(\alpha_2, \beta_2, \gamma_2)$, $(\alpha_3, \beta_3, \gamma_3)$, $(\alpha_4, \beta_4, \gamma_4)$: then the coordinates of the point $\frac{1}{2}AB$ are $\frac{1}{2}(\alpha_1 + \alpha_2)$, $\frac{1}{2}(\beta_1 + \beta_2)$, and so for the points $\frac{1}{2}CD$, &c.; the equations of the line through the points $\frac{1}{2}AB$, $\frac{1}{2}CD$ therefore are

$$\frac{x - \frac{1}{2}(\alpha_1 + \alpha_2)}{\alpha_1 + \alpha_2 - \alpha_3 - \alpha_4} = \frac{y - \frac{1}{2}(\beta_1 + \beta_2)}{\beta_1 + \beta_2 - \beta_3 - \beta_4} = \frac{z - \frac{1}{2}(\gamma_1 + \gamma_2)}{\gamma_1 + \gamma_2 - \gamma_3 - \gamma_4},$$

and I observe in passing that this line passes through the point whose coordinates are

$$\tfrac{1}{4}(\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4), \quad \tfrac{1}{4}(\beta_1 + \beta_2 + \beta_3 + \beta_4), \quad \tfrac{1}{4}(\gamma_1 + \gamma_2 + \gamma_3 + \gamma_4);$$

the other two similar lines pass through the same point, and the above-mentioned property of the general tetrahedron is thus proved.

The condition that the foregoing line may cut at right angles the line AB, the equations whereof are

$$\frac{x - \alpha_1}{\alpha_1 - \alpha_2} = \frac{y - \beta_1}{\beta_1 - \beta_2} = \frac{z - \gamma_1}{\gamma_1 - \gamma_2},$$

is at once seen to be

$$\Sigma\,(\alpha_1 - \alpha_2)(\alpha_1 + \alpha_2 - \alpha_3 - \alpha_4) = 0,$$

where Σ denotes the sum of the corresponding terms in α, β, γ. And so the condition that the same line may cut at right angles the line CD is

$$\Sigma\,(\alpha_3 - \alpha_4)(\alpha_1 + \alpha_2 - \alpha_3 - \alpha_4) = 0.$$

But the conditions $AC = BD$, $AD = BC$ give respectively

$$\Sigma\,[(\alpha_1 - \alpha_3)^2 - (\alpha_2 - \alpha_4)^2] = 0, \quad \Sigma\,[(\alpha_1 - \alpha_4)^2 - (\alpha_2 - \alpha_3)^2] = 0,$$

or writing these in the form

$$\Sigma\,(\alpha_1 - \alpha_3 - \alpha_2 + \alpha_4)(\alpha_1 + \alpha_3 - \alpha_2 - \alpha_4) = 0,$$
$$\Sigma\,(\alpha_1 - \alpha_4 - \alpha_2 + \alpha_3)(\alpha_1 + \alpha_4 - \alpha_2 - \alpha_3) = 0,$$

we obtain, by successively adding and subtracting, the two required equations.

The equations of the line through $\frac{1}{2}AC$, $\frac{1}{2}BD$ are

$$\frac{x-\frac{1}{2}(\alpha_1+\alpha_3)}{\alpha_1+\alpha_3-\alpha_2-\alpha_4}=\frac{y-\frac{1}{2}(\beta_1+\beta_3)}{\beta_1+\beta_3-\beta_2-\beta_4}=\frac{z-\frac{1}{2}(\gamma_1+\gamma_3)}{\gamma_1+\gamma_3-\gamma_2-\gamma_4},$$

and those of the line through $\frac{1}{2}AD$, $\frac{1}{2}BC$ are

$$\frac{x-\frac{1}{2}(\alpha_1+\alpha_4)}{\alpha_2+\alpha_4-\alpha_1-\alpha_3}=\frac{y-\frac{1}{2}(\beta_1+\beta_4)}{\beta_2+\beta_4-\beta_1-\beta_3}=\frac{z-\frac{1}{2}(\gamma_1+\gamma_4)}{\gamma_2+\gamma_4-\gamma_1-\gamma_3};$$

and the condition that these may cut at right angles is

$$\Sigma\,(\alpha_1+\alpha_4-\alpha_2-\alpha_3)(\alpha_1+\alpha_4-\alpha_2-\alpha_3)=0,$$

that is

$$\Sigma\,\{(\alpha_1-\alpha_2)^2-(\alpha_3-\alpha_4)^2\}=0,$$

which is in fact the condition $AB=CD$.

Combining the two theorems we see that if in a tetrahedron the pairs of opposite sides are respectively equal, then the line joining the centres of opposite sides cuts these sides at right angles, and moreover the three joining lines cut each other at right angles.

A tetrahedron of the form in question may be constructed as follows: viz. taking a parallelogram $ABCD$, whereof the diagonals AC, BD are unequal, then bending the parallelogram about its shorter diagonal AC in such manner that in the solid figure BD becomes equal to AC, we have a tetrahedron the opposite sides whereof are respectively equal.

Or it may be constructed even more simply as follows: viz. if $ABCD'$ and $A'B'C'D$ be parallel faces of any rectangular parallelopiped (the angles A and A', B and B', C and C', D and D' being respectively opposite to each other), then $ABCD$ or $A'B'C'D'$ is a tetrahedron of the form in question. The consideration of the rectangular parallelopiped puts in evidence the foregoing geometrical property.

In such a tetrahedron the line joining the centres of a pair of opposite sides is in the language of Bravais, see his "Mémoire sur les polyèdres de forme symétrique," *Liouville*, t. XIV. (1849), pp. 141—180, a binary axis of symmetry: viz. the figure is not altered by turning it round such axis through an angle $=\frac{1}{2}\cdot 360°$. There are thus three such axes at right angles to each other, but the figure has not any centre of symmetry, nor (assuming that it is not further particularised) any plane of symmetry: each of the three axes is a principal axis, and the figure belongs to the sixth of Bravais' twenty-three classes of polyhedra, see the table p. 179. It was in fact by seeking to construct a figure of this class that I was led to the investigation.

383. Problems and Solutions.

[From the *Mathematical Questions with their Solutions from the Educational Times*, vols. I. to IV., 1863 to 1865.]

[Vol. I. (June 1863 to June 1864), pp. 18, 19.]

1373. (By T. T. Wilkinson, F.R.A.S.)—Given a circle (C) and any point A, either within or without the circle: through A draw BAD cutting the circle in B, D. Then it is required to find another point E, such that, if LEM be drawn cutting the circle in L, M, we may always have $AE^2 = LE \cdot EM \pm BA \cdot AD$.

Solution by Professor Cayley.

Consider a circle centre O and radius OA, and in relation thereto a point M either outside or inside the circle, and suppose that

$(OA)^2-(OM)^2$, or the "squared inner potency" of M is denoted by $\square i{\cdot}M$,

and

$(OM)^2-(OA)^2$, or the "squared outer potency" of M is denoted by $\square o{\cdot}M$,

so that, for an outside point, $\square o \cdot M = -\square i \cdot M$, is the square of the tangential distance of M from the circle; and, for an inside point, $\square i \cdot M = -\square o \cdot M$, is the square of the shortest semi-chord through M.

Suppose now that M is a given point; the proposed question is in effect to find the locus of a point P such that $\pm \square o \cdot P \pm \square o \cdot M = (MP)^2$; but we have thus in reality four different questions according as the signs are assumed to be $++$, $+-$, $-+$ or $--$; the case $++$, or when $\square o{\cdot}P+\square o{\cdot}M = (MP)^2$, is perhaps the most inter- esting.

Taking the radius as unity, (α, β) as the coordinates of M, and (x, y) as the coordinates of P, we have here

$$(x^2+y^2-1)+(\alpha^2+\beta^2-1) = (x-\alpha)^2+(y-\beta)^2, \text{ or } \alpha x+\beta y-1=0;$$

that is, the locus of P is a right line, the polar of M in regard to the circle.

It may be remarked, that, when M is an inside point, then throughout the locus P is an outside point; and, replacing the negative quantity $\square o \cdot M$ by its value, $= -\square i \cdot M$, we have $\square o \cdot P - \square i \cdot M = (MP)^2$. If, however, M is an outside point, then in part of the locus P is an outside point, and we have $\square o \cdot P + \square o \cdot M = (MP)^2$, while in the remainder of the locus P is an inside point, and, replacing the negative quantity $\square o \cdot P$ by its value, $= -\square i \cdot P$, we have $-\square i \cdot P + \square o \cdot M = (MP)^2$. For the case $+-$, the locus of P is a right line, but for each of the other two cases $-+$ and $--$ the locus is a circle; the discussion of the several cases presents no particular difficulty.

[Vol. I. pp. 43—45.]

1387. (By W. K. Clifford.)—1. Four common tangents are drawn to a circle and an ellipse which passes through the centre (O) of the circle; if A, B be opposite intersections of the tangents, prove that OA and OB are equally inclined to the tangent at O to the ellipse.

2. If a straight line A join the poles of B with respect to two conics, prove that the lines joining AB to a pair of opposite intersections of common tangents, form, with A, B, an harmonic pencil.

3. If a point A be the intersection of the polars of B with respect to two conics, and AB be cut by a pair of common chords in C, D, prove that $ACBD$ is an harmonic range.

2. Solution by Professor Cayley.

This elegant theorem is included as a particular case in the known theorem, "Given three conics inscribed in the same quadrilateral, the tangents from any point to these conics form a pencil in involution."

Mr Clifford's theorem is in fact as follows: viz., Four common tangents are drawn to a circle and an ellipse which passes through the centre O of the circle; if A, B be opposite intersections of the tangents, then OA, OB are equally inclined to the tangent at O to the ellipse.

This comes to saying that the tangent at O to the ellipse, say OT, is the double or sibi-conjugate line of the involution of the pencil formed by the lines OA, OB, and the lines OI, OJ drawn from O to the circular points at infinity; and if we replace the circle by an arbitrary conic S, and the line at infinity by an arbitrary line IJ, the theorem will be as follows:

Consider a conic S; a line meeting this conic in the points I, J; and the point O, the intersection of the tangents at I, J, or (what is the same thing) the pole of the line IJ in regard to the conic. If through the point O there be drawn any other conic Θ, and if A, B be opposite intersections of the common tangents of the conics S, Θ; then the tangent OT at the point O to the conic Θ is the double or sibi-conjugate line of the involution of the pencil formed by the lines OA, OB, and the lines OI, OJ; or, as we may also express it, the lines OT, OT, the lines OA, OB, and the lines OI, OJ form a pencil in involution.

Now, considering the two points or point-pair (A, B) as a conic inscribed in the quadrilateral formed by the common tangents of the conics S and Θ, the conics S and Θ and the point-pair (A, B) are a system of three conics inscribed in the same quadrilateral; and hence, by the general theorem above referred to, if O' be any point whatever, the tangents from O' to the conic S, the tangents from O' to the conic Θ, and the tangents from O' to the point-pair (that is, the two lines $O'A$, $O'B$) form a pencil in involution. But, if O' coincide with O, then the tangents to the conic S are the lines OI, OJ; and the tangents to the conic Θ are the coincident lines OT, OT;

and we have thence the theorem in question; viz. that the lines OT, OT, the lines OI, OJ, and the lines OA, OB form a pencil in involution.

[Vol. I. pp. 77—79.]

1409. (By W. K. Clifford.)—For every point A on a conic section there exists a straight line BC, not meeting the curve, such that, if through any other point K on the conic there be drawn any two straight lines meeting BC in B, C, and the curve in D, E, the angles BAC, DAE are either equal or supplementary.

Solution by Professor Cayley.

I find that this very elegant theorem depends on the lemma to be presently stated, and that it is intimately connected with Newton's theorem for the organic description of a conic, or, what is the same thing, with the theorem of the anharmonic relation of the points of a conic.

LEMMA.

If AT be the tangent, and AS any other line through a point A of a conic, and if two lines equally inclined to AT and AS respectively meet the conic in the points K and D (viz., if $\angle TAK = SAD$, the two angles being measured in opposite directions from AT, AS respectively); then the line KD meets AS in a fixed point B, that is, a point the position of which is independent of the magnitude of the equal angles.

To prove this, take A for the origin, and the bisectors of the angle TAS for the axes of x and y: then the equation of the conic is

$$ax^2 + 2hxy + by^2 + 2fy + 2gx = 0;$$

the equation of the tangent at the origin, that is, the line AT, is $gx+fy = 0$; and hence the equation of the line AS is $gx - fy = 0$. Taking $y = \alpha x$ for the equation of the line AK, we have, for the coordinates x_1, y_1 of the point K where this meets the conic,

$$(a + 2h\alpha + b\alpha^2)\, x_1 + 2(f\alpha + g) = 0, \quad y_1 = \alpha x_1;$$

and then the equation of the line AD will be $y = -\alpha x$, and we shall have, for the coordinates x_2, y_2 of the point D where this meets the conic,

$$(a - 2h\alpha + b\alpha^2)\, x_2 + 2(f\alpha + g) = 0, \quad y_2 = -\alpha x_2.$$

The equation of the line KD is

$$\begin{vmatrix} x, & y, & 1 \\ x_1, & \alpha x_1, & 1 \\ x_2, & -\alpha x_2, & 1 \end{vmatrix} = 0,$$

that is

$$\alpha x\,(x_1+x_2)+y\,(x_2-x_1)-2\alpha x_1x_2=0;$$

and for the coordinates of the point B where this meets the line AS, the equation whereof is $gx-fy=0$, we have

$$x\,\{f\alpha(x_1+x_2)+g(x_2-x_1)\}-2f\alpha x_1x_2=0,$$

or, as this may be written,

$$x\left\{\frac{f\alpha+g}{x_1}-\frac{-f\alpha+g}{x_2}\right\}-2f\alpha=0.$$

But we have

$$\frac{f\alpha+g}{x_1}=-\tfrac{1}{2}(a+2h\alpha+b\alpha^2),\quad \frac{-f\alpha+g}{x_2}=-\tfrac{1}{2}(a-2h\alpha+b\alpha^2);$$

and hence the equation is

$$x(-2h\alpha)-2f\alpha=0,$$

giving $x=-\dfrac{f}{h}$, and thence $y=-\dfrac{g}{h}$, for the coordinates of the point B; and, these being independent of α, the lemma is seen to be true.

Consider now the points A, K as fixed points on the conic; then, revolving about A the constant angle DAB, and about K the constant (zero) angle DKB, the locus of B is (by the theorem of the anharmonic relation of the points of a conic) given in the first instance as a conic through the points A, K; but, observing that a position of the angle DAB is TAK, and that the corresponding position of DKB is AKA, the line AK is part of the locus; and the locus is made up of this line and a line BC. And, conversely, given the fixed points A, K, and the line BC, the original conic is, by Newton's theorem, described by means of the constant angles DAB, DKB revolving about these points in such a manner that the arms AB, KB generate by their intersections the line BC. This being so, the other two arms AD, KD generate by their intersections the conic.

And then, considering the two positions DAB, EAC of the angle DAB (so that D, B are in a line with K, and E, C are also in a line with K), we have $\angle DAB=\angle EAC$, that is, $\angle DAE=\angle BAC$, which is Mr Clifford's theorem.

It has been seen that, A being given, the same line BC is obtained whatever be the position of the point K; and, taking AK for the normal at A, it at once appears geometrically that (as remarked by Mr Clifford) the line BC is the polar of the point Θ of intersection of all the chords which subtend a right angle at A.

[Professor Cayley's lemma may be otherwise proved, as follows:

The trilinear equation of the conic, referred to two tangents (α at A, β at S) and their chord of contact (γ or AS), is $U = \lambda\alpha\beta - \gamma^2 = 0$; and the equation of two straight lines (AK, AD) equally inclined to α, γ is

$$(\alpha - \mu\gamma)(\mu\alpha - \gamma) = 0, \quad \text{or} \quad V = \alpha^2 + \gamma^2 - (\mu + \mu^{-1})\,\alpha\gamma = 0;$$

also $U + V = 0$ denotes a conic passing through the intersections of U and V; but $U + V$ is resolvable into $\alpha = 0$, or the tangent AT, and

$$\alpha + \lambda\beta - (\mu + \mu^{-1})\,\gamma = 0,$$

which is, therefore, the equation of the chord KD; whence we see that KD meets AS (or γ) in a point B (given by $\gamma = 0$, $\alpha + \lambda\beta = 0$) whose position is independent of μ, that is, of the equal angles SAD, TAK.]

[Vol. I. pp. 125—127.]

1478. (By J. McDowell, M.A.)—(α) Two sides of a given triangle always pass through two fixed points; prove that the third side always touches a fixed circle.

(β) Two sides of a given triangle touch two fixed circles; prove that the third side also touches a fixed circle.

(γ) Two sides of a given polygon touch fixed circles; prove that all the remaining sides also touch fixed circles.

3. Solution by Professor Cayley.

Since the theorem (γ) follows at once from (β), and (α) is included in (β), it is only necessary to prove (β). Consider three given circles, and let it be proposed to construct a triangle the sides whereof touch the given circles, and which is similar to a given triangle; the direction of one side may be assumed at pleasure, and then the triangle is determined. Impose now on the triangle the condition that the area is equal to a given quantity; we obtain for the given area an expression involving the angle θ which fixes the direction of one of the sides, and we have thus an equation for the determination of the angle θ. But, for a properly determined relation between the data of the problem, the expression for the area becomes independent of the angle θ, that is, every triangle, the sides whereof touch the three circles, and which is similar to a given triangle, is of the same area, or say, the area of every such triangle is equal to a given quantity Δ; and, this being so, it is clear that, if we construct a triangle similar to a given triangle and of the given area Δ (that is, a triangle equal to a given triangle), in such manner that two of the sides touch two of the given circles, then the envelope of the remaining side will be the remaining given circle; which is in fact the theorem (β).

It only remains therefore to show that the foregoing porismatic case of the problem exists.

For the first circle, let the coordinates of the centre be a, b, and the radius be c; and suppose in like manner that we have a', b', and c' for the second circle, and a'', b'', and c'' for the third circle. Let λ, λ', λ'' be the inclinations to the axis of x of the perpendiculars on the sides which touch these circles respectively; then the equations of the three sides respectively are

$$\begin{aligned}(x-a)\cos\lambda+(y-b)\sin\lambda-c&=0,\\(x-a')\cos\lambda'+(y-b')\sin\lambda'-c'&=0,\\(x-a'')\cos\lambda''+(y-b'')\sin\lambda''-c''&=0.\end{aligned}$$

If the triangle be similar to a given triangle, then the differences of the angles λ, λ', λ'' will be given angles, or, what is the same thing, we may write

$$\lambda=\theta+\xi,\quad \lambda'=\theta'+\xi,\quad \lambda''=\theta''+\xi,$$

where θ, θ', θ'' are given angles, and ξ is a variable angle. Let Δ be the area of the triangle, then (disregarding a merely numerical factor) we have

$$\begin{aligned}&\sqrt{\Delta}=\sin(\lambda'-\lambda'')\,(a\cos\lambda+b\sin\lambda+c)\\&+\sin(\lambda''-\lambda)\,(a'\cos\lambda'+b'\sin\lambda'+c')+\sin(\lambda-\lambda')\,(a''\cos\lambda''+b''\sin\lambda''+c'');\end{aligned}$$

or, what is the same thing,

$$\begin{aligned}\sqrt{\Delta}=\sin(\theta'-\theta'')\,&\{a\cos(\theta+\xi)+b\sin(\theta+\xi)+c\}\\&+\sin(\theta''-\theta)\,\{a'\cos(\theta'+\xi)+b'\sin(\theta'+\xi)+c'\}\\&\qquad+\sin(\theta-\theta')\,\{a''\cos(\theta''+\xi)+b''\sin(\theta''+\xi)+c''\}.\end{aligned}$$

It is now clear that the right-hand side will be independent of ξ, if only

$$\begin{aligned}\sin(\theta'-\theta'')(a\cos\theta+b\sin\theta)+\sin(\theta''-\theta)(a'\cos\theta'+b'\sin\theta')&\\+\sin(\theta-\theta')(a''\cos\theta''+b''\sin\theta'')&=0,\end{aligned}$$

$$\begin{aligned}\sin(\theta'-\theta'')(-a\sin\theta+b\cos\theta)+\sin(\theta''-\theta)(-a'\sin\theta'+b'\cos\theta')&\\+\sin(\theta-\theta')(-a''\sin\theta''+b''\cos\theta'')&=0;\end{aligned}$$

equations which show that, given the form of the triangle and the centres of two of the circles, the centre of the third circle (in the porismatic case) is a determinate unique point: and the theorem is thus proved.

[Vol. I. pp. 137—141.]

1273. (By the Editor [W. J. Miller, B.A.].)—In a given triangle let three triangles be inscribed, by joining the points of contact of the inscribed circle, the points where the bisectors of the angles meet the sides, and the points where the per- pendiculars meet the sides; then will the corresponding sides of these three triangles pass through the same point; also the triangle formed by the three points of inter- section will be a circumscribed co-polar to the original triangle, and the pole will be on the straight line in which the sides of the given triangle meet the bisectors of its exterior angles.

1. Solution by Professor Cayley.

The theorem is, in fact, included in the following more general

THEOREM.

Let the points O, O', O'', ... lie on a conic circumscribed about a triangle ABC; then first the polars of the points O, O', O'', ... in regard to the triangle (see Note at the end of the Solution) pass through a fixed point Ω. And secondly, if by means of the point O, joining it with the vertices A, B, C, and taking the intersections of these lines with the sides BC, CA, AB, respectively, we form a triangle inscribed in the triangle ABC; and the like for the points O', O'', ...; the corresponding sides of the inscribed triangles meet in three points forming a triangle circumscribed about the original triangle ABC, and such that the lines joining the corresponding vertices of the last-mentioned two triangles meet in the point Ω.

But, in order to see that the proposed theorem 1273 is in fact included under the foregoing more general one, it is necessary to state the following

SUBSIDIARY THEOREM.

Consider a conic inscribed in the triangle ABC, and passing through the points I, J.

Take O the pole of the line IJ in regard to the conic; O' the point of inter- section of the lines joining the vertices of the triangle with the points of contact on the opposite sides respectively; O'' the point of intersection of the lines Al, Bm, Cn, where l is a point on BC such that the lines lA, lBC, lI, lJ form a harmonic pencil, and the like for the points m and n respectively.

Then the points O, O', O'' lie on a conic circumscribed about the triangle ABC.

In fact, if in the subsidiary theorem the inscribed conic be a circle, and the points I, J be the circular points at infinity, the point O will be the centre of the circle, that is, the point of intersection of the interior bisectors of the angles; O' will be the point of intersection of the lines to the points of contact of the inscribed circle; and O'' the point of intersection of the perpendiculars on the sides of the triangle; and, these three points being on a conic circumscribed about the triangle, the general theorem will apply to the three points in question.

I first prove the subsidiary theorem. Taking $x=0$, $y=0$, $z=0$ for the equations of the sides of the triangle and (α,β,γ), (α',β',γ') for the coordinates of the points I, J respectively; the equation of the inscribed conic is

$$\begin{vmatrix} \sqrt{x}, & \sqrt{y}, & \sqrt{z} \\ \sqrt{\alpha}, & \sqrt{\beta}, & \sqrt{\gamma} \\ \sqrt{\alpha}', & \sqrt{\beta}', & \sqrt{\gamma}' \end{vmatrix} = 0,$$

or say

$$a\sqrt{x}+b\sqrt{y}+c\sqrt{z}=0,$$

where

$$a=\sqrt{\beta}\gamma'-\sqrt{\beta}'\gamma=p-p_1, \quad b=\sqrt{\gamma}\alpha'-\sqrt{\gamma}'\alpha=q-q_1, \quad c=\sqrt{\alpha}\beta'-\sqrt{\alpha}'\beta=r-r_1, \text{ suppose}$$

The coordinates of the point of intersection of the lines from the vertices to the points of contact on the opposite sides are

$$x:y:z=\frac{1}{a^2}:\frac{1}{b^2}:\frac{1}{c^2},$$

that is,

$$=\frac{1}{(p-p_1)^2}:\frac{1}{(q-q_1)^2}:\frac{1}{(r-r_1)^2}.$$

The equation of the line IJ is

$$(\beta\gamma'-\beta'\gamma)\,x+(\gamma\alpha'-\gamma'\alpha)\,y+(\alpha\beta'-\alpha'\beta)\,z=0;$$

or, what is the same thing,

$$(p^2-{p_1}^2)\,x+(q^2-{q_1}^2)\,y+(r^2-{r_1}^2)\,z=0.$$

Representing this for a moment by $\lambda x+\mu y+\nu z=0$, the coordinates of the pole of this line, in regard to the inscribed conic $a\sqrt{x}+b\sqrt{y}+c\sqrt{z}=0$, are as

$$c^2\mu+b^2\nu : a^2\nu+c^2\lambda : b^2\lambda+a^2\mu.$$

Now

$$\begin{aligned} c^2\mu+b^2\nu &= (r-r_1)^2(q^2-{q_1}^2)+(q-q_1)^2(r^2-{r_1}^2), \\ &= (r-r_1)(q-q_1)\,[(r-r_1)(q+q_1)+(q-q_1)(r+r_1)], \\ &= 2\,(r-r_1)(q-q_1)\,(qr-q_1r_1), \end{aligned}$$

and we have the like values for $a^2\nu+c^2\lambda$ and $b^2\lambda+a^2\mu$ respectively; hence, omitting the symmetrical factor, we have, for the coordinates of the point in question,

$$x:y:z=\frac{1}{pp_1}:\frac{1}{qq_1}:\frac{1}{rr_1}.$$

Taking the equation of the line Al to be $Qy+Rz=0$, those of the lines lI, lJ will be

$$x=\lambda\,(Qy+Rz),\quad x=\lambda'\,(Qy+Rz),$$

where

$$\lambda=\frac{\alpha}{Q\beta+R\gamma},\quad \lambda'=\frac{\alpha'}{Q\beta'+R\gamma'};$$

and the harmonic condition gives $\lambda+\lambda'=0$, that is,

$$Q\,(\alpha\beta'+\alpha'\beta)+R\,(\alpha\gamma'+\alpha'\gamma)=0;$$

the equation of the line Al is thus found to be

$$(\gamma\alpha'+\gamma'\alpha)\,y=(\alpha\beta+\alpha'\beta)\,z;$$

and, since we have the like forms for the equations of the lines Bm and Cn, we have for the coordinates of the point of intersection of these three lines

$$x:y:z=\frac{1}{\beta\gamma'+\beta'\gamma}:\frac{1}{\gamma\alpha'+\gamma'\alpha}:\frac{1}{\alpha\beta'+\alpha'\beta},$$

that is

$$=\frac{1}{p^2+{p_1}^2}:\frac{1}{q^2+{q_1}^2}:\frac{1}{r^2+{r_1}^2}.$$

The equation of a conic circumscribed about the triangle ABC is

$$\frac{\lambda}{x}+\frac{\mu}{y}+\frac{\nu}{z}=0,$$

where λ, μ, ν are arbitrary coefficients; and the condition for the three points being in the conic is thus found to be

$$\begin{vmatrix}(p-p_1)^2, & (q-q_1)^2, & (r-r_1)^2\\ pp_1, & qq_1, & rr_1\\ p^2+{p_1}^2, & q^2+{q_1}^2, & r^2+{r_1}^2\end{vmatrix}=0,$$

but, in virtue of the relations

$$(p-p_1)^2=-2pp_1+(p^2+{p_1}^2),\ \text{\&c.},$$

this equation is identically true, and the subsidiary theorem is thus proved.

Passing now to the general theorem, I prove the first part of it as follows:

The equation of a conic circumscribed about the triangle $x = 0$, $y = 0$, $z = 0$ is

$$\frac{A}{x} + \frac{B}{y} + \frac{C}{z} = 0;$$

hence, if (α, β, γ), $(\alpha', \beta', \gamma')$, $(\alpha'', \beta'', \gamma'')$ are the coordinates of any three points on the conic, we have

$$\frac{A}{\alpha} + \frac{B}{\beta} + \frac{C}{\gamma} = 0, \quad \frac{A}{\alpha'} + \frac{B}{\beta'} + \frac{C}{\gamma'} = 0, \quad \frac{A}{\alpha''} + \frac{B}{\beta''} + \frac{C}{\gamma''} = 0,$$

and thence

$$\begin{vmatrix} \frac{1}{\alpha}, & \frac{1}{\beta}, & \frac{1}{\gamma} \\ \frac{1}{\alpha'}, & \frac{1}{\beta'}, & \frac{1}{\gamma'} \\ \frac{1}{\alpha''}, & \frac{1}{\beta''}, & \frac{1}{\gamma''} \end{vmatrix} = 0,$$

which is the condition for the intersection in a point of the three lines

$$\begin{gathered} \frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 0, \\ \frac{x}{\alpha'} + \frac{y}{\beta'} + \frac{z}{\gamma'} = 0, \\ \frac{x}{\alpha''} + \frac{y}{\beta''} + \frac{z}{\gamma''} = 0; \end{gathered}$$

and the theorem in question is thus proved. I remark, in passing, that the theorem might also be stated as follows:—The locus of a point O, such that its polar in regard to the triangle ABC passes through a fixed point Ω, is a conic circumscribed about the triangle.

To prove the second part of the theorem, take for the coordinates of the points O, O', O'' respectively (α, β, γ), $(\alpha', \beta', \gamma')$, $(\alpha'', \beta'', \gamma'')$; then

$$\begin{vmatrix} \frac{1}{\alpha}, & \frac{1}{\beta}, & \frac{1}{\gamma} \\ \frac{1}{\alpha'}, & \frac{1}{\beta'}, & \frac{1}{\gamma'} \\ \frac{1}{\alpha''}, & \frac{1}{\beta''}, & \frac{1}{\gamma''} \end{vmatrix} = 0,$$

and if X, Y, Z are the coordinates of the point Ω, then we have

$$\begin{gathered} \frac{X}{\alpha} + \frac{Y}{\beta} + \frac{Z}{\gamma} = 0, \\ \frac{X}{\alpha'} + \frac{Y}{\beta'} + \frac{Z}{\gamma'} = 0, \\ \frac{X}{\alpha''} + \frac{Y}{\beta''} + \frac{Z}{\gamma''} = 0. \end{gathered}$$

The equations of the sides of the inscribed triangle obtained by means of the point O are

$$-\frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=0, \qquad \frac{x}{\alpha}-\frac{y}{\beta}+\frac{z}{\gamma}=0, \qquad \frac{x}{\alpha}+\frac{y}{\beta}-\frac{z}{\gamma}=0,$$

and the like for the triangles obtained by means of the points O' and O'' respectively. Hence, for a set of corresponding sides of the three triangles, we have, e.g.,

$$-\frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=0, \quad -\frac{x}{\alpha'}+\frac{y}{\beta'}+\frac{z}{\gamma'}=0, \quad -\frac{x}{\alpha''}+\frac{y}{\beta''}+\frac{z}{\gamma''}=0,$$

and it is clear that these equations are simultaneously satisfied by the values

$$x:y:z=-X:Y:Z,$$

and we have the like expressions for the other sets of corresponding sides; that is, we have for the coordinates of the vertices of the resulting triangle

$$(-X:Y:Z), \quad (X:-Y:Z), \quad (X:Y:-Z);$$

and hence also the equations of the sides of the triangle in question are

$$\frac{y}{Y}+\frac{z}{Z}=0, \quad \frac{z}{Z}+\frac{x}{X}=0, \quad \frac{x}{X}+\frac{y}{Y}=0,$$

that is, it is a triangle circumscribed about the triangle ABC. The equations of the lines joining the corresponding vertices of the two triangles are

$$\frac{y}{Y}=\frac{z}{Z}, \quad \frac{z}{Z}=\frac{x}{X}, \quad \frac{x}{X}=\frac{y}{Y},$$

and these lines meet in the point $(X:Y:Z)$, which is the point Ω, the intersection of the polars of O, O', O''; the demonstration of the theorem is thus completed.

{The expression Polar of a point in regard to a triangle denotes a line constructed as follows:—viz., O being the point and ABC the triangle, then, taking on BC a point a, the harmonic in regard to the points B and C of the intersection of BC by AO; and in like manner on CA and AB the points b and c respectively, the three points a, b, c lie on a line which is the polar of the point O. If the equations of the sides are $x=0$, $y=0$, $z=0$, and the coordinates of the point are (α,β,γ), then the equation of the polar is $\frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=0$; the equation may also be written $(\alpha\delta_x+\beta\delta_y+\gamma\delta_z)^2\,xyz=0$, and it thus appears that the line just defined as the polar is in fact the second or line polar of the point in regard to the three lines BC, CA, AB considered as forming a cubic curve.}

[Vol. II. July to December 1864, pp. 6—9.]

1505. (Proposed by Professor Cayley.)—If P, Q, 1, 2, 3, 4 be points on a conic, then the four points $P1$, $Q2$; $P2$, $Q1$; $P3$, $Q4$; $P4$, $Q3$ lie on a conic passing through the points P and Q.

Solution by the Proposer.

This is an immediate consequence of the theorem of the anharmonic property of the points of a conic. For if $(P1, P2, P3, P4)$ denote the anharmonic ratio of the lines $P1$, $P2$, $P3$, $P4$, and so in other cases; then

$$(P2, P1, P4, P3) = (P1, P2, P3, P4) = (Q1, Q2, Q3, Q4);$$

that is

$$(P2, P1, P4, P3) = (Q1, Q2, Q3, Q4),$$

which proves the theorem.

In particular, if P, Q are the circular points at infinity, then the conic is a circle. Moreover the points $P1$, $Q2$; $P2$, $Q1$ are the antifocal points of 1, 2; viz., calling these $1'$, $2'$, then 12 and $1'2'$ are lines at right angles to each other, having a common centre O, but such that $1'2' = i.12$, ($i = \sqrt{-1}$, as usual); or, what is the same thing, $O1 = O2 = i.O1' = i.O2'$. And the theorem is as follows: viz., if 1, 2, 3, 4 are points on a circle, and

$1'$, $2'$ are the antifocal points of 1, 2,
$3'$, $4'$ " " 3, 4,

then $1'$, $2'$, $3'$, $4'$ are points on a circle.

As an *à posteriori* proof, take the centre of the given circle as origin, so that (α_1, β_1), (α_2, β_2), (α_3, β_3), (α_4, β_4) being the coordinates of 1, 2, 3, 4, and the radius being taken as unity, we have

$$\alpha_1{}^2 + \beta_1{}^2 = \alpha_2{}^2 + \beta_2{}^2 = \alpha_3{}^2 + \beta_3{}^2 = \alpha_4{}^2 + \beta_4{}^2 = 1.$$

Suppose for a moment that x, y are the coordinates of the antifocal points of 1, 2; we have

$$x - \alpha_1 \pm i\,(y - \beta_1) = 0, \quad x - \alpha_2 \mp i\,(y - \beta_2) = 0,$$

that is

$$x + iy = \alpha_1 + i\beta_1, \quad x - iy = \alpha_2 - i\beta_2,$$

for the coordinates of the one point; and similarly

$$x - iy = \alpha_1 - i\beta_1, \quad x + iy = \alpha_2 + i\beta_2,$$

for the coordinates of the other point.

Hence, taking the new coordinates

$$X = x + iy, \quad Y = x - iy,$$

and similarly $A_1 = \alpha_1 + i\beta_1$, $B_1 = \alpha_1 - i\beta_1$, &c.; the coordinates of the antifocal points $1'$, $2'$ are (A_1, B_2) and (A_2, B_1) respectively; but we have $A_1 B_1 = {\alpha_1}^2 + {\beta_1}^2 = 1$, $A_2 B_2 = {\alpha_2}^2 + {\beta_2}^2 = 1$; so that $B_1 = \dfrac{1}{A_1}$, $B_2 = \dfrac{1}{A_2}$; and the coordinates are $\left(A_1, \dfrac{1}{A_2}\right)$, $\left(A_2, \dfrac{1}{A_1}\right)$ respectively. Similarly the coordinates of the antifocal points $(3', 4')$ are $\left(A_3, \dfrac{1}{A_4}\right)$, $\left(A_4, \dfrac{1}{A_3}\right)$ respectively.

Take as the equation of the circle through the two pairs of antifocal points

$$x^2 + y^2 + 2\lambda x + 2\mu y + \nu = 0,$$

or, what is the same thing,

$$XY + \lambda\,(X + Y) - i\mu\,(X - Y) + \nu = 0,$$

that is

$$XY + LY + MX + N = 0,$$

if

$$L = \lambda + i\mu, \quad M = \lambda - i\mu, \quad N = \nu.$$

We ought then to have

$$\begin{aligned}
\frac{A_1}{A_2} + L\,\frac{1}{A_2} + MA_1 + N &= 0,\\
\frac{A_2}{A_1} + L\,\frac{1}{A_1} + MA_2 + N &= 0,\\
\frac{A_3}{A_4} + L\,\frac{1}{A_4} + MA_3 + N &= 0,\\
\frac{A_4}{A_3} + L\,\frac{1}{A_3} + MA_4 + N &= 0;
\end{aligned}$$

and these will exist simultaneously, if

$$\begin{vmatrix}
\dfrac{A_1}{A_2}, & \dfrac{1}{A_2}, & A_1, & 1\\
\dfrac{A_2}{A_1}, & \dfrac{1}{A_1}, & A_2, & 1\\
\dfrac{A_3}{A_4}, & \dfrac{1}{A_4}, & A_3, & 1\\
\dfrac{A_4}{A_3}, & \dfrac{1}{A_3}, & A_4, & 1
\end{vmatrix} = 0,$$

an identical equation which is easily verified. It, in fact, gives

$$\left(\frac{1}{A_2}-\frac{1}{A_1}\right)(A_3-A_4)-(A_1-A_2)\left(\frac{1}{A_4}-\frac{1}{A_3}\right)+\left(\frac{A_1}{A_2}-\frac{A_2}{A_1}\right)(1-1)+$$
$$(1-1)\left(\frac{A_3}{A_4}-\frac{A_4}{A_3}\right)-\left(\frac{1}{A_3}-\frac{1}{A_1}\right)(A_3-A_4)+(A_1-A_2)\left(\frac{1}{A_4}-\frac{1}{A_3}\right)=0,$$

which is obviously true. The equation may also be written

$$\begin{vmatrix} 1, & A_1, & A_2, & A_1A_2 \\ 1, & A_2, & A_1, & A_1A_2 \\ 1, & A_3, & A_4, & A_3A_4 \\ 1, & A_4, & A_3, & A_3A_4 \end{vmatrix} = 0,$$

and in this form it expresses the known theorem of the equality of the anharmonic ratios of (A_1, A_2, A_3, A_4) and (A_2, A_1, A_4, A_3).

But, in order to actually find the circle, we may write

$$\begin{array}{rll} & XY + LY + MX & = 0, \\ A_1 + L & \qquad + MA_1A_2 + NA_2 & = 0, \\ A_2 + L & \qquad + MA_1A_2 + NA_1 & = 0, \\ A_3 + L & \qquad + MA_3A_4 + NA_4 & = 0, \end{array}$$

and eliminating L, M, N, the equation of the circle is

$$\begin{vmatrix} XY, & Y, & X, & 1 \\ A_1, & 1, & A_1A_2, & A_2 \\ A_2, & 1, & A_1A_2, & A_1 \\ A_3, & 1, & A_3A_4, & A_4 \end{vmatrix} = 0,$$

or, reducing, this is

$$(A_2-A_1)\left[XY(A_3A_4-A_1A_2)+Y\{A_1A_2(A_3+A_4)-A_3A_4(A_1+A_2)\}\right.$$
$$\left.+X(A_1+A_2-A_3-A_4)+(A_3A_4-A_1A_2)\right]=0,$$

or say

$$XY(A_1A_2-A_3A_4)+Y\{A_3A_4(A_1+A_2)-A_1A_2(A_3+A_4)\}$$
$$+X\{A_3+A_4-A_1-A_2\}+(A_1A_2-A_3A_4)=0;$$

that is

$$\begin{vmatrix} XY+1, & X, & Y \\ A_1+A_2, & A_1A_2, & 1 \\ A_3+A_4, & A_3A_4, & 1 \end{vmatrix} = 0,$$

which is the required equation; or, transforming to the original axes, we have $x+iy=X$, $x-iy=Y$, &c., and therefore $XY=x^2+y^2$; and the equation becomes

$$\begin{vmatrix} x^2+y^2+1, & x+iy, & x-iy \\ \alpha_1+\alpha_2+i(\beta_1+\beta_2), & (\alpha_1+i\beta_1)(\alpha_2+i\beta_2), & 1 \\ \alpha_3+\alpha_4+i(\beta_3+\beta_4), & (\alpha_3+i\beta_3)(\alpha_4+i\beta_4), & 1 \end{vmatrix} = 0,$$

which is the equation of the circle through the two pairs of antifocal points.

{Note. The *second* form of the equation of the circle may be otherwise deduced from the *first*, without expanding the determinants, by the following method:

$$\begin{vmatrix} XY, & Y, & X, & 1 \\ A_1, & 1, & A_1A_2, & A_2 \\ A_2, & 1, & A_1A_2, & A_1 \\ A_3, & 1, & A_3A_4, & A_4 \end{vmatrix} = \begin{vmatrix} XY+1, & Y, & X, & 1 \\ A_1+A_2, & 1, & A_1A_2, & A_2 \\ A_1+A_2, & 1, & A_1A_2, & A_1 \\ A_3+A_4, & 1, & A_3A_4, & A_4 \end{vmatrix} =$$

$$\begin{vmatrix} XY+1, & Y, & X, & 1 \\ A_1+A_2, & 1, & A_1A_2, & A_2 \\ 0, & 0, & 0, & A_1-A_2 \\ A_3+A_4, & 1, & A_3A_4, & A_4 \end{vmatrix} = (A_1-A_2) \begin{vmatrix} XY+1, & X, & Y \\ A_1+A_2, & A_1A_2, & 1 \\ A_3+A_4, & A_3A_4, & 1 \end{vmatrix};$$

therefore

$$\begin{vmatrix} XY+1, & X, & Y \\ A_1+A_2, & A_1A_2, & 1 \\ A_3+A_4, & A_3A_4, & 1 \end{vmatrix} = 0. \quad \text{Ed. [W. J. M.]}$$

}

[Vol. II. pp. 22—24.]

1513. (Proposed by the Rev. J. Blissard, B.A.)—Prove the following formulæ:

(1) $$\frac{(x-1)(x-2)\dots(x-n)}{x(x+1)\dots(x+n-1)} =$$

$$1+(-1)^n\left\{n.\frac{1}{x}-\frac{n(n^2-1^2)}{1^2}.\frac{1}{x+1}+\frac{n(n^2-1^2)(n^2-2^2)}{1^2.2^2}.\frac{1}{x+2}-\text{\&c.}\right\}.$$

(2) The above formula expressed as

$$\frac{(\Gamma x)^2}{\Gamma(x-n)\,\Gamma(x+n)} = 1-\frac{n^2}{1}.\frac{1}{x}+\frac{n^2(n^2-1^2)}{1.2}.\frac{1}{x(x+1)}-\frac{n^2(n^2-1^2)(n^2-2^2)}{1.2.3}.\frac{1}{x(x+1)(x+}$$

and show that this equation is subject to the sole restriction that when n is not integral x must not be negative.

Solution by Professor Cayley; and X. U. J.

Let n be a positive integer, and suppose that $[x]^n$ denotes as usual the factorial $x(x-1)\ldots(x-n+1)$; then we have

$$[x+k]^n = (1+\Delta)^k\,[x]^n = \left(1 + k\Delta + \frac{k(k-1)}{1.2}\Delta^2 + \&\text{c.}\right)[x]^n$$
$$= [x]^n + \frac{kn}{1}\,[x]^{n-1} + \frac{k(k-1)\,n(n-1)}{1.2}\,[x]^{n-2} + \&\text{c.};$$

or putting $k=-n$ we have

$$[x-n]^n = [x]^n - \frac{n^2}{1}\,[x]^{n-1} + \frac{n^2(n^2-1^2)}{1.2}\,[x]^{n-2} - \&\text{c.}$$

Writing herein $(x+n-1)$ for x, and dividing by $[x+n-1]^n$, we have

$$\frac{[x-1]^n}{[x+n-1]^n} = 1 - \frac{n^2}{1}.\frac{1}{x} + \frac{n^2(n^2-1^2)}{1.2}.\frac{1}{x(x+1)} - \&\text{c.};$$

or, what is the same thing,

$$\frac{(\Gamma x)^2}{\Gamma(x-n)\,\Gamma(x+n)} = 1 - \frac{n^2}{1}.\frac{1}{x} + \frac{n^2(n^2-1^2)}{1.2}.\frac{1}{x(x+1)} - \&\text{c.},$$

which is the formula (2). The foregoing demonstration applies to the case of n a positive integer; but as the two sides are respectively unaltered when n is changed into $-n$, it is clear that the formula holds good also for n a negative integer. The right hand side is the hypergeometric series $F(n,-n,x,1)$ and the formula therefore is

$$\frac{(\Gamma x)^2}{\Gamma(x-n)\,\Gamma(x+n)} = F(n,-n,x,1),$$

a particular case of the known formula

$$\frac{\Gamma(\gamma)\,\Gamma(\gamma-\alpha-\beta)}{\Gamma(\gamma-\alpha)\,\Gamma(\gamma-\beta)} = F(\alpha,\beta,\gamma,1),$$

which when α or β is a positive integer is a mere identity, true therefore for all values of γ; but if neither α nor β is a positive integer, then the right hand side is an infinite series which is only convergent for $\gamma > \alpha+\beta$. In the particular case we have $\alpha = n$, $\beta = -n$, $\gamma = x$; hence if n be a positive or negative integer, the formula is an identity, but if n be fractional, the condition of convergence is $x > 0$, that is, x must be positive.

To prove the formula (1) it is only necessary to remark, that (n being a positive integer) the quantity $\dfrac{[x-1]^n}{[x+n-1]^n}$ is a rational fraction, the numerator and denominator whereof are of the same degree n, and which becomes $= 1$ for $x = \infty$. Hence, decom- posing it into simple fractions, we may write

$$\frac{[x-1]^n}{[x+n-1]^n} = 1 + S_r.\frac{A_r}{x+r},$$

where the summation extends from $r = 0$ to $r = n-1$ both inclusive. And we have

$$A_r = \left\{ \frac{(x+r)\,[x-1]^n}{[x+n-1]^n} \right\}_{x=-r},$$

or, observing that $[x+n-1]^n = [x+n-1]^{n-r-1}\,(x+r)\,[x+r-1]^r$, we have

$$A_r = \left\{ \frac{[x-1]^n}{[x+n-1]^{n-r-1}\,[x+r-1]^r} \right\}_{x=-r} = \frac{[-r-1]^n}{[n-r-1]^{n-r-1}\,[-1]^r}$$
$$= \frac{(-)^n\,[n+r]^n}{[n-r-1]^{n-r-1}\,(-)^r\,[r]^r} = (-)^{n+r}\,\frac{[n+r]^{n+r}}{[n-r-1]^{n-r-1}\,[r]^r\,[r]^r} = (-)^{n+r}\,\frac{[n+r]^{2r+1}}{[r]^r \cdot [r]^r}.$$

Hence the formula is

$$\frac{[x-1]^n}{[x+n-1]^n} = 1 + (-)^n\,S_r\,(-)^r\,\frac{[n+r]^{2r+1}}{[r]^r\,[r]^r}\,\frac{1}{x+r},$$

or, as this may also be written,

$$\frac{(x-1)(x-2)\dots(x-n)}{x(x+1)\dots(x+n-1)} = 1+(-)^n\left\{ n.\frac{1}{x} - \frac{n(n^2-1^2)}{1^2}.\frac{1}{x+1} + \frac{n(n^2-1^2)(n^2-2^2)}{1^2.2^2}.\frac{1}{x+2}\right.$$

which is the formula in question.

[Vol. II. pp. 51, 52.]

1512. (Proposed by Professor Cayley.)—It is possible to construct a hexagon 123456, inscribed in a conic, and such that the diagonals 14, 25, 36 pass respectively through the Pascalian points (intersections of opposite sides) 23, 56; 34, 61; 45, 12. Given the points 1, 2; 4, 5; to construct the hexagon.

Solution by the Proposer.

Let 12, 45, meet in O, and through O draw at pleasure a line meeting 14 in P, and 25 in Q; let $P2$, $Q4$ meet in 3, and $P5$, $Q1$ in 6; then the line 36 will pass through O, and this being so, the hexagon 123456 satisfies the required conditions.

We have to show that 36 passes through O. Let $Q4$ meet $O12$ in A, and $P2$ meet $O45$ in B; then the points 6, 3, O, are the intersections of corresponding sides of the triangles $A1Q$, $B5P$; and in order that these points may lie in a line, the lines joining the corresponding vertices must meet in a point, that is, we have to show that the lines 15, AB, PQ meet in a point. The property is in fact as follows; viz., given the points 2, 4; and also the points Q, O, P lying in a line; then constructing the points 1, 5, A, B, which are the respective intersections of $P4$, $O2$; $Q2$, $O4$; $Q4$, $O2$; $P2$, $O4$; the lines 15, AB, PQ will meet in a point. Take $x = 0$, $y = 0$, $z = 0$ for the respective equations of $P2$, $Q4$, PQ; then O is an arbitrary point in the line PQ, say that for the point O we have $z = 0$, $ax + by = 0$; also $O2$, $O4$ are arbitrary lines through O: say that their equations are $ax + by + \lambda z = 0$; $ax + by + \mu z = 0$; then we have for the points A and B, respectively, $ax + by + \mu z = 0$, $y = 0$; $ax + by + \mu z = 0$, $x = 0$; hence the equation of AB is $\mu ax + \lambda by + \lambda\mu z = 0$. The equation of $P4$ is $ax + \mu z = 0$, and that of $Q2$ is $by + \lambda z = 0$; the point 1 is therefore given by $ax + \mu z = 0$, $ax + by + \lambda z = 0$; and 5 by $by + \lambda z = 0$, $ax + by + \mu z = 0$; hence the equation of 15 is $\mu ax + \lambda by + (\mu^2 - \mu\lambda + \lambda^2)\, z = 0$; and the equation of PQ being $z = 0$, it is clear that the three lines AB, 15, PQ intersect in the point given by the equations $\mu ax + \lambda by = 0$, $z = 0$.

Obs. 1. By inspection of the figure we see that $3PQ$ is a triangle whereof the sides $3P$, $3Q$, PQ pass respectively through the fixed points 2, 4, O; while the vertices P and Q lie in the fixed lines 14, 25; the locus of the vertex 3 is consequently a conic; and the like as regards the triangle $6PQ$.

Obs. 2. The regular hexagon projects into a hexagon inscribed in a conic and circumscribed about another conic having double contact therewith; in the hexagon so obtained (as appears at once by the consideration of the regular hexagon) the above-mentioned property holds; but the in-and-circumscribed hexagon has the additional property that the three diagonals meet in a point, and it is therefore a less general figure than the hexagon of the foregoing theorem. It would, I think, be worth while to study further the hexagon of the theorem.

{Note. In the solution of Question 1548 it is shown that if two pairs of opposite sides of *any hexagon* intersect each on a diagonal produced, so likewise will the third pair.

A slight variation of Professor Cayley's proof may be obtained by finding the equations of $P5$, $Q1$, and thence of 36, which are respectively

$$ax - (\lambda - \mu)\, z = 0, \quad bx + (\lambda - \mu)\, z = 0, \quad ax + by = 0,$$

showing that 36 passes through O. Ed. [W. J. M.].}

[Vol. II. pp. 70—72.]

1562. (Proposed by F. D. Thomson, M.A.)—Find the locus of the points of contact of tangents drawn from a given point to a conic circumscribing a given quadrangle. The quadrangle being supposed convex, trace the changes of form of the locus for different positions of the given point.

Solution by Professor Cayley; and the Proposer.

Let O be the given point; 1, 2, 3, 4 the vertices of the given quadrangle; A, B, C the centres of the quadrangle, viz., A the intersection of the lines 14, 23; B of 24, 31; C of 34, 12. The polars of O in regard to the several circumscribed conics intersect in a point O'. This being so, the locus is a cubic passing through the nine points 1, 2, 3, 4, A, B, C, O, O', and which is moreover such that the tangents at the four points 1, 2, 3, 4 meet the cubic in the point O, and the tangents at the four points A, B, C, O meet the cubic in the point O'. It is to be remarked that the nine points are so related to each other that a cubic through any eight of these points passes through the remaining ninth point; say a cubic through 1, 2, 3, 4, A, B, C, O passes through O'; the nine points consequently do not determine the cubic; but the cubic will be determined, e.g., by the conditions that it passes through 1, 2, 3, 4, A, B, C, O, and has $O1$ for the tangent at 1. The series of cubics corresponding to different positions of the point O is identical with the series of cubics passing through the seven points 1, 2, 3, 4, A, B, C (which is the series of cubics through the centres of a quadrangle).

Conversely any given cubic curve may be taken to be a cubic of the series; and the points 1, 2, 3, 4 will then be determined as follows, viz., 1, 2, 3, 4 are the points of contact of the tangents to the cubic from an arbitrary point O on the cubic; and then taking as before A, B, C for the intersections of 14, 23, of 24, 31 and of 34, 12, respectively, the points A, B, C will lie on the cubic, and the tangents at A, B, C, O will meet the cubic in a point O'. I call to mind that a cubic curve without singularities is either *complex* or *simplex*; in the simplex kind there can be drawn from any point of the curve two, and only two, real tangents to the curve; in the complex kind, there can be drawn four real tangents or else no real tangent, viz. from any point on a certain branch of the curve there can be drawn four real tangents, from a point on the remaining portion of the curve no real tangent. Hence, in the foregoing construction, in order that the points 1, 2, 3, 4 may be real, the given cubic must be of the complex kind, and the point O must be taken on the branch which has through each of its points four real tangents.

The foregoing results may be established *geometrically* or *analytically*; but for brevity I merely indicate the analytical demonstration. Suppose first, that the points 1, 2, 3, 4 are given as the intersections of the conics $U = 0$, $V = 0$; let α, β, γ be the coordinates of the point O, and write $D = \alpha\delta_x + \beta\delta_y + \gamma\delta_z$, so that $DU = 0$ and $DV = 0$ are the equations of the polars of O in regard to the conics $U = 0$, $V = 0$ respectively. The equation of any conic through the four points is $U + kV = 0$; and the equation of the polar of O in regard thereto is $DU + kDV = 0$; eliminating k from these equations, we have $UDV - VDU = 0$, which is the equation of the given locus. We see at once that it is a cubic curve passing through the points $(U = 0,\ V = 0)$, that is, the points 1, 2, 3, 4; and through the point $DU = 0$,

$DV = 0$, that is, the point O'; it also follows without difficulty that the curve passes through the point O. But for the remaining results it is better to particularize the conics $U = 0$, $V = 0$. Let the equations of 12, 23, 34, 41 be $x = 0$, $y = 0$, $z = 0$, $w = 0$ respectively, (where $x+y+z+w = 0$); and in the same system, let α, β, γ, δ be the coordinates of O ($\alpha+\beta+\gamma+\delta = 0$), then $xz = 0$, $yw = 0$ are each of them a conic (pair of lines) passing through the four points; and we may therefore write $U = yw$, $V = xz$; the equation $UDV - VDU = 0$ thus becomes $yw\,(\alpha z + \gamma x) - xz\,(\beta w + \delta y) = 0$, or, as this equation may also be written,

$$\frac{\alpha}{x} - \frac{\beta}{y} + \frac{\gamma}{z} - \frac{\delta}{w} = 0,$$

which is the equation of the cubic curve; and from this form the several above- mentioned results may be obtained without difficulty.

To give an idea of the form of the curve corresponding to a given convex quadrangle 1234, and given position of the point O, I suppose that O is situate *within* the quadrangle, for instance in the triangle $B12$. The mere inspection of the figure, and consideration of the conditions which are to be satisfied by the cubic curve, is enough to show that this is of the form described by Newton as *anguinea cum ovali*, viz., the oval passes through the points 3, 4, A, B, and the serpentine branch through the points 1, 2, C, O, O'. But the complete discussion of the different cases would be somewhat laborious.

[A geometrical investigation of the locus is given on p. 124 of Cremona's *Teoria Geometrica delle Curve Piane*. Ed. [W. J. M.]]

[Vol. II. pp. 89, 90.]

1533. (Proposed by Professor Cayley.)—If on the sides of a triangle there are taken three points, one on each side; and if through the three points and the three vertices of the triangle there are drawn a cubic curve and a quartic curve, inter- secting in six other points; then there exists a quintic curve passing through each of the three points, and having each of the six points for a double point.

Solution by the Proposer.

Let $P = 0$ be the equation of the quartic curve, $Q = 0$ the equation of the cubic curve, $M = 0$ the equation of the three sides of the triangle; then if we can find A, B, C functions of the orders 0, 1, 2 respectively, and U a function of the fifth order, such that we have identically $MU = AP^2 + BPQ + CQ^2$; we have $MU = 0$, a curve of the eighth order, having a double point at each of the points ($P = 0$, $Q = 0$), which points are the three vertices of the triangle, the three points, and the six points; but the curve $MU = 0$ is made up of the curve $M = 0$ (the three sides of the triangle, being a cubic curve having each of the vertices for a double point, and passing through each of the three points) and of a certain quintic curve $U = 0$; hence the quintic curve must pass through each of the three points, and have a double point at each of the six points; or there exists a quintic curve satisfying the conditions of the theorem.

I take $x = 0$, $y = 0$, $z = 0$ for the equations of the three sides of the triangle, and then (the constants being all of them arbitrary) writing for shortness

$$\begin{array}{lll}
\xi = .\ by + cz, & X = .\ \beta y + \gamma z, & \Theta = \lambda x + \mu y + \nu z, \\
\eta = a'x. \quad + c'z, & Y = \alpha' x. \quad + \gamma' z, & \\
\zeta = a''x + b''y. \quad , & Z = \alpha'' x + \beta'' y. \quad , &
\end{array}$$

I assume that the three points are given by the equations $(x = 0,\ \xi = 0)$, $(y = 0,\ \eta = 0)$, $(z = 0,\ \zeta = 0)$, respectively. This being so, we may write

$$Q = yz\xi\delta + zx\eta\delta' + xy\zeta\delta'' + xyz\epsilon = 0, \quad -P = yz\xi X + zx\eta Y + xy\zeta Z + xyz\Theta = 0,$$

for the equations of the cubic curve and the quartic curve respectively. We have of course $M = xyz = 0$ for the equation of the three sides of the triangle, and the identity to be satisfied is $xyzU = AP^2 + BPQ + CQ^2$.

I was led to the values of A, B, C by considerations founded on the theory of curves in space. We have

$$\begin{aligned}
&A = \delta\delta'\delta'', \\
&B = (\delta'\alpha'' + \delta''\alpha)\,\delta x + (\delta''\beta + \delta\beta'')\,\delta' y + (\delta\gamma' + \delta'\gamma)\,\delta'' z, \\
&C = \alpha'\alpha''\delta x^2 + \beta''\beta\delta' y^2 + \gamma\gamma'\delta'' z^2 + (\gamma\beta'\delta' + \gamma'\beta\delta'')\,yz + (\alpha'\gamma\delta'' + \alpha''\gamma'\delta)\,zx + (\beta''\alpha\delta + \beta
\end{aligned}$$

and with these values it is easy to show that the function $AP^2 + BPQ + CQ^2$ contains the factor xyz; for substituting the values of P, Q, all the terms of $AP^2 + BPQ + CQ^2$ contain explicitly the factor xyz, except the terms

$$\begin{aligned}
A\,(y^2z^2\xi^2X^2 + z^2x^2\eta^2Y^2 + x^2y^2\zeta^2Z^2) - B\,(y^2z^2\xi^2X\delta + z^2x^2\eta^2Y\delta' + x^2y^2\zeta^2Z\delta'') \\
+ C\,(y^2z^2\xi^2\delta^2 + z^2x^2\eta^2\delta'^2 + x^2y^2\zeta^2\delta''^2);
\end{aligned}$$

and these terms will contain the factor xyz, if only the expressions $AX^2 - BX\delta + C\delta^2$, $AY^2 - BY\delta' + C\delta'^2$, $AZ^2 - BZ\delta'' + C\delta''^2$ contain respectively the factors x, y, z. But $AX^2 - BX\delta + C\delta^2$ will contain the factor x, if only the expression vanishes for $x = 0$; and for $x = 0$ we have

$$\begin{aligned}
&AX^2 - BX\delta + C\delta^2 = 0 = \\
&\delta\delta'\delta''\,(\beta y + \gamma z)^2 - [\delta'\delta''\,(\beta y + \gamma z) + \delta\,(\beta''\delta' y + \gamma'\delta'' z)]\,\delta\,(\beta y + \gamma z) + (\beta y + \gamma z)\,(\beta''\delta' y + \gamma'\delta'' z)\,\delta^2;
\end{aligned}$$

that is, $AX^2 - BX\delta + C\delta^2$ contains the factor x; and by symmetry the other two expressions contain the factors y and z respectively. The excepted terms contain therefore the factor xyz; and there exists therefore a quintic function $U = (AP^2 + BPQ + CQ^2) \div xyz$; which proves the theorem.

The values of A, B, C were obtained by considering the surface $w = \dfrac{P}{Q}$, which, as is at once seen, contains upon itself the three lines

$$\left(x = 0,\ w = -\frac{X}{\delta}\right),\quad \left(y = 0,\ w = -\frac{Y}{\delta'}\right),\quad \left(z = 0,\ w = -\frac{Z}{\delta''}\right)$$

or as these equations may be written

$$\begin{aligned} &(x = 0, & &\beta y + \gamma z + \delta w = 0),\\ &(y = 0, \alpha' x & &+ \gamma' z + \delta' w = 0),\\ &(z = 0, \alpha'' x + \beta'' y & &+ \delta'' w = 0); \end{aligned}$$

and then seeking for the equation of the hyperboloid which passes through the three lines, this is found to be $Aw^2 + Bw + C = 0$, where A, B, C have the before-mentioned values.

If in the foregoing theorem the cubic is considered as a given cubic curve, and the three points as three arbitrary points on the cubic, the question then arises to find the triangle; or we have the problem proposed as Question 1607.

[Vol. II. p. 91.]

1542. (Proposed by Professor Cayley.)—If a given line meet two given conics in the points (A, B) and (A', B') respectively; and if (A'', B'') be the sibi-conjugate points (or foci) of the pairs (A, A') and (B, B'), or of the pairs (A, B') and (A', B), then (A'', B'') lie on a conic passing through the four points of intersection of the two given conics.

[Vol. II. pp. 97—100.]

1606. (Proposed by the Editor, [W. J. M.].)—Solve the following problems:

(α) Through three given points to draw a conic whose foci shall lie in two given lines.

(β) Through four given points to draw a conic such that one of its chords of intersection with a given conic shall pass through a given point.

(γ) Through two given points to draw a circle such that its chords of inter- section with a given circle shall pass through a given point.

Solution by Professor Cayley.

(α) Through three given points to draw a conic whose foci shall lie in two given lines.

The focus of a conic is a point such that the lines joining it with the two circular points at infinity (say the points I, J) are tangents to the conic. Hence the question is, in a given line to find a point A, and in another given line to find a point B, such that there exists a conic touching the four lines AI, AJ, BI, BJ (where I, J are any given points) and besides passing through three given points. More generally, instead of the lines from A, B to the given points I, J, we may consider the tangents from A, B, to a given conic Θ; the question then is, in a given line to find a point A, and in another given line to find a point B, such that there exists a conic touching the tangents from A, B to a given conic Θ, and besides passing through three given points. It is rather more convenient to consider the reciprocal question—though it is to be borne in mind that for any two reciprocal questions a solution of the one question by means of coordinates (x, y, z) regarded as point-coordinates is in fact a solution of the other question by means of the same coordinates (x, y, z) regarded as line-coordinates. The reciprocal question is: through a given point to draw a line A, and through another given point to draw a line B, such that there exists a conic passing through the intersections of these lines with a given conic Θ, and besides touching three given lines. The given points may be taken to be $(x = 0,\ z = 0)$, $(y = 0,\ z = 0)$; this determines the line $z = 0$, but not the lines $x = 0$, $y = 0$, so that the point $(x = 0,\ y = 0)$ may without loss of generality be supposed to lie on the conic Θ; the equation of this conic will therefore be

$$(a, b, 0, f, g, h \between x, y, z)^2 = 0.$$

I take $\alpha z + \gamma x = 0$ for the equation of the line A, $\mu y + \nu z = 0$ for the equation of the line B (so that the quantities to be determined are the ratios $\alpha : \gamma$ and $\mu : \nu$); this being so, the required conic passes through the intersections of these lines with the conic Θ; its equation will therefore be

$$(a, b, 0, f, g, h \between x, y, z)^2 + 2(\alpha z + \gamma x)(\mu y + \nu z) = 0;$$

or what is the same thing

$$(a, b, 2\nu\gamma, f + \mu\gamma, g + \nu\alpha, h + \mu\alpha \between x, y, z)^2 = 0;$$

where α, γ, μ, ν have to be determined in such manner that this conic may touch three given lines. It is to be observed that α, γ, μ, ν enter into the equation through the combinations $\alpha\mu$, $\alpha : \gamma$, and $\mu : \nu$, so that there are really only three disposable quantities.

[Vol. III. January to July, 1865, p. 29.]

1607. (Proposed by Professor Cayley.)—In a given cubic curve to inscribe a triangle such that the three sides shall pass respectively through three given points on the curve.

[Vol. III. pp. 60—63.]

1647. (Proposed by Professor Cayley.)—Find the locus of the foci of an ellipse of given major axis, passing through three given points.

{In connexion with the problem the Proposer remarks as follows:

Let A, B, C be the given points; take P an arbitrary point (not in general in the plane of the three given points), then we may find a point Q (not in general in the plane of the three given points) such that $QA+AP = QB+BP = QC+CP =$ given major axis. And this being so, if the locus of P be a given surface, then we shall have a certain surface, the locus of Q; and so if the locus of P be a given curve in space, then we shall have a given curve in space, the locus of Q. In particular, if the locus of P be the plane of the three given points, then the locus of Q will be a certain surface, cutting the plane in a curve which is the locus in the foregoing problem; and when Q is situate on this curve, then also P will be situate on the same curve. Or if the locus of P be the curve in question, then the locus of Q will be the same curve. Say, in general, that the loci of P and Q are reciprocal loci, then the curve in the problem *is its own reciprocal.* And we may propose the following question:

Find the curve or surface, the locus of P, which is its own reciprocal.

We have also analogous to the original problem the following question in Solid Geometry:

Given the four points A, B, C, D in space, to find the locus of the points P, Q such that

$$PA + AQ = PB + BQ = PC + CQ = PD + DQ = \text{a given line.}\}$$

Solution by the Proposer.

In general if a, b, c be the sides of a triangle, and f, g, h the distances of any point from the angles of the triangle (or, what is the same thing, if (a, b, c, f, g, h) be the distances of any four points in a plane from each other), then we have a certain relation

$$\phi\,(a, b, c, f, g, h) = 0.$$

Hence if r, s, t be the distances of the one focus from the angles of the triangle, and the major axis is $= 2\lambda$, then the distances for the other focus are $2\lambda - r$, $2\lambda - s$, $2\lambda - t$; and considering the three angles and the other focus as a system of four points, we have

$$\phi\,(a, b, c, 2\lambda - r, 2\lambda - s, 2\lambda - t) = 0,$$

which is a relation between the distances r, s, t of the first focus from the angles of the triangle, and which, treating these distances as coordinates (of course in a generalised sense of the term "Coordinate"), may be regarded as the equation of the required locus. It is to be observed, that we have identically

$$\phi\,(a, b, c, r, s, t) = 0,$$

and the equation may be expressed in the simplified form

$$\phi\,(a, b, c, 2\lambda - r, 2\lambda - s, 2\lambda - t) - \phi\,(a, b, c, r, s, t) = 0.$$

To develope the solution, I notice that the expression for the equation $\phi\,(a, b, c, f, g, h) = 0$ is

$$\begin{aligned} & b^2c^2\,(f^2 + g^2) + c^2a^2\,(h^2 + f^2) + a^2b^2\,(f^2 + g^2) \\ & \quad + g^2h^2\,(b^2 + c^2) + h^2f^2\,(c^2 + a^2) + f^2g^2\,(a^2 + b^2) \\ & \quad - a^2f^2\,(a^2 + f^2) - b^2g^2\,(b^2 + g^2) - c^2h^2\,(c^2 + h^2) \\ & \quad - a^2g^2h^2 - b^2h^2f^2 - c^2f^2g^2 - a^2b^2c^2 = 0; \end{aligned}$$

see my paper, "Note on the value of certain determinants &c.," *Quart. Math. Journ.* vol. III. (1860), pp. 275—277, [286]. Or, as this may also be written

$$\Sigma\,\{(b^2 + c^2 - a^2)\,(g^2h^2 + a^2f^2) - a^2f^4\} - a^2b^2c^2 = 0,$$

where Σ refers to the simultaneous cyclical permutation of (a, b, c) and of (f, g, h). Hence we have only in this equation to write $2\lambda - r$, $2\lambda - s$, $2\lambda - t$ in place of (f, g, h), and to omit the terms independent of λ, being in fact those which are equal to $\phi\,(a, b, c, r, s, t)$. Observing that we have

$$\begin{aligned} g^2h^2 + a^2f^2 &= [4\lambda^2 - 2\lambda(s + t) + st]^2 + a^2(2\lambda - r)^2 \\ &= 16\lambda^4 - 16\lambda^3(s + t) + 4\lambda^2(s^2 + t^2 + 4st + a^2) - 4\lambda\,[st(s + t) + a^2r] + s^2t^2 + \\ f^4 &= (2\lambda - r)^4 = 16\lambda^4 - 32\lambda^3r + 24\lambda^2r^2 - 8\lambda r^3 + r^4, \end{aligned}$$

the equation becomes

$$\begin{aligned} & 16\lambda^4\,\{\Sigma\,(b^2 + c^2 - a^2) - \Sigma a^2\} \\ & - 16\lambda^3\,\{\Sigma\,(b^2 + c^2 - a^2)\,(s + t) - 2\Sigma a^2r\} \\ & + 4\lambda^2\,\{\Sigma\,(b^2 + c^2 - a^2)\,(s^2 + t^2 + 4st + a^2) - 6\Sigma a^2r^2\} \\ & - 4\lambda\,\{\Sigma\,(b^2 + c^2 - a^2)\,[st(s + t) + a^2r] - 2\Sigma a^2r^3\} = 0, \end{aligned}$$

where the Σ's refer to the simultaneous cyclical permutation of the (a, b, c) and the (r, s, t). The coefficients of λ^4 and λ^3 are, it is easy to see, each $= 0$; and in the coefficient of λ^2 the terms $\Sigma\,(b^2 + c^2 - a^2)\,(s^2 + t^2) - 6\Sigma a^2r^2$ are $= -4\Sigma a^2r^2$; hence dividing the whole equation by 4λ, we find

$$\lambda\,\{\Sigma\,(b^2 + c^2 - a^2)\,(4st + a^2) - 4\Sigma a^2r^2\} - \{\Sigma\,(b^2 + c^2 - a^2)\,[st(s + t) + a^2r] - 2\Sigma a^2r^3\} = 0,$$

which is the required relation between (r, s, t).

It may be noticed that, expressing the distances r, s, t in terms of Cartesian or trilinear coordinates (x, y) or (x, y, z), then r^2, s^2, t^2 are rational and integral functions of the coordinates, and the form of the equation therefore is

$$A_2 + B_2 r + C_2 s + D_2 t + E_2 st + F_2 tr + G_2 rs = 0,$$

where the subscript numbers denote the degrees in regard to the coordinates. Multiply- ing this equation successively by 1, r, s, t, st, tr, rs, rst, we have eight equations linear in the last-mentioned eight quantities, the coefficients being of known degrees respectively; and eliminating the eight quantities, we have the rationalised equation expressed in the form, determinant (of order 8) $= 0$; viz. this is

$$\begin{vmatrix} A_2, & B_2, & C_2, & D_2, & E_2, & F_2, & G_2, & 0 \\ B_2r^2, & A_2, & G_2r^2, & F_2r^2, & 0, & D_2, & C_2, & E_2 \\ C_2s^2, & G_2s^2, & A_2, & E_2s^2, & D_2, & 0, & B_2, & F_2 \\ D_2t^2, & F_2t^2, & E_2t^2, & A_2, & C_2, & B_2, & 0, & G_2 \\ E_2s^2t^2, & 0, & D_2t^2, & C_2s^2, & A_2, & G_2s^2, & F_2t^2, & B_2 \\ F_2t^2r^2, & D_2t^2, & 0, & B_2r^2, & G_2r^2, & A_2, & E_2t^2, & C_2 \\ G_2r^2s^2, & C_2s^2, & B_2r^2, & 0, & F_2r^2, & E_2s^2, & A_2, & D_2 \\ 0, & E_2s^2t^2, & F_2t^2r^2, & G_2r^2s^2, & B_2r^2, & C_2s^2, & D_2t^2, & A_2 \end{vmatrix} = 0.$$

This seems to be of the degree 18 in the coordinates, but it is probable that the real degree is lower.

[Vol. III. pp. 63—65.]

1652. (Proposed by W. K. Clifford.)—Through the angles A, B, C of a plane triangle three straight lines Aa, Bb, Cc are drawn. A straight line AR meets Cc in R; RB meets Aa in P; PC meets Bb in Q; QA meets Cc in r; and so on. Prove that, after going twice round the triangle in this way, we always come back to the same point.

Show that the theorem is its own reciprocal. Find the analogous properties of a skew quadrilateral in space, and of a polygon of n sides in a plane.

Solution by Professor Cayley.

1. The theorem may be thus stated: Given three lines x, y, z, and in these lines respectively the points A, B, C; then there exist an infinity of hexagons, such that the pairs of opposite angles lie in the lines x, y, z, respectively, and that the pairs of opposite sides pass through the points A, B, C, respectively.

2. The demonstration is as follows: We have to show that, starting from an arbitrary point 1 in the line x, and constructing in the prescribed

manner (as shown successively in the figure) the points 2, 3, 4, 5, 6, the last side 61 of the hexagon 123456 will pass through B. By the construction, we have A, 2, 3 in a line, and likewise C, 4, 5; hence, by Pascal's theorem, applied to the six points in a pair of lines, the points of intersection of the lines $(25, 34)$, $(3C, A5)$, $(A4, C2)$, that is, the points B, 6, 1, lie in a line; which is the required theorem.

3. More generally suppose that the points A, B, C are not on the lines x, y, z, respectively. I remark that it is not in general possible to describe a hexagon such that the opposite angles lie in the lines x, y, z, respectively, and the opposite sides pass through the points A, B, C, respectively; but if there exists one hexagon (viz., a proper hexagon, not a triangle twice repeated), then there exists an infinity of such hexagons.

4. In fact, if it be required to find a polygon, the angles whereof lie in given lines respectively, and the sides whereof pass through given points respectively; the problem is either indeterminate or admits of only *two* solutions. If therefore in any particular case there are three or more solutions, the problem is indeterminate, and has an infinity of solutions. Now, in the above-mentioned case of the three lines and the three points, there exist *two* triangles, the angles whereof lie in the given lines, and the sides pass through the given points; and each triangle, taking the angles twice over in the same order 123123, is a hexagon satisfying the conditions of the problem; hence, if we have besides a proper hexagon satisfying the conditions of the problem, there are really *three* solutions, and the problem is therefore indeterminate.

5. Suppose that the three lines x, y, z, and also two of the three points, say the points A and B, are given; we may construct geometrically a locus, such that, taking for C any point of this locus, the problem shall be indeterminate: in fact, starting with the point 4, and constructing successively the points 3, 2; taking an arbitrary direction for the line 21, and constructing successively the points 1, 6, 5; then the intersection of the lines 21 and 54 is a position of the point C: and by taking any number of directions for the line 21, we obtain for each of them a different position of the point C; and so construct the locus.

6. The locus in question is, as will be shown, a line; and if the point A is on the line x, and the point B on the line y, then the locus of C will be the line z; that is, C being any point of the line z, the problem is indeterminate; which is Mr Clifford's theorem.

7. To prove this, consider the lines x, y, z, and also the points A, B, C, as given; the point 1 is an arbitrary point on the line x, linearly determined by means of a parameter u; and for every position of the point 1 we have a corresponding position of the point 4; let u' be the corresponding parameter for the point 4; the series of points 1 is homographic with the series of points 4; that is, the parameters u, u' are connected by an equation of the form $auu' + bu + cu' + d = 0$, (where of course a, b, c, d are functions of the parameters which determine the given lines x, y, z and the points A, B, C).

But if the problem be indeterminate, then starting from the point 1 and constructing the point 4, and again starting from the point 4 and making the very same construction, we arrive at the original point 1, that is, u must be the same function of u' that u' is of u; and this will be the case if $b = c$; hence $b = c$ is the condition in order that the problem may be indeterminate.

8. To effect the calculation, take $x = 0$, $y = 0$, $z = 0$ for the equations of the lines x, y, z respectively; and let (α, β, γ), $(\alpha', \beta', \gamma')$, $(\alpha'', \beta'', \gamma'')$ be the coordinates of the points A, B, C respectively. Let 1 and 4 be given as the intersections of the line $x = 0$ with the lines $y - uz = 0$, $y - u'z = 0$, respectively; and assume that for the point 2 we have $y = 0$, $z - vx = 0$, and for the point 3, $z = 0$, $x - wy = 0$. Then 1, C, 2 are in a line; as are also 2, A, 3; 3, B, 4; hence we obtain

$$v = \frac{\gamma'' u - \beta''}{\alpha'' u}, \quad w = \frac{\alpha v - \gamma}{\beta v}, \quad u' = \frac{\beta' w - \alpha}{\gamma' w};$$

therefore, eliminating v and w, we have

$$(\alpha\gamma'' - \alpha''\gamma)\,\gamma' u u' - \alpha\beta''\gamma' u' - (\alpha\beta'\gamma'' - \alpha'\beta\gamma'' - \alpha''\beta'\gamma)\,u - \beta''(\alpha'\beta - \alpha\beta') = 0.$$

The required condition, therefore, is

$$\alpha\beta'\gamma'' = \alpha\beta'\gamma'' - \alpha'\beta\gamma'' - \alpha''\beta'\gamma, \quad \text{or} \quad \alpha\beta'\gamma'' - \alpha\beta''\gamma' - \alpha'\beta\gamma'' - \alpha''\beta'\gamma = 0;$$

which is linear in regard to each of the three sets (α, β, γ), $(\alpha', \beta', \gamma')$, $(\alpha'', \beta'', \gamma'')$, separately; that is, two of the points A, B, C being given, the locus of the remaining point is a line. In particular, if $\alpha = 0$, $\beta' = 0$; then the equation becomes $\alpha'\beta\gamma'' = 0$, and assuming that neither $\alpha' = 0$ or $\beta = 0$, then the equation becomes $\gamma'' = 0$, that is, A, B being arbitrary points on the lines $x = 0$, $y = 0$ respectively, the locus of C is the line $z = 0$.

9. Mr Clifford's theorem is clearly its own reciprocal. I do not know the *precise* analogues of his special form of the theorem; but the analogue of the more general theorem stated in (6) is as follows: viz., we may have in the plane n lines x, y, z, … and n points A, B, C, …, such that there exist an infinity of $2n$-gons whereof the pairs of opposite angles lie in the given lines respectively; and the pairs of opposite sides pass through the given points respectively; and if the n lines and $n - 1$ of the n points be assumed at pleasure, then the locus of the remaining point is a line. It is moreover clear by the principle of reciprocity, that if the n points and $n - 1$ of the n lines be assumed at pleasure, then the envelope of the remaining line is a point.

There exists also an analogue in space; viz. we may have n lines x, y, z, … and n lines A, B, C, … such that there exist an infinity of (skew) $2n$-gons whereof the pairs of opposite angles lie in the given lines x, y, z, … respectively; and the pairs of opposite sides meet in the given lines

A, B, C, ...respectively. It may be added, that if all but one of the $2n$ lines x, y, z, ... A, B, C ... are given, then the 'six coordi- nates' of the remaining line satisfy a certain linear equation, but I do not stop to explain the geometrical interpretation of this theorem.

[Vol. III. pp. 78, 79.]

1667. (Proposed by Professor Sylvester.)
Show that the discriminant of the form

$$ax^5 + b\lambda x^4 y + c\lambda^2 x^3 y^2 + c\mu^2 x^2 y^3 + b\mu x y^4 + a y^5$$

will be a rational integral function of the quantities a, b, c, $\lambda\mu$, $\lambda^5+\mu^5$, and of the second degree only in respect to the last of them.

Solution by Professor Cayley.

In general

$$\text{Disct.}\,(a,b,c,d,e,f\between \lambda x{+}\mu y,\ \lambda' x{+}\mu' y)^5 = (\lambda\mu' {-} \lambda'\mu)^{20}\,\text{Disct.}\,(a,b,c,d,e,f\between x,y)^5.$$

Hence first, if $(\lambda,\mu,\lambda',\mu') = (0,1,1,0)$, then

$$\text{Disct.}\,(a,b,c,d,e,f\between y,x)^5 = \text{Disct.}\,(a,b,c,d,e,f\between x,y)^5;$$

and secondly, if ω be an imaginary fifth root of unity and $(\lambda,\mu,\lambda',\mu') = (\omega,0,0,1)$, then

$$\text{Disct.}\,(a,b,c,d,e,f\between \omega x,y)^5 = \text{Disct.}\,(a,b,c,d,e,f\between x,y)^5.$$

These two results may also be written,

$$\begin{aligned}&\text{Disct.}\,(a,b,c,d,e,f\between x,y)^5 = \text{Disct.}\,(f,e,d,c,b,a\between x,y)^5,\\&\text{Disct.}\,(a,b,c,d,e,f\between x,y)^5 = \text{Disct.}\,(a,b\omega^4,c\omega^3,d\omega^2,e\omega,f\between x,y)^5;\end{aligned}$$

that is, the discriminant of $(a,b,c,d,e,f\between x,y)^5$ is not altered by taking the coefficients in a reverse order, or by multiplying the several coefficients by the powers ω^5, ω^4, ω^3, ω^2, ω, of an imaginary fifth root of unity. Applying these theorems to the form $(a,b\lambda,c\lambda^2,c\mu^2,b\mu,a\between x,y)^5$, the discriminant is not altered by changing the coefficients into $(a,b\mu,c\mu^2,c\lambda^2,b\lambda,a)$; that is, by the interchange of λ and μ; nor by changing the coefficients into

$$(a,b\omega^4\lambda,c\omega^3\lambda^2,c\omega^2\mu^2,b\omega\mu,a),\quad \text{or}\quad \{a,b(\lambda\omega^4),c(\lambda\omega^4)^2,c(\mu\omega)^2,b(\mu\omega),a\}:$$

that is, the discriminant is not altered by the change of λ, μ into $\lambda\omega^4$, $\mu\omega$ respectively. In virtue of the second property the discriminant is a rational integral function of $(\lambda\mu,\lambda^5,\mu^5)$, and then in virtue of the first property it is a rational integral function of $(\lambda\mu, \lambda^5\mu^5, \lambda^5+\mu^5)$, that is, of $\lambda\mu$, $\lambda^5+\mu^5$. For the general form $(a, b, c, d, e, f \between x, y)^5$, if a term of the discriminant be $a^\alpha b^\beta c^\gamma d^\delta e^\epsilon f^\phi$, then we have $\alpha+\beta+\gamma+\delta+\epsilon+\phi = 8$, $5\alpha+4\beta+3\gamma+2\delta+\epsilon = 20$; hence attending only to the indices α, β, γ we have $5\alpha+4\beta+3\gamma > 20$, and therefore *à fortiori* $3\beta+3\gamma > 20$, so that $\beta+\gamma$ is $= 6$ at most. It follows that for the form $(a, b\lambda, c\lambda^2, c\mu^2, b\mu, a \between x, y)^5$, the sum of the indices of $b\lambda$, $c\lambda^2$ is $= 6$ at most, and therefore, even if the index of $c\lambda^2$ is $= 6$, the index of λ will be only $= 12$, that is, the discriminant contains no power of λ higher than λ^{12}: hence considered as a function of $\lambda\mu$, $\lambda^5+\mu^5$, the highest power of $\lambda^5+\mu^5$ is $(\lambda^5+\mu^5)^2$; which completes the theorem.

[Vol. III. p. 90.]

1687. (Proposed by Professor Cayley.)—To describe a spherical triangle such that the angles thereof and of the polar triangle lie on a spherical conic.

On the sphere, the locus of a point such that the perpendiculars from it upon the sides of a given spherical triangle have their feet on a line (great circle), is in general a spherical cubic; if however the triangle be such as is mentioned in the above Problem, then the locus breaks up into a line (great circle) and into the conic through the angles of the given and polar triangles.

[Vol. III. pp. 92—96.]

1690. (Proposed by W. A. Whitworth, M.A.)—If ABC be the triangle formed by the three diagonals aa', bb', cc' of a complete quadrilateral $aa'bb'cc'$, then a conic can be found having double contact in the chord aa' with the critical conic of the quadri- lateral $bb'cc'$, double contact in the chord bb' with the critical conic of the quadrilateral $cc'aa'$, and double contact in the chord cc' with the critical conic of the quadrilateral $aa'bb'$.

3. Solution by Professor Cayley.

1. The equations of the sides of the quadrilateral may be taken to be respectively $x = 0$, $y = 0$, $z = 0$, $w = 0$, where the implicit constants are so determined that we have identically

$$x+y+z+w = 0;$$

this being so, the equations of the three diagonals are respectively

$$\begin{array}{lllll} x+y=0, & \text{or} & z+w=0, & \text{or} & x+y-z-w=0 \quad (\text{three equivalent forms}) \\ x+z=0, & \text{or} & y+w=0, & \text{or} & x-y+z-w=0 \quad (\quad \text{“} \qquad \text{“} \qquad) \\ x+w=0, & \text{or} & y+z=0, & \text{or} & x-y-z+w=0 \quad (\quad \text{“} \qquad \text{“} \qquad) \end{array}$$

and the equations of the critical conics are respectively

$$xy + zw = 0, \quad xz + yw = 0, \quad xw + yz = 0.$$

Hence we see that the equation of the required conic is

$$\Delta = x^2 + y^2 + z^2 + w^2 - 2yz - 2zx - 2xy - 2xw - 2yw - 2zw = 0.$$

In fact this equation may be written

$$\begin{aligned} \Delta &= (x + y - z - w)^2 - 4(xy + zw) = 0, \\ \Delta &= (x - y + z - w)^2 - 4(xz + yw) = 0, \\ \Delta &= (x - y - z + w)^2 - 4(xw + yz) = 0, \end{aligned}$$

equations which put in evidence the double contact with the three critical conics respectively. We have also, identically,

$$\Delta = (x + y + z + w)(x + y - 3z - w) - 2w(x + y - z - w) + 4(z^2 - xy),$$

and the equation $\Delta = 0$ may therefore be written

$$\Delta = -2w(x + y - z - w) + 4(z^2 - xy) = 0,$$

a form which shows that the conic $z^2 - xy = 0$ meets the line $w = 0$ in the same two points in which it is met by the conic $\Delta = 0$. And it hence appears by symmetry that the conics

$$\Delta = 0,\ x^2 - yz = 0,\ y^2 - zx = 0,\ z^2 - xy = 0 \text{ meet the line } w = 0 \text{ in the same two point}$$

$$\Delta = 0,\ w^2 - yz = 0,\ y^2 - zw = 0,\ z^2 - wy = 0 \text{ meet the line } x = 0 \text{ in the same two point}$$

$$\Delta = 0,\ w^2 - xz = 0,\ x^2 - zw = 0,\ z^2 - wx = 0 \text{ meet the line } y = 0 \text{ in the same two points}$$

$$\Delta = 0,\ w^2 - xy = 0,\ x^2 - yw = 0,\ y^2 - wx = 0 \text{ meet the line } z = 0 \text{ in the same two points}$$

which are the relations constituting the latter part of the proposed theorem.

2. The analogous theorems in space may be briefly referred to. Taking $x = 0$, $y = 0$, $z = 0$, $w = 0$ as the equations of the faces of a tetrahedron $ABCD$, then the implicit constants may be so determined that the coordinates of a given arbitrary point O shall be $(1, 1, 1, 1)$. We may by lines drawn from the vertices of the tetrahedron project the point O on the faces, so as to obtain a point in each of the four faces; and then in each face, by lines drawn from the vertices of the face, project the point in that face upon the edges of the face; the two points thus obtained on each edge of the tetrahedron are (it is easy to see) one and the same point; that is, we have on each edge of the tetrahedron a point; and there exists a quadric surface

$$\Delta = x^2 + y^2 + z^2 + w^2 - 2yz - 2zx - 2xy - 2xw - 2yw - 2zw = 0$$

touching the edges of the tetrahedron in these six points respectively.

The surface in question has plane contact with

the hyperboloid $xy + zw = 0$along the intersection with $x + y - z - w = 0$,
" $xz + yw = 0$" " $x - y + z - w = 0$,
" $xw + yz = 0$" " $x - y - z + w = 0$,

and moreover the surfaces

$\Delta = 0,\ x^2 - yz = 0,\ y^2 - zx = 0,\ z^2 - xy = 0$ meet the line $w = 0,\ x + y + z + w = 0$
in the same two poi
$\Delta = 0,\ w^2 - yz = 0,\ y^2 - zw = 0,\ z^2 - wy = 0$ meet the line $x = 0,\ x + y + z + w = 0$
in the same two poi
$\Delta = 0,\ w^2 - xz = 0,\ x^2 - zw = 0,\ z^2 - wx = 0$ meet the line $y = 0,\ x + y + z + w = 0$
in the same two poi
$\Delta = 0,\ w^2 - xy = 0,\ x^2 - yw = 0,\ y^2 - wx = 0$ meet the line $z = 0,\ x + y + z + w = 0$
in the same two poi

With respect to the construction of the four planes,

$$x+y-z-w = 0, \quad x-y+z-w = 0, \quad x-y-z+w = 0, \quad x+y+z+w = 0,$$

it is to be observed that if through any edge of the tetrahedron, for instance the edge $x = 0$, $y = 0$, we draw the plane $x - y = 0$ through the point O, then the harmonic of this in regard to the planes $x = 0$, $y = 0$ is the plane $x + y = 0$; we have thus six planes, one through each edge of the tetrahedron, viz., these are $y + z = 0$, $z + x = 0$, $x + y = 0$, $x + w = 0$, $y + w = 0$, $z + w = 0$; the six planes being the faces of a hexahedron, which is such that the vertices of the tetrahedron are four of the eight vertices of the hexahedron. The pairs of opposite faces of the hexahedron meet in three lines lying in the plane $x + y + z + w = 0$, and consequently forming a triangle such that through each side of the triangle there pass two opposite faces of the hexahedron; the planes $x + y - z - w = 0$, $x - y + z - w = 0$, $x - y - z + w = 0$ are the harmonics of the plane $x + y + z + w = 0$ in respect of the pairs of opposite faces of the hexahedron; viz., the plane $x + y - z - w = 0$ is the harmonic of the plane $x + y + z + w = 0$ in respect to the planes $x + y = 0$, $z + w = 0$; and the like for the other two planes $x - y + z - w = 0$ and $x - y - z + w = 0$ respectively.

[Vol. IV. July to December, 1865, pp. 17, 18.]

1710. (Proposed by Professor Cayley.)—Trace the curve $y^4 - 2y^2zx - z^4 = 0$, where the coordinates are such that $x + y + z = 0$ is the line *infinity*.

Solution by the Proposer.

We have $x = \dfrac{y^4 - z^4}{2y^2 z}$; or writing $y = \theta z$, then $x = \dfrac{\theta^4 - 1}{2\theta^2}\, z$, that is

$$x : y : z = \theta^4 - 1 : 2\theta^3 : 2\theta^2.$$

Hence, we see that y, z are indefinitely small in comparison of x,

$$\begin{aligned}&\text{if } \theta = \infty, \text{ and then } \quad x : y : z = \theta^4 : 2\theta^3 : 2\theta^2, \text{ that is } y^2 = 2zx;\\ &\text{or, if } \theta = 0, \text{ and then } x : y : z = -1 : 2\theta^3 : 2\theta^2, \text{ that is } z^3 = -2y^2x;\end{aligned}$$

so that in the neighbourhood of the point $(y = 0, z = 0)$ there are two branches coin- ciding with the parabola $y^2 = 2zx$ and with the semicubical parabola $z^3 = -2y^2x$, respectively.

To find the points at infinity we have $x + y + z = 0$, that is $\theta^4 + 2\theta^3 + 2\theta^2 - 1 = 0 = (\theta + 1)(\theta^3 + \theta^2 + \theta - 1) = 0$; and observing that the equation $\theta^3 + \theta^2 + \theta - 1 = 0$ has one real root, say $\theta = k$, if k be the real root of the equation $k^3 + k^2 + k - 1 = 0$ ($k = .505$ nearly),—there are two real points at infinity, viz., corresponding to $\theta = -1$, we have the point $(0, -1, 1)$, and corresponding to $\theta = k$ the point $(-1 - k, k, 1)$.

The equation of the tangent at a point (α, β, γ) is

$$x\,(-\beta^2\gamma) + y\,(2\beta^3 - 2\alpha\beta\gamma) + z\,(-\alpha\beta^2 - 2\gamma^3) = 0,$$

and hence writing $(\alpha, \beta, \gamma) = (0, -1, 1)$ we have the asymptote $x + 2y + 2z = 0$: to find where this meets the curve, we have $\theta^4 + 4\theta^3 + 4\theta^2 - 1 = 0$, that is $(\theta + 1)^2(\theta^2 + 2\theta - 1) = 0$, or at the points of intersection $\theta^2 + 2\theta - 1 = 0$, that is $\theta = -1 \pm \sqrt{2}$, or there are two real points of intersection.

Again writing $(\alpha, \beta, \gamma) = (-1 - k, k, 1)$ we find the asymptote $k^2x - 2y + (k + 1)z = 0$: to find where this meets the curve, we have $k^2(\theta^4 - 1) - 4k\theta^3 + (2k + 2)\,\theta^2 = 0$, that is $k^2\theta^4 - 4k\theta^3 + (2k + 2)\,\theta^2 - k^2 = (\theta - k)^2\,[k^2\theta^2 - 2(k^2 + k + 1)\,\theta - 1] = 0$; or for the points of intersection $k^2\theta^2 - 2(k^2 + k + 1)\,\theta - 1 = 0$, an equation in θ with two real roots, hence the points of intersection are real.

It is now easy to lay down the curve; viz., if, to fix the ideas, the fundamental triangle is taken to be equilateral, and the coordinates x, y, z are considered to be positive for points *within* the triangle, then the curve is as shown in the annexed figure 1.

[Figure 1: The curve $y^4 - 2y^2zx - z^4 = 0$ drawn within an equilateral triangle, showing two infinite branches passing through the triangle with asymptotes. The curve has a branch near the vertex x coinciding with the parabola $y^2 = 2zx$ and another branch coinciding with the semicubical parabola $z^3 = -2y^2x$.]

It may be remarked that the curve is met by every real line in two real points at least, and consequently that it is not the projection of any finite curve whatever. By a modification of the constants of the equation, we might obtain curves which are finite, such as the curve in figure 2; or curves with two or four infinite branches, which are the projections of such a finite curve.

[Figure 2: A finite modification of the curve, forming a closed oval shape.]

[Vol. IV. pp. 32—37.]

1744. (Proposed by W. S. Burnside, B.A.)—It is required to find (x_1, y_1, z_1), functions of (x, y, z), such that we may have identically

$$\frac{{x_1}^3 + {y_1}^3 + {z_1}^3}{x_1 y_1 z_1} = \frac{x^3 + y^3 + z^3}{xyz}.$$

Solution by Professor Cayley.

The Solution is in fact given in my "Memoir on Curves of the Third Order," *Philosophical Transactions*, vol. cxlvii. (1857), pp. 415—446, [146].

Write $\dfrac{x^3 + y^3 + z^3}{xyz} = -6l$; then, taking (X, Y, Z) as current coordinates, (x, y, z) are, it is clear, the coordinates of a point on the cubic curve $X^3 + Y^3 + Z^3 + 6lXYZ = 0$; and if (x_1, y_1, z_1) are the coordinates of any other point on the same cubic curve, then we shall have

$$\frac{{x_1}^3 + {y_1}^3 + {z_1}^3}{x_1 y_1 z_1} = -6l = \frac{x^3 + y^3 + z^3}{xyz},$$

so that (x_1, y_1, z_1) will satisfy the condition in question. Hence, if from a given point (x, y, z) on the cubic curve we obtain by any geometrical construction another point on the curve, the coordinates of this new point will be functions (and, if the construction is such as to lead to a single point only, they will be *rational* functions) of (x, y, z), satisfying the condition in question.

For instance, if the point (x, y, z) be joined with any point (α, β, γ) on the curve, the joining line will again meet the curve in a single point, which may be taken to be the point (x_1, y_1, z_1). This assumes that we know on the cubic curve a point (α, β, γ); but such a point at once presents itself, viz., we may write $(\alpha, \beta, \gamma) = (1, -1, 0)$; which gives only the self-evident solution $(x_1, y_1, z_1) = (y, x, z)$. The point $(1, -1, 0)$ is clearly one of the nine points of inflexion of the cubic curve, and by using these in any manner whatever, viz., joining the point (x, y, z) with any point of inflexion, and then the new point with any other point of inflexion, and so on indefinitely, we obtain in connexion with the given point (x, y, z) seventeen other points on the curve, in all a system of eighteen points: these are

$$\begin{array}{llllll} (x,y,z), & (x,\omega y,\omega^2 z), & (x,\omega^2 y,\omega z), & (x,z,y), & (x,\omega z,\omega^2 y), & (x,\omega^2 z,\omega y) \\ (y,z,x), & (\omega y,\omega^2 z,x), & (\omega^2 y,\omega z,x), & (z,y,x), & (\omega z,\omega^2 y,x), & (\omega^2 z,\omega y,x) \\ (z,x,y), & (\omega^2 z,x,\omega y), & (\omega z,x,\omega^2 y), & (y,x,z), & (\omega^2 y,x,\omega z), & (\omega y,x,\omega^2 z) \end{array}$$

possessing remarkable geometrical properties; and of course each of the seventeen new points furnishes a (self-evident) solution of the given identity.

But we may take $(\alpha, \beta, \gamma) = (x, y, z)$; the point (x_1, y_1, z_1) is here the point of intersection of the cubic by the tangent at the point (x, y, z); or say it is the "tangential" of the point (x, y, z). The values thus obtained for (x_1, y_1, z_1) are

$$(x_1, y_1, z_1) = \{x(y^3 - z^3),\ y(z^3 - x^3),\ z(x^3 - y^3)\},$$

which (excluding the above-mentioned self-evident solutions) is in fact the most simple solution of the proposed identity. In order to verify that the last-mentioned values of (x_1, y_1, z_1) are in fact the coordinates of the tangential of (x, y, z), I observe that this will be the case if only we have

$$(x^3{+}2lxyz)\,x_1{+}(y^3{+}2lxyz)\,y_1{+}(z^3{+}2lxyz)\,z_1 = 0, \quad {x_1}^3{+}{y_1}^3{+}{z_1}^3{+}6lx_1y_1z_1 = 0,$$

the first of which is obviously satisfied by the values in question; and for the verification of the second equation,

$$\begin{aligned} {x_1}^3 + {y_1}^3 + {z_1}^3 &= x^3(y^3 - z^3)^3 + y^3(z^3 - x^3)^3 + z^3(x^3 - y^3)^3, \\ &= -x^3(y^3 - z^3) - y^3(z^3 - x^3) - z^3(x^3 - y^3), \\ &= (x^3 + y^3 + z^3)\,(y^3 - z^3)\,(z^3 - x^3)\,(x^3 - y^3), \\ x_1y_1z_1 &= xyz\,(y^3 - z^3)\,(z^3 - x^3)\,(x^3 - y^3), \end{aligned}$$

therefore

$${x_1}^3{+}{y_1}^3{+}{z_1}^3{+}6lx_1y_1z_1 = (x^3{+}y^3{+}z^3{+}6lxyz)\,(y^3{-}z^3)\,(z^3{-}x^3)\,(x^3{-}y^3) = 0$$

if $x^3 + y^3 + z^3 + 6lxyz = 0$; the same equations verify at once the identity

$$\frac{{x_1}^3 + {y_1}^3 + {z_1}^3}{x_1y_1z_1} = \frac{x^3 + y^3 + z^3}{xyz}.$$

Another solution is as follows: viz., if we take the third intersection with the cubic of the line joining the points (y, x, z) and $\{x(y^3 - z^3),\ y(z^3 - x^3),\ z(x^3 - y^3)\}$, the coordinates of the line in question are

$$\begin{aligned} x_1 : y_1 : z_1 = {} & x^6y^3 + y^6z^3 + z^6x^3 - 3x^3y^3z^3 \\ : {} & x^3y^6 + y^3z^6 + z^3x^6 - 3x^3y^3z^3 \\ : {} & xyz\,(x^6 + y^6 + z^6 - y^3z^3 - z^3x^3 - x^3y^3). \end{aligned}$$

According to a very beautiful theorem of Professor Sylvester's in relation to the theory of cubic curves, the coordinates of a point which depends linearly on a given point of the curve are necessarily rational and integral functions of a *square degree* of the coordi- nates (x, y, z) of the given point; and moreover that (considering as *one* solution those which can be derived from each other by a mere permutation of the coordinates, or change of x into ωx, &c.), there is only one solution of a given square degree m^2; the solutions of the degrees 4 and 9 are given above. The tangential of the tangential, or second tangential of the point (x, y, z), gives the solution of the degree 16; joining this second tangential with the original point (x, y, z), we have the solution of the degree 25; and the same solution is also given as the sixth point of intersection with the cubic, of the conic of 5-pointic intersection at the point (x, y, z). See my memoir "On the conic of 5-pointic contact at any point of a plane curve," *Phil. Trans.* vol. cxlx. (1859), pp. 371—400, [261].

Addition to the foregoing Solution. On a system of Eighteen Points on a Cubic Curve.

Considering the cubic curve $x^3 + y^3 + z^3 + 6lxyz = 0$, we have the nine points of inflexion, which I represent as follows:

$$\begin{array}{lll} a = (0, 1, -1), & d = (-1, 0, 1), & g = (1, -1, 0), \\ b = (0, 1, -\omega), & e = (-\omega, 0, 1), & h = (1, -\omega, 0), \\ c = (0, 1, -\omega^2), & f = (-\omega^2, 0, 1), & i = (1, -\omega^2, 0), \end{array}$$

viz., ω being an imaginary cube root of unity, the coordinates of a are $(0, 1, -1)$, those of b, $(0, 1, -\omega)$, &c.

The points of inflexion lie (as is known) by threes on twelve lines; viz., the lines are

$$\begin{array}{llll} abc, & afh, & bfg, & cei, \\ adg, & bdi, & cdh, & def, \\ aei, & beh, & ceg, & ghi. \end{array}$$

Consider now a point on the curve, the coordinates whereof are (x, y, z), where of course $x^3 + y^3 + z^3 + 6lxyz = 0$; this is one of a system of eighteen points on the curve, which may be represented as follows:

$$\begin{array}{rlrlrl}
A &= (x, y, z), & D &= (x, \omega y, \omega^2 z), & G &= (x, \omega^2 y, \omega z),\\
B &= (y, z, x), & E &= (\omega y, \omega^2 z, x), & H &= (\omega^2 y, \omega z, x),\\
C &= (z, x, y), & F &= (\omega^2 z, x, \omega y), & I &= (\omega z, x, \omega^2 y),\\
J &= (x, z, y), & M &= (x, \omega z, \omega^2 y), & P &= (x, \omega^2 z, \omega y),\\
K &= (z, y, x), & N &= (\omega z, \omega^2 y, x), & Q &= (\omega^2 z, \omega y, x),\\
L &= (y, x, z), & O &= (\omega^2 y, x, \omega z), & R &= (\omega y, x, \omega^2 z),
\end{array}$$

viz., the coordinates of A are (x, y, z); those of B are (y, z, x), &c.

The tangent at A meets the curve in a point, "the tangential of A," the coordi- nates whereof are $x(y^3 - z^3)$, $y(z^3 - x^3)$, $z(x^3 - y^3)$; which point may be called A'. And we have thus the eighteen tangentials

$$A',\ B',\ C',\ D',\ E',\ F',\ G',\ H',\ I',\ J',\ K',\ L',\ M',\ N',\ O',\ P',\ Q',\ R'.$$

The eighteen points A, B, &c., have the following property; viz., the line joining any two of them meets the cubic in a third point, which is either one of the nine points of inflexion, or one of the eighteen tangentials; there are through each point of inflexion 9 such lines, and through each tangential 4 such lines; $(9\times 9)+(18\times 4) = 153 = \frac{1}{2}(18.17)$, the number of pairs of points AB, AC, &c. The lines through the inflexions are the 81 lines obtained by joining any one of the points $(A, B, C, D, E, F, G, H, I)$ with any one of the points $(J, K, L, M, N, O, P, Q, R)$, as shown in the following Table:

	A	*B*	*C*	*D*	*E*	*F*	*G*	*H*	*I*
J	*a*	*d*	*g*	*c*	*f*	*i*	*b*	*e*	*h*
K	*d*	*g*	*a*	*f*	*i*	*c*	*e*	*h*	*b*
L	*g*	*a*	*d*	*i*	*c*	*f*	*h*	*b*	*e*
M	*c*	*f*	*i*	*b*	*e*	*h*	*a*	*d*	*g*
N	*f*	*i*	*c*	*e*	*h*	*b*	*d*	*g*	*a*
O	*i*	*c*	*f*	*h*	*b*	*e*	*g*	*a*	*d*
P	*b*	*e*	*h*	*a*	*d*	*g*	*c*	*f*	*i*
Q	*e*	*h*	*b*	*d*	*g*	*a*	*f*	*i*	*c*
R	*h*	*b*	*e*	*g*	*a*	*d*	*i*	*c*	*f*

Transcription from the original PDF scan.
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concluding with the End of Volume V.

Paper 383 – Problems and Solutions (continued)

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viz., the line AJ passes through a, the line AK through d, &c.; the proof that AJ passes through a depends on the identical equation

$$\begin{vmatrix} x & y & z \\ z & x & y \\ 0 & 1 & -1 \end{vmatrix} = 0,$$

and the like for the other lines AK, AL, &c.

The lines through the tangentials are the 36 lines obtained by joining any two of the points $(A, B, C, D, E, F, G, H, I)$ and the 36 lines obtained by joining any two of the points $(J, K, L, M, N, O, P, Q, R)$; and these 72 lines pass through the tangentials, as shown by the table

$$\begin{array}{ccccccc} ABC, & BDI, & CEG, & \qquad & JKL, & KMR, & LNP, \\ ADG, & BEH, & CFI, & & JMP, & KNQ, & LOR, \\ AEI, & BFG, & DEF, & & JNR, & JOP, & MNO, \\ AFH, & GDH, & GHI, & & JOQ, & LMQ, & PQR, \end{array}$$

viz., in the triad ABC, BC passes through A', CA through B', AB through C'; and the like for the other triads. The proof that BC passes through A' depends on the identical equation

$$\begin{vmatrix} y & z & x \\ z & x & y \\ x(y^2 - z^2) & y(z^2 - x^2) & z(x^2 - y^2) \end{vmatrix} = 0,$$

and the like for the other combinations of points.

If we attend only to the points A, B, C and their tangentials A', B', C'; then we have on the cubic three points A, B, C, such that the line joining any two of them passes through the tangential of the third point. And the figure may be constructed by means of the three real points of inflexion a,

d, g, as follows, viz., joining these with any point J on the cubic, the lines so obtained respectively meet the cubic in three new points which may be taken for the points A, B, C. Or if one of these points, say A, be given, then joining it with one of the three real inflexions, this line again meets the cubic in the point J, and from it by means of the other two real inflexions we obtain the remaining points B and C; it is clear that, A being given, the construction gives three points, say J, K, L, each of them leading to the same two points B and C.

We may consider the question from a different point of view. Let A, B, C be given points, and let there be given also three lines passing through these three points respectively; through the given points, touching at these points the given lines respectively, describe a cubic; and let the given lines again meet the cubic in the points A', B', C' respectively. The equation of the cubic contains three arbitrary parameters; but when two of these are properly determined, the points A, B, C and their tangentials A', B', C' will be related as in the theorem; viz., the line through any two of the points will pass through the tangential of the third point.

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The analytical investigation is as follows:

Let the equations of the three tangents be $x=0$, $y=0$, $z=0$, and suppose that, for the points A, B, C respectively, we have

$$(x=0,\ y=\lambda z),\quad (y=0,\ z=\mu x),\quad (z=0,\ x=\nu y),$$

then the equation of a cubic touching the three lines at the three points respectively will be

$$\begin{aligned}(y-\lambda z)^2(\nu^2By+Cz)+(z-\mu x)^2(\lambda^2Cz+Ax)+(x-\nu y)^2(\mu^2Ax+By)&\\ -\mu\nu^2Ax^3-\nu\lambda^2By^3-\lambda\mu^2Cz^3+Kxyz=0&,\end{aligned}$$

where A, B, C, K are arbitrary coefficients; but if $A:B:C=\lambda:\mu:\nu$, then the equation is

$$\begin{aligned}(y-\lambda z)^2\nu(\mu\nu y+z)+(z-\mu x)^2\lambda(\nu\lambda z+x)+(x-\nu y)^2\mu(\lambda\mu x+y)&\\ -\lambda\mu^2x^3-\mu\nu^2y^3-\nu\lambda^2z^3+Kxyz=0&,\end{aligned}$$

where

A, A'	are the intersections of	$x=0$,	by	$y-\lambda z=0$,	$\mu\nu y+z=0$	respectively,
B, B'	“	$y=0$,	“	$z-\mu x=0$,	$\nu\lambda z+x=0$	
C, C'	“	$z=0$,	“	$x-\nu y=0$,	$\lambda\mu x+y=0$	;

the equations of BC, CA, AB thus are

$$-\mu x+\mu\nu y+z=0,\quad x-\nu y+\nu\lambda z=0,\quad \lambda\mu x+y-\lambda z=0,$$

which pass through A', B', and C' respectively.

If we consider along with the points A, B, C the points J, K, L, and their respective tangentials, then we have inscribed in the cubic a hexagon $ALBJCK$ which has the following properties, viz., the pairs of opposite sides and the three diagonals pass through the three real inflexions in lined, viz.,

AL, JC, BK,	through	g
LB, CK, JA,	“	a
BJ, KA, GL,	“	d.

This shows that the six points A, B, C, J, K, L are the intersections of the cubic by a conic; and moreover, considering the triangles ABC, JKL formed by the alternate vertices, then in each triangle the sides pass through the tangentials of the opposite vertices respectively.

In what precedes we have in effect found the coordinates (z,x,y) of the third point of intersection with the cubic, of the line joining the points

(y, z, x) and $(x(y^2 - z^2),\ y(z^2 - x^2),\ z(x^2 - y^2))$. The coordinates of the same point may be otherwise found by a direct investigation, as follows: Write

$$x_1 : y_1 : z_1 = x(y^2 - z^2) : y(z^2 - x^2) : z(x^2 - y^2); \quad x_2 : y_2 : z_2 = y : z : x.$$

If in the equation of the curve we substitute for x, y, z, the values $ux_1 + vx_2$, $uy_1 + vy_2$, $uz_1 + vz_2$, we find

$$\begin{aligned} u(x_1x_2 + y_1y_2 + z_1z_2 + 2(x_1y_2z_1 + y_1z_2x_1 + z_1x_1y_2)) \\ + v(x_1x_2^2 + y_1y_2^2 + z_1z_2^2) + v(x_2y_2z_2 + y_2z_2x_2 + z_2x_2y_2) = 0, \end{aligned}$$

say $uP + vQ = 0$; we may therefore write $u = Q$, $v = -P$, and the coordinates of the third point are $Qx_1 - Px_2$, $Qy_1 - Py_2$, $Qz_1 - Pz_2$.

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Now

$$\begin{aligned}Qx_1 - Px_2 &= y_1z_2(x_1z_2 - x_2z_1) + z_1x_2(x_1y_2 - x_2y_1) + z_1(x_1y_2z_1 - x_2y_1z_2)\\ &= yz(x^2-y^2)[y^2(z^2-x^2) - zx(y^2-z^2)]\\ &\quad + zx(x^2-y^2)[yz(x^2-y^2) - xy(y^2-z^2)]\\ &\quad + z[yz\cdot yz(z^2-x^2)(x^2-y^2) - x(x^2-y^2)^2zx]\\ &= (x^2y^2 + y^2z^2 + z^2x^2 - 3x^2y^2z^2)z\\ &\quad + xyz(x^2+y^2+z^2 - y^2z^2 - z^2x^2 - x^2y^2)z\\ &\quad - z(x^2y^2 + y^2z^2 + z^2x^2 - 3x^2y^2z^2);\end{aligned}$$

so that we have $Qx_1 - Px_2 = \lambda z$; and in like manner $Qy_1 - Py_2 = \lambda x$, $Qz_1 - Pz_2 = \lambda y$; and therefore $Qx_1 - Px_2 : Qy_1 - Py_2 : Qz_1 - Pz_2 = z : x : y$, which proves the theorem.

I consider in like manner the following question; viz., if (y, x, z) be joined with the tangential of (x, y, z); to find the third point of intersection. We have here

$$x_2 : y_2 : z_2 = x(y^2-z^2) : y(z^2-x^2) : z(x^2-y^2); \quad x_1 : y_1 : z_1 = y : x : z;$$

and P, Q as before; and the coordinates of the third point are $Qx_1 - Px_2$, $Qy_1 - Py_2$, $Qz_1 - Pz_2$; also

$$\begin{aligned}Qx_1 - Px_2 &= xy(z^2-x^2)[y^2(z^2-x^2) - x^2(y^2-z^2)]\\ &\quad + z^2(x^2-y^2)[yz(x^2-y^2) - zx(y^2-z^2)]\\ &\quad + z[yz(z^2-x^2)(x^2-y^2) - x^2(y^2-z^2)^2zx]\\ &= x[y^2(z^2-x^2)^2 - z^2(x^2-y^2)(y^2-z^2)]\\ &\quad + y[z^2(x^2-y^2)^2 - x^2(y^2-z^2)(z^2-x^2)]\\ &\quad + z[(z^2-x^2)(x^2-y^2) - x^2(y^2-z^2)^2],\end{aligned}$$

that is $Qx_1 - Px_2 = (x + y - 2\mu z)(x^2z^2 + y^2x^2 + z^2y^2 - 3x^2y^2z^2)$; similarly

$$Qy_1 - Py_2 = (x + y - 2\mu z)(y^2x^2 + z^2y^2 + x^2z^2 - 3x^2y^2z^2);$$

also

$$Qz_1 - Pz_2 = (x + y - 2\mu z)(x^2 + y^2 + z^2 - yz^2 - zx^2 - xy^2)xyz;$$

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and we hence have the values

$$X:Y:Z = x^2z^2+y^2x^2+z^2y^2-3x^2y^2z^2 : x^2y^2+y^2z^2+z^2x^2-3x^2y^2z^2 \\ : xyz(x^2+y^2+z^2-y^2z^2-z^2x^2-x^2y^2)$$

for the coordinates of the point in question.

[Vol. IV. pp. 38, 39.]

1751. (Proposed by Professor Cayley.) — Let $ABCD$ be any quadrilateral. Construct, as shown in the figure, the points F, G, H, I: in BC find a point Q such that

$$\frac{BQ}{BC}\cdot\frac{CQ}{CD}=\frac{1}{8};$$

and complete the construction as shown in the figure. Show that an ellipse may be drawn passing through the eight points F, G, H, I, α, β, γ, δ, and having at these points respectively the tangents shown in the figure.

(Professor Cayley remarks that if $ABCD$ is the perspective representation of a square, then the ellipse is the perspective representation of the inscribed circle; the theorem gives eight points and the tangent at each of them; and the ellipse may therefore be drawn by hand with an accuracy quite sufficient for practical purposes. The demonstration is immediate, by treating the figure as a perspective representation: the gist of the theorem is the very convenient construction in perspective which it affords.)

[Vol. IV. pp. 70, 71.]

1771. (Proposed by Professor Cayley.) — Given a circle and a line, it is required to find a parabola, having the line for its directrix, and the circle for a circle of curvature.

Solution by the Proposer.

Let $x^2+y^2-1=0$ be the equation of the given circle, $x=-m$ that of the given line. Taking on the circle an arbitrary point $(\cos\theta, \sin\theta)$, we may find a parabola having the given line for its directrix, and touching the circle at the last-mentioned point; viz., the equation of the parabola is found to be

$$y^2-2y\sin\theta(1+2\cos^2\theta-2m\cos\theta)-4x\cos^2\theta(\cos\theta-m)+1+3\cos^2\theta-4m\cos\theta=0.$$

(There is no difficulty in verifying that this parabola has for its directrix the line $x - m = 0$, that the equation is satisfied by the values $x = \cos\theta$, $y = \sin\theta$, and that the derived equation is satisfied by the values $x = \cos\theta$, $y' = \sin\theta$, $y' = -\cot\theta$.)

Representing for a moment the left-hand side of the equation by U, we have identically

$$\begin{aligned}
&U - \cos^2\theta(x^2 + y^2 - 1)\\
&= y^2\sin^2\theta - x^2\cos^2\theta - 2y\sin\theta(1 + 2\cos^2\theta - 2m\cos\theta) - 4x\cos^2\theta(\cos\theta - m)\\
&\quad + 1 + 4\cos^2\theta - 4m\cos\theta,\\
&= (y\sin\theta + x\cos\theta - 1)(y\sin\theta - x\cos\theta - 1 - 4\cos^2\theta + 4m\cos\theta).
\end{aligned}$$

Hence to find the intersections of the parabola with the circle, we have first

$$x^2 + y^2 - 1 = 0, \quad y\sin\theta + x\cos\theta - 1 = 0,$$

giving the point $(\cos\theta, \sin\theta)$ twice, since $y\sin\theta + x\cos\theta - 1 = 0$ is the equation of the tangent to the circle at the point in question; and secondly

$$x^2 + y^2 - 1 = 0, \quad y\sin\theta - x\cos\theta - 1 - 4\cos^2\theta + 4m\cos\theta = 0,$$

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giving the remaining two points of intersection. If the circle be a circle of curvature, one of these must coincide with the point $(\cos\theta, \sin\theta)$, that is the equation

$$y\sin\theta - x\cos\theta - 1 - 4\cos^2\theta + 4m\cos\theta = 0,$$

must be satisfied by the values $x = \cos\theta$, $y = \sin\theta$; this will be the case if $-6\cos^2\theta + 4m\cos\theta = 0$, that is $\cos\theta = 0$, giving for the parabola $y^2 \pm 2y + 1 = 0$, which is not a proper Solution, or else $\cos\theta = \frac{2}{3}$, giving $\sin\theta = \pm(1 - \frac{4}{9}m^2)^{\frac{1}{2}}$, so that there are two parabolas satisfying the conditions of the problem; if to fix the ideas we take the upper sign, the equation of the corresponding parabola is

$$y^2 - 2(1 - \tfrac{4}{9}m^2)^{\frac{1}{2}}y + \tfrac{4}{3}m^2x + 1 - \tfrac{4}{9}m^2 = 0;$$

and it may be added that the coordinates of the focus are

$$x = m - \tfrac{2}{3}m^3, \quad y = (1 - \tfrac{4}{9}m^2)^{\frac{1}{2}}.$$

The equation of the other parabola and the coordinates of the focus are of course found by merely changing the sign of the radical. The parabolas are real if $m < \frac{3}{2}$; if $m = \frac{3}{2}$ we have a single parabola, the point of contact being in this case the vertex of the parabola; and if $m > \frac{3}{2}$ the parabolas are imaginary.

(Professor Cayley states that he was led to the foregoing problem by the consideration of the curve (proposed for investigation in Quest. 1812) which is the envelope of a variable circle having its centre in the given circle and touching the given line. The required curve (which is of the sixth order) has two cusps which, it is easy to see geometrically, are the foci of the parabolas in the problem. Taking $(\cos\theta, \sin\theta)$ for the coordinates of the centre of the variable circle, we shall have

$$x = \tfrac{3}{2}\cos\theta - m\cos 2\theta + \tfrac{1}{2}\cos 3\theta, \quad y = \tfrac{3}{2}\sin\theta - m\sin 2\theta + \tfrac{1}{2}\sin 3\theta,$$

for the coordinates of a point on the envelope.)

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[Vol. IV. pp. 81–83.]

1790. (Proposed by Professor Sylvester.) — (1) If a set of six points be respectively represented by the six permutations of $a:b:c$, show that they lie in a conic, and write down its equation.

(2) Hence prove that if AB, BC, CA be three real lines containing the nine points of inflexion of a cubic curve having an oval, the pairs of tangents drawn to the oval from A, B, C will meet it in six points lying in a conic.

Solution by Professor Cayley.

1. That the six points,

$$\begin{array}{lll} 1=(a,b,c), & 2=(b,c,a), & 3=(c,a,b), \\ 4=(a,c,b), & 5=(b,a,c), & 6=(c,b,a), \end{array}$$

are situate on a conic, appears at once by writing down its equation: viz.,

$$(bc+ca+ab)(x^2+y^2+z^2)-(a^2+b^2+c^2)(yz+zx+xy)=0,$$

which is satisfied by the coordinates of each of the six points.

2. It is interesting to remark that the six points on the conic form, not a general inscribed hexagon, but a hexagon such as is mentioned in Prob. 1512 (vol. II. p. 51), viz., one in which the three diagonals pass respectively through the Pascalian points (intersections of opposite sides) of the hexagon; and that the sides of the hexagon are, in fact, the lines 14, 25, 36 (which are the Pascalian lines of the inscribed hexagon 123456), and the lines 23, 31, 12.

To verify this, if we take $y-\theta x$ for the equation of a tangent from the point $(x=0,\ y=0)$, the equation $(1+\theta^3)x^3+16\theta x^2z+\theta^2z^3=0$ must have a pair of equal roots, giving for θ the equation $(1+\theta^3)^2+32\theta^3=0$; and we then find $z=-\theta(1+\theta^3)/(16)$. The roots of the equation in θ are of the form θ_1, θ_2, θ_3, $\frac{1}{\theta_1}$, $\frac{1}{\theta_2}$, $\frac{1}{\theta_3}$; and assuming that the curve has an oval, there are two real roots θ_1, $\frac{1}{\theta_1}$. Hence, writing $x:y:z=1:\theta_1:\frac{1+\theta_1^3}{16}=a:b:c$, the substitution for θ_1 gives $x:y:z=b:a:c$, that is, the coordinates of the points of contact of the tangents to the oval, from the point $(x=0,\ y=0)$ are (a,b,c) and (b,a,c) respectively; and writing successively (y,z,x) and (z,x,y) in place of (x,y,z), the coordinates for the tangents from $(y=0,\ z=0)$ are (b,c,a), (c,b,a); and those for the tangents from $(z=0,\ x=0)$ are (c,a,b) and (a,c,b); so that the coordinates of the six points of contact are a system of the form in question.

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[Vol. IV. pp. 108, 109.]

1812. (Proposed by Professor Cayley.) — Find the envelope of a series of circles which touch a given straight line and have their centres in the circumference of a given circle. (See Quest. 1771.)

[Vol. IV. pp. 108, 109.]

1816. (Proposed by R. Ball, M.A.) — Express the roots of the equation

$$(ae-4bd+3c^2)(ax^4+4bx^3+6cx^2+4dx+e)^2$$
$$-3\{(ac-b^2)x^4+2(ad-bc)x^3+(ae+2bd-3c^2)x^2+2(be-cd)x+(ce-d^2)\}^2=0,$$

in terms of the roots α, β, γ, δ of $x^4+4bx^3+6cx^2+4dx+e=0$.

Solution by Professor Cayley.

Writing

$$U=(a,b,c,d,e(x,y)^4=a(x-\alpha y)(x-\beta y)(x-\gamma y)(x-\delta y),$$

$$H=(ac-b^2,\ ad-bc,\ ae+2bd-3c^2,\ be-cd,\ ce-d^2(x,y)^2,$$

$$I=ae-4bd+3c^2;$$

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then considering $x : y$ as the unknown quantity, it is required to find the roots of the equation $IU^2 - 3H^2 = 0$ in terms of the roots $(\alpha, \beta, \gamma, \delta)$ of the equation $U = 0$; or, what is the same thing, it is required to find the linear factors of the function $IU^2 - 3H^2$.

The function in question is the product of four quadratic factors, rational functions of $(\alpha, \beta, \gamma, \delta)$; and these being known, the four pairs of linear factors can be determined each of them by the solution of a quadratic equation. In fact, writing

$$\begin{aligned}
\theta_\alpha &= a\{(\beta-\alpha)(x-\gamma y)(x-\delta y) + (\gamma-\alpha)(x-\delta y)(x-\beta y) + (\delta-\alpha)(x-\beta y)(x-\gamma y)\},\\
\theta_\beta &= a\{(\gamma-\beta)(x-\delta y)(x-\alpha y) + (\delta-\beta)(x-\alpha y)(x-\gamma y) + (\alpha-\beta)(x-\gamma y)(x-\delta y)\},\\
\theta_\gamma &= a\{(\delta-\gamma)(x-\alpha y)(x-\beta y) + (\alpha-\gamma)(x-\beta y)(x-\delta y) + (\beta-\gamma)(x-\delta y)(x-\alpha y)\},\\
\theta_\delta &= a\{(\alpha-\delta)(x-\beta y)(x-\gamma y) + (\beta-\delta)(x-\gamma y)(x-\alpha y) + (\gamma-\delta)(x-\alpha y)(x-\beta y)\},
\end{aligned}$$

we have identically $256(IU^2 - 3H^2) = \theta_\alpha\theta_\beta\theta_\gamma\theta_\delta$; so that the quadratic factors of $IU^2 - 3H^2$ are θ_α, θ_β, θ_γ, θ_δ. To show that this is so, it is to be remarked that the product $\theta_\alpha\theta_\beta\theta_\gamma\theta_\delta$ is a symmetrical function of the roots α, β, γ, δ, and consequently a rational and integral function of the coefficients (a, b, c, d, e) of U; moreover θ_α, θ_β, θ_γ, θ_δ being each of them a covariant (an irrational one) of U, the product in question must be a covariant. But a covariant is completely determined when the leading coefficient is given; hence it will be sufficient to show that the leading coefficients, or coefficients of x^4, in the functions $\theta_\alpha\theta_\beta\theta_\gamma\theta_\delta$ and $256(IU^2 - 3H^2)$ are equal to each other. Writing for a moment $\sum\alpha = p$, $\sum\alpha\beta = q$, $\sum\alpha\beta\gamma = r$, $\alpha\beta\gamma\delta = s$, the coefficient of x^2 in $a^{-1}\theta_\alpha$ is $\beta+\gamma+\delta-3\alpha$, which $= p - 4\alpha$; we have thence the product $(p-4\alpha)(p-4\beta)(p-4\gamma)(p-4\delta)$, which is $= p^4 - 4p^3\cdot p + 16p^2\cdot q - 64p\cdot r + 256s$, $= 256s - 64pr + 16p^2q - 3p^4$.

Hence, restoring the omitted factor a^4, and observing that we have p, q, r, s equal to $-4b$, $6c$, $-4d$, e, each divided by a, the coefficient of x^4 in $\theta_\alpha\theta_\beta\theta_\gamma\theta_\delta$ is

$$256(a^4e - 4a^3bd + 6a^2b^2c^2 - 3b^4), \text{ or } 256\{(ae - 4bd + 3c^2)a^2 - 3(ac - b^2)^2\},$$

and is consequently equal to the coefficient of x^4 in $256(IU^2 - 3H^2)$; which proves the theorem.

It may be remarked that the leading coefficient of $IU^2 - 3H^2$ is $= a^{-2}(a, b, c, d, e(b, -a)^4$; and that for a quantic $U = (a, b, \ldots(x, y)^n$ of the order n we have a corresponding covariant of the order $n(n-2)$, the leading coefficient of which is $= a^{-1}(a, b, \ldots(b, -a)^n$.

For $n = 2$, this is the invariant (discriminant) $ac - b^2$; for $n = 3$ it is the cubicovariant $(a^2d - 3abc + 2b^3, \ldots(x, y)^3$; for $n = 4$ it is, as we have seen, the covariant $IU^2 - 3H^2$. For $n = 5$, the leading coefficient $a^4f - 5a^3be + 10a^2cd - 10ab^2d + 4b^2e$ is $= a^2(a^2f - 5abe + 2acd + 4b^2d - 6bc^2) - 2(ac - b^2)(a^2d - 3abc + 2b^3)$, which shows that the covariant in question (of the order 15) is $= \Theta$ (No. 17) $- 2$ (No. 15) $\times$ (No. 18), where the Nos. refer to the Tables of my Second Memoir on Quantics, Phil. Trans., vol. CXLVI. (1856), pp. 101–126, [141; in the notation there explained, the expression for the covariant is $A^2E - 2CF$].

(The roots of $\theta_\alpha = 0$ are readily found to be

$$\frac{\alpha(\beta+\gamma+\delta) - (\beta\gamma+\gamma\delta+\delta\beta) \pm \sqrt{\{[\alpha(\beta-\gamma)]^2 + [\alpha(\gamma-\delta)]^2 + [\alpha(\delta-\beta)]^2\}}}{3\alpha - (\beta+\gamma+\delta)},$$

these then, with three similar pairs, express the eight roots as required.)

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Contents of Problems and Solutions

In the following Contents, the Problems are referred to, each by its No. and the Proposer's name, and the subject is briefly indicated: an Asterisk shows that no solution was given.

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Paper 618 – Notes and References

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302, 305. The theory of curves in space was proposed as the subject of the Prize-question of the Steiner Foundation by the Academy of Sciences of Berlin in the year 1881 " *irgend eine auf die Theorie der höheren algebraischen Raumcurven sich beziehende Frage von wesentlicher Bedeutung vollständig zu erledigen*," and the prize was divided between the two memoirs

Halphen, "Mémoire sur la classification des courbes gauches algébriques." *Jour. École Polyt.* Cah. LII. (1882), pp. 1–200, and

Nother, "Zur Grundlegung der Theorie der algebraischen Raumcurven." *Abh. der Akad. zu Berlin* vom Jahre 1882, pp. 1 to 120; both treating of the classification of curves in space.

We have also the valuable memoir

Valentiner, "Zur Theorie der Raumcurven," *Acta Mathematica*, t. II., 1883, pp. 136–230, which relates less directly to the question of classification.

The three authors all refer to these papers in the Comptes Rendus, and make considerable use of my conception of the monoid surface. It would be out of place to attempt to give any account here of these memoirs: I only refer to such remarks or theorems contained in them as stand in immediate connection with the remarks which follow.

The question of classification is much simplified by excluding from consideration the curves with singular points (that is actual double points and stationary points), and this is in fact done both by Halphen and Nother and in the present Note. The curves considered are thus curves with only apparent double points (adps.) viz. for a curve of the order d (I use Halphen's letters) with h apparent double points, taking an arbitrary point as vertex, the cone through the curve is a cone of the order d, with h nodal lines, each of these meeting the curve in two (real or imaginary) non-coincident points. Such a curve is the partial intersection of the cone in question say $U = (x, y, z)^d = 0$ with a monoid surface $w = \frac{P}{Q} = \frac{(x,y,z)^n}{(x,y,z)^{n+1}}$, where the inferior cone $P = (x, y, z)^n = 0$, of the monoid surface, and the superior cone $Q = (x, y, z)^{n+1} = 0$, of the monoid surface each of them pass through all the h nodal lines of the cone, and besides through θ lines of the cone: the complete intersection of the cone and monoid surface is thus made up of the curve once, the h nodal lines each twice, and the θ lines each once; and if as before the order of the curve is $= d$, then we thus have $(n+1)d = d+2h+\theta$, viz. we must have

$$nd = 2h + \theta,$$

as the condition to be satisfied in order that the curve of the order d may be the partial intersection of the cone and monoid surface.

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In my papers in the Comptes Rendus I endeavoured to find, and Halphen and Nother both endeavour to find, the surfaces of lowest order which have the curve of order d for their complete or partial intersection. This (although, as will presently appear, the theory may be considered in a more complete form) is an important and interesting question; but upon further reflection it appears to me that it is a question beside that which first presents itself and ought to be in the first instance considered, viz. this is the question of the classification of curves in space according to the foregoing representation of any such curve as the partial intersection of a cone and monoid surface. Supposing it effected, and a kind of curve completely defined according to this mode of representation, then there arises the further question to which I have referred (Salmon's *Solid Geometry*, Ed. 3, (1874), p. 285, and Ed. 4, (1882), p. 281), viz. we may have passing through any given curve a complete system of surfaces, that is a system $U=0$, $V=0$, $W=0$, $\ldots$ where these functions are not connected by any such equation as $U=NV+PW+\ldots$, and where every other surface which passes through the curve is expressible in the form $MU+NV+PW+\ldots=0$. It is not easy to prove (but as to this see Hilbert "Zur Theorie der algebraischen Gebilde," Göttingen Nachrichten. 1888, p. 454), but it may be safely assumed that for a curve of any given order whatever, the number of equations in such a complete system is finite, and we have thus the representation of a curve in space by means of a complete system of surfaces passing through it. Obviously the curve is here the partial (or if the system consists of only two surfaces then the complete) intersection say of the two surfaces $U=0$, $V=0$ of lowest order passing through it, which is the question above referred to.

Reverting to the representation by the cone and monoid surface, Halphen gives the capital theorem, that if we have any particular inferior cone $P=0$ passing through the curve, then we may without loss of generality take the equation of the monoid surface to be $w=\frac{P}{Q}$; viz. if instead hereof the equation of the monoid surface is taken to be $w=\frac{P'}{Q'}$, then this equation in virtue of the equation $U=0$ of the cone is always reducible to the first mentioned form $w=\frac{P}{Q}$; that is in virtue of the equation $U=0$, we have $w=\frac{P'}{Q'}=\frac{P}{Q}$, or what is the same thing, $\frac{P'}{Q'}=\frac{P}{Q}$ in virtue of $U=0$, that is $QP'-Q'P=MU$, where M is a rational and integral function $(x,y,z)^k$ of the degree $k=n'+n+1-d$, if k be the degree of P' and n that of P.

It thus appears that if n be the order of the cone of lowest order which passes through the h nodal lines of the cone $U=0$, then we have always functions Q, P

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of the orders $n+1$, n respectively, such that the equation of the monoid surface is $w = \frac{P}{Q}$. Or what is the same thing, we have always a monoid surface of the order $n+1$: we thus arrive at the notion of Halphen's characteristic n.

Instead of the foregoing equation $kd = 2h+\theta$, we thus have $nd = 2h+\theta$, and for given values of d, h there is thus a minimum value of n (viz. nd must be at least $= 2h$); there is also a maximum value of n, viz. this is the least value for which $\frac{1}{2}n(n+3)$ is = or $>$ h, for with such a value of n there is always through the h nodal lines a cone of the order n.

For a given value of d, we have $h =$ at most $\frac{1}{2}(d-1)(d-2)$, and Halphen shows that h must be at least $= [\frac{1}{2}(d-1)^2]$, if we denote in this manner the integral part of the expression within the brackets. And then, h having any value between these limits, for any given values of d, h we have by what precedes a certain number of values of n.

We thus have *primâ facie* curves in space of the several forms (d, h, n): but it may very well be, and in fact Halphen finds that when d is $= 9$ or upwards, then for certain values of h, n as above, there is not any curve (d, h, n): thus $d = 9$, $h = 17$, the values of n are $n = 4$ or 5, but there is not any curve $d = 9$, $h = 17$, for either of these values of n; or say the curves $(9, 17, 4)$ and $(9, 17, 5)$ are non-existent.

And I notice further that in certain cases for which Halphen finds a curve (d, h, n) such curve does not exist except for special configurations of the h nodal lines not determined by the mere definition of n as the order of the cone of lowest order which passes through the h nodal lines: for instance $d = 9$, $h = 16$, $n = 4$ for which Halphen gives a curve, I find that it is not enough that the 16 nodal lines are situate on a quartic cone, but that they must be the 16 lines of intersection of two quartic cones.

I remark moreover that Halphen does not carry out the foregoing principle of classification according to the values of (d, h, n): thus $d = 9$, $h = 22$, the values of n are 6 and 5; viz. the 22 nodal lines are in general on a sextic cone but they may be on a quintic cone; the curves $(9, 22, 6)$ and $(9, 22, 5)$ exist each of them, but he gives only the former of the two forms. The form $(9, 22, 6)$ has a capacity 36 (depends upon 36 constants); the form $(9, 22, 5)$ has a capacity 35f (depends upon $35+\frac{1}{2}$ constants, viz. the nodal lines must be the intersection of a quartic and a quintic cone).

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and this being so we have $w = \frac{P}{Q}$ for the equation of the monoid surface, and consequently $U = 0$ and $w = \frac{P}{Q}$ for the equations of the curve, viz. the cone $U = 0$ and the monoid surface of the order $n+1$ meet in the h lines each twice, in the θ lines, and in the curve of the order d; $(n+1)d = 2h + d + \theta$.

Observe here that the cone $Q = 0$ as a cone of the order $n+1$ subjected only to the conditions of passing through the h lines and the θ lines has in general a capacity $= \frac{1}{2}(n+1)(n+4) - h - \theta$; this number should be $= 3$ at least, for if it were $= 2$, we should have $Q = (x + \lambda y + \mu z)P$ (since $P = 0$ is a cone of the next inferior order through the same $h + \theta$ lines), and thus the curve would be a plane curve.

Observe further that the cone $U = 0$, quâ cone of the order d with h nodal lines has in general a capacity $= \frac{1}{2}d(d+3) - h$; the cone $P = 0$, by what precedes may be regarded as determinate, and the cone $Q = 0$ as just appearing has in general a capacity $= \frac{1}{2}(n+1)(n+4) - h - \theta$; there is a term $+1$ for the implicit constant factor in the function Q, and we thus find for the capacity of the curve the expression

$$\tfrac{1}{2}d(d+3) - h + 1 + \tfrac{1}{2}(n+1)(n+4) - h - \theta,$$

viz. this is

$$= \tfrac{1}{2}d(d+3) + \tfrac{1}{2}(n^2 + 5n) + 3 - nd = \tfrac{1}{2}(d-n)^2 + \tfrac{1}{2}(3d + 5n) + 3,$$

which putting for a moment $d - n = \alpha$ is $= \frac{1}{2}\alpha^2 + \frac{1}{2}(8d - 5\alpha) + 3 = 4d + \frac{1}{2}(\alpha - 2)(\alpha - 3)$; hence restoring for α its value, we find

$$\text{capacity of curve} = 4d + \tfrac{1}{2}(d - 2 - n)(d - 3 - n):$$

in particular if $n = d - 2$ or $d - 3$, the capacity is $= 4d$.

We are thus able in the case where $\frac{1}{2}(n+1)(n+4) - h - \theta = 2$ or more, say $\frac{1}{2}n(n+5) =$ or $> h + \theta + 1$, actually to construct the equation of a curve (d, h, n), having in the case where $n = d - 2$ or $d - 3$ a capacity $= 4d$: the conditions in question for any given value of d, are satisfied by the considerable number of curves which form Halphen's "premier groupe."

For instance $d = 9$, then the complete table of the values of h, n, θ is

d	h	n	θ	Cap.
9	0	0	0	38
9	16	4	4	36
9	17	4	5	0
9	17	5	13	0
9	18	4	0	36
9	18	5	11	36
9	19	5	7	36
9	20	5	5	36
9	21	5	3	36
9	21	6	12	36
9	22	5	1	36
9	22	6	10	35
9	23	6	8	36
9	24	6	6	36
9	25	6	4	36
9	26	6	2	36
9	27	6	0	36
9	28	7	7	36

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and the conditions are satisfied for those values of (d, h, n) against which I have set the capacity $36\frac{1}{2}$. I do not explain the remaining figures of the column of capacities, but remark only that 0 means that the curve is non-existent, and that 35 refers to the curve $(9, 22, 5)$ which is alluded to above as not specified by Halphen.

It is important to remark that if the above-mentioned condition $\frac{1}{2}n(n+5) =$ or $> h+\theta+1$, or restoring it to the original form $\frac{1}{2}(n+1)(n+4)-h-\theta = 3$ at least, is not satisfied, then it by no means follows, and it is not in general the case, that the curve is non-existent: I have said only that the cone $Q = 0$ has in general a capacity $= \frac{1}{2}(n+1)(n+4) - h - \theta$, but the configuration of the $h + \theta$ lines may be such as not to impose on the cone $Q = 0$ which passes through them so many as $h + \theta$ conditions, and the capacity of the cone may thus be greater than $\frac{1}{2}(n+1)(n+4) - h - \theta$, and may thus be $= 3$ at least; moreover supposing that in such a case the curve exists, the capacity of the cone $U = 0$ instead of being $= \frac{1}{2}d(d+3) - h$, may very well have, and presumably has, a greater value, and the reasoning by which the capacity of the curve was found to be $= 4d + \frac{1}{2}(d-2-n)(d-3-n)$ ceases to be applicable. The theory, as depending upon special configurations of the h lines and the θ lines, is a complicated and difficult one, and I do not attempt to enter upon it.

In conclusion I wish to refer to an important theorem given by Valentiner and also by Halphen and Nother. Considering in connexion with the curve of the order d, a surface of the order m, then since the capacity hereof (or number of constants contained in its equation) is $= \frac{1}{2}(m+1)(m+2)(m+3) - 1$ or $\frac{1}{6}m(m^2+6m+11)$, it is obvious that if this be greater than md, the surface can be made to pass through more than md points of the curve, and thus that the curve will lie upon a surface of the order m. But the condition which has really to be satisfied in order that the curve may lie upon a surface of the order m is a less stringent one: if p be the deficiency of the curve, $= \frac{1}{2}(d-1)(d-2) - h$, if as before the curve is without actual singularities, and h be the number of its apparent double points, then the condition is $\frac{1}{6}m(m^2+6m+11)$ greater than $md-p$, viz. the surface of the order m being made to pass through $md+1-p$ points assumed at pleasure on the curve will *ipso facto* pass through p determinate points of the curve, that is in all through $md+1$ points of the curve, or it will contain the curve. The theorem is true subject only to the limitation $m =$ or $> d-2$. The most simple form of statement is perhaps that given by Valentiner, p. 194 (changing only his letters), viz. if m be $=$ or $> d-2$, the intersections of a surface of the order m with a curve of the order d with h apparent double points are determined by means of

$$dm - \tfrac{1}{2}(d-1)(d-2) + h \ (= dm - p)$$

of these intersections.

312. The generalisation which inhere given of Euler's theorem $S+F=E+2$, is a first step towards the theory developed in Listing's Memoir "Census räumlicher Complexe oder Verallgemeinerung des Euler'schen Satzes von den Polyedern." Göttingen Abh. t. X. (1862).

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320. The transcendent $i\,\mathrm{gd}\,(-iu)$, with a pure imaginary argument is the function $\log\tan(\frac{1}{4}\pi+\frac{1}{2}u)$ (hyperbolic logarithm) tabulated by Legendre, *Exer. de Calcul Integral*, t. II. (1816), Table IV. and *Traité des Fonctions Elliptiques*, t. II. (1826), Table IV. at intervals of 30′ from 0° to 90°, to twelve decimals and fifth differences. But the march of the function is somewhat disguised by the argument being taken in degrees and minutes and the function in abstract number. I have in the paper "On the orthomorphosis of the circle into the parabola," *Quart. Math. Jour.* vol. xx. (1885), pp. 213–220, see p. 220, given the table (at intervals of 1° to seven decimals) exhibiting the argument and the function each of them in degrees and minutes and also in abstract number.

335. Besides the 13 numbers mentioned by Gauss it appears by the paper, Perott, "Sur la formation des determinants irréguliers," *Crelle*, t. XCV. (1883), pp. 232–236, that in the first thousand the determinants −468 and −931 are irregular.

341. Consider the equation of a curve as given in the form $y-f(x)=0$; then in the notation of Reciprocants ($t=y'$, $a=\frac{1}{2}y''$, $b=\frac{1}{6}y'''$, $c=\frac{1}{24}y^{\mathrm{iv}}$, $d=\frac{1}{120}y^{\mathrm{v}}$, where the accents denote differentiation in regard to x) the equation of the conic of five-pointic contact at the point (x,y) of the curve is

$$\begin{aligned}&a^2(X-x)^2+a^2b(X-x)[Y-y-t(X-x)]\\&\qquad+(ac-b^2)[Y-y-t(X-x)]^2-a^3[Y-y-t(X-x)]=0,\end{aligned}$$

which I verify as follows: writing $X=x+\theta$, we have

$$Y=y+t\theta+a\theta^2+b\theta^3+c\theta^4+d\theta^5,$$

and thence

$$Y-y-t(X-x)=a\theta^2+b\theta^3+c\theta^4+d\theta^5.$$

Substituting these values and developing as far as θ^5, we find

$$\begin{aligned}&+a^2b\theta(a\theta^2+b\theta^3+c\theta^4)+(ac-b^2)(a^2\theta^4+2ab\theta^5)\\&\qquad\qquad-a^3(a\theta^2+b\theta^3+c\theta^4+d\theta^5)=0,\end{aligned}$$

viz. this is

$$\theta^5+0\cdot\theta^6+0\cdot\theta^7-a(a^2d-3abc+2b^3)\theta^8=0.$$

The equation is thus satisfied as far as θ^4, showing that the conic is a conic of 5-pointic contact; and it will be satisfied as far as θ^5 if only $a(a^2d-3abc+2b^3)=0$.

The value $a = 0$ belongs to an inflexion, and reduces the equation of the conic to $[Y-y-t(X-x)]^2 = 0$, viz. this is the stationary tangent taken twice, which is in an improper sense a conic of six-pointic contact: the other factor determines a sextactic point, viz. we have $a^2d - 3abc + 2b^3 = 0$ as the condition of a sextactic point.

We might from this form, which belongs to the curve as given by the equation $y - f(x) = 0$, pass to the form belonging to the

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equation $U = (x, y, z)^m = 0$, and thus obtain the form given in the memoir, and the process would I can well imagine be a more simple one, but it would certainly be very complicated: as an illustration take the simple case of an inflexion: the condition for this, for the equation $y - f(x) = 0$ of the curve is $a = 0$, that is $\frac{d^2y}{dx^2} = 0$. Passing first to the form $U = (x, y, 1)^m = 0$, we have

$$\frac{\partial U}{\partial x} + \frac{\partial U}{\partial y}\frac{dy}{dx} = 0,$$

and thence

$$\frac{\partial^2 U}{\partial x^2} + 2\frac{\partial^2 U}{\partial x \partial y}\frac{dy}{dx} + \frac{\partial^2 U}{\partial y^2}\left(\frac{dy}{dx}\right)^2 + \frac{\partial U}{\partial y}\frac{d^2y}{dx^2} = 0,$$

viz. substituting for $\frac{dy}{dx}$ its value from the first equation the condition $\frac{d^2y}{dx^2} = 0$, becomes

$$\left(\frac{\partial U}{\partial y}\right)^2 \frac{\partial^2 U}{\partial x^2} - 2\frac{\partial U}{\partial x}\frac{\partial U}{\partial y}\frac{\partial^2 U}{\partial x \partial y} + \left(\frac{\partial U}{\partial x}\right)^2 \frac{\partial^2 U}{\partial y^2} = 0,$$

and we can then make the further transformation to the form $U = (x, y, z)^m = 0$, and so obtain but not very easily the result $H(U) = 0$: but in the transformations for the sextactic point, besides the differential coefficient a of the second order we have the coefficients b, c, d of the orders 3, 4 and 5 respectively; and the complication is thus very much greater.

343, 354, 374. The principal paper is 374; 354 is a mere résumé of this; and 343 relates to the higher singularity which first presented itself, and which is there shown to arise from the coalescence of a node and a cusp, but in 374 (where it is considered more fully) it is shown to be equivalent to a node, a cusp, a double tangent and an inflexion.

On the general subject, and founded on 374, we have

Smith, H. J. S., "On the Higher Singularities of Plane Curves," *Proc. Lond. Math. Soc.* vol. vi. (1875), pp. 153–182. The author refers to the two following enquiries:

(1) It is important to prove that the indices of singularity as defined by Professor Cayley satisfy the equations of Plúcker; and that the "genus" or "deficiency" of the plane curve is correctly given by these indices.

(2) It is also of interest to examine whether any given singularity can be actually formed by the coalescence of the ordinary singularities to which it is regarded as equivalent: in other words whether a singularity of which the indices are δ, τ, κ, ι and which is therefore regarded as equivalent to δ double points, τ double tangents, κ cusps and ι inflexions possesses a penultimate form in which all these singularities exist distinct from one another but infinitely close together.

The paper relates chiefly to the first of these enquiries, the second being reserved for a further communication which was never made.

See also Halphen's "Étude sur les points singuliers des courbes algébriques planes," published as an Appendix, pp. 537–648, to the translation of Salmon's Higher

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Plane Curves, "Traité de Géométrie Analytique," par G. Salmon traduit par O. Chemin, 8vo. Paris, 1884, and the list of Memoirs given, p. 538.

347. I attach some importance to this short paper as giving my own general views of the subject to which it relates, and in particular as to the line of separation between finite and transcendental analysis.

378. I have printed this Report as it was in some measure in connexion therewith that the Royal Society of London undertook the very important work, their Catalogue of Scientific Papers. I do not remember by whom the Report was drafted but some of the recommendations contained in it are due to me. The Catalogue is on a more extensive plan than that recommended in the Report, inasmuch as it is not limited to Physics and Mathematics but extends to all branches of Natural Knowledge—but it is interesting to compare the extent of it with the estimate in the Report—vols. I. to VI. (1800 to 1863) contain together 5743 pages: vols. VII. and VIII. (1864 to 1873) contain together 2357 pages—the number of entries on a page is about = 30: and we thus have, 1800 to 1863, about 173,000 entries, and 1864 to 1873, about 71,000 entries.

END OF VOL. V.

CAMBRIDGE:
PRINTED BY C. J. CLAY, M.A. AND SONS,
AT THE UNIVERSITY PRESS.